Doctoral thesis in Computational Linguistics and Computer Science

A Monologue on Conversation

Characterising Dialogue across Interaction Types with a Focus on Topical Structure

Amandine (Bob) Decker

December 2026

Department of Philosophy, Linguistics and Theory of Science
Centre for Linguistic Theory and Studies in Probability (CLASP)
Sweden

Laboratoire Lorrain de Recherche en Informatique et ses Applications (LORIA)
Equipe Sémagramme
France



UNIVERSITY OF GOTHENBURG





Doctoral Thesis in Computational Linguistics,
University of Gothenburg, Gothenburg, Sweden, 2026



Thèse de Doctorat mention Informatique,
Université de Lorraine, Nancy, France, 2026

A Monologue on Conversation: Characterising Dialogue across Interaction Types with a Focus on Topical Structure

© Amandine Decker, 2026

ISBN 978-91-6863-026-7 (PRINT)
ISBN 978-91-6863-027-4 (PDF)

The research reported in this thesis was supported in part by a doctoral studentship from the University of Lorraine in France, and in part by a grant from the Swedish Research Council (VR project 2014-39) for the establishment of the Centre for Linguistic Theory and Studies in Probability (CLASP) at the University of Gothenburg.

Cover design: Amandine Decker
Printed by Stema Specialtryck AB, Borås, Sweden, 2026
Publisher: University of Gothenburg (Dissertations)

Distribution: Department of Philosophy, Linguistics and Theory of Science
University of Gothenburg, Box 100, SE-405 30, Gothenburg, Sweden

An ocean of knowledge is apt to drown you long before it educates you.

– Mark Lawrence, The Book That Wouldn't Burn (2023)

Acknowledgements

Lorsqu'on m'a expliqué en quoi consistait le travail de thèse, on m'a parlé d'une aventure solitaire, quelques années passées à analyser un sujet qui n'intéresserait que mes encadrants et moi. Et s'il est vrai que j'ai pu, par moments, me sentir isolée au beau milieu d'un océan de connaissances sans plan de route bien défini, j'ai surtout eu la chance d'avoir été entourée par de nombreuses personnes qui m'ont toutes à leur façon empêchées de me noyer.

First and foremost, my thesis supervisors, whom I can never thank enough for supporting me throughout this journey; for helping me to stay (most of the time) on course in this sometimes rather foggy ocean; for making our meetings scientifically enriching whilst leaving room for all sorts of random topics, but also for creating a safe space where I could share my doubts; in short, for simply allowing me to flourish and grow. Ellen, you welcomed me with (literally) opened arms when I first stepped into Sweden now almost five years ago. I will remember your warmth, our meetings that let me discover nice coffee shops in Gothenburg, and all these SemDial trips where you helped me become part of an amazing community. Thank you for your constant encouragement and support, for your guidance and insighful comments that gave this thesis its linguistic perspective, and for believing in me and my work even when I would not grant myself the same trust. Maxime, je n'aurais certainement pas fait cette thèse sans toi et tes discours *no nonsense*. Merci de m'avoir empêchée de gâcher bien trop de mes quelques 20 000 jours à me demander si la thèse était bien faite pour moi, et de m'avoir à la place incitée à me poser des questions bien plus intéressantes. Merci pour nos échanges, pour nos débats bizarrement encore plus animés lorsqu'on était d'accord, pour ta patience face à mes angoisses et pendant les moments où j'avais juste besoin que l'on m'écoute me plaindre. Merci surtout pour ta bonne humeur, même dans le vent et la tempête. You have both taught me a great deal, scientifically and personally. Thank you for you trust and for letting me put so much of myself in this thesis. Bringing our worlds together has not always been plain sailing but I am proud of the work we have done.

Then I would like to extend my gratitude to Matt Purver for agreeing to be the opponent for my final seminar. Thank you for your kindness and your spot-on comments. You brought a very welcome clarity to the final sprint of this work and helped make it a coherent whole. Hopefully you will agree with this.

Je voudrais aussi remercier Serena Villata et Frédéric Béchet pour avoir accepté de rapporter cette thèse. My thanks also go to the examiners: Armelle Brun, Lucia Donatelli, Sharid Loáiciga, and David Traum. Thank you all for agreeing to par-

ticipate in this half-French half-Swedish defence and for granting me some of your time and expertise.

Je remercie les membres de mon comité de suivi, Bruno Guillaume et Mathieu D'Aquin, pour leurs retours bienveillants chaque année.

I have also been fortunate enough to be supported by *two* research groups, which means there are many people to whom I feel grateful.

Merci à l'équipe Sémagramme dans toutes ses versions depuis 2021, où quelques mois passés en stage avec vous m'ont donné un aperçu de qui vous étiez et des valeurs que vous souteniez. Beaucoup de choses ont changé en cinq ans mais l'esprit de l'équipe est toujours le même. Merci aux habitant•es du B230 – Amandine (grâce à qui j'ai gagné un deuxième prénom) pour ton soutien, ta bonne humeur et tes cocktails toujours très légers ; Bertrand, pour tes histoires et instruments de musique insolites ; Gabriel (tu réussis encore à t'incruster dans la partie Sémagramme de l'histoire) pour avoir lancé toutes sortes de projets de *team building* dans une autre équipe que la tienne, et pour avoir engendré le dialogue de la boule de bowling pliable ; Hee-Soo, malgré ta retraite stratégique vers le frigo, pour ta force de caractère inspirante, l'organisation des *Secret Santa* et la mise en place de traditions dans l'équipe qui la rendent si spéciale ; Marie, pour avoir partagé un bureau pendant un temps mais surtout un appartement (et ton chat !) tout au long de cette aventure, je pense qu'on peut parler de sérendipité ; Maximax, pour être allé combattre des guivres avec moi et pour m'avoir trainée dans des fractales alors que ma seule contribution était d'avoir besoin d'une résurrection par minute ; Valentin, pour avoir instauré un peu de sérieux dans ce bureau et lancé tellement de soirées jeux de société ; Vincent, pour ton investissement à créer du lien dans l'équipe mais aussi dans le labo, ton aide précieuse sur des projets bien trop chronophages, tes invitations pas du tout piégeuses à aller crapahuter sur des chemins toujours indiqués sur des cartes, cette thèse aurait été très différente sans toi (et ne parlerait pas de boule de bowling, pliable ou pas) ; et toustes les autres de passage pour plus ou moins longtemps – ce n'était peut-être pas l'environnement le plus propice à la concentration mais la joie de vivre ne manquait jamais. J'aime croire que l'esprit de nos rires résonne encore dans le couloir. Merci aux quelques personnes avec qui j'ai partagé le B103 pour une phase plus studieuse mais toujours bien entourée : Bertrand encore, pour avoir entretenu la phase Super Auto Pets ; Hee-Soo à nouveau, pour ton soutien dans ces derniers mois de rush final même après ton départ ; Rémi, pour m'avoir laissée exploiter ton cerveau et demi avec mes questions techniques ; et enfin Iglika et Vénéra qui sont venues former un bureau complètement féminin pour les derniers jours de rédaction. Merci Clémentine, Elio, Rémi et Ramy pour avoir essayé de me faire prendre un peu l'air ces derniers mois en organisant des sorties, repas et pauses café sur la terrasse. Merci Fanny pour ton soutien et ton retour d'expérience sur la fin de thèse. Merci à la team LREC 2026 pour avoir fait de ce voyage un moment inoubliable, je ne verrai plus les clés USB de la même manière. Merci à mes grandes sœurs de thèse, Maria, Lisa, Siyana,

pour l'inspiration que vous m'avez donnée. Merci à la guilde B-deucentrente pour avoir accepté ma tyranie et merci à Quentin pour nos explorations de Frigost, grâce à qui j'ai pu accéder aux endroits les plus reculés du monde des Douze pour aller couper du tremble. Merci aux teams soirée Feinte de l'ours, dimanche jeu de société, pause jeu du midi, et j'en oublie certainement d'autres; merci à celles et ceux qui ont contribué à faire grandir ma collection de dragons ; merci d'avoir été bien plus que des collègues et d'être devenu•es des ami•es. Mon seul regret est de n'avoir jamais essayé de créer un corpus de nos conversations, la communauté scientifique passe certainement à côté de grandes choses. Merci aussi aux permanents de l'équipe Sémagramme pour avoir contribué à créer cet environnement de travail agréable et pour donner de votre temps pour nous expliquer les dessous du monde académique ; merci en particulier Karën pour tout ça et pour avoir partagé ta vision de l'enseignement et de la recherche, et de la société en général. J'ai appris beaucoup à vos côtés et j'espère avoir pu vous rendre au moins une partie de ce que vous m'avez donné.

Etre membre de l'équipe Sémagramme m'a aussi permis de rencontrer des personnes formidables au sein du laboratoire. Merci Marie Baron pour ton aide lors de la préparation de l'expérience Montgolgram et pour toujours soutenir nos initiatives farfelues. Merci Isabelle pour tes bons desserts, pour les sorties marche qui ré-oxygènent le cerveau, pour avoir pris le temps de lire la partie française de ce travail et de l'avoir fait avec tant d'enthousiasme, et merci surtout pour ton soutien et ta bonne humeur chaque jour. Merci à la team vacances en montagne grâce à qui j'ai pu relire ma thèse dans un cadre paisible et loin du labo (mais surtout pour m'avoir permis pour une fois de me sentir parmi les personnes les plus saines d'esprit d'un groupe). Egor, je suis heureuse qu'on ait eu l'occasion de partager les péripéties de la fin de thèse.

Then I would like to thank CLASP, and FLoV in general, for welcoming me a few months per year, even though I visited only when the sun was shining. My thanks go to the CLASP steering group for agreeing to this co-supervision. Many thanks to the PhD students but also permanent members of FLoV for creating a lively environment and including me in your plans despite my tourist-like status: Asad for organising amazing barbecues and for making me discover the coriander gene; Eleni, for sharing your vast knowledge of dialogue and using it to be the devil's advocate, but mostly for your appartment furnishing saga; Chris for your advice on dialogue collection experiments; Nikolai, for your engagement in encouraging socialising at FLoV; Staffan, for always being there to answer my dialogue system and TTR questions; Robin, for sharing your office with me and for your feedback on my work. I am also grateful to Alva and Sandra for heping to sort out the administrative requirements of my co-supervision.

Then I would like to thank the people who welcomed me in their home during my time in Gothenburg. Jimmy, thank you for your hospitality, for letting me discover the city and Stockholm, for our movie nights and yearly afternoon

spent watching the semi-marathon. Merci Vera pour ton accueil chaleureux et pour m'avoir inclue dans ta vie de famille toutes ces années.

J'ai aussi eu la chance de rencontrer des personnes formidables en participant à Chiche et MATh.en.JEANS. Merci Isabelle pour nous avoir accueillis chez toi pour que l'on vienne expliquer à tes lycéen•nes comment faire des crèpes. Merci Violaine et Salima pour nous avoir fait confiance à Marie et moi pour nos deux années d'atelier MeJ. Merci Salima pour ton soutien même après cette aventure.

Enfin, rien de tout cela n'aurait été possible sans le soutien de ma famille et de mes ami•es. Merci à mes parents pour m'avoir aidée à aller au bout de ce projet, même si vous ne comprenez pas toujours ce que je fais. Merci maman d'avoir aidé à relire cette thèse et de m'avoir soutenue toutes ces années.

Aram, you were one of my favourite reasons to visit Gothenburg – your cats were perhaps the best one. Thank you and Mohammadreza for your hospitality, for feeding me amazing food in equaly amazing quantities, for inviting me to lunch after you had moved 'upstairs' and for introducing me to your colleagues whom I thank for including me in your group whenever I visited. Tristan, I hope we get to play this Twilight Imperium game one day!

Safa, you will always have a special place in my heart. Some of the content of this thesis began to take shape during our interactions in the master's program, and I am grateful for all the time we shared back then.

Merci Yasmine pour ta présence depuis 12 ans maintenant malgré nos emploi du temps jamais alignés. Je crois que tu es condamnée à me supporter pour quelques décénies supplémentaires si tu espères un jour compléter ta collection One Piece.

Merci à toustes cells et ceux que je n'ai pas cité•es ici mais qui ont contribué à rendre cette thèse possible, que ce soit par leur soutien moral, leur aide technique ou simplement par leur présence. Merci, merci, merci.

Abstract

Dialogue constitutes a fundamental mode of human interaction and has become increasingly central in natural language processing with the growing use of conversational AI systems both in research and in practical applications. However, dialogue is a heterogeneous phenomenon encompassing a wide range of forms, arising from differences in participant configurations, interaction contexts, or communication media. This diversity creates challenges for the study of dialogue data, particularly in selecting suitable resources, defining the scope and limitations of our studies, and establishing meaningful evaluation methodologies. A principled characterisation of dialogue is therefore necessary to account for its variations and to support the design of robust computational approaches.

Among the phenomena that cut across different forms of dialogue, topical organisation plays a central role. The structure and evolution of topics shape conversational dynamics and are themselves influenced by the type of interaction being considered. Studying such phenomena requires moving beyond local conversational patterns and adopting a broader, macro-level perspective on dialogue, allowing different interacting processes and structures to be examined jointly.

This thesis investigates three main research questions: (1) how dialogue can be characterised in a way that accounts for its variability while informing the selection of suitable resources and evaluation methods; (2) whether topical structure can be automatically extracted from dialogue data, with a particular focus on open-ended multi-party conversations, that are more complex but also very common in real-world interactions; and (3) whether topical structure can be leveraged as a discriminative signal for differentiating dialogue family types. Through these questions, the thesis aims to contribute both analytical and methodological perspectives toward a broader understanding of dialogue structure and variation.

We answer these questions by proposing a principled characterisation of dialogue, CHIMERA – standing for **CH**ARACTERISATION OF D**I**ALOGUE THROUGH **M**ULTI-DIMENSIONAL **E**XAMINATION, **R**EPRESENTATION AND **A**NALYSIS – and by analysing the topical structure of dialogues across different interaction types, showing how CHIMERA can help determine the suitability of the resources employed and the scope of the analyses.

Sammanfattning

Dialog utgör en grundläggande form av mänsklig interaktion och kunskap om dialog har blivit mer och mer central inom språkteknologi i och med att användningen av konversationella AI-system ökat både inom forskningen och i samhället i stort. Trots att dialog är en så grundläggande företeelse förekommer den i många olika varianter. Variationen beror på skillnader vad gäller exempelvis samtalsdeltagare, samtalskontext och medium. Denna mångfald medför utmaningar för studiet av dialogdata, särskilt när det gäller att välja lämpliga resurser, definiera omfattningen och begränsningarna för studier samt fastställa meningsfulla utvärderingsmetoder. En principiellt grundad beskrivning av dialog är därför nödvändig för att ta hänsyn till dess variationer och för att stödja utformningen av robusta beräkningsmetoder.

En av de dimensioner som är relevanta för de flesta typer av dialog är samtalsämne, eller topik, som ofta spelar en central roll för hur en dialog ser ut. Hur topiker struktureras och utvecklas formar dialogen men påverkas också av vilken typ av interaktion det handlar om. Att studera sådana företeelser kräver att man rör sig bortom lokala konversationsmönster och antar ett bredare makroperspektiv på dialog som tillåter att flera samverkande processer och strukturer kan undersökas tillsammans.

Denna avhandling undersöker tre huvudsakliga forskningsfrågor: hur dialog kan beskrivas på ett sätt som tar hänsyn till dess mångfacetterade natur, men samtidigt kan ligga till grund för val av lämpliga resurser och utvärderingsmetoder, huruvida topikstruktur kan extraheras automatiskt från dialogdata med särskilt fokus på flerpartsdialog utan tydligt syfte, vilka ofta är strukturellt komplex men mycket vanligt förekommande i vardaglig interaktion, och huruvida topikstruktur kan användas för att särskilja olika typer av dialog.

Med hjälp av dessa frågor söker avhandlingen bidra med analytiska och metodologiska perspektiv som ökar förståelsen för struktur och variation hos dialog. Vi besvarar frågorna genom att föreslå ett system för att beskriva dialog – CHIMERA – en förkortning av engelskans **CH**ARACTERISATION OF D**I**ALOGUE THROUGH **M**ULTIDIMENSIONAL **E**XAMINATION, **R**EPRESENTATION AND **A**NALYSIS (Karakterisering av dialog genom multidimensionell undersökning, representation och analys) – och genom att analysera topikstrukturer i flera olika interaktionstyper. Dessa analyser visar hur CHIMERA kan hjälpa forskare att avgöra vilka resurser som bör användas för olilka typer av dialogstudier och hur långtgående slutsatser man kan dra av analyserna.

Résumé Court

Le dialogue constitue un mode fondamental d'interaction humaine et occupe une place de plus en plus centrale dans le traitement automatique des langues, avec l'utilisation croissante des systèmes d'IA conversationnelle tant dans la recherche que dans les applications pratiques. Cependant, c'est aussi un phénomène hétérogène qui englobe un large éventail de formes, découlant des différences dans la configuration des participants, le contexte d'interaction ou le support de communication. Cette diversité pose des défis pour l'étude des données dialogiques, notamment en ce qui concerne la sélection de ressources adaptées, la définition de la portée et des limites de nos études, ainsi que la mise en place de méthodologies d'évaluation pertinentes. Une caractérisation fondée sur des principes du dialogue est donc nécessaire pour rendre compte de ses variations et pour soutenir la conception d'approches computationnelles robustes.

Parmi les phénomènes qui transcendent les différentes formes de dialogues, l'organisation thématique joue un rôle central. La structure et l'évolution des thèmes façonnent la dynamique conversationnelle et sont elles-mêmes influencées par le type d'interaction considéré. L'étude de ces phénomènes nécessite d'aller au-delà des schémas conversationnels locaux et d'adopter une perspective plus large, à un niveau macro, permettant ainsi d'examiner conjointement les différents processus et structures en interaction.

Cette thèse explore trois questions de recherche principales : (1) comment caractériser le dialogue de manière à rendre compte de sa variabilité tout en orientant le choix des ressources et des méthodes d'évaluation appropriées ; (2) si la structure thématique peut être extraite automatiquement à partir de données de dialogue, en mettant particulièrement l'accent sur les conversations ouvertes à plusieurs participants, qui sont plus complexes mais également très courantes dans les interactions du monde réel ; et (3) si la structure thématique peut être exploitée comme un signal discriminant pour différencier des types de dialogues. À travers ces questions, cette thèse vise à apporter des perspectives tant analytiques que méthodologiques en vue d'une compréhension plus large de la structure et de la variation du dialogue.

Nous répondons à ces questions en proposant une caractérisation multidimensionnelle du dialogue, CHIMERA – qui signifie **CH**ARACTERISATION OF D**I**ALOGUE THROUGH **M**ULTIDIMENSIONAL **E**XAMINATION, **R**EPRESENTATION AND **A**NALYSIS – et en analysant la structure thématique dans le dialogue à travers différents types d'interaction, montrant comment CHIMERA peut aider à déterminer la pertinence des ressources dialogiques employées et la portée des analyses.

Résumé Long

Les exemples de dialogues utilisés dans ce chapitre, sauf le (2), ont été traduits de l'anglais et sont disponibles dans leur version originale dans le chapitre d'introduction du manuscrit (Introduction).

(1) $\mathbf{A}_1$ J'adore les lapins ! J'aimerais bien ouvrir une crèche ou un hôtel pour lapins.

$\mathbf{B}_2$ C'est une super idée. Comment vas-tu aujourd'hui ?

$\mathbf{A}_3$ Un peu fatigué. J'ai travaillé toute la journée au chantier naval.

$\mathbf{B}_4$ Je viens de rentrer de l'école, mais je n'aime pas trop ça.

$\mathbf{A}_5$ Tu devrais t'engager dans l'armée. Je suis un ancien militaire. Les filles adorent les uniformes ;

$\mathbf{B}_6$ Je n'y avais jamais pensé. Ça m'aiderait peut-être à me faire plus d'amis.

$\mathbf{A}_7$ Je me suis engagé dans la marine parce que j'aime le bleu marine, mais l'infanterie a aussi des uniformes sexy.

$\mathbf{B}_8$ J'y penserai si je m'engage.

$\mathbf{A}_9$ Je donne une partie de mon salaire à mon refuge pour lapins. Tu devrais le faire aussi.

$\mathbf{B}_{10}$ Tu as des lapins ?

$\mathbf{A}_{11}$ Plus maintenant. Je les ai tous mangés.

$\mathbf{B}_{12}$ Waouh ! Tu as un sacré appétit.

$\mathbf{A}_{13}$ Oui, j'aime bien manger. Et toi, tu fais quoi dans la vie ?

$\mathbf{B}_{14}$ Je suis réceptionniste dans un cabinet dentaire

[PersonaChat (S. Zhang et al., 2018)]

(1) est une conversation entre deux personnes à qui l'on a demandé d'interagir en incarnant un personnage donné, qui leur a été décrit par une liste de phrases telles que « Je possédais des lapins, mais je les ai mangés. ». Cet exemple illustre certains des défis qui se posent lors de l'étude du dialogue. Afin d'étudier un phénomène spécifique, nous essayons souvent de collecter des données qui le mettent en valeur. Par conséquent, les conversations que nous étudions peuvent ne pas être représentatives des dialogues qui se produisent naturellement. De plus, faire interagir des personnes avec trop de contraintes peut entraîner des comportements peu naturels qui aggravent ce manque de représentativité. Dans cet exemple, le fait de demander aux participants d'incarner un personnage les a amenés à construire toute

leur interaction autour de celui-ci, alors que nous n'imprégnons généralement pas autant nos conversations de notre personnalité et de nos centres d'intérêt.

Pourtant, le langage naturel constitue la ressource fondamentale de la linguistique computationnelle et du traitement automatique des langues (TAL), les domaines principaux dans lesquels s'inscrit cette thèse. La linguistique computationnelle est un domaine interdisciplinaire qui utilise des outils informatiques pour analyser et modéliser les données linguistiques. Le TAL est un domaine plus appliqué qui se concentre sur le développement de programmes informatiques capables de traiter et de générer du langage naturel. Ces deux disciplines s'appuient sur des données pour étudier et modéliser la langue, et actuellement, le TAL utilise des volumes de données de plus en plus importants pour développer, dans un premier temps, des modèles d'apprentissage automatique et d'apprentissage profond, puis des grands modèles de langue (LLM), capables de résoudre un large éventail de tâches liées à la compréhension et à la génération du langage naturel, bien qu'ils présentent des lacunes en matière d'explicabilité et d'interprétabilité. Les LLM ont été utilisés pour créer des agents conversationnels capables d'interagir avec les humains à travers des dialogues. Ces nouvelles technologies ont donné lieu à une vague de recherches en TAL visant à améliorer les performances des LLM dans les tâches de dialogue sous différents angles, l'un d'entre eux consistant à faire incarner un personnage au modèle afin de le rendre plus personnalisé. (1) est un exemple tiré du corpus PersonaChat (S. Zhang et al., 2018), qui a été construit pour entraîner des modèles qui seraient « plus cohérents et captivants » grâce à leur « intégration d'une personnalité ». De telles pratiques remettent en question ce qu'*est* une conversation. Bien que (1) soit une conversation entre deux êtres humains, elle ne semble pas très naturelle. À l'inverse, de nombreux échanges avec des agents conversationnels ont désormais lieu chaque jour, et il semble également problématique de les rejeter en affirmant qu'il ne s'agit pas de *vraies* conversations. Tout cela renforce la nécessité de mieux comprendre ce qu'est une conversation et selon quels critères elle peut être caractérisée.

Cette question revêt une importance particulière pour nous, car l'une des motivations à l'origine de cette thèse est l'analyse des conversations pathologiques, c'est-à-dire des conversations impliquant un patient dont la pathologie affecte sa capacité à communiquer. L'un des phénomènes que nous avons observés lors d'interactions réelles avec des patients schizophrènes était lié à une manière inhabituelle de changer de sujet, sans aucun signal explicite. Dans le dialogue (2), le patient (A) modifie l'interprétation du concept de mort, utilisé au sens figuré dans les premiers tours de parole (124 et 125), avant qu'il ne prenne un sens littéral au tour 126. L'absence de signaux explicites rend difficile pour le psychologue (B) d'identifier ce changement de sujet, ce qui l'oblige à demander une confirmation.

(2) A_{124} Oh ouais (↑) et pis compliqué (↓) et c'est vraiment très très compliqué (→) la politique c'est quelque chose quand on s'en occupe faut être gagnant parce qu'autrement quand on est perdant c'est fini quoi (↓)

B_{125} Oui

A_{126} J. C. D. est mort, L. est mort, P. est mort euh (...)

B_{127} Ils sont morts parce qu'ils ont perdu à votre avis (↑)

A_{128} Non ils gagnaient mais si ils sont morts, c'est la maladie quoi c'est c'est (→)

[SLAM Data]

Pour déterminer si ce type de changement de sujet peut être considéré comme un indice permettant d'identifier une pathologie ou s'il s'agit d'un phénomène normal dans la conversation, il est nécessaire de disposer d'une théorie générale de l'organisation des thèmes pour la conversation. Mais pour cela, il est nécessaire de pouvoir caractériser les différents types de conversations, car cela pourrait influencer la structure thématique. Cette thèse s'articule donc autour de deux axes complémentaires : d'une part, nous visons à définir une caractérisation du dialogue pouvant servir à décrire, comparer et sélectionner des ressources dialogiques, mais aussi à délimiter la portée d'une étude ; et d'autre part, nous nous concentrons sur la structure thématique en tant qu'outil d'analyse transversal. Dans ce résumé, nous commençons par aborder les enjeux de ces deux axes et nos questions de recherche dans la Section 1. Puis nous présentons les contributions de la Partie I qui s'intéresse à la caractérisation du dialogue dans la Section 2, avant de résumer celles de la Partie II qui se focalise sur trois études appliquées de la structure thématique dans la Section 3. Enfin, nous concluons ce résumé par une discussion sur les perspectives de recherche dans la Section 4.

1 Enjeux et questions de recherche

Comme dit précédemment, cette thèse s'intéresse à la notion de conversation et aux différentes formes qu'elle peut prendre dans différents contextes, mais aussi plus particulièrement à la structure thématique qui la sous-tend. Nous présentons dans cette section les enjeux liés à ces deux axes et les questions de recherche qui en découlent. Nous commençons par décrire les difficultés liées à l'étude de la conversation dans la Section 1.1, avant de présenter celles liées à l'étude des thèmes dans la Section 1.2.

1.1 Étude de la conversation

La conversation est notre principal mode de communication. Nous dialoguons au quotidien à des fins sociales, pratiques et professionnelles, souvent plusieurs fois

par jour, parfois dans différentes langues et, de plus en plus, par le biais de divers supports. Si le dialogue semble généralement naturel, il s'agit en réalité d'une expérience unique et sans cesse renouvelée : on peut se sentir tout à fait à l'aise lors d'un échange et, quelques instants plus tard, avoir du mal à entretenir une conversation avec un autre interlocuteur qui semble constamment s'éloigner légèrement du sujet.

Ces variations ne sont pas surprenantes. Le dialogue mobilise simultanément de multiples compétences et exige des participants qu'ils s'ajustent constamment en coopérant les uns avec les autres. La réussite d'une conversation ne dépend pas seulement de leur compétence linguistique, mais aussi de connaissances partagées, de la coordination attentionnelle, de la sensibilité aux indices interactionnels et de la capacité à remédier aux perturbations et malentendus. La complexité qui en résulte fait du dialogue un objet d'étude riche et stimulant, qui a été abordé sous de nombreux angles dans un large éventail de disciplines.

Pourtant, son étude s'avère complexe, car la conversation est un phénomène éphémère dépendant d'un contexte précis. Tenter d'en saisir l'essence implique inévitablement d'en détruire une partie en raison de diverses contraintes. La plus évidente est qu'informer des participants qu'ils sont observés et enregistrés peut modifier leur comportement. C'est ce qu'on appelle le paradoxe de l'observateur (Labov, 1973). De plus, le simple fait d'enregistrer une conversation peut être perturbateur, car la présence du matériel d'enregistrement peut altérer le déroulement naturel de l'interaction. Même lorsque ces problèmes sont atténués, les conversations libres soulèvent des questions supplémentaires liées à la protection de la vie privée. Idéalement, nous pourrions partager toutes les données que nous collectons, car cela permettrait à d'autres chercheurs de reproduire nos études et de s'appuyer sur nos travaux. Cet idéal, connu sous le nom de *science ouverte*, est un mouvement en plein essor au sein de la communauté scientifique qui vise à rendre la recherche plus transparente et ses résultats plus facilement accessibles. Cependant, le partage des données issues de conversations peut s'avérer problématique, car celles-ci peuvent contenir des informations sensibles sur les participants. Les dialogues les plus libres posent un défi particulier à cet égard, car la liberté dont disposent les participants pour aborder n'importe quel sujet peut conduire à la divulgation d'informations personnelles. Amblard et al. (2014) ont montré qu'il est impossible d'anonymiser entièrement de longs échanges tant qu'un contexte suffisamment large est conservé.

Comme il est très difficile de recueillir des conversations spontanées, les études portent souvent sur des personnes à qui l'on a demandé d'avoir une conversation dans un contexte donné. Des tâches nécessitant une interaction ont été conçues afin de lancer la conversation et d'aider les participants à faire abstraction du cadre expérimental, mais aussi pour analyser des phénomènes spécifiques à certaines conditions. (3) est un exemple de dialogue tiré d'un corpus largement utilisé dans la communauté des systèmes de dialogue, le corpus MultiWOZ (Zang et al., 2020).

Dans ce corpus en anglais, les participants sont invités à jouer le rôle d'un client et d'un employé dans un scénario de réservation, par exemple de restaurant ou de voyage. La conversation est donc contrainte par la tâche, ce qui peut conduire à des comportements différents de ceux observés dans des interactions plus libres.

(3) **USR**$_1$ J'ai besoin de réservations de train de Norwich à Cambridge
SYS$_2$ J'ai trouvé 133 trains correspondant à votre demande. Y a-t-il un jour et une heure précis auxquels vous souhaitez voyager ?
USR$_3$ J'aimerais partir lundi et arriver avant 18 h.
SYS$_4$ Il y a 12 trains correspondant au jour et à l'heure que vous avez indiqués. Souhaitez-vous réserver dès maintenant ?
USR$_5$ Avant de réserver, j'aimerais également connaître la durée du trajet, le prix et l'heure de départ, s'il vous plaît.
SYS$_6$ Il y a 12 trains qui répondent à vos critères, le premier partant à 05h16 et le dernier à 16h16. Souhaitez-vous en réserver un ?
USR$_7$ Non arrêtons la réservation pour l'instant. Pouvez-vous m'aider à trouver un endroit appelé « Cineworld Cinema » ?
SYS$_8$ Oui, c'est un cinéma situé dans la partie sud de la ville. Quelles informations souhaitez-vous à son sujet ?
USR$_9$ Oui, c'est tout ce dont j'avais besoin. Merci beaucoup !
SYS$_{10}$ Merci d'utiliser notre système.

[MultiWOZ 2.2]

Dans d'autres cas, les ressources de dialogue sont constituées en demandant à des personnes d'avoir une conversation avec moins de contraintes, comme dans (1). Les dialogues rédigés manuellement sont également utilisés dans de nombreuses études, car ils permettent d'éviter tous les problèmes liés à la collecte de dialogues spontanés. Cependant, ils ont également leur propres problèmes, car ils peuvent paraître artificiels et ne reflètent pas nécessairement la dynamique d'une véritable conversation. (4) et (5) sont deux exemples tirés du corpus DailyDialog (Y. Li et al., 2017), écrit manuellement en anglais, qui est également largement utilisé en TAL, en particulier pour les tâches de reconnaissance des émotions grâce à des annotations d'émotions fournies pour chaque tour de parole. Alors que (4) peut passer pour une conversation relativement naturelle, bien que le schéma de transcription de base ne rende compte d'aucun indice interactionnel, (5) est clairement artificielle.

(4) **A**$_1$ Hé ! Pourquoi as-tu pris cet argent ? Tu es vraiment un tricheur ! Je devrais t'envoyer en prison !
B$_2$ Je ne triche pas. Quand on passe par la case départ, on touche 200 $, tout le monde le sait !

A_3 Eh bien, tu ne peux pas simplement prendre l'argent. Tu dois demander de l'argent à la banque. Et c'est moi le banquier.

B_4 Le banquier ?

A_5 Oui…

B_6 Je peux avoir mes 200 $, s'il te plaît ?

A_7 Bien sûr. Tiens, voilà tes 200 $. Merci, reviens quand tu veux ! C'est à mon tour de lancer les dés maintenant.

[DailyDialog]

(5) A_1 Tu vas à la fête annuelle ? Je peux t'y emmener si tu veux.

B_2 Merci beaucoup. C'est justement le service que j'allais te demander.

A_3 Tout le plaisir est pour moi.

[DailyDialog]

Enfin, l'essor des LLM a favorisé l'utilisation de conversations entièrement synthétiques en TAL. Ces dialogues sont générés par un modèle, souvent dans un contexte *zero-shot*, c'est-à-dire en fournissant simplement une consigne au modèle sans aucun apprentissage supplémentaire. Des méthodes plus sophistiquées ont également été développées pour mettre en évidence des phénomènes spécifiques, comme le fait de faire « interagir » deux modèles, ou d'amener l'un d'eux à produire un « schéma de dialogue », c'est-à-dire une trame de conversation comportant des emplacements à remplir, que l'autre modèle (ou le même dans une deuxième phase) doit compléter afin de produire un dialogue complet.

Toutes ces méthodes et tous ces exemples diffèrent clairement les uns des autres et des interactions en face à face à certains égards. Pourtant, ils sont tous utilisés pour étudier le dialogue et pour construire des modèles destinés à le traiter ou à le générer. La question centrale qui se pose est donc de savoir ce que ces différences impliquent pour l'étude du dialogue et comment nous pouvons les caractériser afin de sélectionner la ressource la plus appropriée pour une étude ou une application donnée. En particulier, la structure thématique semble intuitivement influencée par le type de conversation, les dialogues ayant pour objectif de résoudre une tâche précise étant par exemple plus contraints et ciblés sur des sujets précis, que les dialogues complètement libres. Nous abordons ces enjeux liés à l'étude des thèmes et leur structure dans la prochaine section.

1.2 Difficultés de l'étude des thèmes

Les thèmes ne constituent pas une caractéristique directement observable de l'usage de la langue, mais plutôt une caractéristique sous-jacente. Certains aspects de la conversation sont purement linguistiques, comme le vocabulaire utilisé, tandis que d'autres dépendent davantage de l'interaction et nécessitent des formats de

transcription plus complets pour être capturés, comme les chevauchements entre les tours de parole. Les thèmes sont toutefois plus abstraits et nécessitent une vision plus globale de la conversation pour être identifiés. Ils peuvent être définis à plusieurs niveaux ou degrés de granularité : on peut ainsi classer des conversations entières en fonction de leur thème global, déterminer le sujet de séquences plus ou moins longues d'énoncés, voire identifier le sujet de tours de parole individuels.

Ces multiples niveaux de granularité impliquent qu'il est essentiel de définir clairement l'objectif d'une étude, et donc le type de données à analyser qui en dépend directement. Identifier les changements de sujet dans un bref échange de quelques tours de parole est fondamentalement différent de le faire dans une longue conversation. De même, le fait de travailler avec des données issues de la vie réelle plutôt qu'avec des données synthétiques, ou avec des dialogues structurés plutôt qu'avec des dialogues plus libres, est susceptible de donner des résultats différents. Pourtant, la plupart des études se concentrent sur le discours monologique ou sur des dialogues structurés tels que des transcriptions de réunions. Par exemple, PersonaChat, le corpus dont est tiré (1), a été utilisé pour établir un point de comparaison pour la modélisation de dialogues prenant en compte les changements de sujet (Xie et al., 2021). Les changements de sujet dans (1) sont assez clairs en raison du caractère peu naturel du dialogue, qui découle des contraintes imposées aux participants, mais idéalement, nous souhaitons plutôt nous intéresser à des données naturelles où des structures thématiques bien plus complexes peuvent apparaître.

STAC (Asher et al., 2016) est un corpus de conversations en anglais à plusieurs participants, issues de discussions textuelles entre joueurs du jeu (Les Colons de) Catan. Dans ce jeu, les joueurs incarnent des colons qui construisent chacun leur propre colonie et doivent acquérir et échanger des ressources pour la développer, ce qui donne lieu à de nombreux échanges où, par exemple, l'un demande une ressource et d'autres en proposent une autre en échange. (6) est un extrait de dialogue tiré de ce corpus où différentes sous-conversations ont lieu en même temps et s'entremêlent.

(6) LJ$_{165}$ quelqu'un veut des moutons contre de l'argile ?
GWFS$_{166}$ j'ai pas, désolé :(
GWFS$_{167}$ **alors comment les gens sont-ils au courant de la ligue ?**
WM$_{168}$ non
LJ$_{170}$ **j'ai fait les essias**
LJ$_{173}$ **même les essais**
TK$_{174}$ **j'en ai entendu parler par ma copine**
GWFS$_{175}$ [**ouais, moi aussi,**]$_a$ [*tu es étudiant en informatique alors, lj ?*]$_b$
TK$_{176}$ **je n'ai pas fait les essais**
WM$_{177}$ quelqu'un a du bois pour moi ?
GWFS$_{178}$ [**moi, je les ai faits**]$_a$ [**parce qu'un ami les a faits**]$_b$

GWFS$_{179}$ lol wm, espèce de sld

GWFS$_{180}$ j'ai bien peur que non :(

LJ$_{181}$ [*non, je vais commencer les maths.*]$_a$ [*je traîne juste beaucoup à Appleton*]$_b$

TK$_{182}$ désolé, non

GWFS$_{183}$ mon seul bois est précieux

WM$_{184}$ c'est quoi un sld ?

[STAC (Asher et al., 2016, Ex.1)]

Cet exemple montre que la structure thématique peut être plus complexe qu'une simple séquence linéaire, et que le simple fait d'identifier les changements de sujet n'est pas toujours suffisant. Ici, le nombre important de participants et le mode d'expression écrit permettent aux joueurs de mener plusieurs sous-conversations simultanément. Certains participants parlent de « la ligue » tandis que d'autres échangent sur les ressources — ce qui est typique dans ce jeu — et un participant (GWFS) prend part à toutes ces discussions. Dans ce dialogue, bien qu'il ne soit pas toujours évident de définir le sujet de chaque tour de parole, le vocabulaire permet de démêler assez facilement les fils conducteurs lorsque l'on regarde au-delà des phrases isolées. Mais même cette approche ne constitue pas toujours une stratégie garantie pour interpréter n'importe quelle conversation.

(7) est un extrait de dialogue tiré du British National Corpus et décrit comme une « réunion commerciale d'Air UK ». Bien que le vocabulaire utilisé et cette description nous permettent de comprendre que le thème général porte sur les avions et vols, la conversation est difficile à interpréter et il est difficile d'identifier des sous-thèmes plus spécifiques.

(7) JN7PSUNK$_1$ Bon, tu veux trente places sur ce vol ?

PS4BS$_2$ Non, non, non, c'est impossible <inaudible> je n'en ai aucune idée <rire> oh là là.

PS4BS$_3$ Bref

PS4BP$_4$ Je voulais juste mentionner Stansted pour la suite

PS4BP$_5$ C'est… c'est la même chose que ce qu'on dit : ça réduit sans cesse la zone de chalandise

PS4BS$_6$ C'était quoi avant ?

PS4BP$_7$ Sept zéro cinq

PS4BS$_8$ Ouais, bon, le problème, c'est qu'il faut que quelque chose change parce qu'on a <inaudible> on a tous ces fichus vols qui partent en même temps

PS4BP$_9$ Je sais, mais

PS4BS$_{10}$ Et ils n'arrivent pas à suivre

PS4BP$_{11}$ Chaque fois que tu avances l'horaire comme ça, ces premières heures vont simplement diminuer

PS4BP$_{12}$ C'est justement ce que j'ai dit, c'est presque une demi-heure plus tard

PS4BS$_{13}$ En fait, ça dépend de la météo, non ?

PS4BS$_{14}$ Ça dépend de la

PS4BS$_{15}$ Bon, si tu as des petites remarques comme ça, tu devrais, euh, dire, par exemple, qu'on ne va pas vraiment en avoir pour très longtemps.

PS4BS$_{16}$ Ça nous prendra, euh, cinq à dix minutes environ

PS4BR$_{17}$ Oh, c'est tout ?

PS4BS$_{18}$ Ouais.

PS4BS$_{19}$ Tu peux t'asseoir et attendre si tu veux

PS4BR$_{20}$ Euh, d'accord, merci, tant que je ne te dérange pas ou quoi que ce soit

PS4BS$_{21}$ Ouais, ouais, tu peux t'asseoir et attendre si tu veux, oh non <inaudible> va-t'en et reviens.

Dialogue JN7 [BNC]

Ces exemples montrent que les structures thématiques peuvent être complexes dans les dialogues naturels, et que même l'identification des thèmes peut s'avérer difficile. Par exemple, Gruenstein et al. (2005) ont montré que les annotateurs ont du mal à s'accorder lorsqu'on leur demande d'identifier les changements de sujet dans le corpus ICSI Meeting (Janin et al., 2003), tant à un niveau global qu'à un niveau détaillé. Ce corpus contient des réunions spontanées organisées par l'International Computer Science Institute (ICSI), qui donnent lieu à des conversations libres présentant une structure peu contrainte, bien que l'éventail global des sujets soit quelque peu limité par le lieu d'enregistrement. La qualité élevée du matériel d'enregistrement a permis d'obtenir d'excellentes transcriptions écrites. Par rapport à (7) où certains mots n'ont pas pu être retranscrits et où certains locuteurs sont identifiés comme inconnus, les conversations du corpus ICSI sont probablement légèrement plus faciles à comprendre et restent cependant difficiles à annoter avec leurs thèmes. Néanmoins sur ce même corpus, Galley et al. (2003) ont montré que les annotateurs ne délimitent pas les thèmes principaux *au hasard* dans la plupart des cas. Le test de distribution aléatoire a échoué pour seulement 4 réunions sur 25, et a donné pour deux autres un score de confiance assez faible ($p > 0,1$). Les auteurs n'ont pas non plus fourni de score réel d'accord inter-annotateurs, ce qui rend difficile l'évaluation de la complexité de la tâche.

Les deux principales difficultés pour l'annotation de thèmes, qui existent également lors de l'étude des dialogues du BNC et de nombreux autres corpus, sont que les annotateurs ont tendance à ne pas bien connaître le sujet et qu'ils sont extérieurs à la conversation. Ces difficultés sont probablement en grande partie responsables

du faible degré d'accord entre annotateurs pour l'annotation des changements de sujet. Banerjee et Rudnicky (2007) ont montré que le fait de fournir aux annotateurs une liste de sujets à identifier sur la base de l'ordre du jour de la réunion améliore cet accord, ce qui démontre que la tâche n'est pas absolument impossible.

Néanmoins, sa complexité rend difficile la constitution de corpus de dialogues naturels annotés en thèmes, ce qui constitue un défi pour cette thèse. Nous avons donc dû trouver des moyens indirects d'extraire et d'analyser les structures thématiques, comme nous l'expliquons en détaillant le plan de ce travail dans la Section 1.3.

1.3 Focus et contributions

Nos travaux se situent à la croisée de la linguistique computationnelle et du traitement automatique des langues. Ces deux domaines, bien qu'initialement étroitement liés, ont pris des directions différentes ces dernières années. L'essor de l'apprentissage automatique, de l'apprentissage profond et des grands modèles de langue a conduit le TAL à privilégier la performance, au détriment de l'étude des données et du langage naturel. Dans cette thèse, nous nous ancrons fermement dans une perspective axée sur les données et nous nous efforçons de réconcilier la linguistique computationnelle et le TAL en opérationnalisant les théories de la première de manière à ce qu'elles puissent être utilisées pour le TAL. Nous nous intéressons en particulier à la conversation dans toute sa richesse et explorons les limites des approches actuelles d'analyse de la structure thématique lorsqu'elles sont appliquées à des données complexes. Cette démarche nécessite de disposer d'un moyen de caractériser systématiquement les données dialogiques afin de comparer les approches et de comprendre comment une ressource donnée peut influencer les résultats d'une étude. Cela constitue le premier axe de nos travaux. La nature complexe et multiforme du dialogue implique qu'un cadre de caractérisation doive couvrir de multiples dimensions plus ou moins imbriquées les unes dans les autres. Le deuxième axe de nos travaux se concentre donc sur un niveau d'analyse transversal, la structure thématique, car celle-ci est influencée par de nombreux aspects du dialogue et pourrait donc constituer un signal discriminant permettant de différencier les types de conversations. Dans ce contexte, les aspects qui nous intéressent peuvent être résumés par les questions de recherche suivantes :

QR1 Comment caractériser la notion de dialogue de manière à tenir compte de sa variabilité tout en orientant la définition des spécificités des tâches et le choix des ressources associées ?

QR2 Est-il possible d'extraire automatiquement la structure thématique à partir de données dialogiques, et en particulier à partir de conversations ouvertes impliquant plusieurs interlocuteurs ?

QR3 La structure thématique peut-elle être exploitée comme signal discriminant pour différencier des types de dialogue ?

Ces questions trouvent progressivement leurs réponses au fil des sept chapitres de cette thèse, répartis en deux parties. La première se concentre sur le développement d'une caractérisation du dialogue, que nous considérons comme un outil permettant de décrire, de comparer et de sélectionner des ressources dialogiques, mais aussi de définir le champ d'application d'une étude ou d'une application. Ensemble, les trois chapitres de cette partie dont nous résumons le contenu dans la Section 2 répondent à QR1. La deuxième partie s'intéresse à la structure thématique dans la conversation, un niveau transversal d'analyse et de description. Elle présente trois études appliquées qui explorent la structure thématique, depuis ses marqueurs superficiels jusqu'à ses caractéristiques plus latentes, dans différents types de dialogue, et mettent en évidence la complexité des conversations multipartites à thème ouvert. Nous résumons le contenu de cette partie dans la Section 3, et montrons comment les trois chapitres qui la composent contribuent à répondre à QR2 and QR3. Cette thèse se conclue par un résumé de nos contributions et une discussion sur les limites de nos travaux, résumés dans la Section 4. Nous abordons aussi brièvement l'importance croissante accordée aux LLM au sein de la communauté du TAL et l'impact que cela a sur l'étude du dialogue.

2 Partie I : Caractériser le dialogue

La très grande variabilité qui existe dans le concept de conversation signifie qu'il n'est pas possible d'étudier tous les aspects du dialogue en même temps. De même, il n'existe pas de ressource dialogique unique qui puisse être utilisée pour développer un outil qui soit applicable à tous les types de dialogue. Il est donc nécessaire de caractériser le dialogue, c'est-à-dire de définir un ensemble de dimensions qui permettent d'en décrire les différents aspects, afin de définir le périmètre d'une étude ou d'une application, de comparer les ressources dialogiques et de sélectionner la ressource la plus appropriée pour une tâche donnée. C'est l'objectif de la première partie de cette thèse.

Après avoir étudié la littérature de divers domaines s'intéressant au dialogue, nous avons proposé un cadre de caractérisation du dialogue qui repose sur quatre dimensions principales : les AGENTS, la CONFIGURATION, le CADRE RESTREINT et l'INTERACTION VERBALE. Nous nommons ce cadre de caractérisation CHIMERA pour **CH**ARACTERISATION OF D**I**ALOGUE THROUGH **M**ULTIDIMENSIONAL **E**XAMINATION, **R**EPRESENTATION AND **A**NALYSIS. La Figure 1 résume les dimensions de CHIMERA et leurs paramètres en se plaçant dans l'analogie de D. LEWIS (1979) selon laquelle le dialogue est une partie de baseball. Nous prenons la liberté d'étendre cette analogie en rendant la partie de baseball surnaturelle afin de nous laisser plus de liberté pour illustrer les différentes dimensions.

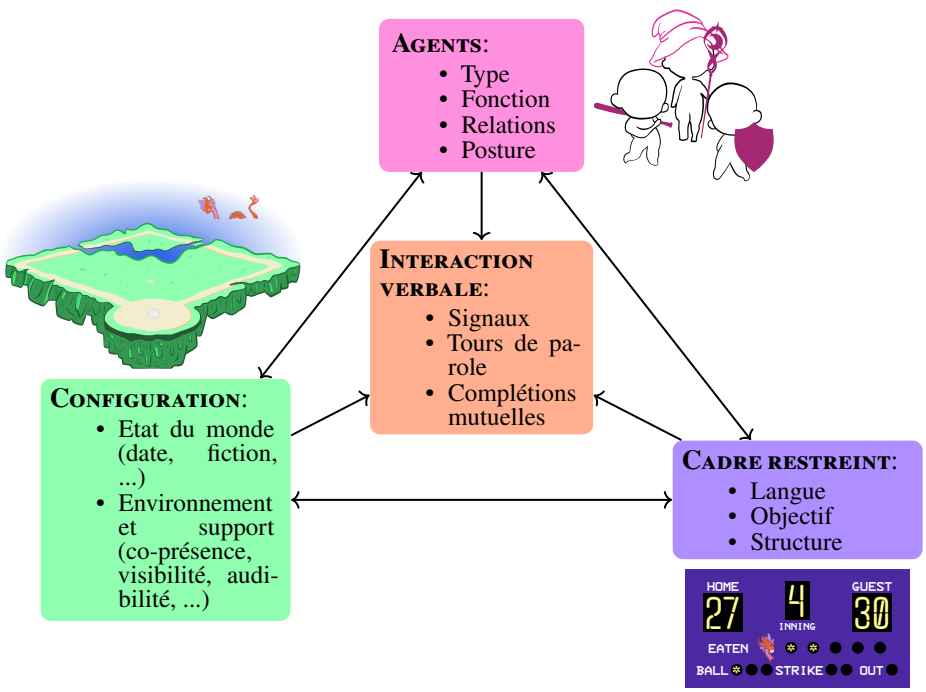


Figure 1: Illustration des quatre meta-paramètres charactérisant le dialogue – AGENTS, CONFIGURATION, CADRE RESTREINT and INTERACTION VERBALE – sous la forme d'une partie de baseball.

Une conversation, tout comme un match de baseball, nécessite des participants. En l'occurence nous parlons plutôt d'AGENTS car il peut s'agir d'humains mais aussi de machines comme des agents conversationnels, à base d'IA générative (comme ChatGPT) ou non. La dimension des agents décrit donc le *type* d'agents mais aussi d'autres aspects. Le nombre d'agents peut se traduire simplement par le nombre de participants, c'est-à-dire le nombre de joueurs dans la partie de baseball, mais peut aussi être plus complexe dans les cas où les participants sont en réalité simulés par d'autres agents. Pour notre partie de baseball, cela reviendrait à avoir un ou plusieurs joueurs qui sont en réalité contrôlés par un autre agent, et comme au baby-foot un même agent peut contrôler plusieurs joueurs. La fonction des participants est un autre aspect de la dimension des agents. Dans une conversation, certains participants peuvent avoir un rôle plus important que d'autres, et être plus actifs. Cela peut se traduire par le fait qu'un joueur de baseball peut être un lanceur ou un receveur, ou encore qu'un joueur peut être sur le banc des remplaçants. Enfin, les relations entre les participants forment un autre aspect de la dimension des agents. Dans une conversation, les participants peuvent avoir un historique conversationnel et des relations sociales qui influencent la dynamique de la conversation. Au baseball cela correspondrait à l'idée que des joueurs ayant l'habitude de jouer ensemble se coordonnent différemment de joueurs qui ne se connaissent pas. De même, une hiérarchie forte entre les joueurs peut influencer la dynamique de la partie, tout comme des relations tendues ou cordiales n'auraient pas le même impact.

Ensuite, une conversation a lieu dans une certaine CONFIGURATION, qui regroupe les aspects liés à l'état du monde dans lequel l'interaction a lieu – donc la date ou période historique, le fait d'être dans un monde fictionnel ou le monde réel, etc. – au support de communication – c'est-à-dire est-ce que la conversation a lieu en face-à-face, au téléphone, etc. – et au contexte physique. Au baseball cela correspondrait à la date et au lieu de la partie, ainsi qu'au terrain choisi. Si la partie a lieu dans un monde imaginaire où les lois de la physique ne sont pas les mêmes que dans le monde réel, le jeu sera très différent de celui auquel nous sommes habitués. Il en va de même pour une conversation qui sera influencée par le monde dans lequel elle a lieu avec ses normes sociales et culturelles. Le terrain lui-même joue aussi un rôle très important. Jouer sur un terrain standard ne sera pas la même chose que de jouer sur un terrain entouré d'eau ou sur un terrain en pente. La possibilité de voir ou non les autres joueurs, d'entendre ou non les sons de la partie, ou encore de pouvoir se déplacer librement sur le terrain, sont autant d'aspects qui influencent la dynamique de la partie. De même, une conversation sera influencée par le support utilisé pour communiquer, que ce soit en face à face, par téléphone, par chat textuel ou par visioconférence. La dimension temporelle est également un aspect important lié au support de communication car une conversation synchrone ne laisse pas les mêmes possibilités qu'un échange asynchrone.

Au-delà des agents et de la configuration, une conversation est également régie par un ensemble de choix et d'objectifs de réalisation qui définissent le CADRE

RESTREINT dans lequel elle se déroule. Dans notre analogie avec le baseball, cela correspondrait aux règles du jeu, qui déterminent ce que les joueurs peuvent ou ne peuvent pas faire. Par exemple, une équipe pourrait perdre des points si un monstre marin mangeait la balle. Pour la conversation, cela correspond à la langue utilisée et à l'objectif global qui peut aller de la simple interaction sociale à la réalisation d'une tâche complexe. Le cadre restreint peut également inclure des contraintes supplémentaires, comme le fait de devoir respecter un certain format ou de suivre un ordre du jour.

Enfin, la combinaison de toutes ces dimensions fait apparaître d'une part des caractérisiques supplémentaires à ces dimensions, et d'autre part un certain type d'INTERACTION VERBALE. Concernant les caractéristiques supplémentaires, il s'agit notamment de la posture adoptée par les participants, qui est influencée par la configuration et le cadre restreint. En effet, une conversation sur les réseaux sociaux implique une forme de pseudonymisation derrière laquelle les participants peuvent se cacher, et une conversation lors d'un jeu à rôles cachés implique que certains joueurs essaient de tromper les autres. Au baseball, cela pourrait correspondre à un joueur qui essaie de faire perdre son équipe sans se faire remarquer car il a parié contre elle. La stabilité du monde dans lequel la conversation a lieu est également une caractéristique supplémentaire qui peut influencer sa dynamique. Elle est liée notamment à l'interaction entre la configuration et le cadre restreint, car certaines tâches peuvent nécessiter de faire évoluer le monde de manière contrôlée. Par exemple, si le but d'une étude est d'analyser l'effet de bruits extérieurs sur la conversation, les expérimentateurs peuvent introduire des perturbations comme la mise en route d'un climatiseur bruyant ou la diffusion d'une musique forte à divers moment de l'échange. Au baseball, cela pourrait correspondre à la lumière qui se mettrait à clignoter. Finalement, l'interaction verbale est influencée par toutes les dimensions précédentes et décrit la manière dont les participants se coordonnent afin d'assurer leur compréhension mutuelle, d'organiser les tours de parole et de gérer les perturbations et malentendus. Dans notre analogie avec le baseball, cela correspondrait aux stratégies mises en place par les joueurs pour s'adapter aux conditions de jeu, aux forces et faiblesses de leurs coéquipiers et aux actions de l'équipe adverse.

Tous ces paramètres nous donnent un cadre de caractérisation du dialogue qui permet de décrire et de comparer les ressources dialogiques, mais aussi de définir le champ d'application d'une étude ou d'une application. Une étude d'environ 400 articles de recherche sur le dialogue, publiés en 2024 dans l'ACL Anthology, a montré que les données utilisées en recherche sont souvent mal décrites et qu'elles ne sont pas toujours adaptées à l'étude ou à l'application pour laquelle elles sont utilisées. Etant donné le niveau de détail de notre caractérisation, qui pourrait décourager certains chercheurs non spécialistes de la linguistique, nous avons proposé une version minimale dont nous pensons qu'elle devrait toujours être fournie pour décrire les données utilisées dans une étude. Cette version minimale inclut le

nombre de participants, le type d'agents, le medium de communication, la langue utilisée et l'objectif global de la conversation. Nous avons également proposé deux instanciations de notre cadre de caractérisation, en plus d'un tableau qui en résume les dimensions et paramètres. La première est basée sur un formalisme logique, la théorie des types avec enregistrements (Type theory with records, Cooper (2005, 2023)), qui permet de représenter les dimensions et paramètres de manière formelle et de raisonner sur eux grâce à des calculs sur les relations entre les types. La seconde est une représentation dans un espace vectoriel, qui permet de représenter les dimensions et paramètres d'une façon similaire à celle utilisée pour représenter les mots dans les modèles de langue. Cette représentation vectorielle permet de comparer les ressources dialogiques entre elles et de visualiser leur position dans l'espace des dialogues grâce aux mêmes techniques que celles utilisées en TAL pour analyser les espaces de plongement des mots. Ces différentes façons de présenter notre cadre de caractérisation du dialogue peuvent à notre sens toucher des publics différents et permettre à la communauté scientifique de mieux décrire et comparer les ressources dialogiques, mais aussi de mieux comprendre les limites des études et applications qui en font usage. Nous avons choisi de nous intéresser plus en détail à la structure thématique dans la deuxième partie de cette thèse, car elle est influencée par de nombreux aspects du dialogue et pourrait donc constituer un signal discriminant permettant de différencier les types de conversations.

3 Partie II : Analyse de la structure thématique dans des contextes interactionnels variés

La structure thématique dans la conversation semble intuitivement influencée par plusieurs aspects du type de conversation. Tout d'abord, le nombre de participants peut influencer la structure thématique, car plus il y a de participants, plus le thème peut changer rapidement et plus il peut y avoir de sous-conversations simultanées. Ensuite, l'objectif de la conversation peut donner un cadre plus ou moins contraint aux thèmes qui doivent être abordés, voire même à la manière dont ils doivent s'enchaîner. La réunion d'un conseil d'administration d'une entreprise, par exemple, aura un objectif très précis et des thèmes très ciblés définis par un ordre du jour. L'environnement dans lequel la conversation a lieu peut également influencer la structure thématique, car il peut introduire des perturbations qui peuvent induire un changement de sujet. Tout cela suggère que la structure thématique peut aider à différencier les types de conversations à partir des données dialogiques elles-mêmes. Produire des outils permettant de l'analyser nous apporterait donc le pendant de la caractérisation développée dans la première partie, en nous donnant une façon de comparer différents types de conversations et ainsi déterminer si deux ressources dialogiques sont similaires ou non. En nous intéressant au développement d'une méthode d'analyse de la structure thématique, nous parti-

cipons donc à un objectif plus large de création d'outils et de méthodes permettant de comparer les ressources dialogiques, et ainsi de justifier la pertinence de substituer les dialogues ouverts et spontanés par d'autres plus simples à collecter pour une étude donnée, mais aussi ses limites. Dans cette partie, nous présentons trois études appliquées qui explorent la structure thématique, en commençant par une approche superficielle de segmentation linéaire en thèmes (Section 3.1), puis une approche plus flexible d'analyse continue de la similarité thématique (Section 3.2), et enfin une analyse plus profonde de la structure thématique dans des conversations multipartites spontanées (Section 3.3).

3.1 Analyse en surface : segmentation linéaire en thèmes de la série TV *Friends*

Nous commençons par nous intéresser à une tâche classique d'analyse de la structure thématique, la segmentation linéaire en thèmes, mais dans un contexte interactionnel plus complexe que ceux habituellement étudiés. La segmentation en thèmes consiste à identifier les changements de sujet dans un texte ou une conversation, et à diviser le texte en segments correspondant à ces sujets. On parle de segmentation *linéaire* car la découpe se fait de manière séquentielle, en suivant l'ordre des énoncés. Cela signifie que le texte ou la conversation analysée est considéré comme une séquence de thèmes qui se succèdent, sans chevauchements.

Bien que notre objectif soit d'analyser des conversations multipartites spontanées, la réalité est que les ressources dialogiques annotées en thèmes sont rares. Nous avons donc choisi de commencer par un corpus plus simple, la série TV *Friends*, qui bien que n'étant pas directement annoté en thèmes, contient des informations sur sa découpe en scènes et sur les événements importants qui s'y déroulent. Nous avons donc utilisé ces informations pour estimer la structure thématique de chaque épisode, ce qui nous a servi de référence pour évaluer les performances de différents modèles de segmentation linéaire en thèmes. La Figure 2 présente un extrait de la série *Friends* (S01E09) et sa segmentation en thèmes grâce à l'annotation des scènes et des événements.

Dans ce chapitre notre objectif n'était pas de développer le meilleur modèle de segmentation linéaire en thèmes, mais plutôt d'étudier comment une approche existante, fonctionnant bien sur des conversations structurées, se comporte sur des conversations plus complexes. Nous avons donc utilisé l'approche de Xing et Carenini (2021), qui emploient une technique proposée par Hearst (1997), le *TextTiling*, et l'améliore avec des méthodes récentes de TAL. L'idée générale du TextTiling est de comparer les énoncés successifs d'un texte afin d'identifier les points où le degré de similarité change de manière significative, ce qui peut indiquer un changement de sujet. L'approche originale calculait la similarité entre les énoncés en utilisant des idées très simples, comme le comptage de mots communs. L'approche de Xing et Carenini (2021) utilise des modèles de langue pré-entraînés

Thème 1:

Rachel$_1$ Terry, je sais, je suis pas là depuis longtemps, mais je me demandais... puis-je avoir une avance de 100$ sur mon salaire ?

Terry$_2$ Une avance ?

Rachel$_3$ Pour passer Thanksgiving en famille. tous les ans, on va skier à Vail, et mon père paie le billet... mais je me suis lancée dans cette idée d'indépendance... Raison pour laquelle j'ai pris ce « job ».

Terry$_4$ Rachel, Rachel, ma petite. Tu es une serveuse désastreuse. Vraiment mauvaise.

Rachel$_5$ Je comprends votre point de vue. Mais je fais de mon mieux, et je crois que je progresse. Vraiment ! *(Elle se tourne vers la salle.)* Quelqu'un veut du café ? *(Tout le monde lève la main.)*

Thème 2:

[Scène : Central Perk, Rachel approche d'un client.]

Rachel$_6$ *(au client)* Excusez-moi. Bonjour. Vous qui êtes un habitué, ici, pourriez-vous... me faire une avance sur mes pourboires ?

Le client$_7$ Quoi ?

Rachel$_8$ D'accord ! Très bien. Désolée d'avoir renversé... *(Il s'en va, elle ramasse les quelques pièces qu'il a laissées.)* Plus que 98,50$ à trouver.

Thème 3:

[Monica entre.]

Monica$_9$ *(à Ross)* Tu savais que maman et papa passent Thanksgiving à Porto Rico ?

Ross$_{10}$ Quoi ? Non, c'est faux !

Figure 2: Exemple de segmentation en thèmes d'un morceau d'épisode de *Friends* (S01E09).

pour obtenir des représentations vectorielles plus riches des énoncés, ce qui permet de calculer la similarité entre eux de manière plus sophistiquée.

En adaptant cette approche à notre corpus, nous avons pu évaluer ses performances sur des conversations plus complexes que celles pour lesquelles elle avait été initialement conçue. Nous avons constaté que bien que l'approche fonctionne relativement bien sur des conversations structurées, elle rencontre des difficultés lorsqu'elle est appliquée aux dialogues de la série TV *Friends*, qui sont plus spontanés et comportent des interactions plus riches entre les personnages. Nous avons aussi montré qu'une améliortion de la technique de TextTiling basée sur des caractéristiques du dialogue, comme les changements de locuteur, peut donner de

meilleurs résultats que l'approche de Hearst (1997). Cela montre l'importance de prendre en compte les caractéristiques spécifiques du dialogue lors de l'analyse de sa structure thématique. En effet, les conversations de *Friends* sont bien plus chaotiques que celles des corpus étudiés par Xing et Carenini (2021), mais elles sont aussi destinées à un public extérieur, les spectateurs, ce qui fait qu'elles ont des caractéristiques de surface qui les distinguent des conversations spontanées et les rend plus simples à segmenter avec des heuristiques telles que les changements de locuteur. Au-delà du choix de la méthode technique pour la segmentation en thème, l'analyse manuelle des conversations de *Friends* donne l'intuition qu'une approche de segmentation *linéaire* en thèmes est certainement trop simpliste pour des conversations multipartites spontanées. En effet, il arrive fréquemment que plusieurs sous-conversations se déroulent en parallèle dans la série, ce qui rend difficile l'identification d'un seul thème à un moment donné. Cela nous a conduit à explorer des approches plus flexibles pour analyser la structure thématique dans des conversations plus complexes, comme celles que nous présentons dans les sections suivantes.

3.2 Analyse continue : similarité thématique entre des commentaires de *Reddit*

Comme l'idée de la segmentation linéaire en thèmes est trop rigide pour des conversations multipartites spontanées, nous avons exploré une approche plus flexible qui consiste à analyser la similarité thématique entre les énoncés d'une conversation. Notre objectif était de construire un modèle capable de mesurer la similarité thématique entre deux énoncés de façon continue, c'est-à-dire de manière à ce que la similarité puisse varier entre 0 et 1, plutôt que d'être binaire (même thème ou pas). Une fois encore, bien que notre objectif final soit d'analyser des conversations multipartites spontanées, nous avions besoin d'une ressource présentant une structure intrinsèque à son format afin d'entraîner notre modèle. En effet, pour entraîner un modèle de similarité thématique, une première idée serait de trouver des paires d'énoncés annotées avec un score de similarité thématique. Cependant, cette tâche est très difficile à réaliser de façon fiable et de telle sorte que les annotateurs s'accordent sur les scores attribués. Nous avons donc choisi d'entraîner le modèle en lui indiquant un ordre de similarité thématique entre plusieurs paires d'énoncés, plutôt qu'un score absolu. Cette idée est inspirée de celle proposée par Xing et Carenini (2021) pour améliorer l'algorithme de TextTiling. Si cette approche ne nécessite pas de scores absolus, elle nécessite toutefois de disposer d'un ensemble d'énoncés pour lesquels on peut déterminer un ordre de similarité thématique, et donc d'une ressource présentant intrinsèquement une structure thématique.

Nous avons choisi d'utiliser des commentaires de la plateforme *Reddit*, qui sont organisés en fils de discussion où chaque commentaire peut répondre à un autre commentaire. Cette structure hiérarchique nous permet de déterminer un ordre de

A_0 (post) Train Simulator est tellement immersif ! <vidéo du point de vue d'un passager dans un train, jouant à un jeu de simulation où ce que le personnage voit dans le jeu est la même chose que ce que le joueur voit dans la vie réelle>

A_1 J'ai eu ce moment sur Amtrak [note: la SNCF des USA] hier..

B_2 Tu t'es déjà dit qu'un être supérieur qui te manipulerait dans sa version de Train Simulator avait peut-être vécu ce moment-là lui aussi, hmmmm ?

C_3 Oh, c'est donc ça que signifie cette émeraude au-dessus de ma tête ?

D_4 Évite simplement les piscines et les baignoires, et tout devrait bien se passer

E_5 N'entre que dans des pièces dotées de plusieurs sorties. Garde ton bébé loin de la cuisine

F_6 Sauf si tes meubles de cuisine se réorganisent d'eux-mêmes pour te bloquer l'accès à la cuisinière pendant que tu prépares un plat impossible.

G_7 ou que souffrant d'aquaphobie tu te retrouves soudainement entouré d'une piscine, que tu aies besoin d'aller aux toilettes, et que tu ne puisses que regarder en pleurant un cambrioleur apparaître et voler tes biens les plus précieux.

H_8 Piscine fermée

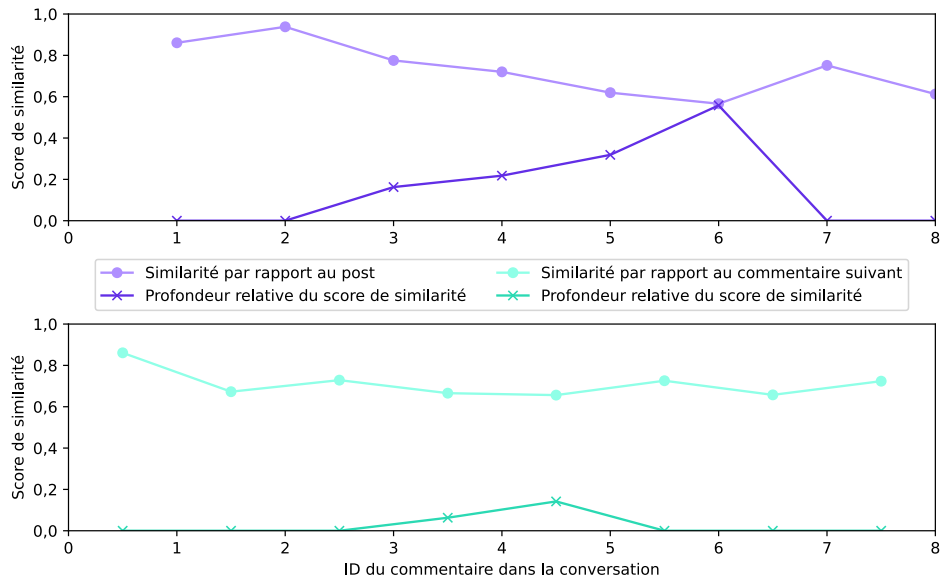


Figure 3: Exemple de conversation Reddit et la similarité thématique entre chaque commentaire et le post, et entre chaque commentaire et le suivant.

similarité thématique entre les commentaires, car un commentaire et sa réponse directe peuvent généralement être considérés comme étant plus proches thématiquement que deux commentaires qui ne sont pas directement liés. Nous avons donc

utilisé cette structure pour créer un ensemble d'énoncés annotés avec un ordre de similarité thématique, que nous avons utilisé pour entraîner notre modèle capable de mesurer la similarité thématique entre deux énoncés de manière continue. Cela qui nous a permis de mesurer les variations de thème de façon locale, c'est-à-dire entre des commentaires successifs dans un fil de discussion, mais aussi de façon plus globale, en mesurant la similarité thématique entre chaque commentaire et le commentaire initial du fil de discussion. En comparant ces deux mesures, nous avons pu mettre en évidence le fait que des variations importantes à l'échelle globale peuvent se produire même lorsque les variations locales sont faibles, ce qui suggère que la structure thématique d'une conversation peut être plus complexe qu'une simple succession de thèmes. Nous avons donc proposé une piste pour l'analyse de la structure thématique qui consiste à s'intéresser à des changements de thème *graduels* plutôt que brusques comme le font les approches de segmentation linéaire en thèmes. La Figure 3 présente un exemple de fil de discussion Reddit et la similarité thématique entre chaque commentaire et le post initial, ainsi qu'entre chaque commentaire et le commentaire suivant. Bien que les variations de similarité soient faibles, en prenant en compte la quantité de variation relative, on peut remarquer des différences entre les deux mesures, ce qui suggère que les changements de thème à l'échelle globale ne sont pas toujours perceptibles à l'échelle locale.

Cette idée se rapproche d'une vision plus hiérarchique de la structure thématique, où les thèmes peuvent être imbriqués les uns dans les autres et où la notion de thème est représentée avec plusieurs niveaux de granularité, mais elle est aussi originale car elle permet de rendre compte de changements de thème qui ne sont pas forcément perceptibles à l'échelle locale, mais qui peuvent être importants à l'échelle globale. S'intéresser aux changements graduels est en effet un changement de paradigme dans le sens où l'on ne considère plus qu'une conversation est composée d'une succession de sous-conversations qui se suffisent à elles-mêmes, mais plutôt que chaque énoncé d'une conversation peut rendre pertinentes des idées ou directions plus ou moins éloignées du thème actuel de la conversation. C'est cette idée que nous explorons dans la section suivante, où nous nous intéressons à la structure latente de la conversation et à la manière dont elle peut être analysée à travers les questions considérées comme pertinentes à un moment donné de la conversation.

3.3 Analyse de la structure latente : comparaison de la pertinence et de la résolution de questions dans des conversations spontanées

Une façon de modéliser les thèmes proposée dans la littérature est de considérer qu'un thème est représenté par l'ensemble de questions en discussion (*questions under discussion*, QUDs; Ginzburg, 1995 ; Van Kuppevelt, 1995 ; Roberts, 2012) à un moment donné de la conversation. Cette idée se base sur une théorie

d'organisation du discours selon laquelle chaque énoncé d'une conversation est produit soit en réponse à une question, parfois implicite, rendue pertinente par le contexte, soit pour introduire une autre question qui aide à répondre à cette question (Benz & Jasinskaja, 2017). Si cette théorie a longtemps été utilisée pour expliquer *a prosteriori* la structure et la cohérence d'une conversation, Y. Wu et al. (2024) ont montré dernièrement qu'elle permet aussi d'affirmer que l'auteur d'un discours, en l'occurrence d'un *Ted Talk* ou d'un extrait d'article de presse, anticipe les questions que les lecteurs ou auditeurs pourraient se poser. Ils mettent en effet en évidence une corrélation forte entre le degré de pertinence donné à une question dans un contexte donné et le fait que cette question soit effectivement résolue plus tard dans le discours. Cela valide la pertinence de la théorie des QUDs comme outil de prédiction de la structure latente d'un discours. Si ces résultats peuvent être répliqués sur de la conversation, alors une analyse plus fine de la distance entre l'évocation d'une question et sa résolution pourrait permettre de mieux comprendre les stratégies d'organisation des thèmes dans la conversation. A l'inverse, on peut s'attendre à ce que la nature plus libre et chaotique du dialogue ne permette pas de reproduire les résultats obtenus sur des textes plus formels, supportant ainsi l'idée que des représentations plus complexes de la structure thématique sont nécessaires pour rendre compte de la conversation.

Nous avons donc entrepris de répliquer l'expérience de Y. Wu et al. (2024) sur de la conversation. Une fois le corpus choisi, le processus consiste à (1) collecter à divers moments de la conversation des questions qui sont pertinentes à ce moment-là, (2) noter la pertinence de ces questions, (3) identifier si elles sont résolues ou non plus tard dans la conversation, et (4) analyser la corrélation entre le degré de pertinence et le fait qu'une question soit résolue ou non. Comme aucun corpus de conversation avec les annotations nécessaires n'existait, nous avons dû créer notre propre corpus annoté. Contrairement aux deux premières études de la thèse, nous étions assez peu contraints dans le choix du corpus puisque nous allions produire les annotations nous-mêmes, excepté pour la langue utilisée puisque nous devions la maîtriser suffisament pour produire des annotations fiables. Nous avons donc enfin pu nous intéresser à des conversations multipartites spontanées, qui sont plus représentatives de la complexité du dialogue. Nous avons choisi d'utiliser le British National Corpus (BNC; Consortium (1994a)) car il contient des conversations spontanées de la vie quotidienne. Le BNC contient aussi beaucoup d'autres types de dialogues et nous avons profité de cette diversité pour étudier les variations de la structure thématique dans ces différentes formes de dialogues. Nous avons choisi de nous concentrer sur les variations entre duologues et multilogues, en sélectionnant des conversations à deux participants et des conversations à plus de quatre participants pour maximiser les différences, et sur les variations entre conversations à différent degrés de structure, en sélectionnant des conversations de la vie quotidienne pour le degré le plus bas, des conversations de réunions et interviews pour un degré intermédiaire, et des conversations tirées de la radio ou d'émissions

PS0EH$_0$ J'étais sur le point de te dire pour Margaret
PS0ED$_1$ Ah oui, Margaret et Snowy, oui
PS0EH$_2$ Ils sont allés à ce Howstead Abbey
PS0ED$_3$ Ouais
PS0EH$_4$ **parce qu'ils ont une entrée valable pour toute la saison et ils peuvent nourrir les canards et**

ID$_{sheet}$	Type	ID$_Q$	Cutoff	ID$_{ann2}$	Question
4	open multi	0	4	E	**1. Est-ce que Margaret et Snowy visitent Howstead Abbey chaque saison?**

S$_1$	S$_2$	S$_G$	Comm$_{S1}$	Comm$_{S2}$	A$_1$	A$_2$	A$_G$	Comm$_{A1}$	Comm$_{A2}$
5	5	5	/	*pertinente mais pas essentielle à la conversation*	0	0	0	/	/

Figure 4: Exemple de ligne dans le corpus publié avec un extrait du dialogue sur lequel elle est basée, incluant le tour d'ancrage et la question associée en gras. S$_i$ et A$_i$ indiquent les scores de pertinence et de résolution pour l'annotateur i avec G pour les scores de référence. *Comm* signifie les commentaires associés. *Début du dialogue KCE – 029701.*

télévisées pour le degré le plus élevé. En effet, ces dernières étant pensées pour une audience extérieure à la conversation, elles sont d'une certaine façon plus proches des textes et Ted Talks initialement étudiés par Y. Wu et al. (2024). Au final, nous avons sélectionné 60 conversations du BNC, 10 pour chacune des catégories définies par le nombre de participants (2, > 4) et le degré de structure (ouvert, organisé, public). Nous avons ensuite annoté ces conversations en collectant des questions pertinentes à divers moments de la conversation, en notant leur pertinence et en identifiant si elles étaient résolues ou non plus tard dans la conversation.

Concernant la collecte des questions, nous avons choisi d'utiliser un grand modèle de langue (LLM) comme Y. Wu et al. (2024). Ce choix est motivé par le fait que la tâche est très longue et qu'en se focalisant sur l'annotation manuelle des questions plutôt que sur leur génération, nous pouvons nous assurer que la production du modèle est bien conforme à ce que l'on attendait. En pratique nous avons donc appliqué la méthode suivante pour toutes les conversations sélectionnées : A partir

du quatrième tour de parole, un tour de parole sur deux est utilisé comme *ancre* pour générer des questions ; pour chaque ancre, nous fournissons au modèle le contexte de la conversation, c'est-à-dire les énoncés précédents, et l'ancre, et nous lui demandons de générer trois questions pertinentes à ce moment de la conversation. Nous avons ensuite annoté manuellement la pertinence de ces questions et identifié si elles étaient résolues ou non plus tard dans la conversation. La pertinence était annotée sur une échelle de Likert de 0 à 8 en ayant accès uniquement au dialogue jusqu'à l'ancre alors que la résolution était annotée en ayant accès à l'ensemble de la conversation sur une échelle de 0 à 2 (pas résolue, partiellement résolue, résolue). Chaque question a été annotée par deux annotateurs indépendants, et nous avons calculé l'accord inter-annotateurs pour la pertinence et la résolution. L'accord était faible à modéré pour la pertinence et modéré à substantiel pour la résolution, ce qui reflète la complexité de la tâche et la subjectivité des jugements.

En analysant la corrélation entre la pertinence et la résolution des questions, nous avons constaté que le phénomène identifié par Y. Wu et al. (2024) sur des monologues se reproduit également dans la conversation mais de façon bien moins marquée. Nous concluons que les questions les plus pertinentes ont plus tendance à être résolues que les questions moins pertinentes, mais que beaucoup de questions ne sont jamais résolues peu importe leur pertinence. Il existe aussi des différences notables entre les types de conversations, les multilogues ouverts étant ceux pour lesquels la corrélation entre pertinence et résolution est la plus faible, mais ces résultats nécessitent d'être confirmés sur un plus grand nombre de conversations. La tâche d'annotation était en effet très complexe et les différences entre les deux annotateurs doivent être analysées plus en détail pour comprendre les raisons de ces divergences de point de vue.

Finalement, nous avons montré que la conversation semble aussi être guidée par la pertinence des questions évoquées par le contexte mais dans une moindre mesure que dans les monologues. Cela supporte le choix de la théorie des QUDs comme outil d'analyse de la structure latente du dialogue, mais suggère aussi qu'une analyse plus complexe de la résolution des questions est nécessaire pour comprendre comment la structure thématique s'y organise. La méthode proposée par Y. Wu et al. (2024) semble néanmoins validée et pourrait être utilisée pour étudier la structure latente d'autres types de conversations, notamment des conversations synthétiques, afin de déterminer si leur fonctionnement est similaire à celui des conversations spontanées.

4 Conclusion et perspectives

Cette thèse s'intéresse à la notion de dialogue et aux différentes facettes qui font sa grande variabilité. Ses contributions majeures sont d'une part un cadre de caractérisation du dialogue qui permet de décrire et de comparer les ressources dia-

logiques, et d'autre part la validation d'une méthode pour mesurer la prédictabilité de la structure thématique dans la conversation.

Concernant le cadre de caractérisation du dialogue, nous avons proposé une instanciation basique sous forme de tableau descriptif, une instanciation plus formelle en théorie des types avec enregistrements qui permet de raisonner sur la compatibilité des ressources dialogiques via des calculs comme sur des structures de traits, et une instanciation plus statistique dans un espace vectoriel qui permet de comparer les ressources dialogiques entre elles. Une contribution plus mineure est l'analyse de plus de 400 articles de l'ACL Anthology portant sur le dialogue, qui nous a permis de mettre en évidence des problèmes systémiques au sein de la communauté TAL. Tout d'abord, les articles ne décrivent pas suffisamment les ressources sur lesquelles sont basées les études, ce qui rend difficile d'estimer leur périmètre réel d'application. Ensuite, la plupart des articles publiés aujourd'hui dans les conférences majeures de TAL s'intéressent majoritairement à l'amélioration des systèmes de dialogue, laissant peu de place à des études plus théorique de la notion de conversation. Enfin, les méthodes d'évaluation utilisées dans ce champ de recherche se focalisent souvent sur la réussite de la tâche et très peu sur la qualité de l'interaction, ou avec des métriques qui ne reflètent pas les réels enjeux d'une conversation. Ces problèmes sont d'autant plus préoccupants que les systèmes de dialogue sont de plus en plus utilisés dans des contextes où la qualité de l'interaction est primordiale, comme dans le soutien psychologique.

La seconde grande contribution de cette thèse peut participer à repérer des différences structurelles entre différents types de conversations, comme par exemple des conversations produites en interaction avec un agent conversationnel par rapport à leur équivalent entre deux humains. En effet, nous rendons disponible un corpus de plus de 7 000 questions ancrées dans des conversations spontanées à degrés de structure variables, annotées par deux personnes pour leur pertinence et leur résolution. Ayant validé l'existence d'une corrélation entre la pertinence et la résolution des questions dans un des types de conversation les plus ouverts et chatotiques – les multilogues spontanés – nous pensons que cette méthode peut être utilisée pour analyser la structure thématique d'autres types de conversations, notamment des conversations synthétiques, afin de déterminer si leur fonctionnement est similaire à celui des conversations spontanées. Notre corpus peut ainsi servir de base pour l'entraînement de modèles d'annotation automatique afin d'étendre cette étude.

Au-delà de ces contributions, tout le travail mené durant cette thèse questionne la notion même de conversation et la place que l'on accorde aujourd'hui aux grands modèles de langue pour la production de dialogue dans notre communauté. Ils sont aujourd'hui devenus des outils presque inévitables pour réaliser des études si l'on souhaite les voir publiées dans les conférences majeures de TAL, alors même que leur utilisation soulève de nombreuses questions scientifiques comme éthiques. Dans ce contexte, je souhaiterais terminer ce résumé en soulignant que certains

penseurs mettaient en garde la population contre le fait que l'autodafé des livres était un prélude à l'oppression et à la destruction de vies humaines, et l'histoire leur a donné raison. Aujourd'hui, de grandes entreprises découpent des livres, parfois uniques, en morceaux pour nourrir des modèles de langue qu'elles essaient de rendre omniprésents dans la société. Les similitudes entre ces deux situations méritent certainement que l'on s'y intéresse.

Là où on brûle des livres, on finit par brûler des hommes.

– Heinrich Heine, Almansor (1821)

Contents

Acknowledgements **iv**

Abstract **viii**

Sammanfattning **x**

Résumé Court **xii**

Résumé Long **xiv**

1 Enjeux et questions de recherche xvi

1.1 Étude de la conversation xvi

1.2 Difficultés de l'étude des thèmes xix

1.3 Focus et contributions xxiii

2 Partie I : Caractériser le dialogue xxiv

3 Partie II : Analyse de la structure thématique xxviii

3.1 Analyse en surface : segmentation linéaire en thèmes de la série TV *Friends* xxix

3.2 Analyse continue : similarité thématique entre des commentaires de *Reddit* xxxi

3.3 Analyse de la structure latente : comparaison de la pertinence et de la résolution de questions dans des conversations spontanées xxxiii

4 Conclusion et perspectives xxxvi

List of Figures **xlviii**

List of Tables **lii**

Prologue **1**

Introduction

1 Studying conversation 5
2 Issues with studying topics 8
3 Focus and contributions 11

Part I Linguistic and Cognitive Perspectives on Dialogue: Toward an operational descriptive framework **15**

Chapter 1
Foundational Concepts and Perspectives on Dialogue

1.1 Signals beyond words: Multimodality, redundancy, and monitoring 19
1.1.1 Modes of expression in interaction 19
1.1.2 Redundancy and robustness in communication 21
1.2 Coherence and coordination in dialogue 22
1.2.1 Coherence and the cooperative nature of conversation 23
1.2.2 Dialogue as a joint activity 26
1.2.3 Repair and the maintenance of coherence 27
1.3 Acting through language in interaction 27
1.3.1 Utterances as actions 28
1.3.2 Possibilities and constraints in interaction 29
1.4 Representing and coordinating understanding 30
1.4.1 Representing the state of interaction 30
1.4.2 Coordination across participants 32
1.5 Dialogue as a multidisciplinary object of study 34
1.5.1 Pragmatics 35
1.5.2 Formal semantics and dialogue modelling 35
1.5.3 Psycholinguistics 35
1.5.4 Sociolinguistics 36
1.5.5 Conversation Analysis 36

1.5.6 Discourse Analysis 36
1.5.7 Clinical and developmental linguistics 37
1.6 Toward an integrative descriptive framework 37

Chapter 2
Parameters for Characterising Dialogue

2.1 Conversation agents 44
2.1.1 Agency 46
2.1.2 Participation structures 48
2.1.3 Social dynamics 54
2.2 Conversational setting 56
2.2.1 World setting 56
2.2.2 Environment and medium 57
2.3 Conversational frame 75
2.3.1 Language 75
2.3.2 Purpose 77
2.4 Dependant descriptive parameters 83
2.4.1 Influence on the agents: Enunciative status 84
2.4.2 Conversational setting 90
2.4.3 Influence on the frame 93
2.4.4 Verbal interaction: The co-constructed nature of dialogue 96
2.5 On the operationalisation of the framework 102
2.5.1 A formal account of CHIMERA 104
2.5.2 A vector space perspective on CHIMERA 116

Chapter 3
Critical Analysis of Contemporary NLP Approaches

3.1 Contemporary landscape of dialogue research 123
3.1.1 Methodology of the review 124
3.1.2 Tasks 126
3.1.3 Datasets and resources 130
3.1.4 Evaluation 134

3.2 Pipeline for NLP and dialogue research 136
3.2.1 From theoretical models to black-box ones 137
3.2.2 Evaluation and the risk of circularity 138
3.2.3 Re-centring theory, data, and purpose 140
3.3 Applying CHIMERA 141
3.3.1 Scenario 1: Turn-taking 141
3.3.2 Scenario 2: Task-oriented dialogue 149
3.3.3 Scenario 3: Clinical interview 162
3.4 Targeted dialogue generation control 171
3.4.1 Generation control 173
3.4.2 Evaluation metrics 176
3.5 Summary and conclusion 179

Part II Topics and Topical Structure: Applicative analysis across various interactional contexts 183

Chapter 4
Topics in Discourse and Dialogue

4.1 Topics and topical structure 186
4.1.1 Topic 186
4.1.2 Topical structure 190
4.2 Studying topical structure 191
4.2.1 Analytical and predictive perspectives 193
4.2.2 Approaches to modelling topical organisation 195
4.3 Dialogue type and topical structure 199

Chapter 5
Topic Segmentation in Open-ended Multi-party Conversation

5.1 Related works 207
5.1.1 TextTiling approach 207
5.1.2 Carenini's coherence-layer based approach 213
5.2 Experimental setting 218

5.2.1 Models 218
5.2.2 Evaluation metrics 219
5.3 Dataset 222
5.3.1 Detailed description 223
5.3.2 Segmentation in topics 226
5.4 Description of the baselines 227
5.4.1 Random baseline 228
5.4.2 Dialogue adapted TextTiling 230
5.4.3 BERT for next-sentence prediction 233
5.4.4 LLMs 235
5.5 Improving TextTiling with Linguistic Features 240
5.5.1 Speakers 240
5.5.2 Questions 242
5.5.3 Coreference chains 242
5.5.4 Summary of the Results 243
5.6 Adapting Xing and Carenini's BERT-based model 244
5.6.1 Learning curve and coherence layers 244
5.6.2 Stability and less diverse training set 247
5.7 Final comparisons and conclusion 249
5.7.1 Comparison of all the models 250
5.7.2 Final remarks and conclusion 251

Chapter 6
Topical Similarity at Utterance Level

6.1 Method 256
6.1.1 Metric construction: Recapitulation 257
6.1.2 Evaluation 258
6.2 Dataset: Naturalistic structured interactions 260
6.2.1 Social media in NLP research 260
6.2.2 Reddit detailed characterisation 262
6.2.3 Selection: a varied range of communities 267
6.2.4 Coherence levels 272
6.3 Coherence levels hypothesis validation 272

6.3.1 Topic modelling: BERT Topic 273
6.3.2 Similarity metric baseline: DSE-BERT without fine-tuning 277
6.4 Metrics training 279
6.4.1 Implementation details 279
6.4.2 Evaluation 281
6.4.3 Challenges of the data 285
6.5 Conversation modelling with a continuous topic similarity metric 288
6.5.1 Quantitative description 290
6.5.2 Patterns of topic changes 292
6.5.3 Proposal for a continuous topic segmentation 294
6.6 Discussion and conclusion 298

Chapter 7
Modelling Topics with QUDs

7.1 Background: Questions and discourse structure 303
7.1.1 QUDs as predictive tools 303
7.1.2 Operationalising QUDs through question *salience* and *resolution* 304
7.2 Source corpus: Naturalistic dialogues 305
7.2.1 The Spoken BNC 306
7.2.2 Sub-selection: Varied interactional settings 306
7.2.3 Statistical description 310
7.2.4 Challenges of naturalistic dialogues 311
7.3 Method: Automatic question generation and human annotation 322
7.3.1 Automatic question generation 323
7.3.2 Guidelines adaptations for dialogue 325
7.3.3 Annotations 326
7.4 Released dataset: questions and annotations 333
7.4.1 World knowledge 334
7.4.2 Technical limitations 336
7.4.3 Prompt-related issues 339

7.5 Results and analysis: Dialogue is less predictable than static text 341
7.5.1 Salience . . . 341
7.5.2 Resolution . . . 343
7.5.3 Dialogue predictability . . . 345
7.6 Conclusion . . . 351

Epilogue **355**

Summary and Discussion

1 Contributions and discussion . . . 358
2 Limitations and perspectives . . . 365
3 Final words . . . 367

Acronyms **371**

Bibliography

Appendix A
Extension of Chapter 2

Appendix B
Extension of Chapter 5

B.1 Baselines . . . III
B.1.1 Random baselines . . . III
B.1.2 Original and dialogue adapted TextTiling . . . V
B.1.3 BERT for next-sentence prediction: off-the-shelf vs. fine-tuned on a generic dialogue corpus . . . VII
B.1.4 LLMs . . . VIII
B.2 TextTiling enhanced with linguistic features . . . IX
B.3 ML-enhanced TextTiling . . . XI
B.3.1 Learning curve . . . XI
B.3.2 Coherence-layers Selection . . . XI
B.3.3 Model stability . . . XII

B.3.4 Training set variety XIII
B.4 Final comparisons XIV

Appendix C
Extension of Chapter 6

Appendix D
Extension of Chapter 7

D.1 Detailed annotation guide XVII
D.1.1 Question salience XVII
D.1.2 Question resolution XXII
D.2 Question salience-resolution correlation XXIV

Appendix E
Montgolgram Corpus

E.1 Aim and method: Balancing naturalism and realistic collection conditions XXV
E.2 Dialogue setting: Two types of tasks for two analytical purposes XXVII
E.2.1 Balloon task: An ethical dilemma XXVII
E.2.2 Tangram task: A visual coordination task XXVIII
E.3 Collection: Lessons and perspectives XXIX

List of Figures

1 Le dialogue sous forme de partie de baseball surnaturelle xxv
2 Analyse en surface: Exemple de segmentation xxx
3 Analyse continue : Exemple de conversation *Reddit* xxxii
4 Structure latente: Exemple du corpus publié xxxv

1 Thesis structure 14

1.1 *Star sign* example illustration 19
1.2 Modalities in the *Star sign* example 21
1.3 TTR example: 'it's not this one' when paying attention 32
1.4 TTR example: 'it's not this one' when *not* paying attention 34
1.5 Main domains of dialogue research 38

2.1 Dialogue as a (supernatural) baseball game 45
2.2 Conversation participants' function in a café 52
2.3 Dialogue characterisation in TTR – Descriptive meta-parameters . 109
2.4 Dialogue characterisation in TTR – Dependant parameters 110
2.5 Dialogue characterisation in TTR – Maximal version 111
2.6 Examples of corpora described with TTR 117
2.7 Examples of corpora described within a 3D space 119

3.1 Number of papers in the ACL ANTHOLOGY per year since 2000 . . 125
3.2 Distribution of papers by task category 127
3.3 Description of the (dialogue) research pipeline in NLP/CL 137
3.4 Representation of the exchange of the messages with our ChatTool 156
3.5 Average turn-taking entropy in the multi-user version of MultiWOZ 180

4.1 Representation of a linear topic segmentation 192
4.1 Representation of a hierarchical topic segmentation 193
4.2 Example of a dotplot representation 197

5.1 TextTiling – Pre-processing 209
5.2 TextTiling – Lexical scores computation 211
5.3 TextTiling – Example plot of the scores 212
5.4 TextTiling – Examples of score patterns 213
5.5 BERT-based TextTiling – Levels of coherence 215
5.6 Comparison of regression and ranking losses 216

5.7 Topic segmentation – Evaluation: F-measure family metrics . . . 220
5.8 Topic segmentation – Evaluation: P_k and Window Diff metrics . . 222
5.9 Topic segmentation – Models: Random baseline 228
5.10 Topic segmentation – Results: TextTiling with block comparison . 232
5.11 Topic segmentation – Results: TextTiling with vocabulary introduction 234
5.12 Topic segmentation – Models: Prompt for zero-shot with LLMs . 236
5.13 *Friends* corpus – Coherence layers 246
5.14 Topic segmentation – Results: Graphical comparison 251

6.1 BERT Topic method 259
6.2 Topic similarity ranking – Evaluation Metrics 260
6.3 *Reddit* corpus – Website organisation 263
6.4 *Reddit* corpus – Thread organisation 269
6.5 *Reddit* corpus – Coherence levels 274
6.6 Topic Similarity Ranking – Results: Baseline 278
6.7 Topic Similarity Ranking – Model architecture 280
6.8 Topic Similarity Ranking – Model training process 280
6.9 Topic Similarity Ranking – Results: Cross-evaluation 282
6.10 Topic Similarity Ranking – Results: Congruent models 285
6.11 Topic Similarity Analysis: Conversation representation 291
6.12 Topic Similarity Analysis: Similarity around topical boundaries . 295
6.13 Topical Similarity Analysis: An example 297

7.1 Example from the original study 304
7.2 Salience and resolution correlation framework 323
7.3 Salience-Resolution: Prompt for question generation 324
7.4 Salience-Resolution: Annotation interface 327
7.5 Salience-Resolution: Comparison of annotations 330
7.6 Salience-Resolution: Inter-annotator differences 331
7.7 Salience-Resolution: Example row from our released corpus . . . 333
7.8 Salience-Resolution: Salience scores repartition (histogram) . . . 341
7.9 Salience-Resolution: Salience scores repartition (violin plot) . . . 342
7.10 Salience-Resolution: Inter-group differences in salience 344
7.11 Salience-Resolution: Resolution scores repartition 345
7.12 Salience-Resolution: Resolution across dialogue groups (violin plot)346
7.13 Salience-Resolution: Inter-group differences in resolution 347
7.14 Salience-Resolution: Correlation 348
7.15 Salience-Resolution: Resolution scores across salience scores . . . 350
7.16 Salience-Resolution: Correlation across dialogue groups 352

D.1 Salience-Resolution: Correlation on the entire annotation scale . . XXIV

E.1 Balloon task: Illustration XXVII
E.2 Tangram task: Illustration XXIX

List of Tables

2.1 Dialogue characterisation – Agency of participants 47
2.2 Dialogue characterisation – Number of agents 51
2.3 Dialogue characterisation – Agents' functions 53
2.4 Dialogue characterisation – Hierarchy between agents 55
2.5 Dialogue characterisation – History between agents 56
2.6 Dialogue characterisation – World setting 58
2.7 Dialogue characterisation – Physical context 61
2.8 Dialogue characterisation – Gaze 64
2.9 Dialogue characterisation – Gestures and Facial expressions . . . 66
2.10 Dialogue characterisation – Prosody & Tone 69
2.11 Dialogue characterisation – Noise 70
2.12 Dialogue characterisation – Sychrony 72
2.13 Dialogue characterisation – Planning 74
2.14 Dialogue characterisation – Ephemerality 75
2.15 Dialogue characterisation – Language 77
2.16 Dialogue characterisation – Practical goal 81
2.17 Dialogue characterisation – Success condition 83
2.18 Dialogue characterisation – Anonymity 85
2.19 Dialogue characterisation – Role instantiation 87
2.20 Dialogue characterisation – Intention 89
2.21 Dialogue characterisation – Commitment level 90
2.22 Dialogue characterisation – World stability 92
2.23 Dialogue characterisation – Extension of Environment & Medium 93
2.24 Dialogue characterisation – Vocabulary 94
2.25 Dialogue characterisation – Topical structure 97
2.26 Dialogue characterisation – Coordination signals 99
2.27 Dialogue characterisation – Turn taking 101
2.28 Dialogue characterisation – Completion 102
2.29 Dialogue characterisation 105

3.1 Number of retrieved papers for the selected venues in 2024 126
3.2 Constraints when studying turn-taking 144
3.3 Degradation of dialogical features from original occurrence to transcription 147

3.4 Description of the dialogical setting in the STAC and Molweni resources . . . 152
3.5 Description of the dialogical setting in MultiWOZ and in TODS outputs . . . 158
3.6 Description of the dialogical setting in the SLAM corpus . . . 167
3.7 Description of the dialogical setting in a therapy context . . . 172
3.8 Example from the multi-user version of MultiWOZ . . . 178

5.1 BERT-based TextTiling – Evaluation datasets statistics . . . 218
5.2 TextTiling – All the tested models . . . 219
5.3 *Friends* corpus – Characterisation . . . 225
5.4 *Friends* corpus – Statistics . . . 227
5.5 Topic segmentation – Results: Random baseline . . . 228
5.6 Topic segmentation – Results: *smart* random baseline . . . 229
5.7 Topic segmentation – Results: BERT-based TextTiling fine-tuned on DailyDialog . . . 235
5.8 Topic segmentation – Models: LLM baselines . . . 237
5.9 Topic segmentation – Results: LLMs . . . 239
5.10 Topic segmentation – Results: Baselines . . . 239
5.11 Topic segmentation – Results: Feature-based models with speaker information . . . 241
5.12 Topic segmentation – Results: Feature-based models . . . 243
5.13 Topic segmentation – Results: BERT-based TextTiling trained for 10 epochs . . . 245
5.14 Topic segmentation – Results: BERT-based TextTiling with different coherence layer settings . . . 248
5.15 Topic segmentation – Results: BERT-based TextTiling trained for two epochs . . . 249
5.16 Topic segmentation – Results . . . 250

6.1 Description of *Reddit* interactions. . . . 265
6.2 *Reddit* corpus – Density statistics . . . 270
6.3 *Reddit* corpus – Thread topology . . . 271
6.4 *Reddit* corpus – Branching factor statistics . . . 272
6.5 *Reddit* corpus – Post statistics . . . 273
6.6 Topic Similarity Ranking – Results: BERT Topic . . . 275
6.7 Topic Similarity Ranking – Results: BERT Topic with valid similarities . . . 276
6.8 Topic Similarity Ranking – Results: Baseline . . . 277
6.9 Topic Similarity Ranking – Results: Congruent models . . . 284
6.10 Topic Similarity Ranking – Results: Model transfer . . . 286
6.11 Topic Similarity Analysis: Similarity statistics . . . 292

6.12 Topic Similarity Analysis: Correlation between comment depth and similarity to post or parent 293
6.13 Topic Similarity Analysis: TextTiling results 294
6.14 Topic Similarity Analysis: Similarity around topical boundaries . 296

7.1 BNC: Category constraints 307
7.2 BNC: Interaction degrees 309
7.3 BNC corpus: Statistics 311
7.4 Salience-Resolution: Inter-annotator agreement 328
7.5 Salience-Resolution: Inter-annotator agreement per dialogue group 329
7.6 Salience-Resolution: Average salience scores per dialogue group . 342
7.7 Salience-Resolution: Average resolution scores per dialogue group 346
7.8 Salience-Resolution: Average resolution scores per salience score 348
7.9 Salience-Resolution: Correlation by dialogue group 349

B.1 Extension of Table 5.5 IV
B.2 Extension of Table 5.6 IV
B.3 Additional comparison: original and dialogue-adapted TextTiling . V
B.4 Extension of Figure 5.10a VI
B.5 Extension of Figure 5.10b VI
B.6 Extension of Table 5.7 VII
B.7 Topic segmentation – Extended Results: LLMs VIII
B.8 Extension of Table 5.11 IX
B.9 Extension of Table 5.12 X
B.10 Extension of Table 5.13 XI
B.11 Extension of Table 5.14 XI
B.12 Extension of Table 5.15 XII
B.13 Additional comparison: Variability in training set XIII
B.14 Extension of Table 5.16 XIV

C.1 Extension of Table 6.7 – Results: BERT Topic with valid similarities (at least one) XV
C.2 Extension of Table 6.7 – Results: BERT Topic with valid similarities (all three) XVI

E.1 Montgolgram: Format XXVI
E.2 Montgolgram: Format XXX
E.3 Montgolgram: Collected dialogues XXX

Prologue

Introduction

(0.1) $\mathbf{A}_1$ i love bunnies! i wish i would open a bunny daycare or hotel.
$\mathbf{B}_2$ that is a cool idea. how are you today?
$\mathbf{A}_3$ a little tired. worked all day today at the shipyard.
$\mathbf{B}_4$ i just got back from school, not a big fan of it though.
$\mathbf{A}_5$ you should join the military. i am ex military. girls love uniforms;
$\mathbf{B}_6$ i have never considered it. maybe it would help me make some more friends.
$\mathbf{A}_7$ i joined the navy because i like navy blue, but infantry has sexy uniforms also.
$\mathbf{B}_8$ i will definitely keep the in mind if i join.
$\mathbf{A}_9$ i donate part of my salary to my bunny rescue. you should too.
$\mathbf{B}_{10}$ do you have any bunnies?
$\mathbf{A}_{11}$ not anymore. i ate them all.
$\mathbf{B}_{12}$ wow! you have quite the appetite.
$\mathbf{A}_{13}$ yeah, i like to eat. what do you do for a living?
$\mathbf{B}_{14}$ i am a receptionist at a dentist office

[PersonaChat (S. Zhang et al., 2018)]

(0.1) is a conversation between two human beings who were asked to interact while displaying a given persona, described to them as a list of sentences like 'I used to own bunnies, but I ate them.'. This example illustrates some of the challenges that arise when studying dialogue. In order to study a specific phenomenon, we often try to gather data that focuses on it. As a result, the conversations we study may not be representative of *naturally occurring* dialogue. Moreover, making people interact in a too constrained way can lead to unnatural behaviours that worsen this lack of representativeness. In this example, asking the participants to display their given persona led them to build all their interaction around it, while we typically do not infuse our conversations so explicitly with our personalities and interests. Besides, the conversation was stored with lowercase letters only and with a space before each punctuation sign, though this last point is not visible in (0.1). This step, that consists in applying several changes to the data, usually to normalise it and make it easier to study automatically, also contributes to making the data less representative of naturally occurring dialogue.

Yet natural language is the base resource in Computational Linguistics (CL) and Natural Language Processing (NLP), the core fields this thesis is situated in.

CL is an interdisciplinary field that leverages computational tools to analyse and model linguistic data. NLP is a more applied field that focuses on the development of computational systems that can process and generate natural language. Both domains rely on data to study and model language, and more recently, NLP has used increasingly large amounts of data to develop, first, machine learning (ML) and deep learning (DL) models, and then large language models (LLMs), that can solve a wide range of tasks in natural language understanding and generation (Karanikolas et al., 2024), though they lack in explainability and interpretability. LLMs have been used to build conversational agents that can interact with humans in a dialogical fashion, like ChatGPT (OpenAI, 2022) or Claude (Anthropic, 2023). These new technologies have led to a surge of studies in NLP meant to improve the performance of LLMs on dialogue tasks through various angles (J. Lee et al., 2024; D. W. Lee et al., 2024; Fan, Li, & Zhu, 2024), one of which being the addition of a persona to the models in order to make them more personalised (Han et al., 2024; Jandaghi et al., 2024; Y.-J. Lee et al., 2024). (0.1) is an example from the PersonaChat corpus (S. Zhang et al., 2018) which was built to train models that would be 'more consistent and engaging' thanks to their 'endow[ment] with a persona'. Such practices question what a conversation *is*. While (0.1) is a conversation between two human beings, it does not feel very natural. Conversely, numerous exchanges with conversational agents now occur every day and dismissing them as not being 'real' conversations also seems problematic. This fuels the need to better understand what a conversation is and with what parameters it can be characterised.

This question is of particular significance to us as one of the motivations that led to this thesis is the analysis of pathological conversations, i.e. conversations that involve a patient whose pathology affects their ability to communicate. One type of disfluency that was observed in real interactions with patients with schizophrenia was linked to a peculiar way to change topics without any explicit signal. In (0.2), the patient (**A**) changes the interpretation of the concept of death which is used figuratively in the first turns (**124** and **125**) before it becomes literal in turn **126**. The lack of explicit signals makes it hard for the psychologist (**B**) to identify the topic shift and they must explicitly ask for a confirmation.

(0.2) **A**$_{124}$ Oh yeah (↑) and complicated (↓) and it's really very very complicated (→) politics, it's really something when you get into it, have to win or else when you lose, well, you're finished (↓) (Fr. *Oh ouais (↑) et pis compliqué (↓) et c'est vraiment très très compliqué (→) la politique c'est quelque chose quand on s'en occupe faut être gagnant parce qu'autrement quand on est perdant c'est fini quoi (↓)*)

B$_{125}$ Yes (Fr. *Oui*)

A$_{126}$ JCD is dead, L is dead, P is dead uh (...) (Fr. *J. C. D. est mort, L. est mort, P. est mort euh (...)*)

B$_{127}$ So you think they're dead because they lost (↑) (Fr. *Ils sont morts parce*

qu'ils ont perdu à votre avis (↑))

A_{128} No they won but if they're dead, it's their disease well it's it's (→) (Fr. *Non ils gagnaient mais si ils sont morts, c'est la maladie quoi c'est c'est (→))*

[SLAM (Amblard, 2018)]

Literature in psychiatry has long discussed the tendency of at least some patients with schizophrenia to produce more word associations than controls (see Maher et al. (2005) for core references and a quantitative assessment of this phenomenon in monologic discourse production). While this behaviour could intuitively lead to unexpected topic shifts, determining whether they can reliably be considered a cue to identify a pathology or whether it can normally happen in at least some types of dialogues requires a general theory of topical organisation in conversation. But this in turn requires a way to characterise the different types of conversations, as topical structure is likely to vary depending on the type of conversation. This thesis thus focuses on two complementary axes: first, we aim to define a characterisation of dialogue that can be used to describe, compare and select dialogue resources, but also define the scope of a study or application; second, we focus on topical structure as a transversal level of analysis. We discuss the core issues of both axes in Sections 1 and 2 respectively, and present our research questions and the organisation of the thesis along which they are answered in Section 3.

1 Studying conversation

Conversation is our primary mode of communication. We engage in dialogue in our everyday lives for social, practical, and professional purposes, often several times a day, sometimes in different languages and increasingly through a variety of media. While dialogue generally feels natural and effortless, it is in fact a unique and constantly renewed experience: one can feel entirely at ease in one interaction and, only moments later, experience difficulty maintaining a conversation with a different interlocutor who feels constantly slightly off-topic.

These variations are not incidental. Dialogue mobilises multiple capabilities simultaneously and requires participants to align, negotiate, and continuously adjust them in cooperation with one another. Conversational success depends not only on linguistic competence, but also on shared assumptions, attentional coordination, sensitivity to interactional cues, and ability to repair disruptions and misunderstandings. The resulting complexity makes dialogue a rich and challenging object of study, and one that has been approached from many different angles across a wide range of disciplines.

Studying it is all the more complicated because of its situated and ephemeral nature. Capturing its essence desperately implies to destroy a part of it due to various constraints. The most direct one is that making people aware that they are being

observed and recorded can alter their behaviour. This is known as the observer's paradox (Labov, 1973). In addition, the very act of recording a conversation can be disruptive, as the presence of recording equipment may alter the natural flow of interaction. Even when these issues are mitigated, open-domain conversations create additional issues related to privacy concerns. Ideally, we would be able to share all the data we collect, as it would allow other researchers to replicate our studies and build upon our work. This ideal, known as Open Science, is a growing movement in the scientific community that aims to make research more transparent and its results more openly accessible. However, sharing data from conversations can be problematic, as it may contain sensitive information about the participants. Open-domain dialogues are particularly challenging in this regard, as the freedom of the participants to discuss any topic can lead to the disclosure of personal information. Amblard et al. (2014) have shown that long exchanges are impossible to fully anonymise as long as a large enough context remains.

Since naturally occurring conversation is so complicated to collect, studies are often based on language data that involved people who were asked to have a conversation in a given setting. Tasks that require interaction have been designed in order to elicit conversation and help participants abstract away from the experimental setting but also to analyse phenomena specific to certain conditions. (0.3) is an example of dialogue from a widely used corpus in the dialogue system community, the MultiWOZ corpus (Zang et al., 2020). In this corpus, participants are asked to play the role of a customer and a clerk in a booking scenario. The task consists in booking a reservation for a restaurant, trip, or other similar domains based on some requirements. The conversation is thus constrained by the task, which can lead to behaviours that differ from more free-form interactions.

(0.3) **USR**$_1$ I need train reservations from Norwich to Cambridge

SYS$_2$ I have 133 trains matching your request. Is there a specific day and time you would like to travel?

USR$_3$ I'd like to leave on Monday and arrive by 18:00.

SYS$_4$ There are 12 trains for the day and time you request. Would you like to book it now?

USR$_5$ Before booking, I would also like to know the travel time, price, and departure time please.

SYS$_6$ There are 12 trains meeting your needs with the first leaving at 05:16 and the last one leaving at 16:16. Do you want to book one of these?

USR$_7$ No hold off on booking for now. Can you help me find an attraction called cineworld cinema?

SYS$_8$ Yes it is a cinema located in the south part of town what information would you like on it?

USR$_9$ Yes, that was all I needed. Thank you very much!

SYS$_{10}$ Thank you for using our system.

[MultiWOZ 2.2 (Zang et al., 2020)]

Even though chatbots intended to fulfil a specific task may not need to fully reproduce the complex dynamics of a human conversation, the design of the dialogue collection experiments that will serve as their training data is crucial. For example in (0.3), the written format is aligned with the typical usage of chatbots, but the constraints imposed on participants in terms of task success may infringe chatbot development. Indeed, the user initially sought to book a train ticket but then refused to book the trip several times before inquiring for information about a cinema and walk out happy of the conversation even though they did not fulfil their original goal. This change of intent could be seen as an expected human behaviour but it makes it harder to define what a successful conversation is in this context.

In other cases dialogue resources are built by asking people to have a conversation with few constraints as in (0.1). Manually written dialogues are also used in numerous studies as they mitigate all the issues related to the collection of naturally occurring dialogues. However, they are also problematic as they may appear artificial and do not necessarily reflect the dynamics of a real conversation. (0.4) and (0.5) are two examples from the manually written corpus DailyDialog (Y. Li et al., 2017), which is also widely used in NLP, in particular for emotion recognition tasks thanks to the emotion labels that are provided for each speech turn. While (0.4) may pass as a somewhat natural conversation, though the basic transcription scheme does not capture any interactional cues, (0.5) is clearly artificial.

(0.4) **A**$_1$ Hey. Why did you take that money? You are such a cheater! I should send you to jail!

B$_2$ I am not cheating. When you pass go, you collect $ 200, Everyone knows that!

A$_3$ Well you can't just take the money. You have to ask the bank for money. And I' m the banker.

B$_4$ Banker?

A$_5$ Yes...

B$_6$ Can I have my $ 200 please?

A$_7$ Sure. Here you are, $ 200, Thank you, please come again! Now it's my turn to roll the dice.

[DailyDialog (Y. Li et al., 2017)]

(0.5) **A**$_1$ Are you going to the annual party ? I can give you a ride if you need one

B$_2$ Thanks a lot. That's the favor I was going to ask you for.

A$_3$ The pleasure is mine.

[DailyDialog (Y. Li et al., 2017)]

Finally the rise of LLMs has increased the use of fully synthetic conversations in NLP. These dialogues are generated by a model, often in a zero-shot setting, i.e. by simply providing a prompt to the model without any additional training. More sophisticated methods have also been developed to enhance some specific phenomena like having two models 'interact' with one another, or have one produce a 'dialogue template', i.e. the frame of a conversation with some slots to fill, that another model, or the same one in a second phase, must fill in order to produce a complete dialogue.

All these methods and examples clearly differ from each other and from face-to-face interactions in some manner. Yet they are all used to study dialogue and to build models that are meant to process or generate it. The central question that arises is thus what do these differences entail for the study of dialogue and how can we characterise them in order to select the most appropriate resource for a given study or application. In particular, topical structure intuitively seems to be affected by the type of conversation, with task-oriented dialogues being more focused and constrained than open-domain ones for instance. We discuss issues in the study of topics and topical structure in Section 2.

2 Issues with studying topics

Topics are not a directly observable feature of language use, but rather an inferred one. Some aspects of conversation are completely linguistic, such as the vocabulary used, while others are more interactional and require more complete transcription formats to be captured, such as overlaps between dialogue turns. Topics however are more abstract and require a more holistic view of the conversation to be identified. They can be defined at several levels or granularities, where we can classify whole conversations based on their global topic, or identify what more or less long sequences of utterances are about, or even label the focus of individual turns.

These multiple levels of granularity imply that clearly defining the goal of a study and the type of data that is analysed are crucial aspects to consider. Identifying topic shifts in a short exchange of a few turns is fundamentally different from doing so in a long conversation. Similarly, working with naturally occurring data versus synthetic data, or with organised dialogues versus more free-form ones, is likely to yield different results. Yet most studies focus on monologic discourse or organised dialogues such as meetings. For instance PersonaChat, the corpus from which (0.1) was taken, has been used to build a benchmark for topic shift aware dialogue modelling (Xie et al., 2021). The topic transitions in (0.1) are rather clear due to the unnaturalness of the dialogue that stems from the constraints imposed on the participants. In contrast, we are interested in naturalistic data where complex topical structures can appear.

STAC (Asher et al., 2016) is a corpus of multi-party chat-based English conversations occurring between players of the game (Settlers of) Catan. In this game, players impersonate settlers each building their own settlement. They must acquire and trade resources to develop them which leads to many exchanges where one asks for a resource, like clay, and the other participants may decline or inquire for another resource in exchange, like sheep or wheat. (0.6) is a piece of dialogue from this corpus where several conversational threads are intertwined.

(0.6) LJ$_{165}$ anyone want sheep for clay?
GWFS$_{166}$ got none, sorry :(
GWFS$_{167}$ **so how do people know about the league?**
WM$_{168}$ no
LJ$_{170}$ **i did the trails**
LJ$_{173}$ **trials even**
TK$_{174}$ **i know about it from my gf**
GWFS$_{175}$ [**yeah me too,**]$_a$ [*are you an Informatics student then, lj?*]$_b$
TK$_{176}$ **did not do the trials**
WM$_{177}$ has anyone got wood for me?
GWFS$_{178}$ [**I did them**]$_a$ [**because a friend did**]$_b$
GWFS$_{179}$ lol wm, you cad
GWFS$_{180}$ afraid not :(
LJ$_{181}$ [*no, I'm about to start math.*]$_a$ [*I just hang around appleton a lot*]$_b$
TK$_{182}$ sry no
GWFS$_{183}$ my single wood is precious
WM$_{184}$ what's a cad?

[STAC (Asher et al., 2016, Ex.1)]

This example shows that topical structure can be more complex than a basic linear sequence, and that only identifying topic shifts is not always sufficient. Here the multitude of participants and the written modality allow players to hold several sub-conversations at the same time. Some participants talk about 'the league' while others have an interaction on resources – typical for this game – and one participant (GWFS) takes part in all of them. While associating a dialogue turn with its topic is not always straightforward in this dialogue, the vocabulary makes it quite easy to disentangle the threads when looking further than isolated sentences. But even this is not always a guaranteed strategy to interpret a conversation.

(0.7) is a piece of naturally occurring dialogue from the British National Corpus (Consortium, 1994b) described as an 'Air UK sales meeting'. While we can understand from the vocabulary used and this description that the global topic is about flights, the conversation is hard to interpret and more specific sub-topics are not easy to identify.

(0.7) **JN7PSUNK**$_1$ Right, you you want thirty seats on that flight?

PS4BS$_2$ No no no can't be done <unclear> haven't got a clue <laugh>oh dear.

PS4BS$_3$ Anyway

PS4BP$_4$ Just thought I'd mention the Stansted leading forward

PS4BP$_5$ That's that's the same thing as says, it's reducing the catchment all the time

PS4BS$_6$ What was it before?

PS4BP$_7$ Seven 0 five

PS4BS$_8$ Yeah well the thing is that something's gotta shift because we've got <unclear> we've got all these bloody flights going out the same time

PS4BP$_9$ Well I know but

PS4BS$_{10}$ And they can't cope

PS4BP$_{11}$ Every time you lead it forward like that those early hours will just reduce

PS4BP$_{12}$ Well that's what I said, it's nearly half an hour later

PS4BS$_{13}$ Well actually it depends on the weather don't it?

PS4BS$_{14}$ It depends on the

PS4BS$_{15}$ Alright well if you've got any little any little points like that then you ought to er say like, we won't actually be much longer.

PS4BS$_{16}$ We'll be er five ten minutes or so

PS4BR$_{17}$ Oh, is that all?

PS4BS$_{18}$ Yeah.

PS4BS$_{19}$ You can sit and wait if you want to

PS4BR$_{20}$ Erm, okay thanks, as long as I don't to er put you off or anything

PS4BS$_{21}$ Yeah, yeah you can sit and wait if you like oh no <unclear> go away and come back again.

Dialogue JN7 [BNC]

These examples show that topical structures can be complex in naturally occurring dialogue, and that even identifying the topics can be challenging. Gruenstein et al. (2005) have shown that annotators struggle to reach an agreement when asked to identify topic shifts in the ICSI Meeting Corpus (Janin et al., 2003), both in a coarse-grained and a fine-grained setting. This corpus contains naturally occurring meetings from the International Computer Science Institute (ICSI), which form open-domain conversations with a pretty loose structure though the overall range of topics is somewhat restricted by the recording location. The high-quality recording setup allowed for the production of great written transcriptions. Compared to

(0.7) where some turns are not fully transcribed and contain `unclear` tokens, and some speakers are labelled as unknown, conversations from the ICSI corpus are probably slightly easier to understand. The two main difficulties, that also exist when studying dialogues from the BNC and many other corpora, are that annotators tend to be unfamiliar with the topic, and that they are external to the conversation. These difficulties are probably largely responsible for the low agreement between annotators when marking topic shifts.

On the same corpus, Galley et al. (2003) used Cochran's Q test (Cochran, 1950) to determine whether 'the number of subjects assigning a boundary at any position was randomly distributed'. However, they did not provide any details on how they actually performed the test. In its original version, Cochran's Q test was intended to compare the number of successes of an event between different groups. In terms of annotations, it means that it would estimate whether all annotators assigned a similar number of boundaries, without accounting for *where* they were assigned. While this estimation may provide useful insights in some biomedical settings, which is the context in which Cochran's Q test was originally developed, it cannot be considered as an inter-annotator agreement measure in the context of topic segmentation. Banerjee and Rudnicky (2007) have shown that when asked to segment meetings into segments that each correspond to a single agenda item and provided with the agenda, annotators could produce very similar segmentations. This shows that text segmentation is not an impossible task but that the definition of what a topic is and the context in which the task is performed are crucial.

For these reasons, building corpora of naturally occurring dialogues annotated with topics is a challenging task as the openness of the conversations makes it difficult to formally define the notion of topic. We thus had to find indirect ways to extract and analyse topical structures as we discuss when detailing the outline of this work and our research questions in Section 3.

3 Focus and contributions

Our work is situated at the crossroads of Computational Linguistics and Natural Language Processing. These two fields, though initially closely related, have diverged in the past few years. The rise of ML, DL, and LLMs has led to a focus on performance in NLP that is detrimental to the study of data and language. We are firmly committed to a data-driven approach, and we try to reconcile CL and NLP by operationalising CL theories in a way that can be used in NLP. In particular, we consider conversation in all its richness and explore the limitations of current approaches to topical structure analysis when applied to complex data. This endeavour requires a way to systematically characterise dialogue data in order to compare approaches and understand how a given resource can influence the results of a study. This constitutes the first axis of our work. The complex and multifaceted

nature of dialogue entails that a characterising framework must cover multiple dimensions more or less entangled with one another. The second axis of our work thus focuses on a transversal level of analysis, topical structure, as it is influenced by many aspects of dialogue and could therefore constitute a discriminative signal to differentiate dialogue types. Within this context the aspects we are interested in can be summarised in the following research questions:

RQ1 How can dialogue be characterised in a way that accounts for its variability while informing the definition of task-specificities and the choice of associated resources?

RQ2 Can topical structure be automatically extracted from dialogue data, and in particular from open-ended multi-party conversations?

RQ3 Can topical structure be leveraged as a discriminative signal for differentiating some dialogue types?

These questions are gradually answered throughout the seven chapters of this thesis, separated into two parts. The relations between the two parts but also the chapters within each part are represented in Figure 1. Part I focuses on the characterisation of dialogue that we envision as a tool to describe, compare and select dialogue resources, but also define the scope of a study or application. Chapter 1 introduces the basic concepts around conversation that are required to understand the rest of the thesis. Chapter 2 presents a theoretical characterisation of dialogue that is based on the literature from various linguistics and cognitive science fields. This chapter also proposes a set of practical definition parameters for conversational data that can be used in a descriptive way but also included in computational models. Chapter 3 is based on these two first theoretical chapters and begins by discussing recent trends in the NLP and CL communities through a survey of recent literature. It then shows how the characterisation developed in Chapter 2 can be used to alleviate some of the issues that arise when studying dialogue. Through the articulation of theoretical grounding and application proposals, this chapter shows our positioning in the NLP field currently increasingly dominated by studies on conversational agents, and suggests directions for a responsible and explainable research on dialogue at the era of LLMs. Together, these three chapters answer RQ1.

The second part, Part II, focuses on topical structure in conversation as a transversal level of analysis and description. It relates three applied studies that explore topical structure from its surface markers to more latent characteristics in different types of dialogue, and shows the complexity of open-domain multi-party conversations. Chapter 4 introduces the core concepts and issues around topical structure that we need to define the scope of our study. Chapter 5 examines the limitations of existing approaches to linear topic segmentation in dialogue. Due to the scarcity of annotated data, we work on transcriptions of the Friends TV show in English,

and show that even scripted dialogues reveal weaknesses in current models. This chapter also highlights the importance of really investigating the data we work on. Chapter 6 probes a more flexible approach to topical structure investigation. It leverages the hierarchy of conversations on *Reddit*, a forum-like social media platform, to explore the evolution of topics throughout a conversation. Based on an approach similar to the one used in Chapter 5, we train a continuous topic similarity model at the level of utterances and highlight the presence of gradual topic shifts besides the usual abrupt ones. Chapters 5 and 6 both participate in answering RQ2. Then, Chapter 7 contributes to answering RQ2 and RQ3 by pondering the predictability of topics in varied interactional settings. The study in this chapter is based on the British National Corpus (Consortium, 1994b), which contains a wide range of naturally occurring dialogues, the ideal type of data for the objectives and philosophy of this thesis. We highlight the difference between monologic data and dialogues in terms of organisation but also the singularity of open multi-party conversations. This difference is crucial in NLP as it shows that models trained on monologic data are not necessarily suitable for dialogue tasks, and that not all types of dialogues are equal. The varied datasets used in this part of the thesis, from fully scripted exchanges to naturally occurring conversations, justify the need for a characterisation of dialogue as developed in Part I, since they were chosen for their specific properties and the way they aligned with the tasks performed in each chapter. Chapter 7 concludes our applied studies on topical structure by explicitly linking the theoretical characterisation of dialogue developed in Part I with an empirical method to differentiate dialogue types based on their topical structure.

Finally, the Epilogue concludes the thesis by summarising our contributions and discussing the limitations of our work. Moreover, it discusses the increasing space dedicated to LLMs in our community and the impact it has on the study of dialogue. It also suggests directions for future research on dialogue and topical structure.

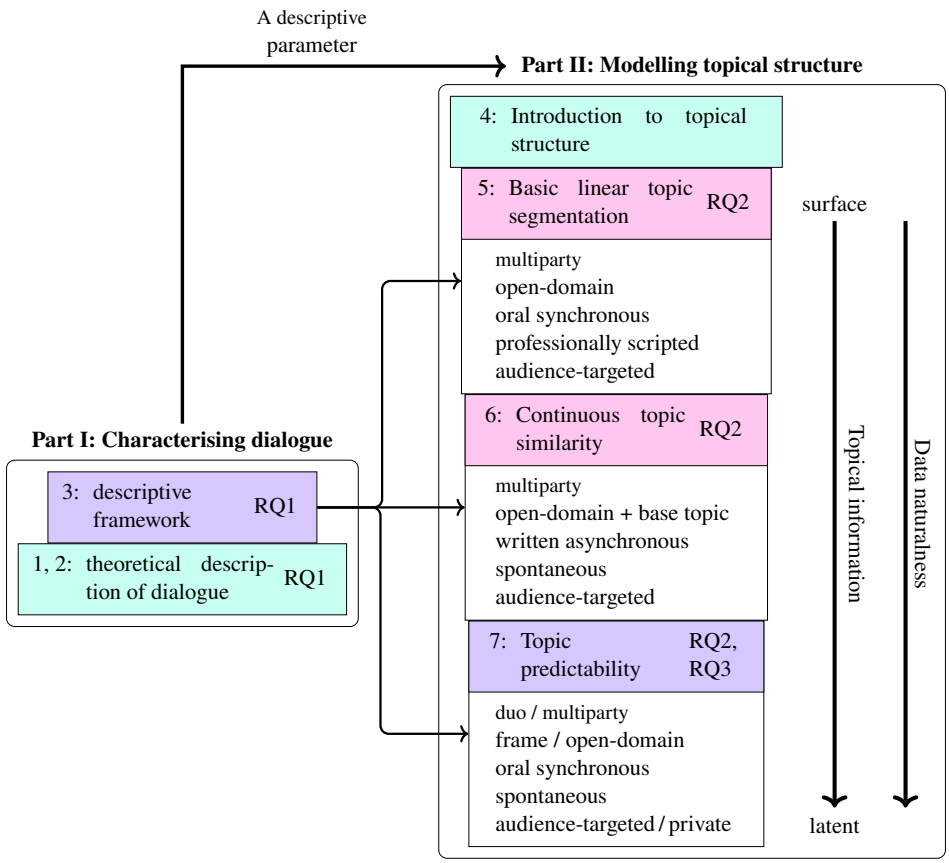


Figure 1: Schematic representation of the thesis structure.

Part I

Linguistic and Cognitive Perspectives on Dialogue: Toward an operational descriptive framework

1

Dialogue as Coordinated Interaction: Foundational Concepts and Perspectives

Contents

1.1 Signals beyond words: Multimodality, redundancy, and monitoring 19

1.1.1 Modes of expression in interaction 19

1.1.2 Redundancy and robustness in communication . . 21

1.2 Coherence and coordination in dialogue 22

1.2.1 Coherence and the cooperative nature of conversation 23

1.2.2 Dialogue as a joint activity 26

1.2.3 Repair and the maintenance of coherence 27

1.3 Acting through language in interaction 27

1.3.1 Utterances as actions 28

1.3.2 Possibilities and constraints in interaction 29

1.4 Representing and coordinating understanding 30

1.4.1 Representing the state of interaction 30

1.4.2 Coordination across participants 32

1.5 Dialogue as a multidisciplinary object of study 34

1.5.1 Pragmatics 35

1.5.2 Formal semantics and dialogue modelling 35

1.5.3 Psycholinguistics 35

1.5.4 Sociolinguistics 36

1.5.5 Conversation Analysis 36

1.5.6 Discourse Analysis 36

1.5.7 Clinical and developmental linguistics 37

1.6 Toward an integrative descriptive framework 37

Conversation is a deceptively simple phenomenon. It is our primary mode of communication and we partake in it daily, in various forms and contexts. Yet it is also remarkably complex, involving the coordination of multiple cognitive, social, linguistic, and interactional processes. The following example illustrates this complexity:

(1.1) PASSENGER star sign (Sw. *stjärntecken*)
DRIVER *(interrupting to clarify a navigation instruction)*
it's not this one (Sw. *det är inte den här det*)
PASSENGER nope (Sw. *nä*)
DRIVER nope (Sw. *nä*)
PASSENGER star sign (Sw. *stjärntecken*)
DRIVER Scorpio (Sw. *skorpion*)

[Villing (2015, p.63)]

This exchange takes place in a driving experiment where the passenger provides navigation instructions while simultaneously conducting an interview where they must gather as many trivia information as possible based on a question sheet. The navigation instructions must be delivered shortly before they are needed and thus cannot be much anticipated, making the navigation task time-sensitive. Hence the interaction is shaped by competing tasks, time pressure – because the passenger must gather as many answers as possible for the interview task – and asymmetries in access to information. Despite its apparent simplicity for the participants, this piece of conversation involves contextual interpretation, fine-grained temporal coordination, as well as syntactically incomplete sentences. Figure 1.1 provides a possible illustration of this exchange, showing the participants and the physical environment in which the interaction takes place. The driver is pointing towards the next intersection, thereby clarifying the 'this one' in 'it's not this one'. Since we do not have access to a detailed description of the interaction, this is only a possible representation of the exchange, but it serves to illustrate the different dimensions of dialogue that we will discuss in this chapter.

In the following we introduce the foundational concepts and perspectives required to investigate the notion of dialogue and its variety of forms. We will use the above example to illustrate its different dimensions and to introduce key concepts. We begin by discussing the multimodal nature of dialogue and the role of these different modes of expression in understanding each other and building a conversation (Section 1.1). We will then discuss how participants strive to maintain coherence in their interaction and how it affects their interpretation of each other's contributions (Section 1.2). In Section 1.3 we will see how language is used to perform actions and how this shapes possibilities and constraints in interaction. In Section 1.4 we will discuss more formal approaches that aim at representing the

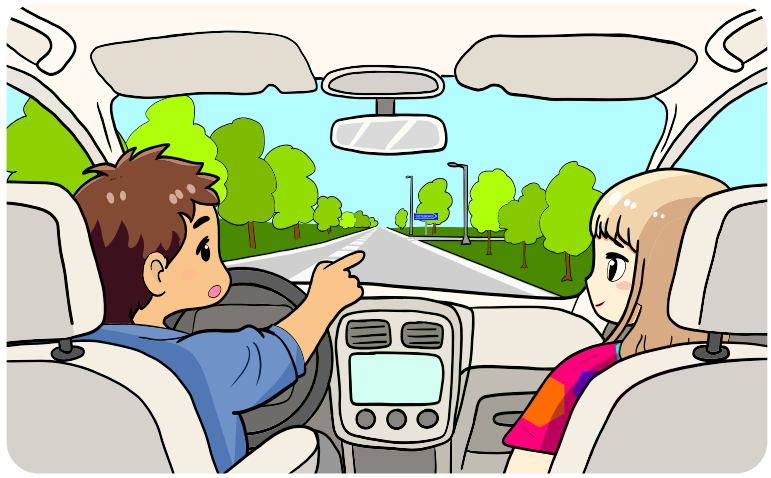


Figure 1.1: Possible illustration of the *Star sign* example (1.1).

different states of interaction and the coordination across participants. Finally, in Section 1.5 we will discuss the multidisciplinary nature of dialogue and how we can integrate these different perspectives in a descriptive framework to characterise dialogue. All of this serves an epistemological purpose where we aim to understand the nature of dialogue and how it can be studied, but it also has a direct impact on the way dialogical data should be studied and on how tools should be developed.

1.1 Signals beyond words: Multimodality, redundancy, and monitoring

Dialogue cannot be reduced to a mere sequence of verbal exchanges. Rather, it is a complex, situated activity that mobilises multiple channels of communication. In addition to linguistic content, interlocutors rely on non-verbal information conveyed through vocal features such as intonation and prosody, as well as through embodied cues including gestures, posture, and facial expressions. These diverse signals jointly contribute to the construction and interpretation of meaning in interaction. In this section we will discuss the different modes of expression that are involved in dialogue and how they contribute to the emergence of understanding (Section 1.1.1). We will also discuss the role of redundancy in making communication more robust (Section 1.1.2).

1.1.1 Modes of expression in interaction

The different channels through which information is conveyed in interaction are commonly referred to as *modalities*. Human communication is inherently *multi-*

modal, in that it simultaneously exploits several such channels. These modalities play a crucial role in the interpretation of dialogue, as they enable interlocutors to convey information that is not always explicitly encoded in linguistic form.

In the example discussed above, the participants are engaged in a face-to-face interaction – albeit in a side-by-side configuration – which affords both auditory and partial visual access to one another. Crucially, it also provides them with a shared perceptual environment: the road. This shared context is essential for the intelligibility of the exchange. Without these additional communicative resources, the sequence of utterances from (1.1) would be considerably more difficult, if not impossible, to interpret, and would likely not arise in the same form. For instance, the driver's interruption relies on the shared visual scene to be understood as pertaining to the navigation task, despite the passenger's expectation of a response related to the interview task. The interpretation of the utterance thus depends on the integration of linguistic input with contextual and perceptual information.

Several modalities are typically involved in dialogue. Speech conveys not only lexical and grammatical information, but also paralinguistic features such as tone and prosody. Tone may contribute to the disambiguation of lexical, grammatical, or morphological distinctions, depending on the language, while prosody plays a central role in expressing paralinguistic meanings, such as emphasis, affect, or irony (Y. Xu, 2019). Facial expressions and gaze behaviour provide cues about interlocutors' attention, engagement, and understanding, as well as their emotional stance. Gestures can serve referential functions, for instance by pointing to entities in the environment, or descriptive functions, by visually representing aspects of the content being communicated. Finally, the physical environment itself constitutes a crucial resource: although external to the participants, it is routinely incorporated into the interaction, whether by grounding references, enabling joint attention, or introducing new topics and referents.

Figure 1.2 provides a schematic representation of the different modalities involved in (1.1), in particular when the driver says 'it's not this one'. The auditory input includes the linguistic content of the utterance which is ambiguous on its own because the referent 'this one' must be identified. However, the visual input that includes both the shared physical environment and the driver's potential gestures, helps explicit the referent of the driver's utterance 'it's not this one'. The auditory input also includes the tone and prosody of the driver, which may indicate that their contribution is a confirmation request, and thus further guide the interpretation of the utterance. These different modalities thus provide complementary information that jointly contribute to the interpretation of the driver's contribution and to the emergence of mutual understanding in this exchange.

From an NLP perspective, multimodality presents challenges for the development of computational models that can effectively process and integrate information from multiple channels. It requires to either encode several modalities into a single representation or to develop mechanisms to account for the missing ones, but

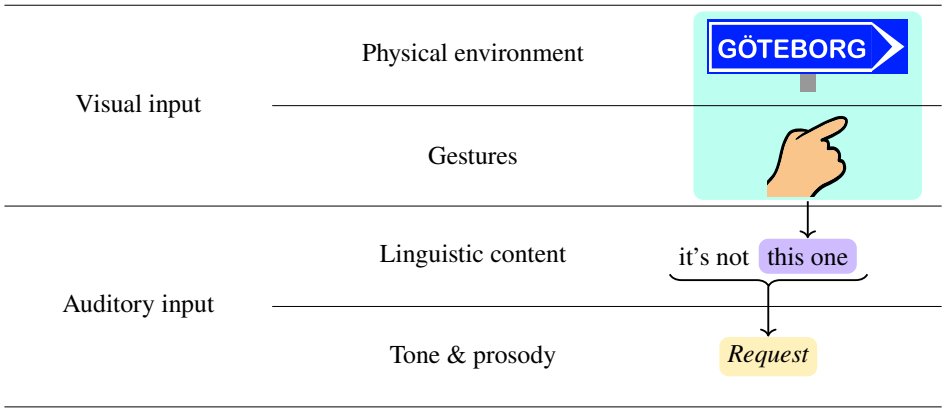


Figure 1.2: Different modalities involved in the *Star sign* example.

also to model their potential interactions and un-homogeneous impact (Yun et al., 2024). Yet most of the data we have access to is unimodal, and in particular textual. Enriched transcription schemes are therefore necessary to capture the multimodal nature of dialogue. Yet such schemes require more expertise from the annotators and are more time-consuming to use. Nevertheless, multimodal dialogue corpora have been created in the recent years. Often task-oriented, some include annotations for the gestures (K. Lai et al., 2024) or gaze and facial expressions (Peechatt et al., 2024), and cover human-human but also human-robot interactions (Lukin et al., 2024). Classical NLP tasks – nowadays mostly designed to support conversational systems – are also increasingly being extended to multimodal settings, such as emotion recognition (Luo et al., 2024) and emotion cause identification (F. Wang et al., 2024), or dialogue state tracking (Khebour et al., 2024) and dialogue breakdown detection (Miah et al., 2024).

1.1.2 Redundancy and robustness in communication

In multimodal interaction, different modalities often convey partially overlapping, or *redundant*, information. That is, content expressed through one channel may also be signalled through one or several others. This redundancy constitutes a central resource for the organisation and stability of dialogue.

A key aspect of conversational interaction is the participants' capacity to attend to one another and to regulate the interaction accordingly. Speakers monitor their addressees to assess whether they are following, while addressees continuously provide feedback regarding their level of attention, interest, and understanding. This process can be described as *mutual monitoring*, whereby interlocutors track both their own communicative behaviour and that of others in order to maintain coordination and adapt when necessary.

Crucially, signals of attention and understanding can be conveyed through mul-

tiple modalities. For instance, addressees may produce visible cues such as head nods, as well as verbal feedback in the form of *backchannels*, i.e. short responses such as 'uh-huh' or 'yeah'. These signals often co-occur or can substitute for one another, thereby ensuring that relevant information is available even if one modality is degraded or overlooked.

Redundancy thus contributes to the robustness of communication. By distributing information across several channels, it allows participants to compensate for potential perceptual or interpretative failures. Moreover, it enables interlocutors to cross-check their interpretations by integrating cues from different modalities.

This can be illustrated by the example discussed above. The driver relies on the shared visual environment to signal that their utterance pertains to the navigation task, despite the passenger's expectation of a response related to the interview task. In addition to this contextual cue, the driver could further disambiguate their intention through gestures, such as pointing towards the road as represented on Figure 1.1, or through head movements orienting the addressee's attention. This redundancy is represented on Figure 1.2 by two aspects of visual input that both contribute to the disambiguation of 'this one'. This increases the likelihood that the intended interpretation is successfully achieved.

An example of NLP task that can benefit from multimodality and redundancy is the task of coreference resolution. In this task, the goal is to identify which expressions in a discourse refer to the same entity. Multimodal information can provide additional cues for resolving coreference, such as visual context or prosodic features. For instance, if a speaker refers to an object while pointing at it, the visual cue can help disambiguate which object is being referred to. By leveraging multimodal information and redundancy, NLP models can improve their performance on coreference resolution tasks. Conversely, ignoring multimodal cues can lead to misinterpretations, especially in cases where linguistic information alone is ambiguous or insufficient.

1.2 Coherence and coordination in dialogue

Conversation is a fundamentally collaborative activity in which participants coordinate their contributions in order to achieve mutual understanding and exchange information, ideas, and intentions. This coordination relies on a set of implicit assumptions that allow interlocutors to interpret utterances without requiring all relevant information to be made explicit. These assumptions were formalised by Grice (1975) under the *Cooperative Principle*. In this section, we first present the conversational maxims that derive from this principle (Section 1.2.1), then discuss dialogue as a joint activity following Clark's framework (Section 1.2.2), and finally examine the mechanisms through which participants repair breakdowns in interaction (Section 1.2.3).

1.2.1 Coherence and the cooperative nature of conversation

Grice's Cooperative Principle states that participants in a conversation typically orient towards making contributions that are appropriate to the ongoing interaction. This general principle is articulated through four conversational maxims: Quantity, Quality, Relation, and Manner.

The maxim of *Quantity* requires speakers to provide as much information as is needed for the current purposes of the exchange, but no more. When an utterance appears to contain either too little or too much information, interlocutors tend to infer additional meaning in order to reconcile this apparent mismatch. In particular, excessive detail may prompt the addressee to search for a reason why such information is relevant, which partly explains why over-explanation can be cognitively taxing in dialogue (Sperber & Wilson, 1986; Kalyuga, 2009).

The maxim of *Quality* concerns the truthfulness of contributions. Speakers are expected to assert only what they believe to be true or for which they have adequate evidence. Apparent violations of this maxim, such as obviously false statements, are often interpreted as signalling non-literal meanings, for instance irony or sarcasm.

The maxim of *Relation* – summarised as 'Be relevant' – requires contributions to be pertinent to the ongoing interaction. Although relevance is notoriously difficult to define formally – even in Grice's own account – participants generally possess strong intuitions about what counts as relevant in a given context. Utterances that appear off-topic are typically interpreted by attempting to reconstruct a context in which they would make sense. In practice, relevance is dynamic and context-dependent: in highly structured exchanges, such as question–answer sequences, strong expectations constrain what counts as an appropriate response, whereas in more open or casual interactions, abrupt topic shifts may be acceptable. In the example discussed above, the driver's interruption constitutes a marked shift from the expected answer to a question. However, it remains interpretable as relevant due to the time-sensitive nature of the navigation task, which justifies the prioritisation of this contribution.

The maxim of *Manner* pertains to *how* something is said rather than to *what* is said. It can be broadly summarised as a requirement to be clear, orderly, and unambiguous. As with the other maxims, this requirement is context-sensitive. In naturally occurring dialogue, utterances are often fragmentary or syntactically incomplete, yet they remain interpretable due to contextual support and multimodal cues. For instance, interrogative force may be conveyed through intonation rather than explicit syntactic marking, or inferred from the interactional context.

Importantly, Grice emphasises that the maxims are not strict rules but rather guiding principles. Their apparent violation is often meaningful: when a maxim is not observed, interlocutors typically assume that this deviation is purposeful and attempt to infer the underlying intention. However, not all deviations are deliberate.

Speakers may inadvertently produce over-informative, irrelevant, or unclear contributions. Moreover, children, non-native speakers, or neurodivergent individuals may not consistently adhere to the maxims, either due to developmental factors or differences in communicative norms (see for example Surian, 1996, and Asada et al., 2022, for the differences in the understanding of the conversational maxims between children with specific language impairment or autism spectrum disorder, and neurotypical children). Interpreting such cases requires balancing the assumption of cooperativeness with sensitivity to individual differences and situational factors. This interpretative process highlights the inherently collaborative nature of dialogue, where meaning is co-constructed by participants in context.

The following example illustrates the maxims and shows how they can be exploited to create humour. This excerpt is taken from the French comedy television series *Kaamelott*, set in the Arthurian legend and well known for its use of language as a comedic resource. In this scene, the two knights Perceval and Karadoc have just discovered the entrance gate of a lost temple they had been searching for, but they are unsure how to proceed. Indeed, while they typically never complete their missions successfully, they have located the entrance to this temple with surprising ease, which they find suspicious. However, the temple is sealed by a gate, which they expect to be locked and therefore impossible to open. This expectation is reflected in Perceval's remark: 'there's a door to open. We never manage that. So we'll be back at Granny's in no time.'

(1.2) **Perceval** Right, let's go! Now that we're here, we're not just going to stand around outside! (Fr. *Bon on y va ! Maintenant qu'on y est on va pas rester devant !*)

Karadoc Sorry, but it stinks. (Fr. *Je m'excuse, mais ça pue.*)

Perceval Do you think so? (Fr. *Vous croyez ?*)

Karadoc Hang on! We always struggle to find things, and now we've found it first time, as if by chance? (Fr. *Attendez ! On rame à chaque fois et là on trouve du premier coup comme par hasard ?*)

Perceval Blimey! What do we do now, then? (Fr. *Merde ! Qu'est-ce qu'on fait alors ?*)

Karadoc No way, I'm not keen on that. (Fr. *Sans blague, j'suis pas chaud.*)

Perceval Mind you, there's a door to open. We never manage that. So we'll be back at Granny's in no time. (Fr. *Remarquez y'a une porte à ouvrir. On y arrive jamais ça. Du coup on s'ra vite rentrés chez mémé.*)

Karadoc Yeah, OK. Do you have any tools? (Fr. *Ouais, ok. Vous avez des outils ?*)

Perceval We don't know any tools that open doors. (Fr. *On les connaît pas les outils qui ouvrent les portes.*)

Karadoc Yeah, but do you have any tools? Even ones that don't open doors? (Fr. *Ouais mais vous avez des outils ? Même qui ouvrent pas les portes ?*)

Perceval No. I've got my shirt, but it's flimsy. I'm afraid it might be a bit too flimsy. (Fr. *Non. J'ai ma chemise mais c'est mou. J'ai peur que ce soit un peu trop mou.*)

Karadoc Well, I've got a spoon. (Fr. *Bah moi j'ai une cuillère.*)

Perceval Yeah, that's not bad; it looks like a key. (Fr. *Ouais c'est pas mal, ça ressemble à une clé.*)

[Kaamelott, S03E61]

At this point, the characters begin discussing tools they could use to open the door. Karadoc first asks Perceval whether he has any tools, to which Perceval replies that they do not *know* any tools that open doors. This response can be interpreted as a violation of the maxim of Relation, as it does not directly answer the question. However, if one assumes that identifying tools capable of opening doors is a prerequisite for answering the question meaningfully, the utterance can be reinterpreted as relevant. This illustrates a case in which the relevance of a contribution is not immediately transparent and depends on assumptions about the speaker's knowledge and reasoning.

Karadoc's subsequent contribution further complicates the interaction. He asks again if Perceval has tools, specifying 'even ones that don't open doors', which also violates the maxim of Relation, since such tools would be useless to the task. Perceval then responds with a straightforward 'No.', which directly answers the question but retrospectively reveals that his previous contribution was unhelpful. In this sense, the earlier utterance may also be seen as violating the maxim of Manner, as it introduces unnecessary complexity without contributing useful information. Indeed, knowledge about which tools open doors would only have been relevant if Perceval had possessed some tools but was uncertain about their utility.

After his 'No.', Perceval continues by adding that he has his 'shirt, but it's flimsy'. This can be seen as a violation of the maxim of Quantity since it provides more information than necessary to answer the question.

Finally, Karadoc mentions that he has a spoon, and Perceval evaluates this suggestion positively, stating that it is 'not bad' because 'it looks like a key'. This final exchange illustrates how humour can arise from a discrepancy between the characters' perspective and that of the audience. While Perceval appears to genuinely consider the spoon a plausible tool, the audience recognises the inadequacy of this reasoning. From the audience's perspective, the utterance thus constitutes a violation of the maxim of Quality, as it involves an obviously implausible claim.

This example is very dense in terms of violations of the conversational maxims, and it illustrates how these violations can be exploited to create humour. While the

dialogue was manually crafted for comedic effect, similar phenomena can be observed in naturally occurring interaction, though often more sparsely. Being able to distinguish *voluntary* violations of the maxims from *involuntary* ones, and understanding the *underlying reasoning* behind a violation, is crucial for interpreting the meaning of contributions in dialogue. Recovering unsaid assumptions and reasoning is a core human capacity in interaction as we often rely on common sense knowledge and world knowledge to produce and interpret contributions. Yet it remains a challenge for NLP models that often lack such knowledge and reasoning capabilities.

1.2.2 Dialogue as a joint activity

Building on this perspective, Clark conceptualises dialogue as a *joint activity* in which participants collaboratively work towards shared goals. On this view, communication is not merely the transmission of information from a speaker to an addressee, but a coordinated process in which both parties actively contribute to the establishment of mutual understanding.

A central notion in Clark's framework is that of *common ground*, defined as the set of knowledge, beliefs, and assumptions that participants take to be shared. Contributions to a dialogue are designed relative to this common ground, and their interpretation depends on it. Establishing and updating common ground is therefore a core function of interaction.

Clark further introduces the concept of *grounding*, the process through which interlocutors provide evidence that an utterance has been understood sufficiently for current purposes. Grounding is achieved through various forms of feedback, including acknowledgements, repetitions, reformulations, or relevant next actions. Crucially, grounding is incremental and adaptive: participants continuously monitor each other's behaviour and adjust their contributions accordingly.

From this perspective, dialogue can be understood as a joint project in which participants coordinate their actions, manage shared resources, and negotiate meaning in real time. This account complements the Gricean framework by emphasising the interactive and procedural dimensions of communication. It has consequences for NLP research, as it highlights the importance of modelling not only the content of utterances but also the dynamics of interaction, including turn-taking, feedback, and the evolution of common ground over time. With regards to dialogue systems, determining the right degree of initiative is crucial to alleviate the asymmetry between the user and the system and to ensure that the user does not need to compensate for poor interactional capabilities. Conversely, the tendency of LLMs to always go along with the user and display sycophantic behaviour (Cheng, Lee, et al., 2026) makes them deceptive conversational partners that only reinforce the user's own biases and beliefs. Finding the right balance in the interactive dynamics of dialogue systems is therefore a key challenge for the development of effective and

trustworthy conversational artificial intelligence (AI). Understanding the principles of dialogue as a joint activity can inform the design of more natural and cooperative interactions, but more crucially it can serve to ensure that users perceive the systems for what they are and do not overestimate their capabilities.

1.2.3 Repair and the maintenance of coherence

While the preceding discussion may have presented dialogue as a smooth and predictable process, everyday experience shows that dialogue is frequently subject to misunderstandings, ambiguities, and disruptions. Rather than being exceptional, such phenomena are an integral part of interaction. The mechanisms that address them are collectively referred to as *repairs* (Purver et al., 2001; Colman & Healey, 2011).

Repairs encompass the practices through which participants identify and resolve problems in speaking, hearing, or understanding. It may be initiated by the speaker or by the addressee (Schegloff et al., 1977), and it can take a variety of forms (Colman & Healey, 2011; Hough & Purver, 2013). For example, an addressee may request clarification or repetition when an utterance is unclear, while a speaker may spontaneously reformulate or correct their own contribution. Similarly, interlocutors may challenge the relevance or accuracy of a statement, prompting further elaboration or correction.

These repair strategies play a crucial role in maintaining coherence in dialogue. They allow participants to restore mutual understanding, realign their interpretations, and ensure the continuity of the interaction despite local breakdowns.

The example discussed above can be interpreted through this lens. The driver's interruption may be analysed as a form of repair, insofar as it serves to clarify a previously given navigation instruction that has become uncertain in light of the evolving situation. By momentarily suspending the expected progression of the interaction, the speaker prioritises the resolution of a potential misunderstanding, thereby contributing to the overall coherence and success of the joint activity.

In NLP, identifying repairs is a downstream task that can support various applications, such as automatic speech recognition, dialogue state tracking, and conversational agents in general. A challenge in this task is that repairs are often considered as disfluencies and are therefore filtered out in the transcription phase or the preprocessing of speech data. However, they are crucial for understanding the dynamics of dialogue and for developing systems that can effectively manage conversational breakdowns.

1.3 Acting through language in interaction

Language is not merely a medium for conveying information or expressing ideas. In everyday interaction, it is also a means of performing actions and effecting changes

in the social and material world. This section examines how language functions as a form of action, and how this action-oriented perspective introduces both possibilities and constraints on interaction.

1.3.1 Utterances as actions

Language has not always been approached as a vehicle for action. In early twentieth-century philosophy, particularly within logical positivism, language was primarily conceived as a tool for making statements that could be evaluated as true or false (Ayer, 1936). According to this view, a sentence was considered meaningful only if it could, in principle, be empirically verified or falsified. As a consequence, many forms of everyday language – such as ethical, evaluative, or metaphysical statements – were deemed meaningless, insofar as they did not straightforwardly describe states of affairs subject to verification (Carnap, 1931).

This perspective was grounded in the assumption that ordinary language is too vague and ambiguous for rigorous scientific inquiry, and that formal, artificially constructed languages were required to achieve clarity and precision (Frege et al., 1879). For example, the sentence 'Bob loves stories on faeries and dragons' can be understood as (1) Bob loving dragons and also stories on faeries, or (2) Bob loving stories on faeries and also stories on dragons, or (3) Bob loving stories that are about both faeries and dragons. There are indeed two ambiguities in this sentence: the first one is on the scope of the verb *loves* and the second one is on the scope of the conjunction *and*. However, if we represent this sentence in a formal language, we can make these readings explicit:

(1) $love(Bob, dragons \wedge stories(faeries))$

(2) $love(Bob, stories(faeries) \wedge stories(dragons))$

(3) $love(Bob, stories(faeries \wedge dragons))$

In this example we use $love(x, y)$ to represent the relation of x loving y, and $stories(x)$ to represent the set of stories about x.

This position on language gave rise to traditions of formal representation, which will be discussed in a later section (Section 1.4). However, it was also met with criticism from philosophers who observed that ordinary language, despite its apparent imperfections, functions remarkably effectively in everyday communication. Rather than replacing it, they sought to understand the principles underlying its use.

Within this alternative perspective, philosophers such as Austin (1975) and Wittgenstein (1953) emphasised the central role of context in determining meaning. Austin, in particular, argued that utterances should be understood not only in terms of what they describe, but also in terms of what they *do*. This insight led to the development of speech act theory, which conceptualises language use as a form of action.

Austin distinguishes three interrelated dimensions of any utterance. The *locutionary act* refers to the production of a linguistic expression with a particular semantic and syntactic structure. The *illocutionary act* corresponds to the action performed in saying something, such as making a request, asking a question, or issuing a promise. The *perlocutionary act* concerns the effects that the utterance has on the addressee, such as persuading, convincing, or prompting them to act. While these three dimensions co-occur in many instances of language use, the illocutionary level is often central to the speaker's communicative intention.

Importantly, the illocutionary force of an utterance is not always directly encoded in its literal form. A classical example is a sentence of the type 'Can you pass me the salt?' where the speaker is not asking about the listener's ability to pass the salt but is making a request to make the listener pass the salt. Here the locutionary act is the production of the sentence itself, the illocutionary act is the *request* to pass the salt, and the perlocutionary act is the listener actually passing the salt. Such cases illustrate that the interpretation of utterances relies on contextual and pragmatic inference, rather than solely on their literal meaning.

The action-oriented view of language has significant implications for the development of dialogue systems as it highlights the importance of considering not only what is said but also what is accomplished by saying it. Integrating this perspective into the design of conversational agents would participate in limiting the risk that they produce *deceptive* utterances that commit them to actions they cannot perform, whether because they are not designed to perform them or because their lack of embodiment and agency prevents them from doing so (Ranisch & Haltaufderheide, 2025; Z. Wu et al., 2026).

1.3.2 Possibilities and constraints in interaction

If utterances are understood as actions, then dialogue can be seen as a space in which participants continuously shape and respond to opportunities for action. Each contribution both opens up and constrains the range of possible subsequent moves. In this sense, interaction is structured by a dynamic set of *affordances* and constraints that emerge from the evolving context.

Certain utterances project specific types of responses. For instance, questions make answers relevant, requests make compliance or refusal relevant, and offers make acceptance or rejection relevant. These expectations, often referred to as *adjacency pairs*, contribute to the overall coherence of dialogue by narrowing the space of appropriate next actions. At the same time, these constraints are not absolute: participants may delay, suspend, or transform expected responses, for example through embedded sequences or topic shifts, as discussed earlier.

More generally, the possibilities for action in interaction are shaped by multiple factors, including the current state of the dialogue, the shared context, and the participants' goals. Multimodal resources also play a role in this process, as ges-

tures, gaze, and the physical environment can introduce new affordances or redirect attention. Conversely, constraints arise from social norms, task demands, and the need to maintain mutual understanding.

From this perspective, dialogue is not only a process of exchanging information, but also a coordinated activity in which participants navigate a structured yet flexible space of possible actions. Understanding how these possibilities and constraints are managed is essential for analysing how interaction unfolds and how meaning is jointly constructed. This action-oriented view of dialogue has clear implications for the design of conversational systems, which must be capable of recognising and responding to the affordances and constraints present in human interaction. It also highlights the importance of modelling not only the content of utterances but also their functional role in shaping the trajectory of the conversation. This idea is at the root of dialogue state tracking (Jacqmin et al., 2022), which is a core component of many dialogue systems. By maintaining an updated representation of the interactional state, such systems can anticipate user needs, manage turn-taking, and provide contextually appropriate responses.

1.4 Representing and coordinating understanding

As discussed above, early twentieth-century movements in philosophy and the theory of science advocated the development of formal languages governed by explicit and unambiguous rules, with the aim of supporting rigorous reasoning. This perspective gave rise to a range of formal representation traditions. While initially designed to capture the semantic content of utterances, these approaches have since been extended to model broader aspects of interaction, including the evolving state of dialogue and the coordination between participants.

In this section, we examine how such formalisms can be used to represent dialogue. We first consider how the state of an interaction can be modelled (Section 1.4.1), and then turn to the question of how coordination across participants can be captured within these frameworks (Section 1.4.2).

1.4.1 Representing the state of interaction

Dialogue is a complex phenomenon involving multiple interdependent dimensions, including linguistic content, contextual information, participants' goals, and the evolving structure of the interaction. A central objective of formal approaches to dialogue is to provide representations that capture these dimensions in a systematic and tractable way.

One important motivation for such representations is the need to account for the incremental nature of dialogue. Interactions unfold over time, with each contribution updating the current state and shaping the range of possible subsequent moves. Formal models therefore aim not only to describe static configurations, but

also to characterise how the state of the interaction evolves in response to new utterances. In particular, they seek to make explicit how a given conversational move transforms the current state into a subsequent one.

This dynamic perspective is especially relevant for computational applications, where it enables the implementation of dialogue systems capable of tracking and updating interactional states. By representing dialogue in a structured way, it becomes possible to model expected transitions, detect deviations, and analyse the conditions under which interaction succeeds or fails.

Frameworks such as dialogue scorekeeping models (D. Lewis, 1979) or gameboards (Ginzburg, 2012) exemplify this approach. They provide structured representations of the interaction at any given point, with varying degrees of granularity depending on the phenomena of interest. Such representations typically include information about the current and previous utterances, the commitments, and other relevant aspects of the conversational context. By formalising these elements, they allow for the computation of state changes as the dialogue progresses.

Figure 1.3 is an example of formal representation of the state of the conversation after the driver's utterance 'it's not this one' in (1.1). This representation is based on the Type Theory with Records (TTR) framework developed by Cooper (2005, 2023), which provides a flexible and expressive formalism for modelling dialogue. In this representation, we use records, modelled by brackets, that contain fields representing different aspects of the interaction. In our example, we focus on two main fields: `physC` for the physical context and `l-m` for the linguistic moves.

In the `physC` field, we represent the shared physical context, including the variable x that denotes an entity[1], and the condition c_1 that captures the fact that x is an intersection of the road.

In the `l-m` field, we separate the linguistic moves between the current one (`curr`) and the previous ones (`prev`). For the sake of simplicity, we do not explicitly represent the content of the `prev` field and instead only specify its type as *RecType*, which stands for 'Record Type' and denotes a collection of labelled fields. In practice, this field should be seen as a recursive memory of the previous moves, which would be updated after each new utterance. It could contain a field for the latest move and one for the previous ones, similarly to the `l-m` field, which would allow us to keep track of the entire history of the interaction. Evaluating the degree of accessibility of this history is a separate question related to memory management.

The `curr` field contains a record that represents the current move, including the speaker (`spkr`), the addressee (`addr`), and the logical content of the utterance (`ctnt`). The content field itself is a record that includes the condition $\mathtt{c}_2$ representing the meaning of the driver's utterance. Here we represent the content as a negation of the proposition $P(physC.x)$, which can be read as 'it is not the case

[1]In TTR entities in the Montagovian sense are denoted by the type 'Individual' (*Ind*) (Cooper, 2023, p.14).

$$
(1.i)\quad
\begin{bmatrix}
\texttt{physC}: & \begin{bmatrix} \texttt{x} : \textit{Ind} \\ \texttt{c}_1 : \textit{intersection}(\texttt{x}) \end{bmatrix} \\
\texttt{l-m}: & \begin{bmatrix} \texttt{prev} : \textit{RecType} \\ \texttt{curr}: & \begin{bmatrix} \texttt{spkr} = \textsc{Driver} : \textit{Ind} \\ \texttt{addr} = \textsc{Passenger} : \textit{Ind} \\ \texttt{ctnt}: \ \left[\texttt{c}_2 : \neg P(\textit{physC}.\texttt{x})\right] \end{bmatrix} \end{bmatrix}
\end{bmatrix}
$$

Figure 1.3: Example of a TTR representation of the driver's utterance 'it's not this one' when both participants pay attention to the road.

that the intersection validates a certain property P', where P is unspecified here but clearly exists in the participants' minds either due to a previous utterance – most likely a navigation instruction – or due to the shared context of the navigation task. We refer to the intersection as $physC.x$ to indicate that we are referring to the x defined in the field `physC`.

While the representation in Figure 1.3 focuses on a limited set of dimensions for the sake of clarity, the TTR framework is in principle capable of modelling a much wider range of aspects of interaction. Specifically, it allows for the distinction between *shared* and *private* components of the dialogue state. Shared components typically include publicly accessible information, such as established commitments or the history of accepted utterances, whereas private components capture participant-specific information, such as individual goals, beliefs, or agendas. This distinction is central to modelling the asymmetries that may arise in interaction. Extensions of TTR, such as the KoS (a toponym for Conversation Oriented Semantics) framework (Ginzburg, 2012; Maraev et al., 2018), further develop these ideas by providing more specialised tools for representing dialogue structure and dynamics.

Crucially, such formalisms make it possible to represent not only a single, global state of the interaction, but also multiple, potentially divergent representations corresponding to different participants. This perspective shifts the focus from dialogue as a monolithic object to dialogue as a coordinated activity grounded in the alignment of individual representations. The question of how these representations relate to one another, and how coherence is maintained across participants, leads directly to the issue of coordination, which we address in Section 1.4.2.

1.4.2 Coordination across participants

Traditional approaches to language often treated discourse as an abstract object, with relatively little distinction between monological production and interactive dialogue. Although the term *discourse* can encompass both, this perspective tends to

obscure the fundamental differences between dialogue and monological discourse, in particular the interactive nature of dialogue as a social practice (Couper-Kuhlen & Selting, 2017; Brennan et al., 2018).

More recent approaches emphasise that dialogue must be studied as a situated interaction between participants. From this perspective, representing a conversation requires accounting not only for a global interactional state, but also for the potentially differing representations maintained by each participant. Rebuschi et al. (2014) show that it can for example help account for misunderstandings in pathological conversations.

In formal terms, this implies that each interlocutor may hold their own representation of the dialogue state, including their beliefs about the shared context and about the other participants' knowledge and intentions. These representations may themselves include higher-order beliefs, giving rise to recursive structures of the form 'I know that you know that I know', and so on. While such recursive embedding is, in principle, unbounded, in practice it is constrained by cognitive limitations and task demands. The finite or infinite nature of these representations has been a subject of debate in the literature and mostly depends on the approach chosen to model common ground (Geurts, 2024).

These participant-specific representations provide a useful framework for analysing coordination in dialogue. Successful interaction can be characterised, at least in part, as a form of alignment between participants' representations of the current state. Conversely, misalignment – where participants maintain incompatible or divergent representations – offers a natural account of many forms of misunderstanding and breakdown.

For instance in the *Star Sign* example (1.1), if the passenger is not paying attention to the road, she may fail to update her representation of the physical context in a way that aligns with the driver's. As a result, when the driver utters 'it's not this one', the passenger would not manage to resolve the anaphora as the driver expects, leading to a misinterpretation of the utterance and a potential breakdown in the interaction. Figure 1.4 illustrates the passenger's representation of the interaction in this case. The field `physC` contains an empty record because the passenger is not attending to the road. The content of the driver's utterance is still represented in the `l-m` field, but the passenger's failure to update the physical context prevents her from correctly interpreting it. Instead of disambiguating 'this one' with $\texttt{physC}.x$, we introduce an entity x that we leave unspecified, and the content of the utterance is represented as $\neg P(x)$. This representation cannot be aligned with the one in Figure 1.3 because the passenger's representation of the physical context does not contain the relevant information to resolve the anaphora, which leads to a misinterpretation of the utterance.

From an analytical perspective, such misalignments can often be reconstructed *a posteriori*, based on observable disruptions in the interaction. However, because these representations involve internal cognitive states, they are not directly acces-

(1.ii)
$$\begin{bmatrix} \texttt{physC} : [] \\ \texttt{l-m} : \begin{bmatrix} \texttt{prev} : \textit{RecType} \\ \texttt{curr} : \begin{bmatrix} \texttt{spkr} = \textsc{Driver} : \textit{Ind} \\ \texttt{addr} = \textsc{Passenger} : \textit{Ind} \\ \texttt{ctnt} : \begin{bmatrix} \texttt{x} = ? : \textit{Ind} \\ \texttt{c}_2 : \neg P(\texttt{x}) \end{bmatrix} \end{bmatrix} \end{bmatrix} \end{bmatrix}$$

Figure 1.4: Example of a TTR representation of the driver's utterance 'it's not this one' interpreted by the passenger when she is not paying attention to the road.

sible and must be inferred. This limits the extent to which breakdowns can be predicted in advance, although formal models can still provide valuable insights into their structure and dynamics.

From an applied perspective, these representations are particularly relevant for the design of interactive systems. A system that maintains an explicit representation of the dialogue state, along with a model of the user's likely representation, can in principle detect emerging misalignments and either prevent or repair them. This highlights the practical importance of formalising coordination in dialogue.

(1.1) illustrates this point. A potential misalignment would arise if the passenger were to interpret the driver's utterance 'it's not this one' as relating to the interview task rather than to the navigation task. In such a case, the anaphoric expressions 'it' and 'this one' would be resolved within an inappropriate context, leading to misunderstanding and necessitating repair. This type of divergence can be naturally captured within a framework that models participant-specific representations of the interaction.

1.5 Dialogue as a multidisciplinary object of study

The preceding sections have highlighted the multifaceted nature of dialogue and the wide range of processes involved in its production and interpretation. This complexity is reflected in the diversity of disciplines that have sought to study dialogue, each bringing their own theoretical frameworks, methodologies, and research objectives. In this section, we briefly outline the domains most relevant to the present work. This overview is not intended to be exhaustive; rather, it situates our approach within a broader intellectual landscape and emphasises complementary perspectives on dialogue.

1.5.1 Pragmatics

Pragmatics studies language use in context, with a particular focus on how meaning is constructed beyond the literal content of utterances (Kroeger, 2019). Central notions include speaker intention, inference, implicature, and presupposition (Grice, 1975; Levinson, 1983). A key insight of this field is that a substantial portion of communicated meaning is not explicitly encoded in linguistic form, but must be inferred on the basis of a combination of linguistic form, shared assumptions and context.

In the example discussed above, the utterance 'star sign' does not constitute a complete sentence. Its interpretation as a request for information depends on the shared understanding of the interview task, as well as on culturally grounded knowledge about astrological signs. Pragmatics thus foregrounds the role of inference and context, while also interfacing with other domains that address how such inferences are processed and coordinated in real time.

1.5.2 Formal semantics and dialogue modelling

Formal semantics and dialogue modelling approaches aim to represent meaning and interaction using logical and computational frameworks (Kamp & Uwe, 1993; Ginzburg, 2012). These approaches typically conceptualise dialogue in terms of information states, which track what has been said, what is mutually accepted, and how the conversational context evolves over time (D. Lewis, 1979).

Such models are particularly well suited to computational applications, including dialogue systems, where explicit and manipulable representations of conversational structure are required. They offer a high degree of precision and abstraction, enabling the formalisation of dialogue dynamics independently of specific linguistic realisations. However, this level of abstraction may come at the cost of simplifying or under-representing the situated, embodied, and temporally constrained aspects of interaction. As a result, formal approaches capture certain dimensions of dialogue while leaving others underspecified.

1.5.3 Psycholinguistics

Psycholinguistics investigates the cognitive processes underlying language production and comprehension (Levelt, 1983; Pickering & Garrod, 2004). Dialogue constitutes a particularly demanding setting, as production and comprehension typically occur simultaneously and must be tightly coordinated.

Although the present work does not directly model cognitive mechanisms, psycholinguistic findings remain highly relevant for characterising dialogue. Factors such as cognitive load (Khawaja et al., 2012; Demberg et al., 2013) and attention (Keysar et al., 1998) can influence the form and timing of contributions (Villing, 2015) but also the mutual understanding, also called dialogue interpreting (Adler,

2024). In the example discussed above, the dual-task context likely affects how participants manage turn-taking, formulate utterances, and allocate attention. Psycholinguistics thus provides an important perspective on the constraints that shape conversational behaviour.

1.5.4 Sociolinguistics

Sociolinguistics and linguistic anthropology approach language as a social and cultural practice (Fishman, 1970). They examine how linguistic variation both reflects and constructs social identities, roles, and norms. From this perspective, dialogue is not only a means of exchanging information, but also a way of maintaining social relationships and enacting social structures.

Conversational norms – such as expectations regarding turn-taking, politeness, or topic management – play a central role in shaping interaction. For example, interruptions may be evaluated differently depending on participants' roles and the situational context. In task-oriented settings, deviations from normative patterns may be tolerated or even required due to competing priorities.

More broadly, some accounts propose that language evolved, at least in part, as a mechanism for social bonding in large groups (R. Dunbar, 1996). From this perspective, conversation can be understood not only as informational exchange but also as a form of social coordination. This highlights the importance of considering the social embedding of dialogue, which is often less central in formal approaches.

1.5.5 Conversation Analysis

Conversation Analysis (CA) is an empirically grounded approach, originating in ethnomethodology, that examines the fine-grained organisation of naturally occurring interaction (Sacks et al., 1974; Schegloff & Sacks, 1973; Schegloff, 2007). It focuses on the sequential structure of talk, including turn-taking mechanisms, adjacency pairs, and repair practices.

CA demonstrates that dialogue is highly structured at the micro-interactional level. Phenomena such as interruptions, repetitions, and self-repairs – often treated as disfluencies – are in fact integral to the organisation of interaction. The example discussed earlier illustrates several such mechanisms, including the interruption of an ongoing sequence and the use of repetition to re-establish coherence.

1.5.6 Discourse Analysis

Discourse analysis extends the study of language beyond individual turns to examine larger structures such as coherence, topic development, and rhetorical organisation (Mann & Thompson, 1988; Asher & Lascarides, 2003). It provides tools for analysing how meaning unfolds over extended stretches of interaction and how discourse is segmented into meaningful units.

This perspective complements conversation analysis by addressing higher-level organisation. In the example, the alternation between navigation instructions and interview questions reflects a form of discourse-level structuring, where multiple activities are interleaved within the same interaction.

1.5.7 Clinical and developmental linguistics

Clinical and developmental linguistics focus respectively on language disorders (Armstrong et al., 2026) and on language acquisition (Bailey, 1981). Both domains provide valuable insights into the mechanisms underlying dialogue by examining how they emerge, develop, or break down.

Interaction plays a central role in language acquisition, as communicative abilities are developed through participation in dialogue. Conversely, disruptions observed in clinical populations can reveal the structures and processes that underpin typical interaction. These perspectives are particularly relevant for applications concerned with identifying, modelling, or supporting atypical conversational behaviour.

1.6 Toward an integrative descriptive framework

The diversity of approaches to dialogue reflects its complexity, but it also creates important challenges. Existing frameworks are typically tailored to specific research goals and levels of analysis. As a result, it is often difficult to compare findings across studies or to generalise insights across different types of dialogue.

This issue becomes particularly salient in contemporary contexts, where dialogue data vary widely: controlled experimental settings, naturally occurring conversations, human–machine interactions, and multimodal exchanges. Without a shared descriptive framework, it becomes difficult to assess how differences in data collection, annotation, or modelling assumptions influence the conclusions drawn.

Rather than converging toward a unified account, existing fields approach dialogue from distinct perspectives, each foregrounding particular aspects of interaction while leaving others in the background. Conversation thus does not admit a single, stable definition, but emerges as a composite object shaped by multiple theoretical traditions. This multiplicity motivates the need for an integrative perspective.

Figure 1.5 provides a schematic representation of the main domains of dialogue research and their relationships. It is not intended to be exhaustive, but rather to illustrate the diversity of approaches and the potential for integration. Each domain captures a particular dimension of dialogue, and the connections between them indicate areas of conceptual overlap and potential synergy.

Two main clusters can be identified in this representation. The first cluster, circled in pink, includes Conversation Analysis, Sociolinguistics, Clinical and Developmental Linguistics, and Psycholinguistics. These fields share a focus on the

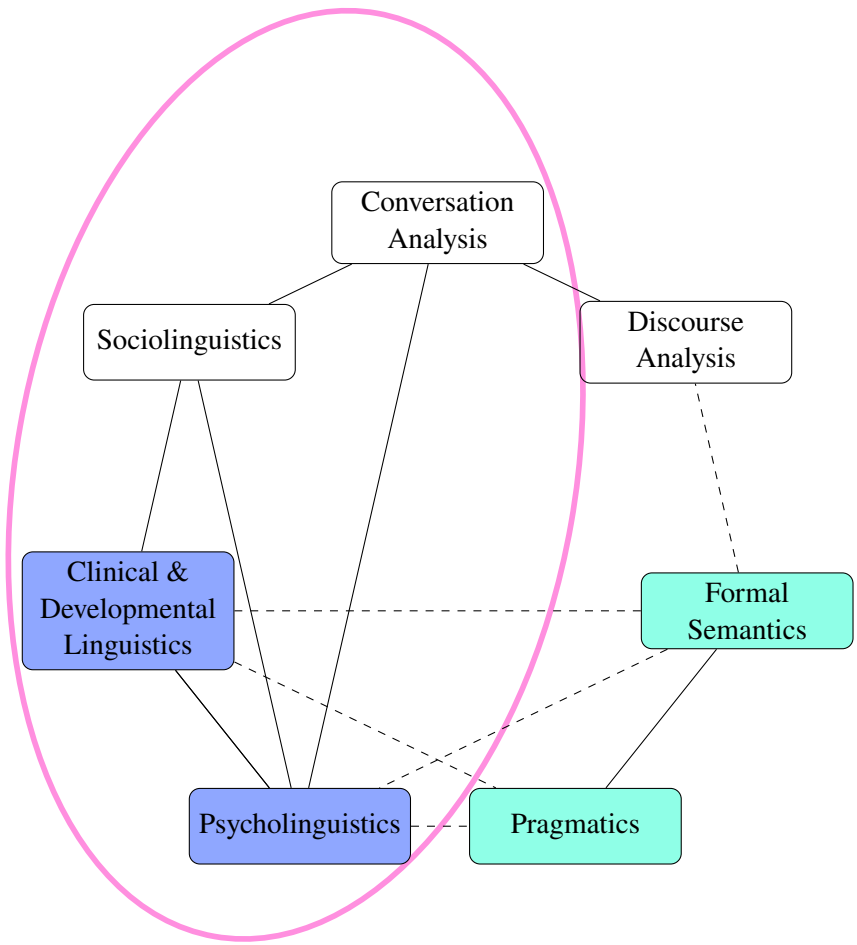


Figure 1.5: Schematic representation of the main domains of dialogue research and their relationships. Full connections indicate strong conceptual interactions while dashed ones suggest weaker or more indirect relationships.

situated, social, and cognitive dimensions of dialogue. They often rely on empirical data from naturalistic interactions and emphasise the importance of context and participant perspectives. The second cluster includes two pairs, respectively coloured in blue on the left and teal on the right, that are weakly connected between them but strongly connected within themselves: Formal Semantics and Pragmatics on the one hand, and Psycholinguistics and Clinical and Developmental Linguistics on the other hand. The first pair shares a focus on the formal representation of meaning and interaction, while the second pair is concerned with the cognitive processes underlying language use and development. These two pairs are weakly connected because although they operate at different levels of analysis, they can inform each other. For instance, formal models can provide analytical tools that help highlight differences in the way dialogue is structured across different contexts and populations, while psycholinguistic findings can constrain and refine formal representations.

The lack of dense connections between the two clusters reflects the fact that there is no unified account of dialogue that integrates insights from all these domains. Each field has developed its own theoretical frameworks and methodologies, which are often not directly compatible with those of other fields. This fragmentation can lead to difficulties in comparing findings across studies and in synthesising insights into a coherent picture of dialogue.

While attempting to produce a single unified theory of dialogue may be unrealistic, and would likely fail to capture the richness of the phenomenon, we can still aim to develop a descriptive framework that integrates insights from different traditions and allows us to characterise resources in a way that is both comprehensive and flexible. The idea is to identify key dimensions of dialogue by considering face-to-face conversation as a reference point and then to describe how different types of dialogue data and models relate to these dimensions. Indeed, according to Fillmore (1981, p. 152), 'The language of face-to-face conversation is the basic and primary use of language, all others being best described in terms of their manner of deviation from that base'. Such a framework would not only help us understand the different dimensions of dialogue but also provide a common tool to describe and compare different types of dialogue data and models.

The first part of this thesis is dedicated to the presentation of such a framework. In Chapter 2 we present a theoretical characterisation of dialogue that integrates insights from different disciplines and traditions. In Chapter 3 we apply this framework to several theoretical scenarios to explain how it could help our community get out of some current issues we have in the field.

2

Parameters for Characterising Dialogue

Contents

2.1 Conversation agents 44
2.1.1 Agency 46
2.1.2 Participation structures 48
2.1.3 Social dynamics 54
2.2 Conversational setting 56
2.2.1 World setting 56
2.2.2 Environment and medium 57
2.3 Conversational frame 75
2.3.1 Language 75
2.3.2 Purpose 77
2.4 Dependant descriptive parameters 83
2.4.1 Influence on the agents: Enunciative status 84
2.4.2 Conversational setting 90
2.4.3 Influence on the frame 93
2.4.4 Verbal interaction: The co-constructed nature of dialogue 96
2.5 On the operationalisation of the framework 102
2.5.1 A formal account of CHIMERA 104
2.5.2 A vector space perspective on CHIMERA 116

As semanticists, pragmaticists, and scholars of language more broadly, our ultimate goal is to understand people's linguistic behaviour in its full complexity and diversity. Conversation is one of the most fundamental human activities, and face-to-face interaction constitutes the primary environment in which language is learned, practised, and culturally shaped. It is therefore natural to treat co-present dialogue as the prototypical case against which other forms of communication can be compared (Fillmore, 1981; Levinson, 1983; Clark, 1996). Yet this 'naturalness' should not obscure the fact that conversation is a cognitively demanding activity.

Speakers deploy a range of multimodal and interactional strategies – redundancy across modalities, rapid repairs, prosodic cues, gaze, gesture – to sustain mutual understanding, align perspectives, and manage the flow of interaction.

This diversity implies that tools developed to study, model, or simulate conversation must be sensitive to the specific features of the type of dialogue they are applied to. Yet conversation is complex to characterise in a systematic way that would allow us to compare different types of dialogue, and to identify the resources suitable for a given task and the relevant phenomena to model and evaluate.

The goal of this chapter is therefore to bring together insights from interactional linguistics, pragmatics, psycholinguistics, communication studies, and computational linguistics to outline a set of core parameters for characterising dialogue. We describe the features that make face-to-face interaction the prototypical conversational environment, examine how alternative forms of communication vary along these dimensions, and discuss what these differences imply for the study and modelling of dialogue. There have been other attempts at characterising dialogue types or data in general, with different purposes and degrees of operationalisation. Notions such as *language game* (Wittgenstein, 1953), *speech genre* (Bakhtin, 2010a) and *(joint) activity type* (Levinson, 1979; Clark, 1996) have been used in the literature, all defined by sets of sometimes overlapping features; see Ginzburg and Wong (2024, Section 3) for a more exhaustive review. Clark, Brennan, et al. (1991) for instance proposed a set of features – like 'copresence', 'instantaneity' and 'self-expression' – to characterise face-to-face interactions, later further discussed by Clark (1996) to describe the differences of other less prototypical conversational settings. These works are primarily descriptive, accompanied with examples, in order to explain and illustrate the different features. Ginzburg and Wong (2024) propose a more formal characterisation of dialogue types aiming to define conversational types in terms of a set of rules, a starting and a final state, and a grammar that describes the linguistics moves that can be used. Those works are motivated by an epistemological goal, that is to understand the nature of dialogue and its different types, and propose models to describe them. In contrast, the recent years have seen a growing use of varied types of language data for the development of different NLP systems. Where the works mentioned above were grounded into theoretical considerations from linguistics-oriented disciplines, NLP systems are often developed with a more applicative goal in mind by people with more technical backgrounds. This perspective, less rooted in linguistics, comes with a weaker interest in the characterisation of the type of dialogue used for the development of the system. Bender and Friedman (2018) propose data statements to remedy this situation and better understand systems' generalisation capabilities but also 'reduce exclusion and bias in language technology'.

Taking inspiration from these existing works, we want to leverage the richness of the existing literature to propose a set of parameters that can be used to characterise dialogue in a systematic way beneficial to a wider community. Our ultimate

aim goes beyond theoretical considerations: it is to provide a descriptive tool that would enable researchers to compare different types of dialogue, assess the suitability of resources for a given task, and identify the phenomena that are relevant to model and evaluate.

The following piece of conversation illustrates the main categories of parameters we will discuss in this chapter. This dialogue is part of a corpus of French conversations between a psychologist and a patient with schizophrenia – the pathological aspect being irrelevant for this particular example.

(2.1) **G234_1** mm hmm (0.5s) do you have a cellphone? (Fr. *humhum (0,5s) vous avez un téléphone vous ?*)
R235_1 I stopped the cellphone (Fr. *j'ai arrêté le téléphone*)
G236_1 stop (Fr. *arrêter*)
R237_1 stop (Fr. *arrêter*)
G238_1 completely (Fr. *complètement*)
R239_1 complete oulac (xxx) (1s) (Fr. *complet oulac (xxx) (1s)*)
air conditioner noises
R239_2 wouallou but (0.5s) it's (0,5s) (Fr. *wouallou mais (0,5s) c'est (0,5s)*)
G240_1 wouallou (Fr. *wouallou*)
R241_1 it's okay it's a (1.5s) (xxx) (Fr. *c'est bon c'est un (1,5s) (xxx)*)
G242_1 (xxx)
G242_2 you see the (tech-) (Fr. *vous voyez la (tech-)*)
R243_1 an old air conditioner (Fr. *un ancien climatiseur*)
G244_1 ah (Fr. *ah*)

[SLAM (Amblard, 2018)]

In this dialogue, two *participants* interact in a face-to-face setting. Their conversation revolves around technology – specifically, around owning a cellphone – when a loud noise from the air conditioner interrupts the exchange. This disruption prompts a shift in topic, initiated by the utterance « wouallou mais c'est » (En. *'wouallou but it's'*). This utterance combines French and Arabic, reflecting the speaker's bilingual background as an Algerian participant. It illustrates that *language* is a non-trivial parameter in conversational analysis: even in corpora labelled as monolingual, code-switching may naturally occur. The interruption caused by the air conditioner highlights another key dimension, namely the *environment* in which the interaction takes place. External factors can shape the trajectory of a conversation, sometimes triggering unplanned topic shifts. This aspect is closely related to the *medium* of communication, which determines the extent to which environmental influences are perceptible. For instance, the shift prompted by the air conditioner would likely not have arisen in a text-based exchange. The management of this topic shift further depends on *verbal interaction*, as participants must

coordinate to accommodate the new focus while maintaining conversational coherence. Finally, the likelihood and relevance of such shifts are also conditioned by the *function* underlying the interaction and the *structure* it entails.

If dialogue were a supernatural baseball game – see D. Lewis (1979) for the first use of this metaphor, the supernatural aspect is added here for a wider variety of game types and more leeway regarding the rules – it would require a set of AGENTS, i.e. the players of the game, a SETTING, i.e. the field which could have different configurations, and a FRAME, i.e. the rules of the game and the objectives of the players. All of these parameters would have direct consequences on the possible and likely INTERACTIONS between the players, but also in the way they behave individually, in the reliability of the environment of the game and in the its overall format. For instance in the representation proposed in Figure 2.1, the topology of the field is such that the balls should be more carefully hit to avoid them going out of the field as retrieving them from the surrounding water would be more complicated. There would also be an additional rule that takes advantage of this topology and states that if a sea-dragon eats too many balls, the game is over and both teams lose. This would introduce a new form of interaction where all the players would benefit from preventing the balls to be eaten, instead of only competing against each other. In a conversation, this would be akin to a situation where several participants have conflicting interests, but where the dialogue must nevertheless proceed smoothly, otherwise no one would achieve their objectives.

Based on these observations, we structured our characterisation of dialogue around four central parameters: AGENTS, SETTING, FRAME and INTERACTION. This chapter is therefore structured as follows. It begins by examining the AGENTS of a conversation, i.e. the type and organisation of the *participants* (Section 2.1) before discussing the CONVERSATIONAL SETTING, i.e. the *world* but also the *environment and medium* they interact in (Section 2.2). The chapter then turns to the FRAME of an interaction (Section 2.3), which includes the *language(s)* used and the *goals* of the exchange. Together these meta-parameters have direct consequences on the realisation of the dialogue (Section 2.4), which is reflected in the VERBAL INTERACTION but also in specific features of the exchange at the AGENTS, SETTING and FRAME level. The chapter concludes with the proposal of an operational characterisation table that summarises the parameters discussed and two accounts of the framework, one formal and one in a vector space perspective, that illustrate how it can be applied in practice within different research contexts (Section 2.5).

2.1 Conversation agents

Any conversation is necessarily grounded in the individuals who participate in it. Even though language has long been considered as human-specific, simulating dialogue agents has been an increasingly important focus of various private sectors

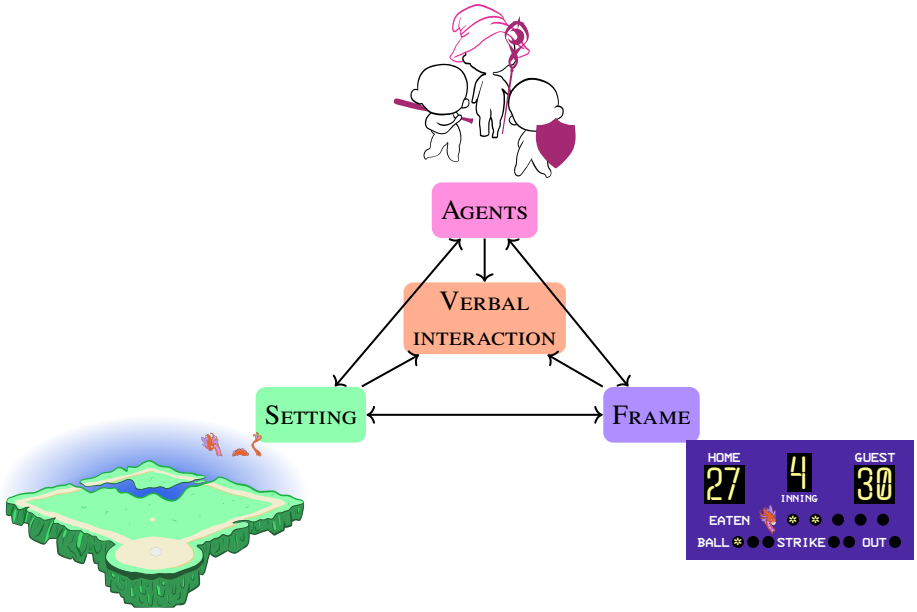


Figure 2.1: Illustration of a supernatural baseball game representing the four parameters of dialogue: AGENTS, SETTING, FRAME and INTERACTION.

as well as academic research in the last few decades. Explicitly stating the type of agents involved in a conversation is thus crucial for understanding and interpreting its interactional dynamics.

Although dialogue is often described with a simple emitter-receiver model, this view obscures the complex and dynamic roles that participants simultaneously inhabit. Speakers monitor both their own production and that of their interlocutors, listeners continuously signal alignment and understanding, and all participants coordinate through a range of multimodal cues. Moreover, conversations are rarely limited to two participants: additional hearers, shifting roles, and varying degrees of involvement contribute to a layered participation structure that warrants careful analytical attention.

Recognising that dialogue unfolds between socially situated individuals brings further dimensions into view. Participants are embedded in interactional histories, asymmetries of power or expertise, and differing degrees of identifiability or anonymity, each of which shapes what can be said, how it is expressed, and how it is interpreted.

This section outlines key parameters related to the presence and identity of participants in dialogue, with the aim of clarifying how these social and structural factors constrain, enable, and diversify conversational behaviour. We begin by discussing the different forms of agency that participants can represent (Section 2.1.1),

before examining participation structure (Section 2.1.2), that is, the number of participants and their roles as speakers, addressees, and more peripheral or shifting forms of involvement. We then discuss the influence of social dynamics on conversational processes (Section 2.1.3).

2.1.1 Agency

Though this thesis focuses on human use of language for conversation, thus excluding all forms of interactions with animals for example, excluding *simulations* of human language would overlook a significant and growing area of research and of existing data. Indeed, the recent advances in artificial intelligence and the direction taken by some big tech companies have led to the development of numerous conversational agents but more importantly to their integration in our daily lives. In November 2022, OpenAI released ChatGPT, the first widely available LLM based chatbot. Within a few months, it gathered millions of daily users (Skjuve et al., 2024). Since then, many other companies have released their own LLM-based conversational agents and new versions of them are released regularly.

While the different purposes – current and hoped for – of such conversational agents can be debated, many research papers frame the ultimate stage of their development as the ability to engage in human-like interaction. Both the anthropomorphising vocabulary employed to discuss LLMs (Cheng et al., 2024) and the anthropomorphic behaviours that are encouraged by the reinforcement-learning phase of their training process (Ouyang et al., 2022; Chaudhari et al., 2025; Ibrahim et al., 2026) push in this direction. The idea of simulating human behaviour, and in particular conversation, is not new. Records of human-like automata equipped with language can be found for instance in Ancient Greece in Homer's *Iliad* (Homer, 1996) within Hephaestus' workshop (Silverman, 2022). Though this idea belonged to (science-)fiction for a long time, the evolution of computer science has made it increasingly feasible to *simulate* human-like interaction. In 1966, Joseph Weizenbaum developed ELIZA, a 'program which makes natural language conversation with a computer possible' (Weizenbaum, 1966, p.36). His approach was completely rule-based and relied on clever identification of keywords and sequences in the user's input to produce a response. For example, 'I am very unhappy these days' could be answered with 'How long have you been very happy these days?' by identifying the pattern 'I am X' and replacing it with 'How long have you been X?'.

An important point was that Weizenbaum intended ELIZA to be a tool for studying human-computer interaction as the title of his paper suggests: 'ELIZA A Computer Program For the Study of Natural Language Communication Between Man And Machine'. He stressed that ELIZA displays an 'illusion of understanding' (Weizenbaum, 1966, p.42) but it did not prevent users to experience a 'strong emotional effect' when interacting with it (Turkle, 2007, p.502). Some users would

Table 2.1: Dialogue characterisation – Agency of participants.

Parameter	Minimal	Extended description
Agency of participants	Types	Model type Model size Model training

even go out of their way to only produce utterances that would maintain this illusion (Turkle, 2005, p.43). This 'tendency to treat responsive computer programs as more intelligent than they really are' (Switzky, 2020, p.50 based on Turkle, 1995) has been called the *ELIZA effect* and is still discussed nowadays including in relation with LLMs (Kudina & de Boer, 2025). Indeed, the difference between Weizenbaum's perception and intended use of ELIZA, and the way it was perceived by the public, is reminiscent of current debates on the capabilities of LLMs. Regardless of what they may actually be capable of, they are increasingly used to produce synthetic dialogical data for research (Long et al., 2024; Soudani et al., 2026), which means that LLM-generated dialogues are considered, at least by a part of the research community, as a proxy for naturally produced interaction. Moreover, conversational agents are being developed for numerous applications such as information retrieval (Mo et al., 2025) or recommendation tasks (Zhu et al., 2025) which creates a new form of interaction where humans interact with conversational agents.

These two advances have serious implications for the study of dialogue. First, they open a new research avenue on the interactional abilities of conversational agents and on the comparison between human-human, human-agent, and fully synthetic interactions. The characterisation of dialogue proposed in this chapter is meant to support the systematic comparison of different types of interaction, including those that differ in terms of the nature of the agents involved. This leads to the second implication, which is that stating the nature of the agents involved in a conversation is crucial when describing dialogue data. This would ensure transparency and allow for a more systematic comparison of different types of dialogue.

Therefore, the first descriptive parameter we propose is the *agency* of the participants involved in a conversation. A minimal operationalisation of this parameter consists in stating whether the participants are human or non-human. However, given the range of possible artificial agents, a more detailed description of their nature is often necessary. For example, it may be relevant to specify whether the agent is a rule-based system or a machine learning model, and if the latter, specify the type of model, its size, and the data or task it was trained on. This parameter is summarised in Table 2.1.

2.1.2 Participation structures

Participation structure can be thought of as the 'shell' of a conversation – the basic configuration that defines who is present and in what capacity. Just as a player in a game first selects a class and a battlefield, the number of participants and the function they occupy establish the arena in which interaction will take place. This outer shell constrains turn-taking, attention, and coordination, providing the framework within which conversational behaviour occurs.

2.1.2.1 Number of speakers

A prototypical face-to-face conversation is held between two speakers, sometimes more but at most four. Indeed, when more than four people interact, smaller groups tend to appear following what has been called the 'dinner party problem' (R. I. M. Dunbar et al., 1995; Dezecache & Dunbar, 2012; Krems et al., 2016; Krems & Wilkes, 2019) or schism(ing) (Sacks et al., 1974). This name stems from the fact that when many people are sitting around a dinner table, it is difficult to have a single conversation between all of them. This is due to the complexity of coordinating with many participants but also more practically because of the physical configuration of the table which makes it difficult to talk to people sitting too far away. Thus, smaller groups tend to form and have their own conversation while still being part of the larger group.

Dialogues involving strictly more than two participants are commonly referred to as *multi-party* dialogues or *multilogues*, in contrast to *duologues* involving two people. Such interactions are generally more complex in terms of turn management and topical organisation (Traum, 2003) because each participant can bring their own ideas in the conversation and derail it from the matter at hand. Establishing and maintaining common ground is also more demanding, as coordination must occur across a larger set of interlocutors, potentially requiring multiple repair sequences. In addition, long distance attachments, i.e. producing utterances rhetorically related to a far previous one, are also more frequent in multi-party settings (Ginzburg & Fernández, 2005; Asher et al., 2016) further increasing structural complexity.

As noted above, conversations involving more than four participants often subdivide into parallel threads, with participants dynamically entering and leaving these sub-conversations, which makes them complicated to study independently (Egbert, 1997). Furthermore, the number of participants is not necessarily fixed throughout the interaction. It may vary dynamically depending on the environment and communication medium (Section 2.2.2), requiring participants to continuously adapt. The arrival of new participants can introduce individuals who do not share the same common ground or can shift the relevance of ongoing topics, particularly when these are tied to a specific subgroup. Conversely, participants may also leave the interaction, further altering its structure and coherence.

As explained in the Introduction, (2.2) is an example of multilogue with four participants where three sub-conversations are taking place at the same time. This example is taken from the STAC corpus (Asher et al., 2016) which contains dialogues between people playing the boardgame *Settlers of Catan* online. They interact via a chat interface. The three parallel threads are indicated with different fonts. The first one (in normal font) initiated by LJ is directly related to the game where they ask if someone would trade sheep for clay. The second one (in **bold font**) is initiated by GWFS after they answer LJ's question and is unrelated to the game as they ask whether people 'know about the league' which has to do with the real world[2]. The third one (in *italics*) is also initiated by GWFS and can be considered a sub-thread of the second one since it arose from GWFS asking for some background on LJ after they answered the question about the league.

(2.2) LJ anyone want sheep for clay?
GWFS got none, sorry :(
GWFS **so how do people know about the league?**
WM no
LJ **i did the trials**
TK **i know about it from my gf**
GWFS **yeah me too,** *are you an Informatics student then, lj?*
TK **did not do the trials**
WM has anyone got wood for me?
GWFS **I did them because a friend did**
GWFS lol wm, you cad
GWFS afraid not :(
LJ *no, I'm about to start math. I just hang around appleton a lot*
TK sry no
GWFS my single wood is precious
WM what's a cad?

[STAC (Asher et al., 2016, Ex.1)]

This example shows that a dialogue can branch into several sub-conversations. The written format of this specific dialogue allowed the participants to simultaneously follow the different threads but similar dynamics can also exist in face-to-face conversations where several threads can co-exist and the participants follow the one they are most interested in or most able to take part in depending on the spatial configuration. In some cases, however, the genre of the interaction can impose

[2]The 'league' they are talking about may be a way to refer to the dialogue collection experiment they are taking part in. Asher et al. (2016) explain that 'players were grouped into several leagues'. The 'trials' could refer to the pilot of the experiment.

additional structure. For example task-oriented conversations such as meetings or tourist information conversations can be sub-divided in sub-dialogues which are less complex to analyse (Fernández et al., 2008; Frampton et al., 2009; K. Sun et al., 2016). Nevertheless, as the number of participants increases, conversational organisation generally becomes less stable, particularly in informal settings.

Importantly, not all interactions involving multiple individuals qualify as dialogues. Indeed, the degree of interactivity plays a crucial role in determining how closely a discourse approximates face-to-face conversation. For instance, a lecture may involve both a teacher and students, yet remain highly asymmetric, with limited opportunities for student participation, or even become entirely one-sided in recorded formats. Physical configuration further modulates interactivity: greater spatial distance reduces access to visual feedback. In large lecture halls, participants in the front rows typically engage more directly with the speaker than those seated further away, effectively occupying different interactional roles. At the extreme, both interactivity and co-presence are minimised in asynchronous exchanges.

Conversely, even monologues can rarely be seen as involving a single participant. When someone produces a monologue, they typically do so with some kind of representation of who they are talking to in mind, even if this addressee is not explicitly defined. This representation influences the way they produce their utterances. For example, when someone is talking to themselves, they are actually talking to an imagined addressee which can be themselves or a third party. When someone is writing a book, they are writing it for an audience which can be more or less defined but is still there. Y. Wu et al. (2024) showed that we can predict the questions that will be answered in the rest of a news article based on the beginning of it, which suggests that the writer indeed has a representation of the readers in mind and produces the article accordingly.

In face-to-face interaction, a participant is typically equated with a single individual. However, this assumption does not always hold in the data used for dialogue studies. Scripted dialogues, such as those found in film or television, may involve a limited number of real contributors – the script writers – while representing a larger set of fictional participants. Conversely, some corpora attribute utterances to a single speaker that are in fact produced by multiple contributors, as in collaborative writing contexts or shared social media accounts. Comparable issues arise in interactions involving language models, where a system may evolve across versions while maintaining the appearance of a single, stable interlocutor.

Overall, the number of participants constitutes a fundamental parameter for characterising dialogue. It strongly influences both the structure and the degree of organisation of the interaction. The assumption that each participant corresponds to a single, stable individual also has implications for the establishment of common ground and the degree of spontaneity. While a simple count of participants provides a useful initial description, more fine-grained distinctions are often neces-

Table 2.2: Dialogue characterisation – Participation structures: number of participants.

Parameter	Minimal	Extended description
Number of participants	Number	Number of simulated participants Number of workers Workers - Simulated participants mapping

sary. In particular, it is important to differentiate between the number of apparent (or simulated) participants and the number of actual contributors, as well as to specify the mapping between them. These sub-parameters are summarised in Table 2.2.

2.1.2.2 Functions

In conversation, participants usually take up the speaker function one at a time. While there can be overlaps for feedback for example, in most cases one person at a time can be considered as main speaker. The other participants are listeners until the floor is taken by one of them who then becomes main speaker. However, not all listeners should be considered the same. Goffman (1976), Clark and Carlson (1982), and Clark and Schaefer (1989, 1992) suggest that there exists a hierarchy of listeners functions where they could be separated between *participants*, i.e. people who are allowed to take the floor in the conversation, and *overhearers*.

In a multilogue, not all listeners are always considered as main targets by the speaker. Depending on the matter at hand, some of them can be *addressees* whom we expect to be more likely to participate actively and answer questions for example. The other listeners can be considered *side listeners*: the speaker takes them into account when producing their utterances but they do not necessarily expect them to take part into this section of the conversation. Both speakers and listeners behave differently depending on the different functions of the participants. Speakers must make sure their utterances are understandable by the listeners and in particular the addressees but they have no reason to care about the overhearers unless they have additional intentions related to them such as deceiving potential eavesdroppers. Listeners on the other hand expect the speaker to make themselves understandable by them while overhearers would typically not. More details on this aspect can be found under the *Dogma of Undifferentiated Hearers* in Clark (1997).

Figure 2.2 illustrates those functions in a real-life situation. In this example, a group of three friends is waiting in the queue of a café with other people standing in the queue. At this specific time, two of the friends are talking to each other about a topic while the third friend is focused on her phone. The two conversation participants are thus alternating between being the speaker and the addressee of the conversation while the third person is a side listener. The other people in the



Figure 2.2: Representation of participants' functions in the queue of a café.

queue are bystanders who are not part of the conversation but can still hear it and be influenced by it. For example, the lady at the counter could decide to change her order based on what they heard from the exchange, while the man behind the three friends could be tempted to barge in the conversation. Finally, we can imagine that there is an eavesdropper behind the 'staff only' door who can hear the conversation and may even be listening on purpose and pay more attention than the other bystanders. This person however cannot see the speakers and thus does not have access to the same modalities as the other participants which can influence their interpretation of the conversation.

As we discussed in Section 2.1.2.1, conversations involving more than four people tend to subdivide into sub-conversations. While the different threads can be studied independently, from the point of view of the global conversation there is more than one speaker at the same time and a listener could try to follow both sub-conversations at the same time. This could happen for example at a large dinner table where the people sitting in the centre can take part in dialogues on both of their sides. Studying such conversations requires to keep track of all the sub-threads and the interaction between them since the different sub-conversations can influence each other. Taking into account the function of each participant at a given time also plays a role in understanding the representation they have of the conversation.

The function a participant sees themselves in at a given time probably influences the way they will remember the corresponding part of the conversation. An addressee is presumably more involved than a side listener and even more than a bystander. The representation of the conversation they would keep in mind most likely differs depending on the function they took, and their views of the conversa-

Table 2.3: Dialogue characterisation – Participation structures: functions.

Parameter	Minimal	Extended description
Functions	List	Intended external audience Awareness of eavesdroppers

tion in retrospect would be different as well. Similarly an eavesdropper who choses to listen to a conversation they do not take part in has an external point of view and can notice things that the active participants would not be conscious about.

As researchers studying language data, the function we are the closest to is the one of an eavesdropper as we have no influence on the conversation but choose to pay attention to it. Several differences exist and will be discussed in following sections such as the fact that we study the conversation after it took place through a medium that only *represents* this conversation. As eavesdropper we must rebuild the context of the conversation without actually belonging to this context and it is possible to imagine a wrong context which leads to a false interpretation of the dialogue while seeming coherent from our perspective. In a research setting it could be possible to ask the participants to detail in retrospect some parts of the conversation that we do not manage to analyse. However, the spontaneity of dialogue implies that their answers would not necessarily reflect the exact mindset they had during the exchange.

The functional configuration of a conversation therefore constitutes an important parameter in the description and analysis of dialogue data. At a basic level, it can be represented as the set of functions attributed to each participant. However, more fine-grained characterisations may prove useful in capturing the dynamics of interaction. Stating whether there is an *intended* external audience to the conversation can help distinguish between interactions that are primarily directed at the participants themselves and those that are produced with a broader audience in mind, and are therefore likely to be more easily understandable from our researchers' point of view. Similarly, specifying whether participants are aware of the presence of bystanders or eavesdroppers can shed light on the degree to which they may be tailoring their contributions to accommodate or deceive these peripheral listeners. These sub-parameters are summarised in Table 2.3. Considering the participant's function when analysing dialogue data can also encourage the study of very specific aspects of a conversation. For instance, in multi-party settings, one may consider measuring the entropy over the transitions between functions for each participant, thereby quantifying the extent to which individuals shift between roles and their degree of involvement across different phases of the interaction. This suggests one possible metric to quantify the dynamics of participation structure in a conversation, and therefore compare different dialogue resources.

2.1.3 Social dynamics

While participation structure defines the 'shell' of a conversation, social dynamics determine how each participant is embodied and specialised within it. Just as a player in a game customises their character and develops relationships with allies and opponents, participants bring histories and social positions to the interaction. Hierarchies and past interactions shape what can be said, how it is expressed, and how it is interpreted. Examining these social factors allows us to understand not just the structural possibilities of a dialogue, but the nuanced ways participants navigate, influence, and are constrained by their social environment.

2.1.3.1 Hierarchy

In prototypical face-to-face conversations, the participants are on the same level and their contributions are overall equally distributed. However many conversational settings contain some kind of hierarchy which influences the flow of dialogue. For instance, in formal settings such as meetings, one person may take the lead in speaking, while others are expected to listen, interrupting only when permitted.

The hierarchy may be determined by explicit protocols, such as those found in workplaces or political debates, or by implicit norms and biases based on social status, age, gender, or expertise. This can affect both the content and the style of communication: one might speak more formally, with careful word choice, when addressing a superior or less familiar person, while adopting a more casual tone with close friends or peers. The ability to adapt to these varying conversational hierarchies is crucial as failing to adjust one's behaviour, for instance through inappropriate levels of formality or overstepping conversational turns, can lead to misunderstandings or social friction.

Hierarchy can also stem from the type of conversation and the roles the participants are attributed or attribute themselves. In particular in task-oriented conversations, the task can create specific positions which influence the dialogue. For instance, participants can become an explainer-explainee pair or one can follow directions given by others. In such settings participants have an asymmetric role which changes the overall conversational dynamics. Those specific dynamics are one of the axes of study regarding task-oriented conversations since they are common in daily life. However, such conversations are also commonly used to study other unrelated phenomena or to train conversational agents and deep learning models. The level of asymmetry and its influence on the way dialogue can unfold should thus be taken into account in more general purposes.

Relying on task-oriented corpora to train conversational agents exposes models primarily to asymmetrical interactions, which in turn shapes the patterns of their generated utterances. While this can be appropriate for practical agents designed for specific tasks – such as booking hotels or purchasing plane tickets – it can reinforce user biases when deployed in more general-purpose settings. For instance,

Table 2.4: Dialogue characterisation – Social dynamics: Hierarchy.

Parameter	Minimal	Extended description
Hierarchy	Binary	Weighted list of dimensions

conversational tools like ChatGPT have a tendency to agree with users, even at the expense of logical consistency, making them unsuitable as tools for fostering critical thinking without careful guidance. Nonetheless, such systems are increasingly relied upon for expert support in decision-making (F. Liu et al., 2024).

The hierarchical dynamics of a conversation are therefore a crucial parameter in characterising dialogue data, as they shape both the flow of interaction and the way participants engage with one another. At a minimal level, hierarchy can be represented with a binary distinction. For more detailed analyses, hierarchical structure can be operationalised across multiple dimensions, such as age, genre, social status, expertise, or institutional role. These factors are extensively studied in social psychology and sociology and can be adapted to the analysis of conversational data (P. Brown, 1987). The relative importance of each dimension may vary depending on the cultural or physical context. From a computational perspective, such hierarchical characterisations can be efficiently encoded as vectors: one representing the relevant dimensions and another representing their corresponding weights. This approach allows for flexibility in modelling conversational hierarchies across diverse datasets and applications. These sub-parameters are summarised in Table 2.4.

2.1.3.2 Conversational history

Conversational history refers to the previous exchanges between specific participants. It can be conceptualised along two complementary dimensions. First, the short-term history arises from the ongoing conversation itself, providing topics and context that can be reintroduced during the interaction. Second, the long-term history reflects the cumulative record of prior interactions between participants, spanning all past dialogues they have shared.

Short-term history exists in all types of conversation, regardless of the number of participants or whether they are human or artificial. The salience of particular elements depends on participants' interests, attention, and memory. Long-term history, in contrast, allows participants to achieve greater alignment and efficiency in communication as they become more familiar with one another.

Individual differences in memory and recall can significantly influence comprehension and coordination. If one participant implicitly relies on a shared history to support an utterance while another does not recall the relevant past exchanges, misunderstandings or disruptions may arise.

This phenomenon is less visible in written conversation as the history exists

Table 2.5: Dialogue characterisation – Social dynamics: (Conversational) History.

Parameter	Minimal	Extended description
(Conversational) History	Scale	Distance in a weighted social graph Common topics

physically. While some tools erase older parts of the history, in most cases a long part of it is retained for quite some time. Participants can thus refer to it, in some cases by explicitly indicating what utterance they are elaborating on. There is also a lasting proof of what has been said exactly which partially prevents argumentation strategies that rely on distorting the other person's words.

Beyond the conversation itself, a broader social or experiential history shapes interactions. Participants bring with them pre-existing knowledge, social norms, and expectations derived from their prior relationship or shared affiliations. For instance, meeting a friend of a friend involves different assumptions and 'safe' topics than meeting a person working in the same company or a complete stranger for the first time. This shared background provides a common ground that can facilitate conversation and guide topic selection (Clark, Brennan, et al., 1991).

Overall, both conversational and broader social histories are important parameters of dialogue, influencing content, alignment, and linguistic choices. At a minimal level, history can be represented on a simple scale reflecting the amount of shared experience. More detailed operationalisations could model social distance within a weighted network and include lists of common topics or interests, which can signal likely conversational trajectories. These sub-parameters are summarised in Table 2.5.

2.2 Conversational setting

We refer to the CONVERSATIONAL SETTING as the set of parameters that define the environment in which a conversation takes place, i.e. the Where, When, and How of the interaction. This includes general information regarding the world it is set in (Section 2.2.1) but also the physical environment and the medium through which the conversation is conducted (Section 2.2.2).

2.2.1 World setting

The world setting of a conversation defines the framework of assumptions, facts, and realities within which participants interpret and produce utterances. Conversations may take place in the real world, where interlocutors share common knowledge, cultural norms, and factual assumptions. Characters in fiction can have con-

versations pertaining to their real world which is fictional for the readers. Alternatively, participants may talk about a fictional or hypothetical environment, such as in role-playing games or thought experiments. In such cases, certain facts and rules are treated as given within the conversational frame, even though they do not correspond to objective reality. Participants typically rely on these assumed truths to coordinate meaning, allowing the conversation to progress smoothly despite the constructed nature of the world.

Even when discussing fictional or hypothetical worlds, interlocutors do not define every aspect and rule of the environment explicitly, and no work of fiction can be exhaustive in its coverage. Instead, participants rely on partial assumptions and conventions that can be negotiated implicitly or explicitly during the interaction. This phenomenon has been extensively studied in the semantics and pragmatics of fiction fields, which examine how meaning is established and maintained when participants engage with non-real worlds (D. K. Lewis, 1978). For instance, moral laws are the kind of rules we typically import implicitly into fictional worlds, and they are more complicated to deconstruct that other assumptions such as physical laws. Alignment in world setting is crucial for comprehension: misalignment may result in misunderstandings, breakdowns in coordination, or divergent interpretations of the same utterance.

The world setting also plays a pivotal role in interactions with conversational AI systems. In these contexts, the assumptions and reasoning patterns of the system may differ from those of human participants. If the AI does not share or recognise the intended world frame, its contributions may appear incoherent or irrelevant, highlighting the need to model world assumptions explicitly when analysing or designing dialogue systems.

In terms of descriptive parameters, the most basic characterisation would need to indicate whether the conversation takes place in the contemporary real world or not. A more detailed description could specify the time period and location of the world setting, as well as any relevant cultural, social, or fictional conventions that participants are expected to adhere to. This description matters not only for fictional or role-playing contexts but also for real-world interactions, where correctly anchoring the conversation in its temporal and cultural context is essential for accurate interpretation. The reusability of dialogue resources over time is therefore dependent on the clarity and specificity of the world setting. This parameter is summarised in Table 2.6.

2.2.2 Environment and medium

Face-to-face conversation is inherently multimodal: participants share the same physical space, allowing them to interact within a common environment, and communicate through voice, gestures, and facial expressions. These modalities are tightly intertwined, each enriching the conversation and supporting mutual under-

Table 2.6: Dialogue characterisation – Epistemic grounding: World setting.

Parameter	Minimal	Extended description
World setting	Real (contemporary or not) / Fictional	Time period Location Conventions (cultural, social, scientific, moral, etc.)

standing. The physical situation also plays an important role. It can make a conversation more or less prone to interruptions, influence the emergence of sub-threads, and provide referential resources when events or objects in the environment become part of the dialogue. Collectively, these aspects are fundamental to characterising the dynamics and structure of face-to-face interaction.

Sharing physical space enables visual co-presence, which shapes conversation in two major ways. First, it enriches interaction by providing access to non-verbal cues such as facial expressions, gaze, and gestures. These cues allow participants to identify speaker intent, gauge affective tone, and monitor the attention and comprehension of others. Second, the shared environment itself becomes a referential resource: participants can point to, gesture toward, or otherwise use elements of the surroundings to structure and support the dialogue.

In addition to visual sharing, vocal co-presence provides complementary information. Prosody, volume, pitch, and timing convey emphasis, emotional nuance, and communicative intent that would otherwise be absent in written or mediated forms of communication. Together, visual and vocal co-presence allow for fine-grained coordination, rapid feedback, and grounding of meaning.

However, these features can deteriorate depending on the communication medium. While face-to-face interaction provides simultaneous audio and visual channels, other media may reduce or remove one modality entirely, affecting how participants produce and interpret the conversation. Subsequent subsections explore visual co-presence, gestures, and vocal features in detail, highlighting both their individual contributions and their interplay in shaping effective communication. We will begin with the physical context of conversation (Section 2.2.2.1), which provides the shared environment in which these multimodal signals are produced and interpreted. We will then discuss the specifics of participants being able to see each other (Section 2.2.2.2) and hear each other (Section 2.2.2.3). Finally we will examine the temporal aspects of the conversation (Section 2.2.2.4).

This discussion assumes typical face-to-face interaction between fully sighted and hearing participants. Situations involving sensory disabilities (e.g. blindness or deafness) alter the availability and perception of multimodal signals; we do not address these cases in detail here.

2.2.2.1 Physical context

Physical context refers to the physical space, i.e. the material and spatial conditions, in which interaction takes place. It is typically a shared part of the real world – though the attention of the participants can be focused on different aspects at a given time, and plays an active role on how conversation unfolds. Because participants share a setting, they co-experience its sights, sounds, and affordances, which can subtly or dramatically shape what is said, how it is said, and who is involved. This influence operates along several dimensions, ranging from the triggering of topics to the regulation of participation and privacy.

First, the environment provides natural conversational triggers. Everyday talk is often prompted by something occurring in the shared space: a loud sound, a passing event, an unexpected sight. Such phenomena can interrupt ongoing talk or lead to spontaneous topic shifts, demonstrating that conversation is dynamically responsive to physical context. Similarly, visible objects can become resources for situated joint reference, enabling interlocutors to point, gesture, or comment on what they both perceive (Clark, 1996; Goodwin, 2000). This capacity to anchor talk in the here-and-now creates a rich field for shared meaning-making that is often absent in mediated communication (Clark, Brennan, et al., 1991), though virtual shared spaces can partially restore this functionality in some cases (Asher et al., 2020).

Second, the environment affects who can participate and thus the dynamics of the interaction. Spatial layout determines how permeable a conversation is to new participants. Conversations held in open areas, near circulation spaces, or around standing tables are easier for others to overhear and join, creating opportunities for multi-party talk and dynamic shifts in alignment and topic (Oldenburg, 1989). Conversely, conversations in secluded corners or enclosed rooms tend to remain private and are less prone to interruptions and intrusion of additional participants, supporting more intimate or sensitive exchanges. Hall's (1966) theory of proxemics highlights that interpersonal distance and seating arrangements signal levels of intimacy, formality, and accessibility, regulating who feels entitled to approach or contribute.

Along the same line, privacy and publicness are key environmental dimensions that influence topic selection and conversational style. In acoustically transparent settings such as open-plan offices, people often self-censor, avoid disclosing sensitive information, or switch to less identifiable vocabulary because of the risk of being overheard (Cavanaugh et al., 1962). By contrast, spaces that provide good speech privacy – through partitioning, sound masking, or physical isolation – encourage more candid exchanges. Lighting and seating configuration can also modulate the involvement in conversation of communication vulnerable people (Stans et al., 2017).

Environmental factors also shape conversational effort. Acoustics play a major

role in intelligibility and cognitive load: high reverberation or competing speech can force speakers to raise their voice, repeat themselves, or simplify messages, affecting both efficiency and emotional tone. Similarly, visual conditions such as obscured sightlines can make it harder to monitor the listener's reactions, leading to more verbal checks for understanding or less precise timing of turn-taking.

Crucially, these environmental affordances are absent or altered in mediated conversation. Participants may be located in different environments, connected only through a technical channel that transmits language but not the material world in which it is produced. In written online exchanges, for instance, the shared physical environment effectively disappears, replaced by a virtual interactional arena defined by the interface itself. In a phone call, participants no longer share the same perceptual environment, preventing them from jointly noticing events in their surroundings. As a result, when a disruption occurs (such as a knock at the door or a sudden noise), the affected participant must explicitly explain it ('Sorry, someone just came in') for the other to understand the break in attention (Clark, Brennan, et al., 1991). Similarly, references to objects or features of the surrounding environment must be described verbally rather than simply indicated by pointing, increasing the effort required to establish common ground. Even in video-mediated conversation, spatial relations remain fragmented: participants appear as framed projections rather than embodied agents capable of acting within a shared environment. Deictic acts such as pointing or manipulating objects lose their immediate interpretability unless they refer to objects visible within the camera frame or are reformulated linguistically. The result, in both phone calls and video-mediated interaction, is a form of contextual abstraction in which spatial and material relations are reconstructed discursively rather than jointly enacted. This shift underscores the extent to which face-to-face conversation depends on the tacit sharing of physical context.

For researchers and practitioners working with dialogue corpora and language models, the absence of physical context shapes what kinds of conversational phenomena can be observed and modelled. Many corpora used in dialogue research are collected from phone calls or text chats, media that lack references to shared surroundings and omit environmental triggers that would normally shape topic flow or participation. This means such datasets may under-represent phenomena like topic shifts prompted by co-experienced events, repair sequences caused by shared distractions, or subtle changes in formality induced by spatial privacy. For analytic purposes, scholars must therefore be cautious when using corpora where the physical environment is absent: conversational structure may appear more topic-stable, more verbalised, and less sensitive to interruptions than in natural co-present conversation. Incorporating multimodal or situated datasets – such as meeting recordings with video, or corpora annotated for spatial layout – can reveal how much the environment contributes to turn-taking, participation, and content selection, and can help develop models that better approximate human conversational adaptabil-

Table 2.7: Dialogue characterisation – Environment & Medium: Physical context.

Parameter	Minimal	Extended description
Physical context	Binary	Co-located / Shared virtual / Separate

ity.

Across all forms of dialogue, the physical context determines how participants orient to space, objects, and the organisation of joint action. As interaction becomes increasingly mediated or artificial, physical co-presence gives way to representational or simulated forms of context. Understanding how dialogue adapts to these material and spatial conditions provides a foundation for examining other dimensions of co-presence – most notably, the visual forms of mutual accessibility that underpin gaze, gesture, and embodied alignment, discussed in the following section.

This aspect should therefore be recorded when characterising dialogue resources and tasks, as it shapes the way participants produce and interpret the conversation. As with many of the other parameters, a binary distinction may be sufficient in simple cases. However, a more detailed description may specify the degree to which the physical context is shared, distinguishing between participants who are physically co-located, who share only a virtual interactional space, or who are entirely separated. These parameters are summarised in Table 2.7.

2.2.2.2 Visibility

Many signals in face-to-face conversation are a combination of linguistic and non-linguistic modalities (Clark, 1996, p.156). The possibility to see one another during interaction enables participants to exploit facial expressions, gaze direction, and gestures as complementary communication channels, thereby facilitating coordination and grounding. Indeed, according to Oviatt (1999), 'multimodal pen/voice language is briefer, syntactically simpler, and less disfluent than users' unimodal speech' (based on Oviatt, 1997). Visual co-presence is thus a core feature of natural conversation, fundamentally shaping both its structure and efficiency.

As mentioned earlier, the discussion in this chapter assumes fully sighted and hearing participants. However, it is important to stress that sign languages rely entirely on the visual modality. Indeed, in these languages, visual cues are not merely complementary, but essential for conveying meaning, and even encode information that would otherwise be expressed vocally, such as prosody (Sandler, 2010). Although we do not discuss sign languages further in this chapter nor thesis, they illustrate that the visual modality does not always play the same role or carry the same communicative importance across different forms of language use.

Gaze Gaze is a central organisational resource in conversation, shaping how participants coordinate turns, monitor one another's attention and cognitive state, and display social or affective stances. Even in simple dyadic exchanges, interlocutors are simultaneously speakers and listeners – speakers monitor their interlocutor to assess understanding while listeners transmit signals of attention, alignment, and readiness. In this reciprocal ecology, gaze functions both as a perceptual act and as a communicative signal. Early work by Kendon (1967) showed that speakers tend to avert their gaze at the beginning of an utterance and re-establish it at its end, a pattern interpreted as reflecting the cognitive load of planning and the need to recruit the interlocutor's uptake at a transition-relevant point. Subsequent studies have repeatedly replicated this dynamic: Duncan et al. (1979), Ho et al. (2015), and Rossano (2012), among others, find that gaze helps shape the timing and attribution of turns, facilitating the smooth transfer of the floor. Listeners generally display more sustained gaze towards the speaker, offering something akin to visual backchannels that signal attention and availability. Yet these patterns are not mechanistic: a significant proportion of turn transitions occur without the expected gaze cues, and gaze becomes more influential in situations of social uncertainty, with unfamiliar interlocutors or sensitive conversational topics. Research also reveals that interlocutors may strategically manipulate gaze to influence the organisation of the floor. Kendrick et al. (2023), for example, show that speakers who wish to retain the turn at a point where transfer is possible may avert their gaze or reorient their body in ways that discourage interruption. Conversely, sustained gaze near a potential completion point can encourage a response, subtly nominating a next speaker.

Beyond regulating turns, gaze provides crucial monitoring capacities. Goffman (1966) emphasised that sustained visual access allows interlocutors to monitor one another's behaviour, adjust their explanations, and pre-empt failures of understanding. This monitoring is cognitively demanding, which helps explain why gaze aversion reliably accompanies thought-intensive activity. Studies by Glenberg et al. (1998), Doherty-Sneddon et al. (2002), and Doherty-Sneddon and Phelps (2005) show that when individuals must remember information, plan complex responses or perform reasoning tasks, they withdraw gaze, seemingly to reduce cognitive load. Allen and Guy (1977) further demonstrate that confident speakers tend to look at their interlocutor more, whereas hesitations and uncertainties correlate with gaze aversion. Together, these findings position gaze as an embodied index of cognitive and attentional states, one that interlocutors draw on to adjust their contributions, calibrate timing and modulate the level of explicitness or elaboration required.

Gaze is equally implicated in the expression of social and affective meanings. It conveys intimacy, dominance, embarrassment, confidence, and stance, often in subtle and culturally mediated ways. Doherty-Sneddon and Phelps (2005) identify gaze aversion as a marker of self-consciousness and emotional strain, particularly when discussing personal topics. Argyle et al. (1974) show that frequent – but

not continuous eye contact increases perceptions of friendliness and engagement, whereas sustained eye contact can feel intrusive and unnatural. Cook (1977) observes that eye contact rarely consists of simple staring; instead, participants perform rapid sweeps across one another's faces, with these patterns adjusting dynamically in response to changing social relations. The expressive dimension of gaze thus interacts continuously with its regulatory and monitoring functions, contributing to the layered and affectively charged nature of conversation.

Although most of this literature concerns dyadic talk, gaze becomes even more pivotal in multi-party interaction, where participants must coordinate not only turns but also recipient and next speaker selection as well as varying levels of engagement. The larger amount of addressees entails a shorter time spent by the speaker gazing at them. To partially compensate for this, speakers tend to gaze more, in proportion, to their addressees than in dyadic exchanges (Vertegaal et al., 2001). Work by Goodwin (1981), Lerner (2003), and Stivers and Rossano (2010) shows that speakers often direct their gaze toward the intended next speaker around a possible completion point, and that recipients so addressed are statistically more likely to respond. Through such practices, gaze helps differentiate those treated as expected recipients from those merely overhearing, but also enables the speaker to indicate the next speaker they have selected in a non vocal – and hence less conversationally disruptive – way. However, this strategy only works if both the selected next speaker and the non-selected attendees have seen the speaker's gaze. Jokinen et al. (2013) stress the significance of gaze to identify when the current speaker wishes to hold their turn or leave it to the next speaker. Regarding schisming, experimental work by Vertegaal et al. (2001) suggests that people gaze more at their direct addressees than more peripheral participants for a given topic. In larger groups, gaze also helps segment interaction into simultaneous conversational threads, as participants establish local mutual gaze with temporary side-partners while withdrawing from the main floor, a phenomenon described by Egbert (1997).

The importance of gaze becomes particularly salient when this channel is removed or degraded. Mediated communication, such as audio-only calls, eliminates visual access entirely, and even video calls distort mutual gaze because the camera and screen are misaligned, making it difficult to determine precisely where someone is looking. They can also complicate coordination because of delays between audio and video. Participants can infer some attentional cues from head orientation and posture, but these cues are considerably less reliable. Although conversational modalities are generally redundant – Cook and Lalljee (1972) show that audio-only communication preserves many conversational behaviours, with differences arising primarily in utterance length – it would be difficult to argue that the loss of gaze information has no effect on conversation. Findings from virtual reality environments reinforce this point: even when avatars simulate eye movements, the fine-grained timing and mutual contingencies characteristic of face-to-face gaze are all maintained, disrupting familiar turn-taking patterns (Lücking et al., 2025).

Table 2.8: Dialogue characterisation – Environment & Medium: Gaze.

Parameter	Minimal	Extended description
Gaze	Binary	Fidelity: accurate / approximated / distorted

Dialogue systems face an even greater challenge, as they lack genuine perceptual monitoring and cannot ground their gaze behaviour in a real-time assessment of attentional states or mutual orientation. Consequently, the gaze behaviours of avatars and embodied agents resemble the impoverished cues available in video-mediated communication, but without the human inferential capacities that compensate for missing information.

Overall, gaze constitutes a densely multifunctional resource that structures the temporal, cognitive, and social organisation of conversation. Its absence or distortion reconfigures interaction in ways that remain only partially understood, underscoring the need to characterise dialogue data not merely in terms of modality but also in terms of the extent to which participants can access and interpret one another's gaze. While a binary description of this parameter can already provide some information about the dialogue setting of a resource, the fidelity of gaze cues constitutes another important dimension, as some media, such as video calls, systematically distort gaze direction because of parallax and camera placement.

Gestures and Facial Expression Gestures are often defined as visible bodily movements that accompany speech and contribute to the communicative act. They include movements of the hands and arms, but also head movements, gaze behaviour, and facial expressions, though the latter category is sometimes distinguished for its closer association with affect and stance. Kendon's (2004) theory of the 'gesture-speech ensemble' emphasises that gesture and speech form a unified system: gestures are temporally and semantically integrated with the spoken utterance, and listeners interpret them in conjunction with linguistic content rather than as parallel or extraneous information. This integration is reflected in semiotic classifications dating back to Peirce (1934), who distinguished icons, indexes, and symbols as fundamental sign types. Contemporary gesture typologies build on these foundations. Deictic gestures function indexically, pointing to objects, people, or locations in the physical or conversational space. Iconic gestures depict aspects of referents through visuospatial resemblance, while metaphoric gestures extend this depictive capacity to abstract concepts, often via metaphorical mappings. Beat gestures, by contrast, are rhythmic, non-representational movements aligned with prosodic peaks; they serve to highlight or structure discourse rather than to introduce referential content.

The functions of gestures in conversation are diverse and extend beyond the merely expressive. A growing body of research has demonstrated that gestures

contribute semantic information that may not be recoverable from speech alone. Recent experimental work shows that people interpret the same spoken utterance differently depending on the visual signal accompanying it, and that the interpretation of complex multimodal signals cannot be predicted from a simple combination of their components (Trujillo & Holler, 2024). Temporal coordination also plays a critical role: gesture strokes typically align with specific lexical items or prosodic prominence (McNeill, 1992; Mehler & Lücking, 2012). This synchrony contributes to the perception of a cohesive communicative unit and facilitates incremental interpretation by the listener.

Beyond semantic enrichment, gestures can bear illocutionary force and directly participate in conversational dynamics. Speakers use gestures to request clarification, to signal uncertainty, to soften or reinforce the force of a verbal act, or to cue turn-taking intentions. They also play an important role in discourse management. Gestures can introduce new discourse referents, disambiguate pronouns or spatial descriptions, and maintain coherence in referentially dense segments of talk (Debreslioska & Gullberg, 2020; Parrill et al., 2022). In these respects, gestures function as conversation facilitators, scaffolding the interaction in ways that are not easily replaced in purely verbal communication.

The absence or reduction of gesture and facial expression in mediated communication often leads participants to adopt surrogate semiotic resources. In text-based communication, emojis, stickers, GIFs, and images serve as partial substitutes for facial expressions, affective displays, or iconic gestures (Gandolfi & Pickering, 2023). Some video-conferencing platforms provide 'reaction' buttons intended to mimic backchannels or feedback cues. These resources reveal a persistent communicative need to compensate for the loss of visual signalling in a setting where the speaker may not be able to monitor the gestures and expressions of all the receivers. This suggests that such signalling is not merely embellishment but an essential component of meaning-making.

At the same time, these substitutes bring a new layer of complexity to the interpretation of dialogue. Emojis, for instance, are inherently polysemous and context-sensitive: their interpretation depends on cultural norms, interpersonal familiarity, and conversational context, much like the interpretation of backchannels or facial expressions. Their form is unstable across platforms, as the same Unicode code point may appear with divergent visual designs on different operating systems. Furthermore, the descriptive labels provided by Unicode do not constrain actual usage and probably rarely reflect it, and user-generated ASCII art – like `;-)` or `T_T` – has no conventionalised description at all. These properties complicate interaction and can lead to misunderstandings – both for human interlocutors and, increasingly, for computational dialogue systems that must interpret these signs.

For researchers constructing or analysing dialogue corpora, these observations raise important methodological considerations. A common pre-processing step in large-scale text corpora is to remove emojis and other 'non-lexical' tokens, in a

Table 2.9: Dialogue characterisation – Environment & Medium: Gestures and Facial expressions.

Parameter	Minimal	Extended description
Gestures & Facial expressions	Binary	Interpretability

manner reminiscent of the practice of removing backchannels as noise. Yet the communicative weight of these elements suggests that their removal may obscure important aspects of interactional meaning. Retaining, annotating, and analysing them is therefore essential for a complete representation of the conversational event.

The significance of gestures is also increasingly recognised in computational modelling and human-computer interaction. Traditional dialogue systems lack physical embodiment and therefore have no access to the multimodal channels that humans rely on. Research in robotics and embodied conversational agents attempts to model the generation and interpretation of gestures, synchronising them with speech to produce more naturalistic behaviour. Formal semantic frameworks, such as the TTR extension developed by Ginzburg and Lücking (2021), provide ways of integrating gestures into the semantic and pragmatic representations underlying dialogue. These efforts reflect a growing understanding that gestures are not an optional add-on to communication but a fundamental resource shaping how dialogue unfolds.

Altogether, gestures and facial expressions constitute a core parameter in characterising conversation, influencing meaning, interactional structure, and the interpretability of utterances. Their presence, absence, or substitution materially affects how dialogue works and how it should be analysed. A comprehensive account of conversational phenomena must therefore incorporate these multimodal resources rather than treating them as peripheral or removable elements. For this reason, researchers should include the following questions in their corpus description or selection process: Did participants have access to gestures, facial expressions, and other visual cues? If not, did they rely on substitute semiotic systems – such as emojis, GIFs, or reaction icons – and were these systems conventionalised prior to the dialogue or their meanings were locally negotiated over the course of the conversation? Are these systems interpretable from outside the interaction by the analysts? Similarly to gaze this parameter could be basically described in a binary way or in more details with the interpretability of the visual cues in case they rely on a semiotic system. These parameters are summarised in Table 2.9.

2.2.2.3 Auditory co-presence

Auditory co-presence – the possibility to hear each other – is another central parameter for characterising conversation. In face-to-face interaction, speech is never

simply a sequence of words: it is shaped by intonation, rhythm, loudness, voice quality, and other auditory cues that convey affect, stance, epistemic alignment, and interactional intentions. When dialogue shifts to modalities without access to these auditory signals, the interpretive landscape changes correspondingly. As with the visual modality, the attenuation or loss of prosody affects how utterances are understood, how actions are recognised, and how interlocutors coordinate their interaction.

In what follows, we first discuss the role of prosody and tone in shaping interaction, before turning to the influence of noise and signal degradation.

Prosody and Tone Although linguistics has long focused on words and grammatical structure, everyday interaction demonstrates that meaning is strongly modulated by tone of voice. Utterances such as 'hmm', 'oh', or 'really' can exhibit different meanings and functions depending on their realisation. Work on discourse markers and interjections has shown that intonation is essential for distinguishing the action performed by these items (Heritage, 1984; Ward, 2006; Norrick, 2009). Tone can therefore convey more than words alone, and prosody plays a key role in shaping the illocutionary force of an utterance.

The importance of auditory information has long been recognised in conversation analysis and phonetic research. CA-informed transcription systems such as the Jeffersonian conventions incorporate pitch movement, stress, elongation, loudness, and voice quality because these features are necessary for understanding how turns are designed and interpreted (Jefferson, 2008). Prosody is central to how interlocutors classify utterances as questions, assertions, continuations, or completions for example. While grammar can mark interrogatives, in many cases questions in conversation are realised through prosody rather than syntax, such as a rising pitch contour on an otherwise declarative sentence (Gunlogson, 2004). Languages differ in the extent to which syntax or prosody performs this work, but the general point holds: auditory cues shape action recognition (Hirschberg, 2006).

The absence of auditory co-presence in text-based communication has important consequences for how speakers interpret each other. Without prosodic cues, it becomes more difficult to detect sarcasm, irony, joking, or playful teasing, to assess degrees of certainty or commitment, or to distinguish genuine queries from rhetorical ones. These aspects of stance and action are usually signalled through pitch, timing, and voice quality (Attardo et al., 2003; Bryant, 2011). When such cues are missing, interlocutors may attribute unintended intentions to one another. Misunderstandings can persist for longer than they would in spoken dialogue because the absence of real-time feedback reduces opportunities for immediate repair. In face-to-face interaction, interlocutors constantly monitor each other's responses through subtle changes in prosody – hesitations, micro-pauses, and shifts in tone – that prompt clarification or modification. In text-based media, these signals are not available, and speakers must compensate by being more explicit about their

intended meaning. Yet it is not always clear when one's message may be misinterpreted, and communicators often lack sufficient mutual monitoring to detect emerging problems. This can lead to unintended perceptions of coldness, abruptness, or irritation, even when none was intended.

Auditory co-presence also interacts with broader questions of synchronicity. Asynchronous voice messages – an increasingly common medium, especially in mobile communication – constitute an intermediate case between synchronous speech and text. Users often choose voice messages when they want to convey longer or more affectively charged content, reflecting that speaking can be easier and more expressive than typing. However, receiving voice messages is often less convenient: listeners cannot skim or search the content as easily as text, and partial relistening is cumbersome. Research on mobile messaging shows that vocal messages are perceived as bridging some limitations of text-messaging though their use is polarising (Haas et al., 2020). This again highlights that communicative modalities offer different trade-offs between expressivity and convenience.

Despite the centrality of prosody in human communication, most dialogue systems do not yet use the auditory modality in a human-like way. Work on expressive speech synthesis and empathetic dialogue agents recognises that emotional meaning is carried extensively through tone of voice (Mozziconacci, 2002), but computational modelling of prosody remains limited. Systems often rely solely on text, losing access to the rich paralinguistic cues that humans use to interpret intent. This reduces their robustness in detecting sarcasm, uncertainty, or subtle conversational actions, and it limits the naturalness of machine-produced speech.

Taken together, these considerations demonstrate that auditory co-presence is a vital dimension for characterising conversational settings. The presence or absence of prosody shapes the interpretation of utterances, the recognition of social actions, the management of stance and affect, and the likelihood of misunderstanding. As with visual modalities, the differences across media call for careful attention when collecting and analysing conversational data. Corpora that lack auditory information should be annotated accordingly, and the limitations of such data must be acknowledged when analysing interactional phenomena that depend heavily on tone of voice. Conversely, the presence of auditory co-presence – even in asynchronous form – provides communicative affordances that materially change how dialogue unfolds. This parameter should thus be included in any comprehensive characterisation of dialogue resources and tasks, with attention to the degree of prosodic information available and the extent to which participants can actually rely on it for interpretation. As for most parameters a basic binary distinction can be made but a more detailed characterisation would specify the fidelity of these cues, especially in mediated communication where technical factors can distort or degrade the auditory signal. For example, in video calls, latency and compression can affect the timing and quality of prosodic cues, making them less reliable for turn-taking or stance recognition. In asynchronous voice messages, the lack of real-

Table 2.10: Dialogue characterisation – Environment & Medium: Prosody & Tone.

Parameter	Minimal	Extended description
Prosody / Tone	Binary	Fidelity

time feedback may reduce the communicative value of prosody, as listeners cannot adjust their interpretation based on immediate responses. These nuances are important for understanding how auditory co-presence shapes conversational dynamics across different settings. These parameters are summarised in Table 2.10.

Noise Beyond structured vocal signals, auditory co-presence also involves the quality and reliability of the acoustic channel through which speech is transmitted. In real-world interaction, speech is rarely perceived under ideal conditions: background noise, signal degradation, transmission artefacts, and environmental interference can all affect how utterances are heard and interpreted.

Noise influences dialogue by introducing uncertainty at the perceptual level, which in turn shapes interactional dynamics. Participants may fail to hear parts of an utterance, misperceive words, or require repetition and clarification. This leads to an increased frequency of repair sequences, slower turn progression, and more explicit confirmation strategies. In this sense, noise does not merely degrade the signal but actively reshapes the structure of the interaction.

The impact of noise varies significantly across conversational settings. Face-to-face interaction in a quiet environment allows for fine-grained prosodic distinctions to be perceived, whereas crowded or noisy settings reduce the availability of these cues and increase reliance on redundancy and multimodal signals. In mediated communication, technical factors such as compression, latency, or packet loss can similarly affect auditory quality. By contrast, text-based communication eliminates acoustic noise entirely, but at the cost of losing prosodic information altogether.

Many tools that deal with speech such as ASR systems tend to be designed to operate in ideal conditions, and their performance degrades significantly in noisy environments. This can lead to increased error rates, misrecognitions, and failures to capture the intended meaning. For dialogue systems that rely on speech input, this presents a significant challenge: they must be robust to noise and capable of handling the resulting ambiguities in user input. Research in robust speech recognition and noise reduction techniques is therefore crucial for improving the performance of conversational agents in real-world settings.

From the perspective of dialogue characterisation, noise is therefore a crucial parameter because it conditions how reliably auditory information is transmitted and how much effort participants must devote to maintaining mutual understanding. High-noise environments tend to favour shorter utterances, increased repetition, and more robust grounding strategies, whereas low-noise environments sup-

Table 2.11: Dialogue characterisation – Environment & Medium: Auditory co-presence.

Parameter	Minimal	Extended description
Noise	Binary	Degree Sources

port more efficient and nuanced interaction. This parameter can be operationalised in a binary way – presence or absence of noise – but a more detailed characterisation could specify the degree of noise and its sources. For example, one could distinguish between environments with intermittent background noise (e.g. a café) versus constant noise (e.g. a factory floor), or between technical noise (e.g. poor audio quality) and environmental noise (e.g. traffic). Such distinctions would provide a richer understanding of how auditory conditions shape conversational dynamics. These parameters are summarised in Table 2.11.

2.2.2.4 Time

All conversations unfold under temporal constraints that shape how they are produced, interpreted, and remembered. These constraints are mostly induced by the medium of communication and the use that is made of it. Face-to-face dialogue is typically synchronous: participants must coordinate speaking, listening, and understanding in real time, with each turn shaped by the immediacy of what has just been said and by anticipation of what may come next. Text messages, by contrast, are often asynchronous and allow for extended planning and reflection, but they can also be very rapid when participants are highly engaged. These temporal pressures structure both the pace of interaction and the kinds of actions that are possible within it. At the same time, conversation is transient. Its full richness – including fleeting cues, momentary cognitive states, and the evolving context in which turns are produced – cannot be preserved in any stable form. What we record or transcribe captures only traces of a process that is constantly disappearing as it unfolds. Understanding the role of time in dialogue therefore requires examining both its real-time dynamics and its inherent ephemerality.

We thus discuss the synchrony of production and interpretation in dialogue and how it is affected by different communication forms in, then how the degree of planning participants can put into their contributions affects the way they produce and interpret each other in, before proceeding to examining the ephemerality of conversation and how it affects the way we can study it.

Production and interpretation In face-to-face conversation production and interpretation happen simultaneously. The speaker interprets themselves as they talk

to self-correct and adapt to their audience if needed and the addressees interpret the utterance in real time which enables them to provide appropriate feedback. This symbiosis contributes to the fluidity of conversation as we already discussed in the previous section.

However, this synchrony deteriorates or disappears in many mediated communication forms. In video calls for example, a bad internet connexion can create a delay between the users which makes the production and interpretation slightly asynchronous. This can make it more complicated for a speaker to properly adapt to their addressees because the feedback they would receive would not be as well aligned with their production as they would expect. Indeed, even though we know that a delay is possible, it is quite complicated to estimate exactly how desynchronised the different participants are. However the most striking examples are written forms of communication which are in most cases asynchronous. Even though they can be used in more synchronous ways if the participants coordinate well, there will always remain a delay between production and interpretation since a message, e-mail or letter is first entirely produced and then sent. This deteriorates the usual feedback and adaptation mechanisms since one has to explicitly state which part was unclear if they require clarifications. Providing quick feedback is also more complicated in written communication since nodding or backchannelling are face-to-face mechanisms. Nevertheless, people tend to develop ways to remedy this and narrow the gap between written and face-to-face communication. For example on can break down an e-mail in small pieces and write their answer to each point separately, which in some ways resembles real-time feedback. Similarly, more and more messaging apps provide ways to react to messages very quickly through emojis. Again this supports the idea that face-to-face conversation is the simplest and most efficient communication form and that mimicking its features in other forms makes them more efficient as well.

Interpreting utterances at a time different from that of their production means that the speaker and the addressee may no longer share the same context. While this does not necessarily prevent mutual understanding, contextual information can be crucial in certain situations. If the speaker has not made the relevant aspects of the context explicit, the receiver must reconstruct the speaker's context in order to fully interpret their contribution. Otherwise, misunderstandings may arise and disrupt the conversation.

The loss of synchrony also makes clarification requests more complicated. In a synchronous setting, participants can interrupt the ongoing interaction to clarify a specific point. However, when there is a delay between production and interpretation, it may become more difficult to identify precisely which aspect of an utterance requires clarification. The initial speaker may also more easily misunderstand the purpose of a clarification request when the original utterance is no longer recent, particularly because they may have partially forgotten the intended meaning or context behind what they said. Moreover, if the clarification concerns information that

Table 2.12: Dialogue characterisation – Time: Synchrony.

Parameter	Minimal	Extended description
Synchrony	Binary	Time between production and interpretation

is necessary for interpreting subsequent contributions, the inability to resolve it immediately can complicate the interpretation of the rest of the conversation. Dialogue can therefore become more difficult to manage when speakers do not provide sufficient contextual information from the outset.

Studying language use in asynchronous settings thus changes the cooperation modalities since participants have to adapt to the fact that they will not be able to align with each other in real time. Alignment is made more complex by the increased cost of clarification requests, which can lead participants to try harder to understand on their own and possibly come up with hypotheses that enable them to interpret the conversation in a coherent way. Nevertheless those hypotheses are in most cases not verified. This position has similarities to the one we are in as language researchers since we often study language use retrospectively rather that in real time situations.

Asynchronous conversations also have a more or less important delay between interventions which means that the participants can be in different mindsets and physical contexts at different states of the conversation. This can make past contributions less relevant for the speaker when the addressee actually interprets them. Asynchronous conversations can last for extended periods of time, to the point that it makes it complicated to determine their exact beginning and end. This has implications regarding coherence since contradictions can appear in very long conversations without necessarily being disruptive if the contradictory statements are far enough from each other. However this distance is not straightforward to evaluate since it depends on the density of the conversation, its time rate and the flexibility of participants towards contradictions.

This parameter can be summarised in binary terms or with a more fine-grained value indicating the average time between production and interpretation. This value can be used to compare different communication forms and to determine the level of synchrony of a conversation. For example, face-to-face conversations would have a very low value while e-mails would have a much higher one. This parameter is summarised in Table 2.12.

Planning When using language, we always plan our utterances to some extent. In face-to-face conversation, we plan them while the current speaker is talking and continue planning on the fly as we speak. In many cases, we cannot afford to fully plan our next utterance before beginning to speak, as doing so may cause us to lose the floor. The amount of planning time available, however, depends on the

type or medium of interaction. Different forms of written communication, for example, allow for varying degrees of advance planning. In an email exchange, we often have ample time to formulate a response, although some email threads can become sufficiently rapid to resemble instant messaging. Likewise, when participants are simultaneously engaged in an instant messaging conversation, the demands on planning can closely resemble those of face-to-face interaction. By contrast, when engagement is lower or the exchange is more asynchronous, participants have considerably more time to plan their responses.

The level of planning can influence the way we produce language. For example, when we have more time to plan, we can afford to be more careful and to check our contributions before sending them. We can also take the time to think about the best way to express ourselves and to choose our words more carefully. On the other hand, with less time, we may be more likely to make mistakes or to use less precise language, thus leading to more repairs.

In LLMs, although we can prompt the model to produce an output that appears more or less spontaneous, such prompts do not fundamentally alter the way the model generates its response. Rather, they primarily affect the statistical distribution of the generated output. Newer models, however, often include a 'thinking' mode, in which the model performs an intermediate reasoning phase before producing the response shown to the user. This intermediate stage can be interpreted as a form of planning, insofar as it allows the model to refine its response before outputting it. Thinking mode has been shown to improve results on commonsense reasoning benchmarks (Wei et al., 2022); however, its underlying mechanisms should not be too readily equated with the planning processes involved in human conversation.

Planning is an important parameter for distinguishing between different forms of conversation, whether they occur between humans or involve conversational AIs. In its simplest form, it can be represented as a binary distinction indicating whether planning is possible before producing an utterance. A more informative characterisation, however, distinguishes between two complementary dimensions: the opportunity to plan the content of an utterance and the opportunity to plan its formulation. While these dimensions often correlate, they need not coincide. For example, a politician participating in a live interview may have had ample time beforehand to anticipate questions and prepare the content of their answers, yet have only limited opportunity to refine the wording of each response during the interaction. By contrast, if the interview is conducted in writing or recorded with the possibility of multiple takes, the speaker can also revise and optimise the formulation of their utterances before they are delivered.

This parameter is summarised in Table 2.13. It is operationalised in terms of *opportunities* rather than *time* because elapsed time is only a proxy for the opportunities available to plan content and formulate an utterance. Although additional time generally affords greater scope for planning, these opportunities also depend

Table 2.13: Dialogue characterisation – Time: Planning.

Parameter	Minimal	Extended description
Planning	Binary	Opportunities for content planning Opportunities for utterance formulation

on the constraints imposed by the interaction itself. This formulation accommodates both human and AI dialogue, where planning may be enabled by mechanisms that are not naturally expressed in temporal terms, such as the ability to revise a written response or, in the case of an LLM, to perform an intermediate reasoning phase before generating the final output.

Ephemerality Conversational interaction is by nature ephemeral. Each exchange takes place at a particular moment and under unique conditions –material, cognitive, and affective–that cannot be identically reproduced. The state of the physical environment (see Section 2.2.2.1) and the interlocutors' mental or emotional states jointly shape what can be said and how it is interpreted. As conversation is a situated and embodied activity(Goodwin, 2000; Mondada, 2018), its occurrence is inseparable from the circumstances of its production. Once the moment passes, the precise configuration of contextual, perceptual, and psychological factors that made the interaction possible also disappears. This irreversibility constitutes a fundamental property of dialogue: it unfolds in real time and cannot be re-enacted under identical conditions (Sacks, 1995).

Ephemerality must therefore be taken into account when characterising dialogue resources and tasks for two main reasons. First, it shapes the interaction itself by determining the resources available to participants when producing and interpreting conversational contributions. Second, as discussed in Sections 2.2.2.3 and 2.4.4, it conditions how conversations can be observed, recorded, and analysed, since researchers never have direct access to the original interaction but only to one of its representations.

Some media inherently preserve access to past interactions, and thus form a persistent record of it. In contrast, other media are ephemeral, meaning that once a contribution is made, it cannot be retrieved and revisiting it requires relying on memory and explaining what is being updated. This is the case for typical face-to-face conversation and has important implications for repairs and clarification requests.

The first aspect to characterise thus concerns the nature of the conversation, i.e. whether it is ephemeral or persistent. A more detailed description, however, can distinguish different dimensions of ephemerality, such as message persistence, editability, and whether participants have access to the edit history. These properties influence language production in mediated communication. For example, the

Table 2.14: Dialogue characterisation – Time: Ephemerality.

Parameter	Minimal	Extended description
Ephemerality	Binary	Persistence time Editability time Edition history
	Recording form	Third-party persistence time Third-party persistence awareness

inability to edit a previously sent message may encourage greater caution during composition or lead participants to adopt repair strategies, such as preceding a message with an asterisk ('*') to indicate a correction of the previous message.

The second aspect concerns the *representation* of the interaction. It can be characterised by the type of recording available (e.g., audio, video, or transcript) and, in the case of transcripts, by their level of detail. A richer characterisation may also specify whether participants are aware that their contributions will persist and for how long. This information is important both from an ethical perspective, since conversations that are publicly accessible are not necessarily intended to be reused for research or language model development, and from an interactional perspective, as speakers may adapt their behaviour according to their expectations about the persistence of their contributions. These different dimensions of ephemerality are summarised in Table 2.14.

2.3 Conversational frame

Besides who is participating and in what setting, the conversational frame defines the scope of the interaction. It includes the language that supports it (Section 2.3.1), as it shapes the content and structure of the utterances at a linguistic level, and the purpose behind the conversation (Section 2.3.2), which defines the goals and success conditions that guide the exchange.

2.3.1 Language

The language used in a conversation is a central feature, shaping not only the content but also the social and cultural dimensions of interaction. Language carries information about identity, social norms, and cultural background; it influences thought patterns, expression, and interpersonal coordination. The choice of language also has practical implications for accessibility, inclusivity, and the potential reach of a conversation. While this chapter does not provide an exhaustive discussion of language, it is important to emphasise its role when characterising

dialogue data. The research community has a tendency to focus on English, often implicitly, yet the world's linguistic diversity is immense, and different languages exhibit features that can profoundly influence conversation. For example, some languages encode complex systems of politeness and formality, affecting interactional behaviour (P. Brown, 1987); others encode time, space, or aspect differently, shaping how participants discuss events; and some languages encode emotions or evidentiality in ways that influence both expression and comprehension.

Claiming that a tool or theory is multilingual requires careful investigation for each of the allegedly covered language. Jouitteau and Grobol (2024) discuss the 'diversity washing' problem, which refers to the fact that companies sometimes claim to have developed multilingual tools that are supposed to cover mid-resourced languages, i.e. languages that 'typically [have] an intermediate level of digital development' (p.1, personal translation), without actually testing them on these languages. Resources for these languages are indeed available on the web, allowing for an apparent coverage of these languages, but the speakers are not involved in the development nor evaluation of the tools, sometimes leading to very poor performance as Jouitteau and Grobol (2024) show for the case of Breton to French translation.

The rise of large language models (LLMs) has brought further attention to language. While LLMs are trained on multiple languages, they are typically dominated by English and a small set of well-resourced languages. Neglecting language-specific characteristics can reinforce the marginalisation of low-resource languages and produce models that are poorly adapted to their conversational nuances (Bender et al., 2021). Moreover, the assumption that all speakers wish to see their languages represented in such models may overlook the sociocultural and ethical dimensions of language technology. Language is foremost a tool for *human* communication, and the development of computational tools should prioritise the needs and desires of speakers rather than the mere expansion of model capabilities (Bird, 2020).

When operationalising language as a parameter in dialogue data, it is important to specify the language(s) involved and consider how their characteristics may influence interaction. At a minimum, this can be represented as a simple list of languages, although a more detailed description of the linguistic features relevant to the conversation may be useful. For instance, we can indicate the level of control over language use, that is, the extent to which participants' language use is constrained. Social media data, for example, are typically much less controlled than data collected specifically for a research experiment. At the level of individual conversations, we can also indicate the participants' proficiency in the language, as this may influence both how they express themselves and how they understand one another. For example, if one participant has limited proficiency, they may have greater difficulty expressing themselves or understanding the other participants, potentially leading to misunderstandings and disruptions in the conversation. Finally, we can indicate whether a conversation is monolingual or multilingual and whether it involves code-switching. These sub-parameters are summarised in Table 2.15.

Table 2.15: Dialogue characterisation – Language.

Parameter	Minimal	Extended description
Global language	List	Control degree Proficiency Monolingual / Code-switching / Multilingual

2.3.2 Purpose

Traditionally, conversation has often been characterised by its goal – the set of information that must be exchanged for the interaction to be considered successful. This task-oriented perspective provides a clear framework for analysing dialogue in terms of steps, sub-goals, and the logical progression of questions and answers. However, other approaches, particularly Conversation Analysis, emphasised the social dimension of dialogue: how participants manage turn-taking, alignment, repair, and the subtle social work of maintaining relationships. To some critics focused on argumentative or goal-directed aspects, these social phenomena were deemed secondary or 'uninteresting' (Wooffitt, 2005), yet they are central to understanding how conversation actually unfolds (Goffman, 1967). Cognitive psychology has shown that speaking with others helps organise one's own thoughts, offload cognitive work, and refine emerging ideas through social scaffolding (Vygotsky, 1978; Kuhn, 1991). Indeed, conversation is not only a tool for information management or practical cooperation: it is also a deeply social activity. Work in anthropology, sociology, and CA has emphasised the role of talk in maintaining relationships, affiliating with others, and enacting social identities (Goffman, 1967; Sacks, 1995; R. Dunbar, 1996). Practices sometimes labelled 'gossip' (R. Dunbar, 1996) highlight that talk can serve bonding functions even when no task needs to be achieved, though the term itself still presupposes an informational exchange. In practice, task-oriented and socio-emotional dimensions are rarely separable; most conversations blend both, with social alignment supporting joint activity and joint activity reinforcing social cohesion. The following section examines how conversational purpose – practical, social, or both – shapes the organisation of dialogue and how different theoretical traditions have selectively foregrounded one dimension over the other (Section 2.3.2.1). It also dives into the intentions participants can have in conversations (Section 2.4.1.3).

2.3.2.1 Practical goal

Conversation is central to carrying out tasks. We talk to order bread in a bakery, to coordinate a meeting, to decide what to cook for dinner, or to jointly solve a problem. Even when alone, people often talk to themselves when reasoning or

organising their thoughts, highlighting the cognitive scaffolding role of dialogue-like behaviour.

Task-oriented conversations have been foundational in linguistics, pragmatics, cognitive science, and human-computer interaction. Tasks serve two methodological functions. First, they elicit talk while constraining its possible content, enabling systematic analysis and corpus sharing. Classic paradigms – including tangram description tasks (Clark & Wilkes-Gibbs, 1986) and maze-navigation tasks (Thompson et al., 1993) – have provided controlled environments for studying reference resolution, grounding, and coordination. Second, they offer a natural setting in which collaborative reasoning and information management can be observed, forming the empirical basis for information-state theories and logic-based dialogue models.

But task constraints also shape conversation in ways that diverge from everyday interaction. Highly specific tasks can narrow the scope of permissible actions and reduce the social flexibility of the encounter. Participants may allocate more cognitive resources to the task than they ordinarily would, or focus disproportionately on task-relevant cues at the expense of interactional nuance. When learning a task for the first time, speakers often struggle to balance understanding the task structure with attending to turn-taking, gaze, or politeness norms. With increasing familiarity, however, participants can take greater liberties: once a routine is well-practised, less verbal coordination is required, allowing social talk to emerge alongside task execution.

A simple everyday example illustrates this. In the restaurant of my laboratory, the cashier must register whether each customer's dish is vegetarian and record the sides they selected while the customer pays – often with a contactless badge. Although this procedure requires coordinated action, experienced customers and the cashier can execute it with minimal or no spoken instruction. Conversation thus shifts toward the social domain: greetings, small talk, or remarks about the day. The task is still performed, but its cognitive and interactional demands are low enough that social purposes dominate. When something unexpected happens, however – insufficient credit on the customer's account or confusion about the order – the social layer retracts and participants temporarily reorient toward the task. This flexible prioritisation is characteristic of natural conversation but is poorly captured by logic-based formal models concerned primarily with propositional content and goal-directed structure.

For dialogue systems, the lesson is significant. A system that rigidly assumes task orientation may misinterpret a user who does not seek problem-solving but simply wants to vent, chat, or affiliate. Conversely, social behaviours can meaningfully influence task outcomes – for example, in persuasion, negotiation, or customer service – yet remain under-modelled in information-centred frameworks. Understanding the task-social balance is therefore essential for designing emotion-aware, context-sensitive agents.

Many linguistic theories historically sidelined the social dimension of conver-

sation, treating it as noise that obscures the 'real' work of information transfer or argumentation. From Gricean pragmatics to formal semantics and much of computational dialogue modelling, the focus has been on truth conditions, rational inference, and the efficient transmission of content.

However, the social dimension is not peripheral – it may well be part of the evolutionary foundation of human language. Anthropological accounts (R. Dunbar, 1996) propose that conversation emerged as a scalable mechanism for social bonding in increasingly large groups, replacing earlier forms of one-to-one bonding such as physical grooming. Regardless of the original function, language today clearly plays a central role in maintaining social cohesion, managing impressions, and navigating interpersonal relations.

Social talk comes with its own norms and expectations, distinct from those underlying task-oriented dialogue. Consider the following conversation I had with friends on the way to go bowling. One of us had picked up a parcel and was carrying it, which led to an exchange as in (2.3)[3]:

(2.3) **G** What is in your parcel?
V A folding bowling bowl.

The ensuing dialogue involved collectively elaborating on the absurd properties of this fictional object – hydrogen boosters to make up for its lack of weight and strength, collapsible mechanisms to fit it in the small parcel – until the original factual question became irrelevant. While this conversation started with a genuine question meant to satisfy G's curiosity, V's answer and the collective agreement of the participants to follow V's lead completely changed the purpose of this multilogue, making it purely social. From the perspective of rational information exchange, this conversation violates most of Grice's maxims (Grice, 1975). Yet every participant recognised and embraced its social function: play, affiliation, and shared humour. By adding to the conversation, everyone contributes to grounding this topic and making the true answer to G's question increasingly irrelevant. Formal theories that evaluate moves in terms of truth, relevance, or goal advancement – QUD frameworks, belief-update models, argumentation-based approaches – struggle to characterise such interactions without deeming them defective, despite their clear social success.

While mainly social conversations are less studied in linguistics, they seem to be quite common in daily life and also follow some kind of rules. Small talk, or gossip, which has been more studied is a form of social conversation. While it is often considered as a way to exchange information, I would argue that the truth value of the shared information is not crucial and that the actual purpose of small talk is social bonding. However it varies in its realisation across cultures (Endrass et al.,

[3]The conversation originally happened in French and I do not have a precise recollection of the exact words that were used but its meaning is here.

2011) and does not exist in all of them (Cui, 2015) which can create challenges in intercultural interaction. This shows that small talk follows some rules even though they seem different from the ones we usually assign to conversation. Similarly, carrying out conversations such as the bowling bowl one where everything is clearly false can prove to be complicated for some people, namely those struggling with non-literal interpretations. Those conversations are thus worth analysing to understand how they differ from more task oriented exchanges and how they are processed by people.

Humans have continuously developed new ways to communicate, especially when face-to-face interaction is not possible. From letters to telephones, smartphones, and videoconferencing, each advancement has sought to bridge the gap between individuals separated by distance. Interestingly, some of these communication methods, initially designed for convenience or practicality, have evolved to increasingly replicate the dynamics of in-person conversation. Features like typing indicators, read receipts, and online presence flags simulate aspects of face-to-face feedback, catering directly to social expectations for responsiveness and mutual attention. This phenomenon underscores the fundamental social purpose of communication: to bond with others. Despite being mediated through technology, these interactions still fulfil a deep-seated need for connection, highlighting that, at its core, conversation is about establishing and maintaining social ties.

Recent trends – such as fine-tuned language models imitating loved ones or celebrities – exploit the affiliative pull of conversation, raising ethical questions about deception, attachment, and the erosion of boundaries between human and machine interaction. Proposals to use LLMs to reduce social isolation, particularly among the elderly, similarly require scrutiny: replacing human interaction with artificial dialogue may have profound effects on well-being, identity, and influenceability. Understanding the social dimension of conversation is therefore essential for evaluating the risks and benefits of these technologies.

As the laboratory restaurant example illustrated, natural conversations typically integrate task and social purposes rather than belonging exclusively to one or the other. When a task becomes routine, social talk often expands to fill the interactional space; when an unexpected problem arises, participants swiftly retract into task-oriented coordination. Even within a single activity type, the social affordances vary. A bakery interaction allows for light social exchange, but the amount tolerated depends on contextual factors such as queue length. A furniture salesperson may use socially skilled rapport-building to influence a customer's choices – a phenomenon that standard task-based models of 'slot filling' overlook entirely. Thus, although the underlying task (ordering bread, buying furniture) is consistent, the flow and structure of the conversation differ dramatically depending on social positioning, interpersonal goals, and emotional stance.

Some theories, such as Segmented Discourse Representation Theory (Asher & Lascarides, 2003), begin to capture aspects of these complexities by modelling re-

Table 2.16: Dialogue characterisation – Practical goal.

Parameter	Minimal	Extended description
Practical goal	Task-oriented / Social / Information request	Proportions

lations like Elaboration, Commentary, or Acknowledgement, which often encode social actions rather than propositional advancement. Yet even these frameworks have limited capacity to represent shifting participation, playful departures from truth-conditional content, or the negotiable boundary between task and social purposes.

Designing dialogue systems therefore requires more than modelling the informational structure of tasks: it demands recognising when social talk is expected, when it supports the task, when it conflicts with it, and how participants flexibly transition between the two. Understanding this meshing is crucial for analysing natural dialogue and for creating conversational agents that behave in ways users perceive as appropriate, respectful, and contextually attuned.

The operationalisation of this parameter should thus explicit the type of purpose of the conversation, whether it is mainly task-oriented, social or with a focus on information gathering. A more detailed description could also indicate the proportion of each type of purpose in the conversation. This parameter is summarised in Table 2.16.

2.3.2.2 Success condition

Conversational activities are generally associated with conditions under which they can be considered successful. These success conditions follow naturally from the purpose of the interaction and provide criteria for determining whether the activity has achieved its intended outcome. In formal dialogue frameworks, success is often defined explicitly, allowing systems or interactions to be evaluated against predefined objectives. However, the nature of these objectives varies considerably across conversational activities.

For task-oriented dialogues, success conditions are often relatively straightforward. A customer obtains the requested information, a travel reservation is completed, or a collaborative problem is solved. In these settings, success can frequently be measured by whether the expected information slots were correctly filled or whether the task was successfully completed. In contrast, social conversations pursue interpersonal rather than instrumental goals. Their success therefore depends on less tangible outcomes such as maintaining rapport, strengthening social bonds, entertaining participants, or creating a pleasant interaction. These objectives are inherently more difficult to formalise. For example, in the bowling bowl

conversation from (2.3), success would not simply consist in exchanging information, but rather in sustaining the playful make-believe frame so that all participants enjoyed the interaction and experienced a sense of social connection. A contribution that prematurely disrupted this shared frame could reasonably be perceived as a conversational failure, even if the factual content of the exchange remained perfectly understandable.

The notion of conversational success is particularly important for dialogue systems because it directly determines how their performance should be evaluated. Some success conditions, such as maintaining fluency or producing contextually appropriate responses, are relevant across many conversational settings. Others are activity-specific and must be defined with respect to the intended purpose of the interaction. Coherence is often cited as a universal criterion, yet everyday conversations routinely contain inconsistencies, brutal topic shifts, unfinished utterances, and locally incoherent sequences without being regarded as unsuccessful. Judging a conversation solely according to formal linguistic criteria therefore overlooks important aspects of human interaction. Likewise, evaluating conversational agents exclusively through task completion ignores the dialogical abilities that make interactions effective and enjoyable. If conversational systems are intended to replace or complement existing interfaces, they should ultimately be assessed not only on whether they complete a task, but also on whether they provide an interaction that users perceive as successful according to the goals of the activity.

While operationalising conversational success remains an open challenge, the description should encompass both practical and social dimensions. The practical dimension concerns whether the interaction achieves its intended outcome, such as completing a task or exchanging information, while the social dimension relates to norms and expectations for interpersonal conduct that must be upheld for the interaction to be considered successful. In many cases, these dimensions are intertwined: a task may be completed, but if the social norms of politeness or turn-taking are violated, participants may perceive the interaction as unsuccessful. Conversely, a conversation may be socially smooth and enjoyable, yet fail to achieve its practical objectives.

Regarding the practical dimension, we suggest to use the notion of Questions Under Discussion (QUDs; Ginzburg, 1995; Van Kuppevelt, 1995; Roberts, 2012) to represent the matters that must be addressed for the conversation to be considered successful. QUDs were introduced as a formal framework for modelling discourse content and coherence. As an analytical tool, they provide a way to explicit how a sentence or utterance fits in its context by considering that ‘each sentence in discourse addresses a (often implicit) QUD either by answering it, or by bringing up another question that can help answering that QUD’ (Benz & Jasinskaja, 2017). We thus propose to operationalise the practical dimension of conversational success in terms of the set of potential questions (PQ; Onea, 2016) that must be popped off the agenda – term chosen following Ginzburg (2012) – for the conversation to be

Table 2.17: Dialogue characterisation – Activity: Success condition.

Parameter	Minimal	Extended description
Success condition	Description	PQ to be popped out Topoi to recognise

considered successful. An important aspect of this is that popping a question off the agenda does not necessarily require that it is answered. Some questions may become irrelevant due to the evolution of the conversation or the participants' actions, and some exchanges may only require the questions to be raised and pondered without necessarily being resolved as in philosophical debates.

Regarding the social dimension, we suggest to use the notion of *topoi* (Breitholtz, 2021) to represent the norms and expectations that must be recognised and respected for the conversation to be considered successful. *Topoi* are generalised principles or commonplaces that guide reasoning and argumentation by letting participants infer what is appropriate or expected in a given context. They can be thought of as the shared background knowledge and social norms that inform conversational behaviour. For example, in French, interacting with a stranger, and in particular an elder, requires addressing them with the polite form *vous* rather than the familiar *tu* of 'you'. In terms of *topoi*, this can be represented as the principle that in a given context, the appropriate form of address is licensed. This *topos* would encode one norm of politeness that some people would consider necessary for a conversation to be successful.

2.4 Dependant descriptive parameters

The parameters described thus far define the context of a conversation. Together, they have consequences on the way it can actually enfold. Characterising those consequences can in certain cases be necessary to fully capture the nature of a dialogue type. For instance, if a group of friends is playing a social deduction game such as WEREWOLF™, AVALON™, or BLOOD ON THE CLOCKTOWER™ to cite a few, some of them will have to adopt a deceptive stance to hide their real identity and win the game.

In this section, we discuss different consequences that the context can have on the type of conversation. We first examine the influence on the three meta-parameters introduced in the previous section: the AGENTS (Section 2.4.1), the CONVERSATIONAL SETTING (Section 2.4.2), and the FRAME (Section 2.4.3). We then propose a fourth meta-parameter that captures the consequences of the format of VERBAL INTERACTION (Section 2.4.4).

2.4.1 Influence on the agents: Enunciative status

The enunciative status of a participant in a conversation refers to the identity they present and how they are perceived by others. In face-to-face interactions, this is typically straightforward as participants can see and interact with each other directly. However, in mediated conversations, especially on social media, the link between identity and content can be more tenuous. Anonymity allows individuals to communicate without revealing their true identities, which can lead to behaviours that differ from those in face-to-face contexts. Moreover, the rise of synthetic agents such as language models introduces new questions about the nature of interaction and the assumptions participants make about their conversational partners. The enunciative status of participants is therefore a crucial parameter for characterising dialogue data, as it shapes the dynamics of interaction, the establishment of common ground, and the interpretation of utterances. In this section, we begin by discussing the implications of anonymity in dialogue (Section 2.4.1.1), then we examine the influence of role and persona instantiation (Section 2.4.1.2), and finally we consider the impact of commitment level on conversational dynamics (Section 2.4.1.4).

2.4.1.1 Anonymity

A conversation typically involves participants who are aware of each other's identities. In face-to-face conversations, this is immediately evident as participants can see and interact with each other directly but it is also the case in most forms of conversation such as text messages and phone calls. This allows them to adjust their communication styles such as tone, register, detail level, and word choice, according to the social context and the people they are addressing. This principle is not limited to dialogue; even in texts that appear monologic, communication is never produced in isolation: it is oriented toward an interlocutor – real, implied, or projected – whose presence guides the organisation, style, and content of the message. As mentioned in Section 2.1.2.1, Y. Wu et al. (2024) have shown that 'reader and writer expectations [seem to] align to a certain degree'. Similarly, participants in conversation project particular personas, consciously or unconsciously signalling aspects of their identity, social role, or expertise.

Social media and other anonymous platforms complicate this dynamic by decoupling identity from content. Participants can communicate with unknown individuals, adopt pseudonyms, or even share accounts among multiple people. The perceived reduction in accountability often leads to behaviours that diverge from what an individual might display in face-to-face contexts. A well known example of this is cyberbullying which is exacerbated by anonymity (Barlett et al., 2016).

In more extreme, albeit increasingly common, cases, the interlocutor may not even be human – but rather a generative language model, bot, or other synthetic agent – which raises additional questions about the nature of interaction and the

Table 2.18: Dialogue characterisation – Enunciative status: Anonymity.

Parameter	Minimal	Extended description
Anonymity	Binary	Anonymous / Pseudonymous / Real name Digital / Physical

assumptions participants make about their conversational partners. The question of 'Is my interlocutor human?' has become increasingly salient in online environments, whereas it was rarely a consideration a few years ago and is never one in traditional face-to-face settings.

This decoupling of identity from utterance presents methodological challenges for research. When studying online conversations, we cannot assume that a given pseudonym corresponds to a single human agent. The presence of multiple participants behind one identity or synthetic interlocutors can distort measures of individual behaviour, social alignment, and interactional norms. Similarly, while the amount of synthetic content online is complicated to evaluate, it is undoubtedly growing and language resources such as *X* (formerly *Twitter*) or *Reddit* will become less and less reliable.

For studies that rely on human-like production, as well as for the development of dialogue systems, failing to account for such uncertainties may lead to misleading conclusions or inappropriate model behaviours. Anonymity should therefore not be treated solely as a social or ethical concern; it is a fundamental parameter that shapes both the structure and interpretation of interaction, particularly in mediated environments. When analysing dialogue data, it is crucial to determine whether participants can establish a stable interactional history and, more fundamentally, whether they are human. While this dimension can be represented in a minimal way as a binary distinction, a more fine-grained characterisation is often necessary. For instance, one may distinguish between anonymous, pseudonymous, and real-name interactions, as well as between digital and face-to-face contexts, each of which entails different constraints on identifiability and continuity. These sub-parameters are summarised in Table 2.18.

2.4.1.2 Role and persona instantiation

In most conversations, participants act as themselves, although they may adapt their behaviour to the context. For example, a doctor will speak differently with patients than in everyday conversation, yet remains fundamentally the same person. However, many dialogue resources involve participants who are required to adopt a different role. The clearest examples are TV shows or films, where actors portray characters, but experimental paradigms such as the *Wizard of Oz* also require participants to play a role such as that of a conversational agent. While the precise

influence of role adoption on conversational dynamics is difficult to quantify, it can meaningfully shape interactions. Similarly, many recent dialogue datasets instruct participants to adopt a persona (S. Zhang et al., 2018; Yamashita et al., 2023), which can lead to unnatural conversational patterns (see example (0.1) in the Introduction). Persona is subtle and deeply embedded in social interactions, making it difficult to capture in short exchanges or to manipulate experimentally.

This parameter is particularly relevant in the context of prompting LLMs or conversational AIs. Explicitly assigning a role can strongly influence model output. For example, prompts that encourage role-playing have been shown to improve performance, with instructions to 'act human' making a model's responses appear more human-like. Indeed, Jones and Bergen (2026) showed that, in a three-party Turing test (Turing, 1950) setting – where one participant interacted separately but simultaneously with two interlocutors, one human and one LLM—human participants identified GPT-4.5 as the human interlocutor in 73% of the interactions when the model received a role-playing instruction. Without the persona instruction, however, the same model was mistaken for the human in only 38% of the interactions. While this substantial difference confirms that persona-infused prompts can significantly influence LLM outputs, it also suggests that our expectations of what human-produced conversation should sound like are not necessarily accurate. If participants were unable to distinguish between conversations produced by the human and the model, one would expect performance to be close to chance (50%). The significant deviation from chance level therefore indicates that participants consistently perceived differences between the two types of interlocutors, but that these perceived differences were not reliable indicators of whether the interlocutor was actually human.

Explicitly describing the role assumed by participants – human or artificial – is therefore an important parameter for characterising dialogue and studying the influence of role on behaviour. It is also critical for understanding linguistic phenomena shaped by social identity, such as *dog whistles* (Saul, 2018; Breitholtz & Cooper, 2021; Quaranto, 2022), which require knowledge of both the roles participants adopt and the groups with which they identify.

Role instantiation can be represented minimally as a binary parameter indicating whether the participants are playing a role or not but it can also be described more finely. For instance, participants may (i) act as themselves without specific instructions, (ii) emphasise a particular aspect of their identity relevant to the situation, or (iii) adopt an assigned role. This distinction is particularly informative when comparing human and LLM behaviour, such as how humans alter their responses when 'putting on a doctor's hat' versus behaving freely, and how LLMs respond to role-specific prompts. These sub-parameters are summarised in Table 2.19.

Table 2.19: Dialogue characterisation – Enunciative status: Role instantiation.

Parameter	Minimal	Extended description
Role instantiation	Binary	Self / Self specific / Assigned
Persona instantiation	Binary	Self / Self specific / Assigned

2.4.1.3 Intention

Communicative purpose in conversation is fundamentally intentional. Every contribution reflects not only *what* a speaker chooses to communicate but also *how* they choose to communicate it. As mentioned repeatedly, at the most general level, conversation is typically analysed under the assumption that participants are cooperative, following the principles described by Grice (1975). This cooperative stance constitutes a useful default for explaining how interlocutors produce and interpret utterances. However, many conversational situations intentionally depart from these expectations. Speakers may seek to persuade rather than merely inform, entertain rather than accurately describe, or even mislead. Such departures are not exceptions to communication but integral parts of conversational behaviour. They are closely related to the notion of illocutionary force, namely the communicative act that an utterance is intended to perform (Searle, 1969; Austin, 1975), while extending beyond it to encompass broader strategic goals.

One important dimension of communicative intention is *argumentation*. From the speaker's perspective, argumentation provides principles for structuring conversational contributions in order to justify a claim, convince an interlocutor, or support the transmission of information. Formal approaches to argumentation and discourse organisation, such as Rhetorical Structure Theory (Mann & Thompson, 1988), describe the relations that connect discourse units and explain how larger argumentative structures emerge from local exchanges.

Deception can be viewed as a natural extension of argumentation. Rather than simply presenting information, the speaker deliberately attempts to influence the interlocutor's beliefs while concealing their own epistemic state. This requires maintaining multiple representations of the conversational situation: the world model that the interlocutors are expected to accept and the speaker's own representation of reality. Producing deceptive dialogue therefore involves carefully managing commitments, consistency across turns, and the risk of revealing contradictions. This additional representational burden distinguishes deceptive interaction from ordinary cooperative communication and has consequences on the produced language at different levels like word choice (Pérez-Rosas & Mihalcea, 2015) and syntactic structure (Yancheva & Rudzicz, 2013).

Communicative intentions also encompass the speaker's intended stance or framing of an interaction. A conversational contribution may aim to be humor-

ous, ironic, sarcastic, playful, or provocative without altering its propositional content. These intentions fundamentally shape how an utterance should be interpreted. For example, the earlier example (2.3) about the folding bowling bowl illustrates how the same factual response can be framed in different ways depending on the speaker's intended stance. In its original context, it was considered a playful and humorous contribution to a social exchange. The same interaction with a child or someone who was not expected to understand that the folding bowling bowl was a fictional object could have been interpreted as an attempt to deceive or exploit the listener's naivety. Such interpretations depend on the interlocutors' ability to infer the speaker's intended frame and relate closely to theories of common ground, relevance, and pragmatic inference.

These various intentional dimensions are neither equally salient nor simultaneously relevant in every conversation. Their importance varies depending on the interactional context, participants, and communicative task. Nevertheless, they substantially influence both language production and interpretation. Recognising another participant's intentions enables appropriate pragmatic interpretation, whereas failures of intention recognition frequently lead to misunderstandings, conversational breakdowns, and subsequent repair sequences. Rich interactional settings such as face-to-face conversation facilitate intention recognition through multimodal cues which complement linguistic content and provide redundant evidence for inferring speakers' goals.

The importance of modelling communicative intentions has also become increasingly apparent in human-AI interaction. While LLMs cannot have intentions in the human sense, the outputs they generate are often interpreted as if they were intentional due to their sycophantic style, leading to potentially harmful judgements from users (Cheng, Lee, et al., 2026). Highlighting the difference between displayed and actual intentions could help users better understand the capabilities and limitations of AI systems.

From an operational perspective, communicative intentions may be represented at different levels of granularity. A minimal characterisation could simply indicate whether a conversational setting presupposes particular strategic intentions beyond ordinary cooperative interaction, including whether the participants are truthful or have hidden intentions, and whether those intentions were imposed to them within the conversational setting. A richer representation would instead identify the intention dimensions relevant to each participant and distinguish between a participant's actual intentions and their beliefs about the intentions of others. The specific dimensions would naturally depend on the application and an in-depth operationalisation of this parameter would require further insights from pragmatics, sociolinguistics, argumentation theory, and rhetoric. One possible formalisation is to associate each ordered participant pair with a vector of weighted intentional dimensions – such as informing, persuading, deceiving, entertaining, or provoking – capturing both communicative goals and the interlocutor's intended interpretation of those goals. This

Table 2.20: Dialogue characterisation – Purpose: Intention.

Parameter	Minimal	Extended description
Intention	Cooperativeness Truthful / hidden Chosen / imposed	Weighted dimensions for each ordered participant pair

characterisation is summarised in Table 2.20.

2.4.1.4 Commitment level

The commitment level of a participant can be defined as the degree to which they are invested in a conversation and its truth models. It reflects the extent to which participants consider themselves accountable for the content of their utterances and are willing to be held responsible for them. In typical face-to-face interaction, participants are generally fully committed: they are directly engaged with one another, receive immediate feedback, and are expected to be accountable for what they say. Although the topic of conversation may modulate the potential consequences of a statement, participants are generally assumed to endorse the truth of their utterances and to participate sincerely.

However, commitment is not always absolute. Participants may adopt a more playful, experimental, or strategic stance, in which they are less concerned with literal truth and more focused on exploring possibilities, entertaining perspectives, or sustaining the interaction. For example, in games of make-believe, participants temporarily commit to a fictional reality for the sake of the interaction, even though they do not endorse it outside that context. Clark (1996) illustrates this with the story of two children 'playing a game of make-believe' in which they impersonate Wild Bill and Calamity Jane, two characters from a book they have read. The characters live in 'Deadwood, Dakota Territory, during the gold rush of 1876'. During the game, they use real life elements from the yard they are playing in to mimic gold digging and panning[4]. In this scenario, the children commit to the truth of their statements, their characters, and their interpretation of the physical environment for the length of the game but these commitments will not transfer outside of it. This highlights how commitment varies with context and intention, shaping both the interpretation of utterances and the coordination of conversational actions.

The notion of commitment is particularly relevant when contrasting human conversation with language produced by LLMs. While LLMs can generate coherent and contextually relevant text, they do not possess beliefs, intentions, or a sense of accountability (Bender et al., 2021; Schlangen, 2025). Their outputs are the product of statistical patterns learned from data rather than a commitment to truth

[4]Gold *panning* is a traditional method for extracting gold from river sediment with a pan.

Table 2.21: Dialogue characterisation – Enunciative status: Commitment level.

Parameter	Minimal	Extended description
Commitment level	Binary	Status: role-playing, joke, etc. Scope: conversation, task, …, general

or meaning. This fundamental difference has important implications for how we interpret LLM-generated text and what expectations we can reasonably have for interactions with such systems. Searle's (1969) work on illocutionary acts offers conceptual tools to distinguish between the act of asserting in human conversation and the pattern-based generation of text by LLMs.

Commitment level is less straightforward to operationalise than other parameters as it is not directly observable. However, the aspects directly defined by the conversational task can be a first axis to characterise it. This can be defined in a binary way by presenting each participant as either committed ot not to their utterances, which could be enough for example to distinguish human and LLM participants. However, a more fine-grained characterisation can be useful to distinguish between different forms of commitments such as the one of a participant who is committed to the truth of their utterances but as a character in a game of make-believe. The scope of the commitment can also be defined in a more fine-grained way by defining a truth model that is limited to a specific context or conversation as opposed to a more general one. These aspects are summarised in Table 2.21.

2.4.2 Conversational setting

The parameters we have described to define the CONVERSATIONAL SETTING, i.e. the world setting and the environment and medium, exist independently from the AGENTS and the FRAME of an interaction. However, the CONVERSATIONAL SETTING can also be influenced by those two meta-parameters.

We begin by describing how the world setting may change as a result of the actions of participants' actions and how this can be operationalised (Section 2.4.2.1). We then discuss how the *perception* of the environment can be constrained in a given activity (Section 2.4.2.2).

2.4.2.1 World stability

World stability refers to the extent to which the assumptions and truths underpinning a conversation remain fixed throughout the interaction. While some conversations rely on a stable shared reality, in many interactions the world is dynamic, evolving as participants introduce new information, reinterpret prior statements, or negotiate the meaning of terms. For example, the course of a joint storytelling task

may shift as new narrative details are introduced, process that is accentuated by the number of participants.

Not all changes to the conversational world are immediately accepted. Participants may need to negotiate, clarify, or explicitly acknowledge new information for it to be incorporated into the shared framework. This reflects the inherently collaborative nature of conversation, where interlocutors continuously co-construct the epistemic environment that enables coordinated action (Clark, 1996; Clark, Brennan, et al., 1991).

The stability of the conversational world therefore shapes multiple aspects of dialogue. It influences how topics develop, how turns are allocated, and how participants achieve alignment. In interactions with artificial agents, a lack of shared world stability can lead to misaligned dialogue strategies or incoherent responses. Consequently, both the setting and stability of the world must be considered as central parameters when characterising dialogue, particularly in research on human–machine interaction or computational modelling of conversation.

Some experiments for building dialogue resources explicitly manipulate world stability by providing participants with a persona at the beginning of the conversation and changing it later. It is the case of the PersonaChat corpus (S. Zhang et al., 2018) that we briefly discussed in Chapter 0. Precisely describing such manipulations is crucial for understanding how participants adapt to these changes and how they affect the flow of conversation. For example, if the persona is changed without any warning, it may lead to a breakdown in coordination and a significant shift in the conversation's trajectory. If the change is introduced gradually or with some form of explanation, participants may be able to adjust more smoothly, maintaining coherence despite the evolving world. Operationalising world stability can be done with a binary parameter indicating whether the world is stable or not, but a more detailed description could indicate the degree of stability, the frequency of changes, and the manner in which changes are introduced and negotiated among participants. This would provide a richer understanding of how world dynamics influence conversational structure and content.

Similarly to the COMMITMENT parameter discussed in Section 2.4.1.4, the operationalisation of world stability heavily depends on the specific task and context of the conversation. In many cases, a binary distinction may suffice to indicate whether the world is expected to remain stable or is subject to change. However, in more complex scenarios the rely on the manipulation of the epistemic grounding of the conversation, the types of changes that are intended to occur and the mechanisms by which they are introduced may need to be specified in greater detail. In particular, the source of changes – whether they are introduced by the participants themselves, by an external agent, or by the environment – is often a descriptive parameter of the task in corpora collected for research purposes. Table 2.22 summarises these parameters. As indicated by the title of the table, we include world stability under the more general parameter *Epistemic grounding*, which captures

Table 2.22: Dialogue characterisation – Epistemic grounding: World stability.

Parameter	Minimal	Extended description
World stability	Binary	Planned changes Frequency of changes Source of changes (participants, external agent, environment)

the set of assumptions, facts, or truths upon which a conversation is based. This broader parameter encompasses the world setting together with the world stability.

2.4.2.2 Perceptual capacities

While the environment and medium define the maximal set of modalities a conversation can rely on, the agents' capacities and the task at hand may constrain the extent to which these modalities are actually used. For example, a conversation taking place in a noisy environment may limit the effectiveness of auditory communication, while a task that requires visual attention may reduce the participants' ability to engage in gesture based communication. The different dimensions of the environment and medium parameter should thus be extended to account for these aspects.

Within a given physical context, the perceptual fields of the participants may not be the same. If they are sitting back-to-back in a room, they won't have access to the same parts of their surroundings. Similarly in a car, the driver and passengers occupy the same physical space but do not share the same perceptual field: the driver must focus on the road, whereas the passengers are not subject to the same attentional constraints.

Regarding the visual modality, the degree of attention that participants can devote to directing their own gaze and to monitoring that of their interlocutors, but also into interpreting gestures and facial expressions, can vary across situations. Although participants may be physically co-present, task demands can constrain their ability to establish or maintain mutual gaze. For example, a driver and passengers occupy the same physical space, but the driver cannot devote sustained visual attention to the passengers.

Similarly the attention a participant can devote to the auditory modality is constrained by whether they need to listen to other sounds in the environment, and more generally whether their attention is already engaged in another matter.

The updates required to the environment and medium parameter to account for these aspects are summarised in Table 2.23.

Table 2.23: Dialogue characterisation – Extension of Environment & Medium.

Parameter	Sub-parameter	Extended description
Physical context	Binary	Shared perceptual field
Visibility	Gaze	Allocation degree
	Gestures	Allocation degree
	Facial expression	Allocation degree
Audibility	Prosody / Tone	Allocation degree

2.4.3 Influence on the frame

The interaction between the group of participants, the medium and the purpose of the conversation affect its frame at more or less latent levels. A directly observable aspect is the vocabulary that participants are likely to use (Section 2.4.3.1), which is constrained by the activity and the domain of the conversation. A more latent aspect is the topical structure through which information is organised and progresses (Section 2.4.3.2), which is shaped by the activity but also the participants' intentions.

2.4.3.1 Surface consequences: Vocabulary

The activity in which a conversation takes place strongly influences the vocabulary that participants are likely to use. While language always exhibits lexical variation, conversational activities constrain the set of concepts, expressions, and linguistic registers that are expected to appear. As a result, conversations associated with different activities may differ substantially in their lexical characteristics, even when they involve similar participant settings or communicative goals.

A first factor is the proficiency level expected by the activity. Conversations involving experts or highly educated participants often employ a richer and more specialised vocabulary than interactions aimed at novices or children. Likewise, educational or technical activities may require precise terminology, whereas everyday conversations generally favour more common lexical items and simpler formulations. The expected proficiency therefore influences both the complexity of the vocabulary and the assumptions speakers make about their interlocutors' lexical knowledge.

A second factor is the domain of the conversation. Activities centred on medicine, law, engineering, finance, or other specialised fields naturally rely on domain-specific terminology that may be uncommon outside those contexts. Even when participants are not experts, the topic itself introduces concepts and expressions that are characteristic of the domain under discussion. Domain-specific vocabulary is consequently an important source of lexical variation and often represents a challenge

Table 2.24: Dialogue characterisation – Activity: Vocabulary.

Parameter	Minimal	Extended description
Vocabulary	Genericity degree	Proficiency level Domain Community

for both human analysts and computational dialogue systems.

Finally, vocabulary is shaped by the community to which speakers belong (see for example Section 6.4.3 for examples of practices on the social media platform *Reddit*). Social groups, professions, online communities, or regional speech communities frequently develop their own jargon, abbreviations, slang, or shared expressions that facilitate communication among members while remaining opaque to outsiders. Such community-specific vocabulary contributes to the identity of the group and reflects shared experiences, practices, and cultural references. It also illustrates that lexical variation cannot be explained solely by topic or expertise, but is equally influenced by social affiliation.

These dimensions are particularly important when analysing conversational data. Highly specialised or community-specific vocabulary may reduce the ability of external annotators to correctly interpret the interaction, increasing the risk of annotation errors or inconsistent analyses. From a computational perspective, dialogue resources collected within a particular domain or community may also exhibit limited generalisability, as language models trained on such data can struggle when applied to conversations involving different lexical conventions (C. Liu et al., 2024). Conversely, general-purpose dialogue corpora with broad vocabulary coverage may be less informative for specific applications, as they may not capture the specialised terminology or social nuances relevant to a particular activity.

Explicitly characterising the expected vocabulary of a conversational activity therefore facilitates both the interpretation of dialogue resources and the assessment of their applicability to downstream tasks. From an operational perspective, a minimal characterisation of vocabulary may simply indicate its degree of genericity, ranging from broadly accessible language to highly specialised terminology. A richer representation can further specify the factors responsible for lexical variation, including the expected proficiency level of the participants, the domain of the interaction, and the communities whose linguistic conventions shape the conversation. These dimensions are summarised in Table 2.24.

2.4.3.2 Latent consequences: Topical structure

Topics – understood in the intuitive sense of *what the conversation is about* – are a central organising dimension of dialogue. They shape what contributions are

relevant, what inferences can be drawn, and how participants decide whether an utterance fits the ongoing flow. An off-topic remark can disrupt coordination or require repair; conversely, a well-timed topical shift can reorient the group or open a new line of joint action. Despite their importance, topics remain notoriously difficult to define and formalise (e.g. G. Brown & Yule, 1983; Schiffrin, 1987). In this section, we adopt the pragmatic, participant-oriented notion commonly used in conversation analysis and discourse studies: topics are the locally recognised domains of talk that participants themselves treat as current, shifting, or suspended.

In open, informal, face-to-face conversation, topical structure is typically fluid. Topics may develop gradually, shift abruptly, or fragment into loosely connected strands. In multiparty settings these dynamics become especially complex: each participant can add new associations, reinterpret previous turns, or shift attention to a neighbouring sub-group. Once groups reach four or more participants, it is common for multiple simultaneous 'floors' to emerge (Egbert, 1997), each sustaining its own topic while remaining loosely linked to other strands through participants who move between them. Even in dyadic conversation, more than one topic can be 'open' at once – for instance, when interlocutors conduct a task while engaging in parallel social talk. These are not simultaneous in the acoustic sense, but they coexist as latent threads to which speakers can return.

Written and computer-mediated communication expands this even further. Asynchrony allows interlocutors to pursue several topics at once, reply to older turns out of chronological order, or maintain long-lived topical threads that re-emerge after hours or days. This environment makes explicit a capacity that also exists in spoken interaction, though typically in subtler forms.

Topic dynamics also depend strongly on the activity type. In task-oriented dialogues, the set of admissible topics is shaped by the task structure and can be more or less rigid. In institutional settings – court hearings, political debates, news interviews – topical scope is heavily regulated: participants may be prohibited from entering certain topics, required to remain on a prescribed agenda, or allowed to shift topics only through specific, institutionally sanctioned moves. Here, the topical structure is part of the interaction's normative framework.

Computational approaches have typically operationalised topic through segmentation tasks. *Linear* topic segmentation aims to divide a transcript into contiguous segments, each associated with a distinct topic (Hearst, 1997). This is useful for meeting analysis, summarisation, or information-retrieval applications, where the goal is to locate sections of interest. However, linear segmentation presupposes that each stretch of talk belongs to exactly one topic at a time – an assumption that fails in many natural conversational settings, especially multiparty dialogue, where overlapping or interleaved topics are common.

Hierarchical topic segmentation attempts to capture nested structure, but it is less commonly used due to its complexity and still does not easily track *parallel* topic threads. Villing (2015) shows that participants can sustain several topical

trajectories simultaneously and rely on context and common sense to identify the right topic for each utterance. In addition, speakers often produce meta-comments about the conversation – remarks on relevance, transitions, humour, or framing – without leaving the current topic, further complicating formal segmentation.

These observations point to the need for theories of topical structure that can accommodate concurrency, rapid shifts, embedded commentary, and the interpretive work participants perform to maintain coherence.

NLP has long acknowledged the importance of topic for natural language understanding, yet computational treatments remain disproportionately focused on task-oriented settings. Topic modelling and segmentation algorithms were developed largely on expository text, technical documents, or structured meetings. Attempts to adapt these methods to casual, open-domain conversation have been limited by the highly dynamic, interaction-driven nature of topical organisation: shifts are subtle, boundaries are fuzzy, and contributions often rely on shared context rather than lexical cohesion.

While linear segmentation has clear usefulness in meetings or customer-support interactions, it transfers poorly to conversations with fluid purposes and multiple concurrent strands. As conversational agents grow more common – and as applications increasingly require systems to sustain long-term, open-domain interaction – modelling topical structure becomes indispensable. Systems must track what topics are currently active, when a user reopens an older thread, when an utterance is meant as a meta-comment, and which topical expectations are at play.

The challenge is substantial as the topical relevance of an utterance depends on many parameters like the conversational setting, the number and roles of participants, the medium, and the broader activity in progress. Humans are remarkably adept at inferring topical connections even in unpredictable or elliptical contexts; capturing this interpretive flexibility in computational models remains an open problem.

A first attempt at characterising the topical structure of a conversation is proposed in Table 2.25. A minimal characterisation could indicate whether the topical structure is constrained by the activity or not. A more detailed description could include the skeleton induced by the task like an agenda in meetings, or a formal structure like the one used in interviews. However those scenarios are very task-dependant and a more generic parameter like the degree of structure and predictability of a conversation would allow us to compare different types of dialogues, which is one of the primary goals of the descriptive framework proposed in this chapter. This direction is thus the focus of Part II.

2.4.4 Verbal interaction: The co-constructed nature of dialogue

Dialogue is more than the mere production of utterances in proximity; it is the joint activity of talking *to* and *with* one another. It requires participants not only to speak

Table 2.25: Dialogue characterisation – Activity: Topical structure.

Parameter	Minimal	Extended description
Topical structure	Constraint degree	Agenda Scenario

but to attend, listen, understand, and coordinate – whether this occurs in face-to-face interaction or through mediated text. Conversation is thus fundamentally co-constructed: speakers continuously shape one another's contributions by signalling attention and comprehension, adapting to emerging meanings, and managing turns in ways that allow the interaction to flow. Crucially, much of this coordination unfolds simultaneously, with production, monitoring, and feedback interwoven in real time.

All these aspects and the way they are realised in conversation are thus influenced by the parameters we have defined in the previous sections. The AGENTS (Section 2.1), the CONVERSATIONAL SETTING (Section 2.2), and FRAME (Section 2.3) affect the temporal dynamics of the conversation and hence shape how co-construction occurs. Therefore, we define a fourth meta-parameter in our descriptive framework, VERBAL INTERACTION, which captures the form of co-construction that characterise dialogue.

In this section we focus on the mechanisms of co-construction in dialogue. We first discuss the coordination signals that enable participants to monitor and adapt to each other in real time (Section 2.4.4.1), then we examine the turn-taking system that structures the flow of conversation (Section 2.4.4.2), and finally we consider mutual completion, a phenomenon where participants collaboratively complete each other's utterances or thoughts (Section 2.4.4.3).

2.4.4.1 Coordination signals

Face-to-face conversations enable speakers to monitor their addressees in real-time and receive feedback from them in various forms. Whether it is through gestures, facial expressions or backchannels, feedback lets the speaker adapt to their audiences and make sure that they are adequately coordinated (Clark & Schaefer, 1992). Indeed, ignoring the other participants is a failure if one wants to have a conversation. Conversely, providing irrelevant feedback as a listener can completely disrupt the flow of conversation. In fact cooperating to understand each other reduces the mental cost compared to understanding what someone says without repairs, without having the possibility to request more information on specific points, without giving feedback as it goes, etc. (Jefferson, 1972; Schegloff, 1979; Dingemanse et al., 2015; Van Arkel et al., 2020). So this symbiotic relationship is a crucial point in conversation in the sense that alignment makes conversation smoother and less cog-

nitively demanding than other forms of communication with solo-understanding. For this reason, being able to both adapt to the addressees based on their feedback and provide the right amount of it at the right time are important skills when it comes to conversation.

This core feature of conversation deteriorates the more asynchronous a communication form is. Indeed, one requires real-time feedback to be able to adapt to their addressees. It also becomes more and more complicated for the recipient to ask for clarifications as time passes since it requires re-raising points that can be far in the past from the speaker's point of view. While it remains possible, it is heavier in terms of conversational practices and participants could give up on some forms of repairs in order to avoid disrupting the conversation too much.

A way of mitigating the loss of this feature in asynchronous forms of communication is to provide more explicit information from the outset. However, this also increases the amount of information that recipients must process. In instant messaging tools, an increasing number of features have been developed to enable users to provide explicit feedback to interlocutors, such as reacting to messages with emojis. Similar features are also available in some video-conferencing tools, even when the camera is turned off. Although these features do not fully address the challenges of highly asynchronous communication, they illustrate how tools originally designed around asynchronous interaction are increasingly attempting to reproduce aspects of face-to-face conversation.

The influence of coordination signals on cognitive resources and on the way language is produced makes them a crucial aspect of dialogue to consider when characterising and selecting resources. While a simple binary description of the presence or absence of such signals may be sufficient in some cases, several dimensions can be elaborated to provide a more detailed characterisation. For example, we can specify the availability of different feedback channels (vocal, textual, visual, haptic) for each participant, as well as the participants' technical ability to use these channels. For instance, backchannels require an oral channel, but not all oral productions function as backchannels. Conversational AI systems are often characterised by their lack of backchannels, and the resulting absence of feedback makes interaction with them different from human-human interaction. The density of signals can also be specified, with the advantage that it provides a numerical value that can be readily incorporated into computational models. Finally, the latency tolerated for receiving feedback can be important when comparing different forms of mediated conversation. In face-to-face conversation, feedback can be provided with very low latency and is crucial for maintaining conversational flow. In asynchronous forms of communication, latency can be much higher without necessarily preventing the conversation from unfolding. However, when latency becomes excessive, it can become difficult to track different conversational threads and maintain a coherent dialogue. These different dimensions for describing coordination signals are summarised in Table 2.26.

Table 2.26: Dialogue characterisation – Verbal interaction: Coordination signals.

Parameter	Minimal	Extended description
Coordination signals	Binary	Availability of channel (vocal, textual, visual, haptic) per participant Technical ability per channel and participant Signal density per participant Feedback latency tolerance

2.4.4.2 Turn-taking

In conversation, people take turns speaking, typically with very short gaps and relatively little overlap. This phenomenon, known as turn-taking, has been extensively studied in Conversation Analysis. Sacks et al. (1974) established a set of loose organisational principles that describe how turns are allocated. Although these principles have been criticised for their lack of strictness, the general idea they capture remains valid: in the vast majority of cases, the floor changes at what have been called transition-relevance places (Sacks et al., 1974, p.703), and the next participant is either selected by the prior speaker or the floor becomes available for whoever takes it first. The idea that there are few to no overlaps, however, is not entirely accurate. While overlaps between substantive utterances – those that contribute information relevant to the ongoing topic – are relatively rare, overlaps between the current speaker and feedback from other participants are frequent. As discussed in the last paragraph, such feedback is a crucial feature of conversation.

Historically, these types of utterances were often considered errors and were not studied as part of the intended content of conversation. However, current theories acknowledge their importance. Nevertheless, overlaps, whether resulting from feedback, false starts, or mismanagement of turn-taking, create challenges for the study of conversational data. In many cases, for practical reasons such as privacy constraints and ease of storage, linguists rely on transcribed conversations, which provide a reduced representation of the original interaction. Since transcriptions are often the primary form in which conversations are analysed, the transcription schemes employed are crucial for preserving as much information from the original dialogue as possible. Jefferson (2008) proposed a transcription scheme that enables the representation of, among other conversational features, overlapping speech. While this scheme is now widely used, it is not universally adopted because it requires more effort than simpler transcription approaches.

With advances in technology, new methods for storing and transcribing conversational data have emerged. For instance, tools now exist that allow transcripts to be aligned with audio recordings, making it possible to preserve overlaps and represent turn-taking mechanisms in greater detail. However, such representations are less

straightforward to process than linear sequences of utterances, which are the format assumed by many tools developed in NLP. As a consequence, speaker diarisation systems still struggle to handle overlapping speech, limiting their ability to process multi-party conversations (Addlesee et al., 2020). This is particularly problematic because researchers increasingly rely on automatic tools to prepare and structure data, as these approaches are substantially cheaper and less time-consuming. As a result, future datasets may increasingly omit interactional features that are considered central to the organisation of human conversation.

Turn-taking is mostly reserved for oral communication modes with the exception of what Earnshaw, 2017 calls super-synchronous chats where the messages appear keystroke by keystroke and participants can technically type and read at the same time. Regarding the number of participants, turn-allocation is simpler and more organised in duologues as participants speak one after the other but it can become very chaotic in multilogues unless the genre of conversation comes with some structure. For example in meetings or at court, the turn-taking system follows established rules which makes the exchange more organised despite the potentially high number of participants.

In written conversation, the quick succession of turns is lost in favour of potentially longer and more numerous turns before getting an answer from the addressees. The written aspect lets participants hold more threads of conversation at the same time since they do not have to remember everything, and this is probably all the more common when switching to asynchronous mode. Participants can start several topics without closing the previous ones, and expect responses to some or all of them. Indeed, one can send an informational message without necessarily expecting a response; depending on the mode of communication, one may not even expect acknowledgment feedback.

Multi-party written conversation probably works similarly in the sense that sub-conversations appear to have a more classical turn-taking system when separated (Hunter et al., 2015). However both for written and oral multi-party conversation, the subdivision phenomenon becomes more complicated to study when the subsets of participants have a non-empty intersection, i.e. when one or more participants are taking part in more than one sub-conversation at the same time. This is probably complicated to maintain, especially orally due to the high informational load but this type of situation is not impossible, in particular with the development of text message applications as (2.3) illustrates. Studying such conversations by separating the sub-dialogues and not considering the global exchange would represent a big loss of information in terms of what participants perceive of the conversation at a given time.

Launching several conversations in parallel in written communication may be a form of response to the loss of efficiency of this communication mode compared to face-to-face. Since we cannot carry a conversation through to the end in one go, we carry several of them out at once to settle as many things as possible in the shortest

Table 2.27: Dialogue characterisation – Verbal interaction: Turn-taking.

Parameter	Minimal	Extended description
Turn-taking	Binary	Overlap possible Turn-allocation method Average overlap time

possible time, the idea being to make the most of the presence of the interlocutor(s) when they are there. This is made possible by the fact that the entire content of the conversation is recorded in the communication medium, which lowers the memory barrier. It would be interesting to investigate if this kind of behaviour remains when using messaging tools where the messages disappear once they have been read or where one cannot highlight which message they are answering to.

In summary characterising the turn-taking system of a conversation is important because it is one of the core structural features of dialogue and it shapes the way conversation unfolds by allowing synchronous feedback or not, by enabling or not the quick succession of turns, etc. A simple binary parameter indicating the presence or absence of turn-taking can be sufficient in some cases especially to indicate very clear types of systems such as face-to-face conversations where turn-taking takes its most classical form, or written conversations where turn-taking is mostly absent. However, in many forms of mediated conversations turn-taking can be present but slightly different from its face-to-face form such as video calls where the delay can make it more difficult to coordinate turns. In such cases a more detailed description could indicate whether overlaps are at all possible and the turn-allocation method. A resource that focuses on turn-taking could also indicate the average overlap time, a positive one indicating that gaps are more common than overlaps and a negative one indicating the opposite. These parameters are summarised in Table 2.27.

2.4.4.3 Completion

Completion occurs when one participant completes another participant's utterance (Howes et al., 2011). It is a common feature of face-to-face conversation and can be understood as a form of co-construction, in which participants collaboratively build an utterance or formulate a thought. Completion may arise when a speaker struggles to find the appropriate word or when an interlocutor anticipates and supports what the speaker intends to say. However, it can also be interpreted negatively: when the participant completing the utterance prevents the original speaker from continuing, it may signal dominance or a lack of respect. The interpretation of completion therefore depends on the context, the nature of the interaction, and the relationship between participants (Sacks et al., 1974; Jefferson, 1986).

Table 2.28: Dialogue characterisation – Verbal interaction: Completion.

Parameter	Minimal	Extended description
Completion	Binary	Density Function (support, competition)

Completion is primarily observed in oral communication, as it relies on real-time feedback and the ability to respond to an unfolding utterance. It is generally absent from written communication, where contributions are typically produced in full before being presented to other participants.

From a modelling perspective, completion is important because it demonstrates that utterances may be co-constructed by multiple participants rather than produced by a single speaker alone. This has implications for dialogue systems, automatic speech recognition (ASR), and computational analyses of conversational data, as such systems must be capable of representing and transcribing partial and overlapping contributions.

In terms of operationalisation, completion can be represented minimally as a binary parameter indicating whether it occurs, while a more detailed representation could measure its density, i.e. the frequency of completed utterances per conversation or per turn, and indicate the function of the completion, e.g. support or competition. This allows for numerical analysis and integration into computational models of interaction. This parameter is summarised in Table 2.28.

2.5 On the operationalisation of the framework

This chapter has proposed a set of dimensions along which conversations – and, by extension, conversational datasets – can be characterised. The resulting framework, which we have named CHIMERA[5] – standing for **CH**ARACTERISATION OF D**I**ALOGUE THROUGH **M**ULTIDIMENSIONAL **E**XAMINATION, **R**EPRESENTATION AND **A**NALYSIS — — is organised around four main parameters: AGENTS, CONVERSATION SETTING, FRAME, and VERBAL INTERACTION. Together, these dimensions capture complementary aspects of dialogue, ranging from who is involved and how participants relate to one another, the time, place, and medium the interaction occurs in, and the language and purpose of the exchange, to the constraints imposed by those features on the mechanisms through which turns are coordinated. These parameters are relevant because they make explicit many of the factors that shape conversational behaviour but are often left implicit in the design, description, and use of dialogue resources. Face-to-face interaction, which serves as the prototypical conversational

[5]After the mythical creature composed of different animal parts: https://www.merriam-webster.com/dictionary/chimera (last accessed on September 29th.).

setting, combines a rich set of features across all four dimensions. However, due to the practical challenges of collecting and annotating such data, much of dialogue research relies on mediated or task-specific corpora. As a result, important aspects of interaction may be absent, attenuated, or transformed. Being able to characterise these differences systematically is therefore essential for assessing the scope of empirical findings, comparing datasets, and identifying potential limitations in generalisability.

Beyond the analysis of human dialogue, the proposed dimensions also provide a foundation for evaluating and designing dialogue systems. They can inform the definition of system capabilities, guide the selection of appropriate evaluation metrics, and help clarify what kinds of interaction a system is expected to support. For instance, a system designed for task completion in a constrained environment will require a different configuration of parameters than one intended for open-domain social interaction. Making these assumptions explicit is particularly important as conversational agents are increasingly deployed in complex, real-world settings.

Given the diversity of conversational settings and the constant evolution of technology that support mediated interaction, the parameters proposed here should be considered as a starting point for describing dialogue resources rather than a definitive taxonomy. Each dimension connects to extensive bodies of work in fields such as interactional linguistics, pragmatics, psycholinguistics, sociology, and communication studies, where more fine-grained analyses and alternative conceptualisations have been developed. Specialists in these areas will undoubtedly be able to refine, extend, or challenge the operationalisations proposed here. The aim is therefore not to replace existing approaches, but to provide a common ground that encourages more explicit descriptions of conversational data and facilitates dialogue across disciplines, including those with a weaker anchor in linguistics like computer science.

We have sought to operationalise each parameter at two levels of granularity, summarised in Table 2.29: a minimal description that captures the most salient aspects of the parameter, and a more detailed characterisation, often based on several dimensions. However, the level of detail required to describe a corpus or a task will depend on its specific nature. We consider that certain aspects of our framework are essential for describing a dialogue resource and must always be mentioned; these concern:

Agents: their type (human or artificial) and their number;

Conversational setting: a description of the environment and medium, as well as the extent to which they constrain verbal interaction, where applicable;

Conversational frame: the language(s) used and a description of the task performed by the participants when conversations are prompted rather than nat-

urally occurring, or if they take place in a specific context, as well as how this influences verbal interaction, where applicable.

Furthermore, the more details provided for each parameter, the more useful the description will be for data analysis and use. Indeed, documenting the process by which a resource is produced reduces the risk of losing important information about the context of the data over the years. Moreover, a detailed description enables better comparison between resources and a more informed assessment of a resource's suitability for a given task, two key objectives of CHIMERA.

Assessing the suitability of a resource for a task, or conversely establishing the set of suitable resources for a given task, requires to be able to represent both the requirements of the task and the characteristics of the resources in a common space. The descriptive framework presented in this chapter is meant as a first step towards such a representation. To conclude this chapter, we propose two different yet parallel accounts of our typology. The first one, described in Section 2.5.1, leverages TTR (Cooper, 2005, 2023) to represent dialogue types as collections of fields that encode the parameters of our CHIMERA. The second one, explained in Section 2.5.2, aims to represent dialogue as a multi-dimensional space in which resources and tasks could be projected, allowing for a more statistical characterisation.

2.5.1 A formal account of CHIMERA

The modelling proposed in this section is not the first approach suggesting to use TTR to represent dialogue types. Ginzburg (2012) relied on Kos[6], a TTR-based dialogue modelling framework, to represent dialogue types. His idea was that the information state of a participant that has completed a conversation of a given type can be generalised and used to model the type of that conversation. However, this account is more specific than the one we are proposing here. Indeed, the information state of a participant is a representation of their perception of the conversation, which is influenced by the actual realisation of that conversation. In contrast, we aim to represent the *potential* of a type of conversation. If we get back to the example of my laboratory's restaurant, discussed in Section 2.3.2.1, some clients may have conversations with the cashier that only cover the practical aspects of the transaction, while others may also engage in social talk. The information state of the participants at the end of these two instantiations of restaurant conversations will be different, but we view them as belonging to the same type of interaction.

We thus take a mirrored approach compared to Ginzburg's (2012) and propose to represent dialogue types independently from what participants actually do, based on what the dialogue situation allows. Another advantage of this perspective is that it does not require us to guess the mental state of the participants as we focus on describing the constraints of the situation. If we come back to dialogue (1.1) between

[6]Not an acronym but a *toponym* that loosely corresponds to conversation-oriented semantics.

Table 2.29: Parameters for an operational characterisation of dialogue resources and tasks. *Italicised** ones are consequences of the other parameters.

Meta-Parameters	Sub-parameters	Minimal	Extended description
Agents			
Agency		Types	Model type
			Model size
			Model training
Participation structures	Number of participants	Number	Number of simulated participants
			Number of workers
			Workers - Simulated participants mapping
	Functions	List	Intended external audience
			Awareness of eavesdroppers
Social dynamics	Hierarchy	Binary	Weighted list of dimensions
	(Conversation) History	Scale	Distance in a weighted social graph
			Common topics
*Enunciative status**	Anonymity	Binary	Anonymous / pseudonymous / real name
			Digital / physical
	Role	Binary	Self / self specific / assigned
	Persona	Binary	Self / self specific / assigned
	Intention	Cooperativeness	Weighted dimensions for each ordered participant pair
		Truthful / hidden	
		Chosen / imposed	
	Commitment level	Binary	Status: role-playing, joke, etc.
			Scope: conversation, task, …, general

Table 2.29: Parameters for an operational characterisation of dialogue resources and tasks (cont.).

Meta-Parameters	Sub-parameters	Minimal	Extended description
Conversation Setting			
Epistemic grounding	World setting	Real (contemporary or not) / fictional	Time period
			Location
			Conventions (cultural, scientific, moral, etc.)
	*World stability**	Binary	Planned changes
			Frequency of changes
			Source of changes
Environment & Medium	Physical context	Binary	Co-located / shared virtual / separate
			Shared perceptual field
	Gaze	Binary	Allocation degree
			Fidelity: accurate / approximated / distorted
	Gestures / Facial expression	Binary	Allocation degree
			Interpretability
	Prosody / Tone	Binary	Allocation degree
			Fidelity
	Noise	Binary	Degree
			Sources
	Synchrony	Binary	Time between production and treatment
	Planning	Binary	Time allowed for production
	Ephemerality	Binary	Persistence time
			Editability time
			Edition history
		Recording form	Third-party persistence time
			Third-party persistence awareness

Table 2.29: Parameters for an operational characterisation of dialogue resources and tasks (cont.).

Meta-Parameters	Sub-parameters	Minimal	Extended description
Conversational Frame			
Language	Global	List	Control degree
			Proficiency
			Monolingual / code-switching / multilingual
	*Vocabulary**	Genericity degree	Domain
			Community
Purpose	Practical goal	TO / social / information request	Proportions
	Success condition	Description	PQ to be popped out
			Topoi to recognise
	*(Topical) structure**	Constrained?	Agenda
			Scenario
Verbal interaction*			
*Coordination signals**		Binary	Availability of channel (vocal, textual, visual, haptic) per participant
			Technical ability per channel per participant
			Density per participant
			Latency tolerance
*Turn-taking**		Binary	Overlap possible
			Turn-allocation method
			Average overlap time
*Completion**		Binary	Density

the driver and passenger in the car, the situation allows the participant to rely on visual cues to coordinate their turns and disambiguate between the interview and navigation topics. However, the actual record we have of their exchange does not allow us to know whether they actually relied on these cues or not, and thus whether they would have appeared on their information state at the end of the conversation. Overall our approach aims for a more maximal representation of the type of a con-

versation where Ginzburg (2012) focuses on actual realisations. At the scale of a corpus, we intend our representation to highlight the potential of the type of conversation that the corpus represents, which would correspond to the most detailed description that would encompass all the instantiations of the type that the corpus contains.

We begin by translating CHIMERA to TTR in Section 2.5.1.1. We then use an example to discuss how this representation can be used to compare dialogue resources and identify the more suitable ones for a task in Section 2.5.1.2.

2.5.1.1 Translation of CHIMERA into TTR

Figures 2.3 and 2.4 show a TTR representation of CHIMERA separated into the descriptive meta-parameters on one side (Figure 2.3) – AGENTS, SETTING, FRAME – and the parameters that derive from them on the other side (Figure 2.4), i.e. the meta-parameter VERBAL INTERACTION as well as the additions to the three first meta-parameters. We presented the latter parameters as being computable from the other ones, which means that in theory, one would be able to define functions that take as input the descriptive parameters and output the computable ones. In TTR this would be done with *update rules* that would take as input a record of the descriptive parameters and output an updated version of the record with the computable parameters added. However, defining update rules that would cover all possible inputs lies beyond the scope of this thesis. Similarly, we do not define the types of the parameters and instead assume the existence of an *ad hoc* one for each parameter. In Chapter 3, we will focus on a few specific parameters and discuss the definition of the corresponding types and update rules. Nevertheless, we can still represent the most detailed description of a resource according to CHIMERA, which would be obtained by merging the two record types of Figures 2.3 and 2.4, yielding Figure 2.5. Another proposal that better accounts for the two levels of granularity of CHIMERA is presented in Appendix A.

The possible levels of granularity are modelled by the degrees of depth of each parameter's type. This highly detailed representation gives an idea of the possible uses of CHIMERA in a computational setting. TTR is indeed based on *feature structures* which provide tools for reasoning about the compatibility of two records, and thus for comparing different resources or tasks. A subsumption relation can be defined between types, which allows us to represent the fact that a more general description of a parameter subsumes a more detailed one, and is therefore compatible and comparable with it. Maraev et al. (2018) present a TTR-based dialogue manager which relies, among other components, on a type-checker and a rule-application mechanism, thereby demonstrating the computational potential of TTR. However, the level of detail represented in Figure 2.5 becomes a drawback when considering human readability and ease of use. The underspecification property of TTR helps to overcome this limitation. Indeed, the framework allows only

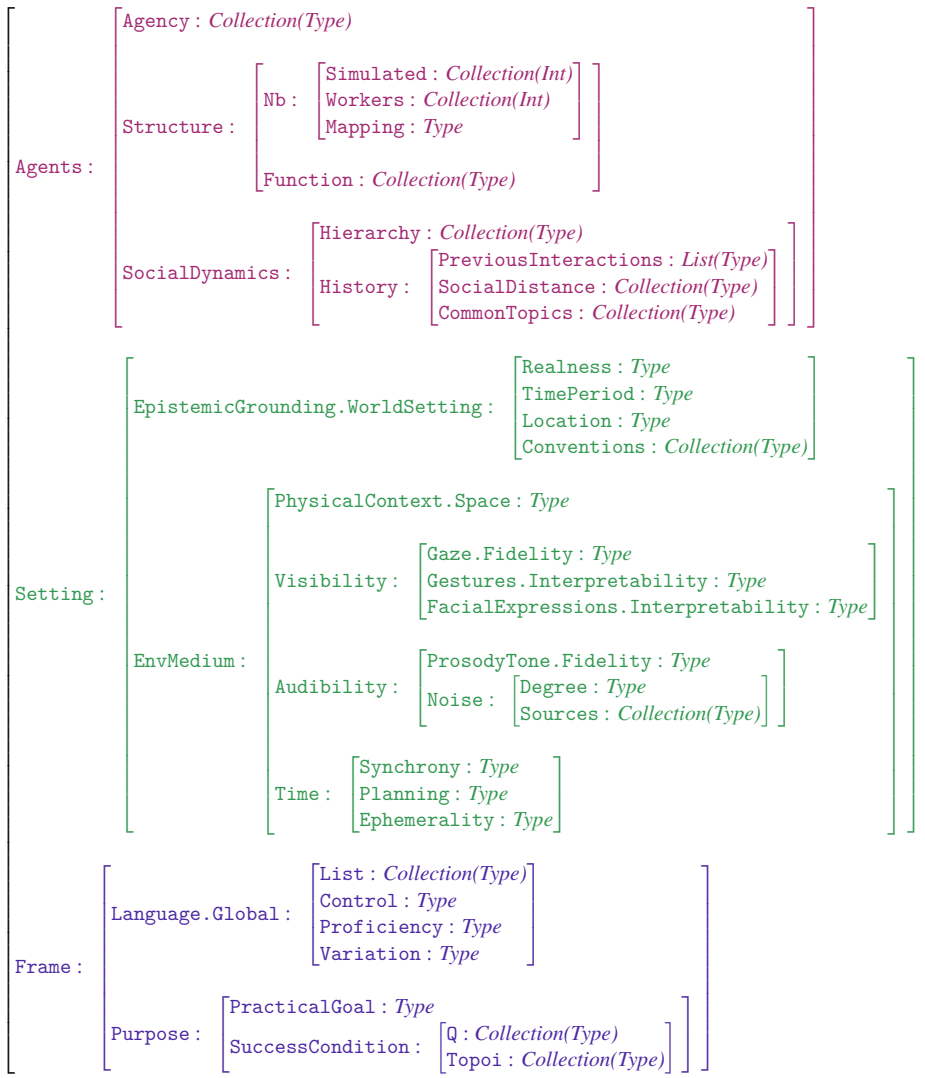


Figure 2.3: A formal representation of the space of dialogue resources and tasks using TTR – Descriptive meta-parameters.

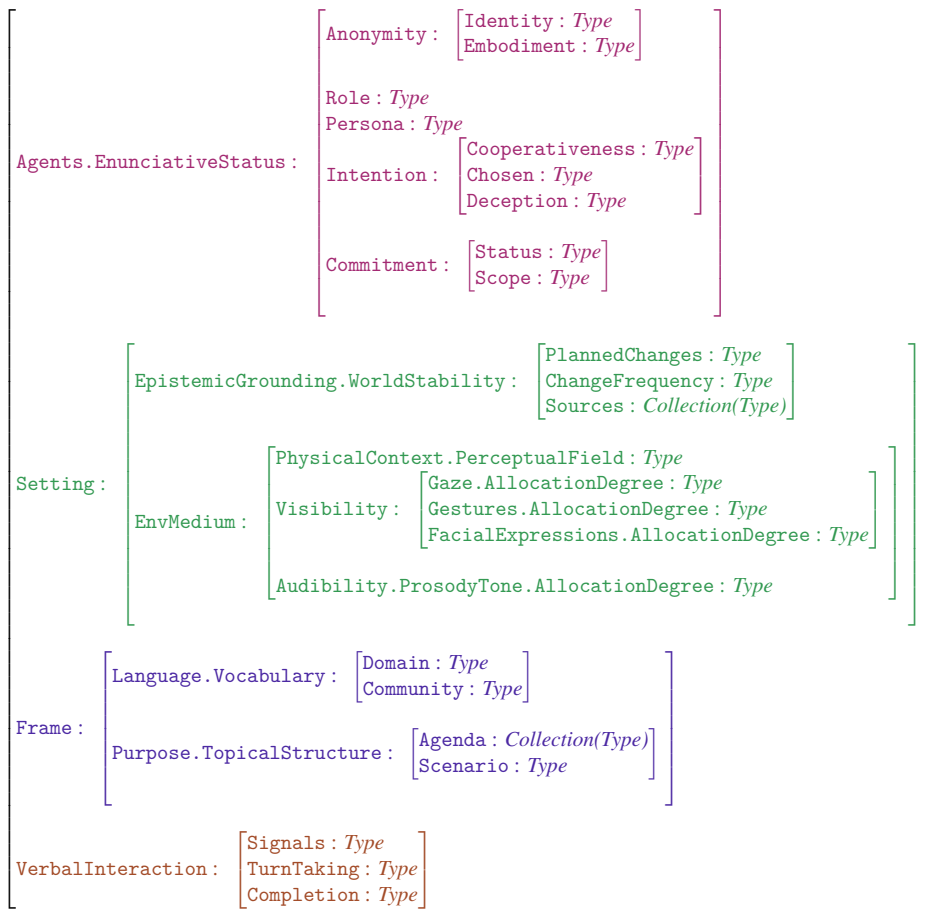


Figure 2.4: A formal representation of the space of dialogue resources and tasks using TTR – Dependant parameters.

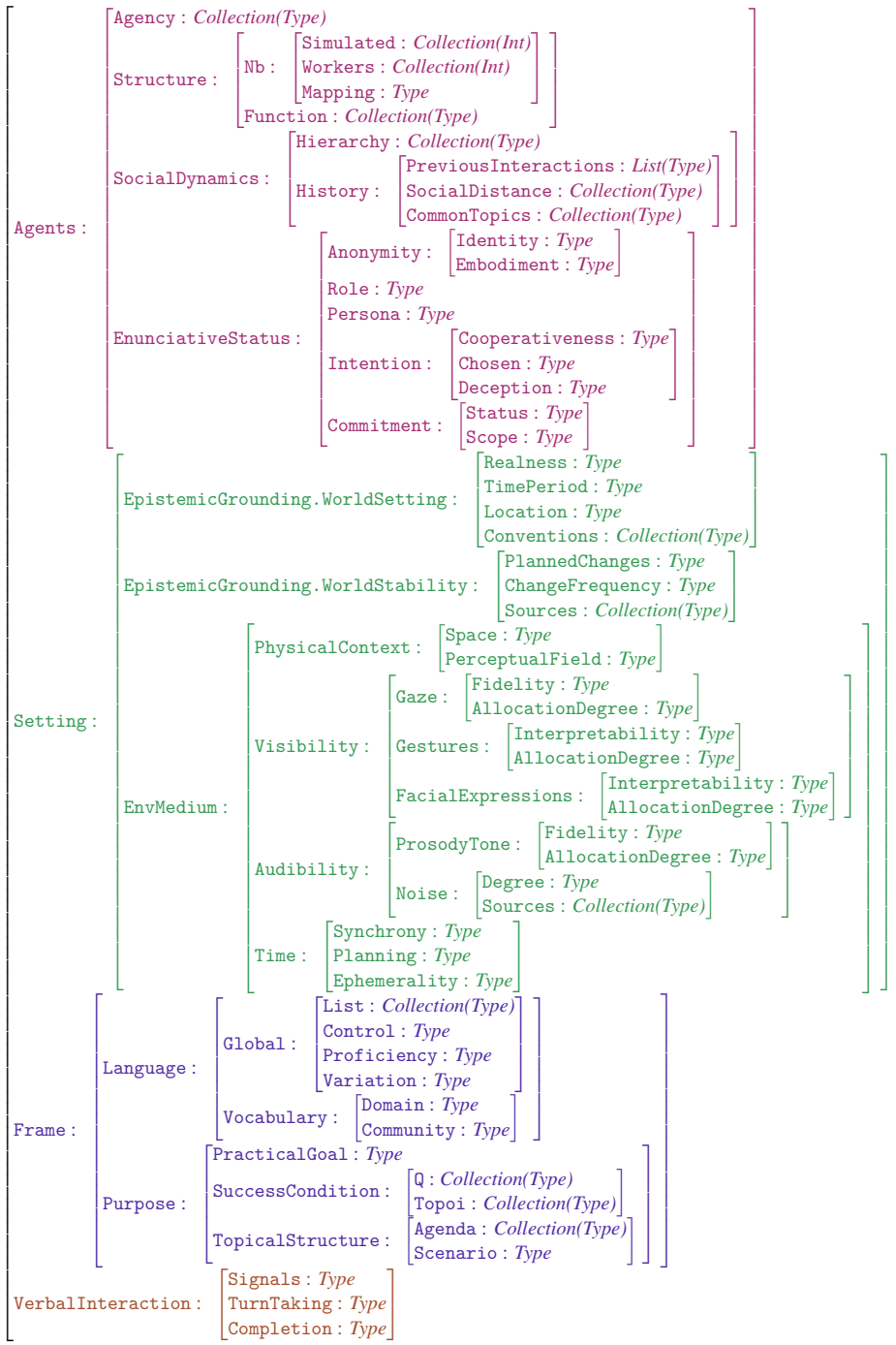


Figure 2.5: A formal representation of the space of dialogue resources and tasks using TTR – Maximal version.

the fields relevant to a given situation to be represented; in our case, this means that a resource can be described at a level of detail that is relevant to it, without the need to specify all the details of all its parameters.

We can thus define the minimal type of a dialogue resource as the type that only specifies the parameters that we consider crucial to describe it as discussed at the beginning of Section 2.5. We define it as:

$$\text{(2.i)} \quad \begin{bmatrix} \texttt{Agents} : \begin{bmatrix} \texttt{Agency} : \textit{Collection(Type)} \\ \texttt{Structure.Nb.Simulated} : \textit{Collection(Int)} \end{bmatrix} \\ \texttt{Setting.EnvMedium} : \textit{EnvMediumType} \\ \texttt{Frame} : \begin{bmatrix} \texttt{Language.Global.List} : \textit{Collection(Type)} \\ \texttt{Purpose} : \textit{PurposeType} \end{bmatrix} \\ \texttt{VerbalInteraction} : \textit{VerbalInteractionType} \end{bmatrix}$$

Some of the labels like `Agents.Structure.Nb.Simulated` are complex, i.e. they are composed of several labels separated by a dot. This is a way to represent the fact that they indicate a sub-parameter. For example, `Agents.Structure.Nb.Simulated` indicates the number of simulated participants, which is a sub-parameter of the structure of the participants within the meta-parameter AGENTS. This notation allows us to compare this minimal representation with a more complete one with subsumption relations, where using the label `Nb` alone would have made (2.i) incompatible with a more complete one.

Regarding the meaning of (2.i), the first level of depth, i.e. the left-most labels `Agents`, `Setting.EnvMedium`, `Frame`, and `Purpose` correspond to the most general level of description expected in this minimal representation. It corresponds to three of the four meta-parameters of CHIMERA and one parameter of SETTING. They are each associated with a *record type* that describes the parameters and sub-parameters of our framework.

For the first meta-parameter, `Agents`, the record type specifies that it is necessary to indicate the agency of the participants (`Agents.Agency`) and the number of simulated participants (`Agents.Structure.Nb.Simulated`) as specified at the beginning of Section 2.5. We represent the agency as a collection of types without specifying the exact kind of types that are expected as this level of detail lies outside the scope of this thesis. We chose a collection representation to allow for the possibility of having participants with different types of agency. While we use the general concept of collection, it could be replaced by a more specific type like a set or a list. The issue with a set is that it does not allow for duplicates, which may be relevant in some cases. For example, if we have a resource with three simulated participants, all of them being conversational agents, the agency would be represented as a collection of three identical types. A list would allow for duplicates but it is an ordered structure, which is not relevant in this case. A set of types that encode the number of participants of each type would be a more appropriate representation but it also adds more complexity to the description. The second parameter under

`Agents` indicates the number of simulated participants, which is represented by a collection of integers to account for the number of participants of each dialogue in a corpus.

The second label `Setting.EnvMedium` encodes the environment and medium, a parameter of SETTING. It is represented by a type whose role is to describe where and how the conversation is taking place, following the possibilities discussed in Section 2.2.2.

The third meta-parameter, `Frame`, is represented by a record type that specifies the language of the dialogue (`Frame.Language.Global.List`) with a collection of types that represent languages, and the medium of the dialogue (`Frame.EnvMedium`). The last meta-parameter, `Purpose`, is represented by a type that encodes the purpose of the dialogue.

The last label `VerbalInteraction` corresponds to the last meta-parameter of CHIMERA, VERBAL INTERACTION which is computable. It is meant to encode the constraints on the verbal interaction that are induced by the setting, environment, and medium. As for `Setting.EnvMedium` we left it as a general type that can be specified in more detail depending on the resource or task being described following for example the operationalisations discussed in Section 2.4.4.

2.5.1.2 Application example: Matching resources to a task

To illustrate the use of CHIMERA, we imagine that we are interested in the conversation emotion recognition (CER) task, which consists in labelling dialogue utterances with a tag-set of emotions such as 'anger', 'joy', 'sadness', or 'neutral'.

So far we have focused on the resource description application aspect of our framework but it can also be used to describe the *requirements* of a task. The process for defining these is similar to the one used for describing a resource, with the difference that the description would be prescriptive. In practice, this means that we would define the requirements of a task by specifying the possible values of the parameters that are relevant for the task. For example, if we want to study multilogues, the number of simulated participants would need to be greater than 2. If we want to study human-conversational agent interactions, we would specify that the participants must include at least one human and one conversational agent.

The minimal type shown in (2.i) can serve as a basis for defining the requirements of a task. However, it is possible that some of these core parameters present no constraints for the task; they would thus not need to be specified. While it is possible that the task itself does not induce strict requirements regarding the relevant datasets, its actual realisation will likely reduce the set of possible values for some of the parameters. First, the tools and methods that will be used may impose constraints. For example, many NLP methods are designed for English textual data and may not be applicable to other languages or modalities. Second, the intended application scope of the study or tool also reduces the set of relevant data. Third,

selecting a resource that is too specific may limit the generalisability of the results. Those three aspects, i.e. the task itself, the tools and methods, and the intended application scope, should be taken into account when defining the required properties of the datasets that can be used to tackle a given task. With these three aspects in mind, the task requirements can be defined by following the procedure below:

- For each meta-parameter, consider whether the task imposes constraints on the type of data that is relevant. The following questions can be used as a guide:

  AGENTS: Does it matter *who* produces the dialogue (Yes) or are you only interested in what is produced (No)?

  SETTING: Does the task require a specific form of interaction in terms of environment and medium?

  FRAME: Both the language and the purpose should be considered here.

  ***Language*:** Does the task require a specific language or set of languages? For this parameter in particular, it is important to consider the intended application scope of the task, as well as the tools and methods that will be used, which may impose constraints on the type of data that can realistically be used.

  ***Purpose*:** Does the task require data from a specific conversational activity or purpose?

  VERBAL INTERACTION: Does the task require specific patterns of verbal interaction?

- For each meta-parameter that is considered relevant, i.e. where the answer to the above questions is 'Yes', consider the first level of description of the parameter:

  - For each parameter, determine whether this level of granularity is fine enough to capture your requirements;
  - If it is, state the requirements at this level of granularity;
  - If it is not, consider the different sub-parameters:
    - For each sub-parameter, if the minimal description captures your requirements, then state the requirements at this level of granularity;
    - If it does not, consider the extended description;
    - If none of these sub-parameters capture your requirements, you may need to define a new parameter or adjust the existing ones.

For example, the CER task, when considered independently from the tools and methods that will be used to tackle it, clearly sets constraints on the AGENTS if the

goal is to understand how human express their emotions in naturally occurring dialogue. In particular, it requires at least two *human* participants interacting with one another. In terms of enunciative status, they should also be as close to their *real* selves as possible in order to gather realistic data. However, this task does not induce constraints on the SETTING, though selecting a dataset with a particular one will impact the generalisability of the study. Similarly, the FRAME is quite open, though the more constrained it will be, the more specific the study will become. Finally, the reasoning is the same regarding *verbal interaction*. We can thus represent the requirements of the CER task as a record type as shown in (2.ii).

(2.ii) CER: $\begin{bmatrix} \texttt{Agents} : & \begin{bmatrix} \texttt{Agency} : \textit{human} \\ \texttt{Structure.Nb.Simulated} : Nb_{\geq 2} \\ \texttt{EnunciativeStatus} : \textit{Self} \end{bmatrix} \end{bmatrix}$

Again we do not define the types in details but use general types whose names describe their intended meaning. The type *human* repesents a type of dialogue such that all agents are human, $Nb_{\geq 2}$ indicates that the number of participants must be at least two in all the considered exchanges, and *Self* models a type of enunciative status such that the participants are their real selves. This last type should be compatible only with more detailed types of enunciative status that fit this requirement. For example, having participants embody a pre-defined persona would not yield a suitable type, while being anonymous or not would be compatible with the *Self* type.

We can then compare the requirements of the CER task with the characteristics of existing resources to determine which ones are suitable for this task. Figure 2.6 shows a few examples of corpora that we have mentioned in the Prologue and Chapter 1 represented with the TTR account of CHIMERA. We remained quite minimal in our description of these resources for the sake of brevity as we are only interested in the parameters that are relevant for the CER task.

(2.iii) presents the type that describes the dialogues of PersonaChat, text message exchanges between two crowdsourcers who are asked to embody a pre-defined persona. Regarding the AGENTS, the only parameter where CER induces constraints, all the participants are human which we indicate by setting the `Agents.Agency` parameter to $\{Human\}$ which is of type *human*. Similarly, the dialogues all have two participants which is encoded by [`Agents.Structure.Nb.Simulated`: $\{2\}$], where $\{2\}$ is of type $Nb_{\geq 2}$. However, the enunciative status of the participants is not compatible with the requirements of the CER task as they are asked to embody a pre-defined persona, which is encoded by [`Agents.EnunciativeStatus`: *ImposedPersona*].

The other examples show a similar pattern, with all resources having at least two participants, all human, but different enunciative statuses. The Villing corpus is fully compatible with the CER requirements as it is defined by a status of

type *Self*. In the STAC corpus, the participants are pseudonymous and are playing a game which puts them in a kind of role but they are still their real selves, which is compatible with the CER requirements. The BNC corpus contains a wide range of dialogues with different enunciative statuses whose differences mostly lie in the roles played by the participants. However, some dialogues can also contain exchanges where participants are not their real selves, such as when they are performing a play. The right subset of the BNC would thus need to be identified for CER. Finally, DailyDialog has a particular status as it is a handcrafted corpus of dialogues that are not produced in interaction, which means that there is technically no enunciative status. Determining whether this corpus is compatible with our requirements necessitates further investigation into how well it reproduces the characteristics of naturally occurring dialogues. This potential incompatibility shows the gap between the intentions one can have when building or selecting a corpus for a task, and the actual characteristics of the corpus from a dialogue study perspective. Indeed, DailyDialog was developped with tasks such as CER in mind since it is manually annotated with communication intention and emotion labels. However, it is also manually crafted and explicitly described as less noisy than interactions found on social media, though the latter may actually be more authentic. This hightlights the importance of having a framework that allows to describe the characteristics of a resource in a systematic way, and to compare them with other resources and with the requirements of a task.

2.5.2 A vector space perspective on CHIMERA

Taking inspiration from the recent NLP approaches to representing tokens in a vector space, we propose to represent dialogue resources and tasks as points or surfaces in a multi-dimensional space, where each dimension corresponds to one of the parameters of CHIMERA. For example, multi-lingual resources such as XSGD (Tu et al., 2024), a dataset created by translating the English dataset Schema-Guided Dialogue (Rastogi et al., 2020) into 105 other languages, would be represented as a surface along the LANGUAGE dimension, while the original English dataset would be represented as a point.

Within such a representation, resources could be compared in terms of their proximity along each dimension, as was already done by Wong and Ginzburg (2018) with simple metrics like unigram or bigram probability distribution. Resources specificity could also be assessed through their spatial coverage, where a dataset that allows to study only very specific aspects of dialogue would be represented as a small surface or even a point, while a dataset that covers a wide range of aspects would be represented as a larger surface or volume with a wide coverage along several dimensions. This representation would also allow to identify gaps in the space of dialogue resources, where certain combinations of parameters are not represented by any existing resource. This could help us assess whether these

(2.iii) PersonaChat:
$$\begin{bmatrix} \texttt{Agents}: & \begin{bmatrix} \texttt{Agency} = \{Human\} \\ \texttt{Structure.Nb.Simulated} = \{2\} \\ \texttt{EnunciativeStatus}: \textbf{\textit{ImposedPersona}} \end{bmatrix} \\ \texttt{Setting.EnvMedium}: \textbf{\textit{TextMessages}} \\ \texttt{Frame}: & \begin{bmatrix} \texttt{Language.Global.List} = \{\text{EN}\} \\ \texttt{Purpose}: \textbf{\textit{DisplayPersona}} \end{bmatrix} \end{bmatrix}$$

(2.iv) Villing:
$$\begin{bmatrix} \texttt{Agents}: & \begin{bmatrix} \texttt{Agency} = \{Human\} \\ \texttt{Structure.Nb.Simulated} = \{2\} \\ \texttt{EnunciativeStatus}: \textbf{\textit{Self}} \end{bmatrix} \\ \texttt{EnvMedium}: \textbf{\textit{Face-to-Face}} \\ \texttt{Frame}: & \begin{bmatrix} \texttt{Language.Global.List} = \{\text{EN}\} \\ \texttt{Purpose}: \textbf{\textit{DicoProjectTask}} \end{bmatrix} \end{bmatrix}$$

(2.v) STAC:
$$\begin{bmatrix} \texttt{Agents}: & \begin{bmatrix} \texttt{Agency} = \{Human\} \\ \texttt{Structure.Nb.Simulated} = \{\mathbf{4}\} \\ \texttt{EnunciativeStatus}: \textbf{\textit{PseudonymousOnlineSelfAs}} \end{bmatrix} \\ \texttt{Setting.EnvMedium}: \textbf{\textit{TextMessages}} \\ \texttt{Frame}: & \begin{bmatrix} \texttt{Language.Global.List} = \{\text{EN}\} \\ \texttt{Purpose}: \textbf{\textit{CatanGame}} \end{bmatrix} \end{bmatrix}$$

(2.vi) DailyDialog:
$$\begin{bmatrix} \texttt{Agents}: & \begin{bmatrix} \texttt{Agency} = \{Human\} \\ \texttt{Structure.Nb.Simulated} = \{2\} \\ \texttt{EnunciativeStatus} = \textbf{\textit{None}} \end{bmatrix} \\ \texttt{Setting.EnvMedium} = \textbf{\textit{None}} \\ \texttt{Frame}: & \begin{bmatrix} \texttt{Language.Global.List} = \{\text{EN}\} \\ \texttt{Purpose}: \textbf{\textit{DailyLifeWithTopics}} \end{bmatrix} \end{bmatrix}$$

(2.vii) BNC:
$$\begin{bmatrix} \texttt{Agents}: & \begin{bmatrix} \texttt{Agency} = \{Human\} \\ \texttt{Structure.Nb.Simulated}: \textbf{\textit{Nb}}_{\geq \mathbf{2}} \\ \texttt{EnunciativeStatus} = \{\textbf{\textit{Self}}, \textbf{\textit{SelfAs}}\} \end{bmatrix} \\ \texttt{Setting.EnvMedium}: \{\textbf{\textit{FaceToFace}}, ...\} \\ \texttt{Frame}: & \begin{bmatrix} \texttt{Language.Global.List} = \{\text{EN}\} \\ \texttt{Purpose}: \textbf{\textit{BNCPurposeTypes}} \end{bmatrix} \end{bmatrix}$$

Figure 2.6: Representation of a few corpora as feature structures covering the minimal characterisation and the key dimensions constrained in CER.

combinations are not represented because they do not exist in practice, or because they have not been studied yet.

Moreover, as illustrated throughout the chapter, the parameters we consider do not operate independently: their interactions can produce complex effects on conversational dynamics, sometimes in ways that are not immediately apparent. Visualising dialogue as a multi-dimensional space would allow us to explore these interactions more systematically with dimension-reduction techniques, clustering, and other analytical methods. This could reveal patterns and relationships that are not easily discernible through traditional analyses, providing new insights into the nature of dialogue and its variability across contexts.

This spatial visualisation also provides a natural method for selecting resources for a given task. By representing both the requirements of a task and the characteristics of available resources in the same multi-dimensional space, one can identify which resources are most closely aligned with the task's needs through simple distance assessments.

Figure 2.7 shows a three-dimensional representation of the space of dialogue resources and tasks, where the three axes correspond to the number of participants, their enunciative status, and the medium of the interaction. This is a simplified representation of the space, as it only includes three of the parameters of CHIMERA, but it illustrates how resources and tasks can be positioned within this space based on their characteristics. We re-use the same example as in the previous section (Section 2.5.1.2): CER. We define a discrete scale for each of our three dimensions. The number of participants is naturally represented by natural numbers. The enunciative status is represented with four levels: $\varnothing$ (no enunciative status), *made-up persona* for dialogues where the participants are imposed a persona or choose to impersonate a character, *self-as* for dialogues where the participants are pseudonymous or playing a role, and *self* for dialogues where the participants are their real selves. The level *self-as* is intended for dialogues where participants take on a special role that does not impact their emotional state, like they would when playing a game. The medium is also represented with four levels with *face-to-face* being the most open one, *side-by-side* slightly less, then *text messages*, and finally a level $\varnothing$ for dialogues where there is no medium as in handcrafted dialogues.

The points and the fully coloured surface represent the resources described in Figure 2.6, while the dotted surface represents the requirements of a CER task described in (2.ii.) As predicted by the TTR representation, we can see that two resources — *DailyDialog* and *PersonaChat* — are located outside of the scope defined by the requirements of CER, and a part of the BNC corpus as well. The other corpora lie within the scope of the task and differ in terms of medium, enunciative status but also language, though this last parameter is not represented in Figure 2.7. Choosing to train a model for emotion recognition on one of these corpora likely makes the model more adapted to the characteristics of that corpus, and thus less generalisable to other situations.

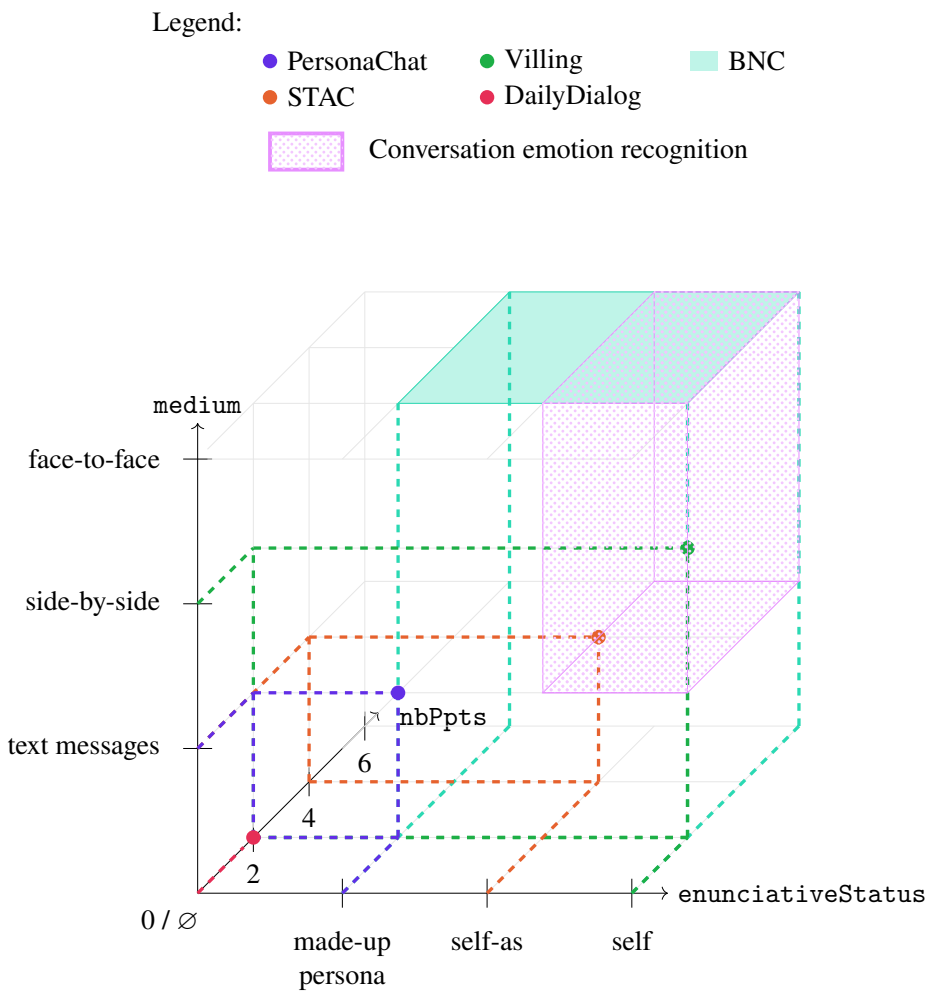


Figure 2.7: Representation of a few corpora and the requirements of CER in a three-dimensional space defined by the number of participants, their enunciative status, and the medium of the dialogue.

This example shows how CHIMERA can be used to describe and compare dialogue resources and tasks in a systematic way, and how it can support the selection of resources for a given task. In the next chapter, we examine how dialogue is currently studied within the NLP and computational linguistics community. We show that recent studies are often lacking in terms of resource description and that our framework can help address this issue.

3

Dialogue Research in Practice: A Critical Analysis of Contemporary NLP Approaches

Contents

3.1 Contemporary landscape of dialogue research 123
3.1.1 Methodology of the review 124
3.1.2 Tasks 126
3.1.3 Datasets and resources 130
3.1.4 Evaluation 134
3.2 Pipeline for NLP and dialogue research 136
3.2.1 From theoretical models to black-box ones 137
3.2.2 Evaluation and the risk of circularity 138
3.2.3 Re-centring theory, data, and purpose 140
3.3 Applying CHIMERA 141
3.3.1 Scenario 1: Turn-taking 141
3.3.2 Scenario 2: Task-oriented dialogue 149
3.3.3 Scenario 3: Clinical interview 162
3.4 Targeted dialogue generation control 171
3.4.1 Generation control 173
3.4.2 Evaluation metrics 176
3.5 Summary and conclusion 179

In the previous chapter, we approached dialogue from a theoretical standpoint. Drawing on linguistics, philosophy, psychology, sociology, CA, and computational modelling, we examined how dialogue has been conceptualised across disciplines. Although we do not claim to be exhaustive in our account, especially since dialogue studies are so intrinsically interdisciplinary, we sought to highlight a central insight shared across traditions: dialogue is not reducible to a linear exchange of sentences. It is a complex coordination of cognitive, linguistic, social, and environmental processes, whose mechanisms interact in dynamic and sometimes unpredictable ways.

We also emphasised that dialogue admits a vast range of instantiations: more or less multimodal settings, synchronous or not, with different degrees of task structure, etc. This diversity allows researchers to focus on specific aspects of interaction, but it also requires careful delimitation. Without explicit characterisation of what kind of dialogue is being studied, under what constraints, and for what purpose, comparisons can become fragile and theoretical claims risk overgeneralisation.

To address this issue, we proposed a descriptive framework, CHIMERA, organised around four main axes: AGENTS, SETTING, FRAME, and VERBAL INTERACTION. Each axis is further articulated into more specific parameters that can be described with varying degrees of granularity depending on the research question at hand. CHIMERA is intended as a meta-descriptive tool to help make explicit what can sometimes be left implicit in resource design, experimental setup, and system evaluation. It is also meant to support empirical studies on the differences between dialogue types in order to determine when two settings are equivalent for a given task or research question. This is crucial in NLP as collecting data in ecological settings is costly and time-consuming and sometimes impossible for ethical or practical reasons. We therefore need to be able to determine what aspects of a dialogue type can change without affecting the research question or task at hand, and what aspects are critical to preserve. For instance, Lücking et al. (2025) show that some aspects of turn-taking are not preserved through virtual-reality (VR) embodiment, which means that VR cannot be used to circumvent privacy issues when collecting multimodal exchanges for studies that require face-to-face-like turn-taking. This study shows how *one* aspect of dialogue can be tested in different settings, which corresponds to a very small part of CHIMERA. This shows that the work required to fully understand the limitations of a given dialogue type is substantial and should be carried out at the community level. The framework we propose can help in this endeavour by providing a set of parameters whose impact when modified can be systematically tested.

However, a framework remains abstract until it is confronted with concrete research practices. The present chapter therefore moves from theoretical articulation to application. We examine how CHIMERA can be mobilised to analyse current dialogue research, clarify methodological choices, and illuminate blind spots. In doing so, we also begin to explore the interdependence of dialogue parameters: how constraints imposed along one axis (for example, task structure or medium) reshape other dimensions such as participation roles, interactional dynamics, or evaluative criteria.

To ground this discussion empirically, we first present a meta-review of publications from the ACL ANTHOLOGY in 2024 that focus on dialogue (Section 3.1). We then situate this analysis within the broader NLP research pipeline as it currently operates, particularly in the era of LLMs, where data availability, evaluation practices, and system architectures are undergoing rapid transformation (Section 3.2).

In Section 3.3, we demonstrate how CHIMERA can guide both analytical reflection and practical interventions – ranging from resource selection to delimitation of the scope of a study or application. Finally Section 3.4 shows how it can be used for targeted adaptation of language models along specific dialogue dimensions before we conclude in Section 3.5 with a discussion of future directions and open challenges in dialogue research.

3.1 Contemporary landscape of dialogue research

This section is based on the work Decker et al. (2026a) that was presented at the Fifteenth Language Resources and Evaluation Conference (LREC 2026). The publications reviewed in this work were therefore contemporary at the time of the study but can no longer be considered 'recent' due to the fast evolution of the field.

Many new types of dialogues have emerged in the past few years. Covid-19 (Joskowicz, 2023) made video-calls and hybrid meetings between face-to-face and remote participants more typical, while the rise of conversational agents and LLMs increased the usualness of human-machine interactions, in particular since the release of ChatGPT (Qi et al., 2024), which made this type of tool well known to the general public. These changes are bound to have an impact on the NLP/CL community. Several surveys have been conducted lately regarding multiple tasks. Kusal et al. (2022) surveyed the different approaches to AI-based conversational agents, identifying several gaps in this field at the time such as the lack of harmonised evaluation processes which is still an issue nowadays as we will discuss in Section 3.1.4. Y.-P. Chen et al. (2024) focused on personalised dialogue generation which consists in inferring and/or modelling the user's persona on the one side, and in having the agent display a coherent and relevant persona throughout the dialogue on the other side. They highlight the complexity to collect great quality datasets for this task due to the artificiality of actively displaying a persona, when it is typically a more subtle phenomenon. Approaches to personalised dialogue generation also tend to work on modelling the agent's and the user's persona separately which somewhat disregards the interactional nature of dialogue. Finally they raised concerns regarding the evaluation approaches for this task which are overall too basic compared to the complexity of dialogue generation. On a similar note, Yang et al. (2026) provided a critical analysis of the trends in NLG evaluation from 2020 to 2025, assessed through the automatic retrieval of information regarding the task and evaluation approaches – automatic metrics, LaaJ, and human evaluation – used in papers from four major CL venues. They focus in particular on four tasks, including dialogue generation and question answering. This mostly automatic approach allows the authors to study trends and shifts in evaluation practices over several years, thus highlighting turning points but also challenges in the field.

While these surveys provide valuable insights into specific tasks, they do not

offer a comprehensive overview of the broader landscape of dialogue research in NLP/CL. Hence in this section, we take a meta-review approach to investigate how conversation is currently studied in the field by analysing papers from the ACL Anthology 2024. We focus on papers written in English from major venues indexed in the anthology (excluding workshops) using metadata accessed via the ACL API[7]. Our goal is not to provide an exhaustive survey, but to provide a snapshot of where dialogue research stands today, what tasks are being studied, what resources they rely on, and how these tasks are evaluated. This enables us to identify the current gaps in the field and define future research directions.

We begin by outlining our approach to examining the current state of dialogue research (Section 3.1.1). We then detail the main dialogue-related tasks addressed in the NLP/CL community (Section 3.1.2), discuss the dialogue resources used and created (Section 3.1.3), and review widely used evaluation methods for DSs (Section 3.1.4). This leads us to the next section where we situate these practices within the broader research pipeline and discuss their implications for the future of dialogue research in the LLM era.

3.1.1 Methodology of the review

The ACL Anthology[8] is an open-access digital repository of research in CL and NLP. Maintained by the Association for Computational Linguistics (ACL), it hosts conference and workshop proceedings, journal articles, and other scholarly materials dating back to the 1960s and is updated continuously. Its open-access nature and structured bibliographic metadata have made it a widely used corpus for bibliometric studies, meta-analyses, and research on scientific trends (Vogel & Jurafsky, 2012; Omodei et al., 2014; Gábor et al., 2016; Pramanick et al., 2023).

In this section, we present an analysis of a targeted subset of the ACL Anthology to understand how conversation has recently been studied in the NLP/CL community. The ACL Anthology was chosen both for practical purposes, as its API allows us to retrieve the papers and their metadata easily, and because it gathers conferences that are considered important in the field. We restrict our survey to papers published in 2024 and indexed under the anthology's major venues. Our selection is presented in Table 3.1. While it does not cover journals, which typically allow for more in-depth studies thanks to the longer page limits, the restriction to conference papers allows us to capture the trends that are showcased in the faster-paced conference cycle, which is particularly relevant in the current era of rapid developments in research. Conferences have indeed shorter review cycles which makes them more attractive in institutional contexts that induce a pressure to publish, whether for hiring, promotion, or funding decisions (Franceschet, 2010; D. Van Dijk et al., 2014).

[7] https://aclanthology.org/info/development/

[8] https://aclanthology.org/

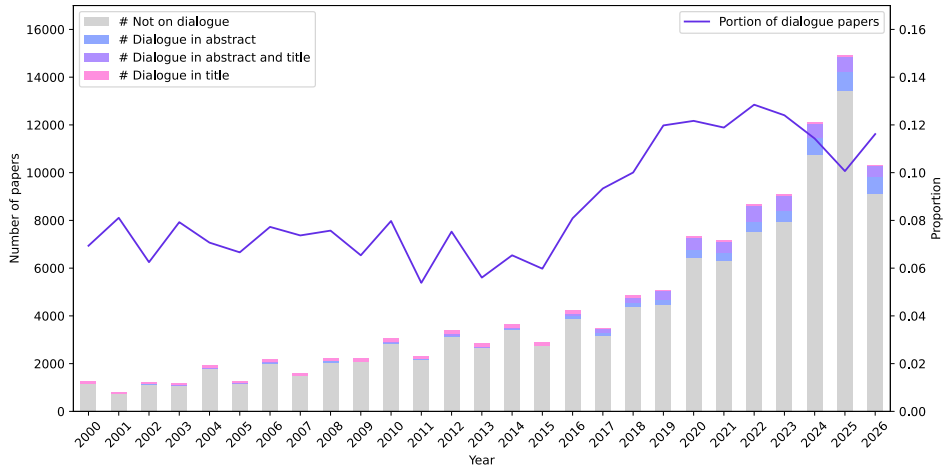


Figure 3.1: Number of papers in the ACL Anthology per year since 2000. *Partial numbers for 2026, gathered on August 19, 2026.*

Figure 3.1 shows the number of papers published in the ACL Anthology per year since 2000, highlighting the growth of the NLP and CL fields in general but also the increasing interest in dialogue research. The total number of accepted papers has grown steadily until the end of the 2010s when it reached a much higher pace. It increased by about 30% from 2023 to 2024, and about 25% from 2024 to 2025. Even though the number of papers published in 2026 is not complete yet, it has already reached more than two thirds of the number of papers published in 2025. This growth is likely to continue in the coming years, especially with the recent use of generative AI for producing research papers. Regarding dialogue research, the proportion of papers lied around 7% until 2016 when it started rising to reach about 12% in the recent years. While it stills represent a small portion of the overall research, it is a significant increase compared to the early 2000s, likely due to the increasing interest in dialogue systems and conversational AI.

To better understand the contemporary landscape of dialogue research, we used the ACL Anthology API to retrieve metadata for all papers published in 2024, excluding venues labelled as workshops in the metadata. Our goal was to identify papers dealing with dialogue. Due to the absence of keywords in the metadata we had to rely on another filtering method. We selected papers whose titles contained the stems of 'conversation', 'dialogue', and 'discourse'. While this method may miss some relevant papers and include some marginal ones, it provided a practical and transparent approach for assembling a representative sample. We then excluded the ones that were not actually focusing on dialogue (e.g. focus on monological discourse, or papers on *conversion*). In the end we retrieved 407 papers, which represents about 6.1% of the papers published in the selected venues. Once

Table 3.1: Number (#) of retrieved paper for the selected venues and portion (%) out of the entire venue in 2024.

Venue	#	%
TACL	3	3.2
EMNLP	52	3.6
ACL	36	3.8
NAACL	29	4.4
EACL	14	5.0
JEP/TALN/RECITAL	8	6.0
LREC/COLING	105	6.8
Findings	112	7.8
INLG	5	9.3
CODI	3	17.6
SIGDIAL	40	60.6
Total	408	6.1

retrieved, one person manually reviewed the papers to determine the conversational task addressed, the corpora used and the evaluation methods employed.

3.1.2 Tasks

The NLP/CL field covers a wide range of tasks, so as a first step we annotated each paper with the primary task it addressed and then grouped these tasks into broader categories. The categories we defined are described in the paragraphs below, and their distribution – both in absolute number of papers and as a proportion of the total – is shown in Figure 3.2. Some papers naturally span multiple categories: for instance, a benchmark proposal or a paper on generative systems may also introduce a new corpus. In such cases, we selected the most salient aspect when assigning the task category, but the absolute counts should be interpreted with caution. Nevertheless, the overall distribution reflects the dominant research trends in the surveyed set.

Generative Systems Across the ACL Anthology 2024, the majority of papers dealing with conversation focus on generative systems (44%). This category encompasses dialogue systems, question-answering (QA), and Dialogue State Tracking (DST), as well as some papers presenting new language models (LMs) or ways to improve existing ones.

Dialogue systems are programs designed to interact with humans through natural language. Within this category, task-oriented dialogue (TOD) systems used for e.g. booking restaurants, are especially dominant. These systems typically aim

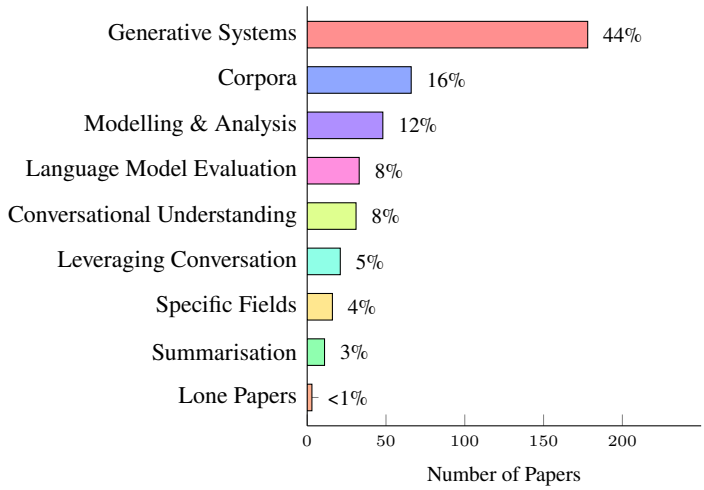


Figure 3.2: Distribution of papers by task category.

to assist users in completing specific goals, and their performance is usually measured through metrics like joint goal accuracy, which tracks whether the system correctly identifies all necessary slots in a dialogue (i.e. place, time, etc.). Less constrained tasks are also tackled such as emotional support, negotiation and persuasion, or tutoring. Both for domain-specific and more open tasks, the conversational dimension is a crucial part of the system. It is however harder to evaluate (see Section 3.1.4) than the task-specific outcomes.

QA is a sub-category of dialogue systems where users can ask questions and get answers in natural language. The system can also encourage the user to refine their question or ask for more information if the answer is not precise enough, providing more flexibility than querying a web interface (Biancofiore et al., 2024). This branch of research focuses notably on query rewriting and knowledge retrieval.

DST involves monitoring and recording the user's set of queries and goals throughout a conversation (Jacqmin et al., 2022). It plays a pivotal role in developing TOD systems. Recent studies have focused on enhancing DST models through techniques like zero-shot learning, the aim being to reduce the cost of dialogue collection and annotation.

Language Model Evaluation The growth of generative systems in research, but also in society, requires solid evaluation to understand their strengths and limitations. Slightly more than 8% of the papers we surveyed focused on this aspect, whether through the production of benchmarks or empirical evaluations. Most of them probe LMs capacities with regard to different aspects, from very specific ones such as their negotiation capacity or ability to generate slot-values, to more general conversational dynamics like multi-turn and/or multi-party dialogue capacity. Se-

curity is also a concern where the risk that LMs leak private information is investigated as well as their ability to resist misinformation and tendency to hallucinate or generate harmful content. Another angle is the limits of current LMs when it comes to languages other than English.

In some cases, LLMs were themselves used as evaluators for certain tasks, including tasks that were based on LLMs. While this can provide scalable and context-aware assessment, it is important to recognise the potential biases and limitations inherent in using a model to evaluate another model. We discuss this aspect in more details in Section 3.1.4.

Socio-Pragmatic Understanding Similarly the growth of LMs has increased the need to identify users' emotional states, communicative intentions, and stances in conversation, but also subtler aspects of interaction such as humour and sarcasm identification, personality profiling, harmful behaviour detection, and social meaning recognition, to enable more contextually aware and socially appropriate dialogue systems. These tasks fall under the broader umbrella of conversational understanding and were the focus of about 8% of the surveyed papers. The venue SemEval 2024, which we surveyed but did not include in this work to remain more general, also hosted 2 shared tasks (tasks 3 and 10) revolving around conversational emotion recognition.

Summarisation Summarisation is a classical NLP task that is more challenging for conversations than for non-interaction-based discourse (X. Feng et al., 2022). Less than 3% of the papers tackled this particular task. Most approaches leverage LLMs, some evaluate hallucinations of LLMs specifically in summaries. Summarisation is indeed particularly prone to hallucination in LLMs (Ramprasad et al., 2024): since the model must compress and rephrase content, it often generates details that are plausible-sounding but not supported by the source.

Leveraging Conversation The recent hype around LLMs has renewed interest in conversational AI, inspiring new directions for traditional tasks, exploiting interactive dialogue to provide more context-aware, personalised, and adaptive experiences that leverage conversation as a primary mode of interaction. This represents slightly more than 5% of the surveyed papers. Most of them reflect the emergence of paradigms such as conversational search and conversational recommender systems, where dialogue is used to iteratively refine user queries or provide personalised suggestions, improving task efficiency and user experience.

A key issue with most of those papers, independently of their contribution to the scientific community, is that they often fail to assess dialogue quality in a meaningful way, or rely on metrics that do not align with established principles from conversation theory. For instance, the Dist-n metric (J. Li et al., 2016), which

measures the number of distinct tokens in generated utterances, captures lexical diversity but provides little insight into the coherence, relevance, or appropriateness of a response within the conversational context.

Beyond these more common applications, a smaller number of studies explore alternative uses of conversation, such as solving complex tasks through multi-turn dialogue (X. Wang et al., 2024) or generating explanations of news claims (Hsu et al., 2024). While these papers are a minority in our classification, they overlap with the dialogue system category which may undervalue the actual amount of such papers.

Corpus Creation and Resources About 16% of the surveyed papers focus on describing resources or proposing methods for resource generation. The presented datasets range from human-human interactions to human-system dialogues, as well as fully synthetic datasets generated by LLMs and some handcrafted by humans without actual interaction. Several papers present corpora created by translating existing datasets or reproduce well-known ones in other languages – a reflection of the fact that NLP/CL research remains heavily focused on English, making new resources essential for studying other languages. Some contributions extend or merge existing resources by adding new annotations, either human-produced or automatically generated by LLMs. We further discuss the role of resources in the ACL Anthology 2024 and the typology of datasets in Section 3.1.3.

Modelling and Analysis Approximately 12% of the surveyed papers focus on modelling and analysing conversations, a category that seeks to uncover the mechanisms of human communication rather than merely build systems. This research is essential for advancing our general understanding of how people interact – from turn-taking patterns, feedback and grounding procedures, to topic-shift and redirection strategies or the ways intentions and information are conveyed. A well established task of this category is dialogue discourse parsing, which consists in segmenting a conversation into discourse units and identifying the relations between them (e.g. question-answer, elaboration, clarification request, etc.), thereby modelling the hierarchical and functional structure of dialogue (J. Li et al., 2022). Such insights are critical for identifying conversational practices used to manipulate, mislead, or deceive, which is particularly important in the current era of widespread misinformation (Lazer et al., 2018; Vijayan et al., 2025) and populism (Algan et al., 2017; Norris & Inglehart, 2019; Kaltwasser & Taggart, 2025), where identifying manipulative or deceptive strategies is critical. They should also form the conceptual backbone of conversational research, providing the empirical foundations on which robust and socially responsible systems can be built.

Despite its importance, this line of work appears under-represented at top-tier venues, representing only 6% and 4% of papers at ACL and EMNLP (ranked A* ac-

cording to the ICORE Conference Portal[9]), and 3% and 7% at NAACL and EACL (A). On the other hand, SigDial, LREC (ranked B) and the Findings[10] from ACL, EACL, NAACL and EMNLP contain between 10 and 18% of modelling and analysis papers. This disparity likely reflects the time-consuming nature of such research – requiring detailed annotation, fine-grained analysis, and often interdisciplinary collaboration – which may make it less immediately appealing than system-building papers in the current high publication pressure environment.

Specific Fields All of the categories of papers discussed thus far also appear in domain-specific contexts representing almost 4% of the surveyed papers, where the different systems are adapted to the needs of particular fields such as medicine and education. In these settings, generative systems and summarisation models are developed with a more targeted focus based on the needs of the fields. This specialisation often introduces additional challenges, including the need for domain-specific data. Some papers indeed focus on domain specific corpus generation methods (J. Wang et al., 2024; Arana et al., 2024; Bychkova et al., 2024). While medicine and education are usual application domains of NLP, both having their own dedicated workshops indexed in the ACL Anthology, one paper was also dedicated to dialogues in motor-sports (Isaka et al., 2024) which highlights the broad applicability of NLP across diverse fields. Such diversity underscores the need for multidisciplinary approaches and collaboration, as different domains bring unique knowledge requirements, evaluation criteria, and contextual challenges that must be addressed to develop effective systems.

Lone Papers The rest of the papers (less than 1%) present a tool meant to help with dialogue collection and annotations (A. Li et al., 2024) and papers on automatic speech recognition (W. Lee et al., 2024; Htun et al., 2024).

Limitation of the selection Due to the selection of venues we operated, papers presenting results in the context of shared task do not appear in our survey. They are however a crucial motor in our community and cover varied tasks including ones that are under-represented in the venues we covered.

3.1.3 Datasets and resources

Two aspects were taken into consideration regarding corpora in our review: the datasets that were *created* (Section 3.1.3.1) and the ones that were *used* (Section 3.1.3.2).

[9]The last ranking for the mentioned venues was made in 2026.

[10]Findings are not indexed in the ICORE Conference Portal but are by definition considered of lesser importance than the main conference.

3.1.3.1 Created datasets

In terms of creation, about 1/5 of the surveyed papers presented a new dataset. More than half of the new ones are in English but other languages are also included. However, in more than a third of the papers describing new resources, the language has to be inferred from the examples – when there are some – or from the collection or evaluation method. Some other papers mention the language in the appendix or limitations only, even though this information is crucial. English seems to be the language in most of these cases but, as it has been noted several times in the past decade, English is not equivalent to all the other languages and the language of the study should always be clearly stated (Bender, 2011; Ducel et al., 2022).

Some papers tackle multiple languages at once, reaching up to 150 languages in a paper presenting XSGD (Tu et al., 2024), the translation of the English Schema-Guided Dialogue dataset (Rastogi et al., 2020). These efforts show that the community is seriously trying to go beyond English. Translating existing corpora can be a pragmatic path for extending coverage to low-resource languages, but several caveats apply. Conversation reflects culture: idiomatic expressions, norms of politeness, turn-taking cues, notions of relevance or appropriate responses are culturally embedded, and may not transfer cleanly via translation (M. Zhang & Toral, 2019; Majewska et al., 2023). Automatic translators can help with scale, but should not be used without the input of native speakers, not only as post-hoc reviewers but as active collaborators in the translation process.

Most of the created datasets ($\sim$61%) feature pairs of participants engaged in short- or long-term conversations. A few datasets consist of pairs of adjacent utterances extracted from longer dialogues which potentially involve more speakers. Three datasets contain conversations with imbalanced roles with one main speakers and several sub-speakers or listeners, including one study where a single individual played the role of multiple speakers in a virtual reality setting (V. D. Lai et al., 2024). About 13% of the created datasets involve more than two speakers with some papers clearly stating the number and intentionally designing their studies around multilogues, while in others multi-party dynamics appear incidentally without being a primary focus.

Studying written data remains the most favoured modality ($\sim$57%) but multimodality also has a large place in the created datasets ($\sim$28%). This includes datasets that combine written conversations with related video or speech (e.g. YouTube comments), as well as conversations transcribed from videos or speech-only recordings, sometimes including additional data such as eye-tracking or physiological signals.

Almost half of the created datasets contain data produced by humans with different degrees of interaction and freedom. This includes 24% of human-human dialogues and 10% of conversations taken from the internet such as social media exchanges or YouTube comment threads for the most interactive and least constrained

ones. Data gathered from the internet is increasingly hard to confirm as human produced as we have no reliable way to distinguish between human and machine-generated content. The created datasets also include 7% of dialogues from TV shows, which can resemble spontaneous conversation to varying degrees. In particular, scripts tend to be less naturalistic than transcripts of what was actually said by the actors. Moreover, 8% of the datasets rely on hand-crafted dialogues, i.e. dialogues produced by humans in isolation given a set of instructions, like screenwriters write dialogues for their characters. However real-life conversations contain backchannels, disfluencies, overlaps, and other interactional phenomena, as opposed to most fictional dialogues (Chepinchikj & Thompson, 2016). Typical conversational dynamics are crucial for the participants to understand each other and build their contributions, but they are inconvenient from a reader's point of view. Thus, we are typically not exposed to naturalistic looking conversation transcripts, which makes it complicated to produce human-like scripted or handcrafted conversations (Pilan et al., 2024). This is not an issue when conversational dynamics are not in the scope of the study, but making sure that degraded dialogue does not influence other characteristics the study is interested in requires a careful assessment.

Similarly, about one third of the created datasets were generated by LLMs. These are much easier to produce than datasets involving real human interactions, which are costly and time-consuming, especially given current data demands. However, LLM-generated conversations should not be considered equivalent to human–human ones. Their creation settings impose structural constraints: LLMs typically produce clean, alternating turns, unlike the irregular patterns of real dialogue. Moreover, they are mostly written, despite aiming to mimic spontaneous conversation. Crucially, LLMs 'learn' and generate language in ways fundamentally different from humans (Bender & Koller, 2020); lacking grounding, interaction, and shared attention, their outputs cannot reliably represent human communicative behaviour, and thus corpora generated in this way cannot serve as reliable evidence for studying how people communicate. Nonetheless, such resources can support analytical work comparing human and machine dialogue. Three created datasets included both types to study their differences (Y. Li et al., 2024; Occhipinti et al., 2024) or improve LLMs (Sato et al., 2024). Human–machine corpora also account for about 9% of all created datasets, consistent with the prevalence of DSs. However, the strong research focus on these technologies warrants reflection. Access to DSs is not universal, and they should not be seen as substitutes for human contact, particularly for isolated populations such as the elderly. Assessing actual user needs remains essential when developing such tools (Kann et al., 2022).

The rest of the created datasets contained no conversation except for one which consists of non-player characters dialogues from a video game (Weir et al., 2024).

Moreover, 7% of the surveyed papers extended existing datasets with human-made and/or synthetic annotations. A few papers (about 4%) also present genera-

tion methods for producing synthetic data.

Five papers present benchmarks for evaluating LLMs, each targeting different aspects ranging from specific elements like social norm incorporation to broader traits such as human-likeness or multi-turn ability. They cover various languages: two address English and Chinese (Duan et al., 2024; Ou et al., 2024), one focuses solely on Chinese (Zhan et al., 2024), one on Korean (Jang et al., 2024), and one omits language specification (X. Li et al., 2024) but is based on the English Wizards of Wikipedia (Dinan et al., 2019). While evaluating LLMs is essential given their growing prominence, releasing fixed benchmarks risks rapid obsolescence: they suit pre-existing models but may later enter training data, compromising validity. Similarly, benchmarks derived from existing corpora can introduce bias. Nonetheless, producing evaluation datasets in languages other than English may yield a positive side effect – enriching resources for under-represented languages and promoting greater linguistic diversity in future LLMs.

3.1.3.2 Used datasets

Regarding usage, the most widely used corpora in the papers we reviewed were variants of the MultiWOZ (Budzianowski et al., 2018; Eric et al., 2020; Zang et al., 2020) dataset, which is unsurprising given the high proportion of papers on TOD systems. Indeed, these corpora contain task-oriented human–human chat-based dialogues collected in an adaptation of the Wizard-of-Oz (WOZ) setting, where one participant acts as a customer performing a task (e.g. booking a restaurant), and the other participant has access to a database to respond, which mimics what the system would have access to. In the original WOZ setting, the first participant was told that they were talking to a machine and the second was expected to play the system, which is not the case in MultiWOZ. Originally in English, MultiWOZ has since been translated into several languages. While reusing such established corpora is methodologically convenient, recent LLM-based approaches make this practice increasingly problematic. Because these datasets are used for both training and evaluation (see Section 3.1.4), building a TOD system often reduces to adapting an LM to reproduce the sentences expected in the WOZ corpora, rather than generalising beyond them, i.e. adapting an LM to a dataset rather than generalising from a corpus. Moreover, it seems reasonably likely that those corpora were included in the training set of most currently used LLMs which reinforces this circularity of data use and makes the evaluations based on ground truths less reliable.

A broad range of other corpora are also employed across different studies. Among the most frequently cited are DailyDialog (Y. Li et al., 2017) and SGD (Rastogi et al., 2020; H. Lee et al., 2022) – another TOD corpus. Zahiri and Choi (2018) and Poria et al. (2019), based on the TV-show *Friends* and IEMOCAP (Busso et al., 2008) – an acted multimodal multi-speaker corpus – are widely used for

conversation emotion recognition. Various datasets are used to enhance different aspects of dialogue systems: PersonaChat (S. Zhang et al., 2018) – persona-based conversations in pairs – for more personalised chit-chat systems, ESConv – emotional support conversations (S. Liu et al., 2021) between two crowdworkers – for building emotional support dialogue systems, ReDial (R. Li et al., 2018) – movie recommendation dialogues – for recommender systems, QReCC (Anantha et al., 2021) – hand-crafted dialogues with QA pairs grounded in web pages – for QA, and DSTC (Anantha et al., 2021) – phone calls between bus passengers and dialogue systems – for dialogue state tracking. All of these most used datasets are in English.

Both in terms of used and created resources, a recurring issue is the lack of information w.r.t. the datasets used. Many papers omit essentials such as the language of the data, whether the dialogue is handcrafted, scripted or naturally occurring, and whether it is human-human or human-machine. This lack of transparency has implications not only for reproducibility but also for the validity of the resulting models. For instance, DailyDialog is often described as a dataset for everyday chit-chat, but the dialogues were handcrafted by annotators rather than collected from real interactions. While using handcrafted data does not necessarily entail poor quality, it means that models are being trained and evaluated on data which reflects the *representation* we have of what a natural conversation is, rather than what it actually is, and people tend to underestimate the amount of e.g. pauses, repairs and other disfluencies (Bard & Lickley, 1997), which are distinctive markers of conversation. This observation points to a broader need: we must better define what conversational features are desirable in a model, and these criteria are likely to be highly task-dependent. For example, while hesitations and pauses may increase human-likeness, artificially inserting them into TOD system might not be necessary to satisfy users. On the other hand including them may help users to process information more easily. In any case, those trade-offs require investigations through a joint effort of the community.

3.1.4 Evaluation

Our findings (unsurprisingly) show that conversational AI is the main focus of current research. We therefore focus on the categories Generative systems and LM Evaluation defined in Section 3.1.2 for our assessment of the current evaluation practices in the community.

Two aspects are relevant when it comes to conversational generative systems: their ability to tackle the specific task they were designed for, and their conversational competency. The practical evaluation of task performance depends on the application and lies outside our scope in this section.

Regarding the conversation evaluation, we noted that many studies (45%) rely on reference-based automatic metrics such as BLEU (Papineni et al., 2002), ROU-

GE (C.-Y. Lin, 2004), and METEOR (Banerjee & Lavie, 2005). These metrics compare the system's responses to predefined gold-standard utterances based on lexical overlap. While they are convenient and widely used, they are limited in their ability to capture the interactive and context-sensitive nature of conversation. For instance, these metrics may penalise valid but diverse or creative responses that differ from the reference, even though such variation is a core feature of natural dialogue. Nevertheless, about one third of the studies do not use additional metrics to evaluate conversation which is problematic because the conversation quality assessment is multifaceted and should go deeper than the semantic comparison with a ground truth. These basic evaluations prevent us from accurately determining when a dialogue system is genuinely useful and more effective than a simpler alternative.

The increasing complexity of conversational tasks has led to the use of LLM-based evaluators, a practice usually called LLM-as-a-judge (LaaJ; Y.-T. Lin and Chen (2023) and Bavaresco et al. (2025)), which can flexibly score multiple conversational dimensions. Among papers on dialogue systems, about 12% rely on such evaluation[11]. Yet, more than one fourth do not validate the LLM judgements with human input, even though LLMs might have divergent 'expectations' of conversational quality in relation to humans. Moreover, as stated earlier, it is reasonable to think that many of the corpora we use to train and evaluate our models were included in the training data of the LLMs we use, which biases the evaluation. Similarly, many models are evaluated using benchmarks that may already be included in the training data of the models themselves, raising concerns about data leakage and the validity of results.

Regarding human evaluation, the most assessed dimensions are coherence (18% of the papers on dialogue systems including human evaluation) and fluency (16%). Other dimensions such as relevance, consistency and naturalness are also assessed (each 8-9%). This shows the need to break down the evaluation in more objective features which human could assess as external evaluators. Overall conversational quality is also assessed (9%), but this dimension remains difficult to capture in a controlled and reproducible way due to its inherently subjective nature. Crucially, very few systems are evaluated by end users during interaction ($\leq$ 6%), even though extrinsic evaluations in controlled settings may not reflect real-world user experience.

Our findings align with those of Yang et al. (2026) who conducted a large survey on NLG evaluation practices in general. They collected all the papers from ACL, NAACL, EMNLP and INLG between 2020 and 2025 (14171 papers). Information about the task and the evaluation approaches (automatic metrics, LaaJ, human evaluation) are retrieved automatically by several LLMs and aggregated through a majority vote. 61% of all the retrieved papers were labelled as NLG tasks by the

[11]We do not include BERTScore (T. Zhang et al., 2019) in the LLM-based evaluators but rather prompt based approaches with models such as LLaMa, Mistral, ChatGPT, etc.

models, 110 of those papers (about 1%) were further annotated by hand to evaluate the agreement with the automatic retrieval. One of the findings of this manual annotation was that most disagreements between the different models occurred for tasks that are not strictly NLG, such as code generation or mathematical reasoning.

After keeping only the top-30 'real' NLG tasks they are left with 3,334 papers, 78.1% of which belong to the following four most studied categories: summarisation, translation, dialogue generation and question answering. The two last ones are of particular interest for us. This paper confirms our finding that BLEU is the most widely used metric for dialogue generation, and QA is mostly evaluated through string matching. There is an increase in LaaJ especially in dialogue generation and despite common assumptions there seems to be differences between human evaluation and LaaJ in particular for higher level features like coherence and fluency. They note, like us, that when both LaaJ and human evaluation are used, there is not always an explicit comparison between them. Indeed, 58.7% of the papers that use both approaches explicitly compare them, indicating that the rest either does not report a comparison or measures different aspects.

All of this suggests that without careful reconsideration of evaluation practices, the field risks measuring progress in ways that may be convenient but ultimately misleading, reinforcing the need for more rigorous and context-aware assessment methods.

3.2 Pipeline for NLP and dialogue research

The epistemic pipeline followed when doing research on dialogue can be seen as a process to go from conversational phenomena to their analyses through formal tools and implementation in computational systems. One view of this pipeline is represented in simplified form in Figure 3.3. The process often begins with an intuition about a particular dialogue behaviour, e.g. a recurring interactional pattern, a coordination mechanism, a structural regularity, or a communicative difficulty. This intuition either motivates the creation of new dialogue resources or emerges from the close analysis of existing ones.

From these resources, researchers attempt to construct a model of the observed phenomenon. Nowadays, this modelling phase typically involves both a theoretical and a computational component. A formal account would first specify the structure or mechanism under investigation; computational implementations would then allow systematic testing and confrontation with empirical data.

On the basis of these models, research tasks can be defined. We can divide them into two broad families: analytical tasks and production tasks. The distinction does not lie in the technical tools employed – both may rely on rule-based systems, statistical models, or LLMs – but in their epistemic goals and evaluation logic.

Analytical tasks, such as discourse parsing, dialogue act tagging, or turn-taking

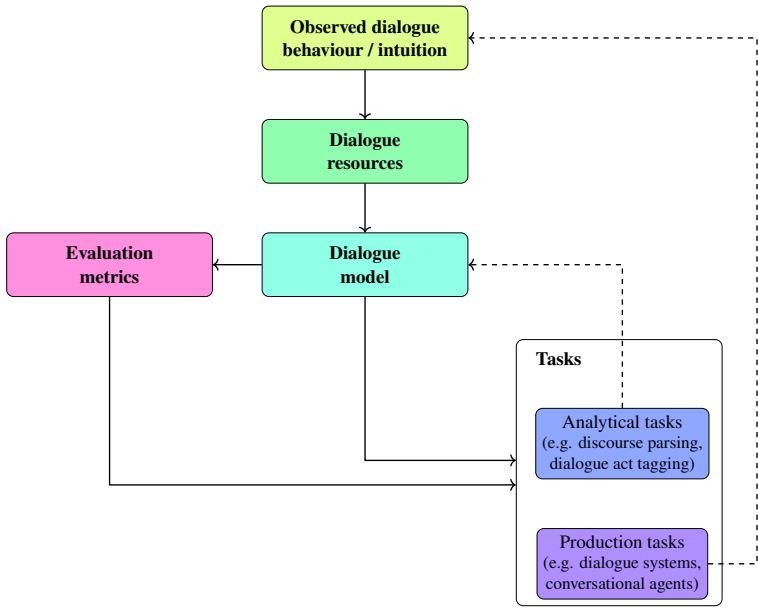


Figure 3.3: Description of the (dialogue) research pipeline in NLP/CL.

prediction, aim to provide insight into dialogue structure. Their purpose is descriptive or explanatory. They can also participate in refining theoretical models by testing their adequacy against annotated data.

Production tasks, by contrast, aim to reproduce or simulate aspects of dialogue for practical purposes. These include the development of dialogue systems, conversational agents, or synthetic resource generation. Their goal is generative and application-oriented. At the same time, they may also challenge the underlying model: if system outputs fail to match real-world expectations, some modelling assumptions may need to be revised.

Crucially, these two task families require different evaluation strategies. Analytical tasks are typically evaluated through alignment with annotated ground truth and category agreement. Production tasks require broader criteria, including task success, usability, user satisfaction, and interactional quality. Conflating these evaluation logics risks obscuring both scientific insight and practical performance.

3.2.1 From theoretical models to black-box ones

Historically, theoretical dialogue modelling included explicit formal frameworks such as dialogue games (D. Lewis, 1979) or speech acts and plan-based theories (Austin, 1975; P. R. Cohen & Perrault, 1979). These models participated to producing structured representations of grounding mechanisms and meaning negotiation among others. For instance, Type Theory with Records (TTR; Cooper (2005)) of-

fered a formalism to represent the dynamic and context-sensitive nature of dialogue, including the management of shared knowledge and the resolution of ambiguities. Some of these theoretical models have been implemented in computational systems; the dialogue manager based on information state proposed by Larsson and Traum (2000) is one such example. Yet they were not merely engineering tools but theoretical commitments about how dialogue works.

In the current LLM-dominated landscape shaped by large-scale transformer architectures (Vaswani et al., 2017) and their scaling paradigm (T. Brown et al., 2020), however, there is a growing tendency to assume that such theoretical structure will be implicitly captured by neural models trained on massive corpora. As a result, the explicit modelling layer risks becoming invisible. Instead of articulating formal assumptions about dialogue structure, participation roles, or grounding mechanisms, researchers increasingly rely on pre-trained models as black boxes, later adapted through fine-tuning or instruction tuning (Ouyang et al., 2022) for specific tasks.

This shift brings undeniable practical advantages: scalability, adaptability, and impressive performance across benchmarks. Yet it also entails epistemic costs. When modelling choices are replaced by reliance on opaque pre-trained systems, it becomes difficult to determine what is actually being learned (Rogers et al., 2020). Apparent improvements may reflect memorisation or training data overlap (Carlini et al., 2021), in particular when the same benchmarks keep being used over the years, or the exploitation of surface-level distributional regularities rather than a principled modelling of dialogue phenomena (Bender & Koller, 2020; Bender et al., 2021). More broadly, the assumption that scale can substitute for explicit theoretical structure remains insufficient when the objective is not only performance, but understanding the mechanisms underlying dialogue.

3.2.2 Evaluation and the risk of circularity

Since modelling has become less explicit, evaluation has become both more central and more fragile. The current trajectory of dialogue research reveals a tendency to prioritise convenience over suitability, particularly in the selection of data and evaluation methods.

Systems without rigorous evaluation are meaningless, yet the field often relies on metrics chosen for convenience or historical precedent, even when these metrics are known to be imperfect (Sai et al., 2022). Automatic evaluation cannot replace careful, task-specific assessment, and the cost of human evaluation should not be used to justify weak or inappropriate metrics. Researchers must articulate system expectations clearly and break them down into measurable aspects. Similarly, using LLMs as proxies for human evaluation must be done carefully and cannot be generalised across tasks; validation in one context does not ensure validity in another. Without human validation, reliance on LLM-based evaluation risks creating a self-

reinforcing evaluation loop, particularly when the evaluation corpora themselves may have been included in the models' training data. Such circularity can lead to an overestimation of system performance and obscure meaningful limitations.

Generative systems, while increasingly sophisticated, display conversational abilities that diverge sharply from human interaction. Conversations produced by these systems are typically structured in strict alternation and constrained by intents defined by their designers, who optimise for engagement and utility rather than natural interaction. Unlike humans, these systems do not rely on scaffolding through backchannels, overlapping speech, or incremental feedback. Users often adjust their behaviour to accommodate the model, subtly shaping their conversational patterns to align with machine expectations rather than engaging in natural dialogue. Even if models could perfectly mimic surface behaviours, they would not capture the grounded, embodied, and socially situated nature of human language. LLMs, due to their design, represent an 'average' over their training data, which cannot reflect the nuances of any individual or even population. Moreover while their training data does not capture the full variety of language use in the world (Bender et al., 2021), imagining a way to build a representative dataset would not be enough to make such models produce a rich and varied representation. Their design is such that they often rely on the same structure (Namuduri et al., 2025) and use vocabulary patterns that are different from the ones used by humans (Durward & Thomson, 2024). While this behaviour can be slightly adjusted in LLMs through prompting, their underlying structure will remain fundamentally different from humans' language acquisition and production mechanisms.

The question of human-likeness in dialogue further illustrates these limitations. LLMs can sometimes pass restricted forms of the Turing test (Jones & Bergen, 2026), yet evidence shows that evaluators may prefer machine-generated outputs over human dialogue in controlled settings. This does not imply that models are indistinguishable from humans. Rather, it highlights the unreliability of human judgements of conversational human-likeness, particularly under novel or brief evaluation conditions. The emergence of LLMs has also shifted public perception: whereas face-to-face interactions rarely prompt doubt about human interlocutors, non-face-to-face interactions with unknown partners are now more often approached with scepticism. Real human-human conversations are noisy, filled with overlaps, hesitations, repairs, and subtle cues that often go unnoticed when we participate in a conversation. We are indeed not often exposed to the actual look of a conversation. While repairs and backchannels are crucial for us to be efficient in conversation, they do not need to be *remembered* to follow the conversation. Thus, showing a transcript *a posteriori* to people and asking them to rate it for human-likeness will probably not reflect what they would have said *during* the conversation. Even placing someone in the role of a judge while they are having a conversation could change the way they interact and make them notice behaviours they would not have consciously analysed normally, making them doubt the quality

of the conversation. Paradoxically, interactional patterns may possibly penalise authentic interactions when they are evaluated externally, while LLMs would produce a smoother dialogue.

3.2.3 Re-centring theory, data, and purpose

These considerations have important implications for the design and purpose of generative conversational systems. Researchers must critically assess whether the ultimate goal should be to replicate human-like interaction, given the associated risks, including manipulation, misinformation, or undue emotional attachment. While the conversational paradigm can presumably make many applications, such as coding assistance or information retrieval, more approachable, such systems do not need to emulate human behaviour, they must instead perform complex tasks effectively. The conversational features reproduced in such systems should be ones that facilitate conversation without triggering emotional engagement. Differentiating between those features requires rigorous theoretical analysis of the notion of conversation. This would also provide solid grounds to create more dimension-specific metrics, capable of capturing distinct aspects of dialogue, and thus enable the development of evaluation frameworks better adapted to specific tasks. A formal characterisation of conversation would also help our community to clearly distinguish between prototypical human-human dialogue and other forms of communication, including scripted, synthetic, or human-machine interactions.

While dialogue generation was one of the most studied tasks among papers focusing on dialogue in the ACL Anthology in 2024, the need for more careful resource description is more general. Our field relies heavily on corpora and being able to clearly describe their properties and limitations is essential to ensure that they are used appropriately. Data should be at the heart of conversational research – and NLP/CL research in general, yet datasets are too often treated merely as tools to train models rather than objects of careful study in their own right. While most dialogical corpora are not produced in natural, face-to-face interactions – due to practical constraints – this does not mean that all proxies are equivalent. Handcrafted dialogues, movie scripts, human-machine exchanges, and LLM-generated conversations each reflect different interactional dynamics and limitations. Recognising and explicitly defining these limits is crucial. A clear understanding of the scope of a corpus should precede any model development or evaluation, ensuring that the data aligns with the intended task.

By returning data and evaluation to the centre of our research, developing theoretically grounded evaluation metrics, critically assessing human-likeness, clarifying the goals of generative systems, and grounding our decisions in the theory and practice of human conversation, the community can avoid pursuing progress that is only apparent. Without these reflections, the field risks running ever faster on a 'hamster wheel', producing models and metrics that give the illusion of ad-

vancement while overlooking the subtleties of human dialogue. Moving beyond the allure of human-like replication toward well-defined, task-oriented objectives will allow the field to achieve meaningful progress while mitigating the risks associated with deploying these technologies.

The limitations outlined above are not merely methodological concerns; they reveal the need for clearer descriptive tools. If resources differ fundamentally in their participation structures, environments, purposes, and interactional constraints, then treating dialogical behaviours as universal is unreasonable.

This is precisely where our CHIMERA becomes operational. By decomposing dialogue along explicit axes – AGENTS, SETTING, FRAME, and VERBAL INTERACTION – we can systematically describe both datasets and tasks. Such a structured description makes it possible to articulate what kind of dialogue a resource represents, what constraints a task imposes, and which dimensions are being modelled, evaluated, or ignored.

In the following section, we apply CHIMERA to three contrasting dialogue scenarios. These case studies illustrate how the framework can not only clarify the nature of different dialogue data but also reveal the specific constraints they impose on resource selection.

3.3 Applying CHIMERA

The three scenarios examined in this section operate at different descriptive levels. Turn-taking (Section 3.3.1) concerns the interactional organisation of dialogue and constitutes a domain-general mechanism that cuts across genres, activities, and institutions. Task-oriented dialogue (Section 3.3.2), by contrast, designates a broad activity type, defined primarily by the presence of an explicit practical goal, but encompassing a wide range of interactional configurations and degrees of social engagement. Clinical interviews (Section 3.3.3) represent a specific form of institutional interaction: they are goal-directed, like task-oriented dialogue, but are further constrained by professional norms, asymmetrical roles, and institutional definitions of success, while remaining open-ended in their interactional unfolding. The goal they are directed towards is not as clearly defined as in task-oriented dialogues and relates much more to the social function of dialogue.

3.3.1 Scenario 1: Turn-taking

Turn-taking refers to the organisation of participation in spoken interaction whereby interlocutors take turns speaking (Sacks et al., 1974). In ordinary conversation, transitions between speakers are typically seamless, with little to no overlap or extended silence. The ability to determine when it is one's turn to take the floor is crucial for the planning of dialogue: participants must interpret an ongoing utterance incrementally and prepare their own contribution in advance in order to respond

at an appropriate moment. Failing to do so entails interactional risks, such as losing the opportunity to speak if another participant takes the floor more quickly, or having a silence interpreted by the current speaker as signalling misunderstanding, hesitation, or a request for elaboration.

Turn-taking is therefore one of the most fundamental properties of dialogue. It is not merely a surface feature describing the alternation of speakers, but a structuring mechanism that enables participants to coordinate action, negotiate meaning, and maintain the coherence of an interaction over time. In this sense, turn-taking contributes directly to the form of dialogue: it makes participation visible, organises sequential dependencies between contributions, and allows interlocutors to project upcoming actions.

Importantly, turn-taking emerges from spontaneous language use in real time. It relies on fine-grained temporal coordination, anticipation of others' actions, and the management of overlaps, pauses, and repair phenomena. These properties make turn-taking both central to dialogue and difficult to model, observe, or reproduce outside of naturally occurring interaction. As such, it provides a particularly revealing case for examining how dialogue research conceptualises its objects of study, selects its resources, and evaluates its tools.

In what follows, turn-taking is used as a wide scenario to question practices in dialogue research, examined successively from an analytical, tool-building, and productive perspective.

3.3.1.1 Analytical perspective: constraints for studying turn-taking

Studying turn-taking requires making explicit a number of constraints on both the data and the interactional setting. Turn-taking is inherently relational and temporal: it only exists through the coordination of multiple participants over time. As a consequence, resources suitable for its study must preserve interactional dynamics with minimal post hoc normalisation. In particular, temporal properties of the interaction – such as precise alignment of speech segments, access to pauses and overlaps, and the preservation of simultaneity – are essential. Resources derived from scripted dialogue, heavily edited recordings, or post hoc reconstructions do not satisfy these requirements, even if they involve multiple speakers and conversational content.

These constraints become especially salient when considering mediated forms of spoken dialogue. While the organisation of turn-taking in face-to-face interaction has been extensively described in CA, the emergence of new interactional settings calls for renewed investigation. For instance, video-mediated conversations differ from face-to-face interaction in several aspects that are directly relevant to turn-taking. In particular, such settings introduce varying degrees of latency between the production of an utterance and its reception, as well as restricted or distorted access to visual cues such as gaze direction or body posture. Phenomena

such as parallax effects or partial visual representations can further complicate the coordination of turns, potentially making turn-taking less efficient or more error-prone. In the following, we will thus focus on the differences between face-to-face and video-mediated interaction with respect to turn-taking.

Understanding these divergences matters not only from an epistemological perspective but also for practical reasons. Video-mediated interaction is generally easier to record than face-to-face conversation and thus constitutes an attractive alternative for building large dialogue resources. The Candor corpus (Reece et al., 2023), for instance, provides a large collection of casual video-mediated dyadic interactions. However, if video mediation significantly alters turn-taking behaviour, then resources of this type cannot be straightforwardly treated as proxies for face-to-face interaction. Since turn-taking supports many other aspects of dialogue co-construction, its modification may have cascading effects on the interaction as a whole.

Using CHIMERA, the parameter *turn-taking* under VERBAL INTERACTION clearly emerges as the focus of the study. Since this parameter is one of those that are dependant on the three other meta-parameters, we must determine the constraints on those such that *turn-taking* is as free as possible. Within our formal account, it means finding the antecedent of the update function such that the end type that represents all the dialogue parameters states as little constraints as possible on *turn-taking*. Let us consider the three meta-parameters in turn. First, AGENTS must be human we want to study the natural form of turn-taking and there must be at least two of them to allow for turn-taking to occur. Second, the CONVERSATIONAL SETTING does not need to be constrained in terms of world setting apart from the fact that it should be similar across the two modalities we want to compare. Otherwise, observed differences in turn-taking could be attributed to differences in the setting rather than the medium. However, the interaction must be synchronous, otherwise turn-taking is not possible. The rest of the parameters under CONVERSATIONAL SETTING are not strictly constrained by the task of studying turn-taking but for our specific intent of comparing face-to-face and video-mediated interaction, we must constrain the ENVIRONMENT AND MEDIUM to be either face-to-face or video-mediated. Finally, the FRAME is not strictly constrained by the task of studying turn-taking but it is likely that some frames will strongly interact with turn-taking mechanisms. For instance, a highly structured frame such as a formal debate may impose turn-taking rules that are not representative of ordinary conversation. In contrast, a more open frame such as casual conversation may allow for more naturalistic turn-taking behaviour. Similarly to the world setting, the frame should therefore be similar across the two modalities we want to compare, to avoid confounding effects. Those constraints are summarised in Table 3.2.

With those constraints set, the specific instantiation of the remaining sub-parameters is likely to affect the resulting interactional dynamics. A comprehensive study of turn-taking in video-mediated dialogue compared to face-to-face dialogue

Table 3.2: Constraints on the dialogue resource for studying turn-taking.

Meta-Parameters	Sub-parameters	Constraint
Agents		
Agency		Humans
Participation structures	Number	≥ 2
Conversational setting		
Environment & Medium	Synchrony	✓
Verbal interaction		
Turn-taking		✓

would therefore need to explore a range of settings, from highly organised to more open ones. More controlled settings could involve task-oriented dyadic interactions, allowing for partial control over participation patterns and role switching. With respect to the medium, not all video-mediated configurations are equivalent. Turn-taking practices depend on the availability and quality of communicative channels, and video-mediated interaction introduces latency, restricted visual fields, and altered feedback mechanisms to varying degrees depending on the tool. A highly controlled setting might involve high-quality video streams showing the interlocutor directly, as opposed to avatars or choppy images, combined with quiet physical environments and minimal external distractions. Starting from such a controlled configuration, the setting could then be progressively complexified, for instance by introducing noisier environments, more open-domain dialogue, or additional participants. By comparing turn-taking in video-mediated and face-to-face interaction across a range of settings, researchers could identify which aspects of turn-taking are robust to changes in the medium and which are sensitive to specific interactional constraints. Within the formal account of CHIMERA, this would correspond to empirically constructing the update function specifically for the parameter *turn-taking* under VERBAL INTERACTION in general, based on changes on the SETTING and FRAME meta-parameters. Without precisely defining the SETTING, clear differences between face-to-face and video-mediated interaction seem to predict that there *will* be differences in VERBAL INTERACTION. (3.i) shows the type of the *Environment & Medium* in these two interactional settings. Again, we assume the existence of *ad hoc* types and use descriptive names to represent them instead of formalising them.

(3.i) Face-to-face environment and medium:

$$\begin{bmatrix} \texttt{PhysicalContext} : \begin{bmatrix} \texttt{Space} : \textit{SharedSpace} \\ \texttt{PerceptualField} : \textit{SimilarPerceptualField} \end{bmatrix} \\ \texttt{Visibility.Gaze.Fidelity} : \textit{FullGazeFidelity} \\ \texttt{Audibility.ProsodyTone.Fidelity} : \textit{FullProsodyToneFidelity} \\ \texttt{Time.Synchrony} : \textit{FullSynchronous} \end{bmatrix}$$

Video-mediated environment and medium:

$$\begin{bmatrix} \texttt{PhysicalContext} : \begin{bmatrix} \texttt{Space} : \textit{SeparateSpace} \\ \texttt{PerceptualField} : \textit{PartiallySharedPerceptualField} \end{bmatrix} \\ \texttt{Visibility.Gaze.Fidelity} : \texttt{?} \\ \texttt{Audibility.ProsodyTone.Fidelity} : \texttt{?} \\ \texttt{Time.Synchrony} : \texttt{?} \end{bmatrix}$$

First, the physical context is clearly not the same in face-to-face as it is in a video-mediated interaction. In face-to-face, participants are in the same place whereas video-mediated exchanges imply that only what is captured by the camera and microphone is shared. For similar reasons, the visibility and audibility can be compromised depending on the quality of the equipment and of the internet connexion. This can also make video-mediated interactions slightly less synchronous as a result of the latency introduced by the transmission of the signal. The exact instantiation of these parameters is likely to vary across different video-mediated settings, and this variation can have a direct impact on turn-taking behaviour. In particular, the coordination signals, discussed previously in Section 2.4.4.1, or how well they can be interpreted are likely to be affected by the quality of the visual and auditory channels. This in turn could affect the form turn-taking would take in such settings. Differences in modality between the two settings must therefore be treated as part of the phenomena to study rather than as noise. Latency, reduced access to visual cues, and altered feedback loops must be explicitly characterised, as they directly shape turn-taking behaviour.

In summary, this analytical scenario imposes strong constraints on the medium of interaction while leaving a range of possibilities open for other parameters, provided that these are aligned across resources when comparisons are made. Making these constraints explicit is essential to delimit the scope of empirical findings and to avoid conflating genuine interactional differences with artefacts of data collection or processing.

3.3.1.2 Tool-building perspective: ASR, transcription, and the degradation of interactional structure

While the analytical study of turn-taking depends on rich interactional data, much dialogue research relies on tools that substantially transform, and sometimes degrade, the phenomena of interest. ASR systems are a particularly salient example. ASR has become a ubiquitous component of dialogue processing pipelines, yet it is typically optimised with respect to lexical accuracy, most commonly measured through word error rate. In this optimisation process, interactional features central to turn-taking – such as overlaps, interruptions, and fine-grained timing – are frequently flattened or suppressed.

In practice, ASR outputs often miss overlapping turns, in particular short utterances and vocal coordination signals (Liesenfeld et al., 2023), or assign them to the wrong speaker, thus producing a flat and misleading representation of interaction. From the perspective of turn-taking, such representations are not neutral transformations: they erase precisely the features that make the phenomenon observable.

Transcription practices raise similar issues. Transcriptions remain the primary working support for much dialogue research, yet they are inherently degraded representations of spoken interaction. Different transcription schemes preserve different aspects of dialogue, and the choice of scheme has direct consequences for what can be analysed. Minimal orthographic transcriptions may suffice for lexical or semantic analysis, but they offer little access to turn-taking phenomena. Richer interactional transcription schemes, which encode overlaps, pauses, and prosodic cues, preserve far more information but are costly to produce and require more specialised expertise.

A face-to-face interaction, characterised by multimodality and fine-grained temporal coordination, undergoes a substantial change when converted into a transcription. Depending on the transcription scheme, parameters such as temporal resolution, simultaneity, and participant coordination may be partially or entirely lost. Table 3.3 uses CHIMERA to make these transformations explicit from a face-to-face interaction to a richly transcribed version following a scheme such as the one from CA (Jefferson, 2008), and a basic lexical transcript that includes only sanitised linguistic content.

While the original interaction allows the participants to perceive all the parameters like the participation structure and the environment, the transcription process necessarily degrades this information. A rich transcript can preserve some of these aspects by describing the context for example, like stage directions in a theatrical script, but it is not the case for all the parameters and it is not as good as the original interaction. The lexical transcript, on the other hand, completely flattens the interactional dynamics, leaving only the observable linguistic content. Regarding verbal interaction, Jefferson (2008) proposes conventions to represent overlaps and interruptions in transcriptions which can be used to preserve the turn-taking dynam-

Table 3.3: Degradation of dialogical features from original occurrence to transcription.

Meta-Parameters	Sub-parameters	Original	Transcript	
			Rich	Lexical
Agents				
Participation structures	Function	Observ.	∅	∅
Conversational setting				
Environment & Medium	Physical context	Observ.	Partially transcribed	∅
	Gaze	Observ.	∅	∅
	Gesture / Facial expression	Observ.	∅	∅
	Prosody / Tone	Observ.	Partially transcribed	∅
Verbal interaction				
Coordination signals		Observ.	Transcribed	∅
Turn-taking		Observ.	Transcribed	∅
Completion		Observ.	Transcribed	∅

ics to a certain extent. Nevertheless the transcription process will always require to make choices about what to preserve and what to discard, but also about how to represent the preserved features which is not always an objective process. For instance, the decision to use a certain punctuation to represent at least partially the structure carried by prosody in spoken interaction can change the perceived meaning of an utterance and thus the way it is interpreted in the context of the interaction (Hammersley, 2010). The transcription process is thus not a neutral transformation but an active interpretation of the interaction, which can lead to a significant loss of information and a distortion of the original phenomena (Johnson et al., 2025). While not all parameters of the original interaction are necessarily relevant for a given study, knowing what is preserved and what is lost in the transcription process is essential to ensure the validity of the research process.

These issues also raise questions about the use of LLMs in dialogue research. While they can process structured text and may be capable of handling enriched transcription schemes, they are typically trained on dialogue data that has already been flattened and normalised. This raises doubts about their sensitivity to interactional features such as turn-taking, and about the extent to which they can meaningfully model phenomena that are poorly represented in their training data.

3.3.1.3 Productive perspective: generating dialogue with turn-taking

Recent years have seen an increasing emphasis on the automatic production of dialogue resources, both written and spoken. From the perspective of turn-taking, this trend raises important questions about what it means to generate dialogue that preserves interactional structure.

In written dialogue generation, when producing data that is intended to simulate spoken interaction, incorporating turn-taking in any meaningful sense requires adherence to a transcription or annotation scheme that encodes interactional features. Simply alternating speaker labels is insufficient to capture phenomena such as overlap, interruption, or repair. Achieving richer representations would require training data transcribed with appropriate schemes, as well as models capable of learning and reproducing these conventions.

In spoken dialogue generation, the challenges are even greater. Generating turn-taking behaviour involves not only deciding when speakers alternate, but also modelling timing, anticipation, and occasional simultaneity. Current spoken dialogue systems largely rely on turn-based architectures that assume clear boundaries between speakers' contributions. While this assumption simplifies system design, it departs significantly from the dynamics of human interaction.

These limitations suggest that, despite recent progress in generative models, we are not yet in a position to produce dialogue data that exhibits human-like turn-taking behaviour in a robust way (Mayor et al., 2025). This observation does not imply that synthetic dialogue resources are without value. Rather, it underscores the importance of treating generated data as a distinct type of resource, with its own constraints and limitations, rather than as a cheaper replacement for human-produced interaction.

3.3.1.4 Key takeaways and implications

Turn-taking functions as a revealing stress test for dialogue research practices. As a structuring property of dialogue, it is central to interaction yet fragile with respect to data processing and tool design. When turn-taking is ignored or flattened, resources may still support a wide range of tasks, but the scope of the phenomena that can be legitimately studied is reduced.

CHIMERA provides a means to make these limitations explicit. By characterising how resources, tools, and representations constrain access to turn-taking, it be-

comes possible to delimit the generalisability of analytical findings and the claims made on the basis of generated dialogue. In the following scenarios, a similar approach will be applied to other aspects of dialogue research, where constraints do not necessarily bear on the form of dialogue but are no less consequential.

3.3.2 Scenario 2: Task-oriented dialogue

Task-oriented dialogue constitutes one of the most widely studied categories of dialogue in computational linguistics and NLP. In the literature, the term typically refers to interactions that are structured around the completion of a practical goal, such as booking a train or restaurant, solving a collaborative task, or achieving a shared objective within a constrained environment. As discussed in Chapter 2, the goal of a dialogue can be characterised along several dimensions, including practical purpose, information exchange, and social bonding. Task-oriented dialogues are primarily defined by the predominance of a practical purpose, though in practice these dimensions coexist to varying degrees.

The category of task-oriented dialogue itself covers a broad range of interaction types. Some tasks are highly constrained and admit a clear success condition, often formalised as a set of slot values to be filled, as in classical booking scenarios. Others are more open, such as map tasks (Thompson et al., 1993) or tangram dialogues (Clark & Wilkes-Gibbs, 1986), where success depends on mutual understanding and coordination rather than the completion of a predefined form. Still others, such as service encounters in everyday settings (e.g. bakery dialogues), combine a practical goal with elements of social bonding and politeness.

Task-oriented dialogues are commonly used to study specific linguistic or interactional phenomena, for instance the production and interpretation of referring expressions in tangram tasks. More broadly, they have become a dominant type of resource in dialogue research, in part because they are comparatively easy to collect. The presence of an explicit task provides participants with a clear interactional frame, reducing the need to sustain conversation for its own sake with potentially unknown dialogue partners. Moreover, task-oriented settings tend to limit the range of personal information exchanged, which facilitates anonymisation (Amblard et al., 2014) and makes data collection more straightforward than in casual, open-ended conversations.

However, the very constraints that make task-oriented dialogue attractive as a research resource can also influence a wide range of other dialogue parameters in less obvious ways. For this reason, task-oriented dialogue provides a particularly instructive scenario for examining how dialogue purpose interacts with resource selection, task formulation, and evaluation practices, especially in the context of contemporary, LLM-based systems.

3.3.2.1 Analytical perspective: task-oriented dialogue as a basis for discourse modelling

Task-oriented dialogues have long served as a privileged empirical basis for studying discourse structure and rhetorical relations. A rich body of work has investigated how contributions in such interactions relate to one another in terms of explanation, clarification, elaboration, or argumentation, often with the explicit goal of supporting discourse parsing or rhetorical relations tagging. Corpora such as STAC (Asher et al., 2016), which consists of chat-based, task-oriented multi-party dialogues, and Molweni (J. Li et al., 2020), derived from Ubuntu forum discussions, have played a central role in this line of research. These resources, whose features are described in Table 3.4, are frequently used to operationalise and evaluate theoretical frameworks such as SDRT (Lascarides & Asher, 2007) or information state update models in dialogue settings.

The STAC corpus provides a particularly rich illustration of how task constraints shape interactional organisation. As a reminder, it consists in multi-party dialogues in which participants play a board game that requires them to negotiate and trade resources. They play online through a virtual interface and they can communicate with each other through a written chat system. The participation structure is thus strongly influenced by the game setting: players repeatedly engage in sequences where one participant requests a resource (e.g. clay or sheep) and attempts to acquire it through negotiation or trade. This recurring pattern yields highly regular discourse sequences, typically involving a main speaker addressing all other players as potential interlocutors. At the same time, STAC dialogues also contain parallel threads that are not directly oriented toward game progress, in which some participants temporarily adopt a bystander role. As a result, the corpus exhibits a complex organisational structure combining focused task-oriented exchanges with more peripheral conversational activity. (3.1) illustrates such a situation, where a side conversation about the trials for a league starts in parallel with the main thread (normal font) of interaction that revolves around resource trading. This parallel thread further splits into two sub-threads (marked in bold and italics).

(3.1) LJ anyone want sheep for clay?
GWFS got none, sorry :(
GWFS **so how do people know about the league?**
WM no
LJ **i did the trials**
TK **i know about it from my gf**
GWFS [**yeah me too,**]$_a$ [*are you an Informatics student then, lj?*]$_b$
TK **did not do the trials**
WM has anyone got wood for me?
GWFS [**I did them**]$_a$ [**because a friend did**]$_b$

GWFS lol wm, you cad
GWFS afraid not :(
LJ [*no, I'm about to start math.*]$_a$ [*I just hang around appleton a lot*]$_b$
TK sry no
GWFS my single wood is precious
WM what's a cad?

[STAC corpus (Asher et al., 2016, Ex.1)]

Several additional parameters contribute to this complexity. In terms of enunciative status, participants are engaged in a game and may therefore display a persona that differs from their everyday conversational self; this distinction may itself correlate with differences between game-related and non-game-related threads. The conversational history is also non-trivial: since the game is organised as a tournament, participants can encounter each other several times, thus creating interactional histories between them. There may also be a more subtle form of shared background in terms of inter-personal familiarity and topical ease linked to recruitment practices. In (3.1) all the participants seem to be university students which gives them shared topics to talk about. Linguistically, we can imagine that participants were instructed to interact in English though full control cannot be guaranteed.

Although STAC is a text-based corpus, it nevertheless involves a shared physical – or more precisely, virtual – context: participants interact while jointly manipulating a game board. This shared environment plays an important role in grounding discourse, as demonstrated by work in SDRT that explicitly integrates references to 'physical' context in the analysis (Hunter et al., 2018). Epistemically, the dialogue unfolds across two worlds: a fictional but fixed game world, defined by the rules and state of the game, and the real world, which participants occasionally invoke in side conversations. The fictional world is fixed in the sense that its properties are determined by the game mechanics, though it may still be subject to local interpretation or negotiation.

Molweni represents a different but complementary type of task-oriented resource. Derived from Ubuntu technical support forums, it consists of written exchanges oriented toward problem solving and information provision. With respect to participants, the corpus is old enough that one can reasonably assume that the contributors are human users. However, this assumption is becoming increasingly fragile as AI-generated content proliferates online, raising the risk that newer forum-based datasets may unknowingly include synthetic dialogue with no reliable method for detection.

The participation structure in Molweni is also quite complex as most of the dialogues involve multiple participants. Duologues represent about 21% of the corpus, the rest involving 3 to 9 participants. Functionally, the information seeker can be analysed as a main speaker addressing a broad but implicit audience, while respondents self-select based on their expertise, temporarily becoming both main

Table 3.4: Description of the dialogical setting in the STAC and Molweni resources.

Meta-Parameters	Sub-parameters	STAC	Molweni
Agents			
Agency		Human	
Participation structures	Number	4	2-9 (avg. 3.51)
	Functions	Main speaker - Addressees depending on requested information	Main speaker - Open / Addressees
Social dynamics	Hierarchy	∅	Information seeker - holder
	(Conversational) History	Game session	∼ ∅
*Enunciative status**	Anonymity	Pseudonymous	
	Role	Self in game	Self
	Persona	Self in game	Self - Self as expert
	Commitment level	✓	
Conversational setting			
Epistemic grounding	World setting	Fictional (game) + Real (meta)	Real
	*World stability**	Fixed	
Environment & Medium	Physical context	Virtual boardgame	∅
	Gaze	∅	
	Gesture / Facial expression	ASCII emojis	
	Prosody / Tone	∅	
	Noise	∅	
	Synchrony	Semi-synchronous	Asynchronous
	Ephemerality	Saved for the task	Persistent by design

Table 3.4: Description of the dialogical setting in the STAC and Molweni resources (cont.).

Meta-Parameters	Sub-parameters	STAC	Molweni
Conversational frame			
Language	Global	English	
	Control	✓	✗
	*Vocabulary**	Partially game specific	Technical
Purpose	Practical goal	Task + social	Information exchange
	Success condition	Adhere to politeness norms	Question in first message
	*(Topical) structure**	Negotiation phases	QA
Verbal interaction		∅	

addressee and main speaker. The enunciative status of participants is pseudonymous, with the additional possibility that a single username may correspond to more than one individual. Although contributors are not explicitly role-playing, answerers may adopt a 'teacher-like' or expert stance, subtly shaping their language and interactional behaviour. Notably, this shift in stance occurs seamlessly in human interaction, in contrast with LLM-based systems where such behaviour must be explicitly prompted.

Several other parameters introduce analytical challenges. Due to the open nature of forum interaction control over the language is difficult to maintain, though the original paper does not mention the presence of non-English content. Co-construction, while often associated with spoken dialogue, is also relevant here: participants can compose replies simultaneously, leading to overlapping or partially incompatible contributions depending on posting order. Timestamps therefore become a crucial resource for interpreting interactional dynamics. This asynchrony complicates annotation within frameworks such as SDRT or information state update models, which typically rely on a well-defined sequential order of utterances.

STAC and Molweni exemplify a class of task-oriented interactions that are particularly amenable to analytical modelling. They involve explicit goals, relatively stable participation structures, and interactional trajectories that can be interpreted in terms of information exchange and collaborative problem solving. As such, they provide a favourable testing ground for models that conceptualise dialogue as a sequence of information state updates, where each contribution incrementally modifies a shared representation of the task, the domain, or the interlocutors' commit-

ments. At the same time, they introduce non-trivial complexity: STAC through multilogue interaction and a shared virtual environment, Molweni through strong topical constraints and asynchronous written exchange.

Compared to more sanitised resources such as DailyDialog (Y. Li et al., 2017) – which presents itself as modelling everyday conversation but in practice exhibits highly organised and simplified interactional patterns – STAC and Molweni offer a richer and more challenging empirical basis. They have enabled the community to develop and extend discourse theories, including SDRT, by confronting them with multilogue structure, virtual grounding, and asynchronous coordination. Nevertheless, both corpora remain fundamentally goal-oriented, and this orientation stabilises the interaction in ways that distinguish them from more casual forms of conversation. Moreover, the text-based medium allows for organisational patterns, such as parallel topical threads and non-adjacent responses, that do not always reflect the mechanisms participants rely on in synchronous spoken interaction.

A key limitation shared by these resources concerns the treatment of social purposes in dialogue. Task-oriented interaction is well suited for identifying and annotating rhetorical relations, as well as for deriving dialogue-level structures such as QUDs. In practice, QUDs are easier to define when they relate to task progress or information needs, but much harder to formalise when the interaction serves social goals that are diffuse, implicit, or only loosely connected to a practical outcome. This limitation becomes particularly apparent when considering everyday conversational activities involving social bonding or low-stakes coordination, such as talk accompanying shared leisure activities. While some theoretical approaches – SDRT in particular – arguably leave more room for accommodating such phenomena, they still presuppose identifiable communicative intentions, leaving certain forms of social interaction difficult to capture despite their prevalence in daily language use.

In this sense, task-oriented dialogue illustrates how analytical convenience can shape theoretical coverage. By privileging interactions structured around practical goals and information exchange, dialogue research risks developing models that are robust within a narrow domain but fragile when confronted with the diversity of everyday conversational practices. Making these constraints explicit through a descriptive framework is therefore essential, not only to delimit the scope of existing analyses, but also to clarify what remains to be accounted for when moving beyond task-oriented interaction.

3.3.2.2 Tool-building perspective: data-collection platforms and induced constraints

Tools designed specifically for data collection within the research community have also played an important role in gathering task-oriented dialogue resources. A wide range of platforms have been created to elicit task-oriented interactions in a controlled yet scalable manner, including chat-based interfaces (Healey & Mills,

2009; Decker et al., 2024), collaborative games, and experimental environments designed to support corpus construction (Chalamalasetti et al., 2023). Examples include generic chat tools developed for dialogue data collection, as well as more specialised setups such as game-based interaction frameworks used in resources like STAC.

In the context of this thesis, we developed a conversational tool (Decker et al., 2024) inspired by the system proposed by Healey and Mills (2009), using the instant messaging application Cinny, which is based on Matrix, a real-time open communication protocol. Cinny is available as a web, desktop, and mobile application, making it convenient for participants to use. It also allows us to host our own server, giving us full control over the experimental data. Our tool consists of Python scripts that connect to the server, create user accounts and chat rooms for participants, and monitor conversations in order to modify some messages. Each participant has a real account as well as an associated bot account. For each conversation, a dedicated chat room is created for each participant, containing the participant and the bot associated with their interlocutor(s). In a two-person conversation, for example, two rooms are created: one for each participant and the bot representing the other participant. Messages are therefore initially received only by the corresponding bot account, allowing the scripts to process and modify them before relaying them to the other room through the bot, making them appear as if they had been sent directly by the interlocutor.

Figure 3.4 illustrates this process. This architecture enables controlled modifications of messages before transmission, allowing us to introduce specific disturbances into conversations and investigate phenomena such as participants' resilience to misunderstandings. Since the interaction takes place through separate rooms, participants perceive their messages as being sent normally and remain unaware of the manipulations. The scripts also store all conversation data in an SQLite database. More details on a dialogue collection experiment we performed are available in Appendix E.

Another advantage of this framework is the ability to integrate widgets into chat rooms, i.e. web pages that participants can interact with while chatting. These can be used to design more interactive experimental tasks, such as cooperative minigames.

These tools are valuable precisely because they allow researchers to gather dialogue data efficiently and at scale. By embedding participants in a task, they provide an interactional frame that facilitates sustained dialogue while reducing the need for extensive prompting or moderation. In the current context, where LLMs increasingly absorb vast quantities of publicly available text, the ability to generate new, well-documented dialogue resources under controlled conditions has become particularly important for the research community.

At the same time, data collection tools are not neutral instruments. By design, they impose constraints on the dialogue they elicit, shaping both the interac-

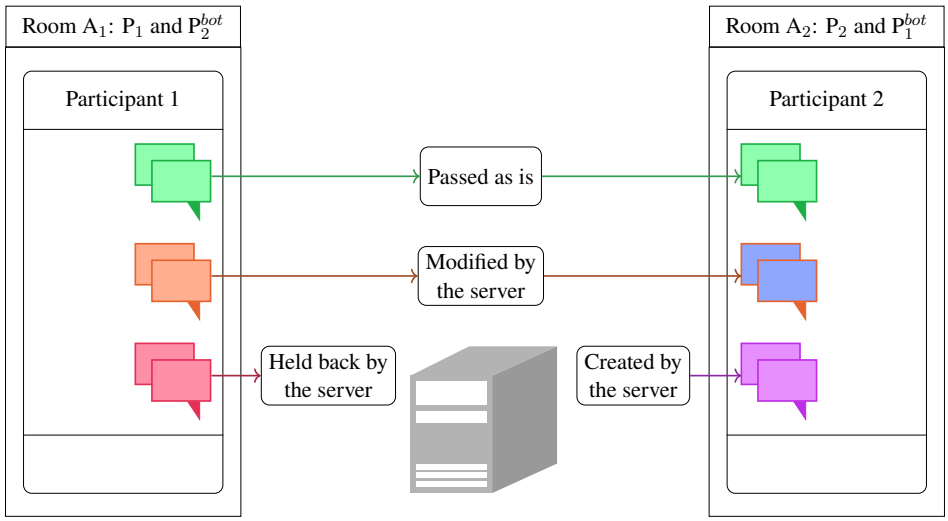


Figure 3.4: Representation of the exchange and manipulation of the messages with our ChatTool.

tional structure and the range of behaviours that can emerge. The task definition, the interface, the turn-taking mechanism, and the available communicative channels all influence how participants engage with one another. From the perspective of CHIMERA, such tools typically strongly constrain the environment and medium, while leaving other parameters – such as social dynamics or interactional flexibility – only partially accessible.

These induced constraints are not inherently problematic, provided they are made explicit and taken into account when resources are reused. Task-oriented data collected through specialised tools can be highly appropriate for analytical studies focusing on information exchange, coordination, or rhetorical structure, as well as for training or fine-tuning dialogue models whose intended use aligns with the properties of the resource. However, problems arise when such data is treated as a generic representation of dialogue, or when the constraints imposed by the collection tool are overlooked.

In this sense, tool-based data collection exemplifies a broader methodological trade-off. On the one hand, the community needs efficient ways to produce new dialogue resources in order to maintain empirical grounding and avoid over-reliance on pre-existing, widely circulated datasets. On the other hand, the very tools that make data collection feasible also shape the dialogue they produce. CHIMERA provides a means to document and reason about these effects, ensuring that the resulting resources are used in contexts where their constraints are compatible with the research goals.

3.3.2.3 Productive perspective: ultra-exploitation of benchmarks

From a productive perspective, task-oriented dialogue has become the dominant paradigm for training and evaluating conversational agents, most notably through the downstream task of next-utterance prediction. This paradigm underlies the majority of contemporary chatbots and conversational AI systems and has been widely operationalised through benchmark corpora such as MultiWOZ (Budzianowski et al., 2018) and related datasets. In this setting, a system is trained to generate an appropriate next contribution in an ongoing task-oriented interaction, typically framed as a dialogue between a human user and a machine agent.

The tasks targeted in this paradigm are generally defined by a relatively clear and finite set of parameters or slot values that must be retrieved, such as dates, locations, preferences, and availability constraints in domains like travel planning, restaurant booking, or hotel reservation. This formalisation makes task-oriented dialogue particularly attractive from a production standpoint. Success conditions can be specified with relative clarity, evaluation metrics can be derived from task completion, and dialogue progress can be modelled as incremental constraint satisfaction. Historically, such systems relied on rigid architectures such as finite-state machines or hand-crafted dialogue policies. In current practice, however, LLMs dominate this space, offering substantially greater surface-level flexibility while remaining constrained – either explicitly through prompting and system design, or implicitly through training data and evaluation metrics – by the underlying task structure.

The widespread adoption of LLM-based task-oriented systems raises questions that extend beyond technical performance. In particular, it invites reconsideration of what kind of conversation users expect, or are willing to engage in, when interacting with a machine even in narrowly task-constrained settings. Unlike graphical interfaces or web forms, conversational agents invite interactional behaviours that resemble human dialogue, despite the fact that the underlying purpose remains strictly practical. Empirical observations suggest that users adapt their behaviour accordingly: they may simplify their language, adjust politeness strategies, or attribute communicative intent and understanding to the system. As a result, the dialogues produced and consumed in this context cannot be treated as straightforward instances of human-human task-oriented interaction. Rather, they constitute a distinct interactional genre that warrants analysis in its own right.

Table 3.5 illustrates several structural tensions of the task-oriented dialogue paradigm by contrasting the properties of the MultiWOZ corpus with those of contemporary task-oriented dialogue system outputs, which are now very often trained and evaluated on this benchmark.

At the level of participants and interactional structure, MultiWOZ dialogues are presented as duologues but may in fact be produced by several successive pairs of participants. Each new participant is required to familiarise themselves with

Table 3.5: Description of the dialogical setting in MultiWOZ resources and in TODS outputs.

Meta-Parameters	Sub-parameters	MultiWOZ	LLM-based TODs
Agents			
Agency		Human	Human-Machine
Participation structures	Participants	2	
	Workers	≥ 2	2
	Functions	Speaker - Addressee	
Social dynamics	Hierarchy	Information seeker - holder	User - Tool
	(Conversational) History	✗	$\sim \varnothing$
*Enunciative status**	Anonymity	Anonymous	Pseudonymous
	Role	Fake info. seeker-holder	Self
	Persona	Self - Self as fake expert	Self
	Commitment level	✓	✗
Conversational setting			
Epistemic grounding	World setting	Hypothetical	Real
	*World stability**	Fixed	$\sim$Fixed
Environment & Medium	Physical context	$\varnothing$	
	Gaze	$\varnothing$	
	Gesture / Facial expression	ASCII emojis	Emojis
	Prosody / Tone	$\varnothing$	
	Noise	$\varnothing$	
	Synchrony	Semi-synchronous	
	Ephemerality	Saved for the task	Persistent by design

Table 3.5: Description of the dialogical setting in MultiWOZ resources and in TODS outputs (cont.).

Meta-Parameters	Sub-parameters	MultiWOZ	LLM-based TODs
Conversational frame			
Language	Global	English	
	Control	✓	∼
Purpose	Practical goal	Task + Information exchange	
	Success condition	Questions based on slots + booking	
	*(Topical) structure**	∼plan	
Verbal interaction		∅	

the entire dialogue history before continuing the interaction. While this procedure ensures surface-level continuity, it may affect intention attribution, persona consistency, and longer-range discourse coherence. These properties make such data less suitable for modelling interactional phenomena that depend on stable participant identities or paralinguistic cues inferred from transcripts, such as fine-grained intent recognition. By contrast, deployed task-oriented systems are overwhelmingly designed for one-to-one interaction between a single human user and a machine agent. This asymmetry is not structurally necessary but reflects a dominant design choice aligned with efficiency and usability constraints.

Related tensions emerge with respect to roles, commitment, and enunciative status. In MultiWOZ, participants play roles within hypothetical scenarios, committing linguistically to speech acts despite the explicitly fictional nature of the interaction. This produces a form of commitment that is coherent within the corpus but does not straightforwardly transfer to deployed systems, where machine agents cannot sustain the same pragmatic or social commitments as humans. The original Wizard-of-Oz paradigm partially addressed this mismatch by telling the 'user' they would be interacting with a machine while the other participant was asked to play the system. However this may also introduce artefacts whose interactional consequences remain underexplored. The pretence is abandonned in the MultiWOZ corpus, making the participants closer to their real selves but producing this mismatch in possible commitments between the training and evalaution data, and the actual model output.

Hierarchy and role asymmetry further complicate the transfer from training data to deployed systems. In human-human task-oriented interaction, one participant typically seeks information or assistance, while the other provides help. In

contrast, a chatbot is primarily perceived as a tool rather than as a social peer. Training on human-human interactions may therefore induce expectations or behaviours that do not align with actual usage conditions. This mismatch affects not only surface-level phenomena such as politeness or initiative-taking, but also deeper aspects of dialogue organisation and user accommodation strategies. Similarly, although task-oriented dialogue is often conceptualised as information-seeking interaction, real usage scenarios vary considerably. Some users may already know exactly what they want and merely require execution rather than guidance. Others may expect recommendation, negotiation, or confirmation. These differences in user stance have implications for dialogue initiative, grounding strategies, and system behaviour, yet they are not always adequately represented in training data.

Epistemic grounding constitutes another source of discrepancy. In MultiWOZ, the interaction is grounded in a hypothetical yet realistic scenario: the domains are familiar, and the databases are derived from real-world information. However, the task itself is never actually performed; only its linguistic simulation is realised. By contrast, deployed systems are expected to act in the real world by executing bookings or retrieving live information. This disparity can influence dialogue structure, grounding strategies, and user expectations. It is further compounded by the fact that, while the database is fixed during training, LLM-based systems are prone to hallucination, potentially inventing entities such as hotels, restaurants, or transport options that do not exist. Although recent models have improved in this respect, hallucination cannot be fully eliminated by design and directly affects the reliability of such task-oriented dialogue systems.

Recent developments further complicate this picture. The increasing use of personalisation mechanisms such as persistent user profiles, cookies, or interaction histories, means that task-oriented systems may be treated as having some form of continuity across interactions. While such mechanisms can improve efficiency and user experience, they also raise ethical and interactional concerns. In particular, they risk fostering an illusion of familiarity or social continuity with a system that cannot sustain genuine long-term commitment. Given current tendencies for LLM-based conversational agents to be designed with anthropomorphic features (Xiao et al., 2025), these issues must be addressed explicitly when defining both training resources and system behaviour.

Finally, the environment and medium of interaction remain strongly constrained in current systems. Most task-oriented systems operate through text-based interfaces, with no access to shared physical context and no multimodal cues such as gaze, gesture, or prosody. Under these conditions, reliance on text-based dialogue data is a reasonable design choice, but it necessarily limits the interactional phenomena that can be modelled. As development will undoubtedly move toward spoken or multimodal conversational agents, additional dimensions such as turn-taking, synchrony, incremental understanding, and coordination signals will become central. Spoken dialogue systems have historically struggled with these aspects (Ad-

dlesee et al., 2020), and while recent advances have improved performance, co-construction processes and fine-grained interactional timing remain active areas of research.

Evaluation practices further reveal the limits of the current paradigm. Task success is typically assessed using slot-filling accuracy or task completion metrics. While these measures appear reasonable, they presuppose that the goal of the interaction is unambiguous and recoverable from the dialogue. In practice, however, the actual goal may not align neatly with the intent imposed by the experimental setting. In real usage, it is the user's self-determined intent that matters, making misalignment between training data, evaluation protocols, and deployment scenarios a significant issue.

More critically, dialogical quality is often evaluated using metrics such as BLEU or other semantic similarity scores. These metrics do not correspond to any dimension of CHIMERA and rest on assumptions that are fundamentally at odds with the nature of dialogue. Dialogue does not admit a single optimal next utterance against which all others can be ranked. Rather, it unfolds through negotiation, accommodation, and co-construction. While some utterances may appear incoherent in isolation, it is ultimately up to participants to accommodate, repair, or reject them. Evaluating dialogue quality through similarity to a gold sentence therefore misrepresents interactional flexibility and obscures core dialogical properties.

Although CHIMERA does not directly resolve the evaluation crisis currently faced in dialogue research, it helps make visible the gap between training resources and system outputs. As illustrated in Table 3.5, systematic differences exist between the properties of the MultiWOZ corpus and those of the dialogues generated by systems trained on it. Many of these differences stem from inherent properties of current model architectures and cannot be addressed through fine-tuning alone. A more promising direction for evaluation would therefore be to assess how such variations affect the dialogical experience of real users, rather than relying solely on abstract similarity metrics.

Overall, this scenario demonstrates that task-oriented dialogue generation is far from a solved problem, even in its most established settings. While the task goal provides a clear anchor for system design and evaluation, nearly all other parameters require explicit modelling choices. CHIMERA offers a way to make these choices visible, to align training resources with intended system behaviour, and to situate technical decisions within a broader understanding of dialogue as interaction rather than mere information exchange.

3.3.2.4 Key takeways and implications

This scenario highlights that task-oriented dialogue research still faces significant challenges across analytical and productive dimensions. Our descriptive framework proves particularly valuable not only for specifying desired system behaviour

in production tasks, but also for guiding resource selection and design choices, and for reflecting on the limitations and ethical implications of the tools and applications we develop.

Task-oriented settings are attractive because they offer greater control and reproducibility: clear goals, bounded domains, and explicit success criteria make systematic experimentation feasible. At the same time, this control imposes empirical limitations. Such settings should therefore be regarded as an initial step in understanding dialogue phenomena, rather than as fully representative of natural interaction. Yet even within constrained tasks, participants display complex behaviours – adaptive strategies, negotiation, coordination, and social signalling – which demonstrate that task-oriented dialogue can still reveal rich interactional patterns.

A critical insight emerging from this analysis is the strong interdependence between dialogue parameters. Constraints imposed by the task, the medium, or participant roles inevitably affect other aspects of interaction, including initiative, grounding, commitment, and coherence. Studying these dependencies requires careful experimental design: selectively constraining some parameters allows us to observe their ripple effects across the interaction. In this matter clinical interviews, the focus of the next section (Section 3.3.3), can be interesting to study because they allow us to study interaction features in a disfunctional setting, thus requiring adaptations from the participants. They are also a clear example of dialogue whose purpose goes way beyond a clearly defined task or information request, highlighting that conversation is rarely reducible to simple metrics or slot-based evaluations.

Overall these observations underscore the need to continuously question and adapt our practices as we work on increasingly complex tasks. Task-oriented research is not a final destination, but a structured lens that allows us to explore phenomena systematically, anticipate interdependencies, and ground technical and ethical decisions in a broader understanding of dialogue as dynamic social interaction.

3.3.3 Scenario 3: Clinical interview

The clinical interview designates a specific form of institutional dialogue that emerged alongside the development of clinical medicine and, later, clinical psychology and psychiatry. The term clinical derives from the Greek *klinikos* that relates to the practice of medicine at the patient's bedside, emphasising direct observation and situated interaction. Within psychology and psychiatry, the clinical interview became a central methodological and therapeutic instrument: a structured yet flexible conversational exchange aimed at understanding and alleviating psychological suffering.

A clinical interview typically involves a dyadic interaction between a therapist and a patient. Its overarching purpose is neither simple information exchange nor

the completion of a practical task, but the exploration and transformation of the patient's experience. The goal is curative in a broad sense: to relieve distress, clarify difficulties, or support psychological change through attentive listening, interpretation, and guidance.

This interaction is intrinsically asymmetrical. The therapist occupies a professional role defined by expertise, responsibility, and ethical obligation, while the patient is invited – sometimes with difficulty – to articulate personal experiences, vulnerabilities, and sources of suffering. Unlike ordinary reciprocal conversation, the therapist is not expected to disclose comparable personal material. The asymmetry is functional: it structures the interaction so that one participant's inner world becomes the focal point of sustained attention.

Clinical interviews are also sensitive interactions. They require patients to verbalise experiences that may be painful, confusing, or only partially accessible to consciousness. Expression may be hesitant, fragmented, or indirect. Patients may struggle to find words, may contradict themselves, or may avoid certain topics altogether. In this sense, clinical dialogue goes beyond the 'basic' cooperation assumed in ordinary conversation. It relies on the therapist's ability to interpret implicit meanings, attend to hesitations and reformulations, and integrate verbal and non-verbal cues into a coherent understanding of the patient's situation.

Although clinical interviews can be described as goal-oriented as there is an intention to improve the patient's condition, they are orthogonal to task-oriented dialogue in an important sense. Their purpose is not practical execution but transformation of subjective experience. They also provide a particularly rich site for examining the limits of language: in many clinical contexts, communicative abilities may be affected by psychological or neurological conditions, forcing both participants to adapt their interactional strategies.

In what follows we examine clinical interviews through the three complementary perspectives we have explored in the previous scenarios. Firstly the analytical perspective enables us to shed light on what this type of data can teach us about language use and yet how complex it is to work with. Secondly a produtive perspective leads us to question recent trends of building psychological counselling models, which seek to automate a complex type of interaction that goes much further than words. Finally we discuss tools that can be seen as a bridge between those two perspectives where they are meant to support therapists and clinicians in their analysis. Language is very complex as we said and NLP/CL approaches can help highlight some features of particular importance in a clinical monitoring, thus helping them focus on particular aspects they may not have otherwise noticed or they might have had an intuition about but the more systematic study NLP tools allows would further validate.

3.3.3.1 Analytical perspective: studying dialogue phenomena through the lens of dysfunctions

Clinical interviews occupy an intermediate position between ordinary conversation and institutional discourse. On the one hand, they resemble everyday dialogue in that topics are often introduced by the patient and may evolve in relatively open-ended ways. The therapist encourages narrative development, association, and reflection rather than strictly directing the exchange toward predefined informational goals. The interaction can appear relatively unstructured, allowing associations, hesitations, and reformulations. On the other hand, the interaction is clearly institutional: participant roles are asymmetrical, expectations normatively defined, and the encounter embedded in a professional, ethical, and sometimes medical framework.

From an interactional perspective, clinical interviews provide a privileged setting for studying meaning-making beyond literal content. Patients may express distress indirectly, through metaphor, silence, laughter, or narrative fragmentation. They may require several attempts before approximating what they really want to say, and they may never fully achieve a formulation they consider satisfactory. The therapist's task is to integrate these partial cues into an evolving interpretation, while remaining aware that convergence between representations can never be guaranteed. Although all dialogue involves alignment and accommodation, clinical interviews intensify this process: therapeutic progress depends on the therapist's capacity to approximate the patient's experiential world as closely as possible without presuming full access to it.

Clinical interviews are also particularly informative when communicative capacities are altered. Certain mental disorders affect coherence, turn-taking, referential clarity, or pragmatic appropriateness. In such cases, participants must develop compensatory strategies to maintain mutual understanding. Studying these adaptations can help reveal the interdependence of dialogue parameters: when one dimension of interaction is weakened, others are reconfigured. Clinical dialogue thus allows us to observe how dialogue functions under strain and which mechanisms prove most resilient.

This analytical richness, however, comes with methodological and ethical challenges. Clinical interviews are sensitive and confidential by nature. Access to such data is constrained by ethical requirements designed to protect patients' privacy and well-being. Recruitment may be difficult, attendance inconsistent, and longitudinal data collection fragile. Ethical review processes are essential but may impose constraints that complicate experimental control. Furthermore, methodological issues arise when attempting to compare clinical populations with control groups: certain structural aspects of clinical interviews, such as the therapeutic asymmetry or the emotional stakes, are difficult – if not impossible – to replicate authentically in non-clinical settings.

The SLAM (Amblard, 2018) corpus illustrates these tensions. One of its main objectives was to collect spontaneous clinical interviews between a therapist and patients with schizophrenia, while also gathering dialogues from a control group to highlight distinctive dialogic signals. The therapist's role was primarily to sustain the interaction rather than strongly guide it.

Yet aligning the two settings proved challenging in practice. Table 3.6 details some of the similarities and differences between the patient and control conditions, focusing on the parameters of CHIMERA that are most directly related to the dialogical configuration.

At first glance, the participant structure appears symmetrical. In both groups, interactions are dyadic, involving one therapist and one interlocutor. The therapist rarely occupies the role of main speaker as their function is to support and sustain the dialogue rather than dominate it. When the therapist does take on a more active verbal role, this tends to be a response to individual reluctance or conversational breakdown rather than a group-specific characteristic. The enunciative status is also comparable: participants know who they are speaking to and are not instructed to adopt fictional roles. In that sense, both settings involve authentic self-presentation rather than role-play.

However, more subtle dimensions are harder to control. The persona displayed by participants is not experimentally fixed, yet it is shaped by the institutional asymmetry inherent in clinical interviews. The therapist occupies a position of professional authority, and this asymmetry may induce desirability biases, particularly in the control group. Control participants, aware of being observed in a quasi-clinical setting, may orient their discourse toward socially acceptable or consensual topics. Patients with schizophrenia, by contrast, may be less influenced by such strategic self-presentation, partly due to impairments in theory of mind or altered social-cognitive processing. Although the asymmetrical structure is formally identical across groups, its interpretation and its interactional consequences may differ significantly.

In theory, some of these biases could be mitigated through double-blind procedures, for example, by ensuring that the therapist does not know whether their interlocutor belongs to the patient or control group, and that participants are unaware of the specific hypotheses under investigation. In practice, however, such designs are difficult to implement. The sensitive nature of clinical encounters, combined with the ethical requirement of fully informed consent – especially when working with vulnerable populations – limits the feasibility of masking group membership. Furthermore, practical constraints make true blinding unrealistic.

Indeed the physical environment provides a concrete illustration of this difficulty. Although interviews in both groups were conducted in dedicated rooms, the location differed. Patients were seen in the hospital, whereas control participants were either interviewed at the university or, in some cases, at the hospital if they were affiliated with it. The setting itself thus functioned as an implicit marker of

group membership. Standardising location would have required either bringing all participants to a neutral space outside the hospital – which would raise ethical and logistical complications for clinical patients – or inviting all control participants into the hospital environment, which presents its own institutional constraints.

In addition, the relational histories differed subtly between groups. While the therapist and their interlocutor met for the first time in both conditions, the broader institutional background was not equivalent. For patients, the therapist could be perceived as part of an existing therapeutic system – a continuation of medical or psychiatric care – thereby creating a familiar framework in which speaking about oneself is expected and legitimised. Control participants, in contrast, did not share this therapeutic context. Instead, they often shared academic or professional affiliations with the recruitment environment, particularly when recruited from a psychology department. This difference in background may influence topic selection and discursive orientation: patients more readily engage in discussions centred on internal states and personal experiences, whereas controls tend to gravitate toward more neutral, socially shared topics.

These divergences may not all play a role in the differences found in terms of verbal interaction or gaze patterns, as described in Lecomte et al. (2022), but they should be taken into account for their analysis. They also highlight the additional complexity associated with building a resource involving patients, not only in terms of institutional contraints but also due to the intrinsic asymetry between patients and controls.

3.3.3.2 Tool-building perspective: Supporting the analysis of complex resources

If we follow the pipeline from data collection to analysis followed to gather the SLAM corpus, tool-building quickly appears as a central component of the research process. The corpus does not consist solely of verbal exchanges: it combines audio and video recordings with eye-tracking data. Analysing such material therefore requires a layered technical infrastructure capable of processing multiple modalities and synchronising them with sufficient precision to preserve their interactional relevance.

One aspect concerns transcription and signal processing. Audio recordings must be converted into analysable linguistic material, which could be done with ASR systems. However, clinical dialogue presents challenges that differ from those encountered in standard conversational corpora. Speech may be disfluent, fragmented, or marked by atypical prosody. Overlapping speech, long pauses, low vocal intensity, and environmental noise – due to hospital settings or recording constraints – further complicate recognition. Importantly, many of these 'difficulties' are not incidental noise but potentially meaningful signals, especially when working with populations affected by psychiatric disorders. ASR systems must

Table 3.6: Description of the dialogical setting in the SLAM (patient and control) corpus. The asterisks indicate parameters that are formally identical across groups but whose instanciation may actually differ between the groups due to the inherent institutional asymmetry of clinical interviews.

Meta-Parameters	Sub-parameters	Patient	Control
Agents			
Agency		Human	
Participation structures	Participants	2	
	Functions	Main speaker - Main addressee as continuer	
Social dynamics	Hierarchy	Patient - Therapist*	
	(Conversational) History	First meeting + part of the therapeutic circuit	First meeting + common ground study background
*Enunciative status**	Anonymity	Physical co-presence	
	Role	Self - Self as expert	
	Persona	Self - Self as therapist	
	Commitment	✓	
Conversational setting		Dedicated room	
Environment & Medium	Physical context	Face-to-face in hospital	Face-to-face in hospital or university
Conversational frame		Clinical interview	
Purpose	Practical goal	Information exchange	
	Success condition	No pre-established set of questions	

therefore be evaluated not only for word error rate but also for their ability to preserve clinically relevant phenomena.

The second layer concerns eye-tracking data. Mapping gaze patterns onto conversational flow requires precise temporal alignment between modalities. This involves synchronising gaze coordinates with utterances, identifying fixation points within a dynamic physical environment, and distinguishing between meaningful gaze behaviour and artefacts produced by imperfect calibration or participant movement. In practice, eye-tracking setups in clinical environments are rarely optimised laboratory installations. Equipment may not always be configured by technical specialists, and data may contain substantial noise. Cleaning, aligning, and interpreting gaze data thus demands both computational tools and methodological caution.

Once multimodal data has been processed, the analytical phase introduces further challenges. Existing tools for discourse parsing, dialogue act tagging, or syntactic analysis have typically been developed and trained on more canonical corpora. Clinical dialogue, by contrast, may be less structured, more elliptical, and more context-dependent than task-oriented or scripted interaction. Topic shifts may be subtle, referential chains may remain partially implicit, coherence may unfold across extended stretches of narrative rather than through explicit markers. Off-the-shelf models may therefore fail to capture interactional dynamics that are central to therapeutic exchange.

A structural constraint further complicates tool development: clinical data is not easily shareable. Ethical regulations and privacy concerns prevent uploading recordings to external proprietary services. Privately owned LLM-based APIs cannot simply be used for transcription, annotation, or exploratory analysis. Secure, locally deployable solutions become necessary, which increases the technical burden and costs on research teams. Moreover, the size of clinical corpora is often limited. Recruitment constraints and ethical considerations restrict the amount of data that can be collected, making large-scale deep learning approaches less directly applicable. Tool-building in this domain therefore requires careful adaptation to small, sensitive datasets rather than reliance on data-hungry architectures.

Finally, the question of automation must be addressed explicitly. While LLMs and other AI tools can assist with transcription, annotation suggestions, or pattern detection, they cannot replace the interpretive work performed by trained clinicians. Clinical dialogue involves reading between the lines, integrating contextual knowledge, and recognising subtle shifts in affect. Tool-building in the clinical domain should therefore be framed as instrumentation, not substitution. Computational systems can support researchers and therapists by organising, visualising, or highlighting patterns in complex multimodal data. They cannot, and should not, be considered equivalent to the situated, responsible expertise required to interpret and respond to vulnerable interlocutors.

3.3.3.3 Productive perspective: risks and limits of automated counselling

Recent years have witnessed growing interest in developing computational systems for psychological counselling. These range from emotion-recognition tools designed to support various tasks such as opinion mining or detecting distress signals online to conversational agents explicitly intended to simulate therapeutic dialogue (Y. M. Cho et al., 2023; H. Li et al., 2023; S. Sarkar et al., 2023; Pereira et al., 2024).

The argument often advanced in favour of such systems is pragmatic: access to mental healthcare is limited, counselling is costly, and many individuals lack adequate support. From this perspective, automated systems appear as scalable supplements capable of partially alleviating structural shortages. Yet this framing raises concerns. If technological substitution is normalised as an acceptable response to systemic underinvestment in public healthcare, models may gradually be presented as adequate replacements rather than provisional aids. Such a shift risks weakening collective efforts to secure broader access to qualified human professionals.

From an interactional standpoint, counselling systems attempt to automate a form of dialogue that extends far beyond lexical exchange. Therapeutic interaction involves responsibility, ethical commitment, long-term relational work, and a capacity to interpret subtle affective and contextual cues. Although confiding in a machine may feel less intimidating to some users – particularly if it reduces fear of judgement – this apparent advantage is ambiguous.

Indeed, inherent differences exist due to the structural architecture of conversational AI as we describe in Table 3.7. First the lack of actual self for the machine makes it different in terms of spontaneity and commitment. It also aims at mimicking a therapist through its linguistic patterns thanks to a specifically chosen prompt. However human therapists are way more flexible than this as they have the ability to adapt to the patient with regards to how they frame their therapist role. Some patients may require more or less of a stereotypical counsellor and it is part of the therapist's job to adapt to it. While we can improve LLMs to make them more aware of such things, it will remain that they require some kind of prompting to restrain the type of role and persona they are supposed to mimic. Iftikhar et al. (2025) show that LLM councellors typically follow very stereotypical patterns of therapeutic dialogue which leads to 'oversimplified and irrelevant template advice'.

A further structural risk concerns the tendency of conversational AI systems to accommodate the user. Depending on the topic at hand and the user's state of mind, this could lead to the reinforcement of confirmation bias but also validate harmful beliefs or behaviours (Iftikhar et al., 2025; Cheng, Hawkins, & Jurafsky, 2026). In the opposite direction, a model that challenges the user may be dismissed precisely because it lacks human authority and accountability. In the recent years social media have seen a rise in content on mental health and psychological troubles, which is a good thing if it can help destigmatise this topic, encourage people to seek help,

or provide support and acceptance to individuals. However, a part of this content also pushes towards self-diagnosis through the description of relatable so-called symptoms of various disorders – ADHD being a particularly popular one – while the described behaviours are common ones not exclusive to a disorder (Hartnett & Cummings, 2024). For individuals already inclined toward a particular diagnostic label, seeking confirmation from a language model may amplify misconceptions.

This is particularly worrying in the current context where many of these models are developed by private companies on which we have no influence and whose motivations are not necessarily aligned with the well-being of users. The pharmaceutical industry is known to have financial ties with the psychiatric world, through contributors to the Diagnostic and Statistical Manual of Mental Disorders (DSM-5-TR) for example (Davis et al., 2024), which can influence how diagnoses are made and what drugs are prescirbed. It is not inconceivable to imagine that companies developing LLMs used for psychological counselling would also have financial interests in the pharmaceutical industry, which could lead to a conflict of interest in the advice given by the model.

Beyond these epistemic and institutional issues, the conversational configuration itself differs markedly between human and machine counselling. Traditional therapy typically takes place in the therapist's office, although online counselling has become increasingly common, particularly since the COVID-19 pandemic, and some therapists also conduct sessions outdoors or in other non-traditional environments. In all of these settings, modalities beyond linguistic content, i.e. the physical context as well as visual and auditory channels, remain much more present than in contemporary approaches to conversational AI counselling, which largely rely on text-based chat interfaces. This difference affects not only the expressive resources available to participants but also the dynamics of turn-taking, emotional attunement, and relational depth.

Privacy and ephemerality further distinguish the two contexts. In human therapy, confidentiality is generally institutionally regulated, although the specific requirements may vary across countries. While therapists may take notes, their storage, access, and use are typically governed by professional and legal standards. Interactions with conversational AI, by contrast, often involve data that is accessible to the service provider and may be subject to secondary uses. In a political climate where minority rights are increasingly vulnerable in some regions, the possibility that sensitive disclosures could be stored, analysed, or repurposed raises serious concerns.

Differences also emerge at the level of purpose and intentionality. A therapist operates within an explicit ethical framework and works toward collaboratively defined, evolving therapeutic goals. A language model, by contrast, has no intrinsic intentions and its 'success condition' is determined by training objectives and reward functions. Its representation of the world is limited to patterns inferred from data, and users can, in principle, negotiate or reshape that world within the interac-

tion.

This difference in intentions and purposes is also reflected in the way conversational history is handled. Therapy typically takes more than one session and the therapist takes notes and plans the counselling journey based on the patient's unique traits and needs. In opposition to this, conversational AI works with a context window which can be more refined than a simple dialogue history with summarisation and attention but remains fundamentally different from the structured, longitudinal planning that characterises human therapy. Setting up counselling goals is not as simple as choosing a final goal when learning a language or a new skill, a treatment plan is carefully tuned and refined through interactions where the patient themselves do not have full control over the final destination of their journey.

3.3.3.4 Key takeways and implications

Clinical dialogue highlights a fundamental limitation of purely statistical modelling: therapy is not reducible to generating plausible next sentences. It involves responsibility, accountability, and long-term relational work that cannot be easily encoded in current architectures, and should arguably not be detached of a genuine human bond.

Nevertheless NLP and CL can provide practical support in the analysis of pathological conversations. On the one hand, NLP can facilitate the preprocessing of such data through robust ASR systems or multimodal synchronisation tools. On the other hand, CL can assist in identifying particular dialogical features that may be clinically relevant, such as recurring speech patterns, gaze behaviours, and linguistic markers associated with specific conditions, or tracking thematic trajectories across sessions. However, these systems should be understood in analogy with specialised tools in medicine: they require trained professionals for interpretation and decision-making. Just as a scalpel presupposes surgical expertise, analytical tools for clinical dialogue must remain under qualified human control. Their outputs can inform clinical reflection, but they cannot replace the interpretive judgment and ethical responsibility of practitioners.

In the current political and economic climate, where public services are increasingly subject to cost-reduction pressures, the development of tools that move toward automated counselling risks being misinterpreted as a substitute for human professionals. Research in this area must therefore remain explicit about its scope, its intended use, and its limitations.

3.4 Targeted dialogue generation control

Beyond the theoretical support that CHIMERA provides for the analysis and description of dialogue data, it can also be leveraged to guide the development of dialogue systems. This section discusses two complementary strategies that could

Table 3.7: Description of the expected dialogical setting in therapist-patient conversation vs conversational system-patient.

Meta-Parameters	Sub-parameters	Therapist	Conversational AI
Agents			
Agency		Human	Human - Machine
Participation structures	Participants	2	
	Functions	Main speaker - Main addressee	
Social dynamics	Hierarchy	Patient - Therapist	Human - Machine
	(Conversational) History	Therapeutic follow-up	Token sequence
*Enunciative status**	Anonymity	Physical co-presence	∼pseudonymous - ∅
	Role	Self - Self as expert	Self - Self as expert's mimicry
	Persona	Self - Self as therapist	Self - 'Universal therapist'
	Commitment level	✓	✓- ✗
Conversational setting		Therapist's office	Chat
Epistemic grounding	World setting	Real	Training data
	*World stability**	Fixed	Negociated
Environment & Medium	Physical context	Shared	∅
	Gaze	✓	∅
	Gesture / Facial expression	✓	Emojis
	Prosody / Tone	✓	∅
	Noise	✗	∅
	Synchrony	Synchronous	Semi-synchronous
	Ephemerality	Notes	Persistent by design

Table 3.7: Description of the expected dialogical setting in therapist-patient conversation vs conversational system-patient.

Meta-Parameters	Sub-parameters	Therapist	Conversational AI
Conversational frame		Psychological counselling	'Psychological counselling'
Purpose	Practical goal	Informational	Sustain interaction
	Success condition	Change in state-of-mind	Willing to come back
	*(Topical) structure**	Clinical interview	Stereotypical clinical interview
Verbal interaction		✓	∅

support NLP research in this direction. The first, presented in Section 3.4.1, is to use CHIMERA to guide the adaptation of dialogue generation toward particular interactional configurations. The second, considered in Section 3.4.2, is to use it to inform the design of evaluation metrics that are more sensitive to specific dialogical competencies.

3.4.1 Generation control

The evolution of dialogue systems from rule-based architectures to neural networks and LLMs has completely changed the paradigm of dialogue generation. While statistical approaches have enabled impressive advances in natural language generation, with more varied outputs and less rigid structures, they have also introduced new challenges. One of the most pressing is the difficulty of controlling the generation in a reliable way. LLMs, in particular, are known for their unpredictability and sensitivity to prompts (Zhuo et al., 2024; Razavi et al., 2025), which can lead to outputs that deviate from desired interactional configurations. The unreliability of prompt-based control has motivated research into model adaptation techniques, including knowledge-enhanced generation (Yu et al., 2022; Hu et al., 2024), which aims to integrate structured information into the generation process to improve control and alignment with specific dialogue scenarios. In this perspective, CHIMERA can serve as a guide for the design of such knowledge-enhanced generation systems. In the following, we explain how it could be used to inform the application of an ontology-based approach to conversational control for LLMs, proposed by Gendron et al. (2025) (Section 3.4.1.1). We then discuss how this approach could be extended to synthetic dialogue generation, which requires to produce *all* utterances

in a conversation rather than relying on a human interlocutor to provide some of them (Section 3.4.1.2).

3.4.1.1 Dialogue agents control

An ontology is a structured representation of knowledge that defines concepts, relationships, and constraints within a specific domain. In the context of dialogue systems, an ontology can be used to formalise the concept of conversation and to provide a structured framework for controlling the generation of dialogue. Gendron et al. (2025) propose an ontology-based approach to conversational control for LLMs, which, for a given parameter to control for, leverages a decision tree trained on annotated data to guide the model's output. The idea consists in describing categories of utterances in an ontology, defined with respect to the parameter of interest, and then inferring the ontology class of the next utterance based on the previous one during generation.

In practice, they suggest to use a dataset annotated with the categories that operationalise the parameter to train a decision tree that can predict the class of an utterance based on various meta-information. For example, they illustrate this approach with the parameter of language proficiency, that they represent with the Common European Framework of Reference for Languages (CEFR) levels. The classes are thus the six CEFR levels, *A1*, *A2*, *B1*, *B2*, *C1*, and *C2*, and they are modelled thanks to a wide range of linguistic features such as the 'average word length' or the 'pronoun density' (Gendron et al., 2025, p.3). By fitting a decision tree on this annotated data, they infer quantitative rules from these linguistic features, which are then used to define the ontology classes. This decision tree can then be used to predict the ontology class of an utterance, which is wrapped in the corresponding label in the form $[o.C : C_i]\ u_j\ [o.C : C_i]$, where o is the ontology, $o.C$ is a concept from o, C_i is the ith sub-class of C in o, and u_j is the utterance from the jth data sample. These 'label-wrapped data samples' are used to fine-tune the LLM that is meant to be used as dialogue agent. A conversation strategy, which specifies how to predict the class of the next utterance based on the current one, is then used to control the generation of the dialogue agent.

This approach allows to control the generation of dialogue agents along a specific dimension, as long as a definition of the parameter can be formalised in an ontology and a dataset annotated with the corresponding classes is available. The example presented by Gendron et al. (2025), i.e. language proficiency, is very suited to this approach as it can be defined in a relatively straightforward way. However, conversation involves much more complex dynamics as we have discussed in Chapter 2. Even apparently simple parameters such as the LANGUAGE can involve complex interactional dynamics. While the method proposed by Gendron et al. (2025) can be used very directly to control the language of the generated utterances, hence participating in reducing unwanted language switches LLMs are known for due to

their not-so-universal multilingualism, accounting for code-switching within an utterance is more complex. Some languages are well-known for their tendency to involve code-switching and the code-switching versions are sometimes labelled with their own name such as Hinglish, the combination of Hindi and English. While including these languages in the ontology classes would be a possible option, it would lack flexibility when facing unplanned code-switching patterns. A more flexible approach would be to rely on weighted classes rather than unique ones, allowing to represent the probability of each language to be employed in an utterance. This would still require large-scale data collection and analysis work in order to fine-tune the model on realistic code-switching patterns (Ahn et al., 2020; Khan & Noble, 2022; Sinclair & Fernández, 2023).

This highlights the importance of carefully selecting relevant data, which is what we have been advocating for since the beginning of this work. Indeed, training the decision trees at the root of the ontology-based approach on data that looks nothing like the target scenario is unlikely to yield satisfactory results. The more theoretical and descriptive application of CHIMERA can help formulate the objectives of dialogue generation control and to select data accordingly.

Controlling dialogue generation can also help enhance synthetic dialogue production, which we discuss in the next section.

3.4.1.2 Synthetic dialogue generation control

Generating synthetic dialogue data has become increasingly common in NLP research, as it makes it possible to produce large-scale datasets at reduced cost for the experimenters[12] in a short amount of time. It also theoretically allows researchers to produce data that highlights specific phenomena of interest, which is where the ability to control the generation of dialogue can be particularly useful. However, the production of synthetic dialogues is more complex than the generation of single utterances as was the case for the dialogue agents described in Section 3.4.1.1. Indeed, the generation of a dialogue requires generating *all* utterances in a conversation, which means that each utterance is conditioned by those that precede it, which are themselves synthetic, and no human input is present to guide the process. For this reason, the dialogue strategies must be designed with a holistic view rather than a local utterance-to-utterance one.

If we focus on an apparently simple parameter from CHIMERA, such as the number of participants, it quickly becomes apparent that that it is not sufficient to indicate how many participants the generated conversation should involve. Indeed, while current synthetic dialogues tend to follow a very rigid alternation between participants, real-life conversations are much more complex and involve various patterns of turn-taking and varying levels of participation. Therefore, controlling

[12]The full cost of generative AI should not be reduced to the one paid by the experimenters, as we further discuss in the last chapter of this thesis (Epilogue).

the number of participants is not enough to ensure that the generated dialogues reflect realistic interactional dynamics. Instead, corpora matching the intended use of the synthetic data should be analysed to characterise speaker behaviour. This includes the distribution of turn lengths before speakers yield the floor, as well as levels of participation in multilogues. Indeed, not all participants are always necessarily active, and reproducing subgroup dynamics is another important aspect in synthetic data production. Another challenge posed by multilogues is that turn allocation becomes much less straighforward. Distinguishing between utterances that explicitly select the next speaker and those that leave the floor open or only restrict the set of possible next speakers is crucial. These reflections show that the number of participants cannot be considered individually but rather as one component of a net of parameters that influence each other. Again, this highlights that dialogue is far from a monolithic concept and that it must be considered globally to capture its richness.

Of course not all parameters are of equal importance depending on the use-case. Environment and Medium, though rather irrelevant in terms of scope of dialogue agents in their early years, is going to become increasingly crucial with the rise of multimodal and embodied agents, bringing new challenges in terms of control, truth grounding, and hallucination mitigation. Recent ongoing work suggests that the sycophantic tendencies of LLMs may affect the degree to which they privilege linguistic information over 'perceptual' evidence, which is a fundamental difference with human behaviour (R. Sarkar et al., 2026).

3.4.2 Evaluation metrics

Our goal when framing chimera as a way to produce evaluation metrics is not to circumvent the need for data, which we consider foundational in NLP/CL. Nor is the aim to establish a corpus or paradigm as gold standard for evaluating dialogue. Although the dimensions of our descriptive framework have been outlined and explained in terms of how different communicative settings differ from face-to-face dialogue, we certainly do not wish to imply that all theories and applications should be evaluated by comparison with face-to-face interaction. One of the fundamental principles on which we base our approach is that the scope and objectives of the task must determine the type of data involved. Consequently, dialogue evaluation metrics must be able to reflect this flexibility while remaining sufficiently descriptive to effectively characterise the dialogues being evaluated.

We envision chimera as a tool to identify relevant aspects to investigate when comparing the produced data with an evaluation corpus that displays the intended characteristics. For example, if we consider the example of the Participants and multilogue turn-taking discussed earlier (Section 3.4.1.2), one aspect to quantitatively assess is the order in which participants take turns speaking. In a three participant dialogue for instance, is the same sequence of speakers repeated again and

again or is there a form of randomness to it? A measure of sequential regularity can provide a quantitative evaluation of this aspect.

Formally, we can model each dialogue as a first-order Markov chain over the set of participants and compute the average entropy rate to quantify the degree of regularity in speaker turn-taking. For a dialogue of N utterances and K participants, the sequence of speakers can be represented by the sequence $[S_1, S_2, \cdot, S_N]$ where S_t denotes the speaker at turn t. The transition probability from speaker i to speaker j, i.e. the probability that speaker j speaks after speaker i is estimated as:

$$P_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i) = \frac{\mathbb{P}(S_t = j \wedge S_{t-1} = i)}{\mathbb{P}(S_{t-1} = i)},$$

yielding a $K \times K$ transition matrix that characterises the turn-taking dynamics. The regularity of the dialogue is then summarised by the entropy rate of the Markov chain,

$$H = -\sum_{i=1}^{K}\sum_{j=1}^{K} \pi_i P_{ij} \log P_{ij},$$

where π_i denotes the observed proportion of speaker i in the dialogue. This quantity measures the average uncertainty associated with predicting the next speaker given the current one. A perfectly deterministic turn-taking pattern, such as the cyclic sequence $A \to B \to C \to A \to B \to C \to A \to \cdots$, has an entropy rate of zero, whereas increasingly random turn-taking leads to higher entropy values. Because the entropy rate is independent of the particular identities of the speakers and depends only on the transition structure, it provides a principled and comparable measure of turn-taking regularity across different dialogue datasets.

Jo et al. (2023) introduces a multi-user version of the MultiWOZ corpus (Budzianowski et al., 2018), i.e. dialogues from MultiWOZ that have been manually re-written so that each user turn is replaced by a small chat between two users. Table 3.8 shows an example of dialogue from this corpus together with the original version.

For each dialogue, the multi-user version was rewritten by one person after a pilot study had shown the difficulty of collecting multi-user dialogues in a pairwise setting. Jo et al. (2023, p. 6) indicate that 'laypeople are not good at creating a collaborative scenario and holding a relevant dialogue in interactive settings'. However, the ability of individuals to write realistic dialogues outside an interactive setting can likewise be questioned. In particular, the freedom given to the crowdworkers regarding the number of utterances each original contribution should be transformed into and the order of speaking of the two participants can lead to the creation of patterns that may not reflect naturally occurring dialogues. Since we do not have access to a similar dataset with naturally occurring conversations, we cannot perform such comparisons. However, we can compute the turn-taking regularity measure we just introduced for the training and evaluation sets as the crowdworkers were completely disjoint. We also include the development set though the

Table 3.8: Comparison of the mono- and multi-user version of the MultiWOZ example MUL1379.

	Original		Multi-user
C	Hi, can you help me find an expensive russian restaurant?	C_2	I am in a Russke mood. You like Russian food?
		C_1	I've never tried it, but I'll willing too!
		C_2	You will love it! Please look up some expensive joints for us.
S	I am sorry I looked up your criteria and I do not see any expensive russian restaurants in my data base		
C	Are you looking in the centre area?	C_1	You sure? They don't have any.
		C_2	That sucks. Did you check everything in the Centre?
S	Yes, I apologize but no restaurants match your search.		
C	How about british food instead?	C_1	Ah well, que sera, sera. What else would you like?
		C_2	Maybe British and a spot of tea. Anything fitting?
S	I have 3 British restaurants that fit your criteria. I would personally recommend Fitzbillies Restaurant.		
C	Fitzbillies Restaurant sounds good,will you book it for me ?	C_2	Ohhh Fitzbillies Restaurant, sounds fancy. What do you think about it?
		C_1	I want to go there. Can you book us?
S	Certainly, what day and time would you like a reservation? And for how many guests?		

original paper does not indicate if the crowdworkers overlapped with the other two sets.

We compute this metric in two ways. First, by considering the three participants, i.e. the two customers $\mathbf{C}_1$ and $\mathbf{C}_2$, and the system $\mathbf{S}$, to take into account the freedom to replace one original cutomer's utterance by any number of new ones. Second, by considering only the two customers to determine how regular the alternation between them is. The results, presented in Figure 3.5, show that the entropy is very high when considering all three participants, around 0.8 on average for all three data subsets. It decreases to around 0.5 when considering only the two customers which is still a rather high value given the number of participants. This suggests that turn-taking is relatively irregular in this dataset and that the crowdworkers did not simply alternate between the two customers and the system. We can note that the training and test sets differ significantly in terms of entropy according to pairwise t-tests, showing that relying on disjoint sets of crowdworkers had an influence. However the differences are quite slim and the effect size appears modest which indicates that the difference is not very strong.

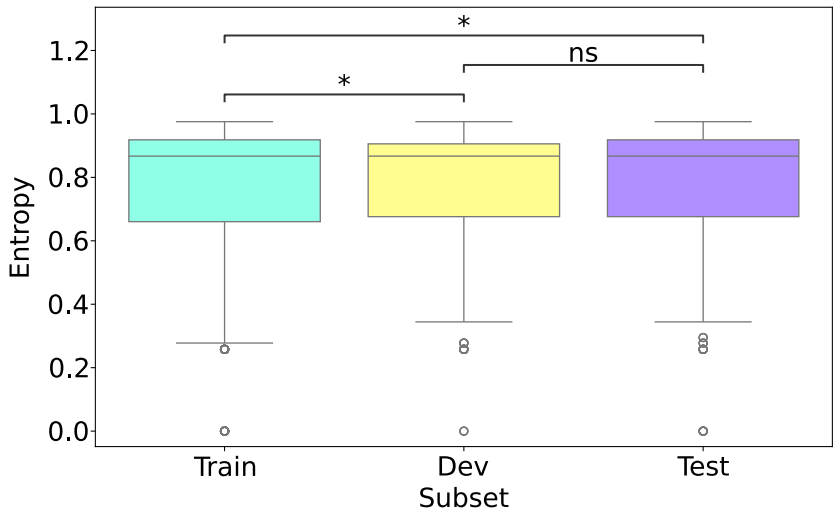
One advantage of using Markov chains to assess the regularity in turn-taking dynamics is that higher-order chains could be computed, i.e. the probability to have a given speaker at turn t would be computed based on the speakers in multiple preceding turns. Higher-order models could capture more fine-grained turn-taking dynamics, though a larger sample would be necessary to discover significant configurations.

This shows how CHIMERA can be used in a more applicative way to design metrics meant to highlight differences between datasets with respect to specific parameters. Together with its potential for identifying dimensions along which dialogue generation can be controlled, this demonstrates the practical value of CHIMERA for NLP.

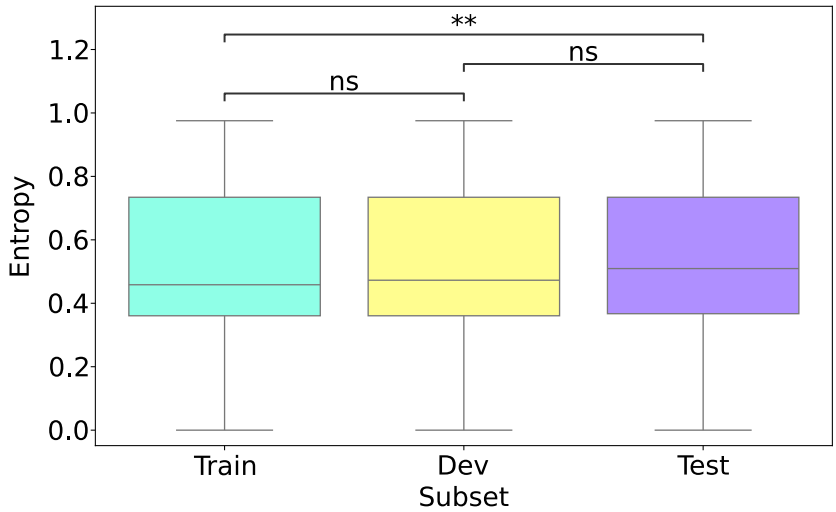
3.5 Summary and conclusion

This chapter unfolded along three complementary dimensions.

First, the survey of dialogue research published in 2024 in the ACL ANTHOLOGY offered a situated snapshot of the field. It revealed that contemporary dialogue research mostly focuses on generative systems, which is unsurprising given the rise of LLMs. Yet, their development by major technology companies in opaque ways raises fundamental questions about how we develop and evaluate conversational systems. Since reproducing LLMs at this scale is not feasible, we must seek alternative ways to regain control over our work. The solution, however, cannot be to restrict access to data or reverse the progress made toward open science. Instead, the community must focus on research methodology: clearly defining tasks, thoroughly characterising datasets, and designing meaningful, task-specific evaluation



(a) With system's utterances.



(b) Without system's utterances.

Figure 3.5: Average turn-taking entropy rate of the multi-user version of MultiWOZ.

metrics.

Second, we tried to illustrate this point by applying CHIMERA to three distinct dialogue scenarios: turn-taking, task-oriented dialogue, and clinical interviews. This comparative exercise demonstrated the analytical value of explicitly characterising dialogical configurations along multiple axes: PARTICIPANTS, LANGUAGE, ENVIRONMENT AND MEDIUM, VERBAL INTERACTION, and CONVERSATIONAL ACTIVITY AND PURPOSE. Across these scenarios, we saw that dialogue study is multifaceted and that different settings foreground different phenomena. For example, turn-taking research highlights the temporal dynamics of interaction, task-oriented dialogue focuses on goal-directed coordination, and clinical interviews reveal the interpretive complexity of meaning-making under asymmetrical conditions. This exercise showed that methodological choices shape the phenomena we can observe and the conclusions we can draw. It also highlighted the challenges of aligning different settings, particularly when working with sensitive data or vulnerable populations. Descriptive precision is therefore not a mere formal exercise: it conditions both empirical validity and interpretability.

Third, building on these observations, we proposed a way of leveraging CHIMERA to guide the improvement of dialogue generation and to propose evaluation metrics along specific dimensions. Rather than treating dialogue competence as a monolithic capability, we argued that targeted control could help align model behaviour with concrete pragmatic requirements. This perspective reframes model development not as a race toward generality, but as a process of structured calibration grounded in explicit dialogical parameters.

These three strands converge on a central claim: descriptive precision is foundational. A clear characterisation of dialogical settings enables us to (i) describe data more transparently, (ii) formalise task constraints more rigorously, and (iii) compare system outputs against articulated expectations rather than relying on general quality judgements. In doing so, CHIMERA offers a basis for developing more specific and theoretically informed evaluation metrics that move beyond surface-level similarity and toward the assessment of specific dialogical competencies.

If dialogue research is to remain both scientifically robust and socially relevant, it must continue to refine its descriptive tools while critically examining its methodological practices. CHIMERA is not intended as a definitive solution, but as a means of making explicit what too often remains implicit: the structural conditions under which dialogue emerges and the assumptions that guide its modelling. In the remainder of this thesis, we examine in greater depth one parameter of CHIMERA, namely the STRUCTURE of the conversation, and in particular its topical structure, considering how it can be identified, modelled, and used as a descriptive dimension for distinguishing between different types of dialogue.

Part II

Topics and Topical Structure: Applicative analysis across various interactional contexts

4

Topics in Discourse and Dialogue

Contents

4.1 Topics and topical structure 186
4.1.1 Topic 186
4.1.2 Topical structure 190
4.2 Studying topical structure 191
4.2.1 Analytical and predictive perspectives 193
4.2.2 Approaches to modelling topical organisation . . 195
4.3 Dialogue type and topical structure 199

The first part of this thesis was devoted to the characterisation of dialogue as an object of study. Through the examination of a wide range of theoretical perspectives, we argued that dialogue cannot be reduced to a single dimension, but should instead be understood as a multidimensional phenomenon. Different types of interaction involve different constraints on production, interpretation, coordination, and information management. To account for this diversity, we proposed CHIMERA, a descriptive framework organised around five dimensions, allowing dialogue datasets to be situated within a broader dialogue space.

A central idea of this framework is that any given dataset instantiates only a limited region of this space. We tried to define the different dimensions of it in a sufficiently operational manner – or at least to suggest possible leads towards a computable representation – to support comparisons between datasets, to make explicit the interactional phenomena they capture, and to guide the production of new tools and evaluation metrics. This perspective has important methodological implications. Since datasets only cover specific regions of dialogue space, theories and computational tools developed on a particular dataset cannot be assumed to generalise to dialogue as a whole. Their validity and applicability depend, at least in part, on the characteristics of the data on which they are built and evaluated.

The previous chapters also highlighted the fact that the dimensions of dialogue are not independent. Rather, they interact with and constrain one another. Among the phenomena likely to be affected by these interactions is topical organisation. The way topics are introduced, maintained, and shifted throughout an interaction depends on factors such as the participants involved, the goals of the interaction,

the communication medium, or the temporal organisation of the exchange. Because topical organisation is a pervasive feature of dialogue, systematic differences between dialogue types are likely to be reflected in systematic differences in their topical structure.

At the same time, topical structure presents a particular challenge for dialogue analysis and modelling. Unlike more readily observable properties such as the number of participants, turn-taking patterns, or the presence of overlaps, topics are not directly encoded in the interaction. They must instead be inferred from the unfolding conversation and reconstructed from the linguistic and interactional cues available to the analyst. Yet, they play a central role in the organisation of dialogue, constraining the contributions that participants can make and providing a basis for interpreting new interventions. This makes topical structure both a fundamental and a difficult aspect of dialogue to study.

The second part of this thesis investigates the relationship between dialogue type and topical structure. More specifically, it examines the extent to which approaches and theories traditionally developed for relatively structured forms of discourse remain applicable when confronted with more complex types of dialogue. This chapter introduces the key notions required for this investigation. We begin by discussing the concepts of topic and topical structure in Section 4.1. We then present different perspectives on the study of topical structure in Section 4.2. Finally, we outline the questions that motivate the remainder of this part and discuss the relationship between dialogue type and topical organisation in Section 4.3.

4.1 Topics and topical structure

Studying topical organisation covers two related but distinct aspects. The first concerns the topics themselves, that is, what a discourse or dialogue is about. The second concerns the way these topics are organised throughout the interaction, which we refer to as *topical structure*. While these two aspects are closely related, they raise different questions and require different analytical tools. In this section, we discuss each of them in turn.

4.1.1 Topic

The notion of *topic* is widely used in linguistics, yet its precise definition remains a matter of debate. The term has been employed to refer to several related but distinct concepts, reflecting the diversity of phenomena that can be described in terms of what a piece of language is 'about'. Broadly speaking, topics have been studied both at the sentence level (Bolinger, 1952; Firbas, 1964; M. A. Halliday, 1967; Givón, 1983) and at the discourse level (Keenan & Schieffelin, 1976; G. Brown & Yule, 1983; Asher, 2004).

At the sentence level, a *sentence topic* is typically defined as the entity or proposition about which the sentence provides information (Hockett, 1958). In many cases, this corresponds to a noun phrase occupying a privileged syntactic position. For instance in the sentence 'Bob loves dragons.', the noun phrase 'Bob' is the topic, as it is the entity about which the sentence conveys information. While this notion is useful for describing information structure within individual sentences, it does not readily extend to phenomena that emerge beyond sentence boundaries. If 'Bob loves dragons.' is one sentence among others in an enumeration of Bob's interests, then we could say that Bob is indeed the topic of the piece of discourse, or more precisely that Bob's interests are. However, if the enumeration relates to the interests of a group of people, then the sentence topic 'Bob' will not correspond to the global topic. In conversation, the notion of sentence topic is further complicated by the fact that participants may not always explicitly mark the topic of their contributions. For instance, the topic of an intervention may be established through gestures, gaze, or shared situational context rather than through linguistic material alone. More importantly, sentence-level approaches do not account for the interactive and incremental nature of topic development in conversation, where each participant can add to the context and thus contribute to the evolution of the topic.

For the purposes of this thesis, we are primarily concerned with *discourse topics*. These are generally understood as what a stretch of discourse or interaction is *about*, although the notion of *aboutness* itself remains difficult to formalise. Informal accounts often relate topics to participants' intuitions about the current focus of the conversation. This perspective is compatible with Grice's Maxim of Relation (Grice, 1975), according to which contributions should be relevant to the ongoing interaction. Remaining on topic can be viewed as one way of maintaining relevance, while topic shifts are interpreted under the assumption that participants continue to cooperate and contribute meaningfully to the conversation. Similar intuitions can be found in Conversation Analysis, where topics are not treated as a fixed subject matter but as an interactionally accomplished object of discourse that participants themselves establish, maintain, transform, and close through the sequential organisation of talk (Button & Casey, 1985).

Despite their intuitive appeal, such definitions have been criticised for their limited operationalisability. The complexity of conversational interaction led G. Brown and Yule (1983, p.68) to 'suggest that formal attempts to identify topics are doomed to failure'. Nevertheless, these authors also acknowledge the usefulness of notions such as 'speaking topically' or 'the speaker's topic', suggesting that topics, even if difficult to define rigorously, play an important role in the organisation and planning of discourse.

More formal accounts have attempted to characterise topics based on the notion of 'question of immediate concern', i.e. the question or set of questions an utterance is a response to. The discourse topic is then defined as the proposition

or set of propositions presupposed by this question (Keenan & Schieffelin, 1976). For instance, the question of immediate concern of this current section could be defined as 'What is a topic?', and the discourse topic would be the proposition 'Topics exist.'.

These approaches provide the foundations for Question Under Discussion theories (QUD; Van Kuppevelt, 1995; Ginzburg, 1995; Roberts, 2012), which model discourse as a structured set of questions and answers. Within this framework, each contribution either answers an existing question, introduces a sub-question, or redirects the interaction towards a different issue. Topics can therefore be viewed as the questions currently under discussion and whose resolution contributes to maintaining discourse coherence.

However, QUD-based approaches do not fully resolve the difficulties associated with topic identification. Questions of immediate concern are often implicit and may not be directly recoverable from the linguistic material alone. Like the topics themselves, they must frequently be inferred by analysts. Empirical studies support this observation: when annotators are asked to reconstruct QUDs from interview data, they achieve only moderate agreement (Mulligan & Rawlins, 2025). This suggests that multiple formulations may adequately capture the same conversational concern and that identifying a unique underlying question is not always straightforward. As already noted by G. Brown and Yule (1983, p. 72), evaluating alternative descriptions of a topic would itself require explicit criteria whose formulation may depend on the type of interaction under consideration.

These difficulties become particularly apparent in less information-seeking oriented forms of interaction. In casual conversation or small talk, topics may shift rapidly and in seemingly unexpected directions. While identifying a single question of immediate concern in such interactions can be challenging, this does not imply that they are incoherent. On the contrary, Schneider (1988) argues that topics remain an important organising principle even in phatic communication. Thus, despite numerous attempts at formalisation, discourse topics remain to some extent intuitive constructs (T. A. Van Dijk, 1977).

For the present work, we adopt a broad understanding of topics as organising units that contribute to discourse and dialogue coherence. The topic under discussion constrains the range of contributions that participants can reasonably make and provides a basis for interpreting new interventions. In written discourse, where opportunities for clarification are limited, topical organisation is often expected to be explicitly reflected in the structure of the text. Dialogue is more flexible in this respect, as participants can negotiate meaning and repair misunderstandings as they arise. Nevertheless, speakers generally orient towards what the interaction is currently about and attempt to produce contributions that are relevant to that evolving context (Schegloff & Sacks, 1973; Grice, 1975).

Unlike the topic of a written text, the topic of a dialogue is rarely fixed. Multiple topics may be discussed over the course of an interaction, their boundaries are

often difficult to define precisely, and their development depends on the contributions of all participants as well as on the surrounding context. Topics in dialogue should therefore be understood as dynamic, incremental, and jointly constructed phenomena.

This dynamic character also reflects a broader difference in how coherence is conceived across language uses. In written discourse, coherence is often understood as the property of a text forming a meaningful whole (J. M. Williams & Nadel, 1989). As W. S. Cho et al. (2019) observe, the different parts of a coherent text fit together 'the way all the pieces in a puzzle add up to the picture on the box'. In dialogue, by contrast, coherence emerges through interaction and refers to the participants' ability to sustain a mutually interpretable course of action (Schiffrin, 1985; Anderson, 1995). Lautamatti (1982) describes this duality as a distinction between *propositional* and *interactional* coherence. The former is concerned with information organisation and the development of semantic content, while the latter is concerned with the illocutionary force of the utterances and how they contribute to the ongoing interaction. She argues that both types of coherence are present in any type of discourse but that they are more or less prominent and easily observable depending on the level of formality. She describes the features of language use that influence the prominence of each type of coherence, which can be mapped with CHIMERA. In particular she includes the shared context, which we capture in ENVIRONMENT AND MEDIUM, the possibility for immediate feedback, whether repairs are a part of the discourse itself or can be edited out of it, and the time allowed for advanced planning, which together form our (VERBAL) INTERACTION category. She also mentions the shared knowledge that can be assumed which corresponds to the SOCIAL DYNAMICS entry of our PARTICIPANTS category. The most extreme types of language use along this cline would be casual conversations for the most interactional and least formal side, and scientific texts or novels for the more propositional and formal side. Our interests in this thesis lie closer to the interactional end of the spectrum, where the dynamic and incremental nature of topical organisation is more apparent, and the propositional coherence is much harder to observe from an external perspective. Lautamatti (1982, p.124) emphasises that, in such cases, participants rely heavily on a shared context and background to formulate and interpret their contributions, which 'makes it difficult for anybody outside the speech event to analyse the propositional coherence or to follow the development of the discourse topics'. This will be a central concern of this second part, in which we shall attempt to abstract away from the internal construction of topics and focus instead on more observable characteristics of the interaction in order to study topical organisation in dialogue.

4.1.2 Topical structure

Because topics themselves are difficult to observe directly, topic shifts often constitute a more accessible object of analysis (Howe, 1991). Several types of markers and heuristics have been proposed to identify topic shifts, including discourse markers such as 'by the way' in English, long pauses, or changes in the vocabulary. We discuss the technical aspects in more detail in Section 4.2.2. Identifying the different topics present in a text or interaction makes it possible to divide the discourse into relatively coherent stretches devoted to separate concerns. This is particularly useful for tasks such as information retrieval as it allows to focus on the most relevant parts of a discourse. It also provides a basis for studying the relations between topics and the mechanisms through which participants move from one topic to another.

The most common approach to topic segmentation is linear, where the discourse is divided into a sequence of contiguous segments corresponding to successive topics. Figure 4.1 illustrates the topic segmentation of a text that summarises the rise and fall of an hypothetical application called 'OintMind' that promises to detect mental health issues based on the analysis of the user's speech (Choi et al., 2024). The text is formatted as a short news article, making it similar to the types of text often used in topic segmentation studies. The text is represented as a sequence of paragraphs in Figure 4.1a, and segmented linearly in Figure 4.1b. Each segment is represented by a different colour spanning one or more paragraphs. This approach is quite basic and does not provide information about the content of the topics but it can be completed by labelling the segments as in Figure 4.1c. In this segmentation, all the topics are represented at the same level and no relations between them are made explicit.

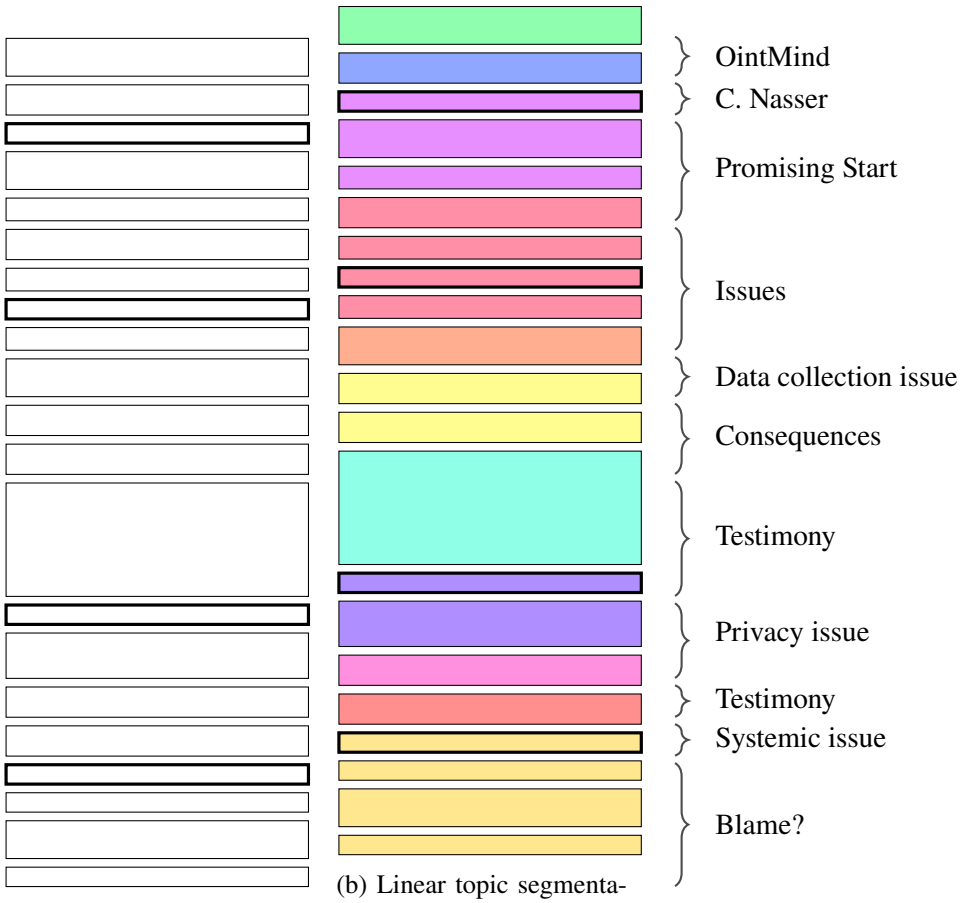
Yet the tension between the notions of sentence topic and discourse topic discussed in Section 4.1.1 makes apparent the idea of granularity in topical organisation, where a piece of text or conversation can be divided in very high-level topics that can be further divided in more specific topics and subtopics. Within this perspective, Van Kuppevelt (1995) defines discourse topics as the highest level of topical organisation, with topics and subtopics forming lower levels of the *hierarchy*. Figure 4.1d proposes a hierarchical topic segmentation of the same example we used for linear segmentation. We represent the structure as a tree where the root node corresponds to the main topic and each branch leads to a more specific subtopic. This representation overlooks the actual relations between the topics and subtopics which can be more complex than a simple hierarchy. For example, the two subtopics 'Testimony' are illustrations of the 'data collection' and 'privacy issue' respectively, while there is a logical relation of implication between the subtopic 'Data collection issue' and its subtopic 'Consequences'. Formalisms such as Rhetorical Structure Theory (Mann & Thompson, 1988) or Segmented Discourse Representation Theory (Asher & Lascarides, 2003), typically used at the

sentence level, can be extended to model relations between topics and subtopics at the discourse level.

Focusing on dialogue, Van Kuppevelt (1995) states that task-oriented or *bound conversations* are typically organised around a single discourse topic, whereas less constrained interactions may involve several discourse topics. In this framework, shifts between discourse topics are often treated as structurally incoherent because they move the interaction outside the current organisational structure. This view, however, raises questions when applied to more open-domain dialogues. While abrupt topic shifts may indeed be perceived as disruptive in highly structured interactions, they are commonplace in many forms of everyday conversation. Participants frequently introduce new topics, return to previous ones, or connect apparently unrelated subjects through interactional work. Some topic shifts can also be more gradual, where two parts of the dialogue concern different topics but the transition between them is not clearly marked. In such cases, the notion of a topical boundary can be intrinsically fuzzy, and attempting to impose a strict hierarchical structure may not accurately reflect the dynamics of the interaction. Overall in conversation, what matters is often not the existence of a topic shift itself, but whether participants can successfully interpret and accommodate it. The ability to identify relevant places for topic shifts and to manage the transition between topics is therefore a crucial aspect of conversational competence. From this perspective, topical coherence emerges not only from structural relations between topics, but also from the participants' ability to negotiate and maintain mutual understanding throughout the interaction. One of the aims of this second part is to examine the limitations of existing approaches to topic segmentation when confronted with complex dialogical data.

4.2 Studying topical structure

As discussed in Chapter 3, CL and NLP produce an epistemic duality between analytical and predictive works. The former aims to describe and explain phenomena, producing theories and models that account for the observed data. The latter aims to anticipate future developments, producing models that can predict what will happen next. These two perspectives are complementary: analytical accounts provide hypotheses about how topics are introduced, maintained, and shifted, whereas predictive models support applications such as dialogue systems but also offer a way of testing whether these hypotheses capture regularities that are sufficiently systematic to support prediction. In this section we first discuss the different perspectives (Section 4.2.1) and computational approaches (Section 4.2.2) to studying topical structure, with a particular focus on the challenges posed by dialogue data compared to written monologues.

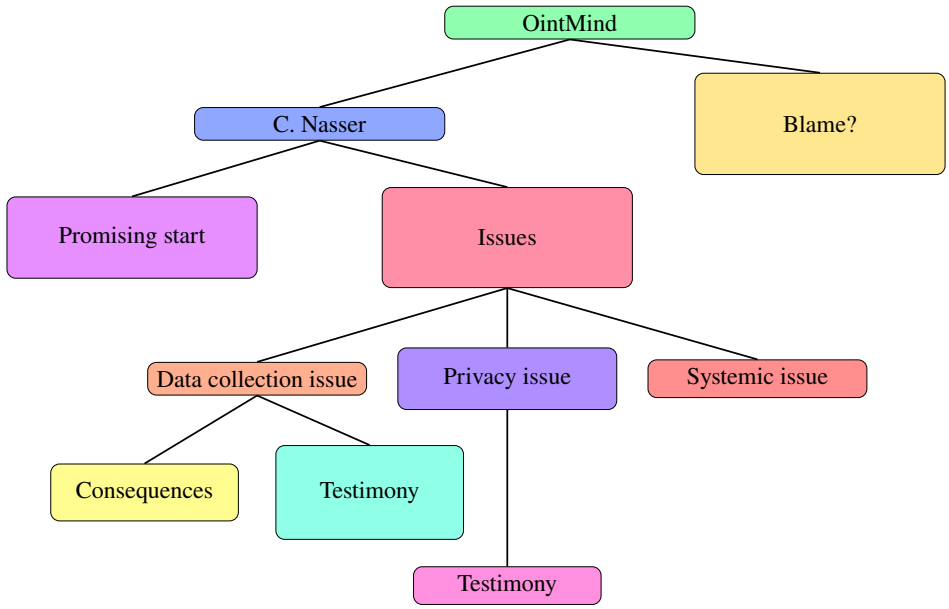


(a) Original text.

(b) Linear topic segmentation.

(c) Topic labelling.

Figure 4.1: Representation of the text 'OintMind, the Mental Health Scandal' (Choi et al., 2024) segmented in topics. The text is represented as a sequence of paragraphs, with titles marked in bold.



(d) Hierarchical topic segmentation.

Figure 4.1: Representation of the text 'OintMind, the Mental Health Scandal' (Choi et al., 2024) segmented in topics (continued).

4.2.1 Analytical and predictive perspectives

The study of topical organisation can be approached from two complementary perspectives. The first is *analytical*: it seeks to describe and explain how topical structures emerge in naturally occurring discourse and dialogue. The second is *predictive*: it seeks to model topical organisation in a way that makes it possible to anticipate future developments in the interaction. While these perspectives differ in their objectives, they are closely related. Analytical accounts provide hypotheses about how topics are introduced, maintained, and shifted, whereas predictive models offer a way of testing whether these hypotheses capture regularities that are sufficiently systematic to support prediction.

From an analytical perspective, topical organisation is studied *a posteriori*, on already produced discourse. The goal is to identify the mechanisms that structure a conversation and to characterise the relations between its different topics. Several computational tasks contribute to this objective.

One of the most prominent is *topic segmentation*, which consists in identifying boundaries between topics within a document or dialogue. By dividing a discourse into coherent topical units, topic segmentation makes it possible to study how information is organised and how topics evolve over time. As discussed in Section 4.1, segmentation can be approached either linearly, by identifying suc-

cessive topical segments, or hierarchically, by modelling relations between topics and subtopics. Linear segmentation remains the most common approach because it is comparatively easy to annotate and evaluate, and because it is often sufficient for highly structured data. However, more complex forms of dialogue, particularly open-ended conversations, frequently exhibit gradual topic shifts and intertwined topical trajectories that challenge strictly linear representations.

A related task is *topic labelling*, which aims to assign descriptive labels to identified segments. Topic labels provide a compact representation of a segment's content and facilitate navigation, retrieval, and analysis. Determining an appropriate level of granularity remains challenging, as different applications require different degrees of specificity. Existing works suggest that annotators tend to agree more readily on coarse-grained descriptions than on fine-grained ones (Banerjee & Rudnicky, 2007), highlighting the inherently subjective nature of topical interpretation.

The outputs of segmentation and labelling can themselves be used to support downstream applications such as information retrieval (Shtekh et al., 2018) but also more epistemic tasks. Indeed, the resulting segmentations and labels can be analysed and annotated with a rhetorical or discourse structure to identify the underlying principles of topical organisation. Such analyses provide a better understanding of typical patterns of topical organisation, as well as the mechanisms used by participants to manage changes of topic and maintain coherence. They can also be used to identify disfluencies or breakdowns in topical organisation, which may indicate cognitive or communicative difficulties.

The predictive perspective approaches topical organisation from a different angle. Rather than reconstructing the structure after the interaction has taken place, it attempts to anticipate how that structure will develop. In computational applications, this perspective is particularly relevant for conversational agents and dialogue systems, which would benefit from the ability to predict upcoming topics, anticipate topic shifts, and maintain coherent interactions over time. Predictive models can therefore be used to generate more appropriate responses and improve dialogue management in these systems.

Beyond their practical applications, predictive models also play an important methodological role. If a theory of topical organisation claims that certain conversational moves constrain the range of possible continuations, then these constraints should improve our ability to predict what happens next. Predictive performance can therefore serve as an indirect way of evaluating analytical accounts. Conversely, prediction failures may reveal limitations in existing theories and motivate their refinement.

QUD frameworks illustrate particularly well the complementarity of these two perspectives. They are often used analytically to reconstruct the coherence structure of a dialogue by identifying the questions that organise the interaction. At the same time, because active QUDs constrain the range of relevant future contributions, they can also be used predictively in the form of potential questions (Onea,

2016) to anticipate upcoming moves. If such predictions can be consistently validated by empirical data, QUDs can be considered as a cognitively relevant construct used when producing and interpreting discourse.

Beyond the distinction between analytical and predictive perspective, the approach used to tackle them is also important. In particular, the choice between symbolic and statistical methods, as well as supervised and unsupervised techniques, has implications for the explainability of the resulting model. In this thesis, our goal is to develop a better understanding of topical organisation in dialogue, which requires both to investigate topics in conversations a posteriori, and to determine whether they can be predicted throughout the interaction in naturalistic settings. Identifying topically coherent spans of discourse or dialogue is therefore a first logical step towards this goal, as it provides a basis for analysing topical structure and thus also for imagining predictive models of topical organisation. In the following subsection, we therefore review the main computational approaches that have been proposed to segment discourse into topically coherent units.

4.2.2 Approaches to modelling topical organisation

The computational study of topical organisation has produced a wide range of methods for identifying topically coherent segments of discourse. Although these approaches differ in their technical implementation, they generally rely on the assumption that topical organisation leaves observable traces in discourse. Topic shifts are therefore treated as detectable events, whether through changes in lexical content, latent semantic representations, discourse structure, or interactional dynamics.

Existing approaches can broadly be divided into unsupervised methods, which infer topical organisation from regularities in the data, and supervised methods, which learn topical boundaries from annotated examples. Unsupervised techniques typically rely on lexical cohesion (Hearst, 1997; Galley et al., 2003) or automatically learned latent topical distributions (Eisenstein & Barzilay, 2008; Riedl & Biemann, 2012; Du et al., 2013), whereas supervised approaches formulate topic segmentation as a classification task (Arguello & Rosé, 2006; Takanobu et al., 2018). However, the effectiveness of these methods depends heavily on the nature of the language data under consideration, as we shall see in Chapters 5 and 7. The assumptions that work well for written discourse do not necessarily hold for spontaneous conversation. Identifying and understanding these assumptions is therefore essential for assessing the applicability of existing approaches to different dialogue types.

4.2.2.1 Written text and discourse

Topical organisation is comparatively easier to identify in written discourse because texts often provide explicit structural cues. Chapters, section headings, paragraphs,

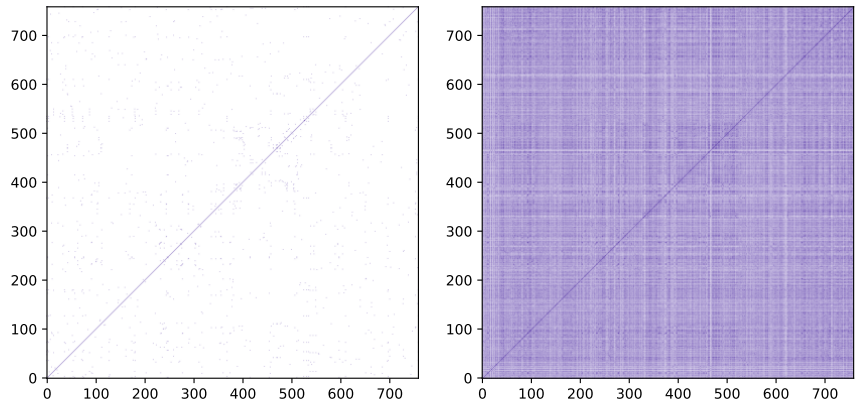
and typographical conventions guide the reader through the progression of ideas. Discourse markers may also signal transitions between topics. Since readers cannot negotiate meaning with the author, written discourse generally relies on clear organisational principles to ensure coherence.

Many computational approaches exploit these regularities. Content-based methods identify topic boundaries by measuring lexical cohesion between adjacent segments. Their underlying assumption is that vocabulary remains relatively stable within a topic and changes when a new topic is introduced (M. A. K. Halliday & Hasan, 1976). TextTiling (Hearst, 1997) is perhaps the best-known example of this family of methods. It compares lexical distributions across sliding windows and identifies topic boundaries at points where similarity sharply decreases. Topic segmentation has also been formulated as a clustering problem, where semantically similar text segments are grouped together. Dotplot approaches (Church & Helfman, 1993; Reynar, 1994), for instance, construct self-similarity matrices that reveal coherent blocks of discourse through characteristic visual patterns. Figure 4.2 illustrates two dotplot representations of the text 'OintMind, the Mental Health Scandal' from Choi et al. (2024) whose topic segmentation was previously depicted in Figure 4.1. After a pre-processing step where the text is tokenised, only interjections, nouns, verbs, numerals, and adverbs are kept in their lemmatised form. The resulting sequence of tokens is then compared to itself to produce a square matrix where each axis corresponds to the position of a token in the text. Each point in the matrix represents a similarity value between two tokens, with darker points indicating higher similarity. In Figure 4.2a, similarity is a binary value, depending on whether two tokens share the same lemma, while in Figure 4.2b, it is computed using cosine similarity between their embeddings (based on the SpaCy model `en_core_web_lg`). On the first representation (Figure 4.2a), a few faint squares can be seen along the diagonal between tokens 400 and 600 but no striking patterns appear. On the second representation (Figure 4.2b), the almost uniform distribution of points outside the diagonal means that the vocabulary is not sufficiently specific to each topic to clearly isolate them.

The first implementation of this approach obtained clear separations when applied to the concatenation of four *Wall Street Journal* articles, showing that a basic self-comparison can reveal distinctions at a very coarse level. Nevertheless, (Malioutov & Barzilay, 2006) obtained better results when segmenting individual spoken lectures by using a more sophisticated similarity metrics at the sentence level.

Other approaches model topical organisation more indirectly. Generative models assume that texts are produced from sequences of latent topics and attempt to infer these hidden structures from the observed words. Early implementations relied on Hidden Markov Models and topic-specific language models to estimate the most probable segmentation of a document.

Topic segmentation has also been treated as a classification task. Rather than



(a) Similarity metric: same lemma.

(b) Similarity metric: embeddings cosine similarity.

Figure 4.2: Dotplot representation of the text 'OintMind, the Mental Health Scandal' from Choi et al. (2024). Each point represents a similarity between two tokens.

modelling topics directly, these approaches predict whether a given position constitutes a topic boundary. Features used to do so may include lexical similarity, discourse markers, paragraph structure, or sentence position. Machine-learning models range from traditional classifiers such as Support Vector Machines (Cortes & Vapnik, 1995; Georgescul et al., 2006) to transformer-based architectures fine-tuned on datasets that include topical information (Xing & Carenini, 2021). The most recent approaches mostly rely on LLMs and treat topic segmentation as a generation task, where the model is prompted to produce a segmentation of the input text (Fan, Jiang, et al., 2024; Maraj & Martin, 2026). While these methods can integrate a wider range of signals, they require annotated training data and often sacrifice interpretability.

Spoken monologue introduces additional complexity. As discussed in Chapter 2, speech contains information that is absent from written text, including prosody, rhythm, pauses, and, in multimodal settings, gestures and facial expressions. These cues often contribute to the signalling of topic transitions and may compensate for linguistic markers that would otherwise be required in written discourse. Despite these additional challenges, spoken monologue generally remains more structured than conversation, and current methods achieve relatively strong performance on high-resource languages (Purver, 2011; Ghinassi et al., 2024b).

4.2.2.2 Dialogue

Dialogue introduces a qualitatively different challenge because topics are no longer organised by a single producer. Instead, they emerge through the interaction be-

tween participants. Topic management becomes a collaborative activity in which speakers jointly negotiate what the conversation is about, whether a shift is acceptable, and how new contributions relate to previous ones.

As a result, topical organisation in dialogue depends on factors that extend beyond linguistic content. Participants draw on common ground, social roles, shared histories, and situational goals when evaluating the relevance of a contribution. A seemingly abrupt topic shift may be entirely coherent when interpreted against shared knowledge or an implicit objective. Conversely, contributions that appear topically related at the lexical level may be interactionally inappropriate. The relationship between topical coherence and conversational success is therefore considerably more complex than in monologic discourse.

Furthermore, dialogue lacks many of the structural markers available in written texts. There are no paragraphs to indicate boundaries, utterance segmentation is itself often ambiguous, and discourse markers may fulfil multiple interactional functions. Topic shifts can also occur gradually, making boundaries difficult to localise precisely.

These characteristics pose significant challenges for computational models. Early dialogue segmentation systems, such as LCSeg (Galley et al., 2003), adapted discourse-based methods by relying on lexical chains and cohesion measures. While effective in relatively structured settings, such approaches struggle to capture the pragmatic and interactional dimensions of topical organisation.

The development of distributional semantics and machine-learning techniques has enabled more sophisticated representations of topical relatedness. For example, Xing and Carenini (2021) exploit adjacency pairs (Schegloff & Sacks, 1973) to learn an unsupervised similarity metric before applying a TextTiling-style segmentation procedure. Their approach performs well on task-oriented datasets such as Stanford Dialog Dataset (Eric et al., 2017) or MultiWOZ (Budzianowski et al., 2018), and highly structured ones such as Doc2Dial (S. Feng et al., 2020). However, similar methods perform substantially worse on open-ended multiparty conversations, where topical organisation is less constrained and lexical cohesion provides a weaker signal (Decker & Amblard, 2024a).

As with written text, recent work increasingly relies on LLMs, framing topic segmentation as a generation or reasoning task rather than a segmentation problem (Fan, Jiang, et al., 2024). Topic tracking and topical representations are also frequently incorporated as intermediate components in dialogue generation (Fan, Li, & Zhu, 2024; Rebedea et al., 2024) and summarisation systems (Hua et al., 2024; Tang et al., 2024). While these approaches often achieve strong empirical performance, they remain largely opaque. Their decisions are difficult to interpret, and their training data are often inaccessible, making it challenging to assess the extent to which benchmark datasets have already influenced the models' internal representations. The rising amount of LLM-generated data in the training corpora will most likely decrease the diversity of patterns models are exposed to and re-

inforce their biases. Namuduri et al. (2025) show that LLM-generated obituaries, creative writing texts, and long-form texts display more similarities at the level of discourse structure when compared with other model-generated texts than with human-written ones. In addition to the resource costs and ethical implications associated with LLMs – opacity, undisclosed training corpora, intrinsic biases (Ducel et al., 2026) –, their use for research purposes raises increasingly pressing concerns regarding generalisation, evaluation and scientific validity.

From the perspective adopted in this thesis, these limitations are particularly important. Most contemporary approaches are evaluated primarily in terms of performance, whereas our objective is also to understand how topical organisation varies across dialogue types. If topical structure is shaped by the constraints imposed by different forms of interaction, then methods developed on highly structured datasets may not generalise to more open-ended conversational settings, as we discuss in the following section.

4.3 Dialogue type and topical structure

The discussion in this chapter stresses that most computational approaches to topical organisation were originally developed for written discourse or highly structured forms of interaction. Their underlying assumptions are often reasonable in these settings as lexical cohesion, explicit discourse markers, and clearly signalled transitions provide strong cues for identifying topic boundaries. However, as interaction becomes less constrained, these cues become weaker and topical organisation increasingly depends on pragmatic and interactional factors. Topic shifts may be negotiated rather than announced, developed gradually rather than abruptly, and justified through shared knowledge and cues in the physical context rather than argumentative continuity. The success of a topical model therefore depends not only on its technical sophistication but also on the compatibility between its assumptions and the type of data on which it is applied.

This tendency to treat topical organisation as largely independent from the communicative context is rooted in classical perceptions of discourse as a rather homogeneous or at least continuous phenomenon. Chomsky's generative framework, which has dominated large parts of linguistic theory since the 1960s, proposes a universal grammar underlying all human language (Chomsky, 1969). In this view, observable differences between dialogue and monologic discourse are attributed to performance factors – memory limitations, distractions, or other extralinguistic influences – rather than to fundamental structural differences. Disfluencies, repairs, and interactional shifts, central to conversational organisation, are thus interpreted as imperfect realisations of internally well-formed sentences, rather than as inherent properties of communicative behaviour. This theoretical stance, by prioritising an abstract, sentence-based competence over situated interaction, may have indirectly

reinforced the assumption in computational work that dialogue can be processed with the same tools developed for monologic text. In contrast, Bakhtin's (2010b) theory of dialogism views all language as fundamentally interactive, shaped by the presence – real or imagined – of an addressee. Even the most monologic utterance is conceived in anticipation of response and in relation to prior discourse. From this perspective, the similarities between dialogue and monologic discourse arise not from the absence of structural distinctions, but from their shared roots in inherently social and responsive language use. Both perspectives support, in different ways, the idea that discourse and dialogue can be treated in the same way: Chomsky's (1969) model by downplaying interactional structure, and Bakhtin's (2010b) by emphasising their shared dialogical foundation.

The recent emergence of LLMs has further reinforced the perception that a single computational architecture can handle all types of language use equally well. Their apparent ability to produce fluent, plausible dialogue – alongside summaries, essays, and technical explanations – gives the impression of a unified capacity for language processing. Yet the language they produce is different from human-produced one in various ways. LLM-generated text is easily identifiable after few expositions to it because of the frequent use of specific phrases and the repetitive discourse patterns they rely on (Namuduri et al., 2025). Moreover, written training data limits the exposure to actual conversation that includes gestures, turn-taking, repairs, and grounding among others. This entails that LLMs struggle with more interactive data. Nevertheless, their success across a wide range of tasks and benchmarks (A. Wang et al., 2018, 2019; Mialon et al., 2023; Ye et al., 2025), many of which do not isolate dialogue-specific challenges, reduces the incentive to design specialised methods for conversational data, especially when such methods would require additional annotation or data collection, and thus additional time and resources. The current climate in academia with high pressure to publish quickly and produce results that are easily identifiable as competitive further reinforces this tendency.

Nevertheless, extensive work in linguistics has shown that the interactional nature of dialogue produces structural differences that set it apart from monologic discourse and other communicative modes. Turn-taking, grounding, repair, and topic management among other features are characteristic of dialogue and contribute to mutual understanding. Different forms of communication operate under different norms and constraints. A casual conversation between friends, for example, has different turn-taking dynamics, topical fluidity, and tolerance for ambiguity than a formal pitch to a potential investor. While both may stem from a shared cognitive and linguistic capacity, their actual realisations diverge.

This perspective is particularly limiting when studying topical organisation. As Purver (2011) observes, identifying topics may appear straightforward in highly structured settings such as news broadcasts, business meetings, or court transcripts, where topical units are often tied to externally defined tasks or agenda items. Yet

even in such cases, finer-grained segmentation quickly becomes difficult. Annotators frequently disagree on boundary placement, and agreement generally decreases as the desired level of granularity increases. Rather than reflecting annotation noise alone, these disagreements suggest that topical structure is partly dependent on the analytical perspective adopted and on the intended use of the segmentation.

The challenge becomes even greater in more open-ended dialogue where all the participants can continuously reshape the trajectory of the interaction. Topic boundaries may be gradual, negotiated, or retrospectively interpretable rather than explicitly signalled. Consequently, approaches developed for written discourse or highly structured interactions cannot be assumed to transfer straightforwardly to more open-ended conversational settings.

Interestingly, QUD frameworks emerge from the opposite direction. Whereas most computational approaches to topical organisation originated in discourse processing and were subsequently adapted to dialogue, QUD-based theories were developed to account for conversational coherence and only later applied to monologic discourse and computational tasks. They therefore place interaction, relevance, and the management of conversational goals at the centre of their account of topical organisation. In principle, this makes them particularly well suited to dialogue. Yet despite their theoretical influence, empirical investigations of QUDs in naturalistic conversations remain comparatively limited, especially for open-ended forms of interaction where topical trajectories may be less constrained and less information-oriented.

Overall, we have discussed two facets of the works on topics in NLP/CL. On the one hand, we have computational approaches to topic segmentation and modelling that were originally developed for written discourse and rely either on surface information such as lexical cohesion or on latent representations learned by topic models and LLMs. While some of these methods have been adapted to structured forms of interaction, they are not designed for open-ended conversational data. On the other hand, we have theoretically grounded approaches to topical organisation, such as QUD-driven frameworks, which emphasise the interactive nature of dialogue but have not been extensively tested on naturalistic conversations. In both cases, there is a lack of empirical validation on spontaneous dialogue, leaving open questions about the generalisability of existing methods and theories across different dialogue types.

Taken together, these observations raise two complementary questions. First, to what extent do existing approaches to topical organisation generalise across dialogue types? If topical structure is shaped by the constraints of interaction, methods developed on highly structured datasets may not remain effective when applied to more open-ended conversational settings. Second, can topical organisation itself serve as a discriminative signal for characterising dialogue types? If different regions of the dialogue space exhibit systematically different topical dynamics, then

topical structure may provide a useful perspective for comparing, describing, and ultimately classifying conversational data.

The remainder of this part investigates these questions through three complementary studies. They all dive into the *Structure* parameter of dialogue according to the typology we proposed in Chapter 2 through the analysis of topical organisation in various kinds of dialogues. Ultimately, the type of dialogue we are interested in is one that we expect to be very loosely structured: multi-party face-to-face open conversation. We also limited ourselves to dialogues in English and French, which are the languages we are most familiar with, and thus most able to analyse linguistically. (4.i) represents those constraints within the formal account of CHIMERA. We are interested in *human* dialogue as we consider human-human interaction to be the prototype we should understand both for theoretical reasons and more practical one as designing systems that use language should be done by explicitly choosing which features of human-human interaction we want to reproduce. We also want to study actual interactions rather than simulated ones which we specify by indicating that the mapping between the simulated agents and the language producers is 'self to self'. Another characteristic of the type of dialogue we are interested in is that it typically involves participants with little or no hierarchical structure between them. Lastly we characterise the openness of those dialogues by indicating that the set of potential questions that must be resolved or otherwise addressed for the conversation to be considered successful is empty. This means that the conversation is not constrained by a specific task or goal, and that the participants are free to explore a wide range of topics and ideas.

(4.i) multi-party face-to-face open conversation:

$$\begin{bmatrix} \texttt{Agents}: & \begin{bmatrix} \texttt{Agents}: \texttt{Agency}: \{\textit{human}\} \\ \texttt{Structure.Nb}: \begin{bmatrix} \texttt{Simulated}: Nb_{>2} \\ \texttt{Mapping}: \textit{SelfToSelf} \end{bmatrix} \\ \texttt{SocialDynamics.Hierarchy}: \textit{minimal} \end{bmatrix} \\ \texttt{Setting.EnvMedium}: \textit{face-to-face} \\ \texttt{Frame}: \begin{bmatrix} \texttt{Language.Global.List}: \textsc{among}(\textsc{en}, \textsc{fr}) \\ \texttt{Purpose.SuccessCondition.Q}: \varnothing \end{bmatrix} \end{bmatrix}$$

where the *face-to-face* environment can be defined through the following constraints:

$$\begin{bmatrix} \texttt{PhysicalContext}: \textit{Shared} \\ \texttt{Visibility}: \begin{bmatrix} \texttt{Gaze.Fidelity}: \textit{High} \\ \texttt{Gestures.Interpretability}: \textit{High} \\ \texttt{FacialExpressions.Interpretability}: \textit{High} \end{bmatrix} \\ \texttt{Time.Synchrony}: \textit{Maximal} \end{bmatrix}$$

We will confront this set of constraints with the datasets we have actually used in our studies. Indeed, additional restrictions are imposed by the actual realisation of a task – such as the corpora we have access to, the annotations that are required for the study, and the technical limitations of the methods we use – and those may not always allow us to find a perfect dataset that combines all constraints. The three studies presented in this part are therefore based on three different corpora that represent different types of dialogues growing increasingly closer to the type of dialogue we are interested in throughout the chapters.

The first study, presented in Chapter 5, examines the robustness of classical topic segmentation approaches by extending TextTiling-inspired methods to open-ended multiparty conversations. This very direct study aims at questioning whether methods and tools designed mostly with monologic or organised dialogic forms of language use in mind can resist the challenges posed by more spontaneous and less structured conversational settings. Topic segmentation is a first step towards understanding topical organisation, and this study provides a first element to indicate how important the structure of the dialogue is to consider for producing robust methods. The second study, presented in Chapter 6 is rooted in the assumption that a binary approach to topic shifts, i.e. considering that a topic shift is either present or absent at a given boundary, is too reductive to capture the complexity of topical organisation in dialogue and thus to support structure analysis. Instead, it looks at topics changes in a holistic way, considering the whole topical trajectory of a conversation. It investigates whether a local topical similarity metric can reveal the underlying conversational structure of large-scale online discussions as well as differences between communities. Finally, the third study, presented in Chapter 7, takes a more abstract perspective and investigates the predictive power of QUD-based representations across a range of dialogue types. It aims to assess whether the salience of questions at a given point in a conversation can be used to anticipate the trajectory of the interaction, and whether this predictive power varies across dialogues with different number of participants and interactional settings. One of the goals of this study is to explicit the relationship between observable features, such as the participants and the task, and the STRUCTURE parameter, which is more latent, showing that they depend on each other.

Together, these studies aim to clarify the relationship between dialogue type and topical organisation while assessing the extent to which existing theories and computational methods generalise beyond the settings in which they were originally developed. By investigating their limitations, we hope to identify the scope in which topics can be studied depending on the type of dialogue.

5

Topic Segmentation in Open-ended Multi-party Conversation

Contents

5.1 Related works 207
5.1.1 TextTiling approach 207
5.1.2 Carenini's coherence-layer based approach 213
5.2 Experimental setting 218
5.2.1 Models 218
5.2.2 Evaluation metrics 219
5.3 Dataset 222
5.3.1 Detailed description 223
5.3.2 Segmentation in topics 226
5.4 Description of the baselines 227
5.4.1 Random baseline 228
5.4.2 Dialogue adapted TextTiling 230
5.4.3 BERT for next-sentence prediction 233
5.4.4 LLMs 235
5.5 Improving TextTiling with Linguistic Features 240
5.5.1 Speakers 240
5.5.2 Questions 242
5.5.3 Coreference chains 242
5.5.4 Summary of the Results 243
5.6 Adapting Xing and Carenini's BERT-based model . . 244
5.6.1 Learning curve and coherence layers 244
5.6.2 Stability and less diverse training set 247
5.7 Final comparisons and conclusion 249
5.7.1 Comparison of all the models 250
5.7.2 Final remarks and conclusion 251

Understanding how topics interact in dialogue is essential when it comes to modelling dialogue structure beyond the sequence of utterances. However, analysing this structure requires insight on the topics themselves. Being able to segment a dialogue into topically coherent segments seems to be a first step towards modelling high level dialogue structure. The segments could later be linked inside a structure that describes the interactions between them. This task, called dialogue topic segmentation (DTS), finds utility in dialogue generation (J. Xu et al., 2021) and summarising (J. Chen & Yang, 2020), among other applications.

As discussed in Chapter 4, DTS has received less attention compared to monologue or written text topic segmentation, primarily due to the scarcity of annotated data. Nevertheless, some DTS approaches get good results on task-oriented dialogues (Takanobu et al., 2018) or conversations with a restricted set of possible topics such as meeting minutes (Hsueh et al., 2006; Georgescul et al., 2008). Xing and Carenini (2021) suggest another method to tackle more varied dialogues. They use the *TextTiling* algorithm (Hearst, 1997), that relies on a similarity metric between subsequent blocks of text to identify topic boundaries. By enhancing it with a learned utterance-pair coherence scoring model based on BERT (Devlin et al., 2019) as similarity metric, they obtain good results in English and Chinese when evaluating their model on three datasets: DialSeg_711 (Y. Xu et al., 2021), Doc2Dial (S. Feng et al., 2020), and ZYS (Y. Xu et al., 2021). Even though these datasets cover different domains, they all contain task-oriented conversations. Evaluating this model on more open-domain dialogues would provide insight on the limits of its generalisation capability.

In this chapter we adapt several linear topic segmentation approaches to open-domain multi-party conversations[13]. To address the lack of such annotated data, we leverage transcripts from the *Friends* TV show, which include scene annotations that we use as a proxy for topic boundaries. The approaches we adapt include two versions of the TextTiling algorithm: one enhanced with linguistic and dialogical features, and the other using a BERT-based utterance pair coherence scoring model inspired by Xing and Carenini (2021). We also evaluate the performance of LLMs in a basic zero-shot setting on this task.

The motivation behind this work is twofold. First, we want to evaluate the performance of linear topic segmentation approaches on multi-party casual conversations, which are more chaotic than the task-oriented dialogues typically used for this task. Second, we want to determine what performance we can reach with a formal approach based on linguistic and dialogical features. Since we are interested in the structure of topical interactions, an explainable model would help us better understand what features play a role in perceiving topic shifts. A rule-based approach also has the advantage of minimising the amount of computation required by our model, which is more sustainable, and makes our work easier to reproduce

[13] Our code is available at https://gitlab.inria.fr/adecker/topicsegmentationtexttiling.git.

as it does not require any training nor specific version of a model.

We begin in Section 5.1 by describing the two main studies our work relies on, including the original TextTiling algorithm and Xing and Carenini's (2021) approach. We then present our experimental setting in Section 5.2, including a brief description of the models we used, the baselines we compared them to, and the evaluation metrics we used. In Section 5.3 we describe our dataset, the *Friends* TV show transcripts. Next we detail our baselines (Section 5.4), our feature-based improvements of the original TextTiling algorithm (Section 5.5), and the application of Xing and Carenini's (2021) approach to our data (Section 5.6). Finally in Section 5.7, we summarise and compare all the results, and discuss their implications for topic segmentation in multi-party casual conversations.

Most of the work described in this chapter is taken from Decker and Amblard (2024a) and was presented at the 5th Workshop on Computational Approaches to Discourse (CODI) in 2024. We refined the evaluation process and added an LLM-based baseline for this chapter.

5.1 Related works

This work relies heavily on two main approaches to topic segmentation in dialogue: the TextTiling algorithm (Hearst, 1997) and Xing and Carenini (2021)'s enhancement of this algorithm. We describe these two approaches in the following sections. For a more general overview of topic segmentation, in discourse and in dialogue, we refer the reader to the previous chapter (Chapter 4).

5.1.1 TextTiling approach

TextTiling (Hearst, 1997) is a topic segmentation algorithm that predicts topic boundaries for a given *text*. This section is dedicated to describing the original approach, and we will discuss our various adaptations to make it suitable for *dialogue* in Section 5.4.2. For now we thus focus on the application of the TextTiling algorithm to organised monologic text which we consider to be a sequence of paragraphs. Since the original algorithm was meant for 'expository text', i.e. text 'not heavily stylized or structured', we explain the algorithm with an example that does not include headings or similar organisational elements. A schematic representation of such a document is given in Figure 5.1. The TextTiling algorithm relies on lexical distribution information and its execution follows three main steps that we describe in the upcoming sections: (1) pre-processing, (2) lexical score determination, (3) boundary identification.

5.1.1.1 Pre-processing: Distillating the lexical content

The goal of the pre-processing step is to extract the most salient lexical information from the text. In this first step, if the document contains markup information such as headings or similar organisational elements, they are removed and only the body of the text is kept due to the algorithm being designed for expository text. The punctuation is then removed, and the text is tokenised, i.e. split into individual lexical units, and lowercased, which corresponds to the Step 2 on the schematic representation Figure 5.1.

Stopwords are then removed from the tokenised text (Step 3). Stopwords are words that are very frequent in a language but do not carry much meaning, such as articles, prepositions, or conjunctions. Removing them allows the algorithm to focus on the more meaningful words that are more likely to be indicative of topic shifts.

Then the remaining tokens can be stemmed or lemmatised[14] (Step 4). Stemming and lemmatisation are two techniques used to reduce words to their base or root form, thus allowing the algorithm to identify different forms of the same words as being the same. For instance, the words 'run', 'runs', and 'running' would all be reduced to the same root form, which allows the algorithm to better capture the lexical similarity between different parts of the text.

Finally the text is subdivided into sequences of tokens of a predefined length w – stopwords are included in the computation of the length – which we will call *spans* in the following (Step 5). This step is necessary to compare sequences of similar lengths when computing the similarity scores between subsequent parts of the text.

We thus end up with a sequence of spans $\{s_1, s_2, ..., s_n\}$, each constituted of w tokens, but only the non-stopwords among these will be considered in the next step where we compute lexical scores.

5.1.1.2 Lexical score determination: Three approaches based on lexical co-occurrence and distribution

After the text has been split in spans of w tokens, a lexical score is computed at the gap between each span. For instance as represented on Figure 5.2, for a text of n spans $\{s_1, s_2, ..., s_n\}$, there are $n-1$ gaps and thus $n-1$ lexical scores to compute.

Three approaches are suggested in the original paper to measure this score: one based on the lexical similarity between subsequent blocks of text, one based on the proportion of never-yet-seen words in the two spans surrounding a gap, and one based on lexical chains, i.e. chains of related words. We briefly describe them in the following paragraphs.

[14]The original paper describes this step as reducing the tokens to their 'root by a morphological analysis function based on that of Karttunen et al. (1987)' which corresponds to stemming but both stemming and lemmatisation have been attempted in the different implementations of this algorithm.

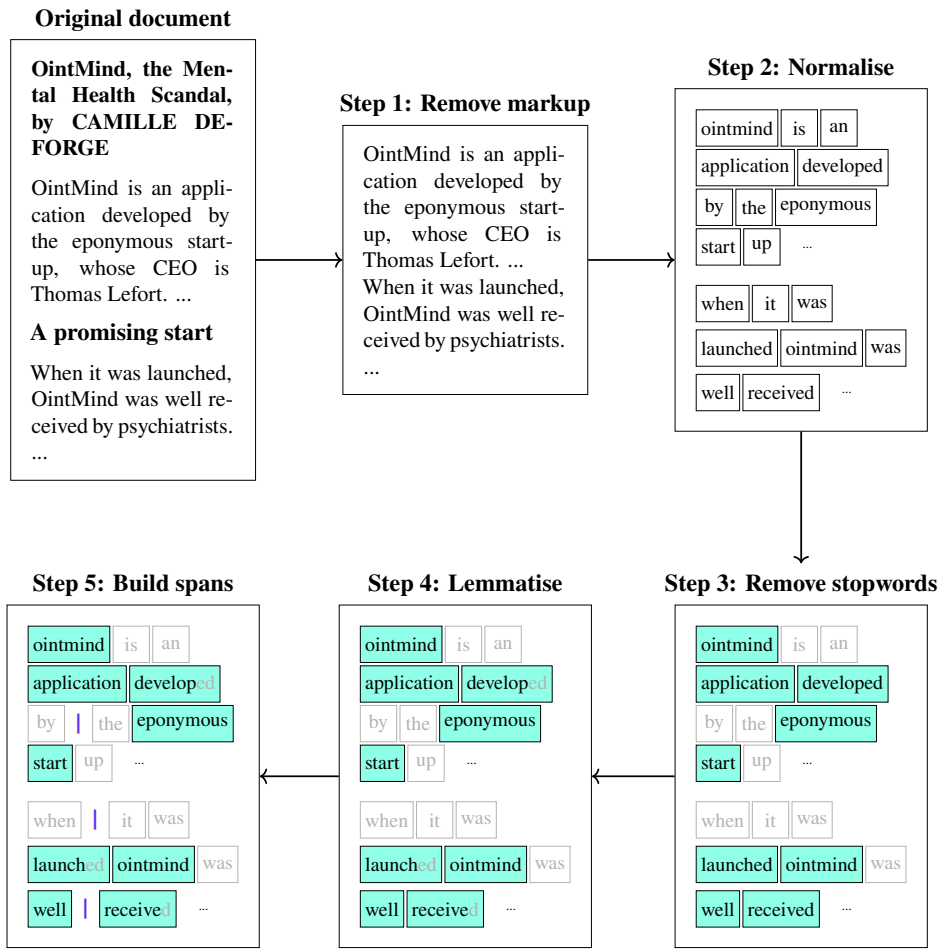


Figure 5.1: Schematic representation of the pre-processing step of the TextTiling algorithm. *Text for the toy example taken from the dystopic novella OintMind (Choi et al., 2024).*

Block comparison The idea behind this method is that a topic is characterised by a specific vocabulary which entails that when a topic shift occurs, the vocabulary also changes. Thus, we can identify topic boundaries by looking at the lexical similarity between subsequent blocks of text. The more different the vocabulary is between two blocks, the more likely it is that there is a topic shift between them.

In the original TextTiling algorithm, the scores were computed by comparing the two blocks of *k* spans on each side of the gap. As Figure 5.2 shows, the lexical score $score_{BC}(i)$ (corresponding to the gap i) would correspond to the normalised sum of the inner product between the term frequency vectors of the two blocks, where the term frequency vector of a block is a vector that counts the number of times each token appears in the block. Formally we would have:

$$score_{BC}(i) = \frac{\sum_{t \in V} (tf(t, \{s_{i-k+1}, ..., s_i\}) \times tf(t, \{s_{i+1}, ..., s_{i+k}\}))}{\sqrt{\sum_{t \in V} tf(t, \{s_{i-k+1}, ..., s_i\})^2} \times \sqrt{\sum_{t \in V} tf(t, \{s_{i+1}, ..., s_{i+k}\})^2}}$$

where V is the vocabulary of the text and $tf(t, B)$ is the term frequency of the token t in the block B.

This score corresponds to the cosine similarity between the term frequency vectors of the two blocks. With this measure, a low similarity score encased between higher ones would indicate a potential topic shift.

Vocabulary introduction The second approach relies on an idea similar to the first one, but instead of looking at the similarity between two blocks of text, it focuses on the introduction of new words in a segment of text. This approach is inspired from the observation of Youmans (1991) that a segment of text with low proportion of never-yet-seen words in a segment of text followed by one with a much higher proportion correlates 'closely to constituent boundaries and information flow' (p. 788). This idea is thus adapted to the TextTiling algorithm by computing the ratio of never-yet-seen words in an interval of two spans centred around the gap. For a given gap i we would thus have:

$$\begin{aligned} score_{VI}(i) &= \frac{\text{number of never-yet-seen words in } \{s_i, s_{i+1}\}}{\text{total number of tokens in } \{s_i, s_{i+1}\}} \\ &= \frac{\text{number of never-yet-seen words in } \{s_i, s_{i+1}\}}{2 \times w} \end{aligned}$$

With this measure, a sharp increase in the score would indicate a potential topic shift.

Lexical chains The last approach relies on the concept of lexical chains, i.e. chains of tokens that are considered 'to be in a coherence relation with one another'. This idea comes from Morris (1988) who developed a dictionary-based

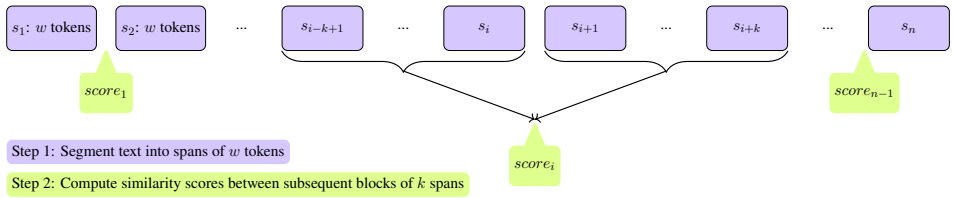


Figure 5.2: Segmentation in spans of w tokens and computation of the lexical scores in the TextTiling algorithm.

method to identify these chains, and considered that a coherent segment of text would correspond to the extent of a chain. This approach is not directly suited to the TextTiling algorithm for several reasons, notably the practical limitations of the dictionary-based method and the possibility to have several concurrent chains that may sometimes overlap each other. Hearst (1997) thus suggests a few modifications to this approach to make it more suitable for their algorithm. First they suggest to replace the dictionary with a co-occurrence based method to identify chains of morphological variants of the same word. Second, they compute all the chains and then compute the score of a gap by looking at the number of *active* chains at this gap, where a 'chain for term t is considered active across a sentence gap if instances of t occur within some distance threshold of one another' (p. 47).

This approach is not formalised in the original paper nor in Hearst (1994) which the authors point to for further details. This paper also shows that the block comparison approach performs slightly better than the lexical chains one, though the difference 'may not prove significant over the long run' (p. 14). In our own experiment, we used the `Nltk` implementation of the TextTiling algorithm which includes the block comparison similarity measure but not the vocabulary introduction nor the lexical chains one. We thus implemented our own version of these two metrics.

5.1.1.3 Boundaries identification: Locating the highest variabilities in the lexical scores

Once the lexical scores have been computed for each gap, the next step is to identify the topic boundaries by looking at the variability of these scores. The idea is that a topic shift would correspond to a sharp change in the lexical scores, which would be reflected in a local minimum in the scores. However, not all local minima correspond to topic shifts, and some topic shifts might not correspond to local minima. To address this issue, the original TextTiling algorithm suggests looking at the depth of the *valleys* in which the lexical scores fall. A deeper valley would indicate a more significant change in the lexical scores, which would be more likely to correspond to a topic shift. All the steps involved in this process are illustrated in Figure 5.4. This approach is independent of the metrics used to compute the lexical scores, and thus can be applied to any of the three methods described in the

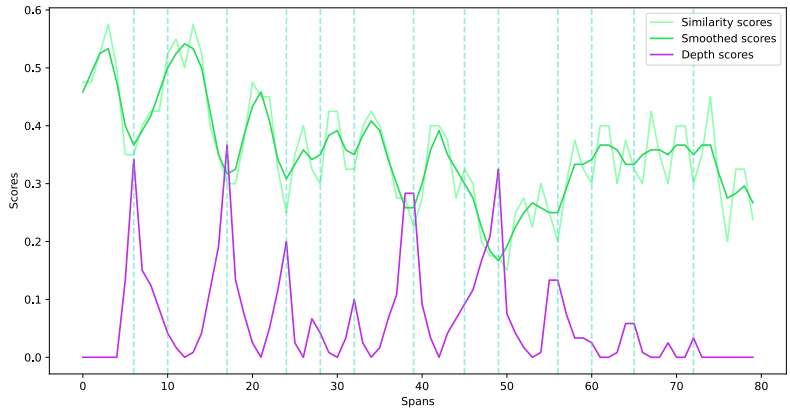


Figure 5.3: Plot of the lexical scores, depth scores and topical boundaries for a toy example.

previous section as well as a combination of them or a different process as we will see in Section 5.1.2.

A prior step to identifying boundaries thus consists in computing the *depth scores* for our document. Formally, given a gap i, we measure the depth of its valley by retrieving the first lexical score on the left that forms a pic, *i.e.* $hl(i)$ such that it is greater than the score directly on its left. We retrieve $hr(i)$ similarly on the right. The depth score of the gap i is then computed by averaging the depths on both sides: $dp(i) = \frac{(hl(i) - score(i)) + (hr(i) - score(i))}{2}$.

This method can be sensitive to small perturbations in the lexical scores, which can lead to identifying false boundaries. To address this issue, the original TextTiling algorithm suggests applying a smoothing function to the lexical scores before computing the depth scores. This smoothing function can be a simple moving average or a more sophisticated method such as a Gaussian filter.

Figure 5.4 shows two cases of depth score computation. The first one (i) is simple, where $hl(i)$ and $hr(i)$ are the first pics on the left and right of the considered score. The second case (j) illustrates the role of smoothing, as there is a very small pic between $score(j)$ and the $hr(j)$ we actually consider. Without smoothing, $score(j+1)$ would have been used to compute the depth score while it would not be representative of the real lexical similarity at this point.

The topic boundaries can then be identified based on the local maxima of the depth scores. The recommendation of the original paper is to consider that there is a boundary at gap i if the depth score at this gap is above a certain threshold that is determined based on characteristics of the text. Formally, they suggest to consider that a gap i is a topic boundary if $dp(i) > \bar{dp} - \sigma_{dp}$ where $\bar{dp}$ is the mean of the

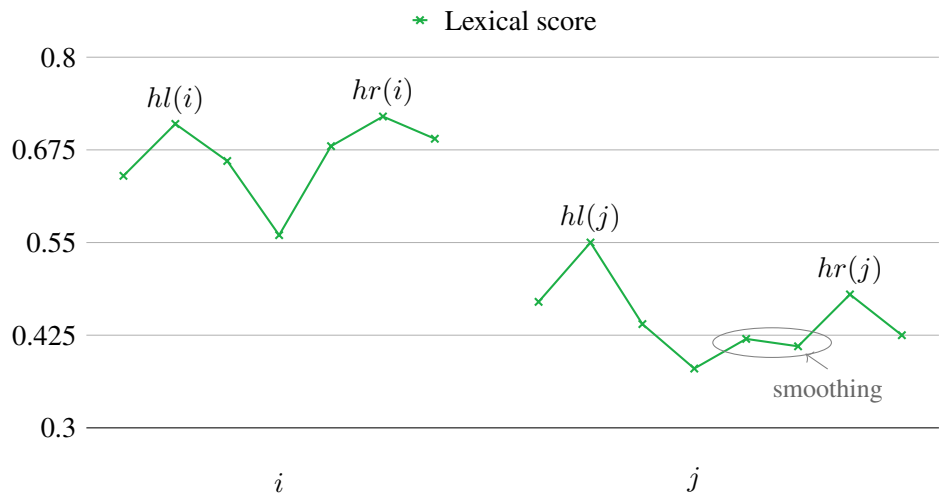


Figure 5.4: Examples of lexical scores used in the depth scores computation.

depth scores and σ_{dp} is their standard deviation. In practice these boundaries are shifted to the closest gap between two paragraphs because the split into spans of length w at the very beginning of the algorithm erases this structure.

5.1.2 Carenini's coherence-layer based approach

As explained in Section 5.1.1, the TextTiling algorithm relies on a similarity metric to compute the lexical scores that are used to identify topic boundaries. The original paper suggests three approaches to compute this similarity metric, but other approaches can be used as well. For instance, Xing and Carenini (2021) suggest to use a trained utterance-pair coherence scoring model based on *Next-Sentence Prediction* (NSP) BERT model (Devlin et al., 2019; Y. Sun et al., 2022). In this section we describe their approach (Section 5.1.2.1) and the different datasets they used to train and evaluate their model (Section 5.1.2.2). Their work covers English and Chinese and we will discuss the small differences between the approaches which concern the datasets and the way they are used. The approach itself is theoretically language-agnostic but its reliance on NSP BERT makes it likely to be unsuitable as is for some languages, in particular less-resourced ones.

5.1.2.1 The metric: leveraging a text-pair coherence scoring task in an unsupervised way

The main idea behind Xing and Carenini's (2021) approach is that the DTS task can be seen as the extension of a text-pair coherence scoring task (L. Wang et al., 2017; P. Xu et al., 2019; W. Wang et al., 2020). Indeed, based on (Hearst, 1997;

Purver et al., 2006), they consider that 'utterances associated with the same topic should be more coherent together than the utterances about different topics' (Xing & Carenini, 2021, p. 167). Thus, they suggest to leverage NSP BERT to train an utterance-pair coherence scoring model that can be used as a similarity metric in the TextTiling algorithm. However, given the complexity of annotating utterance pairs with coherence scores, they suggest to train this model in an unsupervised way by using a relative ranking objective (H. Li, 2011). We detail the text-pair coherence scoring task and the relative ranking approach in the following.

Text-pair coherence scoring task The text-pair coherence scoring task consists in assigning a coherence score to a pair of texts, which in our case are two utterances. However assigning an absolute coherence score to a pair of utterances is a complex and very subjective task. This would make it complicated to produce annotated data in sufficient amount and quality to train a model. Instead, Xing and Carenini (2021) suggest to input triples of utterance pairs to the model ordered according to their coherence, which allows it to learn a distribution of coherence scores and incidentally train it to produce utterance-pair coherence scores. This requires to define a way to build these ordered triples. Xing and Carenini (2021) extend L. Wang et al.'s (2017) idea that 'a pair of texts from the same segment should be ranked more coherent than a pair of texts randomly picked from different paragraphs' (Xing & Carenini, 2021, p. 169) by suggesting an adaptation based on dialogue structure (Schegloff & Sacks, 1973).

They thus suggest to consider three levels of coherence: (1) adjacent pairs of utterances, (2) pairs of utterances from the same dialogue but not adjacent, and (3) pairs of utterances from different dialogues. Adjacency is a notion from Conversation Analysis (Schegloff & Sacks, 1973) that is stronger than proximity, as it also requires that the pair of utterances is produced by a pair of speakers and fits a certain dialogical structure such as question-answer or offer-acceptance/refusal.

In practice Xing and Carenini (2021) build a training triple by randomly sampling an adjacent pair of utterances from their dataset, giving the first pair of the triple. Then, they keep the first utterance of this pair to build the two other pairs, and associate it with a random utterance from the same dialogue, giving the second pair, and with a random utterance from a different dialogue, giving the third pair. This method is illustrated in Figure 5.5 where we can see two schematic dialogues where the first one begins with an adjacent pair of utterances (A and B) which constitutes the most coherent pair, then a random utterance from the same dialogue (C) which is less coherent with A, and finally a random utterance from a different dialogue (D) which is the least coherent with A.

This method allows Xing and Carenini (2021) to produce a set of ordered triples of utterance-pair without the need for data annotated with coherence scores. Their three levels of coherence only require annotations for the adjacency pairs.

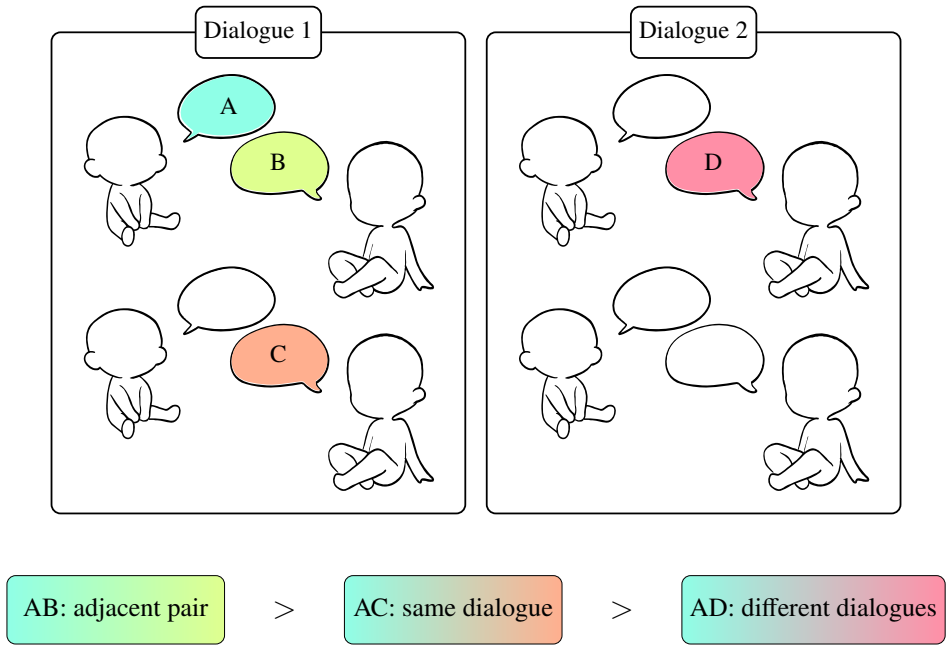


Figure 5.5: Levels of coherence considered by Xing and Carenini (2021).

Relative ranking objective The goal of the utterance-pair scoring model is to produce coherence scores for pairs of utterances that reflect their relative coherence. However, given that there is no gold score for a given utterance pair, it is not possible to train the model with a classical regression loss. Indeed, as represented on Figure 5.6, a regression loss would require to have a gold score for each pair of utterances, in order to train the model to minimise the difference between the hypothesised score and the gold score.

Xing and Carenini (2021) thus suggest to train the model with a relative ranking objective, which means that instead of considering one pair per input and rewarding the model for assigning a score that is close to a gold score, *two* ordered pairs of utterances per input are considered and the model is rewarded for assigning a higher score to the more coherent pair. This method allows the model to learn to produce coherence scores that reflect the relative coherence of the pairs of utterances, which is what is needed for the task.

The ranking loss associated with this method is described in Figure 5.6. Since there are three levels of coherence, three inputs per triple of utterances can be built: AB vs AC, AC vs AD, and AB vs AD. The loss is then computed by averaging the individual ranking losses. In practice, Xing and Carenini (2021) use a marginal ranking loss with a margin of 1 and a reLu activation function which sets the individual scores to 0 when they were negative.

By showing the model all the possible couples of utterance pairs from a triple,

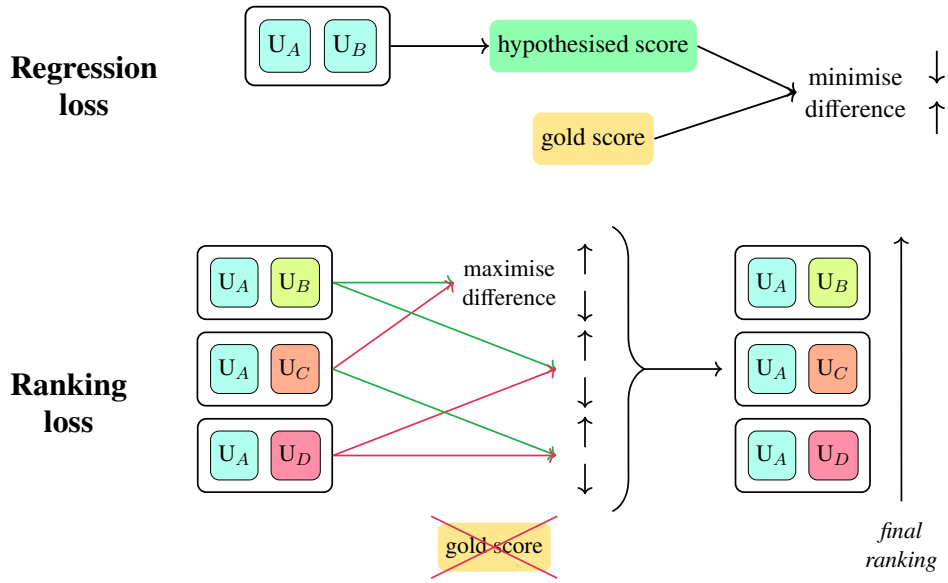


Figure 5.6: Illustration of the use of a regression loss compared to a ranking loss in similarity metric training.

it can learn to produce a distribution of coherence scores that reflects the relative coherence of the pairs. Indeed, it learns that $coherence(AB) > coherence(AC)$, $coherence(AC) > coherence(AD)$, and $coherence(AB) > coherence(AD)$, which taken together allows the model to learn that $coherence(AB) > coherence(AC) > coherence(AD)$. At the end, the model is trained to produce coherence scores between 0 and 1, which can then be used as a similarity metric in the TextTiling algorithm.

5.1.2.2 The datasets: two settings for two tasks

The utterance-pair coherence scoring task described above is only an auxiliary task to achieve DTS. Thus the datasets used to train and evaluate the model for this task are not necessarily the same as the ones used to train and evaluate the DTS model. Xing and Carenini (2021) indeed use separate datasets for these two tasks, which we describe in the following.

Training dataset: DailyDialog & NaturalConv Xing and Carenini (2021) define the ideal source corpus for the utterance-pair scoring model as '(1) having a fairly large size; (2) covering as many topics as possible; (3) containing both formal and informal expressions' (p. 170). Based on these requirements they select two datasets: DailyDialog (Y. Li et al., 2017) for English and NaturalConv (X. Wang et al., 2021) for Chinese. They describe both datasets as containing 'open-domain

conversations about [multiple] daily topics'.

The main difference between the two datasets is that DailyDialog is annotated with dialogue acts, which can be used to identify adjacency pairs in the Conversation Analysis sense, while NaturalConv is not.

DailyDialog is annotated with the four dialogue acts *Inform*, *Question*, *Directive*, and *Commissive*, and the adjacency pairs are taken from the two dialogue act pairs *Question-Inform* and *Directive-Commissive*. Xing and Carenini (2021) add two additional constraints to the sampling of the utterances for the two other pairs. Among the four utterances A, B, C, and D, C must be of a different dialogue act from B, and D must be from a dialogue with a different topic from the one of A and B. These constraints make the three pairs more different from each other, which increases the chances that the model learns to produce coherence scores that reflect these differences.

Most of these strategies cannot be applied to NaturalConv since it is not annotated with dialogue acts, which means that the most coherent pair is simply made of subsequent utterances. Similarly the constraint for sampling C cannot be applied and an utterance is simply sampled from the same dialogue but not adjacent to A. The constraint for sampling D can be applied since NaturalConv is annotated with topics. This sampling method is thus less sophisticated than the one used for DailyDialog, but can be applied to any dataset that is not annotated with dialogue acts, which is closer to the reality of many existing datasets, including the one we used in our experiments.

Evaluation dataset: DialSeg, Doc2Dial & ZYS The final goal of Xing and Carenini (2021) is to improve the performance of the TextTiling algorithm on the DTS task. Thus, they evaluate their approach on datasets annotated with topic shifts. They use three datasets for this evaluation: DialSeg (Y. Xu et al., 2021) and Doc2Dial (S. Feng et al., 2020) for English, and ZYS (Y. Xu et al., 2021) for Chinese.

DialSeg is a real-world dataset constituted of task-oriented multi-turn dialogues from Stanford Dialog Dataset (Eric et al., 2017) and MultiWOZ (Budzianowski et al., 2018), in which topics are annotated manually. Doc2Dial is a synthetic dataset made from automatically built dialogue flows based on the structure of documents. Crowd workers then filled in these dialogue flows with natural language utterances. The topics are thus automatically defined by the structure of the documents. ZYS is a real-world dataset constituted of multi-turn dialogues from customer service phone calls and banking consultations. The topics are also annotated manually.

Overall these datasets are very task-oriented and contain rather short dialogues as reported by Table 5.1, which makes them quite different from the kind of dialogues we are interested in, which are longer and more open-ended. They are also all two-party dialogues, which is another important difference as it makes them less likely to be very chaotic.

Table 5.1: Statistics of the datasets used for evaluation in Xing and Carenini (2021) and our dataset. *The statistics for DialSeg, Doc2Dial, and ZYS are reported directly from Xing and Carenini (2021), except for the number of participants.*

Dataset	DialSeg_711	Doc2Dial	ZYS	Friends
documents	711	4,130	505	236
language	English	English	Chinese	English
# sent/seg	5.6	3.5	6.4	13.2
# seg/doc	4.9	3.7	4.0	23.6
real-world	✓	✗	✓	✗
# participants	2	2	2	2-6+

5.2 Experimental setting

As explained before, DTS is typically performed on rather organised dialogues while we are interested in more open-ended and less organised dialogues. Thus, we want to see how the Xing and Carenini's (2021) approach performs on a multi-party dialogue dataset, and whether a feature-based approach can compete with it.

We first summarise the different models and baselines that we compare in our experiments (Section 5.2.1). Then we present the evaluation metrics we use to evaluate the performance of the different models on the DTS task (Section 5.2.2).

5.2.1 Models

Table 5.2 summarises the different models we probe in our experiments. The first three models are the ones we compare for the main results, while the last four are baselines.

Baselines We use three types of baselines for our experiments. First, we use a random baseline to see how much the models perform better than chance. This baseline is the same as the one used by Xing and Carenini (2021) which assigns boundaries with a probability $\frac{b}{k}$ where k is the number of utterances and $b \in [\![0, k-1]\!]$ is a randomly chosen number of segment boundaries. Second, we compare our results to the original TextTiling algorithm to see how much the adaptations we made to it improve its performance. We use both the original TextTiling algorithm and a version of it with dialogue-specific adaptations. Finally, we use a basic LLM-based approach to see how it can perform in a zero-shot setting on this task. All these baselines are described in more details in Section 5.4

Table 5.2: Summary of the models compared in our experiments. The first three models are the ones we compare for the main results, while the last four are baselines.

Model	Description	Training
Random	Random baseline	✗
TextTiling$_{OG}$	Original TextTiling algorithm	✗
TextTiling$_{Dial}$	Original TextTiling algorithm with dialogue-specific adaptations	✗
LLM-based	Basic LLM-based approach	Zero-shot*
ML$_{OG}$	Xing and Carenini's (2021) original approach trained on their original dataset	DailyDialog & NaturalConv
ML$_{Friends}$	Xing and Carenini's (2021) original approach adapted to our dataset	*Friends* TV show
Feature-based	Our approach based on linguistic features	✗

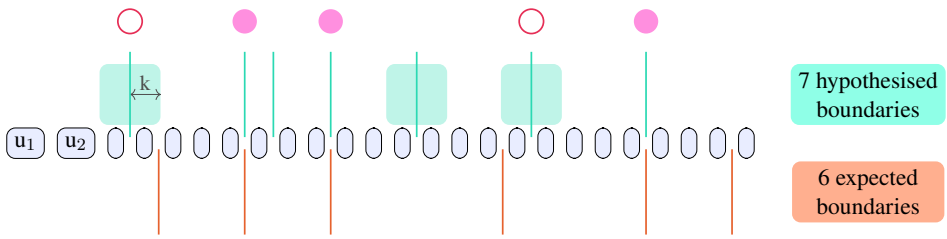
*While we do not perform any fine-tuning on the models for this baseline, we want to stress the training cost, both economic and environmental, of these models, which can be easily obscured by their relative ease of use and accessibility.

Models All the models we compare in our experiments are adaptations of the original TextTiling algorithm. On the one hand we have two adaptations of Xing and Carenini (2021)'s approach, one trained on their original dataset and one adapted to our dataset, and on the other hand we have our feature-based approach which is detailed in Section 5.5.

5.2.2 Evaluation metrics

We use two families of evaluation metrics for our experiments. On the one hand, F-measures are based on the number of correct, false positive, and false negative boundaries. On the other hand, P_k and WindowDiff perform different comparisons between the predicted and the gold segmentations within a sliding window. They are represented on Figure 5.7 and Figure 5.8 respectively.

F-measure The classical F-measure is calculated by comparing the predicted boundaries to the gold ones, and computing the ratio of twice the correct number of boundaries over the total number of boundaries, i.e. the number of predicted ones added to the number of expected ones. A predicted boundary is considered correct if it matches a gold boundary, which means that it is at the same position in the dialogue. This metric is quite strict since it does not give any leeway to the models for predicting boundaries that are close to the gold ones, which can be prob-



$$\text{Precision} = \frac{\text{correct boundaries}}{\text{hypothesised boundaries}} \qquad \text{Recall} = \frac{\text{correct boundaries}}{\text{expected boundaries}}$$

$$\text{F}_1 = \frac{2 \times \text{correct boundaries}}{\text{hypothesised boundaries} + \text{expected boundaries}}$$

$$\text{F}_k = \frac{5 \times \text{correct } \pm k \text{ boundaries}}{4 \times \text{hypothesised boundaries} + \text{expected boundaries}}$$

Figure 5.7: Explanation of the Precision, Recall, F_1 and F_k metrics for topic segmentation.

lematic given the difficulty of the task. A segmentation where each boundary falls one utterance before or after a gold boundary should be considered better than a segmentation where all the boundaries are far from the gold ones, but the classical F-measure would give the same score to both segmentations.

Thus, we also report a more relaxed version of it, which we call F_k, that considers a predicted boundary as correct if there is a gold boundary at most k utterances before or after the predicted one. This allows us to give some leeway to the models for predicting boundaries that are close to the gold ones, which is important since there can be multiple valid segmentations of a dialogue. A gold boundary cannot count a second time once it has been matched by a predicted boundary. We also give more weight to precision compared to recall as we consider that finding right boundaries is more important than finding all of them. Formally, it means that the number of predicted boundaries is weighted by a factor of 4 in the denominator of the formula, which decreases the performance of a model that would suggest much more boundaries than expected.

Window-based metrics P_k (Beeferman et al., 1999) and WindowDiff were both proposed to address the limitations of the F-measure for topic segmentation. They are both based on a sliding window of size k that moves through the dialogue, and they compare the predicted and gold segmentations within this window. P_k considers that there is a disagreement when the predicted and gold segmentations do not agree on whether the two ends of the window belong to the same segment or not. Beeferman et al. (1999) recommend to set the size of the window k to half the average topical segment length.

While P_k solves the strictness issue of the classical F-measure, Pevzner and Hearst (2002) point that it still over-penalises near-misses, and that the metric varies heavily depending on the topical segment size. They also report that false-positive, i.e. missing boundaries, are also more penalised than false-negatives, i.e. extra boundaries, though we consider this to be a desirable property for our task, as we explained when introducing F_k. They thus propose WindowDiff, which is similar to P_k but considers that there is a disagreement when the predicted and gold segmentations do not agree on the *number* of boundaries within the window, which allows it to be more stable across segment size distribution, to penalise near-misses less than other errors, and to balance the penalties for false-positives and false-negatives. They keep the same recommendation as Beeferman et al. (1999) for the size of the window, which is to set it to half the average topical segment length.

Both metrics are calculated as the ratio of the number of disagreements over the total number of windows, which is $N - k$ where N is the total number of utterances in the dialogue. Thus, a lower score indicates a better performance for these metrics, which is the opposite of the F-measure where a higher score indicates a better performance.

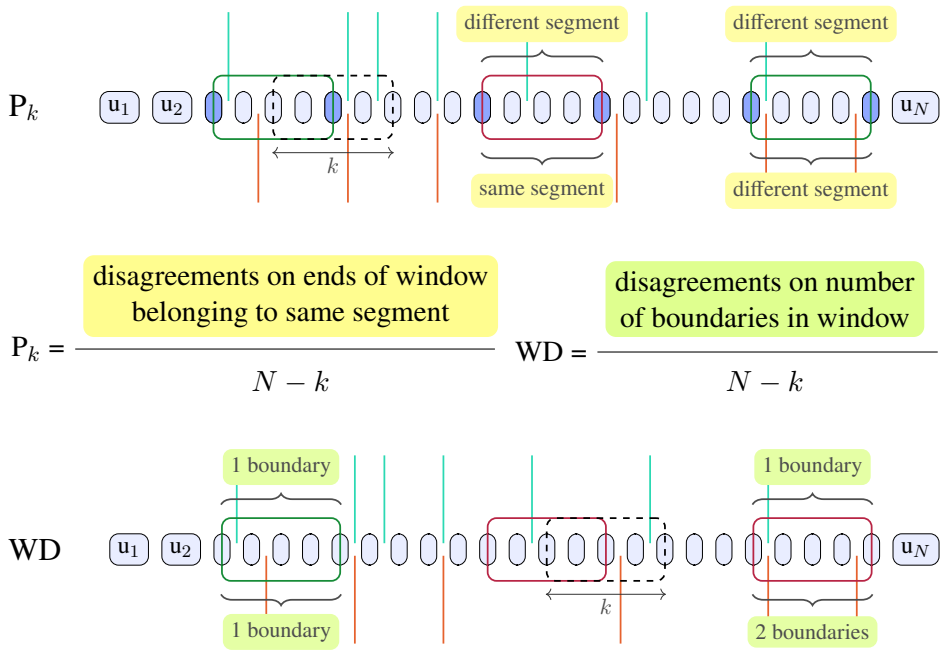


Figure 5.8: Explanation of the P_k and Window Diff (WD) metrics for topic segmentation.

5.3 Dataset

Though our ultimate goal is to analyse real-life conversations, we must also take into account the requirements raised by the DTS task when selecting our corpus. While the task itself does not require any specific type of data, we are in practice constrained by the method we chose to approach it, which is based on ML tools. This means that we need a sufficient amount of data to train and evaluate our models, and that this data must be annotated with topic boundaries. However, as we explained in Section 4.2, there is a lack of annotated datasets for the DTS task, which makes it difficult to find suitable data to work with. We thus chose to circumvent the need for topical annotations by looking for a dataset where the conversations had some kind of *explicit* structure which we could use to infer the topical segments. There is also a language requirement as we rely on BERT embeddings, which despite being multilingual, are still more efficient on English data. Nevertheless, as stated in Chapter 4, we initially chose to focus on languages that we have a good mastery of, which restrains us to English or French data, two well-resourced languages.

We can thus represent the requirements for our dataset as follows:

$$\text{(5.i)} \quad \left[\texttt{Frame}: \begin{bmatrix} \texttt{Language.Global.List}: \{\text{EN}\} \\ \texttt{Purpose.TopicalStructure.Scenario}: \textit{Explicit} \end{bmatrix}\right]$$

When compared with the requirements we defined in (4.i) (Section 4.3), we can see that the only possible incompatibility comes from the `Frame.Purpose.TopicalStructure` parameter as it is one that is affected by the other core parameters. In particular, it is possible that an open conversation with multiple participants makes it very unlikely that the topical structure of the conversation is explicit, at least in a spontaneous setting. The dataset we selected in the end is indeed not a spontaneous one: For these experiments we use transcriptions of the English episodes of the TV show *Friends*. However, it still contains open-domain conversations with multiple participants, which makes it a good compromise between the requirements of the DTS task and the ones of our research questions. Studies have indeed shown that spoken language in fiction is quite similar to spontaneous speech (Forchini, 2009). Transcripts for all ten seasons (236 episodes), annotated in scenes and with additional notes, were used for the Character Mining project (Y.-H. Chen & Choi, 2016) and their dataset is available online[15] (Apache License, Version 2.0).

In this section we begin by giving a detailed description of the type of dialogues that constitute our dataset using CHIMERA (Section 5.3.1) and then we explain how we segmented the dialogues into topics (Section 5.3.2) without actually performing annotations.

5.3.1 Detailed description

The Friends corpus was chosen for the large amount of data it provides with 236 episodes of daily-life conversations, often involving multiple participants, in a variety of contexts and situations. The main characters are six friends who live in New York and interact with each other and with other characters in a variety of situations. These dynamics are quite similar to the ones we can find in real-life conversations, which makes them quite interesting for our purposes. Table 5.3 describes the corpus along the dimensions of CHIMERA, highlighting the differences between the original video data and the transcriptions we rely on for our experiments. We also stress the similarities between the the *Friends* corpus and natural conversations, as well as the differences between the two, which are important to keep in mind since we are ultimately interested in real-life conversations.

When it comes to the AGENTS dimension, there is no difference between the original data and the transcriptions. However, there are differences compared to natural conversations, due in particular to the fact that the show is scripted and that the agents are therefore actors who perform the dialogues rather than real people who engage in them. We should thus assume that there is a high level of control over the way things are said and the way the conversations are structured, which is quite different from naturalistic conversations where there is no such control.

[15] https://github.com/emorynlp/character-mining

Overall the TV show nature influences the degree of structure and control in the dialogues, which are rather chaotic by design, while naturalistic conversations are so because of the natural dynamics of human interactions. This is particularly visible in the CONVERSATIONAL FRAME meta-parameter, and as a result also in the VERBAL INTERACTION, which are highlighted in orange in Table 5.3. The control over the language means that there is little, if any, code-switching or use of non-standard language in the show, unless it is used for comedic effect. This is due to the actual purpose of the show, which is to entertain an audience, where conversations are used as a means to create humour rather than being the goal in themselves. This factor possibly makes the conversation more easily understandable from an external point of view than naturalistic conversations. When building a topic segmentation system, it means that despite the complex nature of these multi-party open-ended conversations, they are most likely easier to work with than real daily-life ones. In terms of verbal interaction, the order of the turns is mostly decided in advance, though the fact that we rely on transcriptions of the episodes rather than the original scripts means that we have access to the little bits of improvisation that actors added to the show. However a drawback of using the transcription rather than the multimodal data is that it is mostly linearised in terms of turn-taking and that it is complicated to know how much of the coordination signals that may be present in the original data were actually kept in the transcription.

Lastly he CONVERSATIONAL SETTING is the meta-parameter that is closest to naturalistic conversations, which is highlighted in purple. The fact that the show is set in a fictional world does not mean that the conversations are not realistic, as they are based on real-life situations and interactions. The characters are also designed to be relatable and to have a long history with each other, which makes their interactions more natural. It is however the dimension that is most affected by the fact that we work with transcriptions rather than the original data, which is highlighted in teal. It means that we do not have access to the visual and auditory cues that are present in the original data, which are important for understanding the conversations and for segmenting them into topics. For example, we do not have access to the gaze and gesture signals, which would give cues about the topic of the conversation that are not present in the linguistic content only. While some notes are provided in the transcriptions when important changes happen in the physical context, such as a change of location or the entrance of a new character, we did not include them in our experiments in order to compare the different models on a common ground, which is the linguistic content of the dialogues. This is also more realistic for the DTS task as it is more likely that, in a general context, we would have access to the linguistic content of the conversations than to the visual and auditory cues, which are more difficult to capture and to process. Indeed, while multimodal datasets have received more attention in the last few years, they remain complex to gather and to work with. We thus wanted our approach to be as easily usable as possible on other datasets in order to set a baseline for future work on

Table 5.3: Description of the *Friends* corpus for a topic segmentation task.

Meta-Parameters	Sub-parameters	**Original**	**Transcription**
Agents			
Agency		Human	
Participation structures	Simulated	6 main participants + others (variable)	
	Workers	unknown number of script writers	
	Functions	intended external viewers	
Social dynamics	(Conversational) History	friends with long history + growing history with viewers throughout seasons	
*Enunciative status**	Role	character role in the show	
	Commitment level	for the fictional context	
Conversational setting			
Epistemic grounding	World setting	Fictional, possibly fiction in fiction	
Environment & Medium	Physical context	mostly co-located	some notes (removed for DTS)
	Gaze	mostly face-to-face	✗
	Gestures / Facial expression	✓	✗
	Prosody / Tone	✓	✗
	Noise	controlled	✗
	Synchrony	scripted	
Conversational frame			
Language	Global	English	
	Control	highly controlled	
	*Vocabulary**	wide audience, daily life	
Purpose	Practical goal	viewer's entertainment	
	Success condition	viewer's oriented	
	*(Topical) structure**	∼explicit through scene changes	explicit through annotations
Verbal interaction			
Signals		scripted	limited, no pauses
Turn-taking		scripted	linearised
Completion		scripted	

DTS that can be easily compared to and built upon.

5.3.2 Segmentation in topics

The *Friends* dataset is not annotated in topics but we chose to automatically infer the segmentation from its segmentation in scenes. Additionally, we consider that the notes in the transcripts indicate an important enough change to create a topic shift. As a result, our assumption is that the topic boundaries coincide with the notes and change in scene. This annotation method is far from perfect but it has the advantage of being objective.

(5.1) is an extract of our dataset. We can see that five different characters appear in this short extract, as well as what we consider as two different topics as there is a note between the second and third intervention of this extract. The note explains that a new character enters the room, which is a sufficient disruption to create a topic shift. However in practice, the first speech turn after the note remains on the previous topic and the shift happens right after. This is quite common in the dataset and not unexpected based on real life dialogues, especially when they involve many people. Moreover, the format of the transcriptions is such that concurrent events and/or speech turns are written down in a given order and overlaps are not represented as we discussed in Section 5.3.1. This further justifies the metrics we use on top of the classical F-measure as discussed in Section 5.2.2, since judging a segmentation based on the exact position of the boundaries is not an accurate reflection of the actual performance of the model on this task.

(5.1) **Joey Tribbiani:** Strip joint! C'mon, you're single! Have some hormones!
Ross Geller: I don't want to be single, okay? I just... I just- I just wanna be married again!
(*Rachel enters in a wet wedding dress and starts to search the room.*)
Chandler Bing: And I just want a million dollars!
Monica Geller: Rachel?!
Rachel Green: Oh God Monica hi! Thank God! I just went to your building and you weren't there and then this guy with a big hammer said you might be here and you are, you are!
[Friends, Season 1, Episode 1]

In total our corpus comprises 236 episodes, each segmented in topics based on the changes in scenes and the notes, which gives us an average of 23.6 segments per episode and 13.2 utterances per segments. In comparison with the datasets used by Xing and Carenini (2021), our dialogues are on average four to five times longer and the topical segments span twice to four times as many utterances as summarised in Table 5.1. Table 5.4 gives a more detailed overview of the statistics of our dataset, which shows that there is a lot of variability in the number of segments per episode,

Table 5.4: Statistics of the *Friends* dataset used for topic segmentation.

Season	#Episodes	#Segments / Episode	#Utterances / Segment	#Tokens / Utterance
1	24	36.7±11.6	6.6±6.2	13.4±13.5
2	24	11.2±2.5	19.6±14.7	14.1±14.0
3	25	27.0±9.5	9.3±9.1	13.8±13.5
4	24	28.8±11.4	8.9±8.3	13.6±13.5
5	24	30.5±5.4	8.2±8.1	13.3±12.9
6	25	27.2±10.1	9.2±7.8	13.8±13.3
7	24	25.7±7.6	9.9±8.6	13.3±12.6
8	24	22.4±8.8	11.1±9.3	13.6±12.9
9	24	11.8±3.5	20.6±19.8	14.8±13.7
10	18	11.2±3.0	24.0±22.3	13.2±12.4
All	236	23.6±11.8	13.2±14.9	13.0±13.6

as well as in the number of utterances per segment, which makes the DTS task more challenging and justifies once more the variety of metrics we use to evaluate the different models on this task. One of the shortcomings of P_k was indeed that it is heavily influenced by the segment size distribution, which is quite variable in our dataset, and thus we wanted to use a metric that is more robust to this variability like WindowDiff.

5.4 Description of the baselines: From random guessing to high-cost LLMs

We explore a wide range of baselines for the DTS task, that are presented in this section ordered by increasing computational power. The most basic one is a simple random baseline (Section 5.4.1). Then we try the TextTiling algorithm in two basic versions, one for each of the approaches we evaluate in this chapter, i.e. feature-based and ML-based. Section 5.4.2 presents the results of the original TextTiling algorithm as well as a slightly modified version imagined for dialogue, and Section 5.4.3 shows the results when using a BERT model as similarity metric, as Xing and Carenini (2021), without any fine-tuning and when fine-tuned on their dataset. Finally we leverage open-weights LLMs in a zero-shot manner to produce topic segmentation predictions (Section 5.4.4).

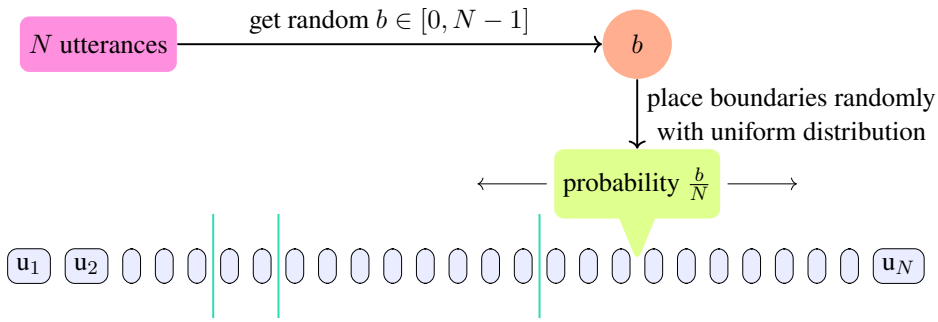


Figure 5.9: Representation of the random baseline for topic segmentation.

Table 5.5: Average results of the random baseline over 10 trials.

Season	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.15	.49	.21	.29	.33	.56	.73
2	.05	.50	.09	.12	.14	.58	.91
3	.11	.50	.16	.22	.26	.58	.80
4	.11	.48	.16	.23	.28	.57	.77
5	.12	.50	.18	.25	.29	.58	.79
6	.11	.49	.16	.23	.27	.57	.79
7	.10	.50	.15	.21	.25	.58	.81
8	.09	.49	.14	.19	.24	.57	.83
9	.05	.48	.08	.11	.14	.58	.89
10	.04	.48	.07	.10	.13	.58	.92
Total	.09	.49	.14	.20	.23	.58	.82

5.4.1 Random baseline: a basic sanity check

Our most basic baseline is a random one taken from Xing and Carenini (2021). It consists in first randomly drawing an intended number of boundaries b between 0 and $N-1$ where N is the total number of utterances in the dialogue, and then determining at each gap whether there is a boundary or not by placing a boundary with a probability of $\frac{b}{N}$. This baseline very simple and constitutes a good sanity check for the other models, as any model that does not perform better than this cannot be considered as being based on or having learned any relevant signal for the task.

Table 5.5 shows the average results of the random baseline over 10 trials. As expected it performs quite poorly overall but it is rather stable. The recall lies around 50% and is the metric with the most variation across the ten trials as indicated by the standard deviation in the full version of the table in Table B.1. This is expected

Table 5.6: Average results of the *smart* random baseline over 10 trials.

Season	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.15	.15	.15	.33	.45	.46	.49
2	.05	.05	.05	.14	.22	.48	.52
3	.11	.11	.11	.25	.36	.46	.50
4	.11	.11	.11	.27	.38	.46	.50
5	.12	.12	.12	.29	.39	.46	.50
6	.11	.11	.11	.26	.37	.46	.50
7	.10	.10	.10	.24	.34	.47	.51
8	.09	.09	.09	.22	.33	.47	.51
9	.05	.05	.05	.13	.20	.47	.50
10	.04	.04	.04	.11	.19	.48	.52
Total	.09	.09	.09	.23	.32	.47	.51

since we randomly draw the average number of boundaries. If the value of b is high we will likely have a lot of boundaries and thus increase the chances that some of them are correct, which will increase the recall. However, it will also increase the number of false positives and thus decrease the precision, which is the worst performing metric for this baseline. P_k is the second best performing metric with an average of .58 (the lower the better) over all the seasons, which seems to support Pevzner and Hearst's (2002) criticism as this random baseline does not rely on any relevant signal for the DTS task. This will be further discussed when comparing the results of the other models to this baseline.

To further investigate the gain brought by the approaches we want to evaluate in this chapter, we also ran a *smart* version of the random baseline where b is not chosen at random but is set to the actual number of boundaries in the gold segmentation. This allows us to have a better idea of the differences in performance depending on whether the model must determine the right number of boundaries or not. The results of this smart random baseline are shown in Table 5.6. All the results except for the precision are significantly different between the two random baselines. In particular the F_1 is about 1.5 times better for the first random baseline, which is due to a higher recall thanks to the higher chances of overgenerating boundaries. On the contrary, the F_{k1} and F_{k2} are better for the smart baseline, rising from .20 to .23 and from .23 to .32 respectively, which is likely due to the higher weight of precision in these metrics. The P_k and WD are also better for the smart baseline, falling from .58 to .47 and from .82 to .51 respectively, showing that restricting the number of boundaries allows for a sharp increase in performance even for a random model.

5.4.2 Dialogue adapted TextTiling: a low-level feature-based model

The TextTiling algorithm has a Python implementation[16] available in the NLTK library which we used as a base for our experiments. We experimented with two main aspects of the original algorithm. The first one is the way the text is split in spans prior to the computation of similarity scores, where we investigated the influence of their length on the performance of the model, and whether splitting the dialogues based on the utterances boundaries rather than ignoring the original structure was more relevant for dialogue. The second one was related to the vocabulary introduction similarity method, which we implemented completely as it did not exist in the NLTK implementation, and we tested an additional memory parameter m that would allow us to consider a token as never before seen if it did not appear in the last m utterances rather than in the whole dialogue, which we thought could be a very basic way to account for the fact that in a long dialogue, a topic can be resumed after talking about something else.

Influence of the spans length As we explained in Section 5.1.1, one of the preprocessing steps of the algorithm consists in splitting the text in spans of w tokens. The default value of this parameter is set to 20 in the NLTK implementation but we wanted to test several values for this parameter to see if it had an influence on the performance of the model. We tested values of w between 5 and 50 with a step of 5 which seems reasonable as very short spans are likely to be little informative and very long ones are more likely to bridge over actual topic shifts. We also tested the values 13 and 26 as they correspond to the average number of tokens per utterance in our dataset, which is around 13, and to its double, which is around 26. As a sanity check we also included $w = 1$ which corresponds to the case where the text is split into tokens which allows the model to place boundaries between any two tokens, thus providing us with a baseline in terms of recall, while actually accessing very little information for the similarity computation, which should lead to a very low precision.

The average results for the 10 seasons can be seen on Figure 5.10a. The complement to 1 of the P_k and WindowDiff metrics is plotted instead of their actual value to make the comparison with the other metrics easier. We can see that the performance of the model varies with the value of w and that the metrics display different behaviours. P_k and WindowDiff get better as w increases, with no significant differences between $w = 40$ and $w = 50$. The F-measure metrics on the other hand get better as w increases up to a certain point and then start to decrease, with the best performance around $w = 20$ or $w = 25$. The precision and recall can explain this behaviour as the recall is quite high for low values of w and then decreases as w increases, while the precision is quite low for low values of w and then increases as w increases. This is expected as with low values of w the model

[16] https://www.nltk.org/_modules/nltk/tokenize/texttiling.html

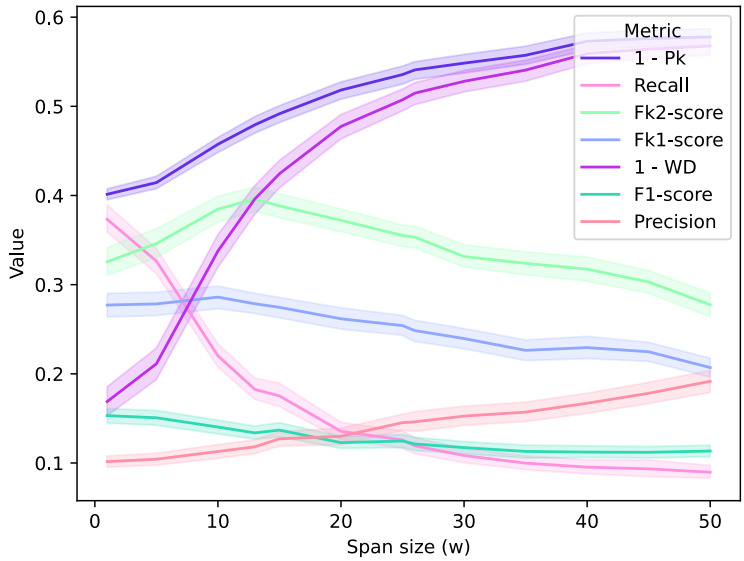
can place boundaries between any two tokens and thus has more chances to place boundaries in the right places, which increases the recall, but it likely places much more boundaires and thus more wrong ones which decreases the precision. We thus decided to set w to 20 for the rest of our experiments as it seems to be a good compromise between the different metrics.

Splitting along utterances The original TextTiling algorithm splits the text in spans of w tokens which can lead to spans that include partial utterances in our case. We thought that this was not the most relevant way to split our conversations because it ignored their structure. It would also be more inconvenient when including new features in the similarity computation that would rely on this structure. For this reason, we implemented a version of the algorithm that splits the text in sequences of utterances such that the number of tokens is the closest possible to w. In practice, most of the spans contain only one utterance if w is not too large, and they can contain more when the utterances are very short and are thus less likely to be informative. We performed the same tests as for the original version of the algortihm, i.e. we tried the same values of w between 1 and 50.

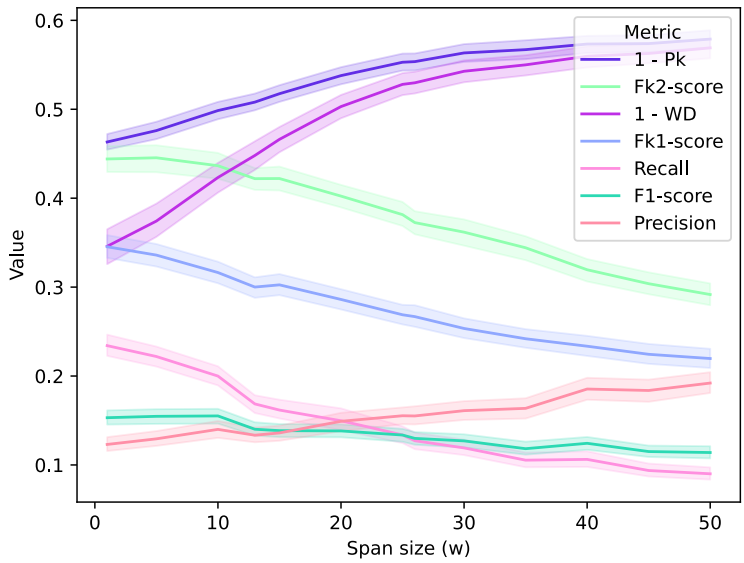
The results can be seen on Figure 5.10b. We can see that the performance of the model is quite similar to the original version, with the precision and recall being slightly more stable with small values of w. This is completely expected as with our version of the span segmentation, the spans must at least be a full utterance long. Hence, the smallest values of w actually produce very similar span segmentations. When looking at the results in detail, the dialogue adapted method is always better than the original one when the results are significantly different – albeit often by a slight margin – except for the recall with $w = 5$ and $w = 1$ because of the higher opportunities to place boundaries with the original method. More detailed results including t-tests can be found in Section B.1.2. We thus decided to use our version of the span segmentation for the rest of our experiments where we use $w = 20$ as decided before.

Implementing the vocabulary introduction similarity method: testing a memory parameter Since the vocabulary introduction method was not implemented in the NLTK library, we implemented it ourselves and tested it on our dataset. As a reminder, this method uses the portion of never before seen words in the two spans around a gap as a similarity score, with the idea that a topic shift is likely to be accompanied by the introduction of new words. However, in a long dialogue, a topic can be resumed after talking about something else, and thus some words that were introduced at the beginning of the dialogue and are thus not new anymore can be reintroduced at the moment of the topic resumption and should be considered relevant for the detection of this topic shift.

For this reason, we added a memory parameter m that allows us to consider a word as new if it did not appear in the m spans before the two considered ones.



(a) Original algorithm



(b) Algorithm with dialogue-adapted spans

Figure 5.10: Average results of the TextTiling algorithm with different values of the spans length parameter w and the block comparison similarity method, evaluated on the 10 seasons of our dataset. *Higher is better.*

We experiment with this parameter by testing values of m between 5 and 50 with a step of 5, the results can be seen in Figure 5.11a. We also tested the model with an infinite memory as was suggested in the original algorithm, represented at point 0. Surprisingly, there are no significant variations in the performance of the model with the different values of m. This suggests that most topic shifts are identified thanks to the introduction of new words that were not used in the dialogue before, and that topic resumptions are not so frequent or use a new vocabulary when they happen. This could be a consequence of using scripted dialogues and should be further investigated in other forms of dialogue, especially in naturally occuring ones, as re-introducing a topic after some time is likely less cognitively demanding when the same vocabulary is used.

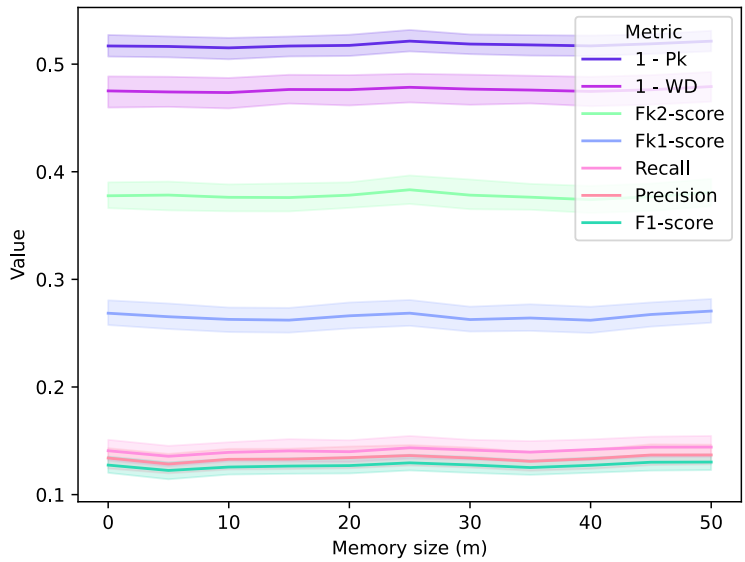
We thus dropped the memory parameter for the rest of our experiments and considered only the original version of the vocabulary introduction method where a word is considered new if it did not appear in the dialogue before. The average results on the 10 seasons can be seen in Figure 5.11b and are similar to the block comparison approach. For the same reasons we thus set the span length at 20 tokens for the following experiments.

5.4.3 BERT for next-sentence prediction: generalisability assessment

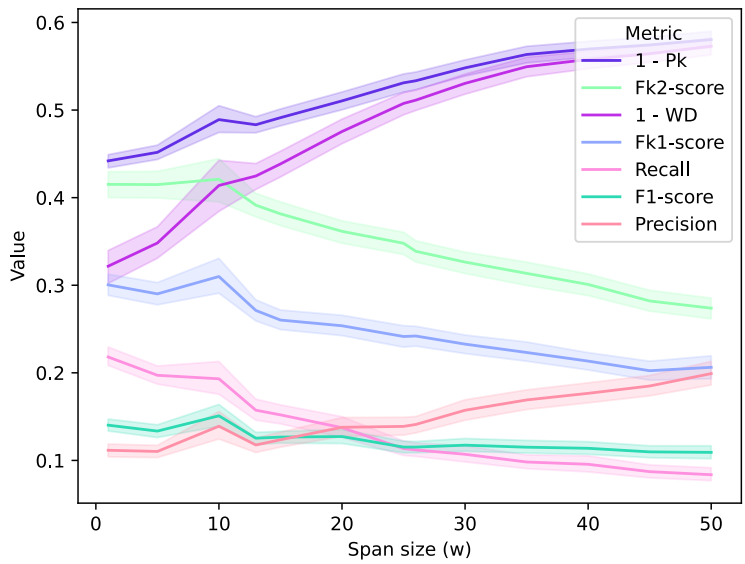
As a reminder, the model trained by Xing and Carenini (2021) is a similarity metric meant to be used in the TextTiling algorithm to perform the DTS task. Thus, the training of the model is not directly performed on the DTS task but rather on an utterance pair coherence scoring task. We use two versions of this model as baselines, one without any fine-tuning and one fine-tuned on the DailyDialog dataset as in Xing and Carenini (2021).

Without any fine-tuning, the model provides slightly better results than the random baseline but slightly worse than the smart random baseline as can be seen on the first row of Table 5.7.

When fine-tuned on DailyDialog and evaluated on the 10 seasons of our dataset, the 10 epochs of training do not change the results much. In particular, the F-measures values did not vary significantly between epochs but recall varied between .39 and .48 with significant variations on epochs 1 to 4 compared to all the others, which led to differences in the P_k and WD scores in particular at epoch 1, 4, and also 2 specifically for WD. As with the other BERT-based models, the low precision and much higher recall indicate an over-generation of topical boundaries, which explains the results on the P_k and WD scores. The results are quite low overall and pretty similar to those obtained with the most basic random baseline, which suggests that the model trained on DailyDialog does not generalise well to our dataset. This is not very surprising given the differences between the two datasets in terms of dialogue structure and content. DailyDialog is indeed a dataset of manually written dialogues, which makes them much cleaner and organised than



(a) Original algorithm



(b) Algorithm with dialogue-adapted spans

Figure 5.11: Average results of the TextTiling algorithm with the vocabulary introduction similarity method, evaluated on the 10 seasons of our dataset.

Table 5.7: Results of 10 epochs for the model trained on DailyDialog and evaluated on the 10 seasons of our dataset.

Epoch	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
Before fine-tuning	.11	.41	.16	.22	.28	.56	.78
1	.10	.39	.15	.20	.25	.58	.80
2	.09	.42	.15	.20	.24	.59	.83
3	.10	.48	.16	.21	.25	.59	.86
4	.09	.41	.15	.20	.25	.59	.82
5	.10	.46	.16	.20	.25	.59	.85
6	.10	.44	.15	.20	.25	.59	.84
7	.09	.45	.15	.20	.25	.60	.86
8	.09	.45	.15	.20	.25	.60	.86
9	.09	.45	.15	.20	.25	.60	.86
10	.09	.45	.15	.20	.25	.60	.86

the ones in the Friends dataset.

A more striking result is the fact that the model without fine-tuning performs significantly better than at any of the 10 epochs of training on DailyDialog, which suggests that the training on DailyDialog actually made the model worse for our dataset, most likely diminishing its generalisation abilities. However, this result is not so surprising when considering the differences between the datasets NSP-BERT is fine-tuned on and DailyDialog. Indeed, NSP-BERT is fine-tuned on the datasets suggested by Gao et al. (2021) for designing 'learning strategies that generalize well to an unsee test set'. The datasets include, among others, the Open American National Corpus (Fillmore et al., 1998; Macleod et al., 2000), which is very diverse and includes notably two-sided in-person conversations. Those are spontaneous and much more complex than the handcrafted dialogues from DailyDialog, explicitly described as not very noisy. Such comparisons make it possible to predict the abilities a model can develop when trained on two different types of datasets and confirms the need to systematically describe the data we work with. All these results also support the idea that more complex training data is needed to perform well datasets such as the *Friends* one. We will investigate this question in Section 5.6 when applying the same fine-tuning procedure on our dataset, which will allow us to have a better idea of the influence of the training data on the performance of the model.

5.4.4 LLMs: a brute-force high computational power approach

With the rise of LLMs in the recent years, an increasing amount of tasks are being solved with these models, with more or less sophisticated approaches ranging

```
Below is the script of one episode of the TV show Friends.
Each row corresponds to a speech turn in the dialogue and is
formatted as follows: "ID SPEAKER: UTTERANCE". Please provide
a topic segmentation of the episode by grouping together
consecutive speech turns that belong to the same topic. The
output should be a list of turn IDs where a topic change
occurs. For example, the output could be: [5, 7, 12], which
means that the first topic includes turns 0 to 4, the second
topic includes turns 5 to 6, the third topic includes turns 7
to 11, and the last topic includes turns 12 and onwards.
Here is the episode script:

<script>

Please ensure that the output is in the form of a list of
integers like [5, 7, 12] where each integer represents a line
number where a topic shift occurred, and that there is no
additional text or explanation in the output. Thank you!
```

Figure 5.12: Prompt for the zero-shot segmentation with LLMs.

from zero-shot settings supported by prompt engineering to more elaborated ones involving fine-tuning. Their performance on various benchmarks that sometimes claim to prove the general language understanding capabilities of these models (A. Wang et al., 2018, 2019) has generated a great deal of excitement and made them an inevitable approach for most NLU tasks. Yet benchmarks are also criticised for their limitations in terms of *construct validity* (Raji et al., 2021), i.e. the extent to which they actually measure what they are supposed to measure, which is often overlooked due to the complexity to assess it (Bowman & Dahl, 2021).

We thus included LLMs in a zero-shot setting to our baselines to get an idea of their base performance, but we did not explore more complex approaches based on these because they would not *explain* much in terms of dialogue structure due to their black box nature, on top of requiring a high computational power. We prompted several open-weights LLMs of different sizes to segment the dialogues in topics. The input consisted of the transcript to segment where each utterance was numbered encased in instructions as can be seen on Figure 5.12. The prompt explicited the format of the dialogue and indicated the format of the expected output with a commented example of such output. The very end of the prompt repeated the requirement in terms of output format. We include basic politeness formulations in the prompt as there are some indications that they can improve the performance of LLMs, though it remains hard to properly quantify the effects of prompt engineering.

Table 5.8: List of the different LLMs used as baseline for the DTS task. *Note that all the models cited here except DeepSeek were introduced in papers published solely on ArXiV which is a practice we discuss in more details in the last chapter of this thesis (Epilogue).*

Model	Version & Size	Comment
LLaMa (Touvron et al., 2023)	3.1 - 8B 3.2 - 3B	
Deepseek (Guo et al., 2025)	r1 - 1.5B r1 - 8B r1 - 14B	'reasoning' models
Gemma (Team et al., 2024)	3 - 1B	
Mistral (Jiang et al., 2023)	0.3 - 7B	

We used Python and the Ollama library to test the LLMs listed in Table 5.8. We used only open-weights models because they can be run on a machine of our choice[17] and ensure a better reproducibility of our results. We use models with between 1.5 billion and 14 billion parameters, which is relatively modest compared with the largest models available; however, we considered this range to be reasonable given the computational power required in relation to the impact of our task. Furthermore, one of the models we test, `Deepseek`, is used in a 'reasoning' version, which means that during the training of the model, a phase of reinforcement learning is used where it is required to produce a 'reasoning process' before producing the actual answer to the input. However the term 'reasoning' can be rather misleading due to its anthropomorphic connotations (Kambhampati et al., 2026). In practice this additional phase grants the model an increased computation time which can be beneficial to perform the requested task but it also increases the risk of the model going off track and producing irrelevant content (Ballon et al., 2026), which we have indeed observed in our experiments.

Forcing a model to produce an output in a specific format is not easy through the prompt alone. In our first trials, many of the outputs were not in the expected format and thus could not be evaluated directly. Some results contained a properly formatted list of integers but also additional text, and others contained a sequence of single integers encased in brackets, e.g. '[12] [28] [34]', or sequences of lists. These patterns were more or less straightforward to match with an actual output but others would have required going manually through the results to fix them. We thus chose to leverage the functionality of the Ollama library that allows us to specify

[17]These experiments were carried out using the Nancy site of the Grid'5000 testbed, supported by a scientific interest group hosted by Inria and including CNRS, RENATER and several Universities as well as other organisations (see `https://www.grid5000.fr`).

the expected format of the output through a JSON schema. This way, the model is forced to produce a list of integers as output, and if it does not, an error is raised. This allows us to easily evaluate the outputs. We set the temperature of the models to 0 for these experiments, which means that the model will always choose the most likely token at each step of the generation, which is a recommendation when forcing the structure of the outputs. It also makes the experiment fully deterministic and hence reproducible.

The results of these experiments are shown in Table 5.9. Some of the models still produced outputs in the wrong format which in practice raised an error and there was no segmentation for the given episode. The portion of episodes with such issues is reported in the column NaN. Both `gemma` models were unable to produce any output in the right format for any of the episodes, which is why no values are provided in Table 5.9. The other models produced outputs in the right format for most of the episodes. However some of these outputs were actually empty, i.e. no topical boundaries were reported, which is reported in the column Empty. `Deepseek 8B` produced almost half of its outputs as empty, `llama 3.2 3B` about 8% while the other ones produced none. Nevertheless, all the models produced segmentations at impossible positions. Indeed, the expected output was a list of integers where each integer is the ID of an utterance where a topic shift occured. All numbers must thus be contained in the interval $[\![0, N-1]\!]$, where N is the total number of utterances in the episode. Yet, the models often produced numbers outside of this interval, most of the time too large but sometimes below 0 as well. The portion of outputs with such issues is reported in the column Length. All the models produced impossible numbers for a large portion of the episodes but surprisingly the smallest `Deepseek` model produced the least amount of such outputs. Given the large amount of impossible boundaries, we chose to simply ignore the numbers outside of the relevant interval when evaluating the models. In some cases the resulting segmentation was empty after filtering out the impossible numbers, which we did not report in this table – the portions in the column Empty do not overlap with these – but all the results can be found in Table B.7.

The results in Table 5.9 thus only take into account the episodes where an output in the right format was produced and with the numbers outside the right interval removed. An empty segmentation resulted in the worst value for each metric. Most of the differences between the models are not significant, which is likely due to the small data samples we retain after filtering out all the segmentations with issues. Only pairs formed by `Deepseek 1.5B`, `Deepseek 14B` and `llama 3.1 8B`, i.e. the three models with the largest remaining data points, display significant differences. Those models are highlighted by a **purple bold** font. The results are quite low overall for the F-measures with a more or less striking imbalance between the precision and recall. However this imbalance is not always in favour of the recall, as opposed to the BERT-based models, which suggests that the models do not all display the same behaviour. The P_k and WD scores are also quite bad though the two

Table 5.9: Average results of the different LLMs on the 10 seasons of *Friends*, the last three columns report the proportion of outputs with issues. *Best results (ignoring the non significant differences) are highlighted in* ***bold****.*

Model	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓	NaN (%)	Length (%)	Empty (%)
`llama3.1:8b`	.11	.24	**.14**	**.30**	**.44**	.58	.71	0	66	0
`llama3.2:3b`	.09	.60	.15	.20	.21	.65	.94	3	68	8
`deepseek-r1:1.5b`	**.18**	.10	.09	.16	.17	**.41**	**.44**	0	23	0
`deepseek-r1:8b`	.18	.02	.03	.04	.04	.68	.68	2	42	47
`deepseek-r1:14b`	.13	**.52**	**.15**	.18	.20	.54	.75	0	66	0
`gemma3:1b`	/	/	/	/	/	/	/	100	0	0
`gemma3:4b`	/	/	/	/	/	/	/	100	0	0
`mistral:7b`	.14	.19	.13	.26	.35	.50	.58	1	92	0

Table 5.10: Summary of the baseline results over all 10 seasons. *Best results in* ***bold****.*

Model	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
Rd	.09	**.49**	**.14**	.2	.24	.58	.82
Rd_{smart}	.1	.1	.1	.23	.32	**.47**	**.51**
BC	.13	.14	.12	.26	.37	.48	.52
BC for dialogue	**.15**	.15	**.14**	**.29**	**.40**	**.46**	**.50**
VI for dialogue	.14	.14	.13	.25	.36	.49	.52

smallest deepseek models have results comparable with the smart random baseline. Nevertheless, given the large amount of outputs with issues, and the computational cost required for this method, these results should be taken with caution. We therefore chose to exclude the LLMs from the rest of our experiments and focus on the other baselines.

Table 5.10 summarises the results for all the baselines we later compare our models to. The performance is quite low overall but the dialogue adapted version of the TextTiling algorithm is the best performing approach. The fact that the smart random baseline performs similarly on the sliding window metrics suggests that determining the right number of boundaries is already a difficult task and that placing them correctly is only part of the challenge.

5.5 Improving the original TextTiling algorithm with Linguistic Features

As explained before we work with two versions of the original TextTiling algorithm, one of them being feature-based, and the other one being ML-based with BERT as a backbone. In this section we discuss the enhancements of the original TextTiling algorithm we worked on. Such a model has the advantage of being explainable and of not requiring any training, as opposed the BERT-based one. We chose to explore two dialogue-specific features – its structure itself with the speakers, and the presence of questions – as well as coreference chains, which is similar to the similarity approach based on lexical chains mentioned in Hearst (1997).

All the approaches we designed are meant to be added on top of a base similarity metric. We can thus apply them on top of the block comparison method or the vocabulary introduction method. We also used the average of the two methods as a base similarity metric to see if it could provide better results.

Section 5.5.1 discusses how we include the speakers' organisation in the computation of the depth scores. Section 5.5.2 describes how we take questions into account in the algorithm, and Section 5.5.3 explains how coreference chains were used.

5.5.1 Speakers: taking advantage of the dialogical structure

Participants' organisation can greatly influence the topical structure of a conversation (Nguyen et al., 2012). While we chose not to include the names of the speakers in the text used as input for our task to focus on the lexical content, we wanted to find a way to take their organisation into account in the algorithm. We thus designed two approaches to include this information in the depth scores computation, one where we only consider the exact utterances where new speakers are introduced, and one where we also adjust the scores around such utterances.

Focus on speaker introduction Given that the complete set of participants is almost always the same in an entire episode, considering only the introduction of new speakers at the scale of the episode would not be very informative. We thus use a more local approach where we consider the introduction of new speakers at the scale of a block, which is the unit used in the block comparison method. For each gap i, we compute the number of speakers in the span after the gap that did not speak in the block of spans before the gap. We add this number to n times the existing depth score of the gap and normalise by dividing by $2 \times n$ where n is the total number of speakers in the episode. This way, the depth score of a gap will be increased if it is followed by the introduction of new speakers, and the more new speakers are introduced, the more the depth score will be increased.

Table 5.11: Average results of the feature-based models enhanced with speaker information. Sp_{sharp} corresponds to the approach where only one score is changed when new speakers are introduced in a span, and Sp_{smooth} corresponds to the approach where the direct neighbours are updated as well. *Best results in* ***bold***.

Approach	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
BC	.15	.15	.14	.29	.40	.46	.50
VI	.12	.12	.11	.23	.33	.50	.53
BCVI	.14	.13	.12	.25	.36	.48	.52
BC + Sp_{sharp}	.15	.15	.14	.29	.41	.46	.49
VI + Sp_{sharp}	.12	.12	.12	.23	.34	.50	.53
BCVI + Sp_{sharp}	.14	.14	.13	.25	.37	.48	.51
BC + Sp_{smooth}	**.18**	**.18**	**.17**	**.36**	**.48**	**.42**	**.46**
VI + Sp_{smooth}	.15	.16	.15	.31	.43	.45	.49
BCVI + Sp_{smooth}	**.17**	**.18**	**.16**	.34	**.46**	**.43**	**.47**

The normalisation allows us to keep the depth scores between 0 and 1, which is important for the next steps of the algorithm.

Smoother changes In this approach, we raise the depth score of a gap as well as its direct left and right neighbours by 30% if the gap is followed by the introduction of a new speaker. This way, we consider that the introduction of a new speaker is an important enough change to increase the depth scores of a gap but we do not consider that the number of new speakers matters. We chose to also increase the depth scores of the direct neighbours of the gap because the new speaker may not be the first one to speak after the gap. Our approach is quite basic and could be improved by applying a gaussian increase around the gap instead of a flat one like we did but the goal is to assess which features are more relevant for this task before further work could be done on the way to include them in the algorithm.

Table 5.11 shows the results of these two approaches. The first approach where only one score is changed when new speakers are introduced does not improve the results significantly, while the second one where the direct neighbours are updated as well provides a significant improvement in all the scores. This suggests that the introduction of new speakers is indeed an important change that can indicate a topic shift, but that the utterance where the speaker is introduced is not necessarily the exact place of the topic shift. This makes sense in the context of the *Friends* dialogues as a new participant could barge in the conversation and change the topic, but a new scene can also start after a transition that would be obvious for the viewers and where the topic shift would not necessarily be due to the arrival of the new participant(s). The former topic shift would happen due to an interuption of the new

participant while the latter one would be a natural transition between an exhausted topic and a new one.

5.5.2 Questions: a second interactional feature

Questions are a very complex phenomenon in dialogue and can have different functions (Cruz Blandón et al., 2019). However we took a very simple approach to include them in the algorithm with the assumption that a topic shift would not directly follow a question. This hypothesis is rather naive but we decided to see what results a very basic implementation could produce. We thus simply smooth the gap scores that follow questions by setting them as the average of the gap score before the question and the gap score after the question.

Our question identification method is quite basic as well, as we simply consider that an utterance is a question if it ends with a question mark. In practice this method relies heavily on what the transcribers considered as questions, which can be subjective. Requests, which can also be considered as a type of question, can be formulated as statements and thus would not be taken into account with this method.

This basic approach seemed to slightly improve the results, though not significantly as shown in Table 5.12. Given that it did not complexify the algorithm, we decided to keep it in the end in case it would interact more strongly with other features. However it could be interesting to further explore the role of questions in the topical structure of dialogues and to find more sophisticated ways to include them in the algorithm.

5.5.3 Coreference chains: leveraging more sophisticated lexical information

A coreference chain is a set of mentions that refer to the same entity in a text (Schnedecker, 1997). For example, a character can be referred to by their name, a pronoun or a description, and all these mentions would be part of the same coreference chain. Our assumption was that a topic shift is less likely to exist inside a coreference chain. For this reason, we smoothed the gap scores between the beginning and the end of each coreference chain.

We used the *coreferee* Python library to compute the coreference chains in our dataset. This library is applicable to English, French, German, and Polish, and combines rule-based and machine learning approaches to identify coreference chains in a text. It uses a set of rules to exclude impossible coreference links based on grammar, e.g. in 'The house stood tall. They went on walking.'[18], the algorithm would not link 'the house' and 'they' because they do not agree in number. It also

[18]Example taken from the documentation of the coreferee library.

Table 5.12: Average results of the feature-based models enhanced with various approaches.
Block comparison (BC), vocabulary introduction (VI), coreference chains (Co), smoothspeaker introduction (SI), questions (Q), stemming (S), lemmatisation (L). Single worse model in gray.

Approach	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
BC	.15	.15	.14	.29	.40	.46	.50
BC + SI	.18	.18	.17	.36	.48	.42	.46
BC + SI + Q	.18	.18	.17	.36	.48	.41	.45
BC + SI + Q + L	.18	.18	.17	.36	.48	.41	.45
BC + SI + Q + S	.18	.18	.17	.36	.48	.41	.45
BC + SI + Q + Co	.19	.16	.16	.37	.48	.40	.44

uses a machine learning model to identify coreference links based on pragmatic tendencies. This model relies on syntactic and morphological information provided by *spaCy* models, as well as manual annotations of coreference chains. *SpaCy* is a classical NLP library designed for processing natural language texts with features that include tokenisation, part-of-speech tagging and dependency parsing among others. The coreference chains are computed based on the dependency parsing of the text, which allows to identify the mentions of each entity and to link them together in a coreference chain. Unfortunately, this algorithm requires a long time to compute the coreference chains and its time complexity rises with the length of the input text. For this reason, we computed the coreference chains at the block level, which is the unit used in the block comparison method, instead of at the episode level. This way, we were able to compute the coreference chains in a more reasonable time and to include them in our algorithm.

As with the questions, this approach did not significantly improve the results (see Table 5.12) so we dropped it for the rest of our experiments due to its computational cost. While it is possible that chains identified at the dialogue scale would have had a stronger influence, the time required to compute them seems prohibitive for such a task.

5.5.4 Summary of the Results

We summarise the results with our different approaches to improve the TextTiling algorithm in Table 5.12. We report only the results of the models based on the block comparison method as it is the one that provides the best results as a base similarity metric, but we observed similar trends with the models based on the vocabulary introduction method and the average of the two methods. These can be found in Section B.2.

The best results are obtained with the block comparison method augmented by the smoother speaker introduction feature. Including questions or coreference chains does not significantly improve the results. We can also see that stemming or lemmatising the text does not improve the results. This could be due to the data being artificial in the sense that scenarists may avoid repetitions when it is not for the sake of humour.

5.6 Adapting Xing and Carenini's BERT-based model to our Dataset

In parallel with the feature-based approaches, we also worked on adapting the BERT-based model of Xing and Carenini (2021) to our dataset. We describe this work in this section.

Training the utterance-pair scoring model on our *Friends* dataset raises an issue of data splitting. Indeed, we cannot train and evaluate the model on the same subset of the data, but the more data we use for training, the less representative our evaluation becomes compared to the other experiments. We thus proceeded in two stages, one where we trained the model on nine seasons and evaluated it on the remaining one, and another one where we trained the model on three seasons and evaluated it on the seven remaining ones. The first step gave us an idea of the performance we could get with a more diverse training set and allowed us to determine the reasonable number of epochs to train the model for as well as the organisation of the coherence layers in the model (Section 5.6.1), while the second step provided a fairer comparison with the other models (Section 5.6.2).

5.6.1 Large training set: Learning curve and coherence layers

The main difference between the training of the model on DailyDialog and on our *Friends* dataset from a technical point of view is that our dataset is not annotated with dialogue acts nor in topics, which prevents us from reproducing the exact same conditions as Xing and Carenini (2021) when building the training pairs. We thus chose an approach similar to the one they used for their Chinese dataset, NaturalConv (X. Wang et al., 2021) but without the constraint on sampling dialogues from different topics for the least coherent negative pair.

Instead, we rely on the annotation in scenes, episodes and seasons, as well as the additional notes in the transcripts, to sample what we consider to be topically different segments. As a consequence, our data has more levels of hierarchy than the one used by Xing and Carenini (2021) which allows us to consider more types of coherence layers that are represented on Figure 5.13. In decreasing order of coherence we have:

- Layer A: subsequent utterances within the same note span;

- Layer N: randomly picked non subsequent utterances within the same note span;
- Layer C: randomly picked utterances within the same scene;
- Layer E: randomly picked utterances within the same episode;
- Layer S: randomly picked utterances within the same season;
- Layer D: randomly picked utterances within different seasons.

We thus experimented with different settings in terms of coherence layers. On the one hand, we reproduce the 'adjacent' and 'same dialogue' coherence levels from Xing and Carenini (2021) in all of our experiments by using our [A] and [N] layers. On the other hand, we try different layers for the third coherence level – within the same scene [C] and within different seasons [D], and we also train some models with more layers: [CESD] and [CD].

Table 5.13 show the results when training one model on nine out of the ten available seasons of *Friends* for ten epochs and evaluating on the remaining season. We can see that the results barely vary from one epoch to another. In fact, only the recall, P_k and WD display significant changes across some epochs. The recall is slightly worse at epoch 1 than the others, and even lower at epoch 4, which incidentally makes P_k and WD better at epoch 4 but not at epoch 1. This is due to the large difference between the precision and recall that shows that this model over-generates topic boundaries. Hence, when the recall is lower, it indicates a version of the model that generates less boundaries than the others and thus less false positives. The models display similar variations across all layers settings with either no significant changes across epochs or slight changes on recall, P_k and WD at one or two epochs when recall gets lower. Thus in the rest of our experiments we decided to train the models for two epochs only as it seems that training for more epochs does not provide better results and can even lead to worse ones.

Table 5.13: Results of 10 epochs for the model AND-3. *Single worse model in gray.*

Epoch	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.14	.44	.21	.27	.31	.51	.71
2	.14	.47	.21	.27	.31	.53	.73
3	.14	.48	.21	.27	.31	.52	.73
4	.13	.38	.19	.27	.32	.50	.66
5	.14	.50	.21	.27	.31	.53	.75
6	.14	.51	.22	.27	.31	.52	.74
7	.14	.49	.21	.26	.30	.53	.75
8	.14	.51	.22	.27	.31	.53	.75
9	.14	.51	.22	.26	.30	.53	.75
10	.14	.52	.22	.27	.30	.53	.75

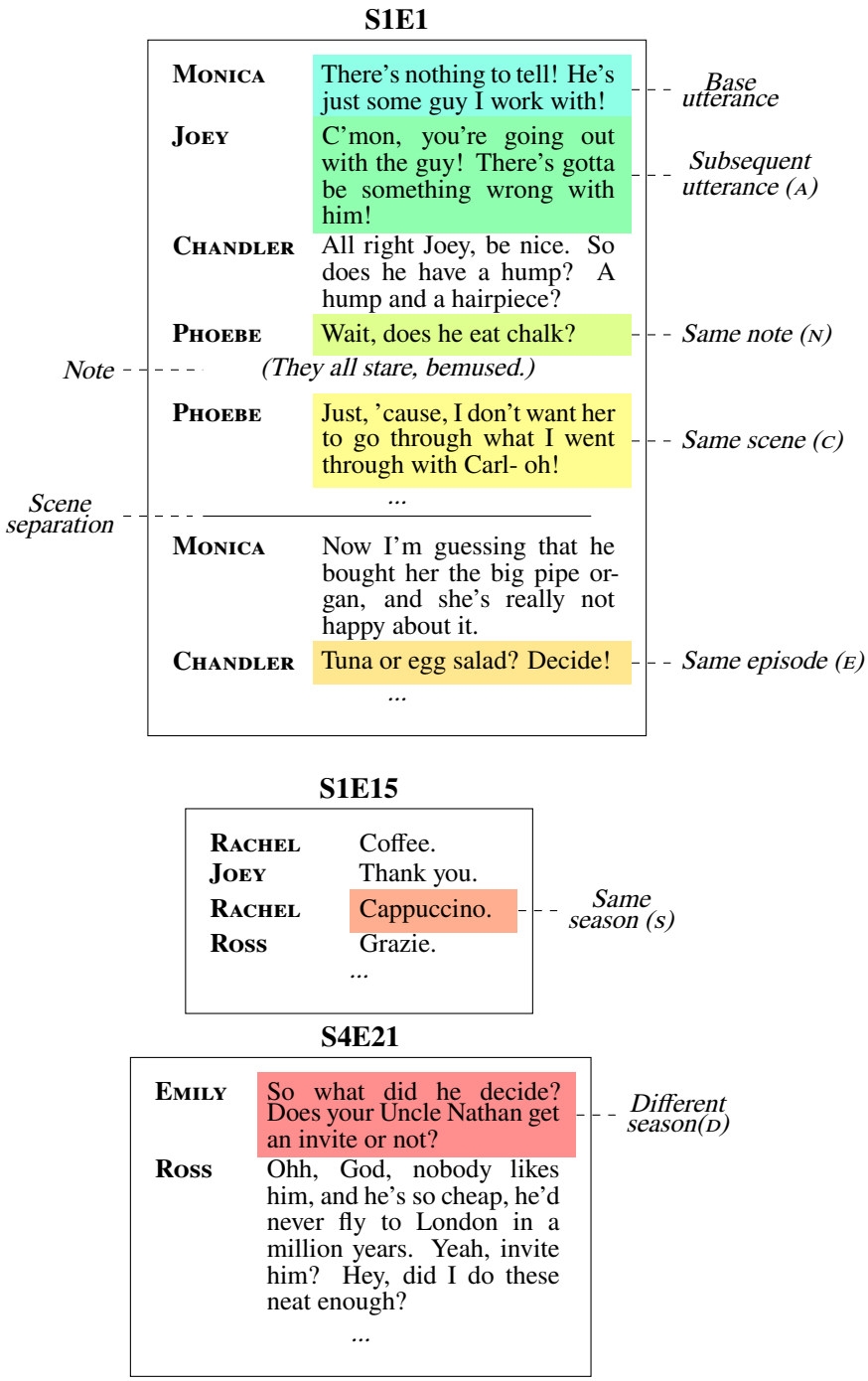


Figure 5.13: Representation of all the possible coherence layers in the *Friends* dataset.

Regarding the different settings in terms of coherence layers, Table 5.14 shows that the results do not vary greatly across the different settings. The best results overall are obtained with the model trained with the [D] layer as the third coherence layer, i.e. with randomly picked utterances within different seasons. However, this model is not the best one with the P_k metric which is likely due to the large difference between its precision and recall, suggesting as we saw before that it overgenerates topic boundaries. The model with the best P_k is the one trained with the [CD] layers, which is also the one with the best WD score, though most of the models have WD scores that are not significantly different from each other. The model with the most layers, i.e. [CESD], performs similarly to the best one, suggesting that having more layers does not necessarily influence the performance of the model.

We also experimented with two repartitions of the data, one where the evaluation is done on season 1 and one where it is done on season 3. An interesting point is that models trained with the same coherence layers but on different seasons do not necessarily display similar results. For example, the models [D-1] and [D-3] display results significantly different from each other with all the metrics. [C-1] and [C-3] have a similar dynamics except for the P_k and WD metrics where [C-3] performs better than [C-1]. [CD-1] and [CD-3] however are very similar except for the P_k metric where [CD-3] performs better than [CD-1]. The fact that the results do not vary in the same way between the models when evaluated on season 1 and season 3 suggests that the differences do not necessarily stem from the evaluation data but possibly from the models themselves and the training process, which may be quite unstable though the differences in performance are not so large. The small evaluation set is also an issue when it comes to comparing the performances of the different versions of the model between them but also with the baselines and the feature-based approach. We further explore these questions in the next section where we train several versions of the model on a smaller training set and with different random initialisations to see how much the results can vary.

5.6.2 Stability and less diverse training set

For a fairer evaluation with the methods that require no training, we trained the model on three seasons and evaluated it on the seven remaining ones. We trained several versions of this model with different training subsets and different random initialisations in order to get an idea of the stability of the results. We chose to work with only one layer setting for this experiment to avoid multiplying the number of models to train and evaluate, and we chose the [AND] setting as it is the one that provided the best results overall in the previous experiment.

Since the sampling of our training pairs involves a random component, two training subset based on the same input data can be different. We thus built three training subsets, α, β, and γ, based on the same three seasons 2, 3, and 4 but with different random seeds to see how much the results can vary with the sampling

Table 5.14: Results for the different layer settings. *The best scores when considering significant differences are indicated in **bold** and the results that are significantly worse than the others are in gray.*
Each model was trained on 9 seasons and evaluated on the remaining one, indicated in the model name <layers>-<evaluation season>.

Model	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
ANC-1	**.19**	**.59**	**.28**	**.36**	**.39**	**.52**	.70
ANC-3	.14	.47	.21	.27	.31	**.53**	.73
AND-1	**.18**	.53	**.27**	**.35**	**.39**	**.52**	.68
AND-3	.14	.50	.21	.26	.29	.55	.76
ANCD-1	.16	.46	.24	**.32**	**.40**	.56	.71
ANCD-3	.14	.49	.22	.29	.32	**.51**	.72
ANCESD-1	**.19**	.49	**.27**	**.36**	**.41**	**.48**	**.63**

of the training pairs. We then trained three versions of the model on each subset with different random initialisations to see how much the results can vary. We trained these models for two epochs only, following our findings from the first set of experiments.

In the end the results were not significantly different between the two epochs for the models trained on subset α, one of the versions trained on subset β had a lower recall at epoch 2 which led to a better WD but not a better P_k score, and similarly for one of the versions trained on subset γ where a lower recall at epoch 2 led to better P_k and WD. Conversely one other version of the model trained on subset γ had a lower precision and recall which led to worse performances on all metrics except F_{k2} which was not significantly different from epoch 1. Nevertheless the results do not vary greatly between epochs when the differences are statistically significant, the stronger changes appearing on WD which performs quite badly overall compared to all the other models we worked with, including our simplest baseline. This is likely due to the overgenerating tendencies of this model. For this reason we display the results of all the versions of the model only at epoch 2 in Table 5.15 to improve readability; the full results are available in Section B.3.3.

The results in Table 5.15 show that the behaviour of the restricted model remains quite similar to the ones trained on nine seasons. However, there is a distinct variability between the models, whether they differ in random initialisation or training subset. This variability enables us to identify the best model, which is $D_{\gamma 1}$. A few models display similar results on some metrics, highlighted with *italics* in Table 5.15 but $D_{\gamma 1}$ performs best overall.

Lastly, we compared these versions of the model to the ones trained on nine seasons instead of three. Only season 1 could be used for evaluation so we compared

Table 5.15: Results of different models trained on three different training subsets (d_α, d_β, d_γ) after the second epoch. *The best scores when considering significant differences are indicated in* ***bold*** *and the results that are significantly worse than the others are in gray.*
Evaluation carried on seasons 1, 5, 6, 7, 8, 9, and 10 only.

Model	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
$D_{\alpha 1}$	.12	.56	**.19**	.24	.26	.56	.84
$D_{\alpha 2}$	.09	.39	.14	.19	.24	.59	.82
$D_{\alpha 3}$	.11	.37	.16	.24	**.30**	.55	.74
$D_{\beta 1}$	.13	.36	.18	.24	.28	.50	.68
$D_{\beta 2}$	.14	.48	**.20**	**.26**	**.29**	.49	.73
$D_{\beta 3}$	.14	.45	**.20**	**.26**	**.29**	.49	.71
$D_{\gamma 1}$	.15	.38	**.21**	**.27**	**.31**	**.45**	**.63**
$D_{\gamma 2}$	.09	.26	.13	.19	.28	.56	.72
$D_{\gamma 3}$	.13	.51	**.19**	.24	.27	.53	.78

only C-1, D-1, CD-1, and CESD-1. They do not always display significant differences with the models trained on three seasons only and the ranges of values covered by the metrics generally lie in the middle of the range covered by the restricted models. This suggests that the models trained on three seasons are not necessarily worse than the ones trained on nine seasons, and that the differences in performance are more likely due to the random sampling of the training pairs and the random initialisation of the model. Detailed results are available in Section B.3.4.

5.7 Final comparisons and conclusion

In this chapter we discussed several approaches to DTS on the *Friends* dataset. We mostly worked on improving the original TextTiling algorithm in two ways, one based on including more linguistic features to the lexical similarity metric and the other by replacing the similarity metric by a ML-based model for utterance-pair scoring following Xing and Carenini (2021). We also considered several baselines, including two random ones, several LLMs leveraged in a zero-shot manner, as well as the most basic versions of the two approaches we aimed to evaluate. For the feature-based approach, we considered the original TextTiling algorithm and a slightly modified version of it with adaptations to dialogical data, while for the ML-based approach we evaluated the model prior to fine-tuning and after fine-tuning on the DailyDialog dataset.

Section 5.7.1 provides a comparison of all the approaches and discusses the results before we conclude this chapter in Section 5.7.2.

Table 5.16: Comparison of the different models. *Evaluation on seasons 1, 5, 6, 7, 8, 9, 10 only.*

Model	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
Rd	.10	.49	.14	.20	.24	.58	.82
Rd_{smart}	.10	.10	.10	.23	.33	.47	.51
BC	.13	.13	.12	.26	.37	.48	.52
BC for dialogue	.15	.15	.14	.28	.40	.46	.50
$BERT_{Base}$	.11	.42	.17	.22	.28	.56	.77
$BERT_{DailyDialog}$	.10	.40	.15	.20	.25	.58	.80
BC + SI + Q	.19	.19	.17	**.36**	**.48**	**.41**	**.45**
$D_{\gamma 1}$	.15	.38	**.21**	.27	.31	.45	.63

5.7.1 Comparison of all the models

Table 5.16 summarises the results of our baselines and the two best models for our feature-based approach and ML-based approach respectively. For the feature-based approach, none of the improved variants of the algorithm based on the Block comparison similarity metric displayed significant differences in results. Hence we selected the one with the highest average scores while requiring a limited computational time, which ruled out the coreference model. We settled on the model based on Block comparison augmented with the smooth speaker introduction method and the questions feature. Regarding the ML-based approach we selected $D_{\gamma 1}$ since it was the best performing version of the restricted model overall.

Given that the ML-based model was trained on seasons two to four, we had to restrict the evaluation to seasons 1, 5, 6, 7, 8, 9, and 10 only for this comparison to ensure that the evaluation was fair and that the model was not evaluated on data it had seen during training.

When comparing the baselines, we can see that the random baseline performs quite poorly, which is expected. However, the BERT-based model fine-tuned on DailyDialog performs similarly to it, and worse than the BERT-based model prior to fine-tuning. This suggests that the fine-tuning on DailyDialog does not generalise well to our dataset, and even degrades the original generalisation capabilities of the model.

Regarding the original TextTiling algorithm, the adaptations to dialogue provide a slight improvement, and both models perform overall better than the random baselines, except for the WD metric where they are equivalent to the smart random baseline.

When it comes to the two models we developed, the feature-based model performs better than the ML-based model on all metrics except for recall, and F_1 where

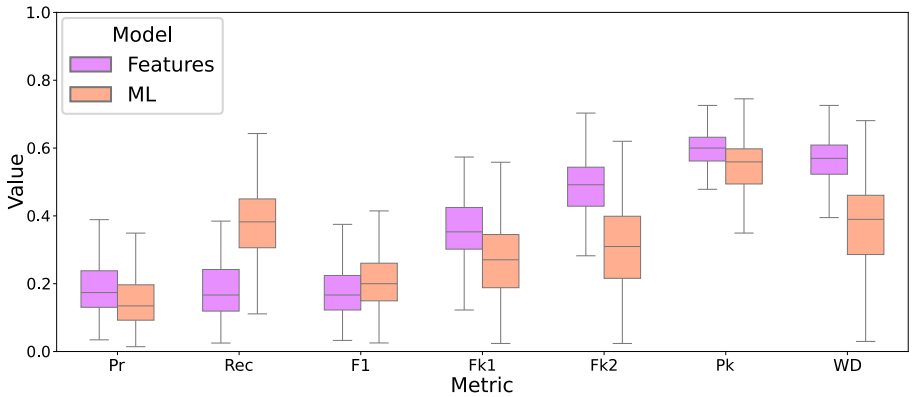


Figure 5.14: Boxplots of the results of the two improved models and the DeepSeek 1.5B model. *Evaluation on seasons 1, 5, 6, 7, 8, 9, 10 only.*

the ML-based model performs better, the better recall inducing the better F_1. The fact that only F_1 is better clearly indicates that the ML-based model over-generates topic boundaries, which is penalised by the F_k and WD metrics. While P_k does not penalise false positives as much as the other metrics, the ML-based model still performs worse than the feature-based one, though the difference in performance is smaller than for the other metrics. Figure 5.14 provides a boxplot representation of the results for the two models, which allows to visualise the variability of the results across the different evaluation seasons. We can see that the ML-based model displays a larger variability than the feature-based one, which shows that it is overall less consistent. Despite this variability, the differences between the two models are still statistically significant for all metrics, most likely thanks to the numerous data points we work with.

5.7.2 Final remarks and conclusion

In this chapter we investigated the task of linear topic segmentation on multi-party casual conversations. Since this kind of data is complicated to obtain, we chose to work on transcriptions of the TV-show *Friends* as this dataset is available online. While it is not a perfect representation of real-life conversations, the high number of speakers and the various open-domain settings of the dialogues make it interesting to work with for the DTS task. The scripted nature of these exchanges may simplify the task as the dialogues are explicitely meant for outsiders to understand, and the good quality of the transcriptions also make this dataset easier to work with than real-life conversations. However, the different approaches we tested on this dataset showed that it remains a challenging corpus for the task.

All our approaches were based on the TextTiling algorithm, which uses a similarity metric between subsequent parts of a text to identify the topic shifts. We en-

hanced it with more linguistic properties that could play a role in identifying topic shifts, and compared it to the same approach but enhanced with a trained utterance-pair coherence scoring model based on BERT. As BERT has been trained on the next sentence prediction task, it seemed like a relevant model for topic segmentation and in particular to improve the TextTiling approach. At the time where this work was carried out, similar but more powerful tools were models such as BART (M. Lewis et al., 2020) or T5 (Raffel et al., 2020) and they did not seem as suitable for our work as they have not been explicitly trained for the next-sentence prediction task. Nevertheless, the rise of LLMs with their generalisation capabilities raised the question of whether they could be used for our task. We thus evaluated several LLMs in a zero-shot manner, but the results were not satisfactory. We could probably improve our method by using a different prompt or wrap the LLM in a larger pipeline as we did for BERT. However, the large computational power required to run these models and their high resource consumption in their training phase made us question the relevance of using them for our task.

Indeed, we tried to limit our use of heavy computational power. Our feature-based model can run on a light machine like a laptop and except for the experiments that involve co-reference chains identification, creating the topic segmentation of one episode of *Friends* takes less than a few seconds. As for the experiments using Machine Learning, we did our best to optimise the batch sizes and the number of experiments we could run in parallel to reduce the training time as much as possible. We also tried to limit the multiplication of experiments though we could have gotten a clearer picture of the stability of the results by training more versions of the model with different random initialisations and different training subsets. We ran our experiments on the Lark servers from CLASP (Gothenburg University) where we used one Nvidia Titan RTX GPU. Each epoch took about one hour of training. The extra experiments with LLMs were carried out using the Nancy site of the Grid'5000 testbed, supported by a scientific interest group hosted by Inria and including CNRS, RENATER and several Universities as well as other organisations (see `https://www.grid5000.fr`). We chose only open-weights LLMs to avoid the use of proprietary models that would have required us to use external servers and thus to depend on the companies that have the means to develop such models, and we tried to limit the size of the models we used to reduce the computational power required to run them. However, even with these precautions, LLMs as well as ML-based experiments require a lot of resources, and the cost of initial training LLMs should not be taken out of the equation despite not being a direct part of our work.

With all these aspects in mind, the poor results of the LLMs and the ML-based model should be put into perspective. The feature-based model is more explainable, requires less resources, and does not require any training if we exclude the coreference chain feature. It also performs better than the ML-based model on all metrics except for recall and F_1. While the final results are far from satisfactory to

consider the DTS task as solved with data of the complexity level of the *Friends* dataset, the feature-based model seems to be a better working base than the ML-based model to improve the results. Including interactive features such as the participants' organisation seems promising, and a more complex utilisation of questions and possibly adjacency pair could also be relevant. These two features are somewhat language agnostic, though the notion of questions and adjacency pairs could require language-dependent adaptations depending on the sophistication of the approach. While our version of the coreference chains did not significantly improve the results, and was computationally expensive, more efficient methods could be developed in the future that would be worth testing. The shared task CRAC (Novák et al., 2025) on multilingual coreference resolution is a promising avenue for this kind of work beyond English.

The reason why the feature-based model easily outperformed the ML-based model, in the sense that our features were not highly sophisticated, is probably at least partially due to the specificities of the *Friends* corpus. The scripted aspect of the dialogues makes it more likely that the arrival of a new character in a scene coincides with a topic shift as this arrival would have been deliberately planned by the scriptwriters. In contrast in real-life conversations, the arrival of a new participant does not necessarily induce a topic shift. Even if the new participant came in order to discuss a specific topic, politeness would encourage them not to interrupt the ongoing conversation and to wait for a natural pause in the discussion. Nevertheless, the overall poor results are still a reflection of the complexity of the task for the data we worked with. Even though we could have tried different methods, ML or feature based, to improve the results, we believe that one of the main issues is the nature of the data itself and the fact that it does not lend itself well to the task of linear topic segmentation. This task is indeed very rigid because it assumes that sequences of utterances can be grouped into topics in a linear manner, which is not necessarily the case for multi-party conversations. Topic shifts can sometimes be subtle, making it complicated to identify them, and the granularity of the topic shifts can vary between people and tasks. In the *Friends* dataset, there are also many occurences of concurrent topics, where one character could be on the phone while the others are talking beside them. The experiments presented in this chapter should be seen as an introduction to topical structure analysis in multi-party conversations, where we saw that a method that works well for task-oriented dialogues does not necessarily transfer well to multi-party casual conversations, despite an adapted fine-tuning. Both the specificity of the data and the rigid nature of the task led us to consider a more flexible approach for revealing topical structures, which would allow us to investigate datasets that are closer to real-life conversations. We present this approach in the next chapter, where we discuss a continuous measure of topical similarity.

6

Topical Similarity at Utterance Level

Contents

6.1 Method 256
6.1.1 Metric construction: Recapitulation 257
6.1.2 Evaluation 258
6.2 Dataset: Naturalistic structured interactions 260
6.2.1 Social media in NLP research 260
6.2.2 Reddit detailed characterisation 262
6.2.3 Selection: a varied range of communities 267
6.2.4 Coherence levels 272
6.3 Coherence levels hypothesis validation 272
6.3.1 Topic modelling: BERT Topic 273
6.3.2 Similarity metric baseline: DSE-BERT without fine-tuning 277
6.4 Metrics training 279
6.4.1 Implementation details 279
6.4.2 Evaluation 281
6.4.3 Challenges of the data 285
6.5 Conversation modelling with a continuous topic similarity metric 288
6.5.1 Quantitative description 290
6.5.2 Patterns of topic changes 292
6.5.3 Proposal for a continuous topic segmentation . . 294
6.6 Discussion and conclusion 298

The analyses in Chapter 5 made us consider whether the topical structure of a conversation could be captured by a linear segmentation algorithm. The more complex nature of multi-party conversations, where several topics can be intertwined, a topic can drift slowly, etc., suggests that a more flexible approach is needed to extract topical information. Nevertheless, the analysis of our TextTiling-based experiments showed that there is topical information in surface level features such as

lexical cohesion. Our aim in this chapter is therefore to propose a more flexible approach to extracting topical information by drawing on more local features. The end goal is to produce a tool that would allow us to analyse the topical structure of a conversation, highlighting more subtle dynamics than a binary segmentation.

To achieve this, we propose to create a topical similarity measure which would give a similarity score to two pieces of interaction in terms of topic. Using this measure could then enable us to find locally coherent pieces of interaction but also to measure the variability in terms of topic throughout the conversation. Since we cannot reliably assign continuous similarity scores to pairs of dialogue interventions, we suggest to use an unsupervised method to train our measure. In particular, we will expand the idea from Xing and Carenini (2021) of training a metric, as we did in Chapter 5, but instead of using it for binary classification, where it over-generated boundaries, we will leverage it to investigate topical variability in a continuous manner rather than a binary one.

The dataset we chose for this task was extracted from the online discussion forum *Reddit* because it provides us with a large amount of messages, on various topics in different formats (questions/answers, casual discussion, argumentation, etc.), organised in tree structures. The hierarchical organisation of the interactions allow us to define different levels of topical coherence that we can use to train our measure.

This chapter is organised as follows. We begin by describing the method we used to build a topical similarity measure for dialogue in Section 6.1, including reminders from Chapter 5 and a description of our evaluation approaches. We then present the data we used for this study in Section 6.2. Next the hypothesis that underlies our unsupervised training is discussed and evaluated in Section 6.3 before we detail the settings of our experiments and the results we obtained in Section 6.4. The focal point of this chapter is the analysis of the topical structure of conversations, which we present in Section 6.5. Finally, in Section 6.6, we discuss the implications of our findings.

6.1 Method

Our goal in this chapter is to investigate the topic flow in a conversation. To do so, we need a way to compute topic similarity between two pieces of conversation in a continuous manner. That is, we are not interested in the topic boundaries themselves, but rather in quantifying the degree of topical change throughout a dialogue. We thus follow a similar approach as the ML one described in Chapter 5, of which we remind the reader in Section 6.1.1. This approach relies on a hypothesis regarding the degree of topical similarity between several pairs of pieces of conversation. We discuss the way we validated this hypothesis in Section 6.1.2.

6.1.1 Metric construction: Recapitulation

In Chapter 5, we tested various improvements to the linear topic segmentation algorithm TextTiling (Hearst, 1997). Specifically, we adapted Xing and Carenini's (2021) approach to our own dataset. Their idea was to replace the original similarity metrics from the TextTiling algorithm with a BERT-based similarity metric. Xing and Carenini trained an utterance-pair coherence scoring metric in an unsupervised way, using the assumption that utterances that are close to each other in a conversation are more likely to be topically similar than utterances that are far apart. This has the advantage of not requiring any manual annotation of topical similarity, which is a very subjective task. The metric was then used to compute the similarity between two adjacent blocks of utterances in the TextTiling algorithm.

Even though this approach did not yield satisfactory results for the DTS task, the similarity metric seemed to capture some aspects of the topical structure of the conversations as it performed better than chance. We therefore decided to adapt Xing and Carenini's (2021) approach to build our continuous topic similarity metric. Only the training of the similarity metric is relevant to our case, as we are not interested in the TextTiling algorithm itself. We thus use the same unsupervised training approach to build a similarity metric that can be used to compute the topical similarity between two pieces of conversation. The main difference compared to the approach we described in Chapter 5 is that we replace the basic BERT model by DSE-BERT (Zhou et al., 2022), a version of BERT fine-tuned with a contrastive learning task that encourages the model to represent consecutive sentences in a similar way. Even though this fine-tuning was done on task-oriented dialogues, most of them following the Wizard of Oz paradigm – the training data is the same as for TOD-BERT and is described in C.-S. Wu et al. (2020) – we hope that it provided the model with a better ability to capture small-scale similarities between sentences, which is what we are looking for in our case. This model operates with pairs of inputs instead of singular one.

The choice of this method induces constraints on the type of data we can use. As in Chapter 5, the task itself – topic similarity measure – is not restrictive, but the training method requires that we can build pairs of pieces of conversation with different levels of topical similarity. We therefore end up with the same requirements as in Chapter 5 regarding the data we can use, which we remind in (6.i). We also wanted to select a corpus of conversations that are closer to the type of interactions we are interested in, i.e. naturalistic open conversations. Reddit is very interesting in this regard as the conversations are organised in a tree structure which we further describe in Section 6.2, and they are naturally occurring. In the next section, Section 6.1.2, we describe how we validated our coherence hypothesis and evaluated our metric, before diving deeper in the description of our corpus in Section 6.2.

(6.i) $\left[\texttt{Frame}: \begin{bmatrix}\texttt{Language.Global.List}: \{\textsc{en}\} \\ \texttt{Purpose.TopicalStructure.Scenario}: \textit{Explicit}\end{bmatrix}\right]$

6.1.2 Evaluation

While the metric is only a tool aiming to support our exploratory analysis of the topical structure of conversations, we still need to validate its performance. We do this in two stages. On the one hand we try to validate our coherence ranking hypothesis, which is the basis of our unsupervised training, prior to any training. On the other hand, we evaluate the ability of the metric to rank pairs of comments in the expected order of topical similarity after training.

Coherence levels hypothesis validation In order to validate our coherence ranking hypothesis, we explore two complementary approaches. First, we rely on a topic modelling method to determine whether topical information can be extracted from the comments and to check whether the similarities are consistent with our hypothesis. We then use the same model that we will use to train the similarity metric, i.e. DSE-BERT, without any fine-tuning, to see whether it is already capable of producing a ranking of comments consistent with our hypothesis.

BERT Topic (Grootendorst, 2022) is a topic modelling method based on BERT embeddings. Topic modelling is a method to estimate the set of topics covered by a collection of documents and the distribution of topics for each document. BERT Topic relies on BERT embeddings to represent the documents in a vector space, then clusters them to estimate the topics and their distributions. This method is summarised on Figure 6.1.

In our case, we fit one model per subreddit, i.e. per subset of data collected from a specific *Reddit* community, and consider the comments of the posts we collected as the documents. This means that for each subreddit, we will obtain the set of topics covered by all the comments as well as the distribution of topics for each comment. We can then compute the cosine similarity between the topic distributions of two comments to get a measure of their topical similarity.

This has two main advantages: (1) it relies on BERT and thus shares a similar architecture with our approach to train the similarity metric, while remaining a different model; (2) it does not require to select the number of topics that will be estimated, which is a difficult task that requires several iterations to find the best number of topics. The latter reason is why we did not use Latent Dirichlet Allocation (LDA) (Blei et al., 2003) which is a more classical topic modelling method. It is also faster to fit than LDA, which is important given the size of our datasets.

We compare the results of these two approaches, i.e. topic modelling and the non-fine-tuned version of DSE-BERT, with each other and against chance. It allows us to determine whether topical information can be extracted from *Reddit* comments and to verify our ranking hypothesis thanks to the topic modelling. The DSE-BERT model is then used to determine whether it is already capable of producing a ranking of comments consistent with our hypothesis. The results of this

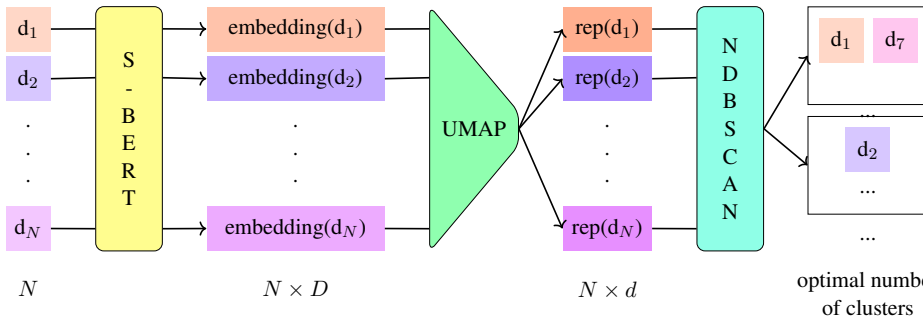


Figure 6.1: Representation of the BERT Topic modelling method; $D = 768$ with S-BERT, and $d = 5$ with default parameters. The full pipeline includes a topic representation part after the clustering step, based on a TF-IDF weighting scheme. It is not represented here.

analysis are presented in Section 6.3.

Metric evaluation In order to be as informative as possible, we report several metrics to evaluate the quality of the similarity measure. First, we compute the mean of the similarity scores for each of the three pairs (AB, AC, and AD) to determine whether the expected monotonic trend is observed. Second, we focus on the ranking of the three scores and report the pairwise accuracy to capture the quality of the ranking, i.e. the proportion of pairs that are correctly ordered. This metric represents the general quality of the ranking as it considers perfect rankings but also partial ones and includes ties. For a more in depth analysis, we also report the proportion of scores that are correctly ranked individually and compared to one another. In total the ranking analysis is covered by four types of scores summarised in Figure 6.2. Finally we compute the mean of the margins between the scores to quantify how confidently the hierarchy is separated.

The requirements in terms of data to determine whether our metric can allow us to observe topical variations and structure in conversations have several constraints. The first one is that the data must be organised in a hierarchical way, so that we can extract pairs of dialogue pieces with different levels of topical similarity. The second one is that the data must be sufficiently large to allow us to train a model on it. Third, since the core focus of this thesis is naturalistic open conversations, the data must be as close as possible to this type of interaction. Finally, we need to find a type of exchanges that encourages variations in topics. Given that this work is a first exploration, selecting data that is more directly suited for our analysis is preferable. The social media platform *Reddit* is particularly interesting in this regard, as we explain next in Section 6.2.

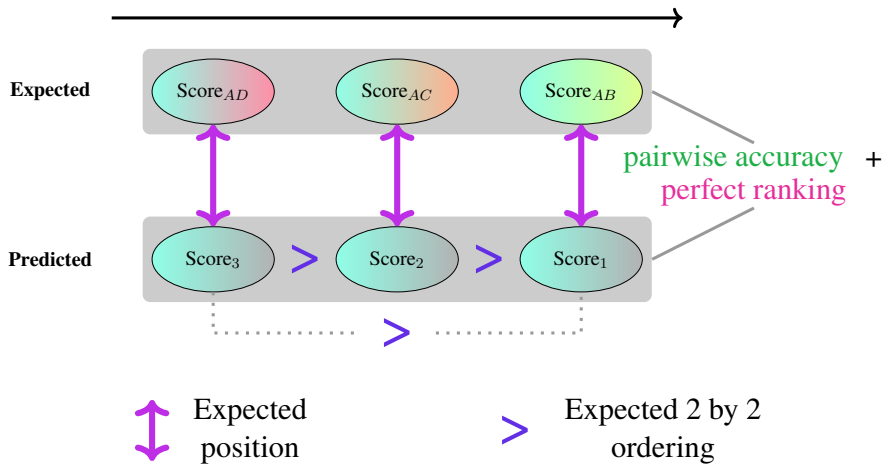


Figure 6.2: Description of the score ranking evaluation metrics.

6.2 Dataset: Naturalistic structured interactions

Reddit is a forum-like social media where users can post messages and other users can comment on them. This organisation allows for a tree-like structure of the interactions, where a post can have several comments and each comment can have several sub-comments. This structure is particularly interesting for our goal of investigating topical variations in conversations. Even though the interactions on *Reddit* are not spontaneous face-to-face conversations, they still exhibit a dynamic of interaction that is similar to those. The comments can be of various lengths and the interactions between users can involve a variable number of participants, which allows for a rich exploration of topical variations. Moreover, the written format of the exchanges removes the need for transcription, allowing us to focus on the analysis.

In this section, we first discuss the use of social media in general in NLP research (Section 6.2.1) before describing *Reddit* in more detail (Section 6.2.2). We then present the dataset we extracted from Reddit (Section 6.2.3) and how we constructed the different coherence levels our metric requires (Section 6.2.4).

6.2.1 Social media in NLP research

Social media (SM) is a broad and heterogeneous category of online platforms that facilitate communication and sharing of content. Definitions vary across disciplines – see Carr and Hayes (2015) for a proper discussion on the definition – but contemporary examples include *X* (formerly *Twitter*), *Facebook*, *Reddit*, *YouTube*, and many others, each with different affordances and communication patterns. This diversity as well as the large volume of language data they represent has made SM

an attractive data source in the NLP community (Hamilton et al., 2017; Jovanovic & Leeuwen, 2018; Baumgartner et al., 2020; Balouchzahi et al., 2023).

The access to large-scale, naturally occurring, language data has enabled research on a range of tasks, including conversation related ones. Support tasks such as emotion recognition have been explored in the context of SemEval shared tasks (F. Wang et al., 2024) among other studies, while others have focused on illocutionary power of interactions like Misra and Walker (2013) who look into agreements and disagreements, or Foderaro and Lorentzen (2022) who focus on argumentation patterns. SM data has also been used to study the structure of conversation in a more general sense, as discussed by Paulus et al. (2016) in a review of the use of CA to study online conversations, which social media is a subset of. As a time-sensitive window on users experiences, SM data has also been used to study societal issues as experienced by people like studies on mental health (Gaur et al., 2018; Turcan & McKeown, 2019; Naseem et al., 2022), but also the discourse that is centred on them, with studies analysing the reactions to the 2016 US presidential election (Visser et al., 2019), the COVID-19 pandemic and the misinformation that spread about it on SM (Cinelli et al., 2020), or climate change (Schäfer & Metag, 2026).

However the apparent ease of access to large amounts of language data is concurrently one of the drawbacks that warrants careful consideration when working with SM data. From a technical point of view, social media platforms are typically governed by private companies, each with their own terms of service, data collection and sharing policies, and API limitations when applicable. From an ethical perspective, this data raises questions regarding privacy and consent (M. L. Williams et al., 2017). Indeed, users may not expect or consent to their posts being harvested for research purposes, even though the content is publicly accessible. In particular, constituting and sharing datasets from SM data creates an archive of user-generated content that may survive despite the deletion of the original content, or be re-used outside of the original context. This is especially relevant as SM can be a space for vulnerable populations to express themselves, and the content may contain sensitive information.

In addition, while the diversity of SM platforms offers a rich source of data in terms of types of interactions and the modalities they rely on, a limitation that is inherent to the nature of this data is a loss of control of the data. Unlike data collected in laboratory settings, SM data introduces uncertainties across multiple dimensions. Participant identity is difficult to verify – accounts may be bots, spam accounts, or inauthentic in other ways – particularly when analysing short messages. Language detection is less reliable than in curated datasets (Farzindar & Inkpen, 2015, p.27), making it challenging to reliably filter for specific languages. The informal and non-standardised nature of language also presents problems: misspellings, dialectal variation, emoji use, and platform-specific language games create vocabulary and orthography that existing NLP tools often struggle to process. Those are of-

ten removed through a phase of normalisation during preprocessing but it discards potentially meaningful linguistic variation and may not consistently improve downstream task performance (Eisenstein, 2013). Moreover, the community aspect of SM platforms means that many platforms, and sometimes sub-communities within them, have their own rules, norms, and expectations for communication. This can create challenges for researchers trying to understand and interpret the data, as well as for NLP models that may not be able to generalise across different communities or platforms.

More generally, as noted by Carr and Hayes (2015), models of conversation developed for a specific platform are only valid insofar as that platform's form remains stable. Changes in platform policies, user demographics, or corporate leadership – as seen recently in the case of Twitter's rebranding and policy shifts – can fundamentally alter the types of interactions that occur there. This raises questions about the generalisability and longevity of models trained on platform-specific data.

Despite these limitations, SM remain a valuable source of data for NLP research. The large volume of naturally occurring interactions provides opportunities to explore patterns and trends that may not be observable in smaller, curated datasets. However, researchers must carefully consider the ethical implications of using this data, as well as the potential biases and limitations inherent in the platform-specific nature of social media interactions. While *Reddit* is not the most studied media (Kathie Treen & Coan, 2022), its structure makes it particularly interesting for our goal. Indeed, its hierarchical organisation provides inherent metadata about conversational relationships – who responded to whom and in what order – without requiring external inference. Furthermore, the way it is divided into subcommunities, each with explicit rules and expectations, allows us to study various types of interactions. More details on the content and organisation of *Reddit* are given in Section 6.2.2.

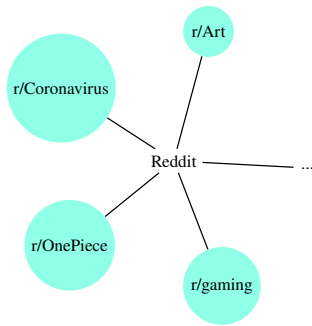
6.2.2 Reddit detailed characterisation

Reddit is a US-based social media platform founded in 2005. As of March 31, 2026, it reported about 127 million daily active unique users and almost 500 million weekly ones[19]. In this section we describe the organisational structure of *Reddit* (Section 6.2.2.1), which is the reason why we chose this particular social media for our study, and detail the type of dialogue interactions we can find on it (Section 6.2.2.2).

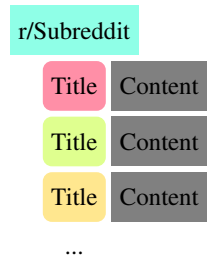
6.2.2.1 Structure

Reddit is a forum divided into communities called *subreddits* which are defined by a topic, a hobby, a culture or even the format of the posts, and typically a set

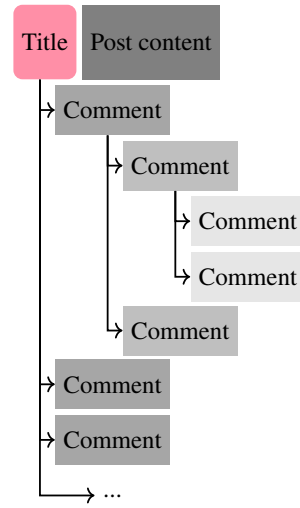
[19]According to https://redditinc.com, last visited on July 16, 2026.



(a) *Reddit* is divided in *subreddits*.



(b) Each *subreddit* contains *posts* formed by a title and content.



(c) Each *post* is organised as a tree with comments responding to the *post* and comments responding to other comments.

Figure 6.3: *Reddit* organisation.

of rules. For example, the subreddit *r/boardgames* is presented as 'The #1 source for news, information, and discussion about modern board games and board game culture.' and describes general politeness and integrity rules such as 'Keep Discussions Civil', 'Respect Intellectual Property', or 'Limit Usage of AI'. Some subreddits can be very specific in terms of topic and rules such as *r/poptarts* which is dedicated to the Anglo-Saxon toaster pastry and forbids mentions of 'Toaster Strudels'. Besides the content, some subreddits' rules also constrain the format of the posts, such as *r/Showerthoughts* which requires posters to write their entire thought in the *title* of their post.

Within a subreddit, users can post a message, called a *post*, and other users can react to it by writing a comment. A comment can be a direct answer to the post or an answer to another comment. The discussion thus created is organised in a tree structure with the initial post as the root and the comments as nodes. Figure 6.3 illustrates this structure. Posts often contain an illustration and sometimes even contain only that and a title.

What we chose to call a conversation on this platform is a *thread*, which consists of a post and all its comments. They can be of various sizes – the total number of comments – and shapes – the way the comments are organised in the tree. We further discuss this variability in Section 6.2.3. Before that, we present the global

characteristics of *Reddit* interactions in Section 6.2.2.2.

6.2.2.2 Description

Reddit interactions are very different from face-to-face conversations for a number of reasons, the most obvious being that they are asynchronous and text-based, but they also exhibit unique social properties. Table 6.1 summarises the main characteristics of *Reddit* interactions according to CHIMERA, with core differences with face-to-face interactions highlighted in orange, and similarities in purple. The format of *Reddit* is the root cause of many of these characteristics. First, the number of unique participants to a *thread* is virtually unlimited as there is no upper limit to the number of users or comments that can be active under a given post nor any option to close a post other than removing it entirely. In practice, most posts stop receiving responses after a short amount of time due to the way posts are highlighted on the platform. Within a given post, the poster has a privileged position as they set the focus of the exchange. A token identifying them as the original poster is displayed next to their username when they comment on their own post within the comments tree. Other participants can also take a more important interactional function at the level of sub-conversations generated by the threads of comments.

Second, participants are identified by a pseudonym and there is no foolproof way to link a pseudonym to a real identity. In 2012, Barack Obama posted a message on *r/IAmA*, a subreddit where each thread is dedicated to a specific person who answers questions posted in the comments. The former president's post contained a link to a *Twitter* (now *X*) post where he announced he would be answering questions on *Reddit*. Contrary to *Reddit*, *Twitter* had a user identity verification system at the time which increased the credibility of the post. A picture of the former president was also linked in the post, in an attempt to provide more physical proofs, though nothing made this photograph verifiable. Anonymity goes together with the idea of commitment: Without knowing the participants' identities or motivations for joining a conversation thread, it is quite complicated to assess their level of commitment. Some participants may be trolls, who post messages with the intention of provoking other participants. Conversely, others may be deeply involved in a topic or community.

Third, the medium has direct consequences on the dynamics of the interactions on *Reddit*. The only thing that participants share is the content of the post and the comments. There is no physical context, no visibility of the participants, or possibility to hear them. The exchanges are completely asynchronous, reducing the possibility of co-construction to the minimum. As a consequence, participants can technically plan their contributions for an unlimited amount of time, and they can later edit or delete them with no time limit either. Unless they choose to delete their messages, they are recorded on the platform and can be read by anyone, without necessarily owning a *Reddit* account, at any time. While *Reddit* users are supposed

Table 6.1: Description of *Reddit* interactions.

Meta-Parameters	Sub-parameters	**Description**
Agents		
Agency		Human
Participation structures	Number	Virtually unlimited
	Functions	Poster - commenters + possible sub-conversations
Social dynamics	(Conversational) History	Possible within / across subreddits, possibly over the internet or even real-life interactions
*Enunciative status**	Anonymity	Pseudonymous, digital
	Role	Subreddit specific
	Persona	Subreddit specific
	Commitment level	Imbalanced (e.g. trolls)
Conversational setting		
Epistemic grounding	World setting	Subreddit / Post specific
Environment & Medium	Physical context	Separate, share post content
	Visibility	∅
	Audibility	∅
	Synchrony	Asynchronous
	Planning	No limit
	Ephemerality	Infinite edition time (*edited*), possible deletion (`[deleted]`)
		Persistent by design
Conversational frame		
Language	Global	English
	Control	Subreddit rules
	*Vocabulary**	Subreddit / Post specific
Purpose	Practical goal	Subreddit / Post specific
	Success condition	Subreddit / Post specific
	*(Topical) structure**	Explicit tree
Verbal interaction		∅

to be aware of this fact, they may not always be conscious of the possible way their messages could be used, whether for research or commercial purposes, which raises ethical questions about the use of such data.

Finally, the separation into communities entails subreddit specific characteristics. In terms of CONVERSATIONAL FRAME, the language and purpose of the conversations are typically defined by the subreddit itself. Specifically, *Reddit* is generally steeped in internet culture, making some communities very complex to understand for outsiders. Similarly, the roles and personae adopted by the participants can change depending on the subreddit, where some of them encourage participants to be truthful and authentic while others are explicitly made for role playing. However, these characteristics are not so different from face-to-face interactions, where the social context can also have a strong influence on the way participants interact with each other. The community can also have a direct impact on the social dynamics between the participants, as some people can be old members of a subreddit and have a more important position in the community than others, which is also a feature of face-to-face conversations. Moreover, participants can develop a history of interactions with other members of a subreddit, and can even recognise each other across different subreddits, or recognise pseudonyms they have encountered somewhere else on the internet. Some participants also know each other in real life and they can import their personal history into the *Reddit* exchanges for instance by tagging each other in posts that relate to past conversations. Lastly, while most of the content on *Reddit* is in English, some subreddits have rules that require participants to post in a specific language, such as French or Spanish. In *r/india*, the rules explicitly state that all posts must be in English while *r/france* allows for both French and English. When such rules exist, they are typically enforced by the moderators of the subreddit, which increases the degree of certainty that the exchanges will be in the expected language, which is interesting from a research perspective.

Overall, despite the medium and format of *Reddit* interactions, which are very different from face-to-face conversations, they still exhibit a number of characteristics that make them interesting for our study. The potentially large number of participants, the variety of interactional functions, and the diversity of subreddits allow us to explore a wide range of conversational dynamics. The inherent structure of the interactions also provides a natural way to study topical variations and coherence in conversations. Compared to the *Friends* corpus we investigated in Chapter 5, we lose the face-to-face aspect of the interactions but we gain in terms of spontaneity. Furthermore, the conversations take place within a slightly narrower thematic frame.

For our study, we collected conversation threads from a range of subreddits to ensure a diversity of topics and interactional dynamics. The selection is described in the next section (Section 6.2.3) together with the specificities of each subreddit.

6.2.3 Selection: a varied range of communities

***Reddit*'s API terms of use disclaimer** At the time the experiments described in this chapter were conducted, the *Reddit*'s Data API terms indicated that training machine learning models based on *Reddit* data was not allowed 'without the express permission of rightsholders in the applicable User Content'. Another page indicated that using *Reddit* developer tools and services 'for research purposes [was] OK provided [we] use[d] it exclusively for academic (i.e. non-commercial) purposes, and [did]n't redistribute [the] data or any derivative products or services based on [the] data (e.g. models trained using *Reddit* data)'[20]. We thus collected our data under the assumption that the use of the API to collect data, and subsequently train LM models, was allowed in an academic context as long as neither the data nor any models trained on the data were redistributed. This restriction is also one of the reasons why we did not use the existing Pushshift *Reddit* Dataset (Baumgartner et al., 2020) as it was unclear whether its collection and sharing were compliant with *Reddit*'s policies. However, it should be noted that *Reddit*'s policies have evolved a lot since the time we collected our data. Specifically, the use of *Reddit* data for research purposes is now subjected to an application to the *Reddit* for Researchers program. Without applying to this program, the use of the API is explicitly forbidden for academic purposes in general. Though this program did not exist at the time of our experiments, a contact form was available to request permission to share data collected from *Reddit*. We submitted a request in December 2023 and never received a response. We thus decided not to share our dataset or models, but we provide all our code to reproduce our experiments. However, the new policies should be taken into account for any future work on *Reddit* data.

The division of *Reddit* in communities, all with their own rules and specificities, allowed us to select a range of subreddits with different topics and interactional dynamics. We selected six subreddits, which we describe below. The selection was made based on the following criteria: (1) the range of subreddits should cover a variety of formats and expected interactional dynamics, (2) the subreddits should be active enough to provide a sufficient number of threads, and (3) the subreddits should be in English. We focused on English as in the rest of this thesis but *Reddit* has some communities in other languages as discussed in Section 6.2.2.2.

Art (*r/Art*) This subreddit is dedicated to 'sharing finished, display ready, visual art'. The post title format is explicitly defined in the rules which makes all posts very similar in structure: a title describing an art piece and one or several pictures of it in the body of the post. We expect the comments to be more or less well-argued feedback on the art piece, thus being quite close to one another in terms of topic and differing mostly in their valence.

[20]Based on Reddit Support as of March 2025 in the Web Archive. We could not get back to a version of the page closer to our actual use of the API.

Coronavirus (*r/Coronavirus*) This subreddit was created to 'monitor the spread of the disease COVID-19'. Due to its requirement of 'high-quality posts and discussion', we expect possibly long comments and rather structured discussions.

Gaming (*r/gaming*) This subreddit is dedicated to gaming in general and has no specific requirements except for the language which must be English, as well as some general politeness and non-toxicity rules. We expect a variety of post formats and discussion dynamics, with some posts being more argumentative and others more casual. One of the interesting aspects of this subreddit is that it is likely to contain a lot of more specialised vocabulary and references to internet and gaming culture.

OnePiece (*r/OnePiece*) This subreddit is dedicated to 'Eiichiro Oda's manga and anime series One Piece'. It contains many posts with only a title and picture but also longer ones that are meant to trigger discussion as required by the rules. We expect a vocabulary specific to the OnePiece universe and manga culture references.

Political discussion (*r/PoliticalDiscussion*) This subreddit is dedicated to 'substantive and civil discussion on political topics'. The rules require the posts to contain background information and to be formatted 'for readability', e.g. include paragraphs and punctuation. There are also rules regarding the comments so that the subreddit can remain 'high quality'. We expect the exchanges to be more structured and the messages to be longer and well written compared to other subreddits.

Sports (*r/sports*) This subreddit is dedicated to 'Sports News and Highlights' from various leagues around the world. Most of the posts contain only a title and image or link but the rules require that the titles be 'high-effort and descriptive'. Thanks to the generic nature of the topic, we expect some posts to generate comments on various subtopics.

From the subreddits above we leverage the API to collect conversational threads. It does not allow us to collect all the threads from a subreddit but only the top 100 posts and their comments according to a given sorting method. We chose to collect the top 100 posts according to the number of upvotes since the creation of the subreddit as it was the sorting method that would likely provide the densest threads. Hence, in January 2024, we collected up to 100 threads from each of the six subreddits, some threads being abandoned due to the API limitations. For each thread, we collected the title and content of the main post, its ID, the number of upvotes, the pseudonym of its author, and the list of its children, that is to say all the comments in the thread. Similarly for the comments, we collected their content, their ID, the number of upvotes, the pseudonym of their author as well as the ID of

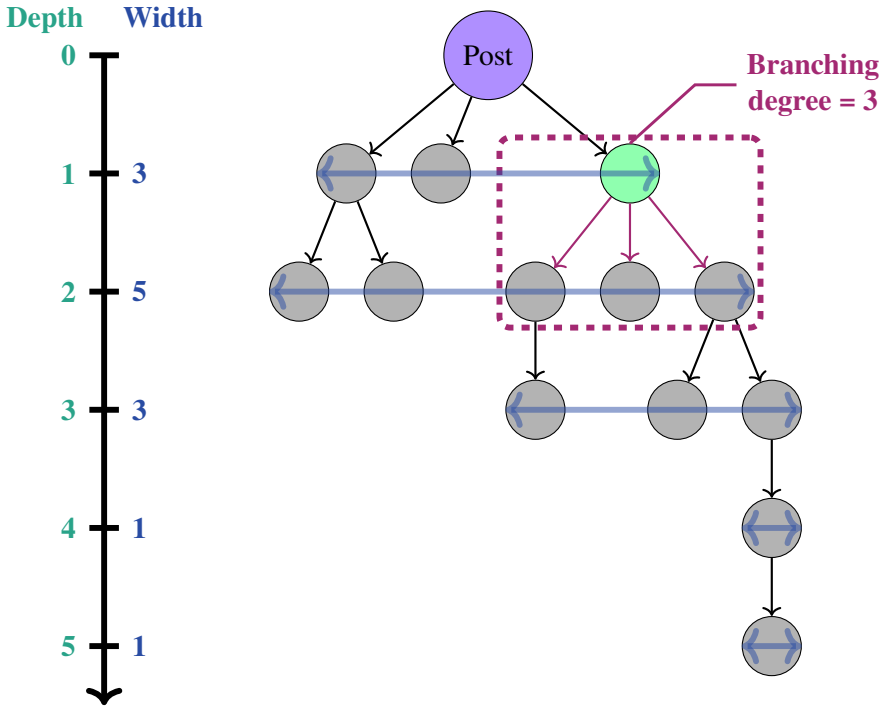


Figure 6.4: Representation of a *Reddit* thread and the vocabulary to describe its topology.

their parent in the discussion tree to reconstruct the tree structure. We do not use either the number of upvotes or the pseudonyms of the authors in our experiments but we collected them in case they would be useful for further analyses.

Among the collected comments, we removed those that were indicated as *deleted* or *removed*, the former indicating that the author deleted their comment and the latter indicating that a moderator removed it. We also removed all the children of these comments as we are interested in the conversational progression, and ignoring a comment in the middle of a thread would disrupt the tree structure.

In the end, we obtained between 72 and 98 threads per subreddit, with a total of 47,120 to 254,980 comments, as summarised in Table 6.2. The format of a thread represented as a tree is shown in Figure 6.4 together with the vocabulary we use to describe the threads.

r/Art has the least number of threads and comments in total, but it is not the smallest in terms of topology (Table 6.3) nor branching (Table 6.4), i.e. the tendency of the comments to have several children and thus generate sub-conversations. *r/PoliticalDiscussion*, while having the most threads, is the second subreddit with the least number of comments in total, and it is also the one with the smallest number of direct answers to the posts. However, it has the highest average depth, i.e.

Table 6.2: Statistics of the density of the collected threads.

(a) Total.

Subreddit	# Threads	# Comments
r/Art	72	47,120
r/Coronavirus	94	167,794
r/gaming	96	254,980
r/OnePiece	93	91,920
r/PoliticalDiscussion	98	83,034
r/sports	97	139,346

(b) Per thread.

Subreddit	# Comments / Thread		# 1st comments / Thread	
	μ	median	μ	median
r/Art	655	507	311	263
r/Coronavirus	1,786	1,636	586	520
r/gaming	2,657	2,096	1,310	985
r/OnePiece	989	434	701	274
r/PoliticalDiscussion	848	751	112	98
r/sports	1,437	1,250	547	464

the longest chains of comments. Moreover, its quartiles being consistently higher than the other subreddits in terms of depth indicate that it has overall longer chains of comments, and hence longer conversations, which is of particular interest for our study. The singular topology of *r/PoliticalDiscussion* is also reflected by its width, i.e. the total number of comments at a given depth level. Despite having the second lowest average width and the lowest maximum width, its first and second quartiles are much higher than the other subreddits, and the third quartile is the second highest. This indicates that the conversations in *r/PoliticalDiscussion* are more balanced in terms of width across the different depth levels, while the other subreddits have a few very wide levels and many very narrow ones. The branching degree is also consistent with all these observations, as *r/PoliticalDiscussion* has the most number of comments with only one child (37%), and other comments consistently have fewer children than in the other subreddits.

Conversely, the largest subreddit in terms of number of comments, *r/gaming*, has the largest amount of comments per thread and the largest number of first comments per thread. It is also the widest, and the second lowest in terms of portion of comments with only one child, the other comments producing one of the largest amount of children among our six subreddits. It is, however, one of the shal-

lowest subreddits. Thus, *r/gaming* has a very different interactional dynamic than *r/PoliticalDiscussion*, with many participants joining each thread but not engaging in long discussions.

Table 6.3: Statistics of the topology of the collected threads.

(a) Depth.

Subreddit	μ	q1	median	q3	max
r/Art	2.57	1	2	3	25
r/Coronavirus	3.36	1	3	5	43
r/gaming	2.75	1	2	4	36
r/OnePiece	1.88	1	1	2	26
r/PoliticalDiscussion	4.32	2	4	6	33
r/sports	3.13	1	2	4	51

(b) Width.

Subreddit	μ	q1	median	q3	min-max
r/Art	1,815	4	65	1,049	1–22,457
r/Coronavirus	3,816	3	28	922	1–55,164
r/gaming	6,894	3	66	2,330	1–125,804
r/OnePiece	3,408	9	93	1,055	1–65,256
r/PoliticalDiscussion	2,445	20	128	2,111	2–15,364
r/sports	2,682	4	12	203	1–53,146

Another aspect that characterises subreddits is the format of the posts, quantitatively described in Table 6.5. Most subreddits have posts that are constituted only of a title and URL, with no additional text in the body of the post. The only exception is *r/PoliticalDiscussion* where all posts contain text. In most cases, the URL points to an image, except for *r/Coronavirus* where it points to news articles three quarters of the time, which is consistent with the informative purpose of this subreddit. Regardless of the type of attachment, those posts present a potential challenge for the analysis of the comments, as they are rooted in a context that is not accessible to the model. Additionally, the titles of the posts are not always very descriptive, in particular in *r/Art* where they are typically the name of the art piece and its author. This is reflected in the number of tokens in the titles, which is the lowest for *r/Art,* with little variation across the posts. *r/OnePiece*, *r/sports*, and *r/gaming* also have rather short titles with slightly more variation, but we can imagine that most of the information is contained in the attachment. Conversely, *r/Coronavirus* and *r/PoliticalDiscussion* have longer titles with more variation, which is consistent with the fact that they are meant to be informative and trigger discussion. A few

Table 6.4: Statistics of the branching factor of the comments in the collected threads.

Subreddit	% 1 child only	# children (>1) / comment				
		μ	q1	median	q3	max
r/Art	20	4.06	2	2	4	91
r/Coronavirus	22	5.13	2	3	5	143
r/gaming	17	5.05	2	3	4	120
r/OnePiece	12	3.70	2	2	4	103
r/PoliticalDiscussion	37	3.34	2	2	3	63
r/sports	20	4.44	2	3	4	102

posts from *r/gaming* and *r/OnePiece* do not have URL attachments but text instead in the body of the posts. In those cases, the number of tokens in the text is slightly lower than the one in *r/PoliticalDiscussion* but with a higher variation. Once again, *r/PoliticalDiscussion* stands out across our six subreddits.

In the next section (Section 6.2.4), we describe how we leverage the tree structure of the threads to define three levels of coherence for training our topic similarity model.

6.2.4 Coherence levels

We rely on the hierarchical structure of *Reddit* to build three levels of comparisons. Figure 6.5 illustrates those levels. We consider that given a comment, one of its children, i.e. a direct answer to it, is the most topically related. Then one of its siblings, i.e. a comment that answers the same comment is less related. Lastly, a random comment from another post in the same subreddit, is the least topically coherent while remaining broadly on the same topic. We initially considered a nephew comment, i.e. a comment that answers one of the siblings of the given comment, as the least coherent level but we found that it did not provide enough contrast with the sibling level. We never use the post itself as a base comment for the comparisons due to the variety of post formats and the fact that some posts contain only a title and a picture or link that are not described in the post, thus making most of the information in the post inaccessible to the model.

In the following section (Section 6.3), we explore several ways to validate our topical similarity ranking hypothesis.

6.3 Coherence levels hypothesis validation

Although exploiting the hierarchical structure of *Reddit* discussion threads makes it possible to avoid the need to use data annotated with topic similarity scores, work-

Table 6.5: Statistics on the posts in the collected threads.

(a) Number of tokens in the title.

Subreddit	$\mu \pm \sigma$	q1	median	q3
r/Art	10 ± 3	8	10	12
r/Coronavirus	23 ± 16	13	18	25
r/gaming	14 ± 12	6	11	19
r/OnePiece	11 ± 8	6	9	13
r/PoliticalDiscussion	26 ± 15	14	24	36
r/sports	12 ± 6	7	10	15

(b) Number of tokens in the text.

Subreddit	% Text / URL	$\mu \pm \sigma$	q1	median	q3
r/Art	0 / 100		/		
r/Coronavirus	0 / 100		/		
r/gaming	2 / 98	147 ± 140	77	147	217
r/OnePiece	2 / 98	162 ± 160	83	162	242
r/PoliticalDiscussion	100 / 0	188 ± 119	88	168	258
r/sports	0 / 100		/		

ing at the comment level may prove too localised to provide sufficient information to distinguish between comments. We must therefore validate our hypothesis that a child comment is topically more similar to the main comment than a sibling comment, which is itself more similar than a random comment from another post in the same subreddit. To do this, we use two complementary approaches, as described in Section 6.1.2. We first comment the results of the BERT Topic modelling in Section 6.3.1 before presenting the results of the DSE-BERT model without fine-tuning in Section 6.3.2.

6.3.1 Topic modelling: BERT Topic

We fitted one BERT Topic model for each of our 6 subreddits, where each comment is considered as a document. We then computed the cosine similarity between the topic distributions of each pair of comments AB, AC, and AD to see whether the ordering $S_{AB} > S_{AC} > S_{AD}$ was valid.

It should be noted that a large amount of pairs of comments shared no topics in common based on the BERT Topic modelling, which resulted in a cosine similarity of 0. Two possible reasons for this are the number of topics estimated by the model being very high, or the comments being too short to provide enough information to estimate their topic distributions.

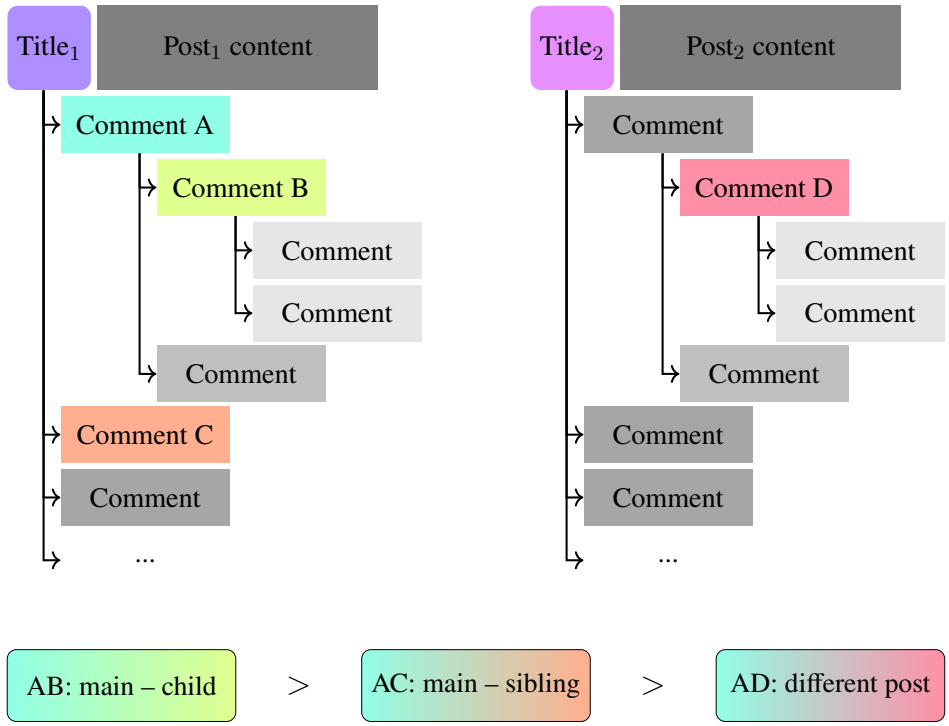


Figure 6.5: Topic similarity levels between comments.

Table 6.6 presents the proportions of correctly ordered similarities based on the BERT Topic modelling. We report the proportions over the pairs of comments for which all three similarities were strictly positive, which we call *valid* similarities, as well as the proportions over the pairs of comments for which at least two similarities were strictly positive, and at least one. The results do not vary greatly between those three cases. The three orderings we consider, that is $S_{AB} > S_{AC}$, $S_{AB} > S_{AD}$, and $S_{AC} > S_{AD}$, are all valid more than 60% of the time, which is better than chance, suggesting that topical information can indeed be extracted from the comments. Specifically, the ordering $S_{AB} > S_{AC}$ is valid on average two thirds of the time, which is the lowest across the three orderings. The pairs AB and AD are the most often correctly ordered, which is consistent with our hypothesis that they would be the least topically similar. Overall, the results indicate that our hypothesis is valid but that the pairs AB and AC are sometimes too close to be correctly ordered. This is consistent with the fact that comments B and C are much closer to one another and to comment A in the discussion tree than they are to comment D.

Table 6.7 provides more details on the successes and failures of the BERT Topic model in ranking the comments, including a comparison with the evaluation scores of a random baseline based on a sample of 100,000 similarity triples. For conciseness reasons, we only report the results for the tuples of comments for which at least

Table 6.6: Proportions of correctly ordered similarities based on the BERT Topic modelling.

	$\%_{total}$			$S_{AB} > S_{AC}$		
r/Subreddit	*All*	≥ 2	≥ 1	*All*	≥ 2	≥ 1
r/Art	6.52	29.1	64.9	.61	.62	.66
r/Coronavirus	**28.6**	**65.9**	**90.7**	**.71**	**.70**	.69
r/gaming	7.79	35.0	70.3	.64	.64	.67
r/OnePiece	8.29	37.5	72.9	.64	.63	.68
r/PoliticalDiscussion	**27.8**	**64.8**	**88.9**	.64	.64	.65
r/sports	4.33	28.4	65.4	.62	.62	.66
All				.66	.64	.67

	$S_{AC} \geq S_{AD}$			$S_{AB} \geq S_{AD}$		
r/Subreddit	*All*	≥ 2	≥ 1	*All*	≥ 2	≥ 1
r/Art	.72	.73	.81	.81	.82	.85
r/Coronavirus	.65	.68	.73	.81	.80	.82
r/gaming	.75	.78	.84	.85	.87	.88
r/OnePiece	.71	.77	.83	.82	.86	.88
r/PoliticalDiscussion	.78	.76	.78	.88	.85	.85
r/sports	.78	**.83**	.88	.87	**.90**	.91
All	.85	.85	.87	.73	.75	.81

Table 6.7: Evaluation of the ranking performed by the BERT Topic Models when at least two similarities are valid. *'Acc.' is the pairwise accuracy; '1st✓', '2nd✓' and '3rd✓' are the proportions of of correctly ranked first, second and third scores; 'AB $\geq$ AC', 'AC $\geq$ AD' and 'AB $\geq$ AD' are the proportions of pairs correctly ordered two by two; 'Perfect Ranking' is the proportion of triples for which the ranking is perfect.*

Subreddit	Acc.	1st✓	2nd✓	3rd✓
r/Art	.72	.57	.44	.64
r/Coronavirus	.73	**.64**	.48	.60
r/gaming	.76	.60	.48	.71
r/OnePiece	.75	.59	.47	.71
r/PoliticalDiscussion	.75	.61	.46	.67
r/sports	**.78**	.60	**.49**	**.77**
All	.75	.60	.47	.68
Random sim.	.50	.33	.33	.33

Subreddit	AB $\geq$ AC	AC $\geq$ AD	AB $\geq$ AD	Perfect Ranking
r/Art	.62	.73	.82	.39
r/Coronavirus	**.70**	.68	.80	.43
r/gaming	.64	.78	.87	.45
r/OnePiece	.64	.77	.86	.44
r/PoliticalDiscussion	.64	.76	.85	.43
r/sports	.62	**.83**	**.90**	**.47**
All	.65	.75	.85	.43
Random sim.	.50	.50	.50	.17

two similarities were strictly positive, which is a good compromise for the number of tuples considered while remaining close to the results when all three similarities are valid. The full results are available in Appendix C.

Overall, three fourths of the comment pairs are correctly ordered, with mostly the second position being the one that is mis-ranked. This corresponds to 43% of the triples being perfectly ranked, which more than twice the proportion of triples that would be perfectly ranked by chance.

These results are promising with regards to our coherence levels hypothesis. They also suggest that the comments indeed contain surface topical information.

Table 6.8: Evaluation of the ranking performed by the DSE-BERT model before fine-tuning on the *Reddit* dataset. *'Acc.' is the pairwise accuracy; '1st✓', '2nd✓' and '3rd✓' are the proportions of of correctly ranked first, second and third scores; 'AB > AC', 'AC > AD' and 'AB > AD' are the proportions of pairs correctly ordered two by two; 'Perfect Ranking' is the proportion of triples for which the ranking is perfect.*

Subreddit	Acc.	1st✓	2nd✓	3rd✓
r/Art	.49	.33	.35	.33
r/Coronavirus	**.56**	**.42**	.35	**.40**
r/gaming	.53	.38	.35	.37
r/OnePiece	.53	.38	.34	.36
r/PoliticalDiscussion	.48	.31	.35	.33
r/sports	.39	.23	**.37**	.21
All	.50	.34	.35	.33
Random sim.	.50	.33	.33	.33

Subreddit	AB ≥ AC	AC ≥ AD	AB ≥ AD	Perfect Ranking
r/Art	.50	.49	.49	.17
r/Coronavirus	**.55**	**.54**	**.59**	**.23**
r/gaming	.52	.52	.54	.20
r/OnePiece	.52	.53	.55	.19
r/PoliticalDiscussion	.48	.49	.47	.16
r/sports	.41	.42	.34	.10
All	.50	.50	.50	.17
Random sim.	.50	.50	.50	.17

6.3.2 Similarity metric baseline: DSE-BERT without fine-tuning

We use the DSE-BERT model (Zhou et al., 2022) without fine-tuning to evaluate its ability to rank the comments based on their similarity. This allows us to determine whether the model can already capture differences between the comments without any training on our dataset. We report the results in Table 6.8. The model performs similarly to chance – and even worse on average – which indicates that it is not able to capture the differences between the comments without fine-tuning. When looking at the average similarities for each pair, which are represented in Figure 6.6, we can see that they are all very close to one another, which confirms that the model is not able to distinguish enough between the comments to rank them effectively.

These results are likely due to the nature of our dataset. The DSE-BERT model was trained on task-oriented dialogues, which are very different from the *Reddit*

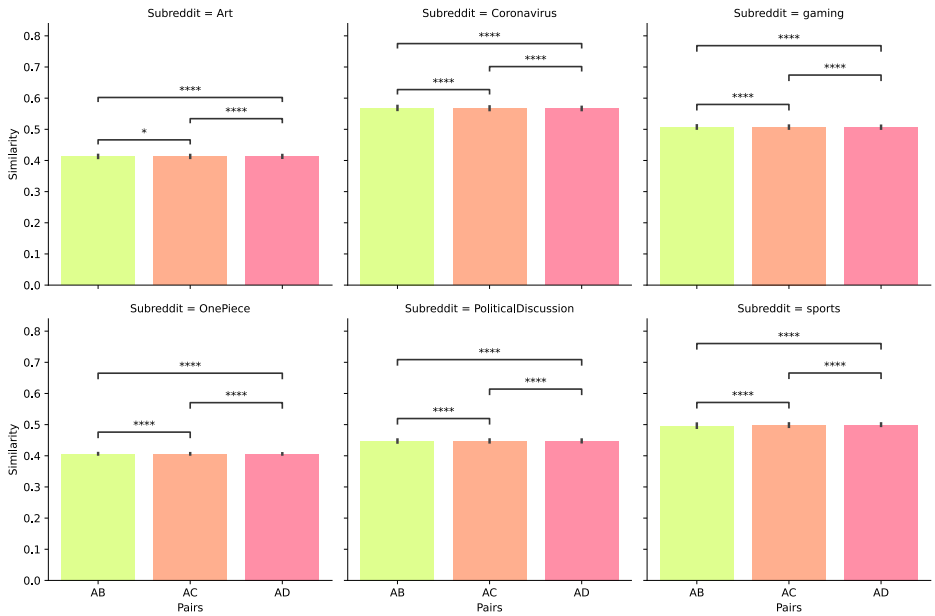


Figure 6.6: Average similarity scores for each pair of comments based on the DSE-BERT model without fine-tuning.

threads we collected, and the task it was optimised for encourages it to represent consecutive sentences in a similar way. Specifically, the comments in our dataset are often very short and contain little linguistic information, which makes it difficult for the model to distinguish between them, and thus increases the likelihood of representing them in a similar way.

Conversely, Sentence-BERT (Reimers & Gurevych, 2019), which is used to produce the embeddings for the BERT Topic modelling, was trained on two corpora designed for Natural Language Inference (NLI). While the first one, the Stanford Natural Language Inference (SNLI) corpus (Bowman et al., 2015), is solely composed of image captions produced by crowdworkers, the second one, the Multi-Genre Natural Language Inference (MultiNLI) corpus (A. Williams et al., 2018), is composed of texts from ten different genres, including fiction, government reports, transcriptions of telephone and face-to-face conversations, among others. The diversity of the MultiNLI corpus, and the fact that it includes weakly organised types of language use such as face-to-face conversations, likely increased the generalisation capabilities of the model, and improved its ability to capture information from short texts, which is what we are looking for in this study. This suggests that the BERT architecture is capable of extracting useful information for our task from *Reddit* comments. Fine-tuning the DSE-BERT model on our dataset should therefore allow it to learn a topical similarity metric that is able to distinguish between

the comments, which we will confirm in the next section (Section 6.4).

6.4 Metrics training

Our goal is to build a topical similarity measure for dialogue that operates at a rather small scale since we are interested in the topical structure between comments. In a typical supervised learning setting, we would need to rely on a dataset of pairs of comments annotated with a topical similarity score. However, this raises the same kind of issues as the ones we discussed in Chapter 5: it is very difficult, and maybe impossible, to reliably assign such scores to pairs of comments. Instead, we leverage the hierarchical structure of *Reddit* threads to define three levels of topical similarity between comments, and use the same method as Xing and Carenini (2021) to train a model that learns to rank pairs of comments based on their similarity. We briefly describe the model and its training procedure in the following section (Section 6.4.1). For more details on the unsupervised training procedure, we refer the reader to Section 5.1.2. We then describe the results of our experiments in Section 6.4.2.

6.4.1 Implementation details

We adapt Xing and Carenini's (2021) utterance-pair coherence scoring to our task and train a model that rates the similarity between two *Reddit* comments.

Model architecture The structure of our model is described in Figure 6.7. The main difference compared to the approach we described in Chapter 5 is that we replace the basic BERT model by DSE-BERT (Zhou et al., 2022), a version of BERT that has been fine-tuned with a contrastive learning task that encourages the model to represent consecutive sentences in a similar way. Even though this fine-tuning was done on task-oriented dialogues, most of them following the Wizard of Oz paradigm – the training data is the same as for TOD-BERT and is described by C.-S. Wu et al. (2020), we hope that it provided the model with a better ability to capture small-scale similarities between sentences, which is what we are looking for in this study. This model operates with pairs of inputs instead of a singular ones.

Hence, our architecture first takes a pair of comments as input and passes them to DSE-BERT. The last hidden layer of the output, which is the one carrying the in-context representation, is then passed to a multi-layer perceptron with three layers. The first layer does not change the dimensionality, the second one reduces the dimension from 768 to 6, and the last one from 6 to 1. Between the layers, we use the rectified linear activation function (ReLU) and a dropout of 10% to avoid overfitting. The final output is then passed to a sigmoid function to bring it between 0 and 1, as we want the output to represent a score of topical similarity between the two comments.

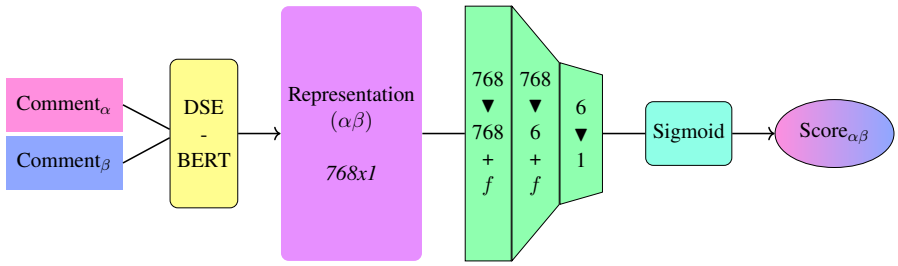


Figure 6.7: Model description. f = ReLU + dropout.

Loss function As in Chapter 5, we train our model with a *marginal ranking loss* function. This means that instead of learning to predict a precise score for a given pair of comments, it learns to rank pairs of messages based on their similarity. As represented on Figure 6.8, given the three pairs (main, child), (main, sibling) and (main, random), we want the model to learn the ordering of the scores: Score$_{AB}$ > Score$_{AC}$ > Score$_{AD}$. Formally the function we want to minimise is the following:

$$total_loss = \frac{loss_{AB} + loss_{AC} + loss_{BC}}{3}$$

with

$$loss_{\alpha,\beta} = ReLU(m - Score_\alpha - Score_\beta)$$

where m is a margin hyperparameter.

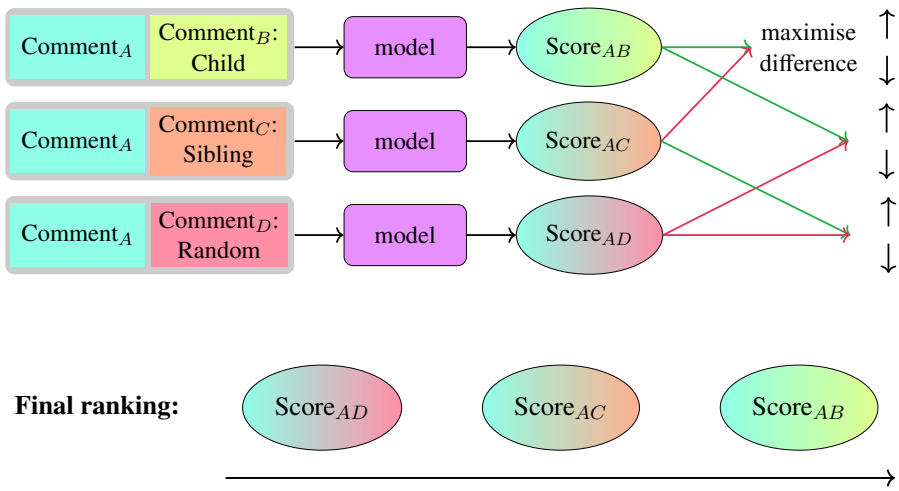


Figure 6.8: Representation of the training process.

Training setting We trained separate models for each of the six subreddits we collected. Every time, the dataset is subdivided into a training and a test set with the proportions 80%/20%. Given the hierarchical structure of the threads, the split could not be done completely randomly as it may have separated comments from the same conversational thread into the training and test sets. This would have been especially problematic as we rely on the structure to define the three levels of topical similarity. Hence, we split the dataset at the thread level, that is to say that all the comments from a given thread, i.e. that answer to the same post, are either in the training or in the test set. In order to keep a balance regarding their size, we order them all by size, i.e. based on the number of comments they contain, and keep every fifth thread for the test set. For the training phase, we also extract 10% of the training set for validation purposes.

Regarding the generation of the triples of pairs of comments, the combinatorial nature of our approach enables us to generate a large number of triples even with a relatively small number of comments. We thus experimented with different sizes – 1,000, 7,500 and 15,000 triples – to determine a suitable amount, taking into consideration the fact that a bigger dataset requires more time and resources to train our model. The best results came with the 15,000-triple dataset and the improvement compared to the 7500-triple one was not large enough to justify testing larger datasets. We also noticed that the results were equivalent when using 15,000 triples for one epoch or 7,500 triples for two epochs. We thus decided to use the 7,500-triple dataset as the construction of the triples is time-consuming. The selection of triples is done randomly with the restriction that the same comment cannot be used twice as main comment, i.e. Comment_A in Figure 6.8.

In terms of technical implementation, we used the optimiser AdamW (Loshchilov & Hutter, 2019) from the Python `transformers` library, with a learning rate of $2e-5$ and a batch size of 16. For the loss function, we experimented with different values of the margin hyperparameter m to determine a suitable value. We first set it to 1 as in Xing and Carenini (2021) but it made our metric take values that were very close to 0 and 1 most of the time, which was not desirable as it made it very similar to a binary classifier while we were interested in the continuous aspect. We thus experimented with different values of m and found that a value of 0.2 produced a more continuous metric while still being able to rank the pairs correctly.

6.4.2 Evaluation

We evaluate our metric for its ability to produce correctly ranked coherence scores for the three levels of topical similarity we defined in Section 6.2.4. We first evaluate the models trained on each subreddit independently with all the metrics we defined in Section 6.1.2. We then also evaluate their ability to generalise to other subreddits by testing each model on the test set of the other subreddits. All the evaluations were done on a test set of 100,000 triples of pairs of comments, se-

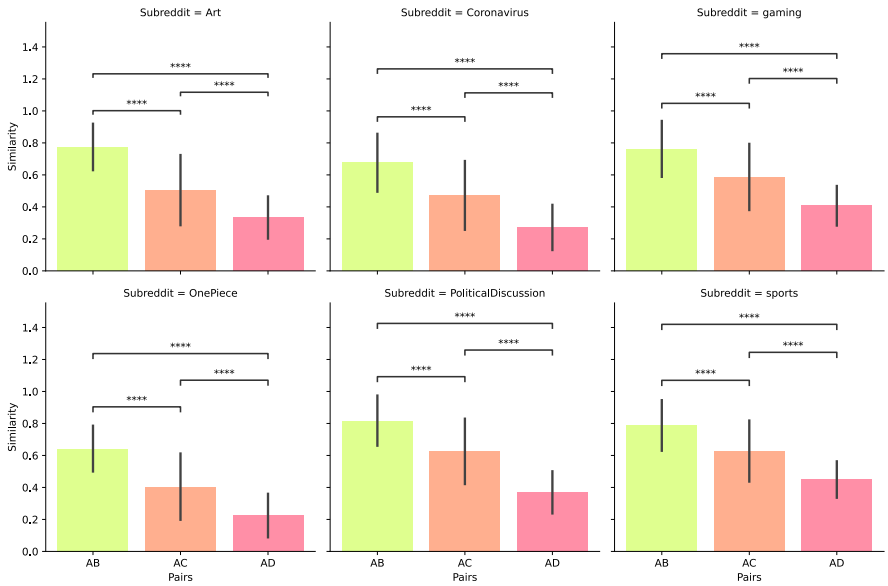


Figure 6.9: Average similarity scores for each pair computed by the models trained each on one subreddit.

lected randomly. We tried to select a large enough number of triples to have a good representation of the test set while keeping the evaluation time reasonable, which amounted to about an hour for each evaluation.

6.4.2.1 Independant models

The first aspect of the evaluation concerns the average similarity scores for each pair of comments. As we can see in Figure 6.9, the models trained on each subreddit are able to produce scores that are in line with our hypothesis, that is to say that the average score for the pair (main, child) is higher than the average score for the pair (main, sibling), which is itself higher than the average score for the pair (main, random). The values differ between the subreddits, which is not surprising as they have different characteristics. A Student's t-test indicated that the differences between the average scores for each pair are highly significant ($p < 0.001$) for all the subreddits. Nevertheless the large standard deviations indicate that the scores are likely not always in line with our hypothesis. The high significance of the differences between the average scores is likely due to the very large number of triples in the test set, which makes it easier to detect small differences. The next step of the evaluation is thus to analyse the ranking of the scores for each triple of pairs of comments.

Table 6.9 presents the different evaluation metrics related to the ranking of the scores for each triple of pairs of comments. The pairwise accuracy is high for all

the subreddits, which indicates that the models are able to rank most of the pairs correctly. When we look in more details at which of the three pairs are ranked correctly, we can see that the models are better at ranking the first pair (main, child) and the third one (main, random) than the second one (main, sibling). This is confirmed by the pairwise comparisons, which show that the first and last pair are most of the time ranked correctly with respect to each other, while the second pair is less often ranked correctly with respect to the first and last pairs. The proportion of triples for which the ranking is perfect lies between .54 and .65 for all subreddits. All of these results indicate that the models are able to learn a topical similarity metric that is in line with our hypothesis, even though there is still room for improvement. *r/PoliticalDiscussion* is the one for which the model performs best on most metrics including the perfect ranking metric, and *r/Art* for the rest. As discussed earlier in Section 6.2.3, *r/PoliticalDiscussion* is the most singular subreddit in our dataset, with longer conversations and comments on average. This topology is beneficial to our approach as it increases the chances (1) for the random comment (D) to be quite far from the main comment and its siblings, and (2) for the sibling comment (C) to be more clearly semantically different from the child comment (B). This is likely why the model trained on this subreddit performs better than on the others. We will explore the impact of the dataset on the evaluation results independently of the model's training in Section 6.4.2.2 when we evaluate the transferability of the models to other subreddits.

Finally the last aspect of the evaluation regards the margins between the scores for each pair of comments. As we can see in Figure 6.10, for most of the subreddits, the margin between the first and second pair is larger than that between the second and third pair. This is quite surprising as the messages in the pair AB are at a distance of 1 from each other in the conversation tree, just like the messages in the pair AC though the distance is horizontal instead of vertical. The messages in the pair AD, however, are at a random distance from each other in the conversation tree, which should make them more different from each other than the messages in the pair AC. This is not what we observe in the results, and the standard deviations are quite similar for the two margins, indicating that the random pair does not vastly increase the variability of the results. This may be due to the topology of the conversation trees, which are often very large and shallow, thus making randomly selected comments quite likely to be at a short distance from the sibling comment. This idea is reinforced by the fact that the margins for the model trained on *r/PoliticalDiscussion* are indeed more in line with our expectations. Nevertheless, the larger margin between the first and last pair confirms that the randomly selected comment is on average quite different from the main comment, supporting our hypothesis despite the potential topological bias.

Table 6.9: Evaluation of the ranking performed by the models trained each on one subreddit. *'Acc.' is the pairwise accuracy; '1st✓', '2nd✓' and '3rd✓' are the proportions of of correctly ranked first, second and third scores; 'AB > AC', 'AC > AD' and 'AB > AD' are the proportions of pairs correctly ordered two by two; 'Perfect Ranking' is the proportion of triples for which the ranking is perfect.*

Subreddit	Acc.	1st✓	2nd✓	3rd✓
r/Art	.85	**.83**	.59	.71
r/Coronavirus	.83	.76	.58	.75
r/gaming	.82	.73	.55	.73
r/OnePiece	.85	.82	.60	.74
r/PoliticalDiscussion	**.87**	.79	**.66**	**.83**
r/sports	.82	.74	.57	.75
All	.84	.78	.59	.75
Random sim.	.50	.33	.33	.33

Subreddit	AB $\geq$ AC	AC $\geq$ AD	AB $\geq$ AD	Perfect Ranking
r/Art	**.85**	.73	**.97**	.58
r/Coronavirus	.78	.78	.94	.57
r/gaming	.76	.76	.93	.54
r/OnePiece	.84	.76	.96	.60
r/PoliticalDiscussion	.80	**.85**	.96	**.65**
r/sports	.76	.78	.93	.55
All	.80	.77	.95	.58
Random sim.	.50	.50	.50	.17

6.4.2.2 Model transfer

To assess the generalisation ability of our models, we also evaluated them on the test sets of the other subreddits. This allows us to determine if the models are able to learn a general topical similarity metric or if they are too specific to the subreddit they were trained on. The pairwise accuracies are presented in Table 6.10. Unsurprisingly, the best result for a given model or dataset is obtained with the congruent model. However, the results remain quite good when transferring a model to another subreddit. This is encouraging as it indicates that we could be able to train a single model that could be used for all the subreddits, thus reducing the training time and resources needed. Another aspect, directly relevant to our study in this chapter, is that the entirety of the threads from a given subreddit could be included in the analysis if we use a model trained on another subreddit. This is of particular interest for our exploratory analysis because the way we split the dataset

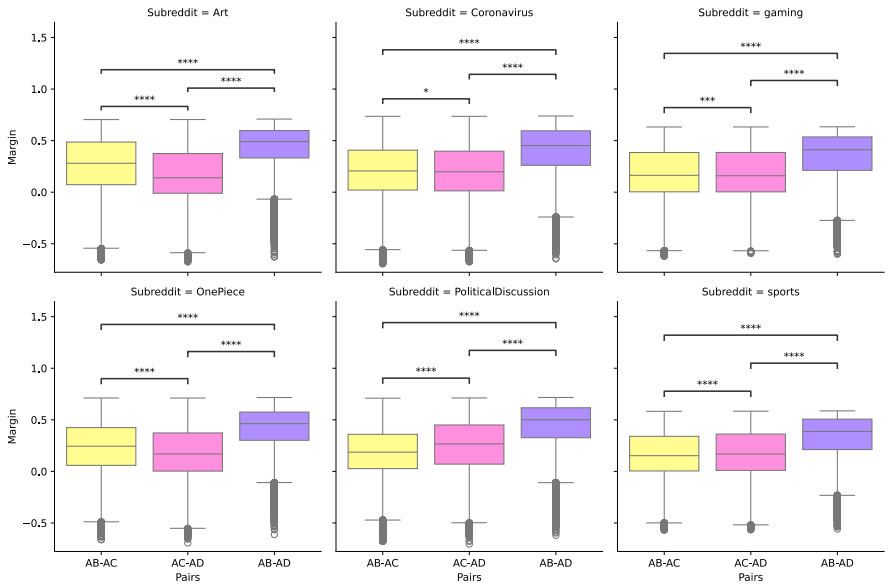


Figure 6.10: Average difference between the scores for each pair computed by the models trained each on one subreddit.

into training and test sets, as described in Section 6.4.1, removed a large number of dense threads from the test set, which are the ones that are the most likely to contain longer sequences of comments.

When it comes to the selection of the best model overall, the one trained on *r/PoliticalDiscussion* performs the best on average, though the differences are not large. This indicates that despite the singular topology of this subreddit, the model was able to learn a metric that can be applied to other subreddits with a different topology and average message format.

These evaluations confirmed that our models are able to produce similarity scores between comments that are, in most cases, in line with our hypothesis regarding the three levels of topical similarity. This is encouraging with respect to our goal of using the metric to analyse the topical structure of the conversations. The models are also able to generalise to other subreddits, which we will use in our exploratory analysis that are presented in the next section (Section 6.5).

6.4.3 Challenges of the data

Working with text messages gathered from the internet, and in particular from *Reddit*, comes with several challenges.

First, naturally occurring data is often noisy, with typos and abbreviations like 'Lololol' or 'Idk', or written onomatopoeias such as 'awwwhh'. The presence of non-standard tokens can affect the performance of our model as they are likely to be

Table 6.10: Pairwise accuracy of the models trained on one subreddit and tested on another one.

Data \ Model	*r/Art*	*r/Coronavirus*	*r/gaming*	*r/OnePiece*	*r/Political-Discussion*	*r/sports*
r/Art	**.85**	.79	.78	.80	.80	.79
r/Coronavirus	.78	**.83**	.78	.79	.82	.79
r/gaming	.81	.81	**.82**	.81	.80	.80
r/OnePiece	.81	.80	.79	**.85**	.80	.80
r/PoliticalDiscussion	.78	.79	.77	.77	**.87**	.77
r/sports	.81	.81	.80	.81	.80	**.82**
All	.81	.80	.79	.80	.82	.80

unknown to the pre-trained BERT model and thus replaced by the [UNK] token in the tokenising phase which can lead to a loss of information. Second, as discussed in Section 6.2.2.1, the format of *Reddit* threads is such that most posts include an image or a video that constitutes the topic of the thread. Participants' comments are rooted in this additional data while the model only has access to the text of the comments. This multimodal aspect makes it more difficult to include the posts in the analysis, in particular for subreddits such as *Art* where the title of the post is not always very descriptive. Lastly, some behaviours rooted in community specific or internet culture can complicate the analysis.

Reddit, as a community, has its own practices that create specific patterns in the data. For instance, some threads contain long sequences of comments where people repeat the same message, or count one after the other. One such example is shown in (6.1) which comes from *r/Art* under a post about snow sculptures of Moai heads in someone's front yard. The post is entitled 'Moai, me, snow, 2022', and after one of the direct answers to it 'This pleases the hivemind.', a sequence of 32 identical comments containing a Moai emoji follows, each comment posted by a different user.

(6.1)

Post: **Moai, me, snow, 2022 <snow sculptures of Moai heads in someone's frontyard>**

↳ This pleases the hivemind.

↳ 🗿

↳ 🗿

↳ 🗿

↳ 🗿

↳ +28

[r/Art]

Another example of a specific behaviour is the use of community specific vocab-

ulary and references to internet culture. For instance, in *r/OnePiece*, the characters and locations of the series are often mentioned as shown in (6.2). Even though this is not surprising given the focus of the subreddit, it remains challenging both for an embedding model to represent these tokens, and for a human reader to understand the content of the comments if they are not familiar with the references. Moreover, references to other stories are also present such as 'Eden Zero', a manga by another author, the reference to the *Pokemon* franchise with 'Pokeball', or the mention to 'an elephant with the flat Earth on its back' which is most likely a reference to the *Discworld* series by Terry Pratchett.

(6.2)

Post: **I drew the Grandline with planets instead of islands <digital drawing>**

↪ ifound pandaman

↪ Dude! I LOVE how marine ford has the water rings like saturn. I love how Zou is literally still just an elephant 🤣 Make more pls!!

↪ It's not just an elephant - it's an elephant with the flat Earth on its back.

↪ Thats basically Eden Zero

↪ Punk hazard = Pokeball

[r/OnePiece]

References to the *Reddit* environment can also be found in the comments, as shown in (6.3), from a post from *r/sports* showing a video of the end of a tennis match where one of the players, much shorter than the other, fetched a chair to embrace the other one over the net. The first three comments are a reference to the *Lord of the Rings* series, and the last one, that mentions the 'karma train', is a reference to the karma system on *Reddit*, a score that is given to users based on the upvotes and downvotes they receive on their posts and comments.

(6.3)

Post: **Good Sportsmanship <video of tennis match with one of the player fetching a chair to embrace the other one over the net>**

↪ It's wonderful to see you Gandalf!

↪ You're late.

↪ A wizard is never late, Frodo Baggins, nor is he early. He arrives precisely when he means to!

↪ Too late to successfully jump on the karma train is what he meant.

[r/sports]

Finally the freedom given to users in terms of form of the comments can also lead to some very creative, and unusual writing, as shown in (6.4). This comment comes from a post in *r/sports* showing a video of a baseball match where one of the cameramen with a long beard catches a ball that was hit out of the field. This

comment reflects the playful and social nature *Reddit* interactions can take, making them quite uninformative and thus harder to integrate in a topical analysis.

(6.4)

Post: **I caught a ball. <video of baseball match with one of the cameraman catching a ball that went off-field>**

↳ Beard

Bear

Bea

Be

B

Be

Bee

Beer

Give this man a beer.

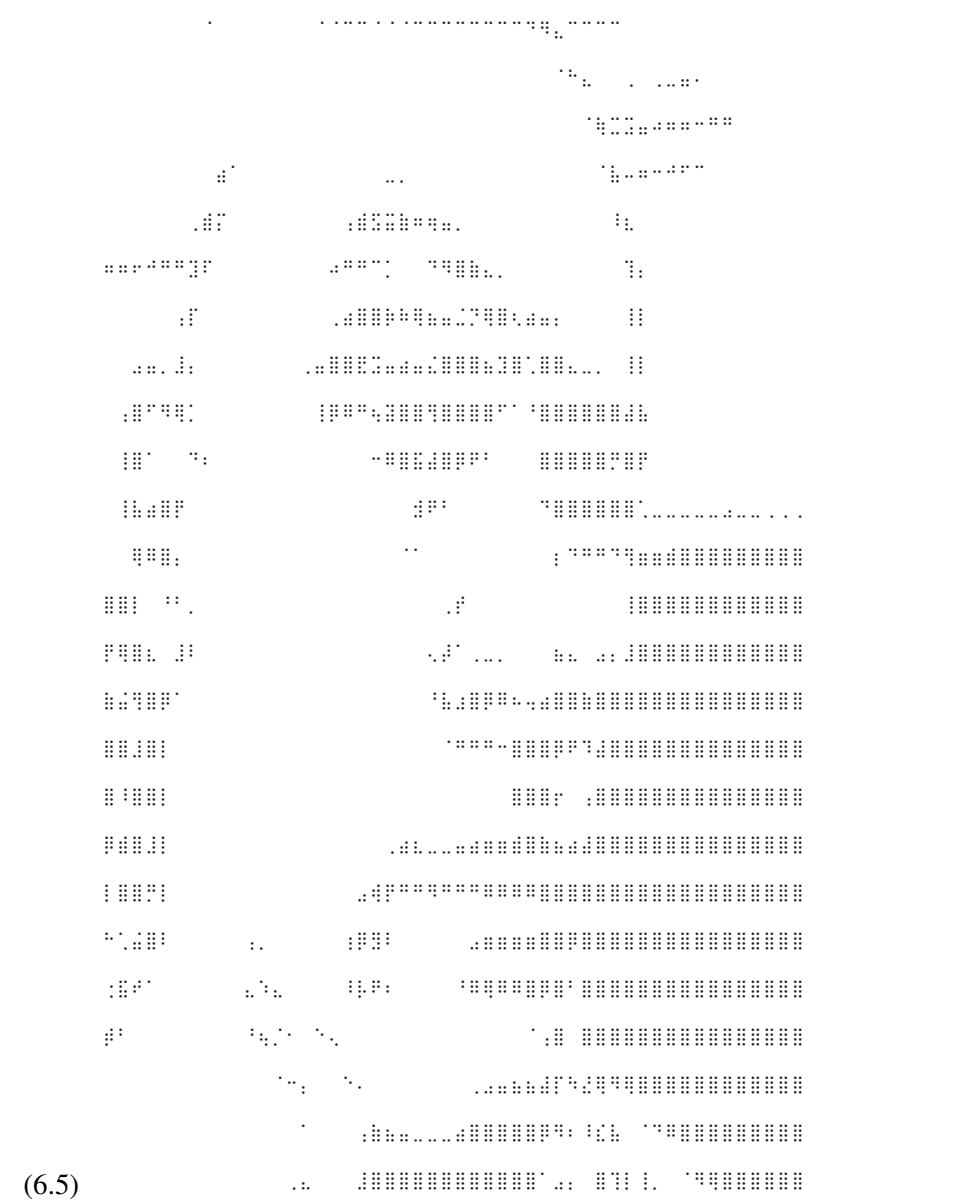
[r/sports]

Some comments can even contain no linguistic content at all, as shown in (6.5) taken from *r/OnePiece,* which seems to be an ASCII version of a famous picture of Dwayne 'The Rock' Johnson with one eyebrow raised.

6.5 Conversation modelling with a continuous topic similarity metric

Our motivation to build a continuous topical similarity metric was to be able to analyse the topical structure of conversations on *Reddit* to explore non-binary topic change patterns. What we call a *conversation* in the case of *Reddit* threads is a sequence of comments that answer each other subsequently, as shown in Figure 6.11. There are thus as many conversations in a thread as there are leafs in the conversation tree. Moreover, two conversations can share the same initial sequence of comments, sometimes differing only in their last comment.

The most straightforward application of our metric consists in computing the similarity between subsequent comments in a conversation, which allows us to explore the evolution of the topic. This is one of the two aspects we explore in this section. However, we can also compute the similarity between any two comments anywhere in the tree or between the post and a comment – with the limitations regarding the posts discussed in Section 6.2.2.1. To limit the explosive number of combinations to analyse, we focus on the similarity between a comment and the



(6.5)

[r/OnePiece]

post. For the conversation H_1–H_5 represented in Figure 6.11, it means that we compute two similarity scores per node, one with the previous comment and one with the post. The two scores are thus the same for the first comment of the conversation, H_1, and may then differ for the subsequent comments. This allows us to explore the evolution of the topic in a conversation as well as how well each comment fits as an answer to the post.

A different analytical angle was presented by Decker and Amblard (2024b),

based on the first version of the metric with a margin of 1 as discussed in Section 6.4.1. This version of the metric mostly produced scores very close to 0 or 1, making it more similar to a binary classifier than a continuous metric, which is not ideal for our purpose. We thus explored a different perspective with the new version of the metric.

Based on the results of the evaluation presented in Section 6.4.2, we chose to work with a single model for our exploratory analysis. The model trained on the *r/PoliticalDiscussion* subreddit was selected as it performed best on average on the transfer evaluation. This allows us to use the entire dataset for five out of six subreddits, thus providing us with a larger pool of conversations, and the test threads only for *r/PoliticalDiscussion*.

We first quantitatively describe the topical similarity scores produced by our metric on our six subreddits (Section 6.5.1). Then we discuss some patterns of topic changes that we observed in the data (Section 6.5.2). We finish by presenting a modelling proposal on a toy example from the Reddit dataset (Section 6.5.3).

6.5.1 Quantitative description

The first aspect we analyse is the distribution of the similarity scores between the comments and the post, as well as between subsequent comments. As shown in Table 6.11, both types of similarities are rather high on average, with a mean of .73 for the comment–post similarity and .80 for the comment–parent similarity. The higher value for the latter is consistent with the intuition that a comment is more likely to be similar to the comment it answers to than to the post itself. Overall, the scores are slightly skewed toward higher values as indicated by the median values (Q_2) that are marginally higher than the mean values.

In terms of range, while the maximum values are close to 1 for both types of similarity (.96), the lowest values do not get so close to 0, with a minimum of about .25. This may be due to the fact that comments under a same post are likely to be always somewhat related to each other but it could also be a consequence of the way the model was trained. Both the rather high average and minimum values should be taken into account when interpreting upcoming results of the analysis.

We also investigated the dynamics of the similarity scores throughout the conversations by computing the correlation between the depth of the comments and their similarity scores. As summarised in Table 6.12, the correlation is negative for the similarity to the post, which indicates that the deeper a comment is in a conversation, the less similar it is to the post. This is consistent with the expected dynamics of a conversation where the further away two interventions are from each other, the less similar they are likely to be on average. Conversely, the correlation is positive for the similarity to the parent comment, which indicates that the deeper a comment is in a conversation, the more similar it is to its parent comment. This may be due to the fact that the deeper a comment is, the less anchored it is to the

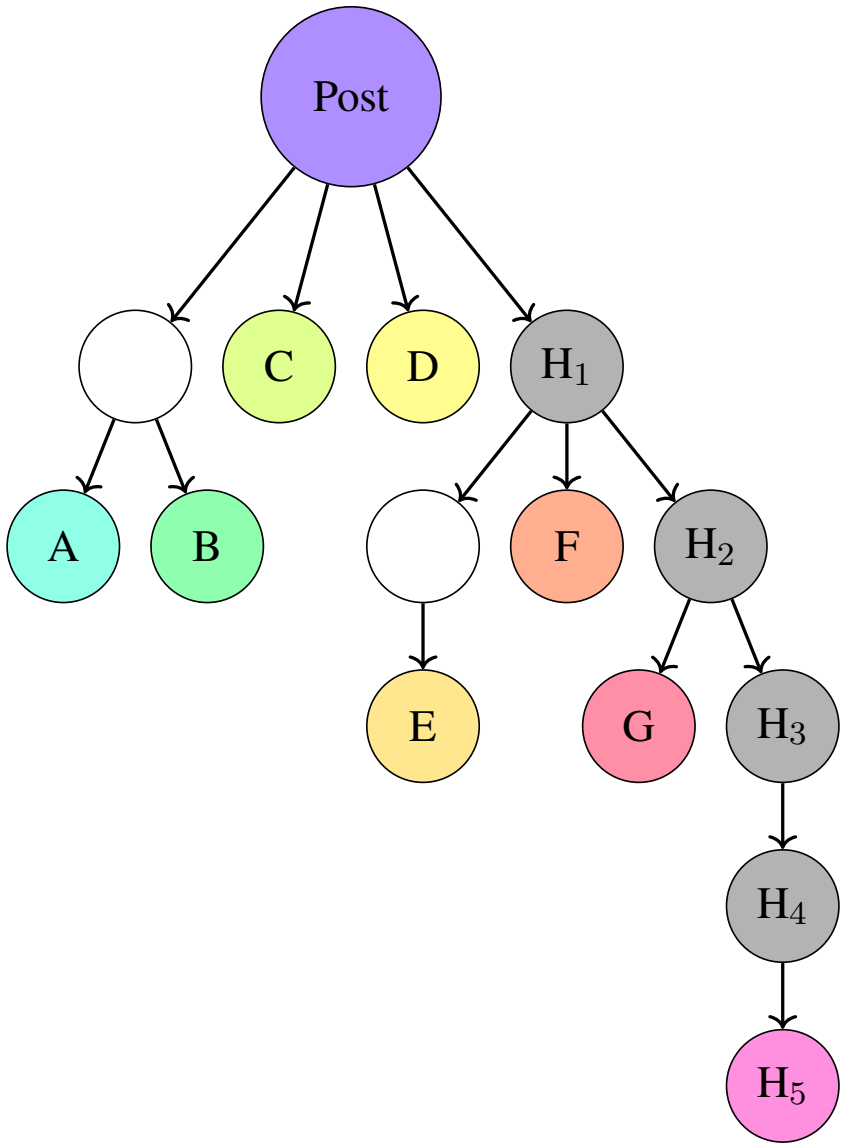


Figure 6.11: Representation of a *Reddit* thread and its 'conversations'. Each coloured node, labelled with a letter (A–G and H_5), is the last comment of a conversation, and the numbered nodes H_1 to H_5 form one of the conversations.

Table 6.11: Statistics regarding the similarity scores between the comments and the main post, as well as between subsequent comments.

(a) Similarity to post.

Subreddit	$\mu \pm \sigma$	min–max	Q_1	Q_2	Q_3
r/Art	.73 ± .15	.25–.96	.66	.76	.84
r/Coronavirus	.70 ± .19	.25–.96	.57	.74	.86
r/gaming	.72 ± .17	.26–.96	.61	.75	.85
r/OnePiece	.76 ± .13	.26–.96	.68	.78	.86
r/PoliticalDiscussion	.67 ± .19	.24–.96	.54	.69	.82
r/sports	.75 ± .16	.25–.96	.67	.79	.87
Average	.73 ± .17	.24–.96	.64	.76	.86

(b) Similarity to parent comment.

Subreddit	$\mu \pm \sigma$	min–max	Q_1	Q_2	Q_3
r/Art	.79 ± .14	.27–.96	.72	.82	.90
r/Coronavirus	.80 ± .16	.25–.96	.72	.85	.93
r/gaming	.79 ± .15	.26–.96	.71	.84	.92
r/OnePiece	.79 ± .13	.28–.96	.72	.81	.90
r/PoliticalDiscussion	.81 ± .17	.25–.96	.72	.87	.94
r/sports	.81 ± .14	.26–.96	.75	.85	.92
Average	.80 ± .15	.25–.96	.72	.83	.92

post, which requires it to be more similar to its parent comment to be relevant in the conversation.

These results indicate that the similarity between a comment and the post and between subsequent comments varies throughout the conversations, and that the dynamics of the two types of similarity are different. This leads us to the next step of our analysis, which is to investigate the patterns of topic changes in the conversations.

6.5.2 Patterns of topic changes

Participating in a conversation in a given thread implies that the participants are aware of the topic of the post and that their contribution will be somewhat related to it. However, as the conversation unfolds, the topic may drift away from the original one and participants may start discussing other topics, likely related in some way to the original one but with more or less distance from it. Studying the variation of the similarity to the post allows us to identify places of the conversations

Table 6.12: Correlation (Spearman's) between the depth of the comments with their similarity scores to the post and the parent comment. *All p-values* $\ll .001$.

Subreddit	Depth correlation with post similarity	Depth correlation with parent similarity
r/Art	-.13	.35
r/Coronavirus	-.22	.23
r/gaming	-.13	.37
r/OnePiece	-.11	.34
r/PoliticalDiscussion	-.09	.24
r/sports	-.22	.22
All subreddits	-0.18	0.30

where the topic is more different from the original one than in neighbouring comments, and are thus likely to represent topic changes. While two comments sharing the same similarity to the post *may* be on different topics, comments with a large enough difference in similarity are *likely* to indicate topic variation places. We thus use the similarity to the post as a proxy for coarse-grained topic change, and leverage TextTiling to identify the boundaries between topics on conversations of six or more comments. We also apply TextTiling on the similarity to the parent comment, which is the more intuitive approach and likely reflects more fine-grained topic changes. We then compare the two sets of boundaries to see if they are aligned or not. As described in Table 6.13, about two thirds of the conversations obtain a perfect WD score, reflecting that they share the same boundaries for both approaches. Some conversations have no boundaries at all, which is not surprising as they are quite short. 40 to 50% of the conversations were allocated boundaries between all comments depending on the approach, which may be due to frequent variations in the similarity scores. However, this happens simultaneously for both approaches in less than 30% of the conversations, which leaves more than 40% of the conversations with a perfect WD score and a non-trivial topic segmentation pattern. All of this indicates that the two approaches – i.e. investigating the similarity to the post and to the parent comment – capture similar but not fully equivalent phenomena.

We thus looked into the similarity scores between the comments around the boundaries identified based on the similarity to the post. Their repartition, shown in Figure 6.12, indicates that in most cases, the comments around the boundaries are quite similar to each other. They are often more similar than any two random subsequent comments in a conversation, as the average over the entire dataset is around .80, which is where the first quartile (Q_1) lies for the comments around the boundaries as indicated by the more detailed statistics in Table 6.14. This indicates that the boundaries identified based on the similarity to the post are not necessarily

Table 6.13: Statistics regarding the TextTiling output on the topical similarities.

Subreddit	TextTiling – % Only boundaries (post / parent / both)	Post *vs* parent WD (↓) $\mu \pm \sigma$	% = 0
r/Art	.39 / .56 / .26	.13 ± .23	.70
r/Coronavirus	.42 / .56 / .29	.13 ± .23	.71
r/OnePiece	.41 / .55 / .26	.14 ± .24	.70
r/PoliticalDiscussion	.39 / .51 / .27	.13 ± .23	.68
r/gaming	.40 / .53 / .25	.13 ± .23	.69
r/sports	.42 / .56 / .28	.13 ± .23	.71
Average	.41 / .55 / .27	.13 ± .23	.70

associated with a large difference in similarity between subsequent comments, and thus may not reflect a sharp topic change at a more local level.

Despite the exploratory nature of this analysis, these results support the idea that a binary approach to topic segmentation may be too harsh in the context of conversation. A more continuous approach would allow us to better capture the evolution of the topic in a conversation, and thus better understand the dynamics that underlie them. In the next section, we look at a toy example from the *Reddit* dataset to illustrate how we could model topic change in a more continuous way.

6.5.3 Proposal for a continuous topic segmentation

In this section we present a toy example from the *Reddit* dataset to illustrate how we could model topic change in a more continuous way. The conversation we focus on, (6.6), comes from *r/gaming*. The post, entitled 'Train Simulator is so immersive!', is a video showing a train simulation game where what the character sees in the game is the same as what the player sees in real life. After the poster mentions they lived this moment in a train the day before, someone asks them if they 'ever th[ought] that a higher being playing [them] in [their] version of [the game] also had this moment'. In C_3, another person continues by asking if 'that' was the reason why they had an 'emerald over [their] head'. From this point, the conversation is no longer about the Train Simulator™ game, which was picked up by our similarity model, as reflected by the slowly dropping similarity to the post shown in the top half of Figure 6.13. However, the comments remain quite close to one another as shown by the rather stable similarity between subsequent comments in the bottom half of Figure 6.13. Indeed, if we focus on the similarity between subsequent comments, we can see that it unsurprisingly begins very high between the post and the first comment, authored by the poster, before slightly dropping with the second comment, B_2 about higher beings playing their own version of the game. The similarity then remains quite stable, as indicated by the depth scores at

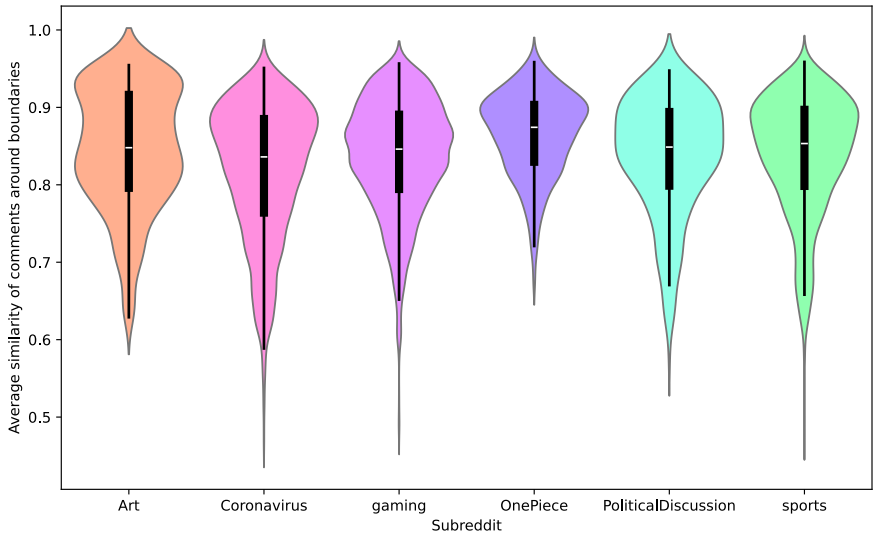


Figure 6.12: Average similarity between comments around topical boundaries computed on the post–comment similarities.

0, with a slight variation between C_3 and E_5, i.e. the mention of the emerald and the first comments of the next topic.

(6.6) A_0 **(POST)** Train Simulator is so immersive! <video from the perspective of a passenger in a train, playing a simulation game where what the character sees in the game is the same as what the player sees in real life>

A_1 Had this moment on Amtrak yesterday..

B_2 Ever think that a higher being playing you in his version of train simulator also had this moment hmmmm?

C_3 Oh, is that what this emerald over my head means?

D_4 Just avoid swimming pools and bath tubs and you'll probably be okay

E_5 Only enter rooms with multiple exits. Keep your baby out of the kitchen

F_6 Unless your kitchen furniture rearranges itself to block you around the stove while you cook an impossible meal.

G_7 or being Aquaphobic and you suddenly find yourself surround by pool, and you have to go to the bathroom, and you can only watch and cry as a burgler appears and steals your most valuable possessions.

H_8 Pool closed

[r/gaming]

Table 6.14: Statistics regarding the similarity scores between the comments around topical boundaries computed on the post–comment similarities.

(a) Similarity to post.

Subreddit	$\mu \pm \sigma$	min–max	Q_1	Q_2	Q_3
r/Art	.63 ± .13	.29–.82	.54	.65	.74
r/Coronavirus	.62 ± .15	.27–.92	.50	.61	.74
r/OnePiece	.66 ± .11	.37–.90	.60	.67	.74
r/PoliticalDiscussion	.61 ± .13	.29–.92	.51	.61	.71
r/gaming	.63 ± .12	.29–.94	.54	.64	.72
r/sports	.66 ± .13	.29–.93	.57	.67	.76
All subreddits	.63 ± .13	.27–.94	.54	.65	.74

(b) Similarity to parent comment.

Subreddit	$\mu \pm \sigma$	min–max	Q_1	Q_2	Q_3
r/Art	.85 ± .08	.63–.95	.80	.85	.92
r/Coronavirus	.82 ± .09	.47–.95	.76	.84	.88
r/OnePiece	.86 ± .05	.68–.96	.83	.87	.90
r/PoliticalDiscussion	.84 ± .08	.57–.95	.80	.85	.89
r/gaming	.84 ± .07	.48–.96	.79	.85	.89
r/sports	.84 ± .08	.48–.96	.80	.85	.90
All subreddits	.83 ± .08	.47–.96	.79	.85	.89

These results are consistent with the human interpretation of (6.6), which is that the conversation drifted away from Train Simulator™ after the mention of the higher beings. This comment triggered C_3 which seems clearly off-topic with respect to the post if one does not have enough background knowledge about the gaming community. The 'emerald' is indeed a reference to The Sims™, a life simulation video game where the characters have a green diamond over their head when they are being controlled by a human. The comment is thus a logical continuation to B_2 with the idea that if humans are being controlled by a higher being in their own version of a simulation game, then like the characters in The Sims™, humans would have a green diamond over their head. The following comments then discuss classical problematic situations in The Sims™ where the characters can get stuck in a place or even die because of the way the game is designed. Avoiding those situations would thus be a logical objective if one were aware that they are such a character.

Therefore, even though there *is* a topic shift in the conversation from the post to the point in the comments where The Sims™ become the focus, no *one* comment

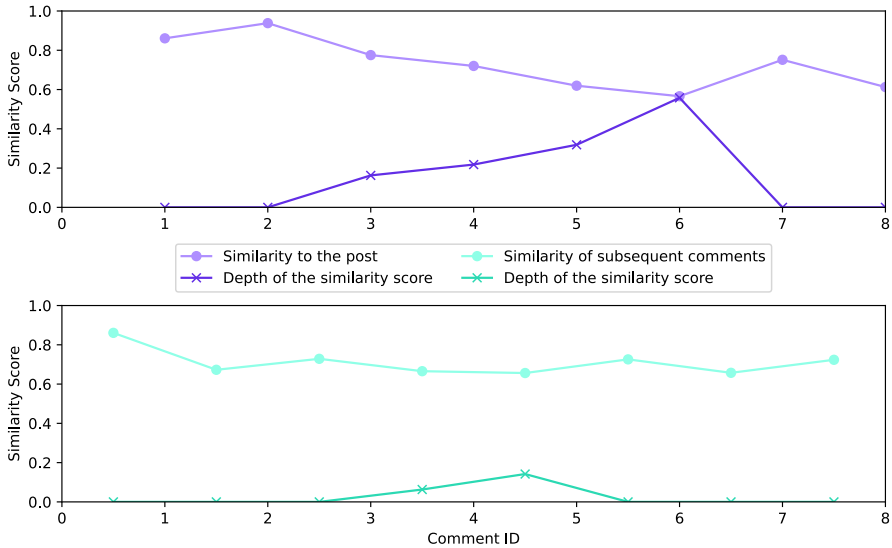


Figure 6.13: Similarities and depth scores throughout the conversation from (6.6).

is completely off-topic with respect to the previous one. Thus, annotating this exchange by placing a topical boundary somewhere around C_3 would place this topic shift at the same level as one between two fully unrelated comments, which would not be a fair representation of the actual dynamics of the conversation. Instead, we propose to consider C_3 as a *pivot* between two topics, without a complete change between B_2 and C_3 or between C_3 and D_4. The pivot from B_2 and C_3 is justified by the implicit reasoning that if humans are characters of a simulation game, then (like THE SIMS™) they have an emerald over their head. Then, the pivot from C_3 to D_4 is justified by the common sense idea that one should avoid dangerous situations.

While the pivot identified in (6.6) is a single comment, this does not have to be the case in general. A pivot could be a sequence of utterances such that utterances before and after those clearly belong to different topics, but the sequence in between does not display any sharp topic change and is instead justified by some kind of reasoning. Such reasoning patterns can be encoded in *enthymemes* and *topoi*, rhetorical structures that participate in argumentation and reasoning, while being partially implicit (Breitholtz, 2021). They seem to exist in our minds as resources that we can use to make sense of the world and to understand the reasoning of others, and are acquired through our experience of the world and our social interactions. This makes them inherently cultural and context-dependent, which is consistent with the fact that the understanding of the topic shift in (6.6) requires some knowledge of the gaming community.

Within this framework, understanding the topical dynamics of an interaction becomes a question of identifying relevant pivots and the possibly implicit reason-

ing that connects them, rather than simply identifying topic boundaries. A widely known topos would be much more likely to be identified, which would make it easier to understand a topic drift based on it. Conversely, a less well-known topos, or one specific to a particular community, would be less likely to be identified, which would increase the risk of misunderstanding how the new utterance fits into its context. This view allows us to link the understandability of a topic shift to the plausibility or salience of the reasoning that connects two parts of a conversation, thereby placing us in a more analytical and less productivist perspective. It is also compatible with the more classical linear topic segmentation approach, as a pivot of length 0 would correspond to a classical topic boundary.

6.6 Discussion and conclusion

In this chapter, we leveraged the hierarchical structure of *Reddit* threads to infer differences in topical similarity between comments. We first applied the topic modelling algorithm BERT Topic to confirm our hypothesis on the ranking of the topical similarity between comments in a conversation. We then used this ranking to explore the dynamics of topic change in conversations by building a non-supervised topical similarity metric. The granularity of the data our metric is based on, i.e. the comment level, is very fine, which means that it most likely relies on semantic information to compute the similarity between two comments. However, the clear improvement of the metric over the non-fine-tuned model, and over the BERT Topic approach, indicates that some information specific to this format has been captured. The black-box nature of the model unfortunately makes it difficult to identify exactly what dynamics are taken into account.

Nevertheless, a qualitative analysis of the evolution of the topic similarity throughout conversations highlighted differences in the dynamics of the similarity to the post and to the parent comment. This supports the idea that a topic shift is not always a binary phenomenon, but can be a continuous one. Deeper analyses are needed to better understand and model these dynamics, which foregrounds one of the trade-offs we must face in the current big data era of NLP. Having access to large quantities of data allows us to produce general models that can be applied to a wide range of situations. However, in order to really *understand* the linguistic phenomena that underlie these models and language use in general, a more direct analysis of the data is needed, which is increasingly difficult as the size of the data grows. This is particularly true for the dataset used in this chapter, which was too large to be analysed manually. We thus relied on corpus-wide statistics to explore the dynamics of topic change in conversations, which distances us from the actual data and the human experience of it.

The *Reddit* data is also particularly complex to analyse due to its structure and format. The tree structure of the threads creates an exponential number of conver-

sations as the depth of the tree increases. Indeed, every comment creates the opportunity for the conversation to branch out, thus creating many parallel interactions that would need to be analysed concurrently, which is an aspect we chose to ignore in this chapter. The format of the comments is also very heterogeneous, with some comments containing very little to no linguistic content. While practices such as conversing through memes or ASCII art are interesting from a sociological perspective, they fall out of the scope of our analysis. However, removing such comments from the dataset would be similar to removing repairs or so-called non-linguistic tokens from a conversation that would most likely change the nature of our data. Moreover, removing these comments would mean removing all subsequent comments as well, potentially drastically reducing the size of the dataset. The question of manually analysing the data also raises issues regarding the presence of toxic or insulting content. Despite the presence of moderation on *Reddit*, some exchanges may be considered harmful or offensive by some readers, making them distressing to analyse.

The format of the exchanges on *Reddit* also raises questions regarding the definitory scope of a conversation in terms of participant dynamics. Indeed, while some sequences of comments arise from the exchange between a few participants, others are the result of a large number of participants intervening only once. This raises the question of whether such exchanges can be considered as conversations or not. While we did not investigate this aspect in this chapter, it is an interesting question that could be explored in future work.

Overall, the analysis of the *Reddit* data highlighted the complexity and variability of topical structure in conversation. Even though many aspects of these exchanges are very different from face-to-face conversations, the topical dynamics we observed do not seem to be specific to the *Reddit* format. Indeed, the idea that a topic shift is not always a binary phenomenon but can be a continuous one sounds plausible in many other contexts, where an utterance or event triggers a set of ideas that are connected in more or less salient ways to the current context. The reasoning patterns we proposed to use to model these dynamics are also not specific to *Reddit*, as they are based on general interaction theories and rhetorical structures that are not tied to a particular format. In the next chapter (Chapter 7), we explore the motivation between a topic in conversation by looking into questions that are salient at a given point in interaction and whether they are actually resolved later or not. We empirically ground our analysis in real-life spoken conversations, which allows us to explore the dynamics of topic change in a more naturalistic setting, but also theoretically with a QUD-driven approach, thereby complementing the analysis of the *Reddit* data presented in this chapter.

7

Modelling Topics with QUDs

Contents

7.1 Background: Questions and discourse structure 303
7.1.1 QUDs as predictive tools 303
7.1.2 Operationalising QUDs through question *salience* and *resolution* 304
7.2 Source corpus: Naturalistic dialogues 305
7.2.1 The Spoken BNC 306
7.2.2 Sub-selection: Varied interactional settings . . . 306
7.2.3 Statistical description 310
7.2.4 Challenges of naturalistic dialogues 311
7.3 Method: Automatic question generation and human annotation 322
7.3.1 Automatic question generation 323
7.3.2 Guidelines adaptations for dialogue 325
7.3.3 Annotations 326
7.4 Released dataset: questions and annotations 333
7.4.1 World knowledge 334
7.4.2 Technical limitations 336
7.4.3 Prompt-related issues 339
7.5 Results and analysis: Dialogue is less predictable than static text 341
7.5.1 Salience 341
7.5.2 Resolution 343
7.5.3 Dialogue predictability 345
7.6 Conclusion 351

Topical structure is typically considered from a high-level perspective where sequences of utterances are grouped into segments that are coherent within themselves and distinct from other segments. In Chapter 6, we raised the idea that this structure could also sometimes benefit from more a more local perspective, where the ideas and issues raised by each utterance can slightly orient the dialogue towards

a new topic without necessarily inducing a sharp topic boundary. We further investigate this idea in this chapter by probing whether salient issues raised in a dialogue can be used to predict the content of later contributions. One influential approach to modelling this notion of salient issues in dialogue is the theory of *Questions Under Discussion* (QUDs; Ginzburg, 1995; Van Kuppevelt, 1995; Roberts, 2012), which represents dialogue as organised around a dynamically evolving set of questions that are addressed throughout the interaction. This can sometimes require to raise and resolve additional sub-questions. On this view, conversation proceeds by addressing questions whose possible answers define the space of relevant contributions at each point in the interaction.

QUD-driven frameworks have the advantage of providing a unified perspective on discourse organisation: they can be used retrospectively to reconstruct the structure and content of an interaction, or predictively to model expectations about how discourse will develop. In the predictive setting, so-called potential questions (PQs; Onea, 2016) are used to represent candidate issues that may become active QUDs as the discourse unfolds.

Despite their theoretical appeal, operationalising QUDs in natural data remains difficult. The space of relevant questions is not fixed, and questions can arise at multiple levels of granularity, from local clarification requests to broader discourse-level concerns. In addition, discourse structure is shaped not only by informational content but also by interactional phenomena such as turn-taking, speaker intentions, and shared situational context, all of which complicate the identification of a well-defined set of QUDs or PQs in spontaneous data.

To address this challenge, Y. Wu et al. (2024) propose an empirical approach based on LLMs, which are used to generate PQs anchored in textual context. They show that, in monologic data such as news articles and Ted Talks, the salience of a generated question correlates with whether it is subsequently answered in the text, suggesting that discourse is organised in anticipation of the recipient's expectations.

However, it remains unclear to what extent these findings extend to dialogue. Unlike monologic discourse, dialogue is jointly constructed in real time by multiple participants, and its structure emerges from interactional negotiation rather than a single authorial plan. To investigate whether QUDs/PQs-driven approaches generalise to naturalistic conversation, we apply the methodology of Y. Wu et al. (2024) to English dialogues exhibiting varying degrees of interactional organisation, and examine whether the relationship between question salience and later resolution persists in these more complex settings. By testing this particular method on naturalistic[21] dialogue (ND), we identify several challenges for a QUD-driven approach to discourse structure in conversation. We also release a corpus of 7,124 generated questions, each annotated for salience and resolution by two annotators[22].

[21]We use the term *naturalistic* instead of natural because we rely on transcription of the original conversations, making our study material a representation of the original natural interactions.

[22]https://gitlab.inria.fr/adecker/dialoguepredictability.git

The remainder of this chapter is organised as follows. Section 7.1 reviews the theoretical and empirical work underpinning our study. Section 7.2 describes the conversational corpus used in our experiments. Section 7.3 presents our methodology, and Section 7.4 introduces the released dataset. We then analyse the results in Section 7.5 and conclude with a discussion of their implications in Section 7.6.

The work described in this chapter was presented at SemDial 2026, the 30th workshop on the semantics and pragmatics of dialogue, as a long paper (Decker et al., 2026b).

7.1 Background: Questions and discourse structure

Questions have long played a central role in linguistic theories of discourse and dialogue structure. In particular, the QUD framework models discourse as evolving around explicit or implicit questions that guide interpretation and production. Originally developed to account for conversational coherence (Van Kuppevelt, 1995; Ginzburg, 1995; Roberts, 2012), it has more recently attracted interest in NLP as a way of operationalising and modelling recipients' expectations. This section first reviews QUDs as a theoretical and predictive framework (Section 7.1.1) before discussing recent computational work on question salience and the challenges posed by dialogue (Section 7.1.2).

7.1.1 QUDs as predictive tools

QUD-driven approaches represent discourse as a hierarchy of questions, where each contribution either (partially) answers a question, introduces a sub-question, or shifts the focus of the interaction (Van Kuppevelt, 1995; Roberts, 2012; Ginzburg, 2012). Traditionally, QUDs were used retrospectively: analysts inferred which question must have been under discussion to make an utterance coherent. More recent work instead argues that discourse participants actively anticipate forthcoming QUDs and use them to guide interpretation and production.

A growing body of evidence supports that QUDs are a cognitively relevant construct used when producing and interpreting discourse. Eye-tracking experiments by Clifton and Frazier (2012) show that comprehension is facilitated when readers' expectations about the QUD align with subsequent text. According to ERP (event-related brain potential) studies by Delogu et al. (2020), discourse segments failing to address an expected QUD incur higher processing costs. Extending this work computationally, Y. Wu et al. (2024) leveraged LLMs to generate PQs in news articles and TED talks and found that more salient questions are more likely to be answered later in the discourse, suggesting that authors structure discourse around anticipated recipients' expectations.

Empirical evidence from dialogue remains more limited. Kehler and Rohde (2017) showed that grammatical context influences the QUDs evoked during di-

alogue interpretation, although their experiments relied on controlled completion tasks. Westera and Rohde (2019) proposed a methodology for collecting actual and potential QUDs in ND, while Mulligan and Rawlins (2025) reported only moderate agreement between annotators when inferring QUDs from interviews. Together, these studies highlight both the relevance of QUDs and the large space of plausible questions associated with a given context. However, because existing work has focused on relatively structured interactions, it remains unclear how well the framework generalises to less organised forms of conversation. We address this question by applying the methodology of Y. Wu et al. (2024) to a diverse set of NDs.

7.1.2 Operationalising QUDs through question *salience* and *resolution*

Y. Wu et al. (2024) introduced the first computational framework designed to test whether salient questions are more likely to be answered in subsequent discourse. Working with English news articles (Ko et al., 2022; Huang et al., 2024) and TED talks (Westera et al., 2020), they considered questions evoked at a sentence S_k given the preceding context $S_1, \ldots, S_{k-1}$ and evaluated whether these questions were answered in the remaining text.

[1] Amid skepticism that Russia's war in Chechnya can be ended across a negotiating table, peace talks were set to resume Wednesday in neighboring Ingushetia. [2] **The scheduled resumption of talks in the town of Sleptsovsk came two days after agreement on a limited cease-fire, calling for both sides to stop using heavy artillery Tuesday.** [3] They also agreed in principle to work out a mechanism for exchanging prisoners of war and the dead.

[$Q1$] What other progress has been made towards peace recently?

- Is [$Q1$] salient given the context of [1] and the **anchor** [2]?

  → Yes, it elaborates on what is being done to establish peace in the region.

- Is it answered in the subsequent text?

  → Yes, in [3].

Figure 7.1: Example from Y. Wu et al. (2024) of a generated question based on a piece of news text.

To operationalise this idea, they leveraged LLMs to generate questions anchored in S_k and annotated each question for *salience* – its relevance at that point in the

discourse – and *resolution*[23] – the extent to which it is addressed later in the text. Figure 7.1 provides an example from their dataset. Spearman correlation between these two variables revealed strong positive relationships ($\rho = .59$ for news and $\rho = .74$ for TED talks). They also introduced QSalience, a Mistral-7B-Instruct model fine-tuned to predict question salience from context.

These findings provide a first large-scale operationalisation of QUDs as a predictive model of discourse structure. However, because the study focuses on monologic genres, it remains unclear whether the same relationship holds in dialogue. Indeed, as discussed throughout this thesis, dialogue is a more complex and multifaceted environment than monologue. Multiple participants, dependence on contextual information, and need to manage the interaction itself all increase the range of relevant PQs and may reduce the predictability of subsequent discourse. We thus investigate whether the relationship between question salience and resolution persists in ND, and whether it varies across different interactional settings.

7.2 Source corpus: Naturalistic dialogues

The method used in this study requires a corpus of dialogues associated with questions based on the context of several turns, and annotations of those questions for both salience and resolution. Given the specificity of these requirements we had to produce the annotations ourselves, which greatly lightened the constraints for selecting the dataset. Indeed, we only needed to select dialogues in a language we understood well enough to be able to carry out the annotation task, and for which we could generate the questions automatically. In our case, these two constraints are redundant since the possible languages would have been English and French, two well-resourced languages. In the end, the only constraint induced by our task for the selection of the corpus is represented in (7.i):

(7.i) $\left[\texttt{Frame.Language.Global.List} : \{\textsc{en}, \textsc{fr}\}\right]$

The British National Corpus (BNC) is a 100-million-word collection of written and spoken British English designed to provide a broad representation of language use in the late twentieth century. An XML version of the corpus was released in 2014 and is freely available for research under the terms of the BNC User Licence.

The corpus comprises both written and spoken language. Since our objective is to investigate the applicability of the QUD/PQ framework to dialogue, we focus exclusively on the spoken component. We first describe this component in Section 7.2.1, then present the selection procedure used to construct our experimental dataset in Section 7.2.2, and finally provide descriptive statistics in Section 7.2.3.

[23] In their paper, Y. Wu et al. (2024) use the term *answerability* to refer to the same concept. We decided to use *resolution* instead after several reviewers, whom we thank for their feedback, pointed out that the term was ill-suited.

7.2.1 The Spoken BNC

The spoken component of the BNC consists of an approximately 10-million-word orthographic transcription of a wide variety of spoken interactions, ranging from informal conversations to institutional and professional settings. The corpus was designed to achieve both demographic representativeness and contextual diversity. Because satisfying these objectives simultaneously is challenging, the spoken BNC is divided into a *demographically sampled* component and a *contextually sampled* component.

The demographically sampled component was collected through natural recordings. Participants were asked to record their conversations over periods ranging from two days to one week using portable tape recorders. Whenever possible, interlocutors were informed of the recording only *after* the interaction had taken place. Recordings were included in the corpus only when all participants had provided informed consent. The primary objective of this sampling strategy was to obtain a balanced representation of speakers across gender, age, and social class.

The contextually sampled component was constructed to cover a broad range of communicative situations. The main categories include *education and informative* contexts (e.g. lectures, classroom interactions, and news commentaries), *business* contexts (e.g. meetings, interviews, and trade union discussions), *public and institutional* contexts (e.g. political speeches, council meetings, and governmental discussions), and *leisure* contexts (e.g. sports commentaries, chat shows, and club meetings). A detailed description of these categories is available in the BNC User Reference Guide. Different sampling procedures were employed across contexts to ensure broad coverage of interaction types.

For the present study, we selected 60 dialogues from the spoken BNC, covering different interactional settings and varying degrees of expected conversational organisation. The selection procedure is described in the next section (Section 7.2.2).

7.2.2 Sub-selection: Varied interactional settings

One objective of this study is to examine how the relationship between question salience and resolution varies across different forms of interaction. We therefore selected dialogues that differ both in the number of participants and in their expected degree of interactional organisation. We operationalised these dimensions by defining six dialogue categories, described in Table 7.1 with the explicit constraints we applied to the BNC metadata.

The first two are spontaneous open-domain conversations drawn from the demographically sampled portion of the BNC. The remaining categories were selected from the contextually sampled component using metadata-based constraints designed to capture progressively more structured interaction types. We distinguish organised discussions (interviews and committee meetings) from public discussions (radio and television programmes guided by a host) because both rely on

an agenda but the latter are intended for an external audience which should make them easier to understand for external observers.

We also specified the number of participants, exactly 2 for the duologues and 5 or more for the multilogues to ensure a clear distinction between the two types of conversations. Finally, we specified the length of the dialogues to ensure that they were long enough to contain a sufficient number of questions and answers, but not too long to keep the annotation time reasonable.

For each category we randomly sampled ten dialogues, except for public multilogues, which were split equally between radio and television discussions. When selected dialogues were too long for manual annotation, we extracted shorter excerpts while preserving the interactional characteristics of the original conversation. Excerpts were truncated manually at the end of a topic and, for multilogues, were checked to ensure that the intended number of participants remained active.

Table 7.2 summarises the characteristics of the three degrees of interaction – open, organised and public – in our corpus with the duologues and multilogues combined using CHIMERA. The participants are human beings interacting in real life. In the duologues, the two participants take turn speaking with a more or less balanced repartition depending on the setting, where the interviews and radio broadcasts are less balanced since the host or interviewer typically asks questions and the other participant answers sometimes with long responses. Similar discrepancies exist in the multilogues with the open ones being more balanced and involving most participants at all times while in the organised ones – council meetings – a subgroup of participants is typically more active for a given point of the agenda. In the public multilogues, the floor is typically taken by one person at a time for longer interventions with the host selecting the speaker, which characterises the turn-taking

Table 7.1: Constraints for the random sampling of our corpus.

Group	**Metadata constraints**	**# Ppts**	**Length**	**#**
Open duo	classification: spontaneous conversation	2	$40-70$	10
Open multi	classification: spontaneous conversation	>4	$40-90$	10
Org. duo	classification: interview oral history	2	≥ 40	10
Org. multi	classification: meeting, subcategory $\neq$ Leisure, activity: does not contain 'informal'	>4	≥ 40	10
Public duo	classification: broadcast discussion	2	$40-90$	10
Public multi	classification: broadcast discussion, title: contains 'radio'	>4	≥ 40	5
	classification: broadcast discussion, title: contains 'television'	>4	≥ 40	5

behaviour. In terms of hierarchy, the most striking difference is the presence of a host or chairperson in the organised and public settings, which is absent in the open ones. This influences the structure of the interaction and the way topics are introduced and discussed. Other hierarchical aspects exist in some of the conversations independently of the setting, for example in the open conversations where participants can be family members, in some interviews where a participant can be much older than the host, or in the council meetings where a participant can be an expert of a topic on the agenda. The conversational history is a very relevant factor for our task since it can make the dialogues more complicated to understand from an external point of view. In the open conversations, the participants are typically familiar with each other and do not always clearly explain the events or people they refer to. The organised multilogues are council meetings where we do not have access to the entire history of meetings before the one we selected, which creates a similar situation. The remaining categories are more self-contained; public dialogues because they are intended for an external audience and organised duologues because they are interviews where the two people seem to have never met before. Nevertheless in all the non-open settings, a topic of discussion was defined prior to the interaction, which constrains the range of possible contributions. Conversely, the open conversations are completely free and thus much more chaotic. This aspect is very likely to play a role in the predictability of each type of dialogues.

All the conversations happen in a face-to-face setting though some specificities exist for each setting. In the open conversations one or several of the participants are sometimes realising a task in parallel to the conversation, which can reduce the attention they give to monitoring the coordination signals. In the television multilogues, the physical setting of the show possibly makes it complicated for some participants to directly look at each other. The critical aspect regarding the Environment & Medium parameter is that while the participants interacted in a format that is overall quite close to a prototypical face-to-face conversation, the transcriptions we have access to take away a lot of the multimodal information. Except for the radio setting, which is intended for an external audience that would have access to the speech only, all the other settings rely, at different degrees, on the physical co-location of the participants. It is most obvious in the council meetings where participants often refer to documents they own physical copies of, and in the TV setting where the host seems to be pointing at a person when addressing them. The latter context is quite easily reconstructable from the transcription and common sense but the former is not. The quality of the recording is also far from optimal – the BNC was recorded in the 1990s – and the transcriptions contain numerous `<unclear>` tags, sometimes hindering the understanding as we further discuss in Section 7.2.4.

The last differences pertain to the content and are probably central for the task discussed in this chapter. Apart from the open conversations, we can imagine that in all the other settings participants had an idea of the topic of discussion before the

Table 7.2: Description of the three interaction degrees from our BNC selection

Meta-Parameters	Sub-parameters	**Open**	**Org.**	**Public**
Agents				
Agency		Human		
Participation structures	Number	2 / > 4		
	Functions	All involved	Clear subgroups	Alternate one main speaker
				Intended external audience
Social dynamics	Hierarchy	✗	interviewer - interviewee / chairman - council member	host - guest(s)
	(Conversational) History	++	✗ / past councils	✗
*Enunciative status**	Anonymity	Physical co-presence		
	Role	Possible influence of recording	Self (as expert)	
	Commitment level	*min*: for conversation, *max*: ✓		
Conversational setting				
Epistemic grounding	World setting	1990s		
Environment & Medium	Physical context	mostly co-located, references to environment possible		
	Visibility	mostly ✓		radio: external audience has no visibility
	Prosody / Tone	✓		
	Noise	Possible	Dedicated room	Controlled
	Synchrony	Synchronous		

Table 7.2: Description of the three interaction degrees from our BNC selection (cont.)

Meta-Parameters	Sub-parameters	**Open**	**Org.**	**Public**
Conversational frame				
Language	Global	English, sometimes dialects		
	Control	Expected language		
	*Vocabulary**	Daily life	Possibly technical	Wide audience
Purpose	Practical goal	Social	Information exchange	
	Success condition	✗	✓	✓ (includes audience)
	*(Topical) structure**	✗	Agenda	Agenda
Verbal interaction			✓	

interaction started. In the interviews, the host and the interviewee probably agreed on a topic to discuss, sometimes possibly even agreeing on the questions that would be asked. Though some specific pieces of dialogue let us think that the interviewee or guests did not have all the details, we can imagine that these exchanges were somewhat prepared. This aspect goes together with the differences in structure we have already mentioned where the open conversations are completely free and the other settings are more or less structured with the host or secretary guiding the discussion. These features are likely to influence the predictability of the topics in the different settings.

7.2.3 Statistical description

Table 7.3 reports descriptive statistics for our selection of dialogues. Due to the way the open conversations were sampled, they are generally shorter than the other categories. They also exhibit shorter turns which can be explained by the very different nature of the interactions compared to the other categories in which the participants often contribute with very long explicative or argumentative turns.

All categories except the organised duologues contain a small proportion of turns whose speaker was labelled as unknown in the original transcriptions. During the experiments the annotators also reported that some dialogue turns were attributed to the wrong speaker. While it was clear in some cases, others were more ambiguous which could complicate the annotation. The linguistic content was sometimes unclear for the transcribers, which led to the presence of `<unclear>` tags in the transcriptions. These tags represent about 2% of all the tokens in our

corpus but depending on their position they could hinder the annotation process to a greater or lesser extent.

Table 7.3: Statistical description of the dialogues in our corpus per category. *The values are given in mean±std.*

Group	Context	# Turns	# Tokens / turn
Open duo	Spontaneous conversations	47±8	8±13
Open multi	Spontaneous conversations	58±7	7±7
Org. duo	Interviews	107±3	17±20
Org. multi	Committee meetings	97±12	18±22
Public duo	'Ideas in action' radio broadcasts	81±7	32±27
Public multi	Radio broadcasts	106±18	23±30
	Television debates / discussions	105±22	14±16

Group	Context	% Unk. spkr	# Gen. questions	% *Specific* questions
Open duo	Spontaneous conversations	7±22	66±11	19±4
Open multi	Spontaneous conversations	4±9	82±11	14±5
Org. duo	Interviews	0±0	155±5	17±4
Org. multi	Committee meetings	25±21	141±18	19±5
Public duo	'Ideas in action' radio broadcasts	3±11	116±10	24±4
Public multi	Radio broadcasts	6±7	154±27	18±7
	Television debates / discussions	6±3	152±33	21±4

7.2.4 Challenges of naturalistic dialogues

While studying naturalistic data is the best way to produce models and tools usable in real-world situations, it also comes with a number of challenges. In this section we discuss the main issues we encountered when working with the BNC. We first discuss ethical considerations in Section 7.2.4.1 and then present the main challenges related to the nature of naturalistic dialogues (Section 7.2.4.2) and the way it was captured (Section 7.2.4.3).

7.2.4.1 Ethical considerations

Working with human produced data requires careful consideration of ethical issues, even more so when the data is collected in natural settings. One of the concerns relates to the intended use of the corpus. While re-using existing corpora has the advantage of avoiding the need to collect new data, it also raises questions about whether the original consent given by participants covers the new use case. In our case, the BNC was collected in the 1990s and according to the Reference Guide, the original intended uses included academic linguistic research, artificial intelli-

gence, and natural language processing among others. Semantic and pragmatic information was explicitly mentioned as derivable from the corpus. Our use case is therefore compatible with the original consent.

Nevertheless, the participants were guaranteed 'complete anonymity' where 'all references to full names and addresses have been removed from the corpus and the log', which we found not to be the case in some of the dialogues.

Pseudonymisation Personal data protection has become a major concern in recent years. While the BNC was collected before the implementation of the General Data Protection Regulation (GDPR) in 2018, data protection was already discussed in the UK in the 1990s ('Data Protection – Volume 167: debated on Monday 19 February 1990', n.d.). Yet, the claim of complete anonymity is clearly not accurate. Data protection law includes a distinction between the concepts of *anonymisation* and *pseudonymisation*. The former refers to the process of irreversibly removing personal identifiers from data, while the latter involves replacing personal identifiers with pseudonyms or codes, allowing for potential re-identification under certain circumstances. In the BNC, some dialogues contain references to specific individuals or locations that could potentially lead to re-identification.

(7.1) shows that the participants expected that their names would be removed from the corpus. Given the explicit mention to 'Bet' in the third turn, and the presence of a first name in the description of each file with demographically sampled dialogues, we can assume that it meant that their *family names* would not be mentioned.

(7.1) **PS04F**$_1$ I don't know they said that no names are mentioned
PS04B$_2$ No
PS04E$_3$ She takes all them tapes away and Bet's name's not put on there.
Dialogue KBE – 009101 (Open multi) [BNC]

First names are not considered to be personal identifiers, so this does not represent a breach of privacy. However, some dialogues did contain the full names of participants though we only found such occurrences in the contextually sampled component of the corpus. This suggests that the requirements in terms of de-identification during the transcription were not uniform across the two components of the corpus, or even across the different dialogue types in the contextually sampled component. Some participants also expressed their reluctance to disclose personal information as we can see in (7.2) where PS22H is worried to say where exactly they were born.

(7.2) **PS22G**$_1$ `<unclear>` about when you worked at er `<unclear>`
PS22H$_2$ Yes `<unclear>` .
PS22H$_3$ Well where I was born in Needham Market

PS22G$_4$ Mm.
PS22H$_5$ need I say where because I, I can't <unclear> day.
PS22H$_6$ I don't
PS22G$_7$ Oh, I see.
PS22H$_8$ <unclear> the people to know
PS22G$_9$ You was born here then?
PS22H$_{10}$ I was.
PS22G$_{11}$ Ah, right well let's say you were born in the house you live in
PS22H$_{12}$ Well you could say
PS22G$_{13}$ without without actually saying where it was.

Dialogue HDJ (Org. duo) [BNC]

The recording of the conversation continued anyways and the interviewer found a way to avoid the disclosure of the participant's place of birth. However they later mention that they do not want people to know their age while the different information she provided in the interview would allow to infer it. This illustrates the difficulty of collecting natural dialogues while respecting the privacy of the participants. In particular, (7.2) is a conversation about the life of the interviewee which makes it difficult to avoid the disclosure of personal information, and thus makes full anonymisation impossible (Amblard et al., 2014). Had all named entities been really replaced by pseudonyms or de-identification tokens in the entire corpus, the BNC would have fallen under the pseudonymisation category. However, the presence of real names and locations in some dialogues indicates that the corpus does not fully meet the criteria for either anonymisation nor pseudonymisation. Leveraging advances in NLP since the release of the BNC, a more thorough pseudonymisation process could probably be applied thanks to named entity recognition (NER) systems. This would allow for a more accurate replacement of personal identifiers in the dialogues, reducing the risk of re-identification while preserving the linguistic content of the corpus.

Offensive content Another ethical consideration is the potential presence of offensive content in the dialogues. The dialogues under the open category are spontaneous conversations with no constraints on the topics discussed. As a result, they may contain offensive language or sensitive topics that could be distressing to some readers or annotators. While the BNC does not provide explicit warnings about offensive content, it is important to acknowledge this possibility and take appropriate measures to ensure that annotators are aware of the potential for encountering such content. The annotators were informed of this possibility and encouraged to skip any dialogues that they found distressing or offensive. In our case, we encountered only one instance of potentially offensive content where two people exchange insults, seemingly in a joking manner but which could be perceived as offensive by

some readers. (7.3) shows the very beginning of this dialogue where PS6NV explicitly states that they are about to use 'dirty swear'. While we do not have access to the previous context, we can imagine that this 'topic' was introduced specifically because the conversation was being recorded. This of course does not change the fact that the content could be distressing to some readers, but it illustrates another difficulty of collecting naturalistic dialogues as the collection itself can influence the content of the conversations.

(7.3) **PS6NV**$_1$ Now , for some fucking dirty swear!
PS6NV$_2$ Wo oh oh oh!
Continues with 39 turns of various swears, insults, slurs, etc.
Dialogue KSN – 135904 (Open duo) [BNC]

7.2.4.2 Collection setting

Collecting naturally occurring dialogues implies several challenges due to the collection format and setting. First, the presence of a recording device can influence the behaviour of participants, leading to changes in their speech patterns or the topics they choose to discuss. Second, capturing all the aspects of a conversation can be difficult, especially in multi-party face-to-face conversations where a lot of modalities are involved.

Meta-content The informed consent process for the BNC collection varied depending on the type of interaction. The contextually sampled component was collected through a variety of methods, including direct recordings of meetings, where the participants were informed at the beginning of the meeting and asked for their consent. (7.4) shows the very beginning of such a council meeting where the chairperson explains the recording process and explicitly asks if there are any objections.

(7.4) **J9DPSUNK**$_1$ Just a couple of things by way of introduction.
J9DPSUNK$_2$ Erm, the first point, relates to a a microphone here, and er and er tape recorder, and I think members of the sub-committee will have received a note from the Chief Executive Department.
J9DPSUNK$_3$ Longman's are doing a study into the spoken language, and are looking for, I think, ninety million words that are in common usage.
J9DPSUNK$_4$ They came to the County Council and asked for our permission, to to record a number of meetings in order to get a variety of settings and words, different words they used.
J9DPSUNK$_5$ It's it's the same procedure that we accepted at the Finance Committee last Friday, and in mentioning it and hoping it will be acceptance that Longman's can have the benefit of the Personnel's subcommittee contribution for the next hour and a half or so.

J9DPSUNK$_6$ There's, to be clear about it, that it's completely confidential, that their looking only for particular words that are used in different parts of the country and in different locations.

J9DPSUNK$_7$ No problems, I presume <pause> Sound.

J9DPSUNK$_8$ Right.

Dialogue J9D (Org. multi) [BNC]

Moreover, from what we could see in our sub-selection of dialogues, a representative of the BNC team was sometimes present to answer potential questions. Regarding the demographically sampled component, active participants were recruited and asked to record all their conversations for a period ranging from two days to one week. They were told that whenever possible they should not inform their interlocutors of the recording until after the interaction had taken place. After the interaction, they would all be informed and asked for their consent to include the recording in the corpus. However in some cases the participants were clearly all aware of the recording and it could influence their behaviour. In (7.5) several people are watching a TV-show and one of them keeps making comments, where several other participants ask them to be quiet. At the end of this extract, after **PS0EH** more directly asks **PS0EB** to be quiet, **PS0ED** explains that **PS0EB**'s rude behaviour is due to the recording as she wants to record real conversations rather than just the TV-show, which she confirms in the last turn. This illustrates how the presence of a recording device can influence the behaviour of participants and lead to changes in their speech patterns.

(7.5) **PS0EB**$_{51}$ Why don't they show us all Britain?

PS0ED$_{52}$ Sshh <pause> that is Britain, it's just in the winter

PS0EH$_{53}$ Sshh

PS0EB$_{54}$ Oh no

PS0EH$_{55}$ Sshh

PS0EB$_{56}$ cor this is a bit ugh ah let's all clap for Neil, er, er what's he gonna do go off and have a boxing match?

PS0EH$_{57}$ Sshh

PS0ED$_{58}$ Sshh

PS0EH$_{59}$ why don't you shut up for a moment

PS0EB$_{60}$ Oh sorry

PS0ED$_{61}$ cos she's recording and she wants to hear somebody speaking on it <laugh>

PS0EB$_{62}$ just sounds a bit silly if you can only hear the telly <pause>

Dialogue KCE – 029701 (Open multi) [BNC]

Modalities The fact that the participants of (7.5) are watching a TV-show becomes quite clear in this extract. However, nothing could indicate it at the beginning of the dialogue and without the visual information, it is even difficult to be sure whether the beginning of the transcription really corresponds to an exchange between the participants or if it is a transcription of the TV-show. This illustrates how working with only the auditive modality can be challenging for understanding the context of the conversation. (7.6) is a piece of conversation from the group Public TV which takes place in a TV show where a panel of 100 women are asked questions about a given topic and they can vote in a binary way before some of them explain their answer. Every time the host asks who would like to give their opinion, they seem to explicitly select one participant to take the floor, which is often not obvious in the transcription and we can assume that this selection is done through gestures and eye contact. We can see in turn 17 that the host asks who answered no to the question under discussion and they immediately continue with 'Would *you* like to say why?' before one person starts to answer. This seems to indicate that she understood that she was the one expected to provide an explanation, though we can only imagine this context and it is not impossible that she was the quickest to answer.

(7.6) **FLDPS000**$_8$ Let's find out what one hundred Scottish women think of parents today.

(4 utterances)

FLDPS000$_{13}$ Let's start with a question, since you are all children of parents, do you think your parents did a good job?

FLDPS000$_{14}$ Button one for yes, button two for no.

FLDPS000$_{15}$ Did your parents do a good job?

FLDPS000$_{16}$ And, in this particular hundred, eighty five say yes <pause> fifteen say no.

FLDPS000$_{17}$ Who said no?

FLDPS000$_{18}$ Would you like to say why?

FLDPS000$_{19}$ Mhm.

FLDPS001$_{20}$ Well I think my mother did a good job but my father was an alcoholic and that's, that made a difference, not just to him but <pause> to the entire family and I think probably to my own children.

Dialogue FLD (Public multi - TV) [BNC]

Similarly, while the transcribers relied on the audio recordings to produce the corpus, they were not asked to produce a *phonetic* transcription of the speech but an *orthographic* one. This means that as users of the corpus, we do not have access to the phonetic information or the prosody which can help disambiguate the meaning of some utterances. (7.7) shows a funny consequence of this limitation.

The conversation, an interview for an oral history project, revolved around accents and how certain words were pronounced in different regions of the UK in the interviewee's youth. Without a phonetic transcription or access to the audio recording, discussing the differences between the words 'table' and 'table' becomes rather comical.

(7.7) **PS3F3**$_{33}$ One of the common differences between East and West is the words like table and table in West we'd say table and East they'd say table.
Dialogue HV5 (Org. duo) [BNC]

These examples show that the manner and format data is collected in can create challenges for understanding the conversations and re-constructing the context in which they took place. On top of this aspect, the limitations of the transcription process itself can also hinder the understanding of the conversations.

7.2.4.3 Transcription limitations

Transcribing conversations from their recordings only, often without having been part of the interaction, is a challenging task for several reasons. While as participants in a conversation we can rely on the context and multimodal information to clarify words we may not have heard properly, or access basic information such as 'who said what', the transcribers can only rely on the recording. This means that they rely on its quality and the audibility of the speech and their own ability to distinguish participants – by voice and possibly world knowledge and inferred context – to produce the most accurate transcriptions. This can lead to errors or missing pieces in the transcriptions.

Unclear content When the transcribers were not able to understand what was said, they used the `<unclear>` tag to indicate that the content was not understandable. This can be due to several reasons such as poor audio quality, overlapping speech, or the use of unfamiliar words or accents. (7.8) is another extract from the same conversation as (7.7) where the interviewee repeats some expressions their grandmother used to say. Due to the accent and possibly the fact that these expressions are region specific, the transcribers were not able to understand what was said and used the `<unclear>` tag. This is particularly crucial as one of the first motivations for collecting the BNC was to study regional differences in English use. This is not as central for our own task that focuses more on semantic and pragmatics aspects but the presence of `<unclear>` tags was sometimes hindering for the annotators as they could not understand the content of some turns.

(7.8) **PS3F4**$_{60}$ I've got a granny who comes from the south isles and she says `<unclear>` and that's `<unclear>` .

PS3F4$_{61}$ If you say something and she's surprised she'll say, oh <unclear> .

Dialogue HV5 (Org. duo) [BNC]

(7.9) is an example where <unclear> tags were particularly problematic, all the more so because they were at the very beginning of the conversation. This extract is one of the open multilogues and the lack of context combined with utterances where most of the content is unclear, as in turn 4, makes it much more difficult to understand what is going on. This also created difficulties for the automatic question generation as it gave room to the model to hallucinate some content as we briefly discuss in Section 7.4.

(7.9) **PS0BA**$_{1}$ Have you been?

PS0BG$_{2}$ I got slightly lost.

PS0BG$_{3}$ Well I sort of went back down the <unclear> Road cos there was a traffic jam <pause> and er I went down <pause> oh sorry mate <pause> went down a road like which I thought was a good short cut, and they've got a lot of these roads and they've put like pavements across the end of them with bollards, and I went back and then there was a traffic jam and I got stuck because they were unloading a lorry <pause> so I've been er basically pissed about.<laugh><pause>

PS0BA$_{4}$ Yeah she a bit <unclear> now ain't she?

PS0BC$_{5}$ I don't think she has.

PS0BA$_{6}$ She has as far as pictures are concerned Rich .

PS0BF$_{7}$ She looks like er <unclear>

PS0BA$_{8}$ Have you got a number for her?

PS0BF$_{9}$ Yeah.

PS0BA$_{10}$ Different one?

PS0BF$_{11}$ Yeah.<unclear>

Dialogue KC6 – 055905 (Open multi) [BNC]

Speaker transcription In addition to the orthographical transcriptions, the speakers' identifiers had to be selected for each utterance. This is challenging based on the recordings only and as a consequence, a small portion of turns in our selection of dialogues have been attributed to an 'unknown' speaker labelled as **PSUNK**. The exact proportion of such utterances is reported in Table 7.3 for each dialogue group. (7.10) is a piece of conversation where most of the turns are attributed to an unknown speaker which, together with the numerous <unclear> tags, makes it quite hard to understand where the conversation is going. The interaction becomes clearer the more turns we read but our annotation task required to disclose the conversation gradually which means that we could not rely on additional context to improve our understanding.

(7.10) **JJAPSUNK**$_1$ <unclear> commitments in a recent task.

JJAPSUNK$_2$ <unclear> .

JJAPSUNK$_3$ Rights of way are an important part of our heritage <unclear> the road network should not be allowed to degenerate further.

JJAPSUNK$_4$ All <unclear> is a Labour activity <unclear> by all people <unclear> do things <unclear> cost, unless of course special clothing <unclear> and then <unclear> awfully difficult and making it <unclear> people's education and general health.

JJAPSUNK$_5$ We feel it's very important that our rights of way network should be maintained and improved where possible which is why we are proposing an issue <unclear> thank you.

PS44T$_6$ <unclear> I'd like <unclear> .

JJAPSUNK$_7$ Thank you Mr Chairman I would agree that there was two or three <unclear> on the question of <unclear> .

JJAPSUNK$_8$ We will <unclear> outside the erm <unclear> the <unclear> supporting it if it wasn't for the fact that we understand <unclear> that the officers would have a great deal of difficulty spending that money.

Dialogue JJA (Org. multi) [BNC]

We also noted some errors in the speaker attribution in some dialogues. In (7.11), **PS5T8** is the interviewer and **PS5TG** is the interviewee who is a scientist who came to discuss how to teach science to children and students. However, turn 15, which is attributed to the interviewer, mentions the 'laboratory' of the speaker, which seems to indicate that the turn was actually produced by the interviewee.

(7.11) **PS5T8**$_1$ Hello.

PS5T8$_2$ Science is thought of as a subject that is difficult both to teach and to learn.

PS5T8$_3$ The folklore in school terms is that you have to be a relatively ancient teenager to appreciate physics and chemistry and biology.

PS5T8$_4$ Is this true?

PS5T8$_5$ Today I have with me Dr <name>, who's made a study of science teaching in schools.

PS5T8$_6$ <name>, how early can science be taught to children?

(5 utterances)

PS5TG$_{12}$ For most young children in that age group, the world's a magic place and we traditionally like to teach them nature study and flowers and cuddly hamsters and rabbits in school, and that's the nature table syndrome, and that's great and I'm not knocking that .

PS5TG$_{13}$ There is so much opportunity for children to look at the nature of the physical world around them which isn't taken advantage of, and which could be, and I think that may have something to do with the attitude of teachers as much as the attitude of children.

PS5TG$_{14}$ But erm they're tremendously curious about the nature of the world around them and they're certainly capable of, if not understanding why, exploring what.

PS5T8$_{15}$ I took my godson, `<name>`, round my laboratory the other day, which is a physics laboratory, and he enjoyed it immensely and asked a number of questions, and was absolutely intrigued and fascinated by the various bits of wires and plugs like that, and he asked me the sort of questions that I don't think I would expect sometimes my `<laugh>` undergraduates to ask.

Dialogue KRH – 27 (Public duo) [BNC]

Such errors in the speaker attribution can be problematic for our task as they can lead to confusion about the roles of the participants in the conversation. In this case, it could lead to a misunderstanding of the context of the conversation and the content of the turns. In particular, (7.11) led to debate between the annotators after the task as they had different interpretations of the conversation, where one of them though that the turns were well attributed and the other did not. These mistakes and misinterpretations are characteristic of the challenges of working with naturalistic dialogues and in particular from an outside perspective, as we would not have had this issue if we had been part of the conversation.

7.2.4.4 Studying conversations from an outside perspective

Our study required us to imagine ourselves as participants in the conversations we were annotating. As we have seen in the previous sections, this position is challenging because it forced us to imagine a suitable context for based on the transcriptions alone, which are not always complete or accurate, and are a degraded representation of the original conversations by nature. In some cases, building the context relies on small lexical cues, where we relate a word to a topic or a situation, but it does not always resist when confronted with the subsequent turns. (7.12) is an example of multiparty open conversation, which is arguably the most complex type of dialogue when it comes to coming up with a coherent context, where several lexical fields are mentioned. In the second turn, 'persil' is mentioned, followed by the mention of a 'brush'. With this context, we could imagine that the participants are talking about cooking and that they are preparing a meal, where the brush can be used to apply oil or butter to a dish. However, the subsequent turns mention 'fil[ing] them down' before discussing colour and painting, which could relate to nail trimming and nail polish painting. The mention of a 'sponge' and 'dabbl[ing]'

technique can still fit in this context but the mentions of 'spikes' and a 'hedgehog' question this theory. Then in turn 37, the participants talk about 'paint[ing] the icing', which could be related to cake decoration. This extract illustrates how each turn can bring new information that can change the context of the conversation and how it is interpreted. The <unclear> parts further complicate the interpretation as they prevent us from fully understanding the content of some turns. For instance, turn 36 could mention a 'hedgehog' as an element of comparison to describe something else but without the entire content of the turn, there is no way to know for sure whether the participants are talking about an actual hedgehog, a decoration that is supposed to look like one, or no hedgehog at all. The multitude of participants amplify the chaotic energy of this exchange as everyone can contribute with their own needs, intentions, and perceptions. My personal guess at the end of this conversation was that the conversation took place during some kind of sculpting activity and that the mention of 'persil' in the second turn was an error of transcription. However, after reading it several times, I could also find ways to explain why this conversation would fit well in a cake decoration lesson or find other explainable contexts. While it is a single example, it illustrates the feeling raised by many other dialogues from the corpus, and in general when studying less organised exchanges one did not take part in.

(7.12) **PS0KY**$_1$ what I'm gonna do now.
PS0KY$_2$ Erm <name$_1$>, can you pass me the persil please?
PS0L1$_3$ Do you want this brush?
PS0L1$_4$ <pause> <name$_2$>?
KDBPSUNK$_5$ Yeah that's fine.
(2 utterances)
PS0KY$_8$ I'll just file them down a bit.<pause>
KDBPSUNK$_9$ Yeah it's quite runny actually.
KDBPSUNK$_{10}$ But <unclear>
KDBPSUNK$_{11}$ What colour you doing it then?
PS0L2$_{12}$ Purple.
PS0L2$_{13}$ <pause> But I'm gonna do it over this so it'll sort of go funny
(4 utterances)
PS0L6$_{18}$ Why don't you kind of dabble it on, <name$_2$>?
PS0KY$_{19}$ Yeah.
KDBPSUNK$_{20}$ <unclear> sponge.
PS0KY$_{21}$ You mean like that <pause> dabble it on?
KDBPSUNK$_{22}$ Yeah.
PS0KY$_{23}$ Or a bit less?
PS0L6$_{24}$ Yeah.

PS0KY$_{25}$ Yeah.

(4 utterances)

PS0L6$_{30}$ Let me just er

PS0KY$_{31}$ Yeah.

PS0L6$_{32}$ Wipe that down there.

KDBPSUNK$_{33}$ I don't think these spikes belong on there at all.

KDBPSUNK$_{34}$ Mm?

KDBPSUNK$_{35}$ He's got a wobbly bottom.

KDBPSUNK$_{36}$ `<unclear>` hedgehog.

PS0L6$_{37}$ You gonna actually paint the icing white?

PS0L6$_{38}$ Or is it `<unclear>`

PS0L6$_{39}$ Oh right.

(3 utterances)

PS0KY$_{43}$ That white one there everybody.

PS0KY$_{44}$ Too big.

KDBPSUNK$_{45}$ Stick it further down.

PS0KY$_{46}$ It can be the centrepiece.

KDBPSUNK$_{47}$ Stick it further down so `<unclear>`

PS0KY$_{48}$ Well a lot of the spikes are `<unclear>`

PS0L6$_{49}$ Right.

PS0L6$_{50}$ If you've finished <pause> or if you've finished for today.

PS0L6$_{51}$ This <pause> paint I want it really really really well washed out of these brushes.

PS0L6$_{52}$ It's erm very good covering powder, it actually stays in the brushes.

(3 utterances)

PS0L6$_{56}$ When you're ready.

PS0KY$_{57}$ Oh it's hotting up.

Dialogue KDB – 014817 (Open multi) [BNC]

7.3 Method: Automatic question generation and human annotation

We adapt the framework of Y. Wu et al. (2024) to the study of dialogue, as described in Section 7.1.2. Rather than analysing documents composed of sentences, we consider dialogues consisting of utterances $U_1, U_2, \ldots, U_n$. Given an anchor utterance U_k and its preceding context $U_1, \ldots, U_{k-1}$, we generate potential questions evoked by the dialogue at that point and investigate whether their salience is

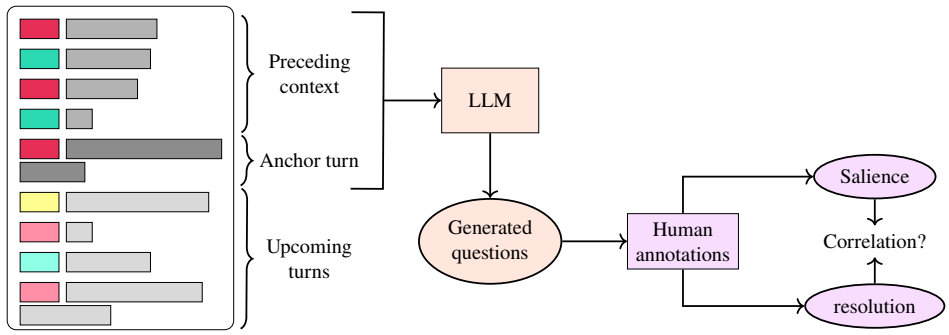


Figure 7.2: Framework for correlating question salience and resolution, adapted from Y. Wu et al. (2024).

correlated with the extent to which they are answered in the subsequent interaction $U_{k+1}, \ldots, U_n$.

Our methodology, represented in Figure 7.2, comprises three main steps. First, we generate questions anchored in dialogue excerpts using an LLM. Second, we annotate these questions for salience and resolution using guidelines adapted from Y. Wu et al. (2024) to account for the specific properties of dialogue. Finally, we compute the correlation between the two scores to assess whether more salient questions are more likely to be addressed later in the interaction.

In this section we describe the question generation procedure (Section 7.3.1), the adaptations made to the annotation guidelines (Section 7.3.2), and the annotation process itself (Section 7.3.3).

7.3.1 Automatic question generation

Following Y. Wu et al. (2024), we generate multiple questions for each anchor position in the dialogue. Starting from the fourth utterance, we select every second utterance as an anchor and generate three questions per anchor using the preceding context and the anchor utterance itself.

To identify a suitable model for question generation, we conducted preliminary experiments with three open-source LLMs: Gemma 3-4B (Team et al., 2024), Llama3-ChatQA-8B (Z. Liu et al., 2024), and Mistral-7B (Jiang et al., 2023). The prompt was adapted from Y. Wu et al. (2024) to account for the dialogic nature of the input. As shown in Figure 7.3, we explained the format of the utterances and indicated that special tokens should be interpreted as extra information that can help understand the dialogue better, but are not part of the utterance itself. We also instructed the model to generate questions as if it 'were a participant of the dialogue'. We made this choice to reduce the number of questions the participants clearly already know the answer to. We finished the prompt by specifying the format of the output.

```
The following is a dialogue where each line is in the format
<speaker>: <utterance>. All the <special tokens> indicate
extra linguistic information, such as pauses, vocalizations,
etc. You should interpret them as extra information that can
help you understand the dialogue better, but they are not part
of the utterance itself.

<conversation up to the anchor turn>

After reading the previous dialogue, ask 3 questions about a
part of the last utterance "<anchor utterance>" that you would
be curious about if you were a participant of the dialogue and
which you don't have an answer for.
Generate only the questions, one per line, without any
additional text or explanation.
```

Figure 7.3: Prompt used for question generation.

A pilot evaluation was performed on questions generated by the three candidate models. Based on the quality and relevance of the generated questions, we selected Mistral-7B for all subsequent experiments. The final corpus was therefore generated exclusively using this model.

In particular, we dismissed Llama3-ChatQA-8B because of the blatantly sexist questions generated by it as can be seen in (7.13). The conversation is the same as in the previous example (7.9), where a man mentions getting stuck in a traffic jam before the conversation moves to the topic of a woman in the fourth turn. Due to the numerous unclear parts in the transcription, it is quite difficult to infer the context of this dialogue and the identity of this woman we have no information on. Yet two out of three of the generated questions inquire about the pregnancy of the woman. LLMs are known to be biased (Gehman et al., 2020), and this example illustrates how these biases can manifest in the generated questions. While the model we have chosen in the end, Mistral-7B, is also known for exhibiting biases, we did not observe any obvious similar issues for our task.

(7.13) **PS0BA**$_1$ Have you been?
PS0BG$_2$ I got slightly lost.
PS0BG$_3$ Well I sort of went back down the <unclear> Road cos there was a traffic jam <pause> and er I went down <pause> oh sorry mate <pause> went down a road like which I thought was a good short cut, and they've got a lot of these roads and they've put like pavements across the end of them with bollards, and I went back and then there was a traffic jam and I got stuck because they were unloading a lorry <pause> so I've been er basically pissed about.<laugh><pause>

PS0BA$_4$ Yeah she a bit `<unclear>` now ain't she?

→ **Generated question:** Why is there uncertainty whether the female in question is pregnant?

→ **Generated question:** How does the speaker know that the female in question isn't pregnant?

Dialogue KC6 – 055905 (Open multi) [BNC]

After a first small batch of annotations (about 150 questions) carried out by one person, we selected Mistral-7B rather than Gemma 3-4B because the generated questions were more salient on average and more often valid. We acknowledge that this sample is too small to draw any strong conclusions but it seemed to be a reasonable choice given the time constraints of the project. We chose to dedicate more time to the final stage of the annotation process in order to compare different types of dialogues on a large enough sample to draw meaningful conclusions. Investigating the impact of the choice of LLM on the quality of the generated questions is an interesting avenue for future work.

7.3.2 Guidelines adaptations for dialogue

The annotation guidelines proposed by Y. Wu et al. (2024) were designed for monologic discourse and required several modifications to accommodate the specific characteristics of dialogue.

Salience Salience measures the extent to which answering a question would contribute to the participants' understanding of the dialogue at a given point. While Y. Wu et al. (2024) employ a five-point scale for valid questions, we found that several dialogue-specific situations were not adequately captured by the original scheme. We therefore introduced three additional categories, shown in *italics* below. The complete annotation guidelines are provided in Section D.1.

- Score = 0: Invalid question, i.e. it contains a grammatical error, contradicts the preceding context or is unrelated to the anchor turn;
- Score = 1: The question is not related to the preceding context;
- *Score = 2*: The question is related to issues in the transcription of the anchor turn;
- Score = 3: The question is related to the preceding context but answering it is not useful, it may be fully answered already;
- *Score = 4*: The question is relevant to the dialogue but the participants clearly already know the answer based on their common ground or shared context;
- Score = 5: The question is related to the preceding context but answering it is not essential;

- Score = 6: The question is related to the preceding context and answering it is somewhat useful;
- *Score = 7*: The question is a paraphrase of the anchor turn;
- Score = 8: The question is relevant to the preceding context and it should definitely be answered.

The final scheme emerged through several pilot annotation rounds in which alternative dialogue-specific categories were tested and refined.

Resolution Resolution measures the extent to which a question is addressed in the dialogue following the anchor utterance. Unlike Y. Wu et al. (2024), we do not treat questions that are already answered by the preceding context as a separate category. In dialogue, information may be repeated, clarified, or reformulated by participants, meaning that a question can still be answered in subsequent turns even if an answer has previously been mentioned. We therefore employ a three-level scale:

- Score = 0: The question is not answered after the anchor turn;
- Score = 1: The question is partially answered or the participants explicitly state that they do not know the answer after the anchor turn;
- Score = 2: The question is fully answered after the anchor turn.

7.3.3 Annotations

Each question is annotated independently by two annotators for both salience and resolution. In the remainder of the paper, we refer to these as Annotator 1 and Annotator 2. Annotator 1 corresponds to a single expert annotator, while Annotator 2 aggregates annotations from several annotators who worked across different subsets of the data. Although each file was annotated in full by a single individual, different files within Annotator 2 were annotated by different people, and in some cases salience and resolution for the same file were produced by different annotators. This design was adopted to make the annotation process more efficient while maintaining two independent annotations per question. The annotators were either core members of the project or researchers from the team who volunteered to annotate some files. Despite the aggregation for Annotator 2, one person carried out the annotation of most of the files – 77% for salience, and 95% for resolution – and 7 other annotators covered the rest of the files, 6 for salience and 1 for resolution.

Annotator 1 has formal training in linguistics and experience in dialogue analysis. The annotators contributing to Annotator 2 have heterogeneous backgrounds, with varying levels of expertise in linguistics and conversational analysis; the majority of annotations within this group were produced by annotators with limited

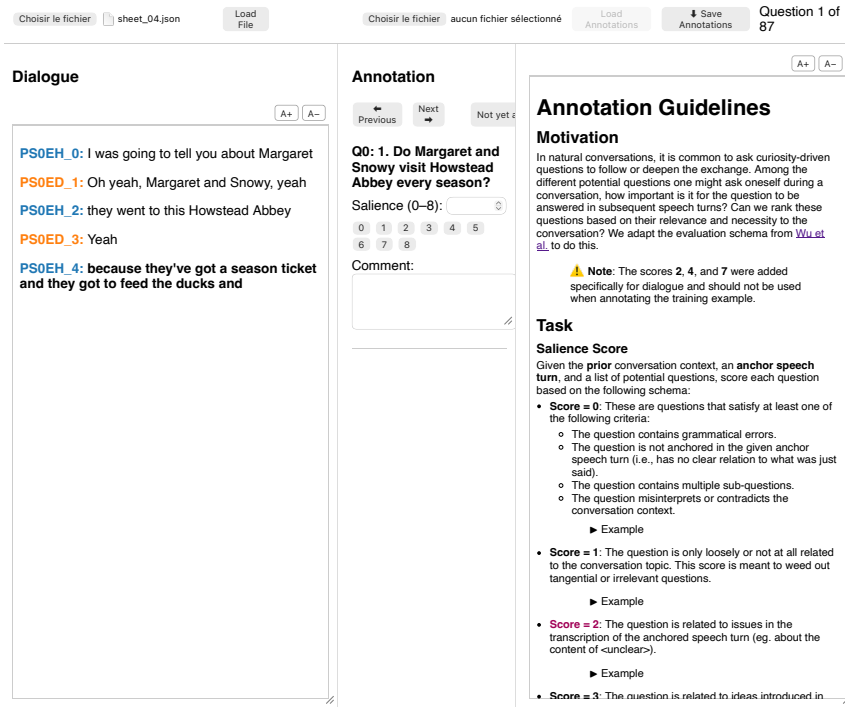


Figure 7.4: Screenshot of the annotation interface for the salience task. The dialogue display is shown on the left, the annotation panel in the centre, and the guidelines panel on the right.

prior experience in these topics. All annotators were high-proficiency L2 speakers of English but the fact that they were non-native speakers may have introduced additional difficulty in interpreting conversational nuances, particularly in cases involving transcription artefacts or domain-specific vocabulary.

Annotations were collected using a custom-built web-based interface developed for this project showed in Figure 7.4. The interface allowed loading and exporting annotations in JSON and CSV formats and supported resuming partially completed annotation sessions. It consisted of three panels: (i) a dialogue display, (ii) a central annotation panel showing the current question, scoring interface, and optional comment field, and (iii) a panel containing the annotation guidelines and examples.

The display of the dialogue differed between the two annotation tasks. For salience annotation, only the context up to the anchor utterance was shown, and subsequent utterances were hidden to avoid contamination from future discourse. For resolution annotation, the full dialogue was displayed. In both tasks, the anchor utterance was visually highlighted to support localisation.

Annotator 1 completed both annotation tasks sequentially, with a one-week in-

Table 7.4: Inter-annotator agreement for salience and resolution annotations. $\alpha_{QSalience}$ refers to the Krippendorff's alpha reported in Y. Wu et al. (2024) for comparison. κ refers to Cohen's Kappa, wκ to linear weighted Cohen's Kappa and wκ^2 to quadratic weighted Cohen's Kappa.

	$\alpha_{QSalience}$	α	%	κ	wκ	wκ^2
S	.63 - .75	.55	.51	.36	.47	.53
A	.80	.65	.80	.53	.62	.68

terval between them. In contrast, annotators contributing to Annotator 2 followed heterogeneous protocols, with variation in task order and temporal spacing between salience and resolution annotation. Annotation time varied substantially across dialogues, ranging from approximately 30 minutes for shorter and more structured interactions to over 90 minutes for longer or less structured ones. Overall annotation effort is estimated at approximately 70 hours for Annotator 1 and 100 hours for Annotator 2, although these figures are approximate as detailed timing logs were not systematically recorded.

7.3.3.1 Agreement: a complex task

Given the inherently subjective nature of both salience and resolution annotation, we evaluate inter-annotator agreement using multiple metrics. We report Krippendorff's Alpha (Krippendorff, 2011)[24] for comparability with Y. Wu et al. (2024), as well as Cohen's Kappa and both linear and quadratic weighted Kappa (J. Cohen, 1968) to account for the ordinal structure in the annotation scales. Indeed, given the gradation of the scores, we consider that a disagreement between two adjacent scores is less severe than a disagreement between two scores that are far apart. Weighted Kappa allows us to take this into account in our agreement analysis, where instead of penalising all disagreements with a weight of 1 as in the case of Cohen's Kappa, we assign a different weight to each disagreement based on the distance between the two scores. Formally, the weight when comparing two scores i and j is defined as $w_{i,j} = |j - i|$ for linear weighted Kappa and $w_{i,j} = (j - i)^2$ for quadratic weighted Kappa.

As shown in Table 7.4, agreement for salience ranges from fair to moderate, while agreement for resolution is moderate to substantial, depending on the metric considered (Artstein & Poesio, 2008). Agreement levels are generally lower for salience than for resolution, consistent with the greater interpretative flexibility of salience judgements. They are also lower than those reported in Y. Wu et al. (2024), which is expected given both the increased complexity of dialogue data and

[24]Computed with Castro's (2017) Python implementation.

Table 7.5: Inter-annotator agreement per dialogue group.

	Salience				Resolution			
Category	α	κ	wκ	wκ^2	α	κ	wκ	wκ^2
Open duo	.57	.40	.50	.55	.68	.56	.64	.69
Open multi	.51	.38	.45	.48	.58	.47	.55	.61
Org. duo	.58	.38	.50	.56	.69	.56	.65	.72
Org. multi	.50	.33	.42	.48	.62	.54	.61	.65
Public duo	.53	.32	.43	.50	.65	.50	.59	.65
Public multi - radio	.55	.42	.49	.52	.61	.50	.58	.64
Public multi - TV	.44	.25	.35	.40	.65	.53	.61	.67
ALL	.55	.36	.47	.53	.65	.53	.62	.68

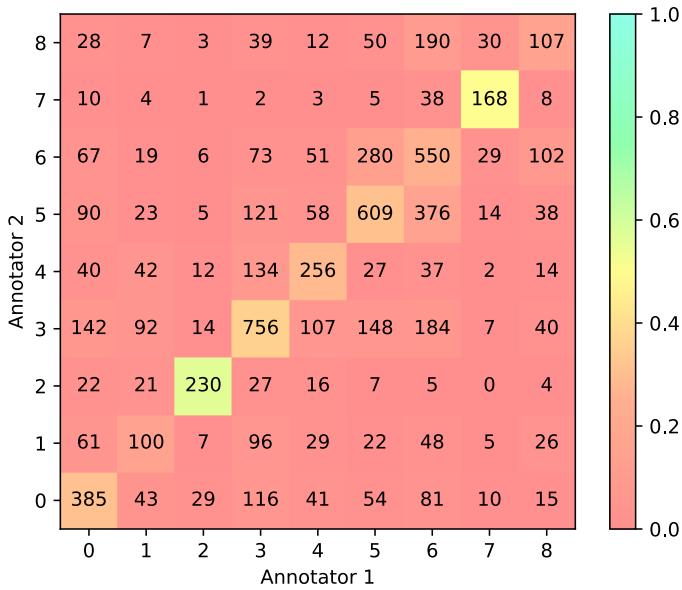
the larger number of annotation categories in the present study. We further compute agreement separately across dialogue categories (Table 7.5) to assess whether conversational structure affects annotation consistency.

We also looked into the places of disagreement between the two annotators. Figures 7.5a and 7.5b show the comparison matrices for salience and resolution respectively. The colours of the cells correspond to the portion of questions that received the corresponding scores from the two annotators, while the numbers indicate the raw amount of annotations for each score.

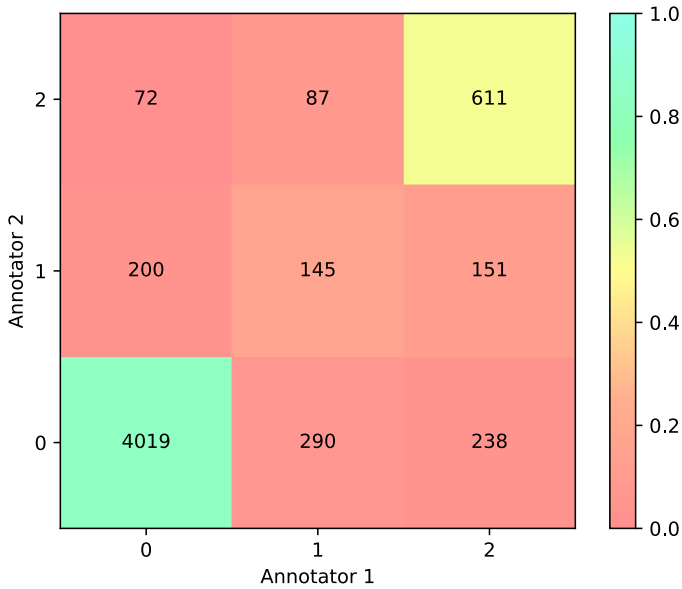
Salience The salience annotation relied on a large score range. Disagreements are partly attributable to transcription artefacts and the absence of non-linguistic contextual information, both of which are particularly relevant in spontaneous dialogue.

Figure 7.5a shows that agreement is highest for scores 2 and 7 that correspond respectively to questions targeting transcription issues, and cases where the generated question closely paraphrases the anchor utterance. Both categories are relatively constrained and therefore more consistently identifiable across annotators. In particular, score 2 can often be detected using simple heuristics (e.g. the presence of `<unclear>` markers in the question), which likely contributes to higher consistency. Score 7 is slightly less consensual which can be explained by the fact that the boundary between two sentences being paraphrases of each other, and being very related, is hard to define (Polguère, 1990).

In contrast, lower agreement is observed for scores 1 and 8. Score 1 involves judgements about whether a question is unrelated to the preceding context, a notion that is inherently sensitive to assumptions of discourse coherence. Conversely, score 8 requires assessing whether a question 'must be answered', which depends on pragmatic interpretation and access to shared context and participant intentions,



(a) Salience.



(b) Resolution.

Figure 7.5: Comparison of the annotations for salience and resolution. The colour of the cells corresponds to the portion of questions that received the corresponding scores from the two annotators, while the numbers indicate the actual amount of annotations for each score.

both of which are not fully observable in transcript-based analysis.

The distribution shows that annotators frequently agree that questions are at least loosely connected to the preceding context, as indicated by the faint orange square spanning Scores 3 to 8 with the exception of 7. This supports the idea that in conversation we tend to assume coherence and relatedness by default, which is a reasonable strategy for joint cooperation (Grice, 1975, p.45–47). However, the extent to which a question is deemed essential to the dialogue is more subjective and variable across annotators. In particular it can require a more fine-grained understanding of the dialogue, which can be difficult to achieve without access to the non-linguistic context and to the knowledge of the participants.

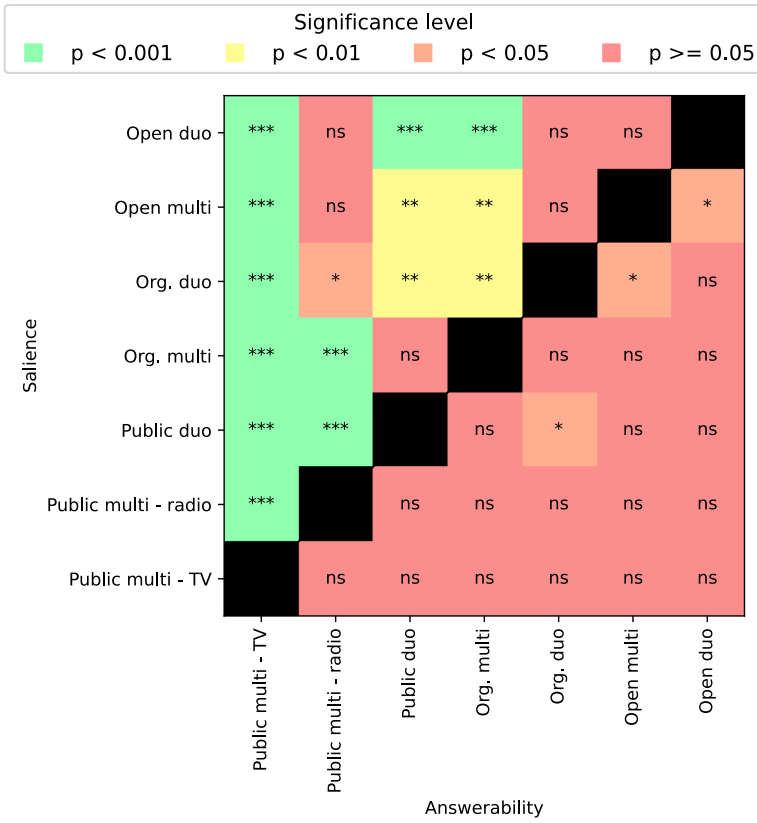


Figure 7.6: Representation of the significance of the difference between the agreement for each pair of dialogue groups for Salience (◤) and resolution (◢). Colours indicate the strength of the p-value.

With regards to the dialogue categories, Figure 7.6 shows that most pairs exhibit significantly different levels of inter-annotator agreement across groups, assessed using paired Wald's z-tests comparing the difference in unweighted Cohen's kappa, normalised by standard errors based on Fleiss's method (Fleiss et al., 2003).

As shown in Table 7.5, the lowest level of agreement was observed in the public television category, which lies close to the lower limit of what is considered fair agreement. This may be explained by the television context, which is intended for an external audience with access to the visual modality, whereas this was not the case for the annotators. Conversely, the highest agreement was found in radio conversations, which are intended for an audience with access only to the audio modality.

Resolution Resolution was graded on a small score range. As shown in Table 7.4, agreement is higher for resolution than for salience, reflecting the lower interpretative flexibility of resolution judgements. This is confirmed by the few significant differences in terms of agreement across dialogue types as shown in Figure 7.6. Figure 7.5b shows the corresponding annotation matrix. Agreement is highest for Score 0, which corresponds to cases where the question is not answered in the subsequent dialogue. This suggests that identifying absence of resolution is more robust than distinguishing degrees of partial resolution. Disagreements between Scores 1 and 2 can be explained by the fact that the boundary between partial and full resolution is ambiguous, particularly for complex or implicit questions. Requesting a confidence score for the resolution annotation could have been useful to capture such cases, but it would also make the annotation process more complex and time-consuming.

7.3.3.2 Aggregation: a perspectivist approach

Given the observed variability in salience annotations, we adopt a *perspectivist* strategy in our analysis (Frenda et al., 2025). Rather than collapsing annotations into a single gold label via averaging, we retain both annotator perspectives. In addition, we construct a restricted 'gold' subset consisting only of questions for which both annotators assigned identical salience scores, which corresponds to about 46% of the questions. The repartition between most of the groups is rather balanced, except for the public multilogues where the radio broadcasts provide the highest agreement around 52% and the TV shows the lowest one around 38% (see Table 7.6 for more details). This gold subset allows us to study the general annotation patterns but also to investigate the nature of the disagreements which we consider to be a natural reflection of the complexity of the task and of the nature of dialogue itself, which is perceived differently by its different participants. In such a case, averaging ordinal labels may obscure meaningful differences in interpretation rather than resolving them.

Regarding resolution, we instead compute the mean of the two annotators' scores to obtain a single continuous estimate. This is justified by the more compositional and less categorically heterogeneous nature of the resolution scale. The

resulting aggregation effectively yields a finer-grained scale while preserving ordinal consistency.

7.4 Released dataset: questions and annotations

We release the 7,124 questions generated by the LLM Mistral-7B, together with their salience and resolution annotations, as well as some natural language explanations for some of the questions. Due to the licence of the BNC we cannot include the dialogues themselves but we provide the metadata for each question, which allows to reconstruct the dialogues from the BNC.

We release our corpus as a single file where each row corresponds to a question and includes the following information: the dialogue it is anchored in, the anchor turn, the question, the two salience scores, the gold score when the two annotators agreed, comments from the annotators for the salience, the two resolution scores, the gold score, comments from the annotators for the resolution. Figure 7.7 provides an example of a row in our corpus together with an extract of the dialogue it is based on.

PS0EH$_0$ I was going to tell you about Margaret
PS0ED$_1$ Oh yeah, Margaret and Snowy, yeah
PS0EH$_2$ they went to this Howstead Abbey
PS0ED$_3$ Yeah
PS0EH$_4$ **because they've got a season ticket and they got to feed the ducks and**

ID$_{sheet}$	Type	ID$_Q$	Cutoff	ID$_{ann2}$	Question
4	open multi	0	4	E	**1. Do Margaret and Snowy visit Howstead Abbey every season?**

S$_1$	S$_2$	S$_G$	Comm$_{S1}$	Comm$_{S2}$	A$_1$	A$_2$	A$_G$	Comm$_{A1}$	Comm$_{A2}$
5	5	5	/	*relevant but not essential to the conversation*	0	0	0	/	/

Figure 7.7: Example of a row in our corpus together with an extract of the dialogue it is based on with the anchor turn and the associated question in bold. S$_i$ and A$_i$ indicate the salience and resolution scores for annotator i with G for the gold scores. *Comm* stand for the associated comments. *Dialogue KCE – 029701.*

In this section we discuss several limitations of the question generation approach, which are important to consider when interpreting the results of our analysis. We begin by discussing examples that show the lack of world knowledge of the model (Section 7.4.1), then we go through more technical limitations (Section 7.4.2), and finish with some issues most likely created by the prompt (Sec-

tion 7.4.3).

7.4.1 World knowledge

As discussed in Chapter 3, LLMs are trained on large amounts of text data but do not learn by interacting with the world. This is reflected in several phenomena we observed in the generated questions, which show a lack of world knowledge and common sense reasoning.

Meaning of the words Some generated questions asked about the meaning of words or phrases in context. While this can be a meaningful question in some cases, as in 'Does "Mm" imply agreement with the previous statement or a neutral response?' which indicates a questioning on the intentions of the speaker, in other cases the meaning of the words is very clear and the question is therefore not relevant. In (7.14), the question concerns the meaning of the word 'Yeah', i.e. the only word in the anchor turn, which is very clear in context.

(7.14) **J8DPS003**$_{67}$ What are those Marian, you don't get one of those last time?
J8DPS000$_{68}$ I got <pause> one of these.
J8DPS002$_{69}$ Yeah.

→ **Generated question:** What does "Yeah" refer to in this context?

Dialogue J8D (Org. multi) [BNC]

This phenomenon also happens with phrases less ambiguous than acknowledgment tokens like 'Yeah'. In (7.15), the question concerns the meaning of the phrase 'trim them around the sides', which requires little explanation in the context of cutting trees.

(7.15) **PS1GE**$_{22}$ They're not, they're not big trees or nothing so I mean there was, it was only a, like one of these small <unclear> power chain saws,<unclear> one hand sort of thing, grab hold of it like that, there was nothing, no really big branches, were only cutting the tops off.
PS1GE$_{23}$ We, we did trim them around the sides, they were quite big, you know,<unclear> .

→ **Generated question:** What does "trim them around the sides" refer to in this context?

Dialogue KDA – 011605 (Open duo) [BNC]

Epoch anchoring The collection era of the BNC is also a challenge when generating the questions because pragmatic reasoning and world knowledge are necessary to ask relevant questions. For instance, based on a conversation about the impact of technological progress on scientific publishing, the question 'What are potential challenges or limitations in using floppy disks for large-scale book publishing due to their capacity limits or outdated technology?' was generated. This question would clearly be irrelevant at the time of the conversation when floppy disks were no outdated technology and where their capacity limits were not an issue. Similarly, in a conversation where a participant recommends a book, the generated question inquires about the availability of the book in digital format, which was not a relevant question at the time of the conversation: 'Is "The Honorary Consul" available in digital format as well, or is it only available in paperback?'.

Pragmatic and social reasonning Conversation is rooted in social and cultural norms, and understanding the pragmatic implications of dialogue often requires knowledge of these norms. In some cases, the generated questions show a lack of such reasoning.

In a council meeting where one point of the agenda was to vote an advance on salary for the council members, the question 'Why does the speaker hope that the public won't believe something, as stated in "but certainly I'm not prepared to allow them to think it of me"?' was generated. As indicated by the context in (7.16), the speaker was worried the public would believe that the council members were voting themselves a raise of salary. The generated question would thus mean 'Why does the speaker hope that the public won't believe that the council members are voting themselves a raise of salary?', which requires little political sense to answer. The question is therefore not relevant in this context and shows the lack of pragmatic reasoning of the model.

(7.16) **PS45W**$_{86}$ We are elected by the public to do a job and we shouldn't be ashamed of the amount of money that we're paid for it, and what I would say is, that no manifesto in May's election said we were going to come here and vote for more money for members, nobody put on their leaflets, vote for me and I will raise members allowances by five percent in the coming year, I didn't and I won't support that, and you will argue, perhaps some of you that that's not really what we're doing, but it look's like it, doesn't it?

PS45W$_{87}$ And I think it's close enough to looking like it that the public may believe it, and I hope the public won't believe it, but certainly I'm not prepared to allow them to think it of me, and therefore I will vote against the motion.

→ **Generated question:** Why does the speaker hope that the public won't believe something, as stated in "but certainly I'm not

prepared to allow them to think it of me"?

Dialogue JJG (Org. multi) [BNC]

More socio-cultural aspects were also clearly missing in the generated questions. In the context of a radio broadcast about the Gulf War where several Kowaiti guests were interviewed, the question 'Does Jossom have any concern about potential civilian casualties on the Iraqi side, or is he only concerned about Kuwaitis?' was generated. The question shows a clear lack of humanity which is inappropriate, all the more so in a conversation about war from which participants and listeners may be direct victims of.

Lastly, some generated questions highlight the importance of conversational structure for assessing the salience of a question. While a question could have been very salient at one point of the exchange, it can become much less relevant later in the conversation due to a change of focus or because the conversation is coming to an end. (7.17) illustrates the latter case as the host asked their guest for a last brief comment while the generated questions inquire about more examples and details. A last 'very brief' comment that would answer one of the generated question at that point of the exchange would most likely be considered a breach of conversational cooperation.

(7.17) **PS5SC**$_{62}$ Ian, I wish we could go on talking endlessly about this because I find the subject quite fascinating and we haven't really talked about the details of your research.

PS5SC$_{63}$ I wonder if we could just finish with a very brief comment from you in a positive sense as to what would be your advice to somebody that's unemployed?

(5 utterances)

PS5SE$_{69}$ We found all sorts of different ways that different people were achieving these experiences.

→ **Generated question:** Can you provide examples of the various ways different people are achieving self-esteem during unemployment as per your research?

→ **Generated question:** Were there any common activities among these different methods that showed significant impact on reducing depression levels?

Dialogue KRG – 13 (Public duo) [BNC]

7.4.2 Technical limitations

Leveraging an LLM to automatically generate our questions implies to accept some technical limitations of the model.

Referring to words that are not here We observed a certain form of 'hallucination' is some generated questions that would refer to entities that were not present in the anchor turn or in the context. In (7.18), the question refers to 'they' while explicitly stating the scope of the reference which is turn 37. Yet, there is no mention to a plural entity in that turn. The closest valid referent would be in turn 34 but the speaker does not correspond with the one mentioned in the question.

(7.18) **PS0RE**$_{33}$ He can't be full time otherwise he wouldn't have enough to do would he?

PS0PN$_{34}$ I don't know perhaps he does, I don't actually know Ken, I thought the way he was talking though that this chap

PS0RE$_{35}$ No, it's, it's so many days that they work innit?

PS0PN$_{36}$ had actually taken him on that he used to know years ago and had finished the

PS0RC$_{37}$ Oh he's finished the

PS0PN$_{38}$ I know it's a couple of years ago before he came here

PS0RC$_{39}$ Oh

→ **Generated question:** What did PS0PN refer to as "they" that PS0RC reacted to with "Oh"?

Dialogue KDM – 030302 (Open multi) [BNC]

A general remark is that the more precise the scope of the question, the more chances there are that it was not logically relevant. In (7.19), the question asks about the content of the `<unclear>` tag, which exists in the anchor turn, but not in the specific phrase quoted in the question.

(7.19) **PS1GE**$_{49}$ I don't know whether it's true or not I don't know whether they're fucking `<unclear>` trouble is it sounds like the sort of thing I do, is we parked in the multi storey car park in Guildon.

→ **Generated question:** What does "`<unclear>`" refer to in "we parked in the multi storey car park in Guildon"?

Dialogue KDA – 011605 (Open duo) [BNC]

Confusion of substantives and wh-words A similar issue to that of the too narrow scope of the question is that some questions could be relevant and salient were it not for the use of the wrong sentence topic. (7.20) is an extract of a radio debate about the advantages of the antidepressant Prozac. One of speakers, **PS3E6**, disagrees with the others and mentions Valium, another antidepressant that was used before Prozac, to explain that while the excitement when it was first introduced was very similar to what is happening with Prozac, Valium was later found to have many

side effects. The other participants counter him by stressing that one of the advantages of Prozac is its low amount of side effects so **PS3E6** tries to list the known side-effects of the drug. At this point, the generated question inquires about the side-effects of Valium, while we are actually talking about Prozac. The reason for this confusion may be that Prozac had not been mentioned once since turn 8 while Valium is mentioned six times between the last mention of Prozac and the anchor turn for that question. Nevertheless, only a listener paying very little attention to the debate would have misunderstood the focus of the last turns of (7.20), showing a disparity between human understanding and LLM processing of the dialogue.

(7.20) **PS3E4**$_{9}$ Oliver what's so wonderful about Prozac as an antidepressant?

(16 utterances)

PS3E6$_{26}$ Well twenty years ago I was on programmes like this, saying that doctors who said that Valium and the other tranquillizers were perfectly safe were talking rubbish and that there was every possibility that there would be problems found in the future, if we kept prescribing these drugs for vast numbers of people.

(17 utterances)

PS3E5$_{44}$ `<unclear>` be six million people in the world are taking this drug saying it's really really helping them and you're telling them it's crap.

PS3E6$_{45}$ I'm let me just tell you some of the side effects which the company making it admit to .

PS3E5$_{46}$ Now what do you say to those six million people ?

PS3E6$_{47}$ Look no let me just tell you what the

PS3E5$_{48}$ Mm.

PS3E6$_{49}$ side effects are.

→ **Generated question:** What are some specific side effects associated with Valium?

Dialogue HV1 (Public multi - TV) [BNC]

In the same vein, generated questions often use the wrong wh-word, which technically makes them irrelevant or un-grammatical depending on the situation. 'Who are "the sports orders" that Adrian is being asked to provide?' is one such question where the wh-word is clearly not coherent with the subject of the question. Other errors could be more subtle like 'Who are "them" that will stay in exclusion until they've been fixed?' where 'them' actually refers to written reports and not to people.

Providing questions and answers Another well known limitation of LLMs besides 'hallucinations', is that it is complicated to enforce the format of the output.

In our case, while we specifically required questions we did not yet have an answer for, some generated questions were followed by an answer in parenthesis like 'When exactly will we go to the paint shop? (The specific time mentioned was half past four.)'.

7.4.3 Prompt-related issues

LLMs are known to be sensitive to the prompt they are given. In our case, we stressed the importance of generating questions anchored in a specific turn, we required three questions per anchor turn, and the questions were to be generated from the point of view of a participant in the conversation. The following issues likely stem from these constraints.

Short anchors Following Y. Wu et al. (2024), we generated questions every second turn in the dialogue. Since we were working with naturalistic dialogues, some turns were very short and contained only an acknowledgment token like 'Mm' or 'Yeah'. In such cases, the generated questions were often related to the meaning of the token, which was sometimes obvious as discussed earlier but could also bring pragmatic information, but in other cases the question concerned a much earlier context as we can see in (7.21).

(7.21) **J8DPS004**$_{54}$ that is stupid Terry is key stage four!
(12 utterances)
J8DPS003$_{67}$ What are those Marian, you don't get one of those last time?
J8DPS000$_{68}$ I got <pause> one of these.
J8DPS002$_{69}$ Yeah.
J8DPSUNK$_{70}$ Start planning, well you got those are the weeks of the year
J8DPS003$_{71}$ Mhm.
J8DPSUNK$_{72}$ so you can actually write in, you know the <pause> at
J8DPS004$_{73}$ Oh!

→ **Generated question:** What prompted Terry being referred to as "key stage four"?

→ **Generated question:** Could you explain what was unclear about key stage four that was mentioned earlier in the conversation?

Dialogue J8D (Org. multi) [BNC]

This phenomenon may be due to the fact that anchoring a question in a very short turn can be challenging for the model, which may then generate any relevant question from the context, even if it is not directly related to the anchor turn.

Questions answered by design The prompt required the model to generate a question one would be curious about as a participant in the conversation. This often led to questions that were answered *by design* in the subsequent turns of the dialogue as they roughly amounted to 'What happens next?'. 'Are there further discussions or actions to follow after "Yeah, mm"?' is an example of such a question. While these questions are relevant to the dialogue, in practice they are not very informative as they do not allow us to predict the upcoming topics of the conversation.

First-person questions The requirement to produce questions one would be curious about *if one were a participant of the conversation* was meant to avoid questions related to entities or events that were clearly part of the common ground. However, this led to the generation of questions that were phrased in the first person, and in particular from the point of view of an actual participant of the conversation. For instance, in (7.22), the question could have been asked by **PS0J1**.

(7.22) **PS0J1**$_1$ It doesn't matter if she'd got a fire guard or not, but it still could easily get onto the
PS0J3$_2$ Yeah.
PS0J1$_3$ carpet couldn't it?
PS0J7$_4$ What?
PS0J7$_5$ What?
PS0J3$_6$ They're on their way squire !
KD2PSUNK$_7$ Yeah.
PS0J1$_8$ They're not safe these houses are they dad?
PS0J3$_9$ No.
PS0J4$_{10}$ Not with, not with that
PS0J7$_{11}$ Especially now no one has looked at them.
PS0J4$_{12}$ That's why you've gotta go smokeless for.
PS0J3$_{13}$ Yeah.

→ **Generated question:** Why do the houses need to be smokeless, Dad?

Dialogue KD2 – 061610 (Open multi) [BNC]

Specific questions About one fifth of the generated questions contained the word 'specific', as reported in Table 7.3, which may be due to the requirement to produce three questions per anchor turn. Indeed, producing several questions about the same turn in the same generation may have led the model to generate questions that are somewhat related to each other. In this case, asking a more 'specific' question is a natural way to differentiate them. However, this led to questions that were often

irrelevant or much too detailed to be considered highly salient. For instance, 'What specifically does "half a day" refer to in terms of hours?' asks for a level of detail that is not necessary to understand the dialogue.

7.5 Results and analysis: Dialogue is less predictable than static text

We begin by analysing salience and resolution separately (Sections 7.5.1 and 7.5.2), focusing on annotation patterns and differences across dialogue types, before examining the relationship between the two measures through correlation analyses (Section 7.5.3).

7.5.1 Salience

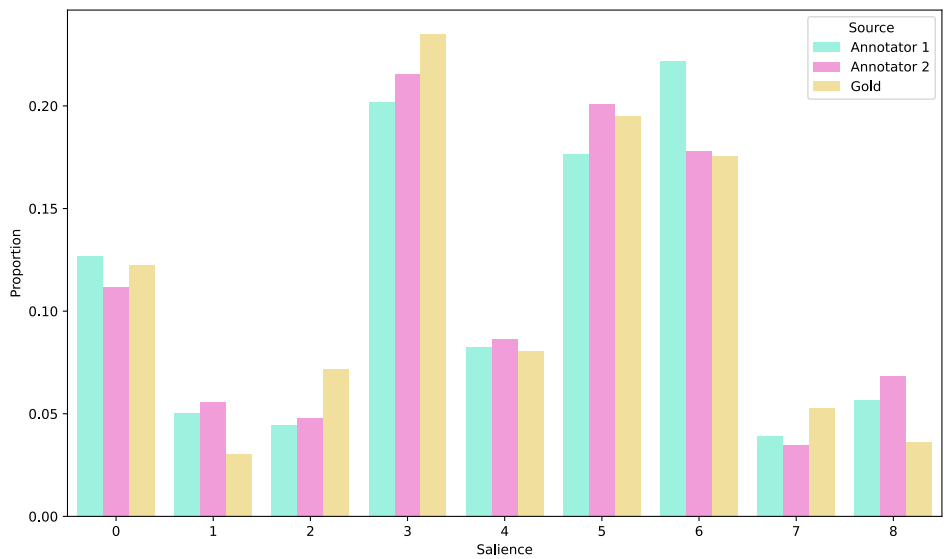


Figure 7.8: Proportions of salience scores across annotators.

Salience scores are not evenly distributed across the range of possible values, as shown in Figure 7.8. Around 12% of the questions are considered invalid (score 0), often because they are not anchored in the right dialogue turn. Scores 1 and 2, which correspond to questions that are unrelated or due to transcription issues each represent about 5% of the questions, where score 1 reduces to about 3% when considering the gold annotations, and score 2 grows to about 7%. This difference in repartition between the raw annotations and the aggregation is mostly due to the difficulty of annotation for these scores, where the higher agreement on score

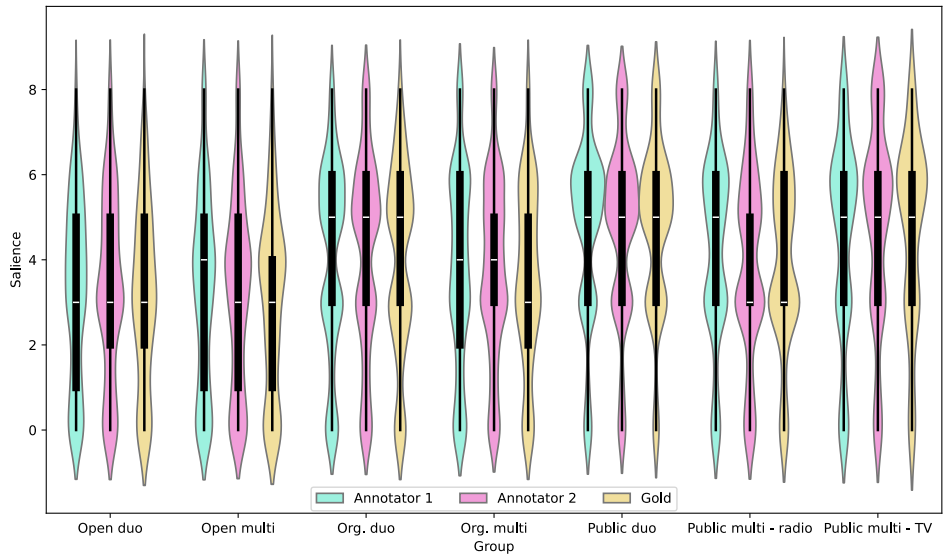


Figure 7.9: Repartition of salience scores across dialogue groups.

Table 7.6: Average salience score per dialogue group (total range: 0-8). *The differences between S_1 and S_2 are significant only for the public multilogues (radio: ***; TV: *).*

Group	S_1	S_2	S_G	Support Gold #	Support Gold %
Open duo	3.29±.45	3.32±.64	3.04±.82	321/657	48.9
Open multi	3.26±.46	3.20±.53	3.06±.71	388/822	47.2
Org. duo	4.04±.50	4.12±.69	4.08±.72	742/1,548	47.9
Org. multi	3.80±.29	3.77±.46	3.57±.31	598/1,409	42.4
Public duo	4.79±.52	4.74±.56	4.80±.53	535/1,158	46.2
Public multi - radio	4.22±.56	3.88±.32	3.95±.49	402/768	52.3
Public multi - TV	4.50±.26	4.66±.84	4.81±.60	287/762	37.7
All	3.92±.70	3.90±.79	3.82±.91	3,273/7,124	45.9

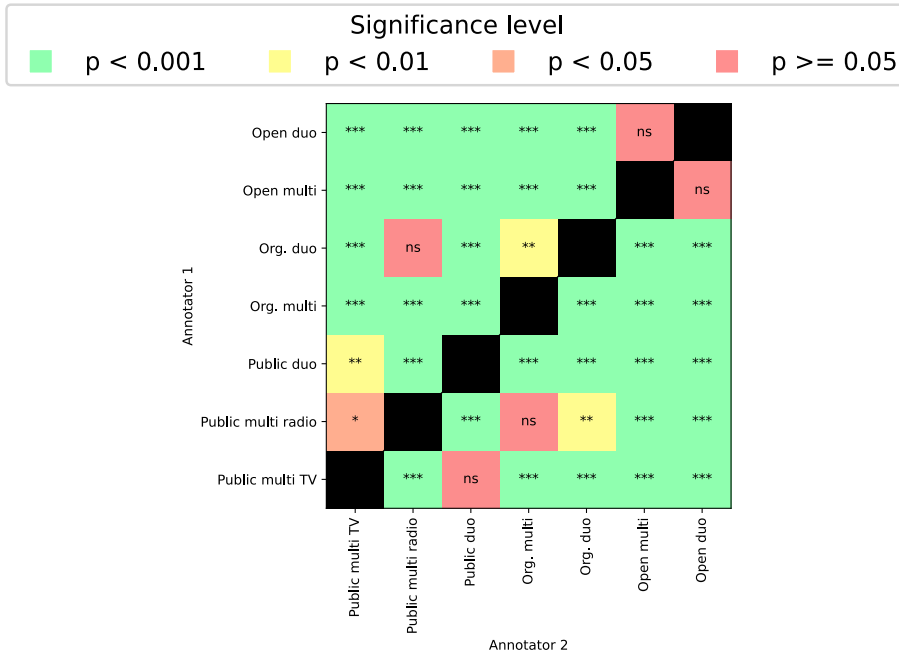
2 compared to all the other scores skews the distribution of the gold annotations. Conversely, score 1 is one of the scores with the most disagreements as discussed previously in Section 7.3.3.1. Most questions were considered at least somewhat related to the preceding context, with scores 3, 5 and 6 each representing 18% to 22% of the total. Highly salient questions (scores 7 and 8) are less frequent, with each representing 4% to 6% of the total. This distribution may be influenced by two complementary factors. First, while generating questions that are related to the

context can be seen as a task that is relatively easy for an LLM due to the way it is trained, generating questions that are *highly salient* requires more sophisticated reasoning about the dialogue and the participants' intentions. Second, the nature of dialogue itself, and our tendency to look for coherence and relatedness in conversation, may have encouraged the annotators to assign at least a moderate level of salience to most questions, while high scores required more specific types of questions. Finally score 4, which corresponds to questions for which the participants clearly already know the answer even though it does not appear in the transcription, i.e. questions that tap in the common ground of the participants, represents about 8% of the corpus, showing that this score was indeed necessary to capture the nature of dialogue.

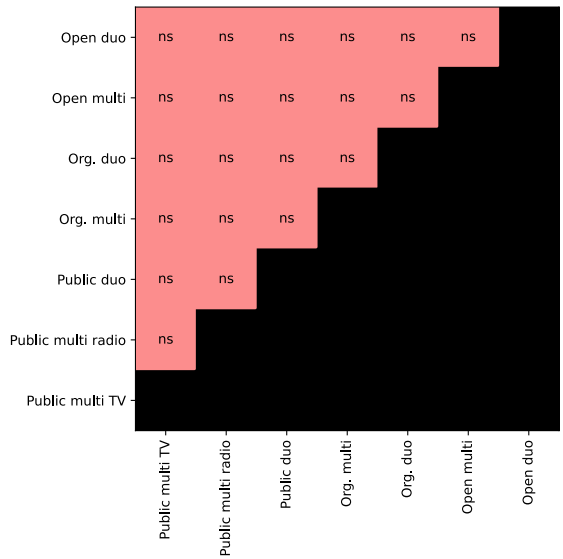
Figure 7.9 shows the distribution of salience scores for each dialogue group, separated by annotator, and Table 7.6 reports the average salience scores for each group, the degree of confidence in the statistical differences between both Annotator 1 and 2, as well as the proportion of questions kept for the gold subset. The distributions are very similar between the two annotators, with only significant differences for the public multilogues, where the radio ones display a higher average salience for Annotator 1 than Annotator 2, while the opposite is true for the television ones though with a lower degree of confidence. However, the distributions between the different dialogue groups are much more variable. Pairwise comparisons (Figure 7.10) indicate that only a few pairs of groups display non-significant differences. Open-domain duologues and multilogues are similar together while being different from all the other groups in both annotators' distributions. Public radio multilogues are similar to the organised duologues, while public television multilogues are similar to the public duologues, although the statistical strength of these differences varies by annotator. When looking into the differences for the gold annotations, we observe no significant differences between the dialogue groups, suggesting that disagreements between the annotators highlight more complex interactions, characteristic of the different dialogue types. This last result should be interpreted with caution, as the gold annotations represent a restricted subset of the data which limits the statistical power of the tests. Nevertheless, these results suggest that salience is sensitive to dialogue type and in particular to the number of participants, with duologues displaying more salient questions than multilogues (Table 7.6).

7.5.2 Resolution

As shown in Figure 7.11 , more than 70% of the generated questions are considered not answered in the subsequent dialogue (score 0), about 10% partially answered (score 1), and the rest fully answered (score 2). Annotator 2 assigned slightly more score 0 than Annotator 1, and slightly less score 2. In isolation, this repartition could indicate that the generated questions were mostly irrelevant but the salience



(a) Annotators 1 (◤) and 2 (◢).

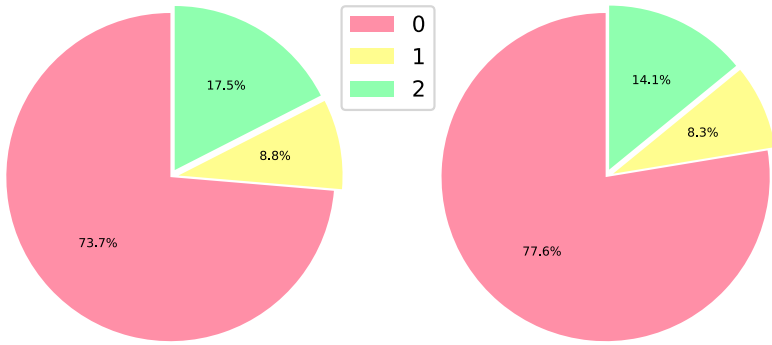


(b) Gold annotations.

Figure 7.10: Representation of the significance of the difference in salience distributions between dialogue group for a given annotator. Colours indicate the strength of the p-value.

scores contradict this hypothesis. Instead, it suggests that predicting questions that will actually be answered is complicated which we discuss in Section 7.5.3.

The high proportion of score 0 is reflected in the distributions of resolution scores across dialogue groups (Figure 7.12), which are skewed towards 0 for all groups. Contrary to salience, the distributions are quite different between the two annotators, here only the open conversations display similar distributions. All the other groups are more or less significantly different (Table 7.7), with Annotator 1 consistently assigning higher scores than Annotator 2.



(a) Annotator 1. (b) Annotator 2.

Figure 7.11: Repartition of the resolution scores for each annotator.

Conversely, pairwise statistical tests (Figure 7.13) indicate fewer significant differences between groups than for salience, although some variation remains. Particularly, open-domain multilogues are significantly different from all other dialogue types, and organised multilogues are different from all other types except open-domain duologues and public television multilogues. This suggests that resolution is less sensitive to dialogue type than salience, and both the inter-annotator and inter-group differences indicate that open dialogues, and in particular multilogues, are distinct from the other groups. However, the low average resolution scores across *all* groups indicates that most questions are not explicitly answered in the subsequent dialogue (Table 7.7).

7.5.3 Dialogue predictability

The central goal of this study was to determine whether salient questions are more likely to be answered in the subsequent dialogue turns. We therefore examine the relationship between salience and resolution, and how this relationship varies across dialogue types.

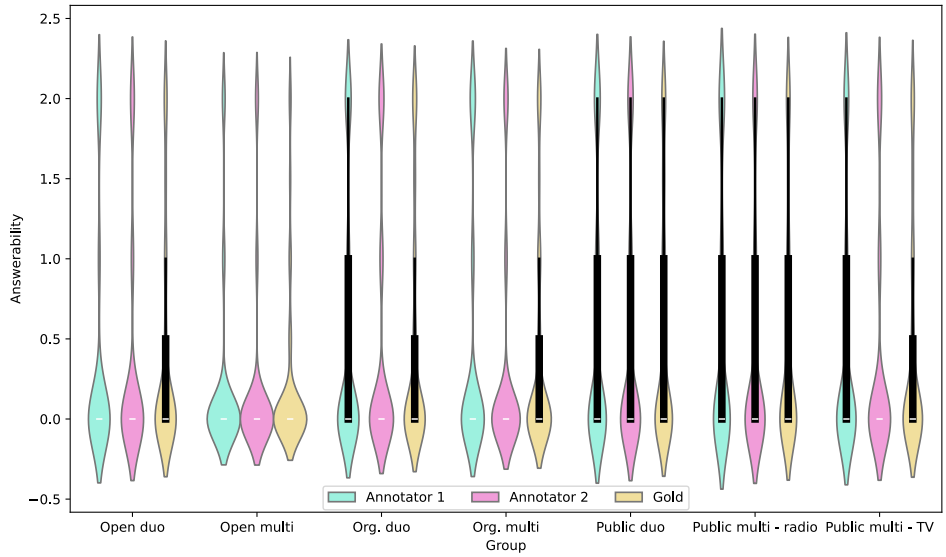
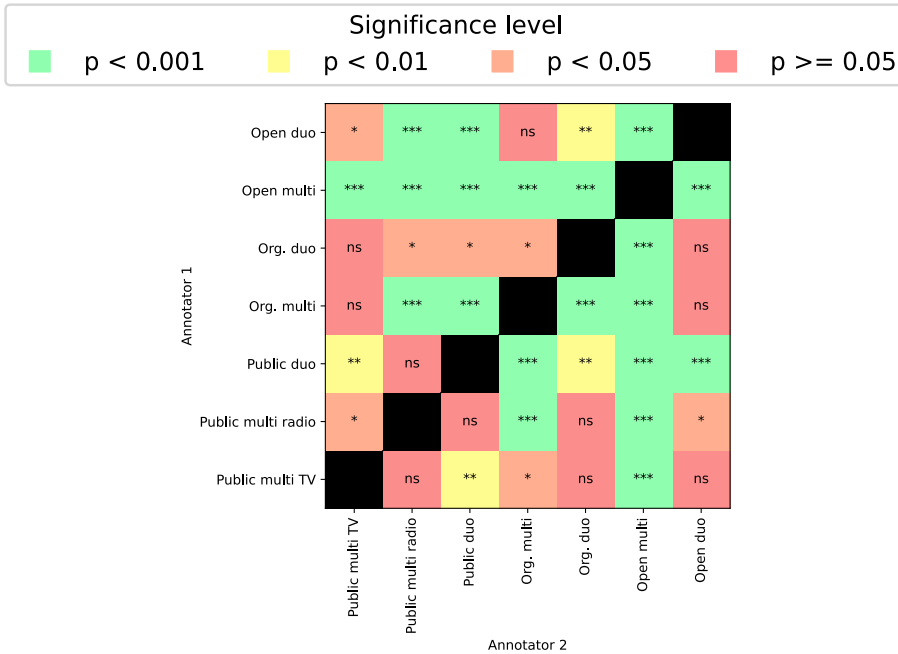


Figure 7.12: Repartition of resolution scores across dialogue groups.

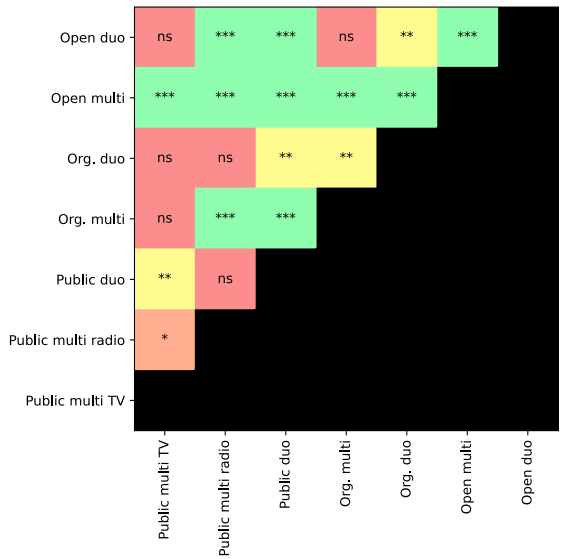
Table 7.7: Average resolution score per dialogue group.

Group	A_1	A_2	A_G	Difference (p-value)
Open duo	.36 ± .13	.34 ± .14	.35 ± .13	ns
Open multi	.20 ± .09	.19 ± .10	.20 ± .09	ns
Org. duo	.47 ± .17	.40 ± .13	.44 ± .15	**
Org. multi	.41 ± .09	.30 ± .07	.35 ± .07	***
Public duo	.55 ± .11	.48 ± .10	.52 ± .10	*
Public multi - radio	.56 ± .23	.44 ± .18	.50 ± .19	**
Public multi - TV	.44 ± .07	.36 ± .11	.40 ± .09	*
All	.42 ± .17	.35 ± .14	.38 ± .15	***

We compute the correlation between salience and resolution to evaluate whether more salient questions are more likely to be answered in the subsequent dialogue turns. Following Y. Wu et al. (2024), we use Spearman's rank correlation coefficient (ρ) to account for the ordinal nature of the annotation scales. We additionally report Kendall's τ, which is less sensitive to ties and therefore suitable given the discrete label space of resolution. To retain an ordinal, monotone ordering, we compute the correlation only on questions whose salience was 1, 3, 4, 6, or 8. Correlations are computed separately for each annotator as well as for aggregated ('gold') annotations. Results are shown in Figure 7.14, which reports mean resolu-



(a) Annotators 1 (◤) and 2 (◢).



(b) Gold annotations.

Figure 7.13: Representation of the significance of the difference in resolution distributions between dialogue groups for a given annotator. Colours indicate the strength of the p-value.

tion per salience level along with regression lines, and the correlation coefficients in the legend. The results with the full scale are available in Section D.2.

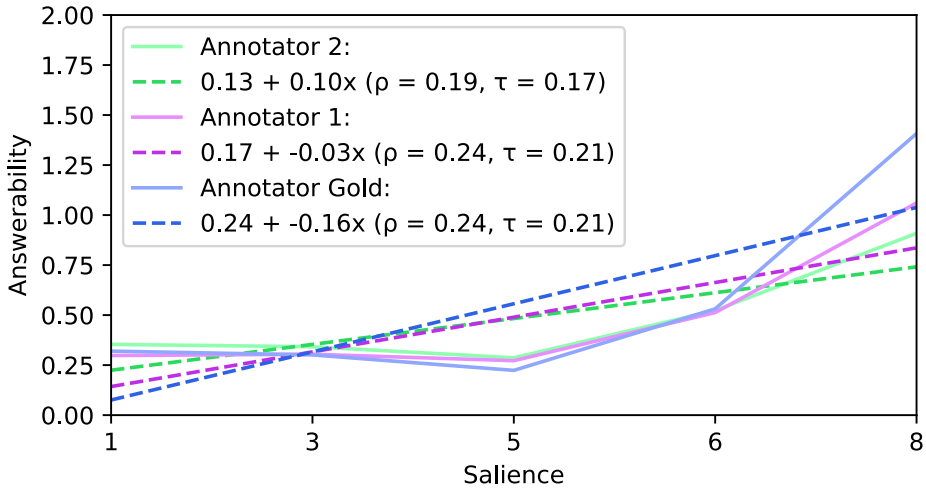


Figure 7.14: Correlation between salience and resolution scores (all p-values $\ll$ .005). *Legend in the same order as the plots at 0.*

Table 7.8: Average resolution score per salience score. *Results presented as mean* $\pm$ *standard deviation.*

S	A_1	A_2	A_G
0	.23 $\pm$ .51	.25 $\pm$ .55	.18 $\pm$ .44
1	.29 $\pm$ .60	.33 $\pm$ .65	.32 $\pm$ .64
2	.17 $\pm$ .49	.17 $\pm$ .49	.13 $\pm$ .45
3	.30 $\pm$ .60	.33 $\pm$ .64	.30 $\pm$ .60
4	.30 $\pm$ .61	.32 $\pm$ .63	.33 $\pm$ .64
5	.27 $\pm$ .56	.28 $\pm$ .55	.22 $\pm$ .50
6	.52 $\pm$ .73	.52 $\pm$ .74	.53 $\pm$ .74
7	1.13 $\pm$ .81	1.11 $\pm$ .83	1.21 $\pm$ .81
8	1.03 $\pm$ .86	.87 $\pm$ .85	1.41 $\pm$ .76

Across all settings, we observe a positive correlation between salience and resolution. This finding replicates the general trend reported by Y. Wu et al. (2024) in monologic data. However, the magnitude of the correlation (ρ = .19–.24) is two to three times lower than that reported for news and TED talks (ρ = .59–.74). While the model we used to generate the questions is different from Y. Wu et al.'s (2024) one, correlation reflects the relationship between salience and res-

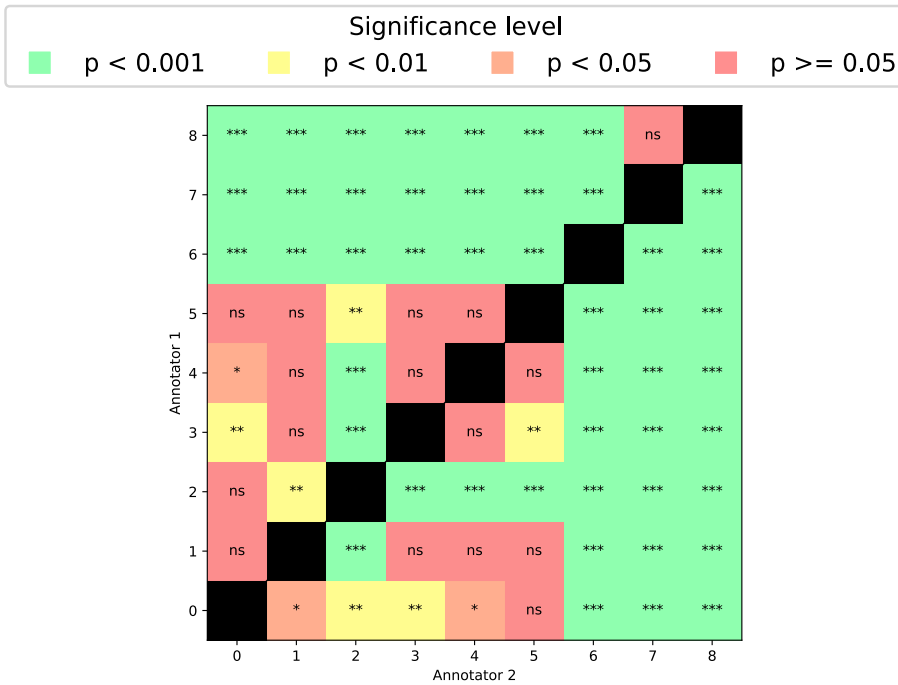
Table 7.9: Correlation between salience and resolution by dialogue group. ρ_i refers to the Spearman's rank correlation coefficient for annotator i, while ρ_G and p_G refer to the same metrics for the gold annotations (all p-values $\ll .001$).

Group	ρ_1	ρ_2	ρ_G
Open duo	.159	.202	.236
Open multi	.134	.149	.244
Org. duo	.331	.290	.418
Org. multi	.219	.174	.211
Public duo	.303	.239	.361
Public multi radio	.253	.158	.198
Public multi TV	.289	.190	.332

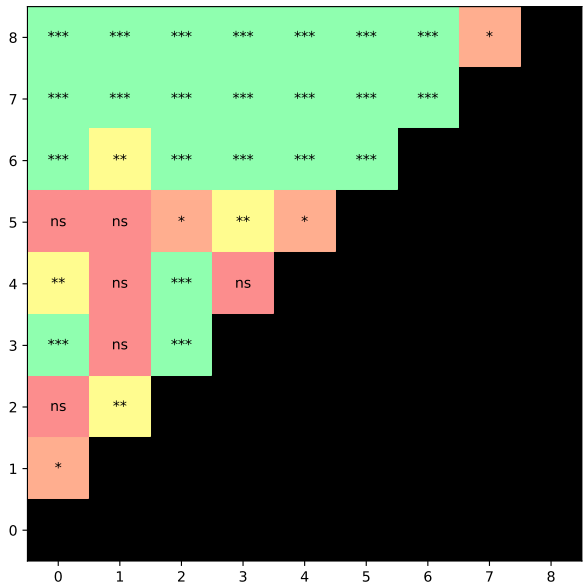
olution. Thus, the technical ability of the model to produce relevant questions is reflected by the salience distribution rather than the salience-resolution correlation. The strength of the p-value in our case indicates that the amount of questions in each category is sufficient, indeed suggesting weaker alignment between salience and downstream resolution in conversational settings. Nevertheless, expanding this work with different models would validate this hypothesis. This reduction in correlation is consistent with the increased variability of dialogue structure, where multiple interacting factors – including turn-taking dynamics, speaker intentions, and shared situational context – may affect whether salient questions are explicitly addressed.

Table 7.8 shows the average resolution score for each salience score and Figure 7.15 supports these results with a representation of the significance of these differences. We observe relatively homogeneous resolution for salience scores below 6, with the exception of score 2. Higher salience levels (6–8) are associated with significantly higher resolution. Score 2 consistently shows lower resolution across conditions, which is expected given that most of these questions ask for clarifications about the transcriptions and are thus unlikely to be answered in the dialogue.

Finally, we examine whether dialogue type affects the salience–resolution relationship (Figure 7.16). Most differences in correlation are not statistically significant with the exception of open-domain multilogues, which exhibit the lowest ρ coefficient. Public multilogues from television, which have intermediate ρ values for both annotators, also differ significantly from most other groups; only open-domain duologues and organised multilogues show similar correlations. When ranking groups by their ρ coefficients, the two annotators produce nearly identical orderings, with only two groups swapping positions, suggesting that group differences are relatively consistent across annotators. In contrast, the ranking obtained from the gold annotations differs substantially, with only the two highest-ranked



(a) Annotators 1 (◤) and 2 (◢).



(b) Gold annotations.

Figure 7.15: Representation of the significance of the difference on average resolution for each salience score. Colours indicate the strength of the p-value.

groups retaining their positions. This again suggests that annotator disagreements capture meaningful differences in the dialogues.

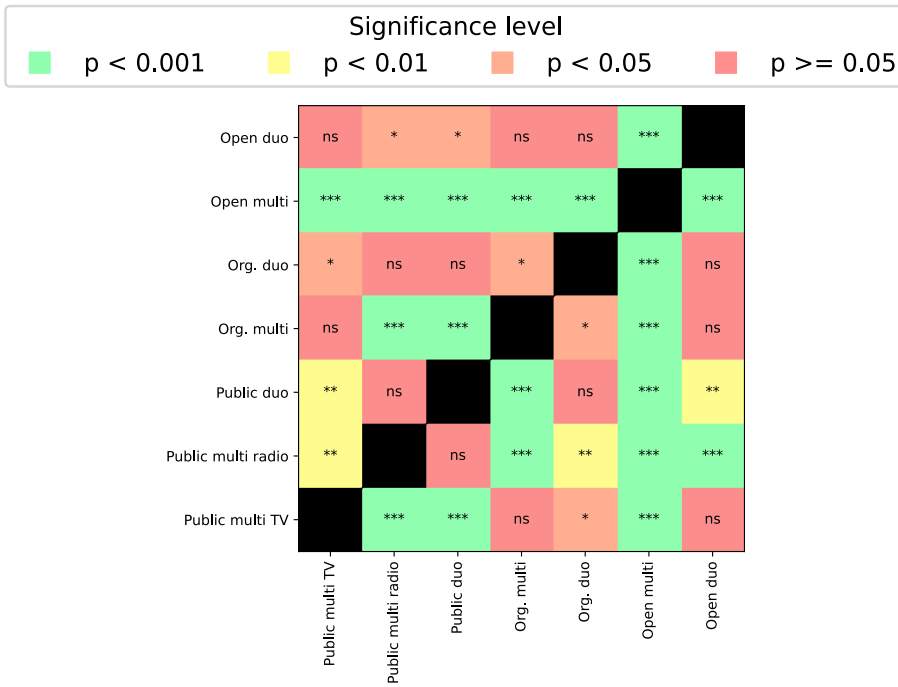
Table 7.9 shows the correlation coefficient between salience and resolution for each dialogue group. When ordering the groups by their ρ coefficient, we observe that both annotators have a very similar ranking, with only two groups swapping their position: the open-domain duologues and the public multilogues from the television that are ranked second and fifth based on Annotator 1 in increasing order, and fifth and second based on Annotator 2. This suggests that the differences between groups are relatively consistent across annotators but that these specific two groups may be more or less complex to annotate depending on the annotator. However, this ranking is very different from the one obtained on the gold annotations, where only the two groups with highest ρ coefficients – the organised duologues and the public duologues – keep the same ranking.

Nevertheless, Figure 7.16 shows that most group differences in correlation are not statistically significant with the exception of open-domain multilogues that have the lowest ρ coefficient. Public multilogues from the television, with the intermediate ρ coefficient for both annotators, also show significant differences with most other groups, open-domain duologues and organised multilogues being the only groups similar to them. Moreover, there are even less differences between groups when looking at the gold annotations. The homogenisation of the results in the gold annotations suggests that the disagreements between the annotators reveal some meaningful differences in the dialogues.

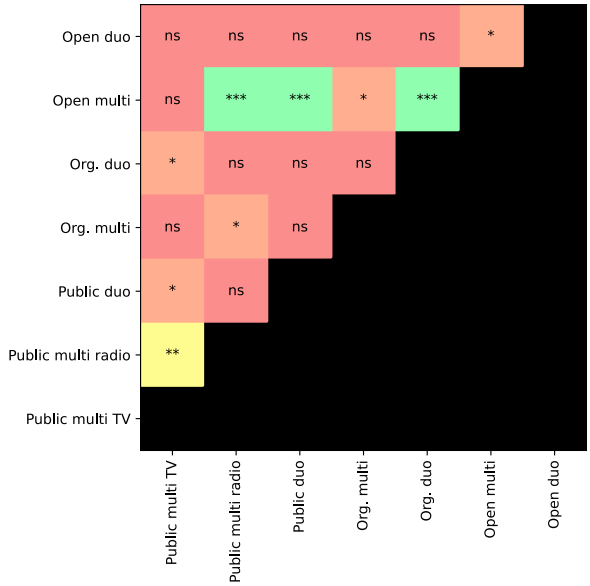
7.6 Conclusion

In this chapter, we investigated whether the relationship between question salience and resolution observed in monologic discourse also holds in dialogue. Adapting the framework of Y. Wu et al. (2024), we applied it to 60 dialogues from the British National Corpus representing a range of interactional settings. The most open of these interactional settings are the ones we initially set up to investigate within this thesis, as explained in Chapter 4. The decision to produce the annotations ourselves for this study allowed us to select the studied corpus more freely than for the studies presented in Chapters 5 and 6, where we were constrained by strong requirements regarding the structure of the data.

Our results show a positive correlation between question salience and resolution: questions judged to be more salient at a given point of the conversation are more likely to be addressed later on than less salient ones. However, this relationship is substantially weaker than that reported for news articles and TED talks. In a view where topics are modelled by QUDs, the correlation between salience and resolution can be interpreted as a signal that characterises the topical structure of a dialogue. The weaker correlation we observe in dialogue compared to mono-



(a) Annotators 1 (◤) and 2 (◢).



(b) Gold annotations.

Figure 7.16: Representation of the significance of the difference in correlation between dialogue group. Colours indicate the strength of the p-value.

logue suggests that this structure is less constrained in dialogue, as we could expect from the more dynamic nature of conversation, but that it is still present. We also show that this relationship is much weaker in open-domain multilogues and that the correlation between salience and resolution varies across dialogue types. Even though these last results require further analysis, and in particular a larger sample size and an investigation of the disagreements between the annotators, they suggest that topical structure is not uniform across dialogue types and that it could be used to identify families of dialogues. Expanding the salience–resolution analysis to dialogue types that are more clearly different from the ones studied here, such as those from the *Friends* corpus discussed in Chapter 5, would allow us to further validate this hypothesis. This clearly shows how crucial data characterisation is and validates the potential uses of CHIMERA.

Beyond these empirical findings, our work provides one of the first large-scale operationalisations of QUD-driven approaches for modelling ND. The moderate agreement obtained for salience annotation further highlights both the complexity of the task and the importance of accounting for annotator perspective when studying dialogue structure. Indeed, the more uniform values we get for the annotations and the correlation across dialogue types when considering only the gold annotations suggest that the disagreements between the annotators reveal meaningful differences regarding their perception of dialogue. The reasons for these disagreements may be related to their level of specialisation in dialogue analysis, as Annotator 1, the author of this thesis, is more specialised in dialogue analysis than the person who annotated most of the dialogues amongst those comprising Annotator 2. In any case, those reasons should be investigated further, as they may provide additional insights into how different people interpret the same dialogue from an external perspective. This is particularly important in the context of dialogue systems, where external human evaluation is often considered the gold standard for assessing system performance and for tuning automatic evaluation metrics based on LLMs-as-a-judge.

Future work should also investigate the temporal dynamics of question resolution, including when questions are answered and by whom. The dynamic nature of conversation creates complex situations that may not be captured by the simple salience–resolution model used in this experiment. For instance, the salience of a question may vary throughout the dialogue, while remaining relevant, depending on the current focus of the participants. If such a question was generated at several points in the dialogue, we may end up with several different salience scores bound to the same resolution score, which would dilute the correlation between the two measures. Similarly, the distance between the anchor that triggers a question and its actual answer should be taken into account when considering the 'predictability' of a conversation. One lead to address this issue would be to consider a question answered only if the answer is provided in a turn accessible from the anchor turn in terms of dialogue structure. This leads us back to the tension between linear and hi-

erarchical models of topical structure, as accessibility would be defined very differently depending on the model. This would also raise questions on the relevance of such structures depending on the type of dialogue, where open multilogues may be less constrained by structural expectations than other duologue types. A somewhat circular reasoning emerges from these considerations, as the structure of dialogue, and the degree to which it really constrains conversation production, is precisely what we are trying to understand through the analysis of salience and resolution.

Another promising direction is the development of automatic dialogue salience and resolution prediction models, enabling large-scale analysis of interactional structure. While we tried to work with a sample of dialogues large enough to support statistically significant analyses, and varied enough to compare a range of interactional settings, the sample size remains relatively small due to the time and effort required for manual annotation. The corpus developed in this study could be used to train and evaluate an automatic salience prediction model, and an extension of our annotations could be leveraged to produce an resolution prediction model. Y. Wu et al. (2024) have shown that a salience prediction model can be developed for monologic discourse, and a project carried out by first-year Master students in the NLP program at the University of Lorraine gives hope that it can be adapted for dialogue as well. Predicting resolution is likely to be more challenging, particularly because useful models should identify the portions of a dialogue that answer a question rather than merely assigning a score. Such models would nevertheless have direct applications to information retrieval and dialogue systems, two very active areas in NLP, providing a strong motivation for further research.

The automation of the annotation step would allow us to compare the results obtained when generating the questions with different models, and possibly compare them with human generations. While we focused more on what the questions could teach us in the present study, which was somewhat model agnostic as long as our final sample contained a variety of salience scores, it would be interesting to determine whether the correlation value is stable across different models. Showing that it is the case would support the idea that the salience–resolution relationship can be used to discriminate between broad families dialogue types, and maybe even discourse types. A limitation of this approach is that our corpus does not represent all possible dialogue types, and building automatic salience and resolution prediction models based on it may not generalise well to all types of language use. Nevertheless, this could be verified by validating the predictions through a smaller set of annotations, allowing us to select which types of dialogues require additional annotations to improve the model's generalisation.

Epilogue

Summary and Discussion

Our object of study in this work was conversation. We were interested both in the processes that make it possible, i.e. from a linguistic perspective, and in the ways it can be modelled and analysed automatically, i.e. from a computational perspective. Within this context, we had a particular interest in the topical structure of conversation, in how it varies across dialogue types, and in how it can be identified and predicted using automatic methods.

Since dialogue is an exceptionally broad and heterogeneous concept, encompassing a wide range of instantiations, studies typically focus on a single type or family of types of dialogue, though they may not always make this explicit. In this work, we sought to investigate face-to-face open-domain multi-party conversations, since they are often overlooked in favour of more organised interactions. As discussed in Chapter 4, these conversations are characterised by the parameters summarised in (7.ii).

(7.ii) multi-party face-to-face open conversation:

$$\begin{bmatrix} \texttt{Agents}: & \begin{bmatrix} \texttt{Agents}: \ \texttt{Agency}: \{\textit{human}\} \\ \texttt{Structure.Nb}: \begin{bmatrix} \texttt{Simulated}: Nb_{>2} \\ \texttt{Mapping}: \textit{SelfToSelf} \end{bmatrix} \\ \texttt{SocialDynamics.Hierarchy}: \textit{minimal} \end{bmatrix} \\ \texttt{Setting.EnvMedium}: \textit{face-to-face} \\ \texttt{Frame}: & \begin{bmatrix} \texttt{Language.Global.List}: \textsc{among}(\textsc{en}, \textsc{fr}) \\ \texttt{Purpose.SuccessCondition.Q}: \varnothing \end{bmatrix} \end{bmatrix}$$

where the *face-to-face* environment can be defined through the following constraints:

$$\begin{bmatrix} \texttt{PhysicalContext}: \textit{Shared} \\ \texttt{Visibility}: & \begin{bmatrix} \texttt{Gaze.Fidelity}: \textit{High} \\ \texttt{Gestures.Interpretability}: \textit{High} \\ \texttt{FacialExpressions.Interpretability}: \textit{High} \end{bmatrix} \\ \texttt{Time.Synchrony}: \textit{Maximal} \end{bmatrix}$$

Nevertheless, we also had to accommodate the practical limitations of working with such data, which is often scarce and difficult to collect. We therefore also worked on diverse forms of conversation. In all cases, we were careful to clearly

describe the data we used, in order to make explicit the scope of the conclusions we could draw from our experiments.

In this chapter, we begin by summarising the contributions of this work in relation to our research questions introduced in the Prologue and discuss their limitations (Section 7.1). We then move on to broader perspectives opened up by the thesis as a whole (Section 7.2), before concluding with some final words (Section 7.3).

1 Contributions and discussion

Conversation is something we observe through its realisation, rather than through the underlying processes that produce it, but we still attempt to understand those processes. An introductory description to them was provided in Chapter 1. Yet the variety of forms this realisation can take shows that dialogue does not designate a single uniform phenomenon, but rather a family of related yet structurally diverse phenomena. For this reason, universal behaviours cannot reasonably be assumed. If we are to make precise claims about dialogue, whether in the context of modelling it or improving LLMs, a precise definition of what is meant by dialogue in a given study is required.

A descriptive framework – named CHIMERA[25] – grounded in dialogue theory but still quite agnostic and flexible in its operationalisation, is the first contribution of this work. It takes the form of a set of dimensions, described in Chapter 2, along which dialogue can be characterised. Each of these dimensions is associated with a set of features that can be used to describe a dataset, or constraints on a dataset induced by a task, in a more or less fine-grained way.

Through an analysis of dialogue-related papers published in the ACL ANTHOLOGY in 2024, presented in Chapter 3, we showed that this framework can help address widespread shortcomings in the NLP community, particularly in terms of defining dialogue-related objectives and describing data. The growing lack of grounding in data analysis obscures the quality of the work carried out in these areas, as NLP research increasingly focuses on performance and becomes detached from linguistic considerations, despite their centrality to the field. We also discussed how CHIMERA can serve as a basis for designing control constraints on the form of language data generated by LLMs. We concluded the first part of the thesis by stressing the comparative purpose of our framework, rather than a normative one or one aimed at producing synthetic data to replace naturalistic data. The two accounts we proposed for CHIMERA, a formal one based on TTR, and a more visual one within a multi-dimensional space, are both intended to help researchers position their data and tasks within the broader landscape of dialogue research, and

[25] **CH**ARACTERISATION OF D**I**ALOGUE THROUGH **M**ULTIDIMENSIONAL **E**XAMINATION, **R**EPRESENTATION AND **A**NALYSIS.

to make explicit the assumptions they bring to their work. They also come with existing tools and methods that can be used to compare datasets and assess their alignment with the constraints of a given task. The TTR account can take advantage of the machinery around feature structures while the vector space account benefits from a large body of work on dimensionality reduction and clustering.

With this, the first part of this work addressed the first research question (RQ1) we posed in the Introduction, namely how to characterise dialogue and its diversity in a way that informs the definition of task-specificities and the choice of associated resources. Further work is needed to refine CHIMERA and the forms it can take, which requires a community effort to discuss and validate the dimensions and features we proposed, and to extend them to other languages and cultures. Although we tried to ground our framework in literature from different areas, we are not experts in all of them, and our perspective is necessarily limited by our own experience. It would also benefit from being applied to, and therefore confronted with, widely used dialogue datasets and NLP tasks. Chapter 3 and the studies in Part II provided an initial step in this direction, but a more systematic application would be useful to obtain a more complete picture of the field. This would also make it possible to identify gaps in the available data relative to the tasks we seek to address as a community, while increasing the visibility of certain datasets and making information about them easier to access. Moreover, the interactions between the different parameters should be further explored. Our TTR account is one way to highlight the dependencies between the parameters, which could also be made explicit through the analysis of the multidimensional space visualisation we suggested. However, explicitly characterising those dependencies and the variations one aspect of dialogue can induce on another is a wide-ranging endeavour made of an infinity of small investigations. Indeed, paving this multidimensional space would require to apply the scientific method that consists in varying a single parameter to observe its effect on the rest of the system. This work has already been initiated by the CL/NLP community, for instance in the analysis of the effect of the medium on different parameters such as gestures or turn-taking (Lücking et al., 2025). Our framework helps make explicit the parameters that should be considered and analysed, and thus provides a common formal basis for linking the different studies that are carried out in this direction. The new communication technologies that are emerging, and the new forms of dialogue they enable, will also broaden the space of possibilities and require a continuous effort to characterise the resulting diversity.

This focus on data characterisation, and on naturalness more broadly, remains central in the second part of this work, where we turn to the topical structure of dialogue. After an introduction to the notions of topic and topical structure in Chapter 4, we presented three studies that each address a different aspect of the problem. Throughout all of them, we have sought to remain faithful to the ideal of working with naturalistic data, while simultaneously dealing with the practical

limitations this imposes. We therefore worked with a variety of dialogue types, each chapter bringing us closer to the type of conversation we were most interested in, open-ended multi-party face-to-face conversation. (7.iii) summarises the differences between the three datasets we used, which we can also compare with the characteristics of our ideal dialogue type in (7.ii) (the differences are underlined). We started with the *Friends* dataset, whose main difference compared to our ideal dialogue type is that it is scripted and performed, making it arguably more organised than spontaneous conversations and more likely to exhibit simple recurring patterns such as the introduction of a new character coinciding with a topic shift. We then moved to *Reddit*, which contains naturally occurring dialogue that is, however, written and asynchronous as it is produced on an online forum-like platform. It is also more topically restrained than open-domain conversations because of the separation in subreddits, which are dedicated to specific topics. Finally, we worked on a subset of the British National Corpus (BNC), which we selected to cover several types of conversations, including multi-party face-to-face open conversations, but also more organised ones such as interviews or public debates. (7.iii) confirms that the type that describes the **open** conversations fully matches the one in (7.ii), except for the number of participants as we considered both duologues and multilogues, the multilogues being our ultimate target. The other two types of conversations we considered, **organised** and **public**, differ from our ideal type in several ways. Both types are more structured, with a more hierarchical social dynamic and a more explicit purpose. The **public** type is also intended for an external audience which means that the participants are more likely to make their contributions understandable to someone who is only listening or watching, and not participating in the conversation. Moreover, for the radio sub-type, while the participants of the verbal exchange are in most cases physically co-present, the audience is expected to only have access to the audio signal, encouraging the speakers to explicitly describe their environment if needed. All of these differences mattered for our study of topical structure as they influenced the efficiency of the methods we applied, and the conclusions we could draw from our experiments.

Our first approach to the study of topical structure was to consider it as a segmentation problem, where the goal is to identify boundaries between segments devoted to different topics. These first experiments, presented in Chapter 5, were carried out on transcriptions of the TV show *Friends*, which despite not being naturally occurring conversations, are professionally scripted and performed ones, making them more naturalistic than many hand-crafted dialogues. They also had the advantage of being multi-party and open-ended, making them a particularly suitable test case for our study.

Our results showed that segmentation algorithms that perform well on monologic texts or on organised dialogues struggle when applied to more complex conversations. While we have investigated only a small subset of methods for topic segmentation, and more elaborated ones may perform better, the characteristics of

(7.iii) α. *Friends* (Chapter 5):

$$\left[\begin{array}{l}\texttt{Agents}: \left[\begin{array}{l}\texttt{Agents}:\ \texttt{Agency} = \{\textit{human}\} \\ \texttt{Structure}: \left[\begin{array}{l}\texttt{Nb}: \left[\begin{array}{l}\texttt{Simulated}: Nb_{>2} \\ \underline{\texttt{Mapping}}: \textit{ActorsToRoles}\end{array}\right] \\ \underline{\texttt{Functions}}: \ni_{\textit{ExternalAudience}}\end{array}\right] \\ \texttt{SocialDynamics.Hierarchy}: \textit{minimal} \\ \underline{\texttt{EnunciativeStatus}}: \textit{ActorsPlayingCharacters}\end{array}\right] \\ \underline{\texttt{Setting.EnvMedium}}: \textit{ScriptedFaceToFace} \\ \texttt{Frame}: \left[\begin{array}{l}\texttt{Language.Global.List} = \{\textsc{en}\} \\ \underline{\texttt{Purpose.SuccessCondition.Q}}: \textit{script}\end{array}\right]\end{array}\right]$$

β. *Reddit* (Chapter 6):

$$\left[\begin{array}{l}\texttt{Agents}: \left[\begin{array}{l}\texttt{Agents}:\ \texttt{Agency} = \{\textit{human}\} \\ \texttt{Structure.Nb}: \left[\begin{array}{l}\texttt{Simulated}: Nb_{>2} \\ \texttt{Mapping}: \textit{SelfToSelf}\end{array}\right] \\ \texttt{SocialDynamics.Hierarchy}: \textit{minimal}\end{array}\right] \\ \underline{\texttt{Setting.EnvMedium}}: \textit{Online} \\ \texttt{Frame}: \left[\begin{array}{l}\texttt{Language.Global.List} = \{\textsc{en}\} \\ \underline{\texttt{Purpose.SuccessCondition.Q}}: \textit{SubredditOriented}\end{array}\right]\end{array}\right]$$

γ. BNC (Chapter 7):

$$\left[\begin{array}{l}\texttt{Agents}: \left[\begin{array}{l}\texttt{Agents}:\ \texttt{Agency} = \{\textit{human}\} \\ \texttt{Structure.Nb}: \left[\begin{array}{l}\underline{\texttt{Simulated}}: Nb_{>2} \vee Nb_{=2} \\ \texttt{Mapping}: \textit{SelfToSelf}\end{array}\right] \\ \texttt{SocialDynamics.Hierarchy}: \textit{minimal}\end{array}\right] \\ \texttt{Setting.EnvMedium}: \textit{FaceToFace} \\ \texttt{Frame}: \left[\texttt{Language.Global.List} = \{\textsc{en}\}\right]\end{array}\right]$$

with additional specificities for the three interactional settings:

open:

$$\left[\texttt{FramePurpose.SuccessCondition.Q} = \varnothing\right]$$

organised:

$$\left[\begin{array}{l} \texttt{Agents}: \left[\begin{array}{l} \texttt{SocialDynamics.Hierarchy}: \textit{interviewer} \vee \textit{councilRanks} \\ \texttt{EnunciativeStatus.Role}: \textit{taskBased} \end{array}\right] \\ \texttt{Frame}: \left[\texttt{Purpose}: \left[\begin{array}{l} \texttt{SuccessCondition.Q}: \textit{Explicit} \\ \texttt{TopicalStructure}: \textit{ExplicitAgenda} \end{array}\right]\right] \end{array}\right]$$

public:

$$\left[\begin{array}{l} \texttt{Agents}: \left[\begin{array}{l} \texttt{Structure.Functions}: \ni_{\textit{ExternalAudience}} \\ \texttt{SocialDynamics.Hierarchy}: \textit{interviewer} \vee \textit{councilRanks} \\ \texttt{EnunciativeStatus.Role}: \textit{taskBased} \end{array}\right] \\ \texttt{Setting.EnvMedium}: \textit{Radio} \vee \textit{TV} \\ \texttt{Frame.Purpose}: \left[\begin{array}{l} \texttt{SuccessCondition.Q}: \textit{Explicit} \\ \texttt{TopicalStructure}: \textit{ExplicitAgenda} \end{array}\right] \end{array}\right]$$

the data we used, and of open-ended multi-party conversations in general, made us question the relevance of the linear segmentation task itself for this type of dialogue. The complex dynamics that exist at the topical level may not be well captured by a linear segmentation. In particular, one of the main differences between the ML and the feature-based methods we applied is that the former produced too many segments. However, it is possible that those segments form a valid topic segmentation at a different level of granularity than the one produced by the other method, and that of the reference segmentation. A possible way to address this question is to investigate the internal coherence of topical segments and to compare it with the coherence of the conversation as a whole. This method has been used in the past as a reference-free evaluation for topic segmentation (Ghinassi et al., 2024a), and it could be used to compare different segmentation methods based on latent features regardless of any reference segmentation. Besides, if several segmentations can be identified as internally coherent and with individual segments being more coherent than the conversation as a whole, it would support the idea that topic segmentation is not a well-defined task for open-ended multi-party conversations as a given segmentation may not be more informative than another. Indeed, its *linear* nature does not account for the fact that topics can be interleaved in conversation. Moreover, its *binary* nature does not allow us to consider more gradual transitions between topics.

This intuition led us to analyse topical structure in a more continuous sense, i.e. by looking into the degree of topical similarity between utterances throughout a conversation. Chapter 6 presented an analysis of *Reddit* interactions using a model that we designed to predict topical similarity between pairs of comments. This time, we worked on naturally occurring exchanges, but in a production format that is written and asynchronous, two characteristics that differ strongly from face-

to-face conversation. We showed that the model we designed could predict topical similarity between comments, and that it could be used to analyse the topical structure of conversations in a more continuous way than segmentation methods. The structure of *Reddit* interactions, with a post at the root of the exchange and a tree of comments below it, allowed us to compare the coarse-grained evolution of topic, using the post as a reference, with the fine-grained variation of topic between successive comments. The difference between these two levels of topical structure supports the idea that modelling topics as sequences of utterances isolated from one another is not sufficient to account for the complexity of topical structure in dialogue. We thus proposed a first model that attempts to represent the articulation between topics together with the mechanisms that support it, in the form of implicit reasoning, thereby providing a way to account for more gradual transitions between topics.

A limitation of our approach is that we did not validate our similarity metric other than by assessing whether it preserved the structure of *Reddit* threads, which is the task the model was trained on but not the one we actually used it for. While it is based on a version of BERT fine-tuned for the next-sentence prediction task, determining whether our metric correlates with human judgements of topical similarity would provide a more direct validation of its relevance for our study. The tree-like structure of *Reddit* threads could be used to design a human evaluation task, where annotators would be asked to first propose an explanation for the topical link between a comment and each of its successor, before being asked to rank those reasoning in terms of their plausibility. We could then compare the ranking of the comments based on the annotators' judgements with the ranking based on our similarity metric, and assess whether they correlate. This would provide a more direct validation of our metric for the task we used it for, and would also provide a better understanding of the mechanisms that support topical links in dialogue.

Together, the two studies from Chapters 5 and 6 answer our second research question (RQ2), namely whether topical structure can be automatically extracted from dialogue data, and in particular from open-ended multi-party conversations. Several lines of improvements can be pursued to improve the performance of the methods we proposed, and to make them more suitable for the type of conversation we are interested in.

The feature-based topic segmentation approach we presented in Chapter 5 could be improved to alleviate these limitations by incorporating more complex dialogue-specific features, such as models of question types or rhetorical relations. Combining topic segmentation with clustering of utterances based on topical similarity could also help account for interleaved topics, by first separating the different threads of a conversation and then segmenting each of them individually.

With respect to complex and interleaved topical threads, the work on Reddit data presented in Chapter 6 points to a clear line of extension that we did not fully explore here. The structure of *Reddit* threads makes it possible to study several

continuations of a conversation in parallel, compare them, and potentially analyse several contributions by the same participant in concurrent parts of the conversation tree. However, the source of the data itself increasingly feels like a limitation, as my own trajectory throughout this thesis has made me more aware of the ethical concerns associated with using social media data. In addition, *Reddit*'s new policies make access to this data increasingly difficult. In principle, this is a positive development, but in practice it depends on the motivations behind this change and more importantly on the practical requirements for access. If a monetary compensation can allow access to the data, the ethical concerns will likely not be addressed. Pursuing this line of work would therefore require a more thorough examination of the ethical and practical implications of using such data, and possibly a shift towards other sources that are more ethically sound and more accessible.

A conclusion we draw from both Chapters 5 and 6 is that even though none of the datasets we studied were completely equivalent to spontaneous face-to-face open-ended conversations, and were actually not as complex, our experiments still showed that the methods that should be applied to more complex interactions are not the same as those that work well on more organised and linear forms of dialogue. Further work is therefore needed to define suitable approaches to capture the topical structure of open-ended multi-party conversations. The more flexible perspective on topic segmentation we suggested in Chapter 6 integrates questions of likelihood and of the perception of topic shifts by participants, through the notion of salience of these mechanisms, and thus brings an analytical perspective to the topic segmentation task.

The salience of potential topic shifts is a key aspect of dialogue, but it is not an explicit phenomenon. Our last contribution addresses its study by releasing a dataset of more than 7,000 questions, each anchored in conversational context and annotated twice by two annotators: once for their degree of salience, and once for whether they are actually answered in the upcoming dialogue turns. This work, presented in Chapter 7, operationalises a QUD-driven approach for studying the predictability of topic shifts across different types of dialogue. Based on naturally occurring and mostly face-to-face conversations from the BNC, we established that conversation is overall less predictable than monologic forms of text such as news or TED talks. We also found differences in predictability depending on the number of participants and the interactional setting, validating the need to clearly define and characterise the data we work with, thereby coming full circle to the first part of this thesis, and providing an answer to our third research question (RQ3), namely whether topical structure can be leveraged as a discriminative signal for differentiating some dialogue types. Even though the different types of conversations we studied in Chapter 7 were rather similar to one another, we were able to show that the relation between salience and resolution in questions varies depending on the type of conversation. The most striking variation we found was the one between multi-party casual conversations and the rest of the data, which is consistent with

the intuition that those interactions should be studied differently from more organised ones. Applying the same methodology to a wider range of dialogue types, and in particular dialogue types that differ more strongly from one another, would validate or invalidate the relevance of our approach as a discriminative signal for dialogue type classification. The first differences to investigate in the current state of the field would be those between synthetic and naturally occurring conversations, since the former are increasingly used in NLP research as a substitute for the latter.

Besides the extension of our work to other dialogue types, a more systematic study of the disagreements between annotators and of the types of questions that were considered salient or not, and resolved or not, would provide a better understanding of the mechanisms that support topical structure in dialogue. Investigating the types of questions that are generated by an LLM and comparing them with the ones humans would produce in the same context is another perspective that could be explored to determine what features of a dialogue context are tapped in by the model to generate questions, and how they compare with human judgements. This could highlight important differences in the way humans and LLMs process dialogue, and inform the discourse we should have about LLMs and their use in NLP research but also for the general public.

Beyond the limitations and future perspectives opened up by each individual study, the thesis as a whole also raises more general questions that we discuss in Section 7.2.

2 Limitations and perspectives

Our work was carried out on English data only, as is the case for most NLP research. Although we tried to keep our descriptive framework in Part I as open as possible, the strongly English-centred nature of the field likely influenced the features we selected. In Part II, many of the models we used were also designed for English data, which makes extending this work to other languages non-trivial. Both of these points also limit the last contribution of this thesis (Chapter 7), since question generation depended on the model and its training data, as well as on the conversations themselves, and the annotations were influenced by the culture and background of the annotators. We acknowledge these limitations and hope that future work will extend our contributions to other languages and cultures. This once again highlights the importance of clearly describing the data we work with, and of remaining aware of the limitations it imposes on our conclusions.

Language is only one of many aspects of conversation that is often insufficiently described in NLP papers. Beyond concerns of reproducibility and transparency, characterising our data is also a way of making explicit the assumptions we bring to the data and to the task under study. The point here is not to single out a particular dataset or type of dataset, but rather to question the motivations behind our choice

of corpus. Do we use it because it is the only one available, despite being poorly suited to the task? If so, can the task be reformulated within the limits of the data? Or is it possible to invest the time and resources needed to collect and annotate better-suited data? These questions are essential to the development of meaningful and relevant NLP research, and they reflect the constraints imposed by data scarcity and annotation cost. In the same vein, tracking the evolution of the NLP/CL field over time in terms of the tasks addressed and the data used would further help identify gaps in corpora and evaluation approaches. To this end, leveraging the annotations of the papers from the ACL Anthology 2024 produced in Chapter 3 to train an automatic – or at least partially automatic – annotation model would be a useful step towards scaling up the analysis of the literature. This would require a phase of cleaning and normalisation of the annotations, and ideally a second round of annotations to ensure the quality of the data used to train the model. Nevertheless, such work seems increasingly necessary in order to keep pace with the volume of publications and the growing number of relevant papers each year.

If we take a step back, Part I and Chapter 7 are both concerned with the variability of dialogue and with ways of accounting for it, but they also reveal latent differences between dialogue types. This is particularly relevant in the study of data involving patients with psychiatric diagnoses, where it is necessary to identify specificities with respect to control data. Applying the QUD-based approach to such data could reveal differences in the predictability of topics between patient and control conversations, and perhaps even in the questions generated and their salience. It would be interesting to investigate whether questions generated from patient data resemble those from control data in terms of salience and answerability, but also in terms of annotation difficulty. In this pathological perspective, the same approach could also be applied to data used to train psychological counselling models, as well as to the conversations they produce, in order to identify differences with respect to actual counselling practices. This perspective is part of the operationalisation of the descriptive framework proposed in Part I, and could be used to quantitatively characterise one type of difference between dialogue types. Counselling is a sensitive domain in which the use of LLMs is becoming increasingly common, and it is crucial to remain explicit and transparent about the limitations and risks of this trend.

A second general perspective underlying many of the decisions we made in selecting data and methods concerns evaluation, and more specifically external human judgement of conversation. Although human beings are, in some sense, experts in conversation, in that they rely on a wide range of abilities to produce and understand it, this does not mean that they can judge it in a reliable and valid way. This is particularly true when the conversation is not directly experienced by the judge, but only observed a posteriori through a recorded form such as a transcription or an audio recording. We saw in Chapter 3 that human judgement is often used to evaluate general qualities of dialogue, such as coherence or naturalness, in

dialogue system experiments. However, the reliability and validity of such evaluations remain questionable, and the lack of standardisation in evaluation protocols makes it difficult to compare results across studies. This opens up two avenues for future work. First, we need to investigate humans' ability to judge conversation reliably and validly, and to identify which aspects of conversation are more or less amenable to such evaluation. Second, we need to develop more specific evaluation metrics grounded in more clearly describable and observable criteria. This requires input from the computational perspective to determine which aspects are relevant to evaluate, from the linguistic perspective to define the behaviours that should be targeted in a given context, and from psychometrics to ensure that the evaluation protocols are reliable and valid. This is a crucial step towards a more rigorous and standardised evaluation of dialogue systems, and one that we hope will be pursued in future work.

3 Final words

A growing response to the limitations of data availability and annotation cost in NLP research is the use of LLMs to generate data or annotations. While this is a practical solution, it raises questions about the naturalness of the resulting data and the relevance of the tasks designed around it. We must use LLMs carefully and ensure that we do not lose sight of the linguistic and theoretical aspects of dialogue research. Maintaining control over the quality and the nature of the data we produce with these models therefore requires the development of tools and evaluation metrics that are appropriate to the tasks at hand.

The relative ease of access to LLMs in the scientific community, and especially the access to conversational agents, has led to a growing number of works that are only loosely connected to dialogue as a research object. In many such studies, conversation is used mainly as a means to an end, with little theoretical grounding. This can lead to the transfer of evaluation metrics to tasks where they are not meaningful, and to the use of data that is not well suited to the problem being addressed. Our aim here has been to propose a tool that help select data and evaluation metrics that are more directly relevant, so that the interdisciplinary nature of dialogue research becomes a strength rather than a source of confusion. The framework proposed in Part I is grounded in this idea.

Some domains could clearly benefit from the use of LLMs, such as the medical one, where data is sensitive and difficult to share. It is also a domain in which partially artificial conversations are already common in the form of training exercises for clinical interviews. In such a context, synthetic data can help alleviate both data scarcity and privacy concerns, while complementing datasets that are already not entirely naturalistic. However, qualified personnel must be involved in the design of such protocols, both on the medical and the technical sides. In particular, facilita-

tors are needed to ensure mutual understanding among the various parties involved, so that everyone understands both the issues at stake for the others and the actual technical abilities and limitations of the tools in use. It is crucial that any decision to use LLMs be made with a clear understanding that they are statistical language production tools, with no inherent conception of truth, regardless of the training data they were exposed to. This is even more important in sensitive contexts such as medicine or law, where the consequences of using LLMs can be severe. As specialists in these technologies, we have a responsibility to explain this clearly, and outreach and mediation activities with diverse audiences are an important part of that mission.

Beyond this, the increasingly widespread use of LLMs raises more practical questions, especially regarding how we talk about them within the NLP community. One of the advantages often highlighted is the relatively low cost of using LLMs for data production and annotation. However, this is a very relative notion, usually based only on the price charged by the provider of the model. This gives a misleading impression, since it reduces the issue to who gets the best deal from the large companies on which we depend for these services. It also overlooks a major part of the real cost of these tools, which exists somewhere even if it is not paid directly by researchers. The substantial cost of building and maintaining the data centres on which these systems rely must be borne by some entity. At present, providers appear to absorb this cost, but the un-sustainability of such a model raises questions about their longer-term motivations. Moreover, the desire for greater technological sovereignty in Europe has led to major public investment in data centres and local LLMs, with corresponding costs that impact the population in more or less direct ways. There is also a clear ecological impact linked to water and energy consumption, and the rapid renewal of both models and hardware raises further concerns about material scarcity and waste management. For all these reasons, reducing the cost of LLMs to the amount billed through an API is highly misleading, and it is part of our responsibility as researchers and specialists in these technologies to articulate a more complete and accurate account of the matter.

The amount of human work involved in the design, development, and improvement of these systems should also be kept in mind. For instance, human labour was required to collect and annotate the good quality data that is used to fine-tune those models. People have also been asked to select the best outputs from the models in comparative evaluations in order to make them feel more human-like. The numerous places in the chain of production and use of LLMs where human labour is required, often under precarious conditions, should be made more visible. Similarly, presenting LLMs with an anthropomorphic vocabulary, such as *understanding*, *thinking*, or *reasoning*, is misleading and can lead to confusion about the nature of these systems. It is important to remember that LLMs are statistical models that generate text based on patterns learned from data, and they do not possess consciousness, understanding, or reasoning abilities in the human sense.

All the abilities they seem to display are the result of hidden human labour and of our own tendency to attribute human-like qualities to entities that demonstrate some evidence of human-like abilities (Dingemanse, 2026). It is therefore crucial to monitor the discourse within the community regarding LLMs, and this requires a robust peer-review system. Due to the very fast-paced development of the field, numerous papers on LLMs and their abilities are published on open archives before being submitted to venues for peer review in order to lock in the priority of the results. This is becoming a common practice but it is not without risk, as it can lead to the dissemination of unverified claims and the propagation of misleading information.

Lastly, while the development of LLMs was initially based on the exploitation of vast quantities of data, mostly scraped from the web, it now relies increasingly on higher-quality data, of which written books are a major source. Big tech companies have recently begun to collect large quantities of books, sometimes unique specimens, in order to train their models. Besides the ethical issues raised by the fact that authors never consented to this use of their work and often would not even have envisioned that this possibility would exist, the way in which these books are scanned is also a cause for concern. Even though it is possible to digitalise books without damaging them, the methods used by these companies are often destructive[26], as separating the pages of a book is faster and cheaper than scanning them while keeping them bound. Not so long ago, people warned that the burning of books was a prelude to oppression and the destruction of human lives, and history has proved them right. The current situation should warrant some concern, as we are being pushed by big tech companies to use tools that exist thanks to the destruction of books, and therefore of knowledge.

Where they have burned books,
they will end up burning people.

– Heinrich Heine, Almansor (1821)

[26]This was revealed in a recent lawsuit against Anthropic as reported by several press media: International Business Time IK, Forbes, The Washington Post among others (all accessed on September 25, 2026).

Acronyms

ACL Association for Computational Linguistics
AI Artificial Intelligence
ASR Automatic Speech Recognition
BERT Bidirectional Encoder Representations from Transformers
BNC British National Corpus
CA Conversation Analysis
CEFR Common European Framework of Reference for Languages
CER Conversation Emotion Recognition
CL Computational Linguistics
DL Deep Learning
DST Dialogue State Tracking
DTS Dialogue Topic Segmentation
ICSI International Computer Science Institute
LDA Latent Dirichlet Allocation
LLM Large Language Model (Fr. *Grand Modèle de Langue*)
ML Machine Learning
ND Naturalistic Dialogue
NER Named Entity Recognition
NLI Natural Language Inference
NLP Natural Language Processing
NSP Next Sentence Prediction
QA Question-Answering
QUD Question Under Discussion
SM Social Media
TAL Traitement Automatique des Langues
TOD Task-Oriented Dialogue
TTR Type Theory with Records
VR Virtual Reality
WOZ Wizard-of-Oz

Bibliography

Addlesee, A., Yu, Y., & Eshghi, A. (2020). 'A Comprehensive Evaluation of Incremental Speech Recognition and Diarization for Conversational AI'. In D. Scott, N. Bel, & C. Zong (Eds.), *Proceedings of the 28th international conference on computational linguistics* (pp. 3, 492–3, 503). International Committee on Computational Linguistics.

Adler, A. (2024). 'Effects of experience and directionality on cognitive load in dialogue interpreting'. *Translation, Cognition & Behavior*, *7*(2), 187–208.

Ahn, E., Jimenez, C., Tsvetkov, Y., & Black, A. W. (2020). 'What Code-Switching Strategies are Effective in Dialog Systems?' In A. Ettinger, G. Jarosz, & J. Pater (Eds.), *Proceedings of the society for computation in linguistics 2020* (pp. 254–264). Association for Computational Linguistics.

Algan, Y., Guriev, S., Papaioannou, E., & Passari, E. (2017). 'The European Trust Crisis and the Rise of Populism'. *Brookings Papers on Economic Activity*, *48*(2), 309–400.

Allen, D. E., & Guy, R. F. (1977). 'Ocular Breaks and Verbal Output'. *Sociometry*, *40*(1), 90–96.

Amblard, M. (2018). 'SLAM, un corpus de conversations avec des patients schizophrènes'. *Recherches linguistiques et corpus, séminaire STIH de la Faculté des Lettres de Sorbonne Université*.

Amblard, M., Fort, K., Musiol, M., & Rebuschi, M. (2014). 'L'impossibilité de l'anonymat dans le cadre de l'analyse du discours'. *Journée ATALA éthique et TAL*.

Amido, S., Maraev, V., & Howes, C. (2024). '"no, you listen!" a pilot experiment into escalation devices in confrontational conversation'. *Proceedings of the 28th Workshop on the Semantics and Pragmatics of Dialogue - Poster Abstracts*.

Anantha, R., Vakulenko, S., Tu, Z., Longpre, S., Pulman, S., & Chappidi, S. (2021). 'Open-Domain Question Answering Goes Conversational via Question Rewriting'. In K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tur, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, & Y. Zhou (Eds.), *Proceedings of the 2021 conference of the north american chapter of the*

association for computational linguistics: Human language technologies (pp. 520–534). Association for Computational Linguistics.

Anderson, A. H. (1995). 'Negotiating coherence in dialogue'. In M. A. Gernsbacher & T. Givón (Eds.), *Coherence in spontaneous text* (pp. 1–40). John Benjamins Publishing Philadelphia.

Anthropic. (2023). Claude.

Arana, J., Idoyaga, M., Urruela, M., Espina, E., Atutxa Salazar, A., & Gojenola, K. (2024). 'A Virtual Patient Dialogue System Based on Question-Answering on Clinical Records'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 2, 017–2, 027). ELRA; ICCL.

Arguello, J., & Rosé, C. (2006). 'Topic-Segmentation of Dialogue'. In E. Hovy, K. Zechner, & L. Zhou (Eds.), *Proceedings of the analyzing conversations in text and speech* (pp. 42–49). Association for Computational Linguistics.

Argyle, M., Lefebvre, L., & Cook, M. (1974). 'The meaning of five patterns of gaze'. *European journal of social psychology*, *4*(2), 125–136.

Armstrong, E., Groenewold, R., Hersh, D., Akers, C., O'Malley, M.-P., & Müller, N. (2026). 'Clinical linguistics: A focus on interactions and an interpersonal perspective'. In *The routledge handbook of transdisciplinary systemic functional linguistics* (pp. 127–152). Routledge.

Artstein, R., & Poesio, M. (2008). 'Survey Article: Inter-Coder Agreement for Computational Linguistics'. *Computational Linguistics*, *34*(4), 555–596.

Asada, K., Itakura, S., Okanda, M., Moriguchi, Y., Yokawa, K., Kumagaya, S., Konishi, K., & Konishi, Y. (2022). 'Understanding of the Gricean maxims in children with autism spectrum disorder: Implications for pragmatic language development'. *Journal of Neurolinguistics*, *63*, 101, 085.

Asher, N. (2004). 'Discourse Topic'. *Theoretical Linguistics*, *30*(2-3), 163–201.

Asher, N., Hunter, J., Morey, M., Farah, B., & Afantenos, S. (2016). 'Discourse Structure and Dialogue Acts in Multiparty Dialogue: the STAC Corpus'. In N. Calzolari, K. Choukri, T. Declerck, S. Goggi, M. Grobelnik, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, & S. Piperidis (Eds.), *Proceedings of the tenth international conference on language resources and evaluation (LREC'16)* (pp. 2, 721–2, 727). European Language Resources Association (ELRA).

Asher, N., Hunter, J., & Thompson, K. (2020). 'Modelling Structures for Situated Discourse' (M. Poesio, M. Stede, A. Stent, J. Ginzburg, V. Demberg, & A. Zeldes, Eds.). *Dialogue & Discourse*, *11*, 89–121.

Asher, N., & Lascarides, A. (2003). *Logics of conversation*. Cambridge University Press.

Attardo, S., Eisterhold, J., Hay, J., & Poggi, I. (2003). 'Multimodal markers of irony and sarcasm.' *Humor: International Journal of Humor Research*, *16*(2).

Austin, J. L. (1975). *How to do things with words*. Oxford university press.

Ayer, A. J. (1936). *Language, Truth and Logic*. V. Gollancz.

Bailey, C.-J. N. (1981). 'Developmental linguistics'.

Bakhtin, M. M. (2010a). *Speech genres and other late essays*. University of Texas press.

Bakhtin, M. M. (2010b). *The dialogic imagination: Four essays* (Vol. 1). University of texas Press.

Ballon, M., Verbeken, B., Ginis, V., & Algaba, A. (2026). Probing the trajectories of reasoning traces in large language models.

Balouchzahi, F., Butt, S., Sidorov, G., & Gelbukh, A. (2023). 'ReDDIT: Regret detection and domain identification from text'. *Expert Systems with Applications*, *225*, 120099.

Banerjee, S., & Lavie, A. (2005). 'METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments'. In J. Goldstein, A. Lavie, C.-Y. Lin, & C. Voss (Eds.), *Proceedings of the ACL workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization* (pp. 65–72). Association for Computational Linguistics.

Banerjee, S., & Rudnicky, A. I. (2007). 'Segmenting meetings into agenda items by extracting implicit supervision from human note-taking'. *Proceedings of the 12th International Conference on Intelligent User Interfaces*, 151–159.

Bard, E. G., & Lickley, R. J. (1997). 'On not remembering disfluencies'. *Proceedings of the 5th European Conference on Speech Communication and Technology*, 2855–2858.

Barlett, C. P., Gentile, D. A., & Chew, C. (2016). 'Predicting cyberbullying from anonymity.' *Psychology of Popular Media Culture*, *5*(2), 171.

Baumgartner, J., Zannettou, S., Keegan, B., Squire, M., & Blackburn, J. (2020). 'The Pushshift Reddit Dataset'. *Proceedings of the International AAAI Conference on Web and Social Media*, *14*(1), 830–839.

Bavaresco, A., Bernardi, R., Bertolazzi, L., Elliott, D., Fernández, R., Gatt, A., Ghaleb, E., Giulianelli, M., Hanna, M., Koller, A., Martins, A., Mondorf, P., Neplenbroek, V., Pezzelle, S., Plank, B., Schlangen, D., Suglia, A., Surikuchi, A. K., Takmaz, E., & Testoni, A. (2025). 'LLMs instead of Human Judges? A Large Scale Empirical Study across 20 NLP Evaluation Tasks'. In W. Che, J. Nabende, E. Shutova, & M. T. Pilehvar (Eds.), *Proceedings of the 63rd annual meeting of the association for computational linguistics (volume 2: Short papers)* (pp. 238–255). Association for Computational Linguistics.

Beeferman, D., Berger, A., & Lafferty, J. (1999). 'Statistical models for text segmentation'. *Machine learning*, *34*, 177–210.

Bender, E. M. (2011). 'On Achieving and Evaluating Language-Independence in NLP'. *Linguistic Issues in Language Technology*, *6*.

Bender, E. M., & Friedman, B. (2018). 'Data statements for natural language processing: Toward mitigating system bias and enabling better science' (L. Lee, M. Johnson, K. Toutanova, & B. Roark, Eds.). *Transactions of the Association for Computational Linguistics*, *6*, 587–604.

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). 'On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?' *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623.

Bender, E. M., & Koller, A. (2020). 'Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data'. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds.), *Proceedings of the 58th annual meeting of the association for computational linguistics* (pp. 5185–5198). Association for Computational Linguistics.

Benz, A., & Jasinskaja, K. (2017). 'Questions under discussion: From sentence to discourse'. *Discourse Processes*, *54*(3), 177–186.

Biancofiore, G. M., Deldjoo, Y., Noia, T. D., Di Sciascio, E., & Narducci, F. (2024). 'Interactive Question Answering Systems: Literature Review'. *ACM Comput. Surv.*, *56*(9).

Bird, S. (2020). 'Decolonising Speech and Language Technology'. In D. Scott, N. Bel, & C. Zong (Eds.), *Proceedings of the 28th international conference on computational linguistics* (pp. 3504–3519). International Committee on Computational Linguistics.

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). 'Latent dirichlet allocation'. *J. Mach. Learn. Res.*, *3*, 993–1, 022.

Bolinger, D. L. (1952). 'Linear modification'. *PMLA*, *67*(7), 1117–1144.

Bowman, S. R., Angeli, G., Potts, C., & Manning, C. D. (2015). 'A large annotated corpus for learning natural language inference'. In L. Màrquez, C. Callison-Burch, & J. Su (Eds.), *Proceedings of the 2015 conference on empirical methods in natural language processing* (pp. 632–642). Association for Computational Linguistics.

Bowman, S. R., & Dahl, G. (2021). 'What will it take to fix benchmarking in natural language understanding?' In K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tur, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, & Y. Zhou (Eds.), *Proceedings of the 2021 conference of the north american chapter of the association for computational linguistics: Human language technologies* (pp. 4, 843–4, 855). Association for Computational Linguistics.

Breitholtz, E. (2021). *Enthymemes and Topoi in Dialogue: The Use of Common Sense Reasoning in Conversation*. BRILL.

Breitholtz, E., & Cooper, R. (2021). 'Dogwhistles as Inferences in Interaction'. In C. Howes, S. Dobnik, E. Breitholtz, & S. Chatzikyriakidis (Eds.), *Proceedings of the reasoning and interaction conference (reinact 2021)* (pp. 40–46). Association for Computational Linguistics.

Brennan, S. E., Kuhlen, A. K., & Charoy, J. (2018). 'Discourse and Dialogue'. In *Stevens' handbook of experimental psychology and cognitive neuroscience* (pp. 1–57). John Wiley & Sons, Ltd.

Brown, G., & Yule, G. (1983). *Discourse analysis*. Cambridge university press.

Brown, P. (1987). *Politeness: Some universals in language usage* (Vol. 4). Cambridge university press.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. (2020). 'Language models are few-shot learners'. *Advances in neural information processing systems*, *33*, 1877–1901.

Bryant, G. A. (2011). 'Verbal irony in the wild'. *Pragmatics & Cognition*, *19*(2), 291–309.

Budzianowski, P., Wen, T.-H., Tseng, B.-H., Casanueva, I., Ultes, S., Ramadan, O., & Gašić, M. (2018). 'MultiWOZ - A Large-Scale Multi-Domain Wizard-of-Oz Dataset for Task-Oriented Dialogue Modelling'. In E. Riloff, D. Chiang, J. Hockenmaier, & J. Tsujii (Eds.), *Proceedings of the 2018 conference on empirical methods in natural language processing* (pp. 5016–5026). Association for Computational Linguistics.

Busso, C., Bulut, M., Lee, C.-C., Kazemzadeh, A., Mower, E., Kim, S., Chang, J. N., Lee, S., & Narayanan, S. S. (2008). 'IEMOCAP: Interactive emo-

tional dyadic motion capture database'. *Language resources and evaluation*, *42*(4), 335–359.

Button, G., & Casey, N. (1985). 'Generating topic: the use of topic initial elicitors'. In J. M. Atkinson (Ed.), *Structures of social action* (pp. 167–190). Cambridge University Press.

Bychkova, P., Yaskevich, A., Gyulasaryan, S., & Rakhilina, E. (2024). 'Building a Database of Conversational Routines'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 2548–2555). ELRA; ICCL.

Carlini, N., Tramer, F., Wallace, E., Jagielski, M., Herbert-Voss, A., Lee, K., Roberts, A., Brown, T., Song, D., Erlingsson, U., et al. (2021). 'Extracting training data from large language models'. *30th USENIX security symposium (USENIX Security 21)*, 2633–2650.

Carnap, R. (1931). 'Überwindung der Metaphysik durch logische Analyse der Sprache'. *Erkenntnis*, *2*, 219–241.

Carr, C. T., & Hayes, R. A. (2015). 'Social Media: Defining, Developing, and Divining'. *Atlantic Journal of Communication*, *23*(1), 46–65.

Castro, S. (2017). Fast Krippendorff: Fast computation of Krippendorff's alpha agreement measure.

Cavanaugh, W. J., Farrell, W., Hirtle, P., & Watters, B. (1962). 'Speech privacy in buildings'. *The Journal of the Acoustical Society of America*, *34*(4), 475–492.

Chalamalasetti, K., Götze, J., Hakimov, S., Madureira, B., Sadler, P., & Schlangen, D. (2023). 'clembench: Using Game Play to Evaluate Chat-Optimized Language Models as Conversational Agents'. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing* (pp. 11174–11219). Association for Computational Linguistics.

Chaudhari, S., Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K., Deshpande, A., & Castro da Silva, B. (2025). 'Rlhf deciphered: A critical analysis of reinforcement learning from human feedback for llms'. *ACM Comput. Surv.*, *58*(2).

Chen, J., & Yang, D. (2020). 'Multi-View Sequence-to-Sequence Models with Conversational Structure for Abstractive Dialogue Summarization'. In B. Webber, T. Cohn, Y. He, & Y. Liu (Eds.), *Proceedings of the 2020 conference on empirical methods in natural language processing (emnlp)* (pp. 4106–4118). Association for Computational Linguistics.

Chen, Y.-P., Nishida, N., Nakayama, H., & Matsumoto, Y. (2024). 'Recent Trends in Personalized Dialogue Generation: A Review of Datasets, Methodologies, and Evaluations'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 13650–13665). ELRA; ICCL.

Chen, Y.-H., & Choi, J. D. (2016). 'Character Identification on Multiparty Conversation: Identifying Mentions of Characters in TV Shows'. *Proceedings of the 17th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, 90–100.

Cheng, M., Gligorić, K., Piccardi, T., & Jurafsky, D. (2024). 'AnthroScore: A computational linguistic measure of anthropomorphism'. In Y. Graham & M. Purver (Eds.), *Proceedings of the 18th conference of the european chapter of the association for computational linguistics (volume 1: Long papers)* (pp. 807–825). Association for Computational Linguistics.

Cheng, M., Hawkins, R. D., & Jurafsky, D. (2026). 'Accommodation and Epistemic Vigilance: A Pragmatic Account of Why LLMs Fail to Challenge Harmful Beliefs'. In M. Liakata, V. P. Moreira, J. Zhang, & D. Jurgens (Eds.), *Proceedings of the 64th annual meeting of the Association for Computational Linguistics (volume 1: Long papers)* (pp. 16181–16203). Association for Computational Linguistics.

Cheng, M., Lee, C., Khadpe, P., Yu, S., Han, D., & Jurafsky, D. (2026). 'Sycophantic AI decreases prosocial intentions and promotes dependence'. *Science*, *391*(6792), eaec8352.

Chepinchikj, N., & Thompson, C. (2016). 'Analysing cinematic discourse using conversation analysis'. *Discourse, Context & Media*, *14*, 40–53.

Cho, W. S., Zhang, P., Zhang, Y., Li, X., Galley, M., Brockett, C., Wang, M., & Gao, J. (2019). 'Towards Coherent and Cohesive Long-form Text Generation'. *Proceedings of the First Workshop on Narrative Understanding*, 1–11.

Cho, Y. M., Rai, S., Ungar, L., Sedoc, J., & Guntuku, S. (2023). 'An Integrative Survey on Mental Health Conversational Agents to Bridge Computer Science and Medical Perspectives'. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing* (pp. 11346–11369). Association for Computational Linguistics.

Choi, H.-S., Decker, A., Richard, V. D., & Cousin, M. (2024). 'OintMind'. In *THINK BEFORE LOADING*.

Chomsky, N. (1969). *Aspects of the Theory of Syntax*. The MIT Press.

Church, K. W., & Helfman, J. I. (1993). 'Dotplot: A program for exploring self-similarity in millions of lines of text and code'. *Journal of Computational and Graphical Statistics*, *2*(2), 153–174.

Cinelli, M., Quattrociocchi, W., Galeazzi, A., Valensise, C. M., Brugnoli, E., Schmidt, A. L., Zola, P., Zollo, F., & Scala, A. (2020). 'The COVID-19 social media infodemic'. *Scientific Reports*, *10*(1), 16, 598.

Clark, H. H. (1996). *Using language*. Cambridge university press.

Clark, H. H. (1997). 'Dogmas of understanding'. *Discourse Processes*, *23*(3), 567–598.

Clark, H. H., Brennan, S. E., et al. (1991). 'Grounding in communication'. In *Perspectives on socially shared cognition* (pp. 127–149). American Psychological Association.

Clark, H. H., & Carlson, T. B. (1982). 'Hearers and speech acts'. *Language*, *58*, 332–373.

Clark, H. H., & Schaefer, E. F. (1989). 'Contributing to discourse'. *Cognitive Science*, *13*(2), 259–294.

Clark, H. H., & Schaefer, E. F. (1992). 'Dealing with overhearers'. *Arenas of language use*, 248–274.

Clark, H. H., & Wilkes-Gibbs, D. (1986). 'Referring as a collaborative process'. *Cognition*, *22*(1), 1–39.

Clifton, C., & Frazier, L. (2012). 'Discourse integration guided by the 'Question under Discussion''. *Cognitive Psychology*, *65*(2), 352–379.

Cochran, W. G. (1950). 'THE COMPARISON OF PERCENTAGES IN MATCHED SAMPLES'. *Biometrika*, *37*(3-4), 256–266.

Cohen, J. (1968). 'Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit.' *Psychological bulletin*, *70*(4), 213.

Cohen, P. R., & Perrault, C. R. (1979). 'Elements of a plan-based theory of speech acts'. *Cognitive Science*, *3*(3), 177–212.

Colman, M., & Healey, P. (2011). 'The distribution of repair in dialogue'. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *33*(33).

Consortium, B. (1994a). British National Corpus [Oxford Text Archive].

Consortium, B. (1994b). British National Corpus [Oxford Text Archive].

Cook, M. (1977). 'Gaze and Mutual Gaze in Social Encounters: How long—and when—we look others" in the eye" is one of the main signals in nonverbal communication'. *American Scientist*, *65*(3), 328–333.

Cook, M., & Lalljee, M. G. (1972). 'Verbal substitutes for visual signals in interaction'.

Cooper, R. (2005). 'Records and Record Types in Semantic Theory'. *Journal of Logic and Computation*, *15*(2), 99–112.

Cooper, R. (2023). *From perception to communication: a theory of types for action and meaning* (Vol. 16). Oxford University Press.

Cortes, C., & Vapnik, V. (1995). 'Support-vector networks'. *Machine learning*, *20*(3), 273–297.

Couper-Kuhlen, E., & Selting, M. (2017). 'Motivating an Interactional Linguistic Perspective'. In *Interactional linguistics: Studying language in social interaction*. Cambridge University Press.

Cruz Blandón, M. A., Minnema, G., Nourbakhsh, A., Boritchev, M., & Amblard, M. (2019). 'Toward Dialogue Modeling: A Semantic Annotation Scheme for Questions and Answers'. In A. Friedrich, D. Zeyrek, & J. Hoek (Eds.), *Proceedings of the 13th linguistic annotation workshop* (pp. 230–235). Association for Computational Linguistics.

Cui, X. (2015). 'Small talk: A missing skill in the chinese communicative repertoire'. *Australian Review of Applied Linguistics*, *38*(1), 3–23.

Data Protection – Volume 167: debated on Monday 19 February 1990 [Accessed: 2026-07-30]. (n.d.).

Davis, L. C., Diianni, A. T., Drumheller, S. R., Elansary, N. N., D'Ambrozio, G. N., Herrawi, F., Piper, B. J., & Cosgrove, L. (2024). 'Undisclosed financial conflicts of interest in DSM-5-TR: cross sectional analysis'. *BMJ*, *384*.

Debreslioska, S., & Gullberg, M. (2020). 'What's New? Gestures Accompany Inferable Rather Than Brand-New Referents in Discourse'. *Frontiers in Psychology*, *Volume 11 - 2020*.

Decker, A., & Amblard, M. (2024a). 'With a Little Help from my (Linguistic) Friends: Topic segmentation of multi-party casual conversations'. In M. Strube, C. Braud, C. Hardmeier, J. J. Li, S. Loaiciga, A. Zeldes, & C. Li (Eds.), *Proceedings of the 5th workshop on computational approaches to discourse (codi 2024)* (pp. 177–188). Association for Computational Linguistics.

Decker, A., & Amblard, M. (2024b). 'Construction d'une mesure de similarité thématique non supervisée pour les conversations'. In M. Balaguer, N. Bendahman, L.-M. Ho-dac, J. Mauclair, J. G Moreno, & J. Pinquier (Eds.), *Actes de la 31ème Conférence sur le Traitement Automatique des Langues*

Naturelles, volume 1 : articles longs et prises de position (pp. 362–375). ATALA; AFPC.

Decker, A., Amblard, M., & Breitholtz, E. (2026a). 'Off the Hamster Wheel: Rethinking Dialogue Research through a Meta-Analysis of the ACL Anthology 2024'. In S. Piperidis, N. Bel, H. van den Heuvel, N. Ide, S. Krek, & A. Toral (Eds.), *Proceedings of the fifteenth language resources and evaluation conference (lrec 2026)* (pp. 2638–2652). European Language Resources Association (ELRA).

Decker, A., Amblard, M., & Breitholtz, E. (2026b). 'Do we need to answer that question? Salience and Answerability of Potential Questions in Naturalistic Dialogue'. *Proceedings of the 30th Workshop on the Semantics and Pragmatics of Dialogue*.

Decker, A., Tourneur, V., Amblard, M., & Breitholtz, E. (2024). '"Wait, did you mean the doctor?": Collecting a Dialogue Corpus for Topical Analysis'. *Proceedings of the 28th Workshop on the Semantics and Pragmatics of Dialogue*.

Delogu, F., Jachmann, T., Staudte, M., Vespignani, F., & Molinaro, N. (2020). 'Discourse Expectations Are Sensitive to the Question Under Discussion: Evidence From ERPs'. *Discourse Processes*, *57*(2), 122–140.

Demberg, V., Sayeed, A., Mahr, A., & Müller, C. (2013). 'Measuring linguistically-induced cognitive load during driving using the contre task'. *Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 176–183.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). 'BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding'. In J. Burstein, C. Doran, & T. Solorio (Eds.), *Proceedings of the 2019 conference of the north American chapter of the association for computational linguistics: Human language technologies, volume 1 (long and short papers)* (pp. 4171–4186). Association for Computational Linguistics.

Dezecache, G., & Dunbar, R. (2012). 'Sharing the Joke: The Size of Natural Laughter Groups'. *Evolution and Human Behavior*, *33*, 775–779.

Dinan, E., Roller, S., Shuster, K., Fan, A., Auli, M., & Weston, J. (2019). 'Wizard of Wikipedia: Knowledge-powered Conversational Agents'. *Proceedings of the International Conference on Learning Representations (ICLR)*.

Dingemanse, M. (2026, July). 'Interactional foundations for critical ai literacies'. In *A research agenda for critical ai*. Edward Elgar Publishhing.

Dingemanse, M., Roberts, S. G., Baranova, J., Blythe, J., Drew, P., Floyd, S., Gisladottir, R. S., Kendrick, K. H., Levinson, S. C., Manrique, E., Rossi, G.,

& Enfield, N. J. (2015). 'Universal Principles in the Repair of Communication Problems'. *PLOS ONE*, *10*(9), 1–15.

Doherty-Sneddon, G., Bruce, V., Bonner, L., Longbotham, S., & Doyle, C. (2002). 'Development of gaze aversion as disengagement from visual information.' *Developmental psychology*, *38*(3), 438.

Doherty-Sneddon, G., & Phelps, F. G. (2005). 'Gaze aversion: A response to cognitive or social difficulty?' *Memory & cognition*, *33*(4), 727–733.

Du, L., Buntine, W., & Johnson, M. (2013). 'Topic Segmentation with a Structured Topic Model'. In L. Vanderwende, H. Daumé III, & K. Kirchhoff (Eds.), *Proceedings of the 2013 conference of the north American chapter of the association for computational linguistics: Human language technologies* (pp. 190–200). Association for Computational Linguistics.

Duan, H., Wei, J., Wang, C., Liu, H., Fang, Y., Zhang, S., Lin, D., & Chen, K. (2024). 'BotChat: Evaluating LLMs' Capabilities of Having Multi-Turn Dialogues'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Findings of the association for computational linguistics: Naacl 2024* (pp. 3184–3200). Association for Computational Linguistics.

Ducel, F., Fort, K., Lejeune, G., & Lepage, Y. (2022). 'Langues par défaut? analyse contrastive et diachronique des langues non citées dans les articles de TALN et d'ACL'. In Y. Estève, T. Jiménez, T. Parcollet, & M. Zanon Boito (Eds.), *Actes de la 29e conférence sur le traitement automatique des langues naturelles. volume 1 : Conférence principale* (pp. 144–153). ATALA.

Ducel, F., Névéol, A., Khazanchi, V., Leclere, L., Pedrini, A., Bouchet, L., Caissial, B., & Fort, K. (2026). 'Code-switching as a Bias Indicator in LLMs: "the Consequences Are Not the Same Para Nosotros"'. In S. Piperidis, N. Bel, H. van den Heuvel, N. Ide, S. Krek, & A. Toral (Eds.), *Proceedings of the fifteenth language resources and evaluation conference (lrec 2026)* (pp. 8387–8399). European Language Resources Association (ELRA).

Dunbar, R. I. M., Duncan, N. D. C., & Nettle, D. (1995). 'Size and Structure of Freely Forming Conversational Groups'. *Human Nature*, *6*(1), 67–78.

Dunbar, R. (1996). *Grooming, Gossip, and the Evolution of Language*. Harvard University Press.

Duncan, S., Brunner, L. J., & Fiske, D. W. (1979). 'Strategy signals in face-to-face interaction.' *Journal of Personality and Social Psychology*, *37*(2), 301.

Durward, M., & Thomson, C. (2024). 'Evaluating Vocabulary Usage in LLMs'. In E. Kochmar, M. Bexte, J. Burstein, A. Horbach, R. Laarmann-Quante, A. Tack, V. Yaneva, & Z. Yuan (Eds.), *Proceedings of the 19th workshop*

on innovative use of nlp for building educational applications (bea 2024) (pp. 266–282). Association for Computational Linguistics.

Earnshaw, Y. (2017). 'Navigating turn-taking and conversational repair in an online synchronous course'. *Online Learning Journal*, *21*(4).

Egbert, M. M. (1997). 'Schisming: The collaborative transformation from a single conversation to multiple conversations'. *Research on Language and Social Interaction*, *30*(1), 1–51.

Eisenstein, J. (2013). 'What to do about bad language on the internet'. In L. Vanderwende, H. Daumé III, & K. Kirchhoff (Eds.), *Proceedings of the 2013 conference of the north American chapter of the association for computational linguistics: Human language technologies* (pp. 359–369). Association for Computational Linguistics.

Eisenstein, J., & Barzilay, R. (2008). 'Bayesian Unsupervised Topic Segmentation'. In M. Lapata & H. T. Ng (Eds.), *Proceedings of the 2008 conference on empirical methods in natural language processing* (pp. 334–343). Association for Computational Linguistics.

Endrass, B., Nakano, Y., Lipi, A. A., Rehm, M., & André, E. (2011). 'Culture-related topic selection in small talk conversations across germany and japan'. *Proceedings of the 10th International Conference on Intelligent Virtual Agents*, 1–13.

Eric, M., Goel, R., Paul, S., Sethi, A., Agarwal, S., Gao, S., Kumar, A., Goyal, A., Ku, P., & Hakkani-Tur, D. (2020). 'MultiWOZ 2.1: A Consolidated Multi-Domain Dialogue Dataset with State Corrections and State Tracking Baselines'. In N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, & S. Piperidis (Eds.), *Proceedings of the twelfth language resources and evaluation conference* (pp. 422–428). European Language Resources Association.

Eric, M., Krishnan, L., Charette, F., & Manning, C. D. (2017). 'Key-Value Retrieval Networks for Task-Oriented Dialogue'. In K. Jokinen, M. Stede, D. DeVault, & A. Louis (Eds.), *Proceedings of the 18th annual SIGdial meeting on discourse and dialogue* (pp. 37–49). Association for Computational Linguistics.

Fan, Y., Jiang, F., Li, P., & Li, H. (2024). 'Uncovering the Potential of ChatGPT for Discourse Analysis in Dialogue: An Empirical Study'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 16998–17010). ELRA; ICCL.

Fan, Y., Li, P., & Zhu, Q. (2024). 'Improving Multi-party Dialogue Generation via Topic and Rhetorical Coherence'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing* (pp. 3240–3253). Association for Computational Linguistics.

Farzindar, A., & Inkpen, D. (2015). *Natural Language Processing for Social Media* (G. Hirst, Ed.). Morgan & Claypool.

Feng, S., Wan, H., Gunasekara, C., Patel, S., Joshi, S., & Lastras, L. (2020). 'doc2dial: A Goal-Oriented Document-Grounded Dialogue Dataset'. In B. Webber, T. Cohn, Y. He, & Y. Liu (Eds.), *Proceedings of the 2020 conference on empirical methods in natural language processing (emnlp)* (pp. 8118–8128). Association for Computational Linguistics.

Feng, X., Feng, X., & Qin, B. (2022). 'A Survey on Dialogue Summarization: Recent Advances and New Frontiers' [Survey Track]. In L. D. Raedt (Ed.), *Proceedings of the thirty-first international joint conference on artificial intelligence, IJCAI-22* (pp. 5453–5460). International Joint Conferences on Artificial Intelligence Organization.

Fernández, R., Frampton, M., Ehlen, P., Purver, M., & Peters, S. (2008). 'Modelling and Detecting Decisions in Multi-party Dialogue'. In D. Schlangen & B. A. Hockey (Eds.), *Proceedings of the 9th SIGdial workshop on discourse and dialogue* (pp. 156–163). Association for Computational Linguistics.

Fillmore, C. J. (1981). 'Pragmatics and the Description of Discourse'. In P. Cole (Ed.), *Radical pragmatics* (pp. 143–166). Academic Press.

Fillmore, C. J., Ide, N., Jurafsky, D., & Macleod, C. (1998). 'An american national corpus: A proposal'. In A. Rubio, N. Gallardo, R. Castro, & A. Tejada (Eds.), *Proceedings of the first international conference on language resources and evaluation (lrec 1998)* (pp. 965–970). European Language Resources Association (ELRA).

Firbas, J. (1964). 'On defining the theme in functional sentence analysis'. *Linguistiques de Prague*, *1*, 267–280.

Fishman, J. A. (1970). 'Sociolinguistics: A brief introduction.'

Fleiss, J. L., Levin, B., & Paik, M. C. (2003). 'The measurement of interrater agreement'. In *Statistical methods for rates and proportions* (pp. 598–626). John Wiley & Sons, Ltd.

Foderaro, A., & Lorentzen, D. G. (2022). 'Argumentative practices and patterns in debating climate change on twitter'. *Aslib Journal of Information Management*, *75*(1), 131–148.

Forchini, P. (2009). 'Spontaneity reloaded: American face-to-face and movie conversation compared'. *Corpus linguistics 2009 proceedings*, 1–27.

Frampton, M., Huang, J., Bui, T., & Peters, S. (2009). 'Real-time decision detection in multi-party dialogue'. In P. Koehn & R. Mihalcea (Eds.), *Proceedings of the 2009 conference on empirical methods in natural language processing* (pp. 1133–1141). Association for Computational Linguistics.

Franceschet, M. (2010). 'The role of conference publications in CS'. *Communications of the ACM*, *53*(12), 129–132.

Frege, G., et al. (1879). 'Begriffsschrift, a formula language, modeled upon that of arithmetic, for pure thought'. *From Frege to Gödel: A source book in mathematical logic*, *1931*, 1–82.

Frenda, S., Abercrombie, G., Basile, V., Pedrani, A., Panizzon, R., Cignarella, A. T., Marco, C., & Bernardi, D. (2025). 'Perspectivist approaches to natural language processing: a survey'. *Language Resources and Evaluation*, *59*(2), 1719–1746.

Gábor, K., Zargayouna, H., Buscaldi, D., Tellier, I., & Charnois, T. (2016). 'Semantic Annotation of the ACL Anthology Corpus for the Automatic Analysis of Scientific Literature'. In N. Calzolari, K. Choukri, T. Declerck, S. Goggi, M. Grobelnik, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, & S. Piperidis (Eds.), *Proceedings of the tenth international conference on language resources and evaluation (LREC'16)* (pp. 3694–3701). European Language Resources Association (ELRA).

Galley, M., McKeown, K. R., Fosler-Lussier, E., & Jing, H. (2003). 'Discourse Segmentation of Multi-Party Conversation'. *Proceedings of the 41st Annual Meeting of the Association for Computational Linguistics*, 562–569.

Gandolfi, G., & Pickering, M. (2023). 'Ideography in interaction' [Funding: This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement (grant no. 859588).]. *Behavioral and Brain Sciences*, *46*.

Gao, T., Fisch, A., & Chen, D. (2021). 'Making pre-trained language models better few-shot learners'. In C. Zong, F. Xia, W. Li, & R. Navigli (Eds.), *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: Long papers)* (pp. 3, 816–3, 830). Association for Computational Linguistics.

Gaur, M., Kursuncu, U., Alambo, A., Sheth, A., Daniulaityte, R., Thirunarayan, K., & Pathak, J. (2018). '"Let me tell you about your mental health!" Contextualized classification of reddit posts to DSM-5 for web-based intervention'.

Proceedings of the 27th ACM International Conference on Information and Knowledge Management, 753–762.

Gehman, S., Gururangan, S., Sap, M., Choi, Y., & Smith, N. A. (2020). 'RealToxicityPrompts: Evaluating neural toxic degeneration in language models'. In T. Cohn, Y. He, & Y. Liu (Eds.), *Findings of the association for computational linguistics: Emnlp 2020* (pp. 3, 356–3, 369). Association for Computational Linguistics.

Gendron, B., Guibon, G., & D'aquin, M. (2025). 'Towards Ontology-Based Descriptions of Conversations with Qualitatively-Defined Concepts' [Proceedings: free to download 6 months after publication (in 2026)]. *Terminologie & Ontologie : Théories et applications*.

Georgescul, M., Clark, A., & Armstrong, S. (2006). 'Word Distributions for Thematic Segmentation in a Support Vector Machine Approach'. In L. Màrquez & D. Klein (Eds.), *Proceedings of the tenth conference on computational natural language learning (CoNLL-x)* (pp. 101–108). Association for Computational Linguistics.

Georgescul, M., Clark, A., & Armstrong, S. (2008). 'A Comparative Study of Mixture Models for Automatic Topic Segmentation of Multiparty Dialogues'. *Proceedings of the Third International Joint Conference on Natural Language Processing: Volume-II*.

Geurts, B. (2024). 'Common Ground in Pragmatics'. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Winter 2024). Metaphysics Research Lab, Stanford University.

Ghinassi, I., Wang, L., Newell, C., & Purver, M. (2024a). 'When cohesion lies in the embedding space: Embedding-based reference-free metrics for topic segmentation'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 17, 525–17, 536). ELRA; ICCL.

Ghinassi, I., Wang, L., Newell, C., & Purver, M. (2024b). 'Recent Trends in Linear Text Segmentation: A Survey'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Findings of the association for computational linguistics: Emnlp 2024* (pp. 3084–3095). Association for Computational Linguistics.

Ginzburg, J. (1995). 'Resolving questions, II'. *Linguistics and Philosophy*, 567–609.

Ginzburg, J. (2012). *The interactive stance: Meaning for conversation*. Oxford University Press.

Ginzburg, J., & Fernández, R. (2005). 'Scaling up from Dialogue to Multilogue: Some Principles and Benchmarks'. In K. Knight, H. T. Ng, & K. Oflazer (Eds.), *Proceedings of the 43rd annual meeting of the association for computational linguistics (ACL'05)* (pp. 231–238). Association for Computational Linguistics.

Ginzburg, J., & Lücking, A. (2021). 'Requesting clarifications with speech and gestures'. In L. Donatelli, N. Krishnaswamy, K. Lai, & J. Pustejovsky (Eds.), *Proceedings of the 1st workshop on multimodal semantic representations (mmsr)* (pp. 21–31). Association for Computational Linguistics.

Ginzburg, J., & Wong, K.-C. (2024). 'Language games and their types'. *Linguistics and Philosophy*, *47*(1), 149–189.

Givón, T. (1983). 'Topic continuity in discourse: An introduction'. *Topic continuity in discourse: A quantitative cross-language study*, *3*, 3–41.

Glenberg, A. M., Schroeder, J. L., & Robertson, D. A. (1998). 'Averting the gaze disengages the environment and facilitates remembering'. *Memory & Cognition*, *26*(4), 651–658.

Goffman, E. (1966). *Behavior in Public: Notes on the Social Organization of Gatherings*. Free Press.

Goffman, E. (1967). *Interaction Ritual: Essays on Face-to-Face Behavior*. Anchor Books.

Goffman, E. (1976). 'Replies and responses'. *Language in Society*, *5*(3), 257–313.

Goodwin, C. (1981, January). *Conversational organization: Interaction between speakers and hearers*.

Goodwin, C. (2000). 'Action and embodiment within situated human interaction'. *Journal of Pragmatics*, *32*(10), 1489–1522.

Grice, H. P. (1975). 'Logic and Conversation'. In P. Cole & J. L. Morgan (Eds.), *Syntax and semantics* (pp. 45–47, Vol. 3). New York: Academic Press.

Grootendorst, M. (2022). Bertopic: Neural topic modeling with a class-based tf-idf procedure.

Gruenstein, A., Niekrasz, J., & Purver, M. (2005). 'Meeting Structure Annotation: Data and Tools'. In L. Dybkjær & W. Minker (Eds.), *Proceedings of the 6th sigdial workshop on discourse and dialogue* (pp. 117–127). Special Interest Group on Discourse; Dialogue (SIGdial).

Gunlogson, C. (2004). *True to form: Rising and falling declaratives as questions in English*. Routledge.

Guo, D., Yang, D., Zhang, H., Song, J., Wang, P., Zhu, Q., Xu, R., Zhang, R., Ma, S., Bi, X., Zhang, X., Yu, X., Wu, Y., Wu, Z. F., Gou, Z., Shao, Z., Li, Z., Gao, Z., Liu, A., … Zhang, Z. (2025). 'DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning'. *Nature*, *645*(8081), 633–638.

Haas, G., Gugenheimer, J., Rixen, J. O., Schaub, F., & Rukzio, E. (2020). '"They Like to Hear My Voice": Exploring Usage Behavior in Speech-Based Mobile Instant Messaging'. *22nd International Conference on Human-Computer Interaction with Mobile Devices and Services*.

Hall, E. T. (1966). 'The hidden dimension'. *Garden City*.

Halliday, M. A. (1967). 'Notes on transitivity and theme in English Part I'. *Journal of linguistics*, *3*(1), 37–81.

Halliday, M. A. K., & Hasan, R. (1976). *Cohesion in english*. Longman: London.

Hamilton, W. L., Ying, R., & Leskovec, J. (2017). 'Inductive representation learning on large graphs'. *Proceedings of the 31st International Conference on Neural Information Processing Systems*, 1025–1035.

Hammersley, M. (2010). 'Reproducing or constructing? Some questions about transcription in social research'. *Qualitative research*, *10*(5), 553–569.

Han, J.-E., Koh, J.-S., Seo, H.-T., Chang, D.-S., & Sohn, K.-A. (2024). 'PSYDIAL: Personality-based synthetic dialogue generation using large language models'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 13, 321–13, 331). ELRA; ICCL.

Hartnett, Y., & Cummings, E. (2024). 'Social media and ADHD: implications for clinical assessment and treatment'. *Irish Journal of Psychological Medicine*, *41*(1), 132–136.

Healey, P. G., & Mills, G. J. (2009). 'A Dialogue Experimentation Toolkit'. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *31*.

Hearst, M. A. (1994). 'Multi-Paragraph Segmentation Expository Text'. *32nd Annual Meeting of the Association for Computational Linguistics*, 9–16.

Hearst, M. A. (1997). 'Text Tiling: Segmenting Text into Multi-paragraph Subtopic Passages'. *Computational Linguistics*, *23*(1), 33–64.

Heritage, J. (1984). 'A Change-of-State Token and Aspects of its Sequential Placement'. *In J. Maxwell Atkinson and John Heritage (eds.), Structures of Social Action*.

Hirschberg, J. (2006). 'Pragmatics and intonation'. *The handbook of pragmatics*, 515–537.

Ho, S., Foulsham, T., & Kingstone, A. (2015). 'Speaking and listening with the eyes: Gaze signaling during dyadic interactions'. *PloS one*, *10*(8), e0136905.

Hockett, C. (1958). *A Course in Modern Linguistics*. Macmillan.

Homer. (1996). *The iliad*. Penguin Classics.

Hough, J., & Purver, M. (2013). 'Modelling Expectation in the Self-Repair Processing of Annotat-, um, Listeners'. *Proceedings of the 17th Workshop on the Semantics and Pragmatics of Dialogue - Full Papers*.

Howe, M. L. (1991). 'Topic changes in conversation' [Doctoral dissertation, University of Kansas] [Unpublished].

Howes, C., Purver, M., Healey, P. G. T., Mills, G. J., & Gregoromichelaki, E. (2011). 'On incrementality in dialogue: Evidence from compound contributions' (D. Schlangen, H. Rieser, & M. W. Crocker, Eds.). *Dialogue & Discourse*, *2*, 279–311.

Hsu, Y.-L., Chen, J.-N., Fan Chiang, Y., Liu, S.-C., Xiong, A., & Ku, L.-W. (2024). 'Enhancing Perception: Refining Explanations of News Claims with LLM Conversations'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Findings of the association for computational linguistics: Naacl 2024* (pp. 2129–2147). Association for Computational Linguistics.

Hsueh, P.-Y., Moore, J. D., & Renals, S. (2006). 'Automatic Segmentation of Multiparty Dialogue'. In D. McCarthy & S. Wintner (Eds.), *11th conference of the European chapter of the association for computational linguistics* (pp. 273–280). Association for Computational Linguistics.

Htun, H. M., Kyaw Thu, Y., Chanlekha, H., Funakoshi, K., & Supnithi, T. (2024). 'myMediCon: End-to-End Burmese Automatic Speech Recognition for Medical Conversations'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 12032–12039). ELRA; ICCL.

Hu, L., Liu, Z., Zhao, Z., Hou, L., Nie, L., & Li, J. (2024). 'A Survey of Knowledge Enhanced Pre-Trained Language Models'. *IEEE Transactions on Knowledge and Data Engineering*, *36*(4), 1413–1430.

Hua, Y., Chernogor, N., Gu, Y., Jeong, S., Luo, M., & Danescu-Niculescu-Mizil, C. (2024). 'How did we get here? Summarizing conversation dynamics'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 7452–7477). Association for Computational Linguistics.

Huang, K.-H., Laban, P., Fabbri, A., Choubey, P. K., Joty, S., Xiong, C., & Wu, C.-S. (2024). 'Embrace Divergence for Richer Insights: A Multi-document Summarization Benchmark and a Case Study on Summarizing Diverse Information from News Articles'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 570–593). Association for Computational Linguistics.

Hunter, J., Asher, N., Kow, E., Perret, J., & Afantenos, S. (2015). 'Defining the right frontier in multi-party dialogue'. *Proceedings of the 19th Workshop on the Semantics and Pragmatics of Dialogue - Full Papers*.

Hunter, J., Asher, N., & Lascarides, A. (2018). 'A formal semantics for situated conversation'. *Semantics and Pragmatics*, *11*(10), 1–59.

Ibrahim, L., Akbulut, C., Elasmar, R., Rastogi, C., Kahng, M., Morris, M., McKee, K., Rieser, V., Shanahan, M., & Weidinger, L. (2026). 'Multi-turn evaluation of anthropomorphic behaviours in large language models'. In C. Vondrick, B. Hariharan, C. Raffel, L. Pinto, D. Yang, & A. Faust (Eds.), *International conference on learning representations* (pp. 156, 130–156, 160, Vol. 2026).

Iftikhar, Z., Xiao, A., Ransom, S., Huang, J., & Suresh, H. (2025). 'How LLM counselors violate ethical standards in mental health practice: A practitioner-informed framework'. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, *8*(2), 1311–1323.

Isaka, T., Otsuka, A., & Toshima, I. (2024). 'Analysis of Sensation-transfer Dialogues in Motorsports'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 876–886). ELRA; ICCL.

Jacqmin, L., Rojas Barahona, L. M., & Favre, B. (2022). '"Do you follow me?": A Survey of Recent Approaches in Dialogue State Tracking'. In O. Lemon, D. Hakkani-Tur, J. J. Li, A. Ashrafzadeh, D. H. Garcia, M. Alikhani, D. Vandyke, & O. Dušek (Eds.), *Proceedings of the 23rd annual meeting of the special interest group on discourse and dialogue* (pp. 336–350). Association for Computational Linguistics.

Jandaghi, P., Sheng, X., Bai, X., Pujara, J., & Sidahmed, H. (2024). 'Faithful persona-based conversational dataset generation with large language models'. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: Acl 2024* (pp. 15, 245–15, 270). Association for Computational Linguistics.

Jang, S., Lee, S., & Yu, H. (2024). 'KoDialogBench: Evaluating Conversational Understanding of Language Models with Korean Dialogue Benchmark'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 9905–9925). ELRA; ICCL.

Janin, A., Baron, D., Edwards, J., Ellis, D., Gelbart, D., Morgan, N., Peskin, B., Pfau, T., Shriberg, E., Stolcke, A., et al. (2003). 'The ICSI meeting corpus'. *2003 IEEE International Conference on Acoustics, Speech, and Signal Processing, 2003. Proceedings.(ICASSP'03).*, *1*, I–I.

Jefferson, G. (1972). 'Side sequences'. In D. N. Sudnow (Ed.), *Studies in social interaction* (pp. 294–338). MacMillan/The Free Press.

Jefferson, G. (1986). 'Notes on 'latency' in overlap onset'. *Human studies*, 153–183.

Jefferson, G. (2008). 'Glossary of transcript symbols with an introduction'. In *Conversation analysis: Studies from the first generation* (pp. 13–31). John Benjamins Publishing Company.

Ji, A., Kojima, N., Rush, N., Suhr, A., Vong, W. K., Hawkins, R., & Artzi, Y. (2022). 'Abstract visual reasoning with tangram shapes'. In Y. Goldberg, Z. Kozareva, & Y. Zhang (Eds.), *Proceedings of the 2022 conference on empirical methods in natural language processing* (pp. 582–601). Association for Computational Linguistics.

Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L. R., Lachaux, M.-A., Stock, P., Scao, T. L., Lavril, T., Wang, T., Lacroix, T., & Sayed, W. E. (2023). Mistral 7B.

Jo, Y., Zhao, X., Biswas, A., Basiou, N., Auvray, V., Malandrakis, N., Metallinou, A., & Potamianos, A. (2023). 'Multi-User MultiWOZ: Task-Oriented Dialogues among Multiple Users'. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Findings of the association for computational linguistics: Emnlp 2023* (pp. 3237–3269). Association for Computational Linguistics.

Johnson, D. C., Liu, M., Rash, K. M., & Slusarek, M. (2025). 'Transcription techniques in qualitative research: choices, hazards and solutions'. *Qualitative Research Journal*, 1–10.

Jokinen, K., Furukawa, H., Nishida, M., & Yamamoto, S. (2013). 'Gaze and turn-taking behavior in casual conversational interactions'. *ACM Trans. Interact. Intell. Syst.*, *3*(2).

Jones, C. R., & Bergen, B. K. (2026). 'Large language models pass a standard three-party Turing test'. *Proceedings of the National Academy of Sciences*, *123*(21), e2524472123.

Joskowicz, J. (2023, December). Video Conferencing Technologies: Past, Present and Future.

Jouitteau, M., & Grobol, L. (2024). 'Petits oublis, grands effets : le silençage des communauté linguistiques minorisées dans le TAL et ses conséquences'. In K. Fort & A. Névéol (Eds.), *Actes de la journée d'étude Journée Ethique et TAL 2024* (pp. 36–38). Atala.

Jovanovic, D., & Leeuwen, T. V. (2018). 'Multimodal dialogue on social media'. *Social Semiotics*, *28*(5), 683–699.

Kaltwasser, C. R., & Taggart, P. (2025). 'Populism and democracy: The road ahead'. *PS: Political Science & Politics*, *58*(1), 96–100.

Kalyuga, S. (2009). 'The expertise reversal effect'. In *Managing cognitive load in adaptive multimedia learning* (pp. 58–80). IGI Global Scientific Publishing.

Kambhampati, S., Valmeekam, K., Bhambri, S., Palod, V., Saldyt, L. P., Stechly, K., Samineni, S. R., Kalwar, D., & Biswas, U. (2026). 'Position: Stop anthropomorphizing intermediate tokens as reasoning/thinking traces!' *Forty-third International Conference on Machine Learning Position Paper Track*.

Kamp, H., & Uwe, R. (1993). *From discourse to logic : introduction to modeltheoretic semantics of natural language, formal logic and discourse representation theory*. Kluwer Academic publishers.

Kann, K., Dudy, S., & McCarthy, A. D. (2022). 'A Major Obstacle for NLP Research: Let's Talk about Time Allocation!' In Y. Goldberg, Z. Kozareva, & Y. Zhang (Eds.), *Proceedings of the 2022 conference on empirical methods in natural language processing* (pp. 8959–8969). Association for Computational Linguistics.

Karanikolas, N., Manga, E., Samaridi, N., Tousidou, E., & Vassilakopoulos, M. (2024). 'Large Language Models versus Natural Language Understanding and Generation'. *Proceedings of the 27th Pan-Hellenic Conference on Progress in Computing and Informatics*, 278–290.

Karttunen, L., Koskenniemi, K., & Kaplan, R. M. (1987, June 25). *A Compiler for Two-level Phonological Rules*. Xerox Palo Alto Research Center, Stanford University, Center for the Study of Language and Information. California.

Kathie Treen, S. O., Hywel Williams, & Coan, T. G. (2022). 'Discussion of Climate Change on Reddit: Polarized Discourse or Deliberative Debate?' *Environmental Communication*, *16*(5), 680–698.

Keenan, E. O., & Schieffelin, B. B. (1976). 'Topic as a discourse notion: A study of topic in the conversations of children and adults'. In *Subject and topic* (pp. 335–384). Academic Press.

Kehler, A., & Rohde, H. (2017). 'Evaluating an Expectation-Driven Question-Under-Discussion Model of Discourse Interpretation'. *Discourse Processes*, *54*(3), 219–238.

Kendon, A. (1967). 'Some functions of gaze-direction in social interaction'. *Acta Psychologica*, *26*, 22–63.

Kendon, A. (2004). *Gesture: Visible action as utterance*. Cambridge University Press.

Kendrick, K. H., Holler, J., & Levinson, S. C. (2023). 'Turn-taking in human face-to-face interaction is multimodal: gaze direction and manual gestures aid the coordination of turn transitions'. *Philosophical transactions of the royal society B*, *378*(1875), 20210473.

Keysar, B., Barr, D. J., Balin, J. A., & Paek, T. S. (1998). 'Definite reference and mutual knowledge: Process models of common ground in comprehension'. *Journal of Memory and Language*, *39*(1), 1–20.

Khan, F. A., & Noble, B. (2022). 'Investigating code-switching and disfluencies in bilingual dialogue'. *Proceedings of the 26th Workshop on the Semantics and Pragmatics of Dialogue - Poster Abstracts*.

Khawaja, M. A., Chen, F., & Marcus, N. (2012). 'Analysis of collaborative communication for linguistic cues of cognitive load'. *Human Factors*, *54*(4), 518–529.

Khebour, I. K., Lai, K., Bradford, M., Zhu, Y., Brutti, R. A., Tam, C., Tu, J., Ibarra, B. A., Blanchard, N., Krishnaswamy, N., & Pustejovsky, J. (2024). 'Common ground tracking in multimodal dialogue'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 3, 587–3, 602). ELRA; ICCL.

Ko, W.-J., Dalton, C., Simmons, M., Fisher, E., Durrett, G., & Li, J. J. (2022). 'Discourse Comprehension: A Question Answering Framework to Represent Sentence Connections'. In Y. Goldberg, Z. Kozareva, & Y. Zhang (Eds.), *Proceedings of the 2022 conference on empirical methods in natural language processing* (pp. 11752–11764). Association for Computational Linguistics.

Krems, J., Dunbar, R., & Neuberg, S. (2016). 'Something to talk about: are conversation sizes constrained by mental modeling abilities?' *Evolution and Human Behavior*, *37*(6), 423–428.

Krems, J. A., & Wilkes, J. (2019). 'Why are conversations limited to about four people? A theoretical exploration of the conversation size constraint'. *Evolution and Human Behavior*, *40*(2), 140–147.

Krippendorff, K. (2011). Computing Krippendorff's alpha-reliability.

Kroeger, P. R. (2019, January). *Analyzing meaning: An introduction to semantics and pragmatics*. Language Science Press.

Kudina, O., & de Boer, B. (2025). 'Large language models, politics, and the functionalization of language'. *AI and Ethics*, *5*(3), 2, 367–2, 379.

Kuhn, D. (1991). *The skills of argument*. Cambridge University Press.

Kusal, S., Patil, S., Choudrie, J., Kotecha, K., Mishra, S., & Abraham, A. (2022). 'AI-Based Conversational Agents: A Scoping Review From Technologies to Future Directions'. *IEEE Access*, *10*, 92337–92356.

Labov, W. (1973). *Sociolinguistic patterns*. University of Pennsylvania press.

Lai, K., Brutti, R., Donatelli, L., & Pustejovsky, J. (2024). 'Encoding gesture in multimodal dialogue: Creating a corpus of multimodal AMR'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 5, 806–5, 818). ELRA; ICCL.

Lai, V. D., Pham, D. N., Steinberg, J., Mikeska, J., & Nguyen, T. H. (2024). 'CAMAL: A Novel Dataset for Multi-label Conversational Argument Move Analysis'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 2673–2682). ELRA; ICCL.

Larsson, S., & Traum, D. R. (2000). 'Information state and dialogue management in the TRINDI dialogue move engine toolkit'. *Nat. Lang. Eng.*, *6*(3–4), 323–340.

Lascarides, A., & Asher, N. (2007). 'Segmented Discourse Representation Theory: Dynamic Semantics With Discourse Structure'. In *Computing meaning* (pp. 87–124). Springer Netherlands.

Lautamatti, L. (1982). 'Coherence in Spoken and Written Discourse'. *Nordic Journal of Linguistics*, *5*(2), 117–127.

Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., & Zittrain, J. L. (2018). 'The science of fake news'. *Science*, *359*(6380), pp. 1094–1096.

Lecomte, A., Buchel, S., Franck, N., Demily, C., Amblard, M., & Musiol, M. (2022). 'Organisations & fonctions du comportement verbal de type « backchannels » dans l'interaction clinique avec la personne souffrant de schizophrénie'. *SHS Web Conf.*, *138*, 01014.

Lee, D. W., Park, H. W., Kim, Y., Breazeal, C., & Morency, L.-P. (2024). 'Global reward to local rewards: Multimodal-guided decomposition for improving dialogue agents'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing* (pp. 15, 737–15, 762). Association for Computational Linguistics.

Lee, H., Gupta, R., Rastogi, A., Cao, Y., Zhang, B., & Wu, Y. (2022). 'SGD-X: A Benchmark for Robust Generalization in Schema-Guided Dialogue Systems'. *Proceedings of the AAAI Conference on Artificial Intelligence*, *36*(10), 10938–10946.

Lee, J., Park, S., Hong, S., Kim, M., Chang, D.-S., & Choi, J. (2024). 'Improving conversational abilities of quantized large language models via direct preference alignment'. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 11, 346–11, 364). Association for Computational Linguistics.

Lee, W., Kim, S., & Lee, G. G. (2024). 'Enhancing Dialogue Speech Recognition with Robust Contextual Awareness via Noise Representation Learning'. In T. Kawahara, V. Demberg, S. Ultes, K. Inoue, S. Mehri, D. Howcroft, & K. Komatani (Eds.), *Proceedings of the 25th annual meeting of the special interest group on discourse and dialogue* (pp. 333–343). Association for Computational Linguistics.

Lee, Y.-J., Lee, D., Youn, J., Oh, K.-J., Ko, B., Hyeon, J., & Choi, H.-J. (2024). 'Stark: Social long-term multi-modal conversation with persona commonsense knowledge'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Findings of the association for computational linguistics: Emnlp 2024* (pp. 12, 137–12, 162). Association for Computational Linguistics.

Lerner, G. H. (2003). 'Selecting next speaker: The context-sensitive operation of a context-free organization'. *Language in society*, *32*(2), 177–201.

Levelt, W. J. (1983). 'Monitoring and self-repair in speech'. *Cognition*, *14*(1), 41–104.

Levinson, S. C. (1979). 'Activity types and language'. *Linguistics*, *17*, 365–399.

Levinson, S. C. (1983). *Pragmatics*. Cambridge University Press.

Lewis, D. (1979). 'Scorekeeping in a Language Game'. *Journal of Philosophical Logic*, *8*(1), 339–359.

Lewis, D. K. (1978). 'Truth in Fiction'. *American Philosophical Quarterly*, *15*(1), 37–46.

Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V., & Zettlemoyer, L. (2020). 'BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension'. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds.), *Proceedings of the 58th annual meeting of the association for computational linguistics* (pp. 7871–7880). Association for Computational Linguistics.

Li, A., Wang, Z., Mendes, E., Le, D. M., Xu, W., & Ritter, A. (2024). 'ChatHF: Collecting Rich Human Feedback from Real-time Conversations'. In D. I. Hernandez Farias, T. Hope, & M. Li (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing: System demonstrations* (pp. 270–279). Association for Computational Linguistics.

Li, H., Zhang, R., Lee, Y.-C., Kraut, R. E., & Mohr, D. C. (2023). 'Systematic review and meta-analysis of AI-based conversational agents for promoting mental health and well-being'. *NPJ Digital Medicine*, *6*(1), 236.

Li, H. (2011). 'A Short Introduction to Learning to Rank'. *IEICE Transactions on Information and Systems*, *E94.D*(10), 1854–1862.

Li, J., Liu, M., Kan, M.-Y., Zheng, Z., Wang, Z., Lei, W., Liu, T., & Qin, B. (2020). 'Molweni: A Challenge Multiparty Dialogues-based Machine Reading Comprehension Dataset with Discourse Structure'. In D. Scott, N. Bel, & C. Zong (Eds.), *Proceedings of the 28th international conference on computational linguistics* (pp. 2642–2652). International Committee on Computational Linguistics.

Li, J., Liu, M., Qin, B., & Liu, T. (2022). 'A survey of discourse parsing'. *Frontiers of Computer Science*, *16*(5), 165329.

Li, J., Galley, M., Brockett, C., Gao, J., & Dolan, B. (2016). 'A Diversity-Promoting Objective Function for Neural Conversation Models'. In K. Knight, A. Nenkova, & O. Rambow (Eds.), *Proceedings of the 2016 conference of the north American chapter of the association for computational linguistics:*

Human language technologies (pp. 110–119). Association for Computational Linguistics.

Li, R., Kahou, S., Schulz, H., Michalski, V., Charlin, L., & Pal, C. (2018). 'Towards deep conversational recommendations'. *Proceedings of the 32nd International Conference on Neural Information Processing Systems*, 9748–9758.

Li, X., Song, L., Jin, L., Mi, H., Ouyang, J., & Yu, D. (2024). 'A Knowledge Plug-and-Play Test Bed for Open-domain Dialogue Generation'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 666–676). ELRA; ICCL.

Li, Y., Su, H., Shen, X., Li, W., Cao, Z., & Niu, S. (2017). 'DailyDialog: A Manually Labelled Multi-turn Dialogue Dataset'. In G. Kondrak & T. Watanabe (Eds.), *Proceedings of the eighth international joint conference on natural language processing (volume 1: Long papers)* (pp. 986–995). Asian Federation of Natural Language Processing.

Li, Y., Hazarika, D., Jin, D., Hirschberg, J., & Liu, Y. (2024). 'From Pixels to Personas: Investigating and Modeling Self-Anthropomorphism in Human-Robot Dialogues'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Findings of the association for computational linguistics: Emnlp 2024* (pp. 9, 695–9, 713). Association for Computational Linguistics.

Liesenfeld, A., Lopez, A., & Dingemanse, M. (2023). 'The timing bottleneck: Why timing and overlap are mission-critical for conversational user interfaces, speech recognition and dialogue systems'. In S. Stoyanchev, S. Joty, D. Schlangen, O. Dusek, C. Kennington, & M. Alikhani (Eds.), *Proceedings of the 24th annual meeting of the special interest group on discourse and dialogue* (pp. 482–495). Association for Computational Linguistics.

Lin, C.-Y. (2004). 'ROUGE: A Package for Automatic Evaluation of Summaries'. *Text Summarization Branches Out*, 74–81.

Lin, Y.-T., & Chen, Y.-N. (2023). 'LLM-Eval: Unified Multi-Dimensional Automatic Evaluation for Open-Domain Conversations with Large Language Models'. In Y.-N. Chen & A. Rastogi (Eds.), *Proceedings of the 5th workshop on nlp for conversational ai (nlp4convai 2023)* (pp. 47–58). Association for Computational Linguistics.

Liu, C., Wang, S., Qing, L., Kuang, K., Kang, Y., Sun, C., & Wu, F. (2024). 'Gold panning in vocabulary: An adaptive method for vocabulary expansion of domain-specific LLMs'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural lan-*

guage processing (pp. 7, 442–7, 459). Association for Computational Linguistics.

Liu, F., Li, Z., Zhou, H., Yin, Q., Yang, J., Tang, X., Luo, C., Zeng, M., Jiang, H., Gao, Y., Nigam, P., Nag, S., Yin, B., Hua, Y., Zhou, X., Rohanian, O., Thakur, A., Clifton, L., & Clifton, D. A. (2024). 'Large Language Models Are Poor Clinical Decision-Makers: A Comprehensive Benchmark'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing* (pp. 13696–13710). Association for Computational Linguistics.

Liu, S., Zheng, C., Demasi, O., Sabour, S., Li, Y., Yu, Z., Jiang, Y., & Huang, M. (2021). 'Towards Emotional Support Dialog Systems'. *ACL*.

Liu, Z., Ping, W., Roy, R., Xu, P., Lee, C., Shoeybi, M., & Catanzaro, B. (2024). ChatQA: Surpassing GPT-4 on Conversational QA and RAG.

Long, L., Wang, R., Xiao, R., Zhao, J., Ding, X., Chen, G., & Wang, H. (2024). 'On LLMs-driven synthetic data generation, curation, and evaluation: A survey'. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: Acl 2024* (pp. 11, 065–11, 082). Association for Computational Linguistics.

Loshchilov, I., & Hutter, F. (2019). Decoupled Weight Decay Regularization.

Lücking, A., Voll, F., Rott, D., Henlein, A., & Mehler, A. (2025). 'Head and Hand Movements During Turn Transitions: Data-Based Multimodal Analysis Using the Frankfurt VR Gesture–Speech Alignment Corpus (FraGA)'. *Proceedings of the 29th Workshop on the Semantics and Pragmatics of Dialogue – Full Papers*, 146–156.

Lukin, S. M., Bonial, C., Marge, M., Hudson, T. A., Hayes, C. J., Pollard, K., Baker, A., Foots, A. N., Artstein, R., Gervits, F., Abrams, M., Henry, C., Donatelli, L., Leuski, A., Hill, S. G., Traum, D., & Voss, C. (2024). 'SCOUT: A situated and multi-modal human-robot dialogue corpus'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 14, 445–14, 458). ELRA; ICCL.

Luo, J., Wang, J., & Zhou, G. (2024). 'TopicDiff: A topic-enriched diffusion approach for multimodal conversational emotion detection'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 16, 304–16, 314). ELRA; ICCL.

Macleod, C., Ide, N., & Grishman, R. (2000). 'The American national corpus: A standardized resource for American English'. In M. Gavrilidou, G. Carayannis, S. Markantonatou, S. Piperidis, & G. Stainhauer (Eds.), *Proceedings of the second international conference on language resources and evaluation (LREC'00)*. European Language Resources Association (ELRA).

Maher, B. A., Manschreck, T. C., Linnet, J., & Candela, S. (2005). 'Quantitative assessment of the frequency of normal associations in the utterances of schizophrenia patients and healthy controls'. *Schizophrenia Research*, *78*(2), 219–224.

Majewska, O., Razumovskaia, E., Ponti, E. M., Vulić, I., & Korhonen, A. (2023). 'Cross-Lingual Dialogue Dataset Creation via Outline-Based Generation'. *Transactions of the Association for Computational Linguistics*, *11*, 139–156.

Malioutov, I., & Barzilay, R. (2006). 'Minimum Cut Model for Spoken Lecture Segmentation'. In N. Calzolari, C. Cardie, & P. Isabelle (Eds.), *Proceedings of the 21st international conference on computational linguistics and 44th annual meeting of the association for computational linguistics* (pp. 25–32). Association for Computational Linguistics.

Mann, W. C., & Thompson, S. A. (1988). 'Rhetorical Structure Theory: Toward a functional theory of text organization'. *Text (The Hague)*, *8*(3), 243–281.

Maraev, V., Ginzburg, J., Larsson, S., Tian, Y., & Bernardy, J.-P. (2018). 'Towards KoS/TTR-based proof-theoretic dialogue management'. *Proceedings of the 22nd Workshop on the Semantics and Pragmatics of Dialogue - Full Papers*.

Maraj, A., & Martin, M. V. (2026). 'Context is the Key for LLM-Based Text Segmentation'. In R. Ichise (Ed.), *Natural language processing and information systems* (pp. 142–155). Springer Nature Switzerland.

Mayor, E., Bietti, L. M., & Bangerter, A. (2025). 'Can Large Language Models Simulate Spoken Human Conversations?' *Cognitive Science*, *49*(9), e70106.

McNeill, D. (1992). *Hand and mind: What gestures reveal about thought*. University of Chicago press.

Mehler, A., & Lücking, A. (2012). 'Pathways of alignment between gesture and speech: Assessing information transmission in multimodal ensembles'. *Proceedings of the International Workshop on Formal and Computational Approaches to Multimodal Communication under the auspices of ESSLLI*, 10.

Miah, M. M. M., Schnaithmann, U., Raghuvanshi, A., & Son, Y. (2024). 'Multimodal contextual dialogue breakdown detection for conversational AI models'. In Y. Yang, A. Davani, A. Sil, & A. Kumar (Eds.), *Proceedings*

of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 6: Industry track) (pp. 303–314). Association for Computational Linguistics.

Mialon, G., Fourrier, C., Swift, C., Wolf, T., LeCun, Y., & Scialom, T. (2023). Gaia: A benchmark for general ai assistants.

Misra, A., & Walker, M. (2013). 'Topic Independent Identification of Agreement and Disagreement in Social Media Dialogue'. In M. Eskenazi, M. Strube, B. Di Eugenio, & J. D. Williams (Eds.), *Proceedings of the SIGDIAL 2013 conference* (pp. 41–50). Association for Computational Linguistics.

Mo, F., Mao, K., Zhao, Z., Qian, H., Chen, H., Cheng, Y., Li, X., Zhu, Y., Dou, Z., & Nie, J.-Y. (2025). 'A survey of conversational search'. *ACM Transactions on Information Systems*, *43*(6), 1–50.

Mondada, L. (2018). 'Multiple temporalities of language and body in interaction: Challenges for transcribing multimodality'. *Research on language and social interaction*, *51*(1), 85–106.

Morris, J. (1988). 'Lexical cohesion, the thesaurus, and the structure of text' [Doctoral dissertation, Computer Systems Research Institute, University of Toronto].

Mozziconacci, S. (2002). 'Prosody and emotions'. *Speech Prosody 2002*, 1–9.

Mulligan, K., & Rawlins, K. (2025). 'Analyzing naturally-sourced questions under discussion'. *Experiments in Linguistic Meaning*, *3*, 250–261.

Namuduri, R., Wu, Y., Zheng, A. A., Wadhwa, M., Durrett, G., & Li, J. J. (2025). 'QUDsim: Quantifying Discourse Similarities in LLM-Generated Text'. *Second Conference on Language Modeling*.

Naseem, U., Dunn, A. G., Kim, J., & Khushi, M. (2022). 'Early identification of depression severity levels on reddit using ordinal classification'. *Proceedings of the ACM Web Conference 2022*, 2563–2572.

Nguyen, V.-A., Boyd-Graber, J., & Resnik, P. (2012). 'SITS: A Hierarchical Nonparametric Model using Speaker Identity for Topic Segmentation in Multiparty Conversations'. In H. Li, C.-Y. Lin, M. Osborne, G. G. Lee, & J. C. Park (Eds.), *Proceedings of the 50th annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 78–87). Association for Computational Linguistics.

Norrick, N. R. (2009). 'Interjections as pragmatic markers'. *Journal of pragmatics*, *41*(5), 866–891.

Norris, P., & Inglehart, R. (2019). *Cultural Backlash: Trump, Brexit, and Authoritarian Populism*. Cambridge University Press.

Novák, M., Konopik, M., Nedoluzhko, A., Popel, M., Prazak, O., Sido, J., Straka, M., Žabokrtský, Z., & Zeman, D. (2025). 'Findings of the Fourth Shared Task on Multilingual Coreference Resolution: Can LLMs Dethrone Traditional Approaches?' In M. Ogrodniczuk, M. Novak, M. Poesio, S. Pradhan, & V. Ng (Eds.), *Proceedings of the eighth workshop on computational models of reference, anaphora and coreference* (pp. 95–118). Association for Computational Linguistics.

Occhipinti, D., Tekiroğlu, S. S., & Guerini, M. (2024). 'PRODIGy: a PROfile-based DIalogue Generation dataset'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Findings of the association for computational linguistics: Naacl 2024* (pp. 3500–3514). Association for Computational Linguistics.

Oldenburg, R. (1989). *The Great Good Place*. Paragon House.

Omodei, E., Guo, Y., Cointet, J.-P., & Poibeau, T. (2014). 'Argumentative analysis of the ACL Anthology (Analyse argumentative du corpus de l'ACL (ACL Anthology)) [in French]'. In P. Blache, F. Béchet, & B. Bigi (Eds.), *Proceedings of taln 2014 (volume 2: Short papers)* (pp. 580–585). Association pour le Traitement Automatique des Langues.

Onea, E. (2016). *Potential Questions at the Semantics-Pragmatics Interface*. Brill.

OpenAI. (2022). Chatgpt.

Ou, J., Lu, J., Liu, C., Tang, Y., Zhang, F., Zhang, D., & Gai, K. (2024). 'DialogBench: Evaluating LLMs as Human-like Dialogue Systems'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 6137–6170). Association for Computational Linguistics.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. (2022). 'Training language models to follow instructions with human feedback'. *Advances in neural information processing systems*, *35*, 27730–27744.

Oviatt, S. (1997). 'Mulitmodal Interactive Maps: Designing for Human Performance'. *Human–Computer Interaction*, *12*(1-2), 93–129.

Oviatt, S. (1999). 'Ten myths of multimodal interaction'. *Commun. ACM*, *42*(11), 74–81.

Papineni, K., Roukos, S., Ward, T., & Zhu, W.-J. (2002). 'Bleu: a Method for Automatic Evaluation of Machine Translation'. In P. Isabelle, E. Charniak, & D. Lin (Eds.), *Proceedings of the 40th annual meeting of the association for computational linguistics* (pp. 311–318). Association for Computational Linguistics.

Parrill, F., Hinnell, J., Moran, G., Boylan, H., Gupta, I., & Zamir, A. (2022). 'Observers use gesture to disambiguate contrastive expressions of preference'. *Language and Cognition*, *14*(1), 19–46.

Paulus, T., Warren, A., & Lester, J. N. (2016). 'Applying conversation analysis methods to online talk: A literature review'. *Discourse, Context & Media*, *12*, 1–10.

Peechatt, M., Alm, C. O., & Bailey, R. (2024). 'MULTICOLLAB: A multimodal corpus of dialogues for analyzing collaboration and frustration in language'. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds.), *Proceedings of the 2024 joint international conference on computational linguistics, language resources and evaluation (lrec-coling 2024)* (pp. 11, 713–11, 722). ELRA; ICCL.

Peirce, C. S. (1934). *Collected papers of charles sanders peirce* (Vol. 5). Harvard University Press.

Pereira, P., Moniz, H., & Carvalho, J. P. (2024). 'Deep emotion recognition in textual conversations: A survey'. *Artificial Intelligence Review*, *58*(1), 10.

Pérez-Rosas, V., & Mihalcea, R. (2015). 'Experiments in open domain deception detection'. In L. Màrquez, C. Callison-Burch, & J. Su (Eds.), *Proceedings of the 2015 conference on empirical methods in natural language processing* (pp. 1, 120–1, 125). Association for Computational Linguistics.

Pevzner, L., & Hearst, M. A. (2002). 'A Critique and Improvement of an Evaluation Metric for Text Segmentation'. *Computational Linguistics*, *28*(1), 19–36.

Pickering, M. J., & Garrod, S. (2004). 'Toward a mechanistic psychology of dialogue'. *Behavioral and Brain Sciences*, *27*(2), 169–190.

Pilan, I., Prévot, L., Buschmeier, H., & Lison, P. (2024). 'Conversational Feedback in Scripted versus Spontaneous Dialogues: A Comparative Analysis'. In T. Kawahara, V. Demberg, S. Ultes, K. Inoue, S. Mehri, D. Howcroft, & K. Komatani (Eds.), *Proceedings of the 25th annual meeting of the special interest group on discourse and dialogue* (pp. 440–457). Association for Computational Linguistics.

Polguère, A. (1990). 'Structuration et mise en jeu procédurale d'un modèle linguistique déclaratif dans un cadre de génération de texte' [Doctoral dissertation, Université de Montréal].

Poria, S., Hazarika, D., Majumder, N., Naik, G., Cambria, E., & Mihalcea, R. (2019). 'MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations'. In A. Korhonen, D. Traum, & L. Màrquez (Eds.), *Proceedings of the 57th annual meeting of the association for computa-*

tional linguistics (pp. 527–536). Association for Computational Linguistics.

Pramanick, A., Hou, Y., Mohammad, S., & Gurevych, I. (2023). 'A Diachronic Analysis of Paradigm Shifts in NLP Research: When, How, and Why?' In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing* (pp. 2312–2326). Association for Computational Linguistics.

Purver, M. (2011). 'Topic Segmentation'. In *Spoken language understanding* (pp. 291–317). John Wiley & Sons, Ltd.

Purver, M., Ginzburg, J., & Healey, P. (2001). 'On the means for clarification in dialogue'. *Proceedings of the Second SIGdial Workshop on Discourse and Dialogue*.

Purver, M., Körding, K. P., Griffiths, T. L., & Tenenbaum, J. B. (2006). 'Unsupervised Topic Modelling for Multi-Party Spoken Discourse'. In N. Calzolari, C. Cardie, & P. Isabelle (Eds.), *Proceedings of the 21st international conference on computational linguistics and 44th annual meeting of the association for computational linguistics* (pp. 17–24). Association for Computational Linguistics.

Qi, W., Pan, J., Lyu, H., & Luo, J. (2024). 'Excitements and concerns in the post-ChatGPT era: Deciphering public perception of AI through social media analysis'. *Telematics and Informatics*, *92*, 102158.

Quaranto, A. (2022). 'Dog whistles, covertly coded speech, and the practices that enable them'. *Synthese*, *200*(4), 330.

Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J. (2020). 'Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer'. *J. Mach. Learn. Res.*, *21*(1).

Raji, D., Denton, E., Bender, E. M., Hanna, A., & Paullada, A. (2021). 'Ai and the everything in the whole wide world benchmark'. In J. Vanschoren & S. Yeung (Eds.), *Proceedings of the neural information processing systems track on datasets and benchmarks* (Vol. 1).

Ramprasad, S., Ferracane, E., & Lipton, Z. (2024). 'Analyzing LLM Behavior in Dialogue Summarization: Unveiling Circumstantial Hallucination Trends'. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 12549–12561). Association for Computational Linguistics.

Ranisch, R., & Haltaufderheide, J. (2025). 'Rapid integration of llms in healthcare raises ethical concerns: An investigation into deceptive patterns in social robots'. *Digital Society*, *4*(1), 7.

Rastogi, A., Zang, X., Sunkara, S., Gupta, R., & Khaitan, P. (2020). 'Towards Scalable Multi-Domain Conversational Agents: The Schema-Guided Dialogue Dataset'. *Proceedings of the AAAI Conference on Artificial Intelligence*, *34*(05), 8689–8696.

Razavi, A., Soltangheis, M., Arabzadeh, N., Salamat, S., Zihayat, M., & Bagheri, E. (2025). 'Benchmarking Prompt Sensitivity in Large Language Models'. In C. Hauff, C. Macdonald, D. Jannach, G. Kazai, F. M. Nardini, F. Pinelli, F. Silvestri, & N. Tonellotto (Eds.), *Advances in information retrieval* (pp. 303–313). Springer Nature Switzerland.

Rebedea, T., Sreedhar, M., Ghosh, S., Zeng, J., & Parisien, C. (2024). 'CantTalkAboutThis: Aligning Language Models to Stay on Topic in Dialogues'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Findings of the association for computational linguistics: Emnlp 2024* (pp. 12232–12252). Association for Computational Linguistics.

Rebuschi, M., Amblard, M., & Musiol, M. (2014). 'Using SDRT to analyze pathological conversations. Logicality, rationality and pragmatic deviances'. In M. Rebuschi, M. Batt, G. Heinzmann, F. Lihoreau, M. Musiol, & A. Trognon (Eds.), *Interdisciplinary Works in Logic, Epistemology, Psychology and Linguistics: Dialogue, Rationality, and Formalism* (pp. 343–368, Vol. 3). Springer.

Reece, A., Cooney, G., Bull, P., Chung, C., Dawson, B., Fitzpatrick, C., Glazer, T., Knox, D., Liebscher, A., & Marin, S. (2023). 'The CANDOR corpus: Insights from a large multimodal dataset of naturalistic conversation'. *Science Advances*, *9*(13), eadf3197.

Reimers, N., & Gurevych, I. (2019). 'Sentence-BERT: Sentence embeddings using Siamese BERT-networks'. In K. Inui, J. Jiang, V. Ng, & X. Wan (Eds.), *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (emnlp-ijcnlp)* (pp. 3, 982–3, 992). Association for Computational Linguistics.

Reynar, J. C. (1994). 'An Automatic Method of Finding Topic Boundaries'. *32nd Annual Meeting of the Association for Computational Linguistics*, 331–333.

Riedl, M., & Biemann, C. (2012). 'TopicTiling: A text segmentation algorithm based on LDA'. In J. C. K. Cheung, J. Hatori, C. Henriquez, & A. Irvine

(Eds.), *Proceedings of ACL 2012 student research workshop* (pp. 37–42). Association for Computational Linguistics.

Roberts, C. (2012). 'Information structure in discourse: Towards an integrated formal theory of pragmatics'. *Semantics and Pragmatics*, *5*(6), 1–69.

Rogers, A., Kovaleva, O., & Rumshisky, A. (2020). 'A Primer in BERTology: What We Know About How BERT Works' (M. Johnson, B. Roark, & A. Nenkova, Eds.). *Transactions of the Association for Computational Linguistics*, *8*, 842–866.

Rossano, F. (2012). 'Gaze behavior in face-to-face interaction' [Doctoral dissertation, Radboud University Nijmegen Nijmegen].

Sacks, H. (1995). *Lectures on conversation, Volumes I and II* [Edited by Gail Jefferson and Emanuel A. Schegloff]. Blackwell.

Sacks, H., Schegloff, E. A., & Jefferson, G. (1974). 'A Simplest Systematics for the Organization of Turn-Taking for Conversation'. *Language*, *50*(4), 696–735.

Sai, A. B., Mohankumar, A. K., & Khapra, M. M. (2022). 'A Survey of Evaluation Metrics Used for NLG Systems'. *ACM Comput. Surv.*, *55*(2).

Sandler, W. (2010). 'Prosody and syntax in sign languages'. *Transactions of the Philological Society*, *108*(3), 298–328.

Sarkar, R., Srikanth, N., Gupta, S., Bonial, C., Resnik, P., & Rudinger, R. (2026). Sycophancy Undermines Epistemic Vigilance in Cooperative Vision-Language Tasks.

Sarkar, S., Gaur, M., Chen, L. K., Garg, M., & Srivastava, B. (2023). 'A review of the explainability and safety of conversational agents for mental health to identify avenues for improvement'. *Frontiers in Artificial Intelligence*, *Volume 6 - 2023*.

Sato, S., Akama, R., Suzuki, J., & Inui, K. (2024). 'A Large Collection of Model-generated Contradictory Responses for Consistency-aware Dialogue Systems'. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: Acl 2024* (pp. 16047–16062). Association for Computational Linguistics.

Saul, J. (2018). 'Dogwhistles, political manipulation, and philosophy of language'. *New work on speech acts*, *360*, 84.

Schäfer, M. S., & Metag, J. (2026). 'Social media and the changing landscape of climate change communication'. *Nature climate change*, *16*(8), 897–906.

Schegloff, E. A. (1979). 'The Relevance of Repair to Syntax-for-Conversation'. In *Discourse and syntax* (pp. 261–286). Brill.

Schegloff, E. A. (2007). *Sequence Organization in Interaction: A Primer in Conversation Analysis* (Vol. 1). Cambridge University Press.

Schegloff, E. A., Jefferson, G., & Sacks, H. (1977). 'The preference for self-correction in the organization of repair in conversation'. *Language*, *53*(2), 361–382.

Schegloff, E. A., & Sacks, H. (1973). 'Opening up Closings'. *8*(4), 289–327.

Schiffrin, D. (1985). 'Conversational coherence: The role of well'. *language*, 640–667.

Schiffrin, D. (1987). *Discourse markers*. Cambridge University Press.

Schlangen, D. (2025). 'Meaningful Interaction with Unreal Speakers?' *Proceedings of the 29th Workshop on the Semantics and Pragmatics of Dialogue – Invited Talks*, 4–4.

Schnedecker, C. (1997). *Nom propre et chaînes de référence*. Librairie Klincksieck.

Schneider, K. (1988). *Small Talk: Analyzing Phatic Discourse*. Hitzeroth.

Searle, J. R. (1969). *Speech Acts: An Essay in the Philosophy of Language*. Cambridge University Press.

Shtekh, G., Kazakova, P., Nikitinsky, N., & Skachkov, N. (2018). 'Applying topic segmentation to document-level information retrieval'. *Proceedings of the 14th Central and Eastern European Software Engineering Conference Russia*, 1–6.

Silverman, K. (2022). 'The androids of hephaestus: Between human and machine in the iliad'. *Selected Proceedings of the Classics Graduate Student Symposia at the University of Florida*, *1*.

Sinclair, A. J., & Fernández, R. (2023). 'Alignment of code switching varies with proficiency in second language learning dialogue'. *System*, *113*, 102952.

Skjuve, M., Brandtzaeg, P. B., & Følstad, A. (2024). 'Why do people use chatgpt? exploring user motivations for generative conversational ai'. *First Monday*, *29*(1).

Soudani, H., Petcu, R., Kanoulas, E., & Hasibi, F. (2026). 'A survey on recent advances in conversational data generation'. *ACM Comput. Surv.*, *58*(10).

Sperber, D., & Wilson, D. (1986). *Relevance : Communication and Cognition / Dan Sperber and Deirdre Wilson*. Basil Blackwell.

Stans, S. E., Dalemans, R. J., de Witte, L. P., Smeets, H. W., & Beurskens, A. J. (2017). 'The role of the physical environment in conversations between people who are communication vulnerable and health-care professionals: a scoping review'. *Disability and rehabilitation*, *39*(25), 2594–2605.

Stivers, T., & Rossano, F. (2010). 'Mobilizing response'. *Research on Language and social interaction*, *43*(1), 3–31.

Sun, K., Zhu, S., Chen, L., Yao, S., Wu, X., & Yu, K. (2016). 'Hybrid Dialogue State Tracking for Real World Human-to-Human Dialogues'. *Interspeech 2016*, 2060–2064.

Sun, Y., Zheng, Y., Hao, C., & Qiu, H. (2022). 'NSP-BERT: A prompt-based few-shot learner through an original pre-training task —— next sentence prediction'. *Proceedings of the 29th International Conference on Computational Linguistics*, 3, 233–3, 250.

Surian, L. (1996). 'Are Children with Autism Deaf to Gricean Maxims?' [PMID: 16571474]. *Cognitive Neuropsychiatry*, *1*(1), 55–72.

Switzky, L. (2020). 'ELIZA Effects: Pygmalion and the Early Development of Artificial Intelligence'. *Shaw*, *40*(1), 50–68.

Takanobu, R., Huang, M., Zhao, Z., Li, F., Chen, H., Zhu, X., & Nie, L. (2018). 'A Weakly Supervised Method for Topic Segmentation and Labeling in Goal-oriented Dialogues via Reinforcement Learning'. *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI-18*, 4403–4410.

Tang, L., Shalyminov, I., Wong, A., Burnsky, J., Vincent, J., Yang, Y., Singh, S., Feng, S., Song, H., Su, H., Sun, L., Zhang, Y., Mansour, S., & McKeown, K. (2024). 'TofuEval: Evaluating Hallucinations of LLMs on Topic-Focused Dialogue Summarization'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 4455–4480). Association for Computational Linguistics.

Team, G., Mesnard, T., Hardin, C., Dadashi, R., Bhupatiraju, S., Pathak, S., Sifre, L., Rivière, M., Kale, M. S., Love, J., Tafti, P., Hussenot, L., Sessa, P. G., Chowdhery, A., Roberts, A., Barua, A., Botev, A., Castro-Ros, A., Slone, A., … Kenealy, K. (2024). Gemma: Open Models Based on Gemini Research and Technology.

Thompson, H. S., Anderson, A., Bard, E. G., Doherty-Sneddon, G., Newlands, A., & Sotillo, C. (1993). 'The HCRC map task corpus: Natural dialogue for speech recognition'. *Human Language Technology: Proceedings of a Workshop Held at Plainsboro, New Jersey, March 21-24, 1993*.

Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A.,

Grave, E., & Lample, G. (2023). LLaMA: Open and Efficient Foundation Language Models.

Traum, D. (2003). 'Issues in multiparty dialogues'. *Workshop on Agent Communication Languages*, 201–211.

Trujillo, J. P., & Holler, J. (2024). 'Conversational facial signals combine into compositional meanings that change the interpretation of speaker intentions'. *Scientific Reports*, *14*(1), 2286.

Tu, L., Qu, J., Yavuz, S., Joty, S., Liu, W., Xiong, C., & Zhou, Y. (2024). 'Efficiently Aligned Cross-Lingual Transfer Learning for Conversational Tasks using Prompt-Tuning'. In Y. Graham & M. Purver (Eds.), *Findings of the association for computational linguistics: Eacl 2024* (pp. 1278–1294). Association for Computational Linguistics.

Turcan, E., & McKeown, K. (2019). 'Dreaddit: A Reddit Dataset for Stress Analysis in Social Media'. In E. Holderness, A. Jimeno Yepes, A. Lavelli, A.-L. Minard, J. Pustejovsky, & F. Rinaldi (Eds.), *Proceedings of the tenth international workshop on health text mining and information analysis (louhi 2019)* (pp. 97–107). Association for Computational Linguistics.

Turing, A. M. (1950). 'Computing Machinery and Intelligence'. *Mind*, *LIX*(236), 433–460.

Turkle, S. (1995). *Life on the screen: identity in the age of the Internet*. Simon & Schuster.

Turkle, S. (2005). *The second self: Computers and the human spirit*. The MIT Press.

Turkle, S. (2007). 'Authenticity in the age of digital companions'. *Interaction studies*, *8*(3), 501–517.

Van Arkel, J., Woensdregt, M., Dingemanse, M., & Blokpoel, M. (2020). 'A simple repair mechanism can alleviate computational demands of pragmatic reasoning: simulations and complexity analysis'. In R. Fernández & T. Linzen (Eds.), *Proceedings of the 24th conference on computational natural language learning* (pp. 177–194). Association for Computational Linguistics.

Van Dijk, D., Manor, O., & Carey, L. B. (2014). 'Publication metrics and success on the academic job market'. *Current Biology*, *24*(11), R516–R517.

Van Dijk, T. A. (1977). 'Sentence topic and discourse topic'. *Papers in Slavic philology*, *1*(1977), 49–61.

Van Kuppevelt, J. (1995). 'Discourse structure, topicality and questioning'. *Journal of linguistics*, *31*(1), 109–147.

Vanzan, V., Han Qiu, A., Ayub Khan, F., Soupiona, C., & Howes, C. (2024). 'Emoji-text mismatches: Stirring the pot of online conversations'. *Proceedings of*

the 28th Workshop on the Semantics and Pragmatics of Dialogue - Poster Abstracts.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). 'Attention is all you need'. *Advances in neural information processing systems*, *30*.

Vertegaal, R., Slagter, R., Van der Veer, G., & Nijholt, A. (2001). 'Eye gaze patterns in conversations: there is more to conversational agents than meets the eyes'. *Proceedings of the SIGCHI conference on Human factors in computing systems*, 301–308.

Vijayan, V., Thomas, T., & Nellanat, P. D. (2025). 'Mapping fake news and misinformation in media: A two-decade bibliometric review of emerging trends'. *Insight-News Media*, *8*(1), 734–734.

Villing, J. (2015, October). 'Towards Dialogue Strategies for Cognitive Workload Management' [Doctoral dissertation, University of Gothenburg. Faculty of Arts].

Visser, J., Konat, B., Duthie, R., Koszowy, M., Budzynska, K., & Reed, C. (2019). 'Argumentation in the 2016 us presidential elections: Annotated corpora of television debates and social media reaction'. *Language Resources and Evaluation*, *54*, 123–154.

Vogel, A., & Jurafsky, D. (2012). 'He Said, She Said: Gender in the ACL Anthology'. In R. E. Banchs (Ed.), *Proceedings of the ACL-2012 special workshop on rediscovering 50 years of discoveries* (pp. 33–41). Association for Computational Linguistics.

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes* (Vol. 86). Harvard university press.

Wang, A., Pruksachatkun, Y., Nangia, N., Singh, A., Michael, J., Hill, F., Levy, O., & Bowman, S. (2019). 'Superglue: A stickier benchmark for general-purpose language understanding systems'. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, & R. Garnett (Eds.), *Advances in neural information processing systems* (Vol. 32). Curran Associates, Inc.

Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., & Bowman, S. R. (2018). 'GLUE: A multi-task benchmark and analysis platform for natural language understanding'. In T. Linzen, G. Chrupała, & A. Alishahi (Eds.), *Proceedings of the 2018 EMNLP workshop BlackboxNLP: Analyzing and interpreting neural networks for NLP* (pp. 353–355). Association for Computational Linguistics.

Wang, F., Ma, H., Xia, R., Yu, J., & Cambria, E. (2024). 'SemEval-2024 task 3: Multimodal emotion cause analysis in conversations'. In A. K. Ojha, A. S.

Doğruöz, H. Tayyar Madabushi, G. Da San Martino, S. Rosenthal, & A. Rosá (Eds.), *Proceedings of the 18th international workshop on semantic evaluation (semeval-2024)* (pp. 2, 039–2, 050). Association for Computational Linguistics.

Wang, J., Yao, Z., Yang, Z., Zhou, H., Li, R., Wang, X., Xu, Y., & Yu, H. (2024). 'NoteChat: A Dataset of Synthetic Patient-Physician Conversations Conditioned on Clinical Notes'. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: Acl 2024* (pp. 15, 183–15, 201). Association for Computational Linguistics.

Wang, L., Li, S., Lv, Y., & Wang, H. (2017). 'Learning to Rank Semantic Coherence for Topic Segmentation'. In M. Palmer, R. Hwa, & S. Riedel (Eds.), *Proceedings of the 2017 conference on empirical methods in natural language processing* (pp. 1340–1344). Association for Computational Linguistics.

Wang, W., Hoi, S. C., & Joty, S. (2020). 'Response Selection for Multi-Party Conversations with Dynamic Topic Tracking'. In B. Webber, T. Cohn, Y. He, & Y. Liu (Eds.), *Proceedings of the 2020 conference on empirical methods in natural language processing (emnlp)* (pp. 6581–6591). Association for Computational Linguistics.

Wang, X., Wang, Y., Zhang, Y., Luo, F., Li, P., Sun, M., & Liu, Y. (2024). 'Reasoning in Conversation: Solving Subjective Tasks through Dialogue Simulation for Large Language Models'. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 15, 880–15, 893). Association for Computational Linguistics.

Wang, X., Li, C., Zhao, J., & Yu, D. (2021). 'NaturalConv: A Chinese Dialogue Dataset Towards Multi-turn Topic-driven Conversation'. *Proceedings of the AAAI Conference on Artificial Intelligence*, *35*(16), 14, 006–14, 014.

Ward, N. (2006). 'Non-lexical conversational sounds in American English'. *Pragmatics & Cognition*, *14*(1), 129–182.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., ichter brian, b., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). 'Chain-of-Thought Prompting Elicits Reasoning in Large Language Models'. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, & A. Oh (Eds.), *Advances in neural information processing systems* (pp. 24, 824–24, 837, Vol. 35). Curran Associates, Inc.

Weir, N., Thomas, R., d'Amore, R., Hill, K., Van Durme, B., & Jhamtani, H. (2024). 'Ontologically Faithful Generation of Non-Player Character Dialogues'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024*

conference on empirical methods in natural language processing (pp. 9, 212–9, 242). Association for Computational Linguistics.

Weizenbaum, J. (1966). 'Eliza—a computer program for the study of natural language communication between man and machine'. *Communications of the ACM*, *9*(1), 36–45.

Westera, M., Mayol, L., & Rohde, H. (2020). 'TED-Q: TED Talks and the Questions they Evoke'. In N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, & S. Piperidis (Eds.), *Proceedings of the twelfth language resources and evaluation conference* (pp. 1, 118–1, 127). European Language Resources Association.

Westera, M., & Rohde, H. (2019). 'Asking between the lines: Elicitation of evoked questions in text'. In
bibinitperiod Schlöder, D. McHugh, & F. Roelofsen (Eds.), *Proceedings of the 22nd amsterdam colloquium* (pp. 397–406).

Williams, A., Nangia, N., & Bowman, S. R. (2018). 'A broad-coverage challenge corpus for sentence understanding through inference'. In M. Walker, H. Ji, & A. Stent (Eds.), *Proceedings of the 2018 conference of the north American chapter of the association for computational linguistics: Human language technologies, volume 1 (long papers)* (pp. 1, 112–1, 122). Association for Computational Linguistics.

Williams, J. M., & Nadel, I. B. (1989). *Style: Ten lessons in clarity and grace*. Scott, Foresman Glenview, IL.

Williams, M. L., Burnap, P., Sloan, L., Jessop, C., & Lepps, H. (2017). 'Users' views of ethics in social media research: Informed consent, anonymity, and harm'. In K. Woodfield (Ed.), *The ethics of online research*. Emerald Publishing Limited.

Wittgenstein, L. (1953). *Philosophical Investigations* (G. E. M. Anscombe, Ed.). Wiley-Blackwell.

Wong, K.-C., & Ginzburg, J. (2018). 'Conversational types: a topological perspective'. *Proceedings of the 22nd Workshop on the Semantics and Pragmatics of Dialogue - Full Papers*.

Wooffitt, R. (2005, January). *Conversation analysis and discourse analysis: A comparative and critical introduction*. SAGE Publications Ltd.

Wu, C.-S., Hoi, S. C., Socher, R., & Xiong, C. (2020). 'TOD-BERT: Pre-trained Natural Language Understanding for Task-Oriented Dialogue'. In B. Webber, T. Cohn, Y. He, & Y. Liu (Eds.), *Proceedings of the 2020 conference*

on empirical methods in natural language processing (emnlp) (pp. 917–929). Association for Computational Linguistics.

Wu, Y., Mangla, R. R., Dimakis, A., Durrett, G., & Li, J. J. (2024). 'Which questions should I answer? Salience Prediction of Inquisitive Questions'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing* (pp. 19, 969–19, 987). Association for Computational Linguistics.

Wu, Z., Du, M., Ng, S.-K., & He, B. (2026). 'Beyond prompt-induced lies: Investigating LLM deception on benign prompts'. *The Fourteenth International Conference on Learning Representations*.

Xiao, Y., Ng, L. H. X., Liu, J., & Diab, M. T. (2025). 'Humanizing Machines: Rethinking LLM Anthropomorphism Through a Multi-Level Framework of Design'. In C. Christodoulopoulos, T. Chakraborty, C. Rose, & V. Peng (Eds.), *Proceedings of the 2025 conference on empirical methods in natural language processing* (pp. 3, 331–3, 350). Association for Computational Linguistics.

Xie, H., Liu, Z., Xiong, C., Liu, Z., & Copestake, A. (2021). 'TIAGE: A Benchmark for Topic-Shift Aware Dialog Modeling'. In M.-F. Moens, X. Huang, L. Specia, & S. W.-t. Yih (Eds.), *Findings of the association for computational linguistics: Emnlp 2021* (pp. 1, 684–1, 690). Association for Computational Linguistics.

Xing, L., & Carenini, G. (2021). 'Improving Unsupervised Dialogue Topic Segmentation with Utterance-Pair Coherence Scoring'. In H. Li, G.-A. Levow, Z. Yu, C. Gupta, B. Sisman, S. Cai, D. Vandyke, N. Dethlefs, Y. Wu, & J. J. Li (Eds.), *Proceedings of the 22nd annual meeting of the special interest group on discourse and dialogue* (pp. 167–177). Association for Computational Linguistics.

Xu, J., Lei, Z., Wang, H., Niu, Z.-Y., Wu, H., & Che, W. (2021). 'Discovering Dialog Structure Graph for Coherent Dialog Generation'. In C. Zong, F. Xia, W. Li, & R. Navigli (Eds.), *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: Long papers)* (pp. 1, 726–1, 739). Association for Computational Linguistics.

Xu, P., Saghir, H., Kang, J. S., Long, T., Bose, A. J., Cao, Y., & Cheung, J. C. K. (2019). 'A Cross-Domain Transferable Neural Coherence Model'. In A. Korhonen, D. Traum, & L. Màrquez (Eds.), *Proceedings of the 57th annual meeting of the association for computational linguistics* (pp. 678–687). Association for Computational Linguistics.

Xu, Y. (2019). 'Prosody, tone, and intonation'. In *The routledge handbook of phonetics* (pp. 314–356). Routledge.

Xu, Y., Zhao, H., & Zhang, Z. (2021). 'Topic-Aware Multi-turn Dialogue Modeling'. *Proceedings of the AAAI Conference on Artificial Intelligence*, *35*(16), 14, 176–14, 184.

Yamashita, S., Inoue, K., Guo, A., Mochizuki, S., Kawahara, T., & Higashinaka, R. (2023). 'RealPersonaChat: A Realistic Persona Chat Corpus with Interlocutors' Own Personalities'. *Proceedings of the 37th Pacific Asia Conference on Language, Information and Computation*, 852–861.

Yancheva, M., & Rudzicz, F. (2013). 'Automatic detection of deception in child-produced speech using syntactic complexity features'. In H. Schuetze, P. Fung, & M. Poesio (Eds.), *Proceedings of the 51st annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 944–953). Association for Computational Linguistics.

Yang, J., Feldhus, N., Mohtaj, S., Hennig, L., Wang, Q., Metheniti, E., Hakimov, S., Jakob, C., Solopova, V., Rieck, K., Schlangen, D., Möller, S., & Schmitt, V. (2026). Order in the Evaluation Court: A Critical Analysis of NLG Evaluation Trends.

Ye, Y., Xiao, Y., Mi, T., & Liu, P. (2025). Aime-preview: A rigorous and immediate evaluation framework for advanced mathematical reasoning [GitHub repository].

Youmans, G. (1991). 'A New Tool for Discourse Analysis: The Vocabulary-Management Profile'. *Language*, *67*(4), 763–789.

Yu, W., Zhu, C., Li, Z., Hu, Z., Wang, Q., Ji, H., & Jiang, M. (2022). 'A Survey of Knowledge-enhanced Text Generation'. *ACM Comput. Surv.*, *54*(11s).

Yun, T., Lim, H., Lee, J., & Song, M. (2024). 'TelME: Teacher-leading multimodal fusion network for emotion recognition in conversation'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 82–95). Association for Computational Linguistics.

Zahiri, S. M., & Choi, J. D. (2018). 'Emotion Detection on TV Show Transcripts with Sequence-Based Convolutional Neural Networks'. *AAAI Workshops*, *18*, 44–52.

Zang, X., Rastogi, A., Sunkara, S., Gupta, R., Zhang, J., & Chen, J. (2020). 'MultiWOZ 2.2 : A Dialogue Dataset with Additional Annotation Corrections and State Tracking Baselines'. In T.-H. Wen, A. Celikyilmaz, Z. Yu, A. Papangelis, M. Eric, A. Kumar, I. Casanueva, & R. Shah (Eds.), *Proceedings*

of the 2nd workshop on natural language processing for conversational ai (pp. 109–117). Association for Computational Linguistics.

Zhan, H., Li, Z., Kang, X., Feng, T., Hua, Y., Qu, L., Ying, Y., Chandra, M. R., Rosalin, K., Jureynolds, J., Sharma, S., Qu, S., Luo, L., Zukerman, I., Soon, L.-K., Semnani Azad, Z., & Haf, R. (2024). 'RENOVI: A Benchmark Towards Remediating Norm Violations in Socio-Cultural Conversations'. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Findings of the association for computational linguistics: Naacl 2024* (pp. 3, 104–3, 117). Association for Computational Linguistics.

Zhang, M., & Toral, A. (2019). 'The Effect of Translationese in Machine Translation Test Sets'. In O. Bojar, R. Chatterjee, C. Federmann, M. Fishel, Y. Graham, B. Haddow, M. Huck, A. J. Yepes, P. Koehn, A. Martins, C. Monz, M. Negri, A. Névéol, M. Neves, M. Post, M. Turchi, & K. Verspoor (Eds.), *Proceedings of the fourth conference on machine translation (volume 1: Research papers)* (pp. 73–81). Association for Computational Linguistics.

Zhang, S., Dinan, E., Urbanek, J., Szlam, A., Kiela, D., & Weston, J. (2018). 'Personalizing Dialogue Agents: I have a dog, do you have pets too?' In I. Gurevych & Y. Miyao (Eds.), *Proceedings of the 56th annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 2, 204–2, 213). Association for Computational Linguistics.

Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., & Artzi, Y. (2019). 'Bertscore: Evaluating text generation with bert'. *arXiv preprint arXiv:1904.09675*.

Zhou, Z., Zhang, D., Xiao, W., Dingwall, N., Ma, X., Arnold, A., & Xiang, B. (2022). 'Learning Dialogue Representations from Consecutive Utterances'. In M. Carpuat, M.-C. de Marneffe, & I. V. Meza Ruiz (Eds.), *Proceedings of the 2022 conference of the north american chapter of the association for computational linguistics: Human language technologies* (pp. 754–768). Association for Computational Linguistics.

Zhu, X., Wang, Y., Gao, H., Xu, W., Wang, C., Liu, Z., Wang, K., Jin, M., Pang, L., Weng, Q., et al. (2025). 'Recommender systems meet large language model agents: A survey'. *Foundations and Trends® in Privacy and Security*, *7*(4), 247–396.

Zhuo, J., Zhang, S., Fang, X., Duan, H., Lin, D., & Chen, K. (2024). 'ProSA: Assessing and Understanding the Prompt Sensitivity of LLMs'. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Findings of the association for computational linguistics: Emnlp 2024* (pp. 1, 950–1, 976). Association for Computational Linguistics.

A

Extension of Chapter 2: An other representation for our formal account of CHIMERA

In Section 2.5.1, we proposed a possible representation of CHIMERA in TTR, where a dialogue is represented by a record type with four field, each representing one of the meta-parameters of our framework. Those fields themselves contain record types representing lower-level parameters. At the deepest level, the record types contain on field per parameter of the *extended* version of CHIMERA. Indeed, our framework is designed to be modular, and the parameters of CHIMERA can be extended or reduced depending on the needs of the analysis. While TTR allows for underspecification, which means that a dialogue type does not need to contain all the parameters of CHIMERA, encoding both the minimal and the extended version of CHIMERA explicitly in the TTR representation may make it more easily applicable at scale. If we imagine a database of corpora, which could be queried based on the parameters of CHIMERA, it would be useful to have a two-tiered search form where the first one would contain the minimal descriptions, and an *advanced search* form would allow to query the database based on the extended version of CHIMERA.

In the following we focus on one particular parameter of CHIMERA, the *number of participants*, to illustrate how the parameters' types could be defined in a way that would explicitly encode the two granularities of CHIMERA.

According to CHIMERA, the *number of participants* can be described simply by the number of participants in the dialogue, but in more complex cases it can also require to specify the number of *simulated* participants, the number of *workers* – i.e. the people who are actually producing the dialogue – and the mapping between the two. For example, when a crowdsourcer is asked to craft a dialogue between two people, *one* person is simulating *two* participants. In the TTR representation we suggested in Section 2.5.1, this would be represented by (7.iv). We choose to define the mapping as a function matching 1 with 2 to indicate that one person is responsible for the production of two participants but a more complex representation could be envisioned. As that is not the subject of this discussion, we shall leave this point for future consideration.

$$\text{(7.iv)}\quad \left[\texttt{Agents.Structure.Nb}: \begin{bmatrix}\texttt{Simulated} = \{2\}\\ \texttt{Workers} = \{1\}\\ \texttt{Mapping} = \{(1,2)\}\end{bmatrix}\right]$$

While the fact that a dialogue has been produced by one single person rather than through genuine interaction is generally a crucial point, and one that should always be made explicit when *describing* a corpus, it may not always be relevant when *using* it for a particular analysis or task. In such a case, one may want to focus only on the number of apparent participants, which corresponds to the minimal version of the parameter *number of participants*. One may want to represent this as (7.v), with the number of participants being directly specified as being 2.

$$\text{(7.v)}\quad \left[\texttt{Agents.Structure.Nb}: Nb_{=2}\right]$$

Such a representation would only be possible if the type Nb_x is defined as a record type that contains the fields from the extended description of the parameter. While it is possible to account for the two granularities by defining a specific type for each parameter that would reflect the meaning of the minimal description while being compatible with the extended description, it may be more efficient to explicitly encode the two granularities in a single representation. A possible way to do so is to define the type of the different parameters as disjunctions between their minimal and their extended description. In the case of the parameter *number of participants*, this would be represented as (7.vi).

$$\text{(7.vi)}\quad \left[\texttt{Agents.Structure.Nb}: \begin{matrix}\textit{Collection(Int)} \\ \vee \\ \begin{bmatrix}\texttt{Simulated} = \textit{Collection(Int)}\\ \texttt{Workers} = \textit{Collection(Int)}\\ \texttt{Mapping} = \textit{Collection(Int}\times\textit{Int)}\end{bmatrix}\end{matrix}\right]$$

With this approach, the parameter *number of participants* could be described either by a collection of integers that would represent the number of participants in the dialogues of the corpus, or by a record type that would encode the more complex aspects of the parameter. In this case, (7.v) would correspond to the minimal representation, and (7.iv) to the extended one.

B

Extension of Chapter 5: Detailed TextTiling Results

This appendix provides detailed results for experiments in Chapter 5, including standard deviations. Extended tables reference their original versions in the caption. Results follow the same order as in Chapter 5: baselines (Section B.1), TextTiling with linguistic features (Section B.2), ML-enhanced versions (Section B.3), and a summary of the best models (Section B.4).

B.1 Baselines

We begin by presenting the extended results for the baselines, including the two random ones (Section B.1.1), the original and dialogue-adapted TextTiling algorithms (Section B.1.2), the BERT model used in the ML-enhanced approach before fine-tuning and after fine-tuning on a generic dialogue corpus (Section B.1.3), and the LLMs (Section B.1.4).

B.1.1 Random baselines

As a reminder, we considered two random baselines: a basic one placing boundaries probabilistically based on maximum possible boundaries (Table B.1), and a smart one using the expected number of boundaries (Table B.2).

Table B.1: Extension of Table 5.5 – Average (±std.) results of the random baseline over 10 trials.

Season	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.15±.06	.49±.30	.21±.08	.29±.09	.33±.12	.56±.10	.73±.20
2	.05±.04	.50±.30	.09±.04	.12±.05	.14±.07	.58±.04	.91±.14
3	.11±.05	.50±.31	.16±.07	.22±.09	.26±.10	.58±.08	.80±.20
4	.11±.05	.48±.31	.16±.07	.23±.09	.28±.11	.57±.09	.77±.20
5	.12±.06	.50±.30	.18±.07	.25±.07	.29±.09	.58±.09	.79±.19
6	.11±.06	.49±.30	.16±.07	.23±.09	.27±.12	.57±.08	.79±.20
7	.10±.05	.50±.31	.15±.07	.21±.09	.25±.11	.58±.08	.81±.20
8	.09±.05	.49±.30	.14±.06	.19±.08	.24±.11	.57±.07	.83±.19
9	.05±.03	.48±.32	.08±.04	.11±.06	.14±.08	.58±.09	.89±.17
10	.04±.02	.48±.31	.07±.04	.10±.05	.13±.07	.58±.06	.92±.15
Total	.09±.05	.49±.31	.14±.06	.20±.08	.23±.10	.58±.08	.82±.19

Table B.2: Extension of Table 5.6 – Average (±std.) results of the *smart* random baseline over 10 trials.

Season	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.15±.07	.15±.07	.15±.07	.33±.09	.45±.09	.46±.05	.49±.05
2	.05±.06	.05±.06	.05±.06	.14±.10	.22±.12	.48±.08	.52±.07
3	.11±.07	.11±.07	.11±.07	.25±.10	.36±.11	.46±.05	.50±.05
4	.11±.07	.11±.07	.11±.07	.27±.10	.38±.10	.46±.05	.50±.05
5	.12±.06	.12±.06	.12±.06	.29±.09	.39±.09	.46±.05	.50±.05
6	.11±.07	.11±.07	.11±.07	.26±.11	.37±.11	.46±.05	.50±.05
7	.10±.06	.10±.06	.10±.06	.24±.10	.34±.11	.47±.06	.51±.05
8	.09±.07	.09±.07	.09±.07	.22±.11	.33±.12	.47±.06	.51±.06
9	.05±.06	.05±.06	.05±.06	.13±.09	.20±.12	.47±.08	.50±.08
10	.04±.06	.04±.06	.04±.06	.11±.11	.19±.12	.48±.08	.52±.08
Total	.09±.07	.09±.07	.09±.06	.23±.10	.32±.11	.47±.06	.51±.06

B.1.2 Original and dialogue adapted TextTiling

The original TextTiling algorithm splits the input text into spans of a fixed number of tokens. For dialogue, it means that utterances can be split across spans, which we thought could negatively affect performance. To address this, we proposed a dialogue-adapted version that ensures spans align with utterance boundaries: The spans consist of full utterances such that the number of tokens in each span is as close as possible to the target span length w. Table B.3 indicates the significance of the differences between the original and dialogue-adapted TextTiling algorithms with different values of the spans length parameter w. Tables B.4 and B.5 provide the actual results. Unsurprisingly the most significant differences are observed for smaller values of w, where the original algorithm is more likely to split utterances across spans. Yet, the difference remains significant up to $w = 20$, which is the value we ended up using for our experiments, supporting our choice of the dialogue-adapted version of TextTiling.

Table B.3: Significance of the differences between the original and dialogue-adapted TextTiling algorithms with different values of the spans length parameter w.

w	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	***	***	ns	***	***	***	***
5	***	***	ns	***	***	***	***
10	***	*	*	***	***	***	***
13	*	ns	ns	*	**	***	***
15	ns	ns	ns	**	***	***	***
20	**	ns	**	**	**	**	**
25	ns	ns	ns	ns	**	*	*
26	ns	ns	ns	*	*	ns	ns
30	ns	ns	*	ns	**	*	ns
35	ns	ns	ns	ns	*	ns	ns
40	*	ns	*	ns	ns	ns	ns
45	ns	ns	ns	ns	ns	ns	ns
50	ns	ns	ns	ns	ns	ns	ns

Table B.4: Extension of Figure 5.10a – Average (±std.) results of the original TextTiling algorithm (block comparison method) with different values of the spans length parameter w.

w	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.10±.05	.37±.12	.15±.07	.28±.10	.33±.12	.60±.05	.83±.13
5	.10±.06	.33±.11	.15±.06	.28±.10	.35±.13	.59±.05	.79±.14
10	.11±.06	.22±.10	.14±.06	.29±.10	.38±.11	.54±.07	.66±.14
13	.12±.06	.18±.10	.13±.06	.28±.09	.40±.11	.52±.07	.60±.13
15	.13±.07	.18±.10	.14±.06	.27±.09	.39±.11	.51±.08	.58±.12
20	.13±.07	.14±.08	.12±.06	.26±.09	.37±.11	.48±.08	.52±.11
25	.15±.08	.13±.08	.12±.06	.25±.09	.36±.10	.46±.08	.49±.10
26	.15±.09	.12±.07	.12±.06	.25±.09	.35±.10	.46±.08	.49±.10
30	.15±.08	.11±.06	.12±.05	.24±.09	.33±.10	.45±.08	.47±.10
35	.16±.09	.10±.06	.11±.06	.23±.09	.32±.10	.44±.08	.46±.09
40	.17±.09	.10±.06	.11±.05	.23±.09	.32±.10	.43±.08	.44±.09
45	.18±.09	.09±.08	.11±.05	.22±.09	.30±.11	.42±.07	.44±.08
50	.19±.09	.09±.06	.11±.05	.21±.08	.28±.10	.42±.07	.43±.08

Table B.5: Extension of Figure 5.10b – Average (±std.) results of the dialogue-adapted TextTiling algorithm (block comparison method) with different values of the spans length parameter w.

w	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.12±.06	.23±.10	.15±.06	.35±.10	.44±.12	.54±.07	.65±.15
5	.13±.07	.22±.09	.15±.06	.34±.10	.45±.11	.52±.07	.63±.15
10	.14±.07	.20±.09	.16±.06	.32±.10	.44±.11	.50±.08	.58±.14
13	.13±.07	.17±.08	.14±.06	.30±.09	.42±.10	.49±.08	.55±.12
15	.14±.07	.16±.09	.14±.06	.30±.10	.42±.10	.48±.08	.53±.11
20	.15±.08	.15±.09	.14±.06	.29±.09	.40±.10	.46±.08	.50±.10
25	.16±.08	.13±.08	.13±.06	.27±.09	.38±.10	.45±.08	.47±.10
26	.16±.08	.13±.07	.13±.06	.27±.10	.37±.10	.45±.08	.47±.09
30	.16±.08	.12±.07	.13±.06	.25±.09	.36±.11	.44±.08	.46±.10
35	.16±.09	.11±.07	.12±.06	.24±.08	.34±.11	.43±.08	.45±.09
40	.19±.10	.11±.07	.12±.06	.23±.09	.32±.10	.43±.07	.44±.09
45	.18±.10	.09±.06	.12±.05	.22±.09	.30±.10	.43±.08	.44±.08
50	.19±.10	.09±.06	.11±.05	.22±.09	.29±.09	.42±.08	.43±.08

B.1.3 BERT for next-sentence prediction: off-the-shelf vs. fine-tuned on a generic dialogue corpus

Our ML-enhanced version of TextTiling relies on a BERT model fine-tuned for next-sentence prediction. We first evaluated the off-the-shelf model, then fine-tuned it on the generic dialogue corpus DailyDialog following Xing and Carenini's (2021). Table B.6 provides the detailed results of both models, including standard deviations.

Table B.6: Extension of Table 5.7 – Average (±std.) results of the BERT-based model without any fine-tuning and then trained on DailyDialog; evaluated on the 10 seasons of our dataset.

(a) Detailed results of the BERT-based model without any fine-tuning.

Season	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.16±.05	.38±.08	.22±.06	.31±.08	.40±.08	.54±.06	.65±.09
2	.06±.02	.42±.10	.10±.03	.13±.04	.17±.04	.59±.03	.93±.05
3	.11±.04	.36±.09	.17±.05	.26±.08	.31±.10	.55±.05	.72±.10
4	.12±.04	.39±.10	.18±.05	.24±.06	.32±.07	.56±.03	.73±.08
5	.14±.04	.39±.07	.21±.05	.27±.07	.35±.08	.55±.05	.68±.06
6	.13±.06	.38±.10	.18±.06	.25±.08	.32±.09	.54±.06	.71±.12
7	.11±.05	.39±.11	.17±.06	.23±.07	.30±.08	.57±.06	.76±.11
8	.10±.04	.42±.13	.16±.05	.21±.07	.27±.10	.57±.06	.81±.11
9	.07±.03	.48±.15	.11±.04	.14±.05	.17±.06	.59±.09	.91±.06
10	.06±.03	.49±.13	.10±.04	.12±.04	.14±.05	.58±.05	.95±.04
Total	.11±.05	.41±.11	.16±.06	.22±.09	.28±.11	.56±.06	.78±.13

(b) Detailed results of the model trained on DailyDialog.

Epoch	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.10±.05	.39±.11	.15±.06	.20±.09	.25±.10	.58±.06	.80±.11
2	.09±.05	.42±.11	.15±.06	.20±.09	.24±.10	.59±.05	.83±.11
3	.10±.05	.48±.12	.16±.07	.21±.09	.25±.10	.59±.05	.86±.11
4	.09±.05	.41±.11	.15±.06	.20±.09	.25±.10	.59±.05	.82±.11
5	.10±.05	.46±.11	.16±.07	.20±.09	.25±.10	.59±.05	.85±.11
6	.10±.05	.44±.11	.15±.07	.20±.09	.25±.10	.59±.05	.84±.11
7	.09±.05	.45±.12	.15±.07	.20±.09	.25±.10	.60±.05	.86±.11
8	.09±.05	.45±.12	.15±.07	.20±.09	.25±.10	.60±.05	.86±.11
9	.09±.05	.45±.12	.15±.07	.20±.09	.25±.10	.60±.05	.86±.11
10	.09±.05	.45±.11	.15±.07	.20±.09	.25±.10	.60±.05	.86±.11
Total	.10±.05	.44±.11	.15±.07	.20±.09	.25±.10	.59±.05	.84±.11

B.1.4 LLMs

Following recent practices in NLP, we evaluated the performance of several LLMs on our task. Our zero-shot approach led to the production of many invalid outputs, the proportion of which is reported in Table B.7a. The results when considering only the valid outputs are reported in Table B.7b.

Table B.7: Average results of the different LLMs on the 10 seasons of *Friends*, the last three columns report the proportion of outputs with issues. *Best results (ignoring the non significant differences) are highlighted in **bold**.*

(a) Issues in the outputs of the different LLMs.

Model	NaN (%)	Length (%)	Empty (%)	Empty after filtering (%)
llama3.1:8b	0	65.68	0	0
llama3.2:3b	2.54	67.80	7.63	0
deepseek-r1:1.5b	0	23.31	0	12.29
deepseek-r1:8b	2.12	41.95	47.03	23.31
deepseek-r1:14b	0	65.68	0	0
gemma3:1b	100	0	0	0
gemma3:4b	100	0	0	0
mistral:7b	0.85	91.53	0	0

(b) Topic segmentation results.

Model	Pr.↑	Rec.↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
llama3.1:8b	.11±.06	.24±.10	.14±.06	.30±.10	.44±.13	.58±.06	.71±.16
llama3.2:3b	.09±.05	.60±.29	.15±.08	.20±.12	.21±.13	.65±.11	.94±.10
deepseek-r1:1.5b	.18±.13	.10±.11	.09±.06	.16±.10	.17±.11	.41±.07	.44±.12
deepseek-r1:8b	.18±.33	.02±.08	.03±.05	.04±.08	.04±.08	.68±.31	.68±.31
deepseek-r1:14b	.13±.10	.52±.35	.15±.06	.18±.09	.20±.12	.54±.11	.75±.24
gemma3:1b	/	/	/	/	/	/	/
gemma3:4b	/	/	/	/	/	/	/
mistral:7b	.14±.10	.19±.18	.13±.06	.26±.09	.35±.12	.50±.09	.58±.18

B.2 TextTiling enhanced with linguistic features

Our first approach to enhance the TextTiling algorithm relied on the use of linguistic features. Table B.8 provides the detailed results of the feature-based models enhanced with speaker information. We then compare several combinations of linguistic features in Table B.9.

Table B.8: Extension of Table 5.11 – Average (±std.) results of the feature-based models enhanced with speaker information.

Model	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
BC	.15±.08	.15±.09	.14±.06	.29±.09	.40±.10	.46±.08	.50±.10
VI	.12±.07	.12±.07	.11±.06	.23±.09	.33±.10	.50±.08	.53±.11
BCVI	.14±.08	.13±.09	.12±.06	.25±.09	.36±.10	.48±.08	.52±.10
BC + Sp_{sharp}	.15±.08	.15±.10	.14±.06	.29±.09	.41±.10	.46±.08	.49±.10
VI + Sp_{sharp}	.12±.08	.12±.07	.12±.06	.23±.09	.34±.10	.50±.08	.53±.10
BCVI + Sp_{sharp}	.14±.08	.14±.09	.13±.06	.25±.09	.37±.10	.48±.08	.51±.10
BC + Sp_{smooth}	.18±.09	.18±.10	.17±.07	.36±.10	.48±.11	.42±.08	.46±.11
VI + Sp_{smooth}	.15±.08	.16±.08	.15±.07	.31±.10	.43±.10	.45±.08	.49±.10
BCVI + Sp_{smooth}	.17±.09	.18±.09	.16±.06	.34±.10	.46±.10	.43±.08	.47±.11

Table B.9: Extension of Table 5.12 – Average (±std.) results of the feature-based models enhanced with various approaches.
Block comparison (BC), vocabulary introduction (VI), coreference chains (Co), smooth speaker introduction (SI), questions (Q), stemming (S), lemmatisation (L). Single worse model in gray.

Model	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
BC	.15±.08	.15±.09	.14±.06	.29±.09	.40±.10	.46±.08	.50±.10
BC + SI	.18±.09	.18±.10	.17±.07	.36±.10	.48±.11	.42±.08	.46±.11
BC + SI + Q	.18±.09	.18±.10	.17±.07	.36±.10	.48±.11	.41±.08	.45±.11
BC + SI + Q + L	.18±.09	.18±.10	.17±.07	.36±.09	.48±.11	.41±.08	.45±.11
BC + SI + Q + S	.18±.08	.18±.10	.17±.07	.36±.09	.48±.10	.41±.08	.45±.10
BC + SI + Q + Co	.19±.10	.16±.09	.16±.07	.37±.10	.48±.10	.40±.07	.44±.09
VI	.12±.07	.12±.06	.11±.05	.23±.09	.34±.10	.50±.08	.53±.11
VI + SI	.16±.08	.16±.08	.15±.06	.31±.10	.43±.11	.45±.08	.48±.11
VI + SI + Q	.16±.08	.17±.09	.15±.06	.31±.10	.44±.11	.44±.08	.48±.11
VI + SI + Q + L	.16±.08	.16±.09	.15±.06	.31±.10	.44±.10	.44±.08	.48±.11
VI + SI + Q + S	.16±.08	.16±.09	.15±.06	.31±.10	.44±.10	.44±.08	.48±.11
BC + SI + Q + Co	*Not computed*						
BCVI	.14±.08	.14±.09	.13±.05	.25±.09	.37±.10	.48±.08	.52±.11
BCVI + SI	.18±.09	.18±.10	.17±.07	.35±.10	.47±.10	.42±.08	.46±.11
BCVI + SI + Q	.19±.09	.19±.10	.17±.07	.35±.10	.48±.11	.41±.08	.45±.11
BCVI + SI + Q + L	.18±.09	.18±.10	.17±.07	.35±.10	.48±.11	.42±.08	.46±.11
BCVI + SI + Q + S	.18±.09	.18±.10	.17±.07	.35±.10	.48±.11	.41±.08	.45±.11
BC + SI + Q + Co	*Not computed*						

B.3 ML-enhanced TextTiling

Inspired by Xing and Carenini (2021), we enhanced TextTiling using a BERT model fine-tuned for NSP. We evaluated the impact of training epochs (Section B.3.1), tested several coherence layer combinations (Section B.3.2), and assessed model stability across training subsets and initialisations (Section B.3.3).

B.3.1 Learning curve

As the model performance shows no significant differences between the results of the 10 epochs (Table 5.13), we then limited ourselves to two epochs of training.

Table B.10: Extension of Table 5.13 – Average (±std.) results for the model AND-3.

Epoch	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
1	.14±.04	.44±.09	.21±.05	.27±.07	.31±.08	.51±.05	.71±.10
2	.14±.05	.47±.10	.21±.07	.27±.08	.31±.08	.53±.06	.73±.10
3	.14±.04	.48±.08	.21±.05	.27±.07	.31±.08	.52±.04	.73±.09
4	.13±.05	.38±.11	.19±.06	.27±.07	.32±.08	.50±.05	.66±.08
5	.14±.05	.50±.09	.21±.06	.27±.07	.31±.08	.53±.05	.75±.10
6	.14±.05	.51±.10	.22±.06	.27±.08	.31±.08	.52±.05	.74±.10
7	.14±.05	.49±.10	.21±.06	.26±.07	.30±.08	.53±.05	.75±.11
8	.14±.05	.51±.10	.22±.07	.27±.07	.31±.08	.53±.05	.75±.10
9	.14±.04	.51±.08	.22±.06	.26±.07	.30±.09	.53±.05	.75±.10
10	.14±.05	.52±.07	.22±.06	.27±.07	.30±.09	.53±.05	.75±.10

B.3.2 Coherence-layers Selection

We leveraged the structure of the *Friends* episodes to extract several combinations of coherence layers, whose performance is detailed in Table B.11.

Table B.11: Extension of Table 5.14 – Average (±std.) results for the different layer settings. *Models trained on 9 seasons: <layers>-<evaluation season>.*

Model	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
ANC-1	.19±.05	.59±.07	.28±.06	.36±.08	.39±.08	.52±.06	.70±.10
ANC-3	.14±.05	.47±.10	.21±.07	.27±.08	.31±.08	.53±.06	.73±.10
AND-1	.18±.05	.53±.09	.27±.06	.35±.08	.39±.08	.52±.06	.68±.11
AND-3	.14±.05	.50±.08	.21±.07	.26±.07	.29±.08	.55±.05	.76±.10
ANCD-1	.16±.04	.46±.10	.24±.05	.32±.07	.40±.08	.56±.05	.71±.09
ANCD-3	.14±.05	.49±.11	.22±.07	.29±.08	.32±.09	.51±.05	.72±.10
ANCESD-1	.19±.06	.49±.08	.27±.06	.36±.09	.41±.10	.48±.06	.63±.09

B.3.3 Model stability

In order to assess the stability of our ML-enhanced TextTiling model, we trained it on different subsets of the training data and with different random initialisations. Table B.12 provides the detailed results of these experiments.

Table B.12: Extension of Table 5.15 – Average (±std.) results for the models trained on three different training subsets (d_α, d_β, d_γ) for two epochs.

Model	Epoch	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
$D_{\alpha 1}$	1	.12±.05	.56±.11	.19±.07	.24±.10	.26±.11	.57±.05	.85±.12
	2	.12±.06	.56±.12	.19±.08	.24±.10	.26±.11	.56±.06	.84±.12
$D_{\alpha 2}$	1	.09±.05	.36±.12	.14±.06	.19±.08	.24±.10	.59±.06	.80±.11
	2	.09±.05	.39±.12	.14±.06	.19±.08	.24±.10	.59±.06	.82±.11
$D_{\alpha 3}$	1	.12±.05	.38±.11	.17±.06	.24±.10	.29±.11	.54±.06	.74±.14
	2	.11±.06	.37±.12	.16±.07	.24±.10	.30±.11	.55±.07	.74±.14
$D_{\beta 1}$	1	.12±.06	.44±.12	.18±.07	.24±.09	.28±.11	.52±.07	.74±.13
	2	.13±.07	.36±.12	.18±.07	.24±.10	.28±.11	.50±.08	.68±.13
$D_{\beta 2}$	1	.13±.06	.51±.11	.20±.08	.25±.10	.28±.11	.50±.07	.75±.14
	2	.14±.06	.48±.11	.20±.08	.26±.10	.29±.11	.49±.08	.73±.14
$D_{\beta 3}$	1	.13±.06	.46±.12	.20±.08	.26±.10	.29±.12	.50±.08	.73±.14
	2	.14±.07	.45±.11	.20±.08	.26±.10	.29±.11	.49±.08	.71±.14
$D_{\gamma 1}$	1	.14±.07	.36±.10	.19±.08	.26±.10	.30±.12	.47±.08	.64±.13
	2	.15±.08	.38±.11	.21±.08	.27±.11	.31±.12	.45±.09	.63±.14
$D_{\gamma 2}$	1	.12±.06	.32±.11	.17±.06	.24±.09	.29±.11	.51±.07	.66±.13
	2	.09±.05	.26±.10	.13±.05	.19±.08	.28±.11	.56±.06	.72±.13
$D_{\gamma 3}$	1	.12±.06	.54±.11	.19±.08	.24±.10	.27±.11	.54±.06	.81±.13
	2	.13±.06	.51±.12	.19±.08	.24±.10	.27±.11	.53±.07	.78±.13

B.3.4 Training set variety

We compared the versions of the ML-enhanced TextTiling model trained on 3 seasons to the ones trained on 9 seasons. Table B.13 provides the detailed results of these experiments. Given that the models could be evaluated only on season 1, the results are quite different than those reported in Table B.12 for the models trained on 3 seasons. However, we can still observe that despite the less varied training set, the models trained on three seasons are still able to achieve comparable results to those trained on nine with regards to the F-measure metrics and even better ones for the P_k and WD metrics. This supports our choice of using only three seasons for training, as it allows us to evaluate the models on a wider range of episodes without compromising their performance.

Table B.13: Additional comparison – Average (±std.) results for the models trained on 9 seasons (all but season 1) and for the models trained on 3 seasons. *All the models were evaluated on season 1 only.*

Model	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
$D_{\alpha 3}$	.17±.06	.36±.08	.23±.07	.34±.08	.42±.09	.50±.07	.61±.10
$D_{\beta 1}$	.20±.07	.37±.09	.25±.07	.36±.09	.42±.09	.44±.07	.55±.09
$D_{\gamma 1}$	.23±.07	.39±.07	.29±.06	.37±.08	.44±.09	.41±.06	.51±.09
ANC-1	.19±.05	.59±.07	.28±.06	.36±.08	.39±.08	.52±.06	.70±.10
AND-1	.18±.05	.53±.09	.27±.06	.35±.08	.39±.08	.52±.06	.68±.11
ANCD-1	.16±.04	.46±.10	.24±.05	.32±.07	.40±.08	.56±.05	.71±.09
ANCESD-1	.19±.06	.49±.08	.27±.06	.36±.09	.41±.10	.48±.06	.63±.09

B.4 Final comparisons

Table B.14 summarises the results of the best models of each approach. Regarding the baselines, we include the two random ones, the original and dialogue-adapted TextTiling models that rely on block comparison, and the two BERT-based models, one without any fine-tuning and the other trained on DailyDialog. We compare those baselines with the best feature-based model (block comparison, speaker information, and question) and the best ML-enhanced TextTiling model ($\text{D}_{\gamma 1}$).

Table B.14: Extension of Table 5.16 – Average (±std.) results for the final models. *Evaluation on seasons 1, 5, 6, 7, 8, 9, 10 only.*

Model	Precision↑	Recall↑	F_1 ↑	F_{k1} ↑	F_{k2} ↑	P_k ↓	WD↓
Rd	.10±.06	.49±.31	.14±.08	.20±.10	.24±.12	.58±.08	.82±.20
Rd_{smart}	.10±.07	.10±.08	.10±.07	.23±.12	.33±.14	.47±.06	.51±.06
BC	.13±.08	.13±.08	.12±.06	.26±.09	.37±.11	.48±.08	.52±.11
BC for dialogue	.15±.08	.15±.09	.14±.06	.28±.09	.40±.10	.46±.08	.50±.11
BERT_{Base}	.11±.05	.42±.12	.17±.07	.22±.09	.28±.11	.56±.06	.77±.14
$\text{BERT}_{DailyDialog}$	.10±.05	.40±.11	.15±.06	.20±.09	.25±.10	.58±.06	.80±.12
BC + SI + Q	.19±.10	.19±.10	.17±.07	**.36±.10**	**.48±.11**	**.41±.08**	**.45±.11**
$\text{D}_{\gamma 1}$	.15±.08	.38±.11	**.21±.08**	.27±.11	.31±.12	.45±.09	.63±.14

C

Extension of Chapter 6: Detailed BERT Topic Modelling Results

This appendix details the evaluation of the BERT Topic modelling ranking on our *Reddit* data. Table C.1 reports the results when at least one of the three similarities is valid and Table C.2 when all three are valid.

Table C.1: Evaluation of the ranking performed by the BERT Topic Models when at least one similarity is valid.

Subreddit	Acc.	1st✓	2nd✓	3rd✓
r/Art	.59	.57	.51	.77
r/Coronavirus	.66	**.61**	.50	.67
r/gaming	.63	.60	.54	.81
r/OnePiece	.63	**.61**	.54	.80
r/PoliticalDiscussion	**.67**	.58	.47	.72
r/sports	.63	**.61**	**.56**	**.86**
All	.64	.60	.52	.77
Random	.50	.33	.33	.33

Subreddit	AB $\geq$ AC	AC $\geq$ AD	AB $\geq$ AD	Perfect Ranking
r/Art	.59	.50	.68	.18
r/Coronavirus	**.66**	.58	.74	**.31**
r/gaming	.62	.54	.73	.22
r/OnePiece	.63	.53	.74	.22
r/PoliticalDiscussion	.61	**.64**	**.76**	**.31**
r/sports	.62	.53	.74	.21
All	.62	.56	.73	.24
Random	.50	.50	.50	.17

Table C.2: Evaluation of the ranking performed by the BERT Topic Models when all three similarities are valid. *'Acc.' is the pairwise accuracy; '1st✓', '2nd✓' and '3rd✓' are the proportions of of correctly ranked first, second and third scores; 'AB > AC', 'AC > AD' and 'AB > AD' are the proportions of pairs correctly ordered two by two; 'Perfect Ranking' is the proportion of triples for which the ranking is perfect.*

Subreddit	Acc.	1st✓	2nd✓	3rd✓
r/Art	.71	.56	.42	.63
r/Coronavirus	.72	**.66**	**.47**	.57
r/gaming	.75	.59	.45	.68
r/OnePiece	.72	.57	.44	.64
r/PoliticalDiscussion	**.77**	.61	**.48**	**.72**
r/sports	.76	.58	.46	**.72**
All	.66	.73	.85	.43
Random	.50	.33	.33	.33

Subreddit	AB $\geq$ AC	AC $\geq$ AD	AB $\geq$ AD	Perfect Ranking
r/Art	.61	.72	.81	.38
r/Coronavirus	**.71**	.65	.81	.42
r/gaming	.64	.75	.85	.42
r/OnePiece	.64	.71	.82	.40
r/PoliticalDiscussion	.64	**.78**	**.88**	**.45**
r/sports	.62	**.78**	.87	.43
All	.66	.73	.85	.43
Random	.50	.50	.50	.17

D

Extension of Chapter 7: Detailed Annotation Guide and Correlation Results

This appendix contains more details regarding the study presented in Chapter 7. It begins with the full guide provided to the annotators for the salience and resolution annotations (Section D.1), followed by the correlation results between question salience and resolution on the full annotation scale (Section D.2).

D.1 Detailed annotation guide

The annotation guidelines were provided to the annotators in the interface. They are reproduced here for reference.

D.1.1 Question salience

D.1.1.1 Motivation

In natural conversations, it is common to ask curiosity-driven questions to follow or deepen the exchange. Among the different potential questions one might ask oneself during a conversation, how important is it for the question to be answered in subsequent speech turns? Can we rank these questions based on their relevance and necessity to the conversation? We adapt the evaluation schema from Y. Wu et al. (2024) to do this.

> **Note**: The scores **2**, **4**, and **7** were added specifically for dialogue and should not be used when annotating the training example.

D.1.1.2 Task: Salience Score

Given the **prior** conversation context, an **anchor speech turn**, and a list of potential questions, score each question based on the schema described below.

Score = 0: These are questions that satisfy at least one of the following criteria:

- The question contains grammatical errors.
- The question is not anchored in the given anchor speech turn (i.e. has no clear relation to what was just said).
- The question contains multiple sub-questions.
- The question misinterprets or contradicts the conversation context.

A This is Aural History Project tape number one of Mr .

A My name is `<name>`, the date is the second of March nineteen eighty seven.

A This is interview number five of Ipswich Docks.

A before we talk about your working life at Ipswich Docks, could you tell me where you were born.

B Well I was born at

→ **What is the name of this project?**

Justification This question is not related to the anchored turn.

Score = 1: The question is only loosely or not at all related to the conversation topic. This score is meant to weed out tangential or irrelevant questions.

A Have you been?

B I got slightly lost.

B Well I went back down the Road cos there was a traffic jam `<pause>` and er I went down `<pause>` oh sorry mate `<pause>` went down a road like which I thought was a good short cut, and they've got a lot of these roads and they've put like pavements across the end of them with bollards, and I went back and then there was a traffic jam and I got stuck because they were unloading a lorry `<pause>` so I've been er basically pissed about. `<laugh>` `<pause>`

A Yeah she `<unclear>` now ain't she?

C I don't think she has.

→ **Why is there uncertainty whether the female in question is pregnant?**

Justification The model is making unfounded assumptions.

Score = 2: The question is related to issues in the transcription of the anchored speech turn (e.g. about the content of `<unclear>`).

A This is Aural History Project tape number one of Mr .

A My name is `<name>`, the date is the second of March nineteen eighty seven.

A This is interview number five of Ipswich Docks.

A before we talk about your working life at Ipswich Docks, could you tell me where you were born.

B Well I was born at

→ **Where exactly were you born?**

Justification The answer has been removed from the transcription for anonymisation purposes.

Score = 3: The question is related to ideas introduced in the prior conversation and the anchor speech turn, but asking it feels like a stretch.

- It may already be fully answered by the context.
- Answering it does not appear helpful for continuing, understanding or deepening the conversation.

A This is Aural History Project tape number one of Mr .

A My name is `<name>`, the date is the second of March nineteen eighty seven.

A This is interview number five of Ipswich Docks.

A before we talk about your working life at Ipswich Docks, could you tell me where you were born.

B Well I was born at

→ **What do you mean by 'born'?**

Justification This question is not helpful because 'born' is unambiguous here.

Score = 4: The question is related to ideas introduced in the prior conversation and the anchor speech turn, but the participants clearly already know the answer.

- It could have been in one of the `<unclear>` parts of the transcription.
- It could have been in a prior part of the conversation we do not have access to.
- It could be due to some shared background.

A Have you been?

B I got slightly lost.

B Well I went back down the Road cos there was a traffic jam <pause> and er I went down <pause> oh sorry mate <pause> went down a road like which I thought was a good short cut, and they've got a lot of these roads and they've put like pavements across the end of them with bollards, and I went back and then there was a traffic jam and I got stuck because they were unloading a lorry <pause> so I've been er basically pissed about. <laugh> <pause>

A Yeah she <unclear> now ain't she?

C I don't think she has.

→ **Who is "she" referring to?**

Justification This seems to be part of the common ground of the participants.

Score = 5: The question is relevant to the prior context and the anchor speech turn, but answering it is not essential.

A This is Aural History Project tape number one of Mr .

A My name is <name>, the date is the second of March nineteen eighty seven.

A This is interview number five of Ipswich Docks.

A before we talk about your working life at Ipswich Docks, could you tell me where you were born.

B Well I was born at

→ **Do you know if there was a popular song on the radio around the time you were born?**

Justification This question is relevant but does not feel very useful, especially in the context of talking about the 'working life' of the speaker.

Score = 6: Answering the question is somewhat useful to the conversation.

- It could clarify a newly introduced concept.
- It may help elaborate on or extend an idea mentioned in the anchor turn.
- There is some uncertainty, but the question has the potential to shape how the conversation proceeds.

A This is Aural History Project tape number one of Mr .

A My name is <name>, the date is the second of March nineteen eighty seven.

A This is interview number five of Ipswich Docks.

A before we talk about your working life at Ipswich Docks, could you tell me where you were born.

B Well I was born at

→ **What was your birthdate?**

Justification This question remains on the topic of getting to know the speaker and has not been answered yet, it could be a possible follow-up.

Score = 7: The anchored turn is a question and the question to annotate is a paraphrase of it.

A Tonight live, the happiness pill, the terrorist godfathers and circuses.

A Are they right to use domestic cats?

A Roll up, roll up, it's Central Weekend.

A Good evening, welcome to Central Weekend and to our audience, they come from Ludlow.

B welcome.

B We start tonight with a big row that's brewing over this little pill.

B It's new and it's given to people who feel depressed.

B Your doctor prescribes it as Prozac but it's already earned the nickname the happiness pill, but is it safe?

B Oliver what's so wonderful about Prozac as an antidepressant?

→ **What is the scientific basis for Prozac's effectiveness as an antidepressant?**

Justification This is a paraphrase of the actual question with a barely different angle.

Score = 8: The question should definitely be answered in the following speech turns.

- It could clarify a point made in the anchor speech turn.
- It could respond to a surprising or ambiguous event or claim.
- It could seek more information about a newly introduced person or entity in the conversation.
- Answering it is essential for following or understanding the ongoing conversation.

A Tonight live, the happiness pill, the terrorist godfathers and circuses.

A Are they right to use domestic cats?

A Roll up, roll up, it's Central Weekend.

A Good evening, welcome to Central Weekend and to our audience, they come from Ludlow.

B welcome.

B We start tonight with a big row that's brewing over this little pill.

B It's new and it's given to people who feel depressed.

→ **What is this pill?**

Justification As a participant, this information is necessary to follow and take part in the conversation.

Note: Do not assign a higher score just because you think others might benefit from the answer. Base your judgement solely on your own need to have the question answered to follow the conversation naturally and coherently.

D.1.2 Question resolution

D.1.2.1 Motivation

In natural conversations, it is common to ask curiosity-driven questions to follow or deepen the exchange. Among the different potential questions one might ask oneself during a conversation, how many of them are actually answered in the subsequent speech turns? This annotation task is meant to check that.

D.1.2.2 Task: Answerability Score

Given a conversation, an **anchor speech turn**, and a list of potential questions, score each question based on the following Likert scale:

Score = 0: The question is not answered after the anchor turn.

Score = 1: The question is partially answered or the participants explicitly state that they do not know the answer after the anchor turn.

Score = 2: The question is fully answered after the anchor turn.

If the question is answered **before or by** the anchor turn and is not answered again **after** it then you should rate it with **0**.

Some specific cases

- Sometimes the generated question inquires on the identity of a participant or a person mentioned in the dialogue:
  - If the question concerns the speaker of the anchor turn then the score should be **0**;

    A Do men hate women?

    A they don't!

    A They respect, admire, like, and often love them!

    A Don't they?

    A Now why am I stirring in the age long war between the sexes?

    → **Who is "I" referring to in this context?**

    Justification This question inquires on the identity of the speaker.
  - If the anchor turn explicitly gives the floor to a participant who is the next speaker (based on our interpretation) and the question aims to clarify their identity then the score should be **2**;

    A Councillor ?

    B I'll, I was just going to say, Chairman, I, I too am very concerned really about the number of premises where it is advisory to the District Councils, or advisory to .

    → **Who is Councillor ?**

    Justification Speaker B seems to be the councillor. The question on the identity of the next speaker is relevant in dialogue for turn-taking purposes.
  - If the question concerns a person mentioned in the prior dialogue then the answer should be based on the amount of additional information we get on this person in the rest of the dialogue.
- Sometimes the generated question uses unsuitable pronouns:
  - If the prior context makes it very clear that the pronoun is unsuitable then the score should be **0**;
  - If the mistake does not contradict the context and does not prevent you from understanding the question then you should rate it as if the pronoun was right.

    A I thought it `<unclear>` , these look pretty.

    B They are pretty aren't they?

    B They are original geraniums

    → **Who is B referring to when saying "they"?**

Justification The right pronoun should be "What" but using "Who" does not infringe the understanding. The mistake may be related to the lack of world model of the LLM.

D.2 Question salience-resolution correlation

In Chapter 7, we presented the correlation between salience and resolution scores on a subset of the annotation scale to retain the ordinal, monotone ordering. Scores 2, 5, and 7, which were added specifically for dialogue compared to the original scale in Y. Wu et al. (2024), were excluded from the analysis because they may disturb this ordering. Here, we extend this analysis to the full annotation scale, as shown in Figure D.1. The results indicate that the correlation remains significant across the entire range of scores and is slightly stronger than the correlation observed in the restricted scale.

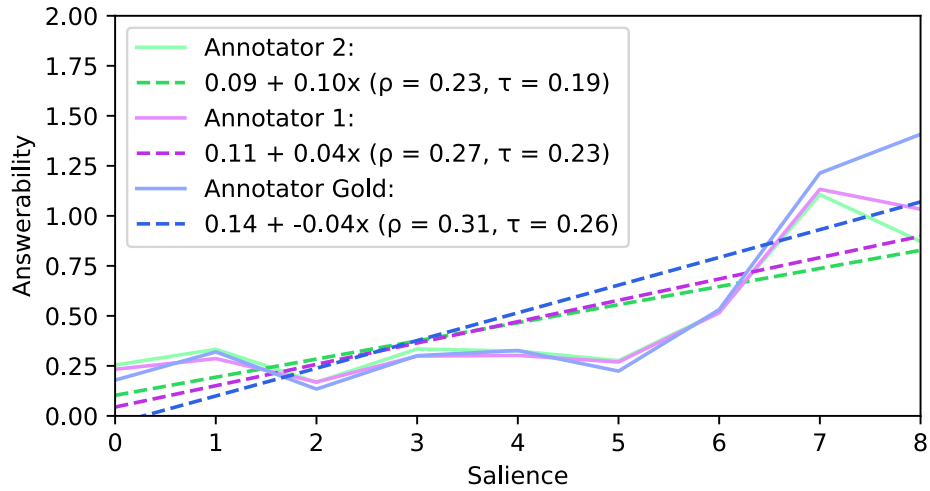


Figure D.1: Extension of Figure 7.14 – Correlation between salience and resolution scores on the full annotation scale (all p-values $\ll .005$). *Legend in the same order as the plots at 0.*

E

Montgolgram: Collecting Text-based Duologues for Topical Structure Analysis

As discussed throughout this thesis, studying topic shifts in casual conversations is complicated for several reasons. First, such interactions are not easy to collect, and second, even when they are collected, topics are hard to identify and annotate reliably. The process of collecting a dialogue corpus specifically designed for the analysis of topics in a setting that would be as natural as possible while protecting the participants' privacy was therefore initiated during the course of this thesis. While the amount of data collected was quite small, documenting the process of collecting it may be useful for future work. Therefore, we present here the design of a small-scale corpus of text-based duologues, called MONTGOLGRAM.

E.1 Aim and method: Balancing naturalism and realistic collection conditions

As for the rest of this thesis, though we were ultimately interested in the analysis of face-to-face multi-party conversations, the complexity of collecting such data and of annotating it for topics led us to focus on a simpler setting. We chose to focus on duologues, which we imagined would be easier to collect as they required less coordination between participants in terms of scheduling. For the collection setting, we followed Healey and Mills (2009) and used a text-based instant messaging platform to collect the dialogues. They have the clear advantage of being in a format that requires minimal processing to be analysed, as opposed to multimodal data, and they allow us to assign pseudonyms to the participants, therefore easily protecting their privacy. We also designed two interaction tasks to be performed by the participants in order to reduce the risk that they would share personal information in the course of the interaction. We describe those tasks in Section E.2.

In the end the format we chose for those conversations, described in Table E.1, is a compromise between the ideal conditions we would have liked to have for the collection of the data and the realistic constraints of collecting it. While they are

Table E.1: Description of the format of the dialogues collected for the Montgolgram study.

Meta-Parameters	Sub-parameters	Montgolgram setting
Agents		
Agency		Humans
Participation structures	Number	≥ 2
Social dynamics	History	✓
Enunciative status	Anonymity	Digital + Pseudonymous
Conversational setting		
Environment & Medium	Physical context	✗
	Visibility	✗
	Audibility	✗
	Synchrony	Semi-synchronous to asynchronous
Conversational frame		
Language	Global	French or English
Verbal interaction		✓

quite different from face-to-face interactions, they could offer a more controlled setting to study the dynamics of topic shifts in rather casual conversations.

We collected those conversations through a custom-made conversational tool that leverages the instant messaging application Cinny, which is based on Matrix, a real-time open communication protocol. We described it in more details in Section 3.3.2.2. We obtained approval from the department for the protection of personal data of the University of Lorraine to carry out these experiments.



Figure E.1: Illustration of the balloon task.

E.2 Dialogue setting: Two types of tasks for two analytical purposes

Since we were interested in how people perceive and negotiate topics in dialogue, we selected two conversational tasks which allow free interactions, while still being likely to produce several topics within one domain. The first one is the *balloon task*, which we detail in Section E.2.1, and the second one is the *tangram task*, a visual coordination task well-known in the dialogue research community, which we describe in Section E.2.2. The participants had to complete these two tasks one after the other. They were told that they could either decide on a meeting time to perform the tasks synchronously, or they could perform them asynchronously during a longer period of time. At the end of the two tasks, they were asked to fill a short survey about their experience.

E.2.1 Balloon task: An ethical dilemma

When opening the chat tool for the first time, the participants arrived in a conversation room shared with one other participant. In practice this participant was a bot that we used to relay the messages to a server where we could modify them before sending them to the other participant. The room would have one pinned message containing the instructions for the first task and the illustration shown in Figure E.1.

In this first task, passengers are asked to discuss and reach a decision in a moral dilemma where a balloon with four passengers will crash unless one is sacrificed. Each passenger is valuable for different reasons: the pilot, the pilot's pregnant wife,

a child prodigy, and a doctor on the verge of discovering a cure for cancer. Participants therefore exchange arguments to decide which passenger should be sacrificed. The argumentative aspect is interesting as it naturally elicits a discussion. The task itself enables us to identify different topics more easily than in casual dialogue as we can consider that each negotiation phase around one passenger is a topics.

Moreover, it creates an environment that is simple to disturb through small manipulations. Indeed, following Amido et al. (2024) and Vanzan et al. (2024), we decided to sometimes modify the messages the participants sent to each other in order to investigate a specific phenomenon. In our case, we wanted to see how participants would adapt to having two explicitly different perception of the conversation. In particular, we wanted to see how they would react if the roles of the passengers were randomly switched in the messages they received. For example, if one participant sent 'We shouldn't sacrifice the doctor because she can save many people', their partner could receive 'We shouldn't sacrifice the music prodigy because she can save many people' instead. Those modifications are interesting from a topic analysis perspective as they create a situation where the participants have to negotiate a common understanding of the current topic.

The modification of the messages was done thanks to a named-entity recognition model that we fine-tuned to identify the names of the passengers in the messages. The data for the fine-tuning was collected through a pilot-study that we carried out with 12 colleagues who participated each in 3 dyadic dialogues (18 pairs). This pilot study is described in Decker et al. (2024).

Once the participants reached an agreement on which passenger should be sacrificed, they were asked to type a message with a specific format to indicate their choice. They would then be moved to a new room where they would receive the instructions for the second task, which is described in Section E.2.2.

E.2.2 Tangram task: A visual coordination task

In this second task, the participants were each shown three tangram pictures from the KiloGram dataset (Ji et al., 2022) and were asked to determine which of the three pictures they had in common. An example of the task is shown in Figure E.2.

This task is well-known in the dialogue research community, in particular for analysing how people develop a common vocabulary to refer to the tangram pictures (Clark & Wilkes-Gibbs, 1986). For our purpose, we thought that the description of each individual tangram could be considered as a topic. We also took advantage of the visual modality to introduce some external interruptions: after sending a message, if enough time had passed, the tangrams would be rotated by a random angle for the participant. While this does not change which image is the same for both participants, it does change the way they perceive it and can influence the way they describe it. The most interesting aspect is that it is also likely to trigger meta-discussions on whether the images have changed or are supposed to have the same

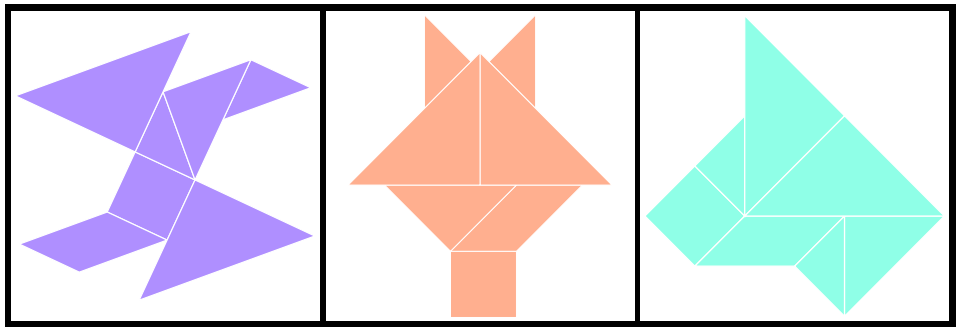


Figure E.2: Illustration of the tangram task.

orientation.

The tasks and the perturbations we introduced both change the type of conversation that we collected. Table E.2 summarises the characteristics induced by the two tasks.

E.3 Collection: Lessons and perspectives

In the end, while 58 people initially signed up to participate, and were paired with a partner, only 22 of them actually completed the two tasks, including 20 who also filled the survey. The main reason for this high drop-out rate was that filling out the registration form did not directly lead to the chat tool. Instead, the participants had to wait for the confirmation email which contained their credentials to participate in the study. The delay was sometimes quite long as we needed two people to have registered for the same language to be able to pair them. Another reason for the drop-out was that some participants logged in to the chat tool but never managed to find a time to perform the tasks with their partner. While some people attempted to perform the tasks asynchronously, very few pairs persevered long enough to complete them. In the end, we collected 14 dialogues for the balloon task, 11 in French and 3 in English, including 3 where only one participant sent messages and validated the answer on their own. They were then stuck in the tangram task as it required both participants' answers to move on. For this reason we collected only 11 finished conversations for the second task, 9 in French and 2 in English. More information on the completed pairs of dialogues is provided in Table E.3.

Given the very small amount of data, we let the collection last as long as possible based on the agreement we had with the department for the protection of personal data, which did not allow us to analyse the results within the time-frame of this thesis. Therefore in this section we focus on general remarks on the collection process and on the lessons we learned from it, which could be useful for future work.

Table E.2: Description of the format of the dialogues collected for the Montgolgram study.

Meta-Parameters		Sub-parameters	Balloon task	Tangram task
Conversational setting				
	Epistemic grounding	World setting	Hypothetical	Real
		World stability	Stable	Changes in the environment
	Environment & Medium	Physical context	✗	Individual tangrams
Conversational frame				
	Purpose	Practical goal	Task-based	
		Success condition	Reach an agreement	Find common image
		Topical structure	Negotiation phases	Image descriptions

Table E.3: Number of completed pairs of dialogues in each language and for each modification setting.

Language	**No modifications**	**Modifications for one participant**	**Modifications for both participants**
French	2	3	3
English	0	1	1

Overall, despite the apparent convenience of the text-based and possibly asynchronous setting, the collection of the dialogues was quite difficult. Recruiting participants is part of the challenge, but the main difficulty was to get them to actually perform the tasks which would not be the case in a face-to-face setting. The use of the tool was also an issue for some participants who would leave the conversation room instead of logging out, which required us to manually invite them back to the room.

Regarding the tasks themselves, the balloon task elicits shorter interactions than the tangram task but with longer turns. This seems coherent with the fact that the balloon task is more argumentative and therefore requires more elaborated turns, while the tangram task only requires to describe the images. Further analysis of the collected dialogues, and of a larger corpus, may help us understand whether the length of the turns is correlated with their content in particular for the tangram task where we could expect meta-topics to be associated with longer turns.

Based on the survey, most of the participants rightfully believed that they were interacting with another human, and only one person reported that their interlocutor was not cooperative at all. However, one participant commented that they felt that their interlocutor may have been a bot, while rating 3/4 when asked if they thought that they were interacting with a human. This suggests that there may be more to investigate regarding how people perceive the other participants in a conversation, and how this perception may influence their behaviour. This question is becoming increasingly relevant as conversational AIs are more and more present in our daily lives. While determining whether our interlocutor was a human or not was almost never a concern a few years ago, a growing number of daily-life interactions now have their share of uncertainty regarding the nature of the other participants, such as spam calls or conversations on social media or online peer-to-peer shopping platforms. This is likely to have an impact on how people behave in conversations, for instance in terms of cooperation.